Active Calculus

2nd Edition

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Acknowledgements

The first edition of this text began as my sabbatical project in the winter semester of 2012, during which I wrote most of the material for the first four chapters. For the sabbatical leave, I am indebted to Grand Valley State University for its support of the project, as well as to my colleagues in the Department of Mathematics and the College of Liberal Arts and Sciences for their endorsement of the project. I'm also grateful to the AIM Open Textbook Initiative[1], sponsored by the American Institute of Mathematics, for their support of open textbooks generally and their support of this one in particular. Since August 2013, *Active Calculus - Single Variable* has been endorsed by the American Institute of Mathematics and its Open Textbook Initiative.

The .html version of the text is possible only because of the amazing work of Rob Beezer and his development of PreTeXt[2] that's been ongoing for more than a decade. The PreTeXt community has also been invaluable: Oscar Levin and Steven Clontz developed the CLI and related tools that make writing and producing the text far easier; Alex Jordan's work made live WeBWorK exercises possible, plus he has contributed some WeBWorK versions of Preview Activities; and Mitch Keller brings wide-ranging expertise across many aspects of PreTeXt work to his role as production editor. Due to PreTeXt, one or more .html versions of the text have been available since 2017.

For the graphics in the text, I'm grateful for the support of David Austin, first through his Python library that uses Bill Casselman's PiScript program[3], and now through his latest fantastic project, PreFigure[4].

David Austin and Steve Schlicker each wrote a chapter to support the completion of the material on integral calculus early in the life of the textbook late in 2012. David wrote Chapter 7 and Steve wrote the version of Chapter 8 that appears in the first edition. The new Chapter in this second edition is primarily my own contribution, but with substantial critical feedback and support from Spencer Bagley and Chrissy Safranski. Chrissy has also authored a considerable number of new WeBWorK exercises and Preview Activities, the latter of which especially support the Runestone implementation of the text. Ted Sundstrom and Steve Schlicker each contributed a large number of exercise and activity solutions and answers.

For the 2018 first edition, Kathy Yoshiwara of the AIM Editorial Board read the entire text

[1]`https://textbooks.aimath.org/`
[2]`pretextbook.org`
[3]`gvsu.edu/s/bi`
[4]`prefigure.org`

and contributed editorial suggestions for every section, making the prose cleaner, more direct, and simply better. Mike Shulman is a long-time user of the first edition and has frequently provided substantive feedback that has improved many aspects of the text. Especially through their in-class work with students and the text, Chrissy, Mike, and Spencer have been invaluable partners in strengthening *Active Calculus* in both its first and second editions.

Over my more than 27 years at GVSU, many of my colleagues have shared with me ideas and resources for teaching calculus. I am particularly indebted to David Austin, Will Dickinson, Paul Fishback, Marcia Frobish, Jon Hodge, Lauren Keough, and Steve Schlicker for their contributions that improved my teaching of and thinking about calculus, including materials that I have modified and used over many different semesters with students. Parts of their ideas can be found throughout this text.

Many other people have written me with comments, suggestions, and corrections, and I am indebted to everyone who has taken the time to share their feedback. Overall, I'm grateful for how the work of these friends and colleagues has made the text so much better.

Finally, I am thankful for all that my students have taught me over the years. Their responses and feedback have helped to shape me as a teacher, and I appreciate their willingness to wholeheartedly engage in the activities and approaches I've tried in class, to let me know how those affect their learning, and to help me learn and grow as an instructor.

Any and all remaining errors or inconsistencies are mine. I will gladly take reader and user feedback[5] to correct them, along with other suggestions to improve the text.

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[5] gvsu.edu/s/0vK

Active Calculus: Our Goals

Several fundamental ideas in calculus are more than 2000 years old. As a formal subdiscipline of mathematics, calculus was first introduced and developed in the late 1600s, with key independent contributions from Sir Isaac Newton and Gottfried Wilhelm Leibniz. The subject has been understood rigorously since the work of Augustin Louis Cauchy and Karl Weierstrass in the mid 1800s when the field of modern analysis was developed. As a body of knowledge, calculus has been completely understood for at least 150 years. The discipline is one of our great human intellectual achievements: among many spectacular ideas, calculus models how objects fall under the forces of gravity and wind resistance, explains how to compute areas and volumes of interesting shapes, enables us to work rigorously with infinitely small and infinitely large quantities, and connects the varying rates at which quantities change to the total change in the quantities themselves.

While each author of a calculus textbook certainly offers their own creative perspective on the subject, it is rarely the case that many of the ideas they present are new. Indeed, the mathematics community broadly agrees on what the main ideas of calculus are, as well as their justification and their importance. In the 21st century and the age of the internet, no one should be required to purchase a calculus text to read, to use for a class, or to find a coherent collection of problems to solve. Calculus belongs to humankind, not any individual author or publishing company. Thus, a primary purpose of this work is to present a calculus text that is *free*. See https://activecalculus.org[6] for links to both the .html and .pdf versions of the text. In addition, instructors who use a calculus text should have the opportunity to download the source files and make modifications that they see fit; thus this text is *open-source*. See GitHub[7] for the PreTeXt source.

In *Active Calculus*, we actively engage students in learning the subject through an activity-driven approach in which the vast majority of the examples are generated by students. Where many texts present a general theory followed by substantial collections of worked examples, we instead pose problems or situations, consider possibilities, and then ask students to investigate and explore. Following most activities, the text usually includes some overall perspective and a brief synopsis of general trends or properties, followed by formal statements of rules or theorems. While we often offer plausibility arguments for such results, rarely do we include formal proofs. It is not the intent of this text for the instructor or author to demonstrate to students that the ideas of calculus are coherent and true, but rather for students to encounter these ideas in a supportive, leading manner that enables them to begin to under-

[6] `activecalculus.org`

[7] `github.com/active-calculus/active-calculus-single-mbx`

stand calculus for themselves. This approach is consistent with the scholarly consensus[8] that calls for students to be interactively engaged in class.

Moreover, this approach is consistent with the following goals:

- To have students engage in an active, inquiry-driven approach, where learners construct solutions and approaches to ideas, with appropriate support through questions posed, hints, and guidance from the instructor and text.
- To build in students intuition for why the main ideas in calculus are natural and true. Often we do this through consideration of the instantaneous position and velocity of a moving object.
- To challenge students to acquire deep, personal understanding of calculus through reading the text and completing preview activities on their own, working on activities in small groups in class, and doing substantial exercises outside of class time.
- To strengthen students' written and oral communicating skills by having them write about and explain aloud the key ideas of calculus.

[8] launchings.blogspot.com/2011/07/the-worst-way-to-teach.html

Features of the Text

Students and instructors alike will find several consistent features in the presentation, including:

Motivating Questions
: Each section begins with 2–3 *motivating questions* that provide impetus for why the following material is of interest to us. One overall goal of each section is to answer each of the motivating questions.

Preview Activities
: Each section of the text begins with a short introduction, followed by a *preview activity*. This brief reading and the preview activity are designed to foreshadow the upcoming ideas in the remainder of the section; both the reading and preview activity are intended to be accessible to students (and completed by them) *in advance* of the class meeting in which the particular section is to be considered.

Activities A typical section in the text has at least three *activities*. These are designed to engage students in an inquiry-based style that encourages them to construct solutions to key examples on their own, working individually or in small groups.

Exercises There are dozens of calculus texts with (collectively) tens of thousands of exercises. Rather than repeat standard and routine exercises in this text, we recommend the use of WeBWorK with its access to the Open Problem Library and around 20,000 calculus problems. In this text, each section includes a small number of anonymous WeBWorK exercises, as well as 3–4 challenging problems per section. The WeBWorK exercises can be completed in the .html version of the text, as this provides students with immediate feedback without penalty. Almost every non-WeBWorK exercise has multiple parts, requires the student to connect several key ideas, and expects that the student will do at least a modest amount of writing to answer the questions and explain their findings. For instructors interested in a more conventional source of exercises, consider the freely available *APEX Calculus* text by Greg Hartmann et al., available from www.apexcalculus.com. In addition, see the Google group for instructors[9] for information about using WeBWorK sets hosted on your own server, in the Runestone version of the text, or alternate free or low-cost options such as Edfinity or MyOpenMath.

[9]activecalculus.org/instructors/

Graphics We strive to demonstrate key fundamental ideas visually, and to encourage students to do the same. Throughout the text, we use full-color[10] graphics to exemplify and magnify key ideas, and to use this graphical perspective alongside both numerical and algebraic representations of calculus.

Links to interactive graphics
Many of the ideas of calculus are best understood dynamically; interactive graphics offer an often ideal format for investigations and demonstrations. Relying primarily on the work of David Austin of Grand Valley State University and Marc Renault of Shippensburg University, each of whom has developed a large library of such (free) interactives for calculus, we frequently point the reader (through active links in the electronic versions of the text) to resources that are relevant for key ideas under consideration.

Summary of Key Ideas
Each section concludes with a summary of the key ideas encountered in the preceding section; this summary normally reflects responses to the motivating questions that began the section.

[10] To keep cost low, the graphics in the print-on-demand version are in black and white. When the text itself refers to color in images, one needs to view the .html or .pdf electronically.

Students! Read this!

This book is different.

The text is available in three different formats: HTML, PDF, and print, each of which is available via links on the landing page at activecalculus.org[11]. The first two formats are free. If you are going to use the book electronically, the best mode is the HTML[12]. The HTML version looks great in any browser, including on a smartphone, and the links are much easier to navigate in HTML than in PDF. Some particular direct suggestions about using the HTML follow among the next few paragraphs; alternatively, you can watch this short video from the author[13]. It is also wise to download and save the PDF, since you can use the PDF offline, while the HTML version requires an internet connection. A print copy of the second edition will be available for purchase starting in January 2026; until that time, only the first edition is available in print (for about $25 via Amazon[14]).

This book is intended to be read sequentially and engaged with, much more than to be used as a lookup reference. For example, each section begins with a short introduction and a Preview Activity; you should read the short introduction and complete the Preview Activity prior to class. Your instructor will likely require you to do this. Most Preview Activities can be completed in 15-20 minutes and are intended to be accessible based on the understanding you have from preceding sections. There are not answers provided to Preview Activities, as these are designed simply to get you thinking about ideas that will be helpful in work on upcoming new material.

As you use the book, think of it as a workbook, not a worked-book. There is a great deal of research that shows people learn better when they interactively engage and struggle with ideas themselves, rather than passively watch others (even though many people think they prefer watching someone else do the work). Thus, instead of reading worked examples or watching an instructor complete examples, you will engage with Activities that prompt you to grapple with concepts and develop deep understanding. You should expect to spend time in class working with peers on Activities and getting feedback from them and from your instructor, and likely to do some additional follow-up work outside of class. It is best if you use a print version of the workbook that you can write in directly; your instructor will direct you to the best way to acquire the workbook, including as a free PDF if you wish to work on an iPad or tablet. Your goal should be to do all of the activities in the relevant sections of the text and keep a careful record of your work. You can find answers to the

[11]activecalculus.org
[12]activecalculus.org/single
[13]gvsu.edu/s/14i
[14]gvsu.edu/s/14n

activities in the back matter.

Each section concludes with a Summary. Reading the Summary after you have read the section and worked the Activities is a good way to find a short list of key ideas that are most essential to take from the section. A good study habit is to write similar summaries in your own words.

At the end of each section, you'll find two types of Exercises. First, there are several anonymous WeBWorK exercises. These are online, interactive exercises that allow you to submit answers for immediate feedback with unlimited attempts without penalty; to submit answers, you have to be using the HTML version of the text (see this short video[15] on the HTML version that includes a WeBWorK demonstration). You should use these exercises as a way to test your understanding of basic ideas in the preceding section. If your institution uses WeBWorK, you may also need to log in to a server as directed by your instructor to complete assigned WeBWorK sets as part of your course grade. The WeBWorK exercises included in this text are ungraded and not connected to any individual account. Following the WeBWorK exercises there are 3-4 additional challenging exercises that are designed to encourage you to connect ideas, investigate new situations, and write about your understanding. There are answers to most of the non-WeBWorK exercises in the back matter.

You can find additional support for your work in learning calculus from the GVSU Math 201 YouTube Channel[16] and GVSU Math 202 YouTube Channel[17] where there are several short video tutorials for each section of the text, numbered in alignment with the textbook sections in the first edition of the text[18]. Math 201 corresponds to Chapters 1-4 and Math 202 to Chapters 5-8; there are about 90 videos for each, totaling more than 180. The numbering of those videos aligns with the first edition of the text.[19]

The best way to be successful in mathematics generally and calculus specifically is to strive to make sense of the main ideas. Human beings make sense of ideas by asking questions, interacting with others, attempting to solve problems, making mistakes, revising attempts, and writing and speaking about our understanding. This text has been designed to help you make sense of calculus; we wish you the very best as you undertake the large and challenging task of doing so.

[15] `gvsu.edu/s/14i`

[16] `gvsu.edu/s/pG`

[17] `gvsu.edu/s/Nb`

[18] `activecalculus.org/single1e/`

[19] In some places in Chapter 3, the numbering in the videos will not exactly correspond to the numbering here in the second edition. In addition, there are not yet videos for Chapter 8 of the second edition.

Instructors! Read this!

This is the Second Edition of Active Calculus. Some of what follows regarding how to use this text mentions differences between the first and second editions. You can see a detailed summary of the biggest changes at https://activecalculus.org/acs1e-vs-acs2e/[20]. The first edition remains available at https://activecalculus.org/single1e/[21].

This book is different. Before you read further, first read "Students! Read this!" More information for instructors can also be found at the home page for the text[22], at activecalculus.org generally[23], and especially on the instructors page[24].

Chapters 1-4 are designed to correspond to what is often called first semester differential calculus. Chapters 5-8 correspond roughly to what is often called second semester integral calculus, including chapters on differential equations and Taylor series.

Among the three formats (HTML, PDF, print), the HTML is optimal for display in class if you have a suitable projector. The HTML is also best for navigation, as links to internal and external references are much more obvious. We recommend saving a downloaded version of the PDF format as a backup in the event you don't have internet access.

It is ideal (even essential) for each student to have a printed version of the Activities Workbook; free PDFs can be found at the home page for the second edition[25] (scroll down in the left panel and look for links to "download PDF 1-4" and "download PDF 5-8"); many instructors use the PDF to have coursepacks printed for students to purchase from their local bookstore. Print versions of the workbook for this second edition of the text will be available for purchase beginning in January 2026.

The text is written so that, on average, one section corresponds to two hours of class meeting time. A typical instructional sequence when starting a new section might look like the following:

- Students complete a Preview Activity in advance of class. Class begins with a short debrief among peers followed by all class discussion. (5-10 minutes)
- Brief all-class discussion to build on the preview activity and set the stage for the next activity. (5-10 minutes)

[20] activecalculus.org/acs1e-vs-acs2e/
[21] activecalculus.org/single1e/
[22] https://activecalculus.org/acs/
[23] https://activecalculus.org/
[24] https://activecalculus.org/instructors/
[25] activecalculus.org/acs2e/

- Students engage with peers to work on and discuss the first activity in the section. (15-20 minutes)
- Brief all-class discussion and possibly lecture to reach closure on the preceding activity, followed by transition to new ideas. (Varies, but 5-15 minutes)
- Possibly begin next activity.

The next hour of class would be similar, but without the Preview Activity to complete prior to class: the principal focus of class will be completing the other 2 activities. Then rinse and repeat.

We recommend that instructors use appropriate incentives to encourage students to complete Preview Activities prior to class. Having these be part of completion-based assignments that count 5% of the semester grade usually results in the vast majority of students completing the vast majority of the previews. If you'd like to see a sample syllabus for how to organize a course and weight various assignments, you can request one via email to the author.

Note that the WeBWorK exercises in the HTML version are anonymous and there's not a way to track students' engagement with them. These are intended to be formative for students and provide them with immediate feedback without penalty. If your institution has a WeBWorK server, we have existing sets of .def files that correspond to the sections in the text; these are available upon request to the author.

In the back matter of the text, you'll find answers to the Activities and to non-WeBWorK Exercises. Instructors interested in solutions to these should contact the author directly.

You and your students can find additional resources in the GVSU Math 201 YouTube Channel[26] and GVSU Math 202 YouTube Channel[27] where there are short video tutorials for every section of the text. Math 201 (GVSU's Calculus I) corresponds to Chapters 1-4 and Math 202 (GVSU's Calculus II) to Chapters 5-8 of the first edition; for this second edition, some of the video numbering in Chapter 3 won't match exactly, and there are not yet videos for Chapter 8[28] of the second edition.

The PreTeXt source code for the text can be found on GitHub[29]. If you find errors in the text or have other suggestions, you can file an issue on GitHub or email the author directly. To engage with instructors who use the text, we maintain a Google group and a blog[30]; you can request to join the Google group via the link at the instructors page[31].

Thank you for considering *Active Calculus* as a resource to help your students develop deep understanding of the subject. I wish you the very best in your work and hope to hear from you.

[26] `gvsu.edu/s/pG`
[27] `gvsu.edu/s/Nb`
[28] `activecalculus.org/single2e/C-8.html`
[29] `github.com/active-calculus/active-calculus-single-mbx`
[30] `activecalculus.org/blog`
[31] `https://activecalculus.org/instructors/`

Contents

CHAPTER 1

Understanding the Derivative

1.1 How do we measure velocity?

Motivating Questions

- How is the average velocity of a moving object connected to the values of its position function?
- How do we interpret the average velocity of an object geometrically on the graph of its position function?
- How is the notion of instantaneous velocity connected to average velocity?

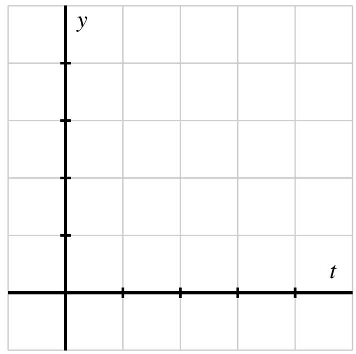
1.1.1 Introduction

Calculus can be viewed broadly as the study of change. A natural and important question to ask about any changing quantity is "how fast is the quantity changing?"

We begin with a simple problem: a ball is tossed straight up in the air. How is the ball moving? Questions like this one are central to our study of differential calculus.

Preview Activity 1.1.1. Suppose that the height s of a ball at time t (in seconds) is given in feet by the formula $s(t) = 64 - 16(t-1)^2$.

(a) Construct a graph of $y = s(t)$ on the time interval $0 \le t \le 3$. Label at least six distinct points on the graph, including the three points showing when the ball was released, when the ball reaches its highest point, and when the ball lands.



(b) Describe the behavior of the ball on the time interval $0 < t < 1$ and on time interval $1 < t < 3$. What occurs at the instant $t = 1$?

(c) Consider the expression

$$AV_{[0.5,1]} = \frac{s(1) - s(0.5)}{1 - 0.5}.$$

Compute the value of $AV_{[0.5,1]}$. What does this value measure on the graph? What does this value tell us about the motion of the ball? In particular, what are the units on $AV_{[0.5,1]}$?

1.1.2 Position and average velocity

Any moving object has a *position* that can be considered a function of *time*. When the motion is along a straight line, the position is given by a single variable, which we denote by $s(t)$. For example, $s(t)$ might give the mile marker of a car traveling on a straight highway at time t in hours. Similarly, the function s described in Preview Activity 1.1.1 is a position function, where position is measured vertically relative to the ground.

On any time interval, a moving object also has an *average velocity*. For example, to compute a car's average velocity we divide the number of miles traveled by the time elapsed, which gives the velocity in *miles per hour*. Similarly, the value of $AV_{[0.5,1]}$ in Preview Activity 1.1.1 gave the average velocity of the ball on the time interval $[0.5, 1]$, measured in feet per second.

In general, we make the following definition:

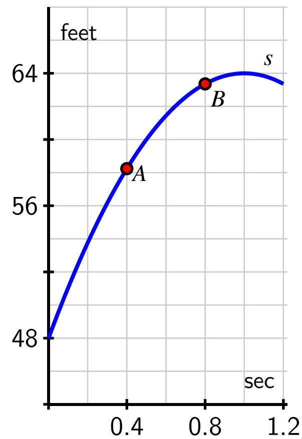
Average Velocity.

For an object moving in a straight line with position function $s(t)$, the *average velocity of the object on the interval from $t = a$ to $t = b$*, denoted $AV_{[a,b]}$, is given by the formula

$$AV_{[a,b]} = \frac{s(b) - s(a)}{b - a}.$$

Note well: the units on $AV_{[a,b]}$ are "units of s per unit of t," such as "miles per hour" or "feet per second."

Activity 1.1.2. The following questions concern the position function given by $s(t) = 64 - 16(t-1)^2$, considered in Preview Activity 1.1.1, and plotted in the given figure.



(a) Compute the average velocity of the ball on each of the following time intervals: $[0.4, 0.8]$, $[0.7, 0.8]$, $[0.79, 0.8]$, $[0.799, 0.8]$, $[0.8, 1.2]$, $[0.8, 0.9]$, $[0.8, 0.81]$, $[0.8, 0.801]$. Include units for each value.

(b) On the graph provided in (a), sketch the line that passes through the points $A = (0.4, s(0.4))$ and $B = (0.8, s(0.8))$. What is the meaning of the slope of this line? In light of this meaning, what is a geometric way to interpret each of the values computed in the preceding question?

(c) Use a graphing utility to plot the graph of $s(t) = 64 - 16(t-1)^2$ on an interval containing the value $t = 0.8$. Then, zoom in repeatedly on the point $(0.8, s(0.8))$. What do you observe about how the graph appears as you view it more and more closely?

(d) What do you conjecture is the velocity of the ball at the instant $t = 0.8$? Why?

1.1.3 Instantaneous Velocity

Whether we are driving a car, riding a bike, or throwing a ball, we have an intuitive sense that a moving object has a velocity at any given moment — a number that measures how fast the object is moving *right now*. For instance, a car's speedometer tells the driver the car's velocity at any given instant. In fact, the velocity on a speedometer is really an average velocity that is computed over a very small time interval. If we let the time interval over which average velocity is computed become shorter and shorter, we can progress from average velocity to *instantaneous* velocity.

Informally, we define the *instantaneous velocity* of a moving object at time $t = a$ to be the value that the average velocity approaches as we take smaller and smaller intervals of time containing $t = a$. We will develop a more formal definition of instantaneous velocity soon, and this definition will be the foundation of much of our work in calculus. For now, it is fine to think of instantaneous velocity as follows: take average velocities on smaller and smaller time intervals around a specific point. If those average velocities approach a single number, then that number will be the instantaneous velocity at that point.

Activity 1.1.3. Each of the following questions concern $s(t) = 64 - 16(t-1)^2$, the position function from Preview Activity 1.1.1.

(a) Compute the average velocity of the ball on the time interval $[1.5, 2]$. What is different between this value and the average velocity on the interval $[0, 0.5]$?

(b) Use appropriate computing technology to estimate the instantaneous velocity of the ball at $t = 1.5$. Likewise, estimate the instantaneous velocity of the ball at $t = 2$. Which value is greater?

(c) How is the sign of the instantaneous velocity of the ball related to its behavior at a given point in time? That is, what does positive instantaneous velocity tell you the ball is doing? Negative instantaneous velocity?

(d) Without doing any computations, what do you expect to be the instantaneous velocity of the ball at $t = 1$? Why?

At this point we have started to see a close connection between average velocity and instantaneous velocity. Each is connected not only to the physical behavior of the moving object but also to the geometric behavior of the graph of the position function. We are interested in computing average velocities on the interval $[a, b]$ for smaller and smaller intervals. In order to make the link between average and instantaneous velocity more formal, think of the value b as $b = a + h$, where h is a small (non-zero) number that is allowed to vary. Then the average velocity of the object on the interval $[a, a+h]$ is

$$AV_{[a,a+h]} = \frac{s(a+h) - s(a)}{h},$$

with the denominator being simply h because $(a+h) - a = h$. Note that when $h < 0$, $AV_{[a,a+h]}$ measures the average velocity on the interval $[a+h, a]$.

To find the instantaneous velocity at $t = a$, we investigate what happens as the value of h approaches zero.

Example 1.1.1 Computing instantaneous velocity for a falling ball. The position function for a falling ball is given by $s(t) = 16 - 16t^2$ (where s is measured in feet and t in seconds).

a. Find an expression for the average velocity of the ball on a time interval of the form $[0.5, 0.5+h]$ where $-0.5 < h < 0.5$ and $h \neq 0$.

b. Use this expression to compute the average velocity on $[0.5, 0.75]$ and $[0.4, 0.5]$.

c. Make a conjecture about the instantaneous velocity at $t = 0.5$.

Solution.

a. We make the assumptions that $-0.5 < h < 0.5$ and $h \neq 0$ because h cannot be zero (otherwise there is no interval on which to compute average velocity) and because the function only makes sense on the time interval $0 \leq t \leq 1$, as this is the duration of time during which the ball is falling. We want to compute and simplify

$$AV_{[0.5,0.5+h]} = \frac{s(0.5+h) - s(0.5)}{(0.5+h) - 0.5}.$$

We start by finding $s(0.5+h)$. To do so, we follow the rule that defines the function s, and then expand and simplify. Doing so,

$$\begin{aligned} s(0.5+h) &= 16 - 16(0.5+h)^2 \\ &= 16 - 16(0.25 + h + h^2) \\ &= 16 - 4 - 16h - 16h^2 \\ &= 12 - 16h - 16h^2. \end{aligned}$$

Now, returning to our computation of the average velocity, we find that

$$\begin{aligned} AV_{[0.5,0.5+h]} &= \frac{s(0.5+h) - s(0.5)}{(0.5+h) - 0.5} \\ &= \frac{(12 - 16h - 16h^2) - (16 - 16(0.5)^2)}{0.5 + h - 0.5} \\ &= \frac{12 - 16h - 16h^2 - 12}{h} \\ &= \frac{-16h - 16h^2}{h} \\ &= \frac{h(-16 - 16h)}{h}. \end{aligned}$$

At this point, we note two things: first, the expression for average velocity clearly depends on h, which it must, since as h changes the average velocity will change. Further, we note that since h can never equal zero, we may remove the common factor of h from the numerator and denominator. It follows that

$$AV_{[0.5,0.5+h]} = -16 - 16h.$$

b. From this expression we can compute the average for any small positive or negative value of h. For instance, to obtain the average velocity on $[0.5, 0.75]$, we let $h = 0.25$, and the average velocity is $AV_{[0.5,0.75]} = -16 - 16(0.25) = -20$ ft/sec. To get the average velocity on $[0.4, 0.5]$, we let $h = -0.1$, and compute the average velocity as

$$AV_{[0.4,0.5]} = -16 - 16(-0.1) = -14.4 \text{ ft/sec}.$$

c. We can even explore what happens to $AV_{[0.5,0.5+h]} = -16 - 16h$ as h gets closer and closer to zero. As h approaches zero, $-16h$ will also approach zero, so it appears that the instantaneous velocity of the ball at $t = 0.5$ should be -16 ft/sec.

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Activity 1.1.4. For the function given by $s(t) = 64 - 16(t-1)^2$ from Preview Activity 1.1.1, find the most simplified expression you can for the average velocity of the ball on the interval $[2, 2+h]$. Use your result to compute the average velocity on $[1.5, 2]$ and to estimate the instantaneous velocity at $t = 2$. Finally, compare your earlier work in Activity 1.1.3.

1.1.4 Summary

- For an object moving in a straight line with position function $s(t)$, the *average velocity of the object on the interval from* $t = a$ *to* $t = b$, denoted $AV_{[a,b]}$, is given by the formula

$$AV_{[a,b]} = \frac{s(b) - s(a)}{b - a}.$$

- The average velocity on $[a, b]$ can be viewed geometrically as the slope of the line between the points $(a, s(a))$ and $(b, s(b))$ on the graph of $y = s(t)$, as shown in Figure 1.1.2.

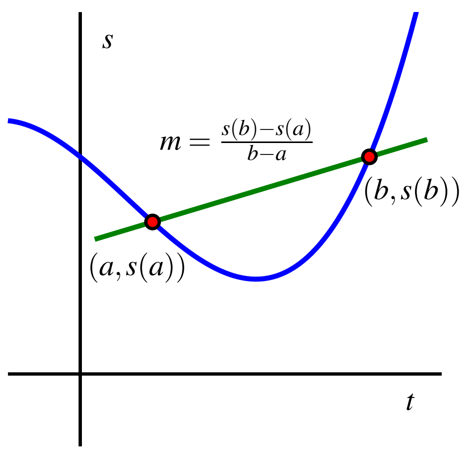


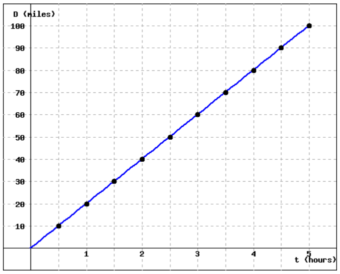
Figure 1.1.2 The graph of position function s together with the line through $(a, s(a))$ and $(b, s(b))$ whose slope is $m = \frac{s(b)-s(a)}{b-a}$. The line's slope is the average rate of change of s on the interval $[a, b]$.

- Given a moving object whose position at time t is given by a function s, the average velocity of the object on the time interval $[a, b]$ is given by $AV_{[a,b]} = \frac{s(b)-s(a)}{b-a}$. Viewing the interval $[a, b]$ as having the form $[a, a+h]$, we equivalently compute average velocity by the formula $AV_{[a,a+h]} = \frac{s(a+h)-s(a)}{h}$.

- The instantaneous velocity of a moving object at a fixed time is estimated by considering average velocities on shorter and shorter time intervals that contain the instant of interest.

1.1.5 Exercises

1. The graph below shows the distance traveled, D (in miles) as a function of time, t (in hours).



(Click on the graph to get a larger version.)

a) For each of the intervals, find the values of ΔD and Δt between the indicated start and end times. Enter your answers in their respective columns in the table below.

Time Interval	ΔD	Δt
t = 1.5 to t = 4	________	________
t = 1 to t = 4	________	________
t = 0.5 to t = 2.5	________	________

b) Based on your results from (a) it follows that the average rate of change of D is constant, it does not depend over which interval of time you choose. What is the constant rate of change of D?

$\frac{\Delta D}{\Delta t}$ = ____________

c) Which of the statements below CORRECTLY explains the significance of your answer to part (b)? Select ALL that apply (more than one may apply).

A It is the average velocity of the car over the first two hours.

B It is the acceleration of the car over the five hour time interval.

C It is how far the car will travel in a half-hour.

D It is the slope of the line.

E It is the total distance the car travels in five hours.

F It represents the car's velocity.

G None of the above.

2. A ball is thrown into the air by a baby alien on a planet in the system of Alpha Centauri with a velocity of 26 ft/s. Its height in feet after t seconds is given by $y = 26t - 13t^2$.

a.) Find the average velocity for the time period beginning when $t_0 = 1$ second and lasting for the given time.

$t = .01$ sec:______________________________

$t = .005$ sec:______________________________

$t = .002$ sec :______________________________

$t = .001$ sec:______________________________

b.) Estimate the instantaneous velocity when $t = 1$.

Answer: __________

NOTE: For the above answers, you may have to enter 6 or 7 significant digits if you are using a calculator.

3. Let $f(x) = 16 - x^2$.

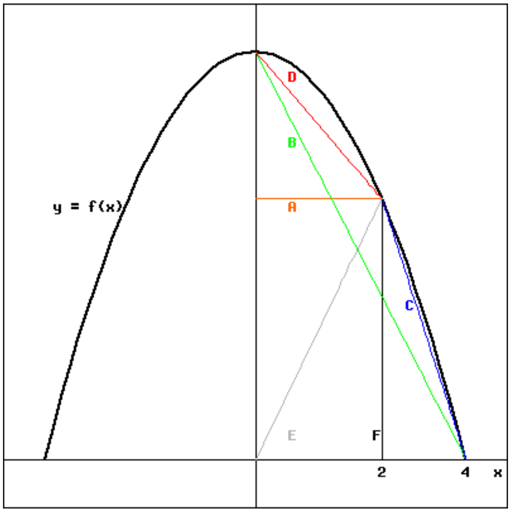
a) Compute each of the following expressions and interpret each as an average rate of change:

(i) $\frac{f(2)-f(0)}{2-0}$ = ____________________

(ii) $\frac{f(4)-f(2)}{4-2}$ = ____________________

(iii) $\frac{f(4)-f(0)}{4-0}$ = ____________________

b) Based on the graph sketched below, match each of your answers in (i) - (iii) with one of the lines labeled A - F. Type the corresponding letter of the line segment next to the appropriate formula. Clearly not all letters will be used.



(click on image to enlarge)

______	$\frac{f(2)-f(0)}{2-0}$
______	$\frac{f(4)-f(2)}{4-2}$
______	$\frac{f(4)-f(0)}{4-0}$

4. The table below gives the average temperature, T, at a depth d, in a borehole in Belleterre, Quebec.

d, depth (m)	T, temp (C°)
25	5.50
50	5.20
75	5.10
100	5.10
125	5.30
150	5.50
175	5.75
200	6.00
225	6.25
250	6.50
275	6.75
300	7.00

Evaluate $\Delta T/\Delta d$ on the following intervals

a) $75 \le d \le 125$ $\Delta T/\Delta d =$ ______________________________

b) $50 \le d \le 175$ $\Delta T/\Delta d =$ ______________________________

c) $125 \le d \le 275$ $\Delta T/\Delta d =$ ______________________________

d) Which of the statements below correctly explains the significance of your answer to part (c)? Select all that apply (more than one may apply).

A 0.0097 is the slope of the graph of at $d = 125$.

B The temperature changes by a total of 0.0097 degrees Celsius when moving from a depth 125 meters to 275 meters.

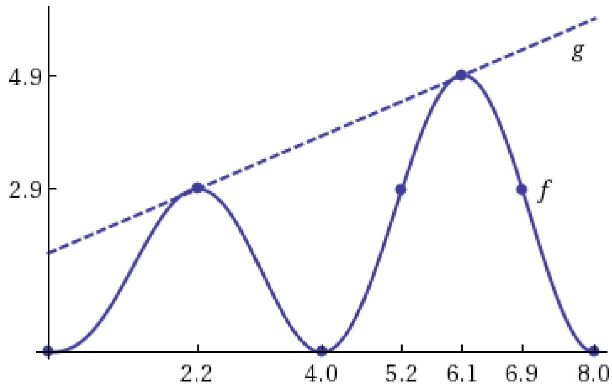
C On average, the temperature is changing at a rate of 0.0097 degrees Celsius per minute over the interval $125 \le d \le 275$.

D Over the interval from 125 meters to 275 meters, the temperature changes on average at a rate of 0.0097 degrees Celsius per meter.

E The temperature is changing at a rate of 0.0097 degrees Celsius per minute when the depth is 125 meters.

F None of the above

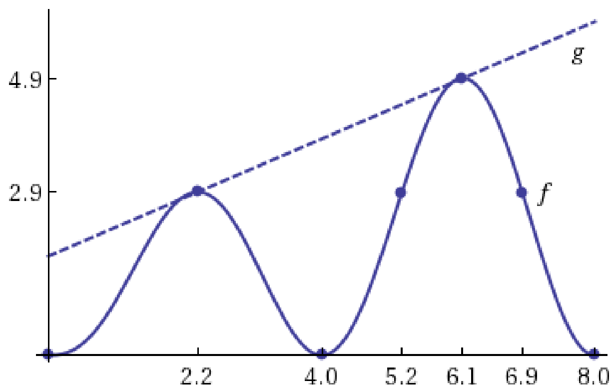
5. Consider the graphs of $f(x)$ and $g(x)$ below:



For each interval given below, decide whether the average rate of change of $f(x)$ or $g(x)$ is greater over that particular interval.

Interval	Which function has GREATER average rate of change?
$2.2 \le x \le 6.9$	(□ ? □ f □ g □ both have an equal rate of change)
$0 \le x \le 2.2$	(□ ? □ f □ g □ both have an equal rate of change)
$2.2 \le x \le 4$	(□ ? □ f □ g □ both have an equal rate of change)
$5.2 \le x \le 6.9$	(□ ? □ f □ g □ both have an equal rate of change)
$5.2 \le x \le 6.1$	(□ ? □ f □ g □ both have an equal rate of change)

6. Consider the graphs of $f(x)$ and $g(x)$ below:



For each interval given below, decide whether the average rate of change of $f(x)$ is positive, negative, or zero over that particular interval.

Interval	Sign of Average Rate of Change of $f(x)$
$2.2 \le x \le 6.1$	(□ ? □ positive □ negative □ zero)
$5.2 \le x \le 6.9$	(□ ? □ positive □ negative □ zero)
$5.2 \le x \le 6.1$	(□ ? □ positive □ negative □ zero)
$2.2 \le x \le 4$	(□ ? □ positive □ negative □ zero)
$0 \le x \le 8$	(□ ? □ positive □ negative □ zero)

7. Consider the function $f(x) = 5x - 4$ and find the following:

a) The average rate of change between the points $(-1, f(-1))$ and $(1, f(1))$.

b) The average rate of change between the points $(a, f(a))$ and $(b, f(b))$.

c) The average rate of change between the points $(x, f(x))$ and $(x + h, f(x + h))$.

8. A car is driven at a speed that is initially high and then decreases, starting at noon. Which of the following could be a graph of the distance the car has traveled as a function of time past noon?

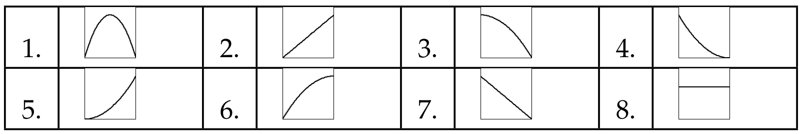


figure _____.

9. A bungee jumper dives from a tower at time $t = 0$. Her height h (measured in feet) at time t (in seconds) is given by the graph in Figure 1.1.3. In this problem, you may base your answers on estimates from the graph or use the fact that the jumper's height function is given by $s(t) = 100\cos(0.75t) \cdot e^{-0.2t} + 100$.

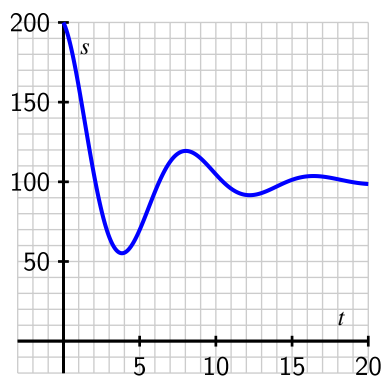


Figure 1.1.3 A bungee jumper's height function.

a. What is the change in vertical position of the bungee jumper between $t = 0$ and $t = 15$?

b. Estimate the jumper's average velocity on each of the following time intervals: $[0, 15]$, $[0, 2]$, $[1, 6]$, and $[8, 10]$. Include units on your answers.

c. On what time interval(s) do you think the bungee jumper achieves her greatest average velocity? Why?

d. Estimate the jumper's instantaneous velocity at $t = 5$. Show your work and explain your reasoning, and include units on your answer.

e. Among the average and instantaneous velocities you computed in earlier questions, which are positive and which are negative? What does negative velocity indicate?

10. A diver leaps from a 3 meter springboard. Their feet leave the board at time $t = 0$, they reaches a maximum height of 4.5 m at $t = 1.1$ seconds, and enter the water at $t = 2.45$. Once in the water, the diver coasts to the bottom of the pool (depth 3.5 m), touches bottom at $t = 7$, rests for one second, and then pushes off the bottom. From there they coast to the surface, and take a breath at $t = 13$.

a. Let $s(t)$ denote the function that gives the height of the diver's feet (in meters) above the water at time t. (Note that the "height" of the bottom of the pool is -3.5 meters.) Sketch a carefully labeled graph of $s(t)$ on the provided axes in Figure 1.1.4. Include scale and units on the vertical axis. Be as detailed as possible.

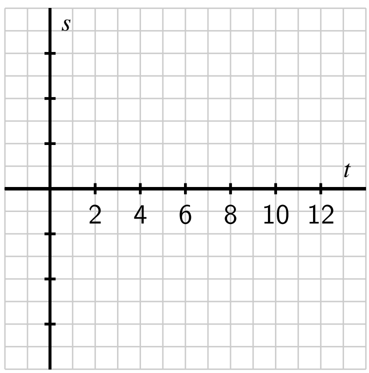


Figure 1.1.4 Axes for plotting $s(t)$ in part (a).

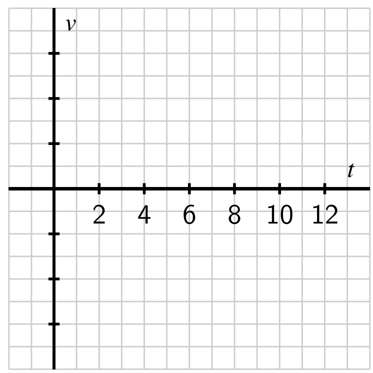


Figure 1.1.5 Axes for plotting $v(t)$ in part (c).

b. Based on your graph in (a), what is the average velocity of the diver between $t = 2.45$ and $t = 7$? Is their average velocity the same on every time interval within $[2.45, 7]$?

c. Let the function $v(t)$ represent the *instantaneous vertical velocity* of the diver at time t (i.e. the rate at which the height function $s(t)$ is changing; note that velocity in the upward direction is positive, while the velocity of a falling object is negative). Based on your understanding of the diver's behavior, as well as your graph of the position function, sketch a carefully labeled graph of $v(t)$ on the axes provided in Figure 1.1.5. Include scale and units on the vertical axis. Write several sentences that explain how you constructed your graph, discussing when you expect $v(t)$ to be zero, positive, negative, relatively large, and relatively small.

d. Is there a connection between the two graphs that you can describe? What can you say about the velocity graph when the height function is increasing? decreasing? Make as many observations as you can.

11. According to the U.S. census, the population of the city of Grand Rapids, MI, was 181,843 in 1980; 189,126 in 1990; and 197,800 in 2000.

a. Between 1980 and 2000, by how many people did the population of Grand Rapids grow?

b. In an average year between 1980 and 2000, by how many people did the population of Grand Rapids grow?

c. Just like we can find the average velocity of a moving body by computing change in position over change in time, we can compute the average rate of change of any function f. In particular, the *average rate of change* of a function f over an interval $[a, b]$ is the quotient

$$\frac{f(b) - f(a)}{b - a}.$$

What does the quantity $\frac{f(b)-f(a)}{b-a}$ measure on the graph of $y = f(x)$ over the interval $[a, b]$?

d. Let $P(t)$ represent the population of Grand Rapids at time t, where t is measured in years from January 1, 1980. What is the average rate of change of P on the interval $t = 0$ to $t = 20$? What are the units on this quantity?

e. If we assume the population of Grand Rapids is growing at a rate of approximately 4% per decade, we can model the population function with the formula

$$P(t) = 181843(1.04)^{t/10}.$$

Use this formula to compute the average rate of change of the population on the intervals $[5, 10]$, $[5, 9]$, $[5, 8]$, $[5, 7]$, and $[5, 6]$.

f. How fast do you think the population of Grand Rapids was changing on January 1, 1985? Said differently, at what rate do you think people were being added to the population of Grand Rapids as of January 1, 1985? How many additional people should the city have expected in the following year? Why?

1.2 The notion of limit

Motivating Questions

- What is the mathematical notion of *limit* and what role do limits play in the study of functions?
- What is the meaning of the notation $\lim_{x\to a} f(x) = L$?
- How do we go about determining the value of the limit of a function at a point?
- How do we manipulate average velocity to compute instantaneous velocity?

1.2.1 Introduction

In Section 1.1 we used a function, $s(t)$, to model the location of a moving object at a given time. Functions can model other interesting phenomena, such as the rate at which an automobile consumes gasoline at a given velocity, or the reaction of a patient to a given dosage of a drug. We can use calculus to study how a function value changes in response to changes in the input variable.

Think about the falling ball whose position function is given by $s(t) = 64 - 16t^2$. Its average velocity on the interval $[1, x]$ is given by

$$AV_{[1,x]} = \frac{s(x) - s(1)}{x - 1} = \frac{(64 - 16x^2) - (64 - 16)}{x - 1} = \frac{16 - 16x^2}{x - 1}.$$

Note that the average velocity is a function of x. That is, the function $g(x) = \frac{16-16x^2}{x-1}$ tells us the average velocity of the ball on the interval from $t = 1$ to $t = x$. To find the instantaneous velocity of the ball when $t = 1$, we need to know what happens to $g(x)$ as x gets closer and closer to 1. But also notice that $g(1)$ is not defined, because it leads to the quotient $0/0$.

This is where the notion of a *limit* comes in. By using a limit, we can investigate the behavior of $g(x)$ as x gets arbitrarily close, but not equal, to 1. We first use the graph of a function to explore points where interesting behavior occurs.

Preview Activity 1.2.1. Suppose that g is the function given by the graph below. Use the graph in Figure 1.2.1 to answer each of the following questions.

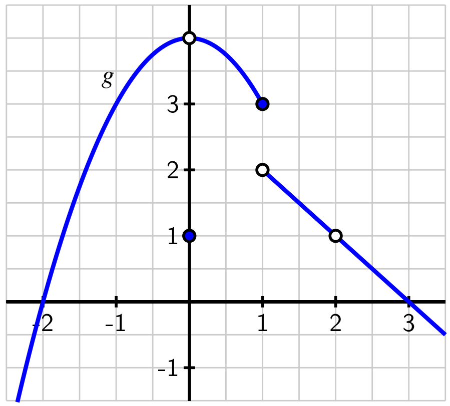


Figure 1.2.1 Graph of $y = g(x)$ for Preview Activity 1.2.1.

(a) Determine the values $g(-2)$, $g(-1)$, $g(0)$, $g(1)$, and $g(2)$, if defined. If the function value is not defined, explain what feature of the graph tells you this.

(b) For each of the values $a = -1$, $a = 0$, and $a = 2$, complete the following sentence: "As x gets closer and closer (but not equal) to a, $g(x)$ gets as close as we want to ________."

(c) What happens as x gets closer and closer (but not equal) to $a = 1$? Does the function $g(x)$ get as close as we would like to a single value?

1.2.2 The Notion of Limit

Limits give us a way to identify a trend in the values of a function as its input variable approaches a particular value of interest. We need a precise understanding of what it means to say "a function f has limit L as x approaches a." To begin, think about a recent example.

In Preview Activity 1.2.1, we saw that as x gets closer and closer (but not equal) to 0, $g(x)$ gets as close as we want to the value 4. At first, this may feel counterintuitive, because the value of $g(0)$ is 1, not 4. But limits describe the behavior of a function *arbitrarily close to* a fixed input, and the value of the function *at* the fixed input does not matter. More formally,[1] we say the following.

Definition 1.2.2 Given a function f, a fixed input $x = a$, and a real number L, we say that f **has limit** L **as** x **approaches** a, and write

$$\lim_{x \to a} f(x) = L$$

provided that we can make $f(x)$ as close to L as we like by taking x sufficiently close (but not

[1]What follows here is not what mathematicians consider the formal definition of a limit. To be completely precise, it is necessary to quantify both what it means to say "as close to L as we like" and "sufficiently close to a." That can be accomplished through what is traditionally called the epsilon-delta definition of limits. The definition presented here is sufficient for the purposes of this text.

equal) to a. If we cannot make $f(x)$ as close to a single value as we would like as x approaches a, then we say that f **does not have a limit as x approaches a.** ◊

Example 1.2.3 For the function g pictured in Figure 1.2.1 from Preview Activity 1.2.1, what can we say about the values of

$$\lim_{x \to -1} g(x), \ \lim_{x \to 0} g(x), \ \lim_{x \to 1} g(x), \text{ and } \lim_{x \to 2} g(x)?$$

Solution. When working from a graph, it suffices to ask if the function *approaches* a single value from each side of the fixed input. The function value *at* the fixed input is irrelevant. Based on the graph of g in Figure 1.2.1, we can conclude that

$$\lim_{x \to -1} g(x) = 3, \ \lim_{x \to 0} g(x) = 4, \text{ and } \lim_{x \to 2} g(x) = 1,$$

because at each of those x-values, the function approaches the same height from both sides of the point of interest.

However, g does not have a limit as $x \to 1$ due to the jump in the graph at $x = 1$. If we approach $x = 1$ from the left, the function values tend to get close to 3, but if we approach $x = 1$ from the right, the function values get close to 2. There is no single number that all of these function values approach. This is why the limit of g does not exist at $x = 1$. ◊

For any function f, there are typically three ways to answer the question "does f have a limit at $x = a$, and if so, what is the limit?" The first is to reason graphically as we have just done with the example from Preview Activity 1.2.1. If we have a formula for $f(x)$, there are two additional possibilities:

1. Evaluate the function at a sequence of inputs that approach a on either side (typically using some sort of computing technology), and ask if the sequence of outputs seems to approach a single value.

2. Use the algebraic form of the function to understand the trend in its output values as the input values approach a.

The first approach produces only an approximation of the value of the limit, while the latter can often be used to determine the limit exactly.

Example 1.2.4 Limits of Two Functions. For each of the following functions, we'd like to know whether or not the function has a limit at the stated a-values. Use both numerical and algebraic approaches to investigate and, if possible, estimate or determine the value of the limit. Compare the results with a careful graph of the function on an interval containing the points of interest.

a. $f(x) = \frac{4-x^2}{x+2}$; $a = -1$, $a = -2$

b. $g(x) = \sin\left(\frac{\pi}{x}\right)$; $a = 3$, $a = 0$

Solution. a. We first construct a graph of f along with tables of values near $a = -1$ and $a = -2$.

x	$f(x)$
-0.9	2.9
-0.99	2.99
-0.999	2.999
-0.9999	2.9999
-1.1	3.1
-1.01	3.01
-1.001	3.001
-1.0001	3.0001

Table 1.2.5 Values of f near $x = -1$.

x	$f(x)$
-1.9	3.9
-1.99	3.99
-1.999	3.999
-1.9999	3.9999
-2.1	4.1
-2.01	4.01
-2.001	4.001
-2.0001	4.0001

Table 1.2.6 Values of f near $x = -2$.

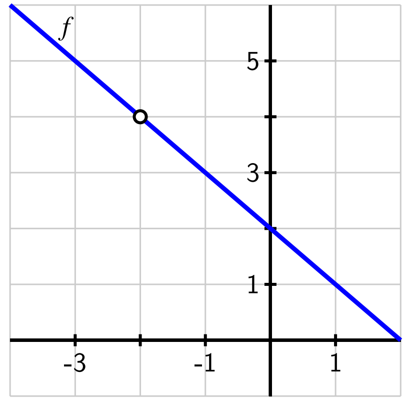


Figure 1.2.7 Plot of $f(x)$ on $[-4, 2]$.

From Table 1.2.5, it appears that we can make f as close as we want to 3 by taking x sufficiently close to -1, which suggests that $\lim_{x\to-1} f(x) = 3$. This is also consistent with the graph of f. To see this a bit more rigorously and from an algebraic point of view, consider the formula for f: $f(x) = \frac{4-x^2}{x+2}$. As $x \to -1$, $(4 - x^2) \to (4 - (-1)^2) = 3$, and $(x + 2) \to (-1 + 2) = 1$, so as $x \to -1$, the numerator of f tends to 3 and the denominator tends to 1, hence $\lim_{x\to-1} f(x) = \frac{3}{1} = 3$.

The situation is more complicated when $x \to -2$, because $f(-2)$ is not defined. If we try to use a similar algebraic argument regarding the numerator and denominator, we observe that as $x \to -2$, $(4 - x^2) \to (4 - (-2)^2) = 0$, and $(x + 2) \to (-2 + 2) = 0$, so as $x \to -2$, the numerator and denominator of f both tend to 0. We call $0/0$ an *indeterminate form*. This tells us that there is more work to do to try to determine the limit's value. From Table 1.2.6 and Figure 1.2.7, it appears that f should have a limit of 4 at $x = -2$.

To see algebraically why this is the case, observe that

$$\begin{aligned}\lim_{x\to-2} f(x) &= \lim_{x\to-2} \frac{4-x^2}{x+2}\\ &= \lim_{x\to-2} \frac{(2-x)(2+x)}{x+2}.\end{aligned}$$

It is important to observe that, since we are taking the limit as $x \to -2$, we are considering x values that are close, but not equal, to -2. Because we never actually allow x to equal -2, the quotient $\frac{2+x}{x+2}$ has value 1 for every possible value of x. Thus, we can simplify the most recent expression above, and find that

$$\lim_{x\to-2} f(x) = \lim_{x\to-2} 2 - x.$$

This limit is now easy to determine, and its value clearly is 4. Thus, from several points of view we've seen that $\lim_{x\to-2} f(x) = 4$.

b. Next we turn to the function g, and construct two tables and a graph.

x	$g(x)$
2.9	0.88351
2.99	0.86777
2.999	0.86620
2.9999	0.86604
3.1	0.84864
3.01	0.86428
3.001	0.86585
3.0001	0.86601

Table 1.2.8 Values of g near $x = 3$.

x	$g(x)$
-0.1	0
-0.01	0
-0.001	0
-0.0001	0
0.1	0
0.01	0
0.001	0
0.0001	0

Table 1.2.9 Values of g near $x = 0$.

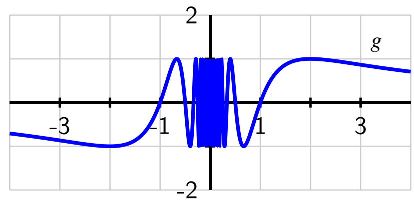


Figure 1.2.10 Plot of $g(x)$ on $[-4, 4]$.

First, as $x \to 3$, it appears from the table values that the function is approaching a number between 0.86601 and 0.86604. From the graph it appears that $g(x) \to g(3)$ as $x \to 3$. The exact value of $g(3) = \sin(\frac{\pi}{3})$ is $\frac{\sqrt{3}}{2}$, which is approximately 0.8660254038. This is convincing evidence that

$$\lim_{x \to 3} g(x) = \frac{\sqrt{3}}{2}.$$

As $x \to 0$, we observe that $\frac{\pi}{x}$ does not behave in an elementary way. When x is positive and approaching zero, we are dividing by smaller and smaller positive values, and $\frac{\pi}{x}$ increases without bound. When x is negative and approaching zero, $\frac{\pi}{x}$ decreases without bound. In this sense, as we get close to $x = 0$, the inputs to the sine function are growing rapidly, and this leads to increasingly rapid oscillations in the graph of g betweem 1 and -1. If we plot the function $g(x) = \sin\left(\frac{\pi}{x}\right)$ with a graphing utility and then zoom in on $x = 0$, we see that the function never settles down to a single value near the origin, which suggests that g does not have a limit at $x = 0$.

How do we reconcile the graph with the righthand table above, which seems to suggest that the limit of g as x approaches 0 may in fact be 0? The data misleads us because of the special nature of the sequence of input values $\{0.1, 0.01, 0.001, \ldots\}$. When we evaluate $g(10^{-k})$, we get $g(10^{-k}) = \sin\left(\frac{\pi}{10^{-k}}\right) = \sin(10^k\pi) = 0$ for each positive integer value of k. But if we take a different sequence of values approaching zero, say $\{0.3, 0.03, 0.003, \ldots\}$, then we find that

$$g(3 \cdot 10^{-k}) = \sin\left(\frac{\pi}{3 \cdot 10^{-k}}\right) = \sin\left(\frac{10^k\pi}{3}\right) = \frac{\sqrt{3}}{2} \approx 0.866025.$$

That sequence of function values suggests that the value of the limit is $\frac{\sqrt{3}}{2}$. Clearly the function cannot have two different values for the limit, so g has no limit as $x \to 0$. ◊

An important lesson to take from Example 1.2.4 is that tables can be misleading when determining the value of a limit. While a table of values is useful for investigating the possible value of a limit, we should also use other tools to confirm the value.

Activity 1.2.2. Estimate the value of each of the following limits by constructing appropriate tables of values. Then determine the exact value of the limit by using algebra to simplify the function. Finally, plot each function on an appropriate interval to check your result visually.

(a) $\lim_{x \to 1} \frac{x^2 - 1}{x - 1}$

(b) $\lim_{x \to 0} \frac{(2 + x)^3 - 8}{x}$

(c) $\lim_{x \to 0} \frac{\sqrt{x + 1} - 1}{x}$

Recall that our primary motivation for considering limits of functions comes from our interest in studying the rate of change of a function. To that end, we close this section by revisiting our previous work with average and instantaneous velocity and highlighting the role that limits play.

1.2.3 Instantaneous Velocity

Suppose that we have a moving object whose position at time t is given by a function s. We know that the average velocity of the object on the time interval $[a, b]$ is $AV_{[a,b]} = \frac{s(b)-s(a)}{b-a}$. We define the *instantaneous velocity* at a to be the limit of average velocity as b approaches a. Note particularly that as $b \to a$, the length of the time interval gets shorter and shorter (while always including a). We will write $IV_{t=a}$ for the instantaneous velocity at $t = a$, and thus

$$IV_{t=a} = \lim_{b \to a} AV_{[a,b]} = \lim_{b \to a} \frac{s(b) - s(a)}{b - a}.$$

Equivalently, if we think of the changing value b as being of the form $b = a + h$, where h is some small number, then we may instead write

$$IV_{t=a} = \lim_{h \to 0} AV_{[a,a+h]} = \lim_{h \to 0} \frac{s(a + h) - s(a)}{h}.$$

Again, the most important idea here is that to compute instantaneous velocity, we take a limit of average velocities as the time interval shrinks.

Activity 1.2.3. Consider a moving object whose position function is given by $s(t) = t^2$, where s is measured in meters and t is measured in minutes.

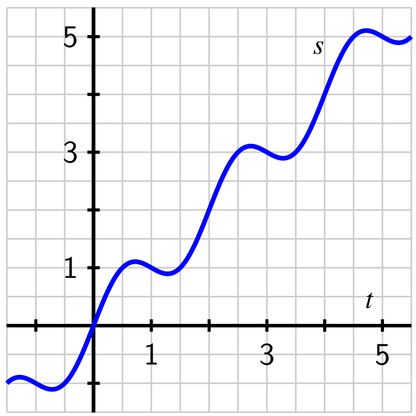
(a) Determine the most simplified expression for the average velocity of the object on the interval $[3, 3 + h]$, where $h > 0$.

(b) Determine the average velocity of the object on the interval $[3, 3.2]$. Include units on your answer.

(c) Determine the instantaneous velocity of the object when $t = 3$. Include units on your answer.

The closing activity of this section asks you to make some connections among average velocity, instantaneous velocity, and slopes of certain lines.

Activity 1.2.4. For the moving object whose position s at time t is given by the graph provided, answer each of the following questions. Assume that s is measured in feet and t is measured in seconds.



(a) Use the graph to estimate the average velocity of the object on each of the following intervals: $[0.5, 1]$, $[1.5, 2.5]$, $[0, 5]$. Draw each line whose slope represents the average velocity you seek.

(b) How could you use average velocities or slopes of lines to estimate the instantaneous velocity of the object at a fixed time?

(c) Use the graph to estimate the instantaneous velocity of the object when $t = 2$. Should this instantaneous velocity at $t = 2$ be greater or less than the average velocity on $[1.5, 2.5]$ that you computed in (a)? Why?

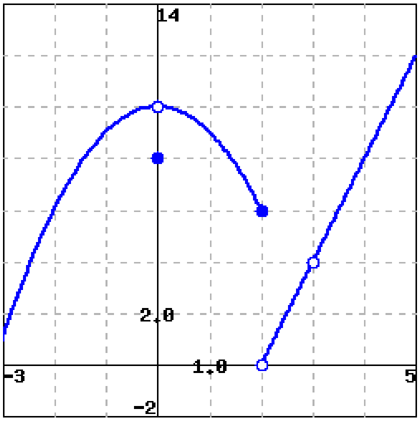
1.2.4 Summary

- Limits enable us to examine trends in function behavior near a specific point. In particular, taking a limit at a given point asks if the function values nearby tend to approach a particular fixed value.
- We read $\lim_{x \to a} f(x) = L$, as "the limit of f as x approaches a is L," which means that we can make the value of $f(x)$ as close to L as we want by taking x sufficiently close (but not equal) to a.

- To find $\lim_{x\to a} f(x)$ for a given value of a and a known function f, we can estimate this value from the graph of f, or we can make a table of function values for x-values that are closer and closer to a. If we want the exact value of the limit, we can work with the function algebraically to understand how different parts of the formula for f change as $x \to a$.
- We find the instantaneous velocity of a moving object at a fixed time by taking the limit of average velocities of the object over shorter and shorter time intervals containing the time of interest.

1.2.5 Exercises

1. Use the figure below, which gives a graph of the function $f(x)$, to give values for the indicated limits. If a limit does not exist, enter ***DNE***.



(a) $\lim\limits_{x\to -2} f(x) =$ ________________

(b) $\lim\limits_{x\to 0} f(x) =$ ________________

(c) $\lim\limits_{x\to 2} f(x) =$ ________________

(d) $\lim\limits_{x\to 3} f(x) =$ ________________

2. Evaluate the limit, if it exists. If a limit *does not exist*, type "DNE".

$$\lim_{t\to -1} (t-2)^3(t^2+1)^3$$

Limit: ____________________

3. Evaluate the limit

$$\lim_{y\to 1} \frac{y^3-y}{y^2-1}.$$

(If the limit does not exist, enter "DNE".)

Limit = ________________________

4. Evaluate the limit, if it exists. If a limit *does not exist*, type "DNE".

$$\lim_{h \to 0} \frac{\sqrt{4+h}-2}{h}$$

Limit: ____________________

5. Use algebra to evalute the following limit.

$\lim_{h\to 0} \frac{(3+h)^2-9}{h}$ = ___________________________________

6. Consider the function $f(x) = \frac{4^x-1}{x}$.

(a) Fill in the following table of values for $f(x)$:

$x =$	-0.1	-0.01	-0.001	-0.0001	0.0001	0.001	0.01	0.1
$f(x) =$	_____	_____	_____	_____	_____	_____	_____	_____

(b) Based on your table of values, what would you expect the limit of $f(x)$ as x approaches zero to be?

$\lim_{x\to 0} \frac{4^x-1}{x}$ = __________

(c) Graph the function to see if it is consistent with your answers to parts (a) and (b). By graphing, find an interval for x near zero such that the difference between your conjectured limit and the value of the function is less than 0.01. In other words, find a window of height 0.02 such that the graph exits the sides of the window and not the top or bottom. What is the window?

__________ $\leq x \leq$ __________,

__________ $\leq y \leq$ __________.

7. Evaluate the limit

$$\lim_{x\to 6}\left(\frac{1}{x+6}+\frac{x-6}{x^2-36}\right)$$

If the limit does not exist enter DNE.

Limit = _________________________

8. Consider the function whose formula is $f(x) = \frac{16-x^4}{x^2-4}$.

a. What is the domain of f?

b. Use a sequence of values of x near $a = 2$ to estimate the value of $\lim_{x\to 2} f(x)$, if you think the limit exists. If you think the limit doesn't exist, explain why.

c. Use algebra to simplify the expression $\frac{16-x^4}{x^2-4}$ and hence work to evaluate $\lim_{x\to 2} f(x)$ exactly, if it exists, or to explain how your work shows the limit fails to exist. Discuss how your findings compare to your results in (b).

d. True or false: $f(2) = -8$. Why?

e. True or false: $\frac{16-x^4}{x^2-4} = -4 - x^2$. Why? How is this equality connected to your work above with the function f?

f. Based on all of your work above, construct an accurate, labeled graph of $y = f(x)$ on the interval $[1, 3]$, and write a sentence that explains what you now know about $\lim_{x\to 2} \frac{16-x^4}{x^2-4}$.

9. Let $g(x) = -\frac{|x+3|}{x+3}$.

a. What is the domain of g?

b. Use a sequence of values near $a = -3$ to estimate the value of $\lim_{x\to -3} g(x)$, if you think the limit exists. If you think the limit doesn't exist, explain why.

c. Use algebra to simplify the expression $\frac{|x+3|}{x+3}$ and hence work to evaluate $\lim_{x\to -3} g(x)$ exactly, if it exists, or to explain how your work shows the limit fails to exist. Discuss how your findings compare to your results in (b). (Hint: $|a| = a$ whenever $a \geq 0$, but $|a| = -a$ whenever $a < 0$.)

d. True or false: $g(-3) = -1$. Why?

e. True or false: $-\frac{|x+3|}{x+3} = -1$. Why? How is this equality connected to your work above with the function g?

f. Based on all of your work above, construct an accurate, labeled graph of $y = g(x)$ on the interval $[-4, -2]$, and write a sentence that explains what you now know about $\lim_{x\to -3} g(x)$.

10. For each of the following prompts, sketch a graph on the provided axes of a function that has the stated properties.

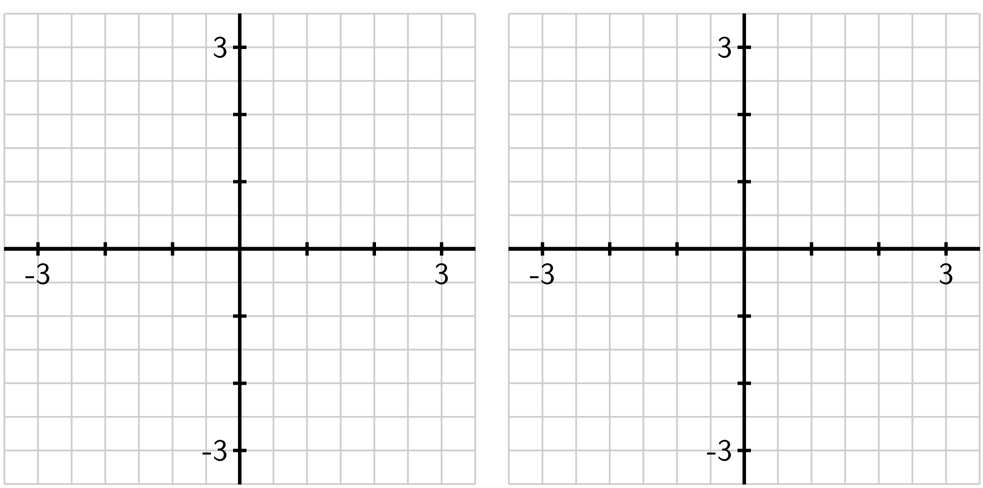


Figure 1.2.11 Axes for plotting $y = f(x)$ in (a) and $y = g(x)$ in (b).

a. $y = f(x)$ such that

- $f(-2) = 2$ and $\lim_{x\to -2} f(x) = 1$
- $f(-1) = 3$ and $\lim_{x\to -1} f(x) = 3$
- $f(1)$ is not defined and $\lim_{x\to 1} f(x) = 0$
- $f(2) = 1$ and $\lim_{x\to 2} f(x)$ does not exist.

b. $y = g(x)$ such that

- $g(-2) = 3$, $g(-1) = -1$, $g(1) = -2$, and $g(2) = 3$
- At $x = -2, -1, 1$ and 2, g has a limit, and its limit equals the value of the function at that point.
- $g(0)$ is not defined and $\lim_{x\to 0} g(x)$ does not exist.

11. A bungee jumper dives from a tower at time $t = 0$. Her height s in feet at time t in seconds is given by $s(t) = 100\cos(0.75t)\cdot e^{-0.2t} + 100$.

a. Write an expression for the average velocity of the bungee jumper on the interval $[1, 1+h]$.

b. Use computing technology to estimate the value of the limit as $h \to 0$ of the quantity you found in (a).

c. What is the meaning of the value of the limit in (b)? What are its units?

1.3 The derivative of a function at a point

Motivating Questions

- How is the average rate of change of a function on a given interval defined, and what does this quantity measure?
- How is the instantaneous rate of change of a function at a particular point defined? How is the instantaneous rate of change linked to average rate of change?
- What is the derivative of a function at a given point? What does this derivative value measure? How do we interpret the derivative value graphically?
- How are limits used formally in the computation of derivatives?

1.3.1 Introduction

The *instantaneous rate of change* of a function is an idea that sits at the foundation of calculus. It is a generalization of the notion of instantaneous velocity and measures how fast a particular function is changing at a given input. If the original function represents the position of a moving object, this instantaneous rate of change is precisely the instantaneous velocity of the object. In other contexts, instantaneous rate of change could measure the number of cells added to a bacteria culture per day, the number of additional gallons of gasoline consumed per mile by increasing a car's velocity one mile per hour, or the number of dollars added to a mortgage payment for each percentage point increase in interest rate. The instantaneous rate of change can also be interpreted geometrically on the function's graph, and this connection is fundamental to many of the main ideas in calculus.

Recall that for a moving object with position function s, its average velocity on the time interval $t = a$ to $t = a + h$ is given by the quotient

$$AV_{[a,a+h]} = \frac{s(a+h) - s(a)}{h}.$$

In a similar way, we make the following definition for an arbitrary function $y = f(x)$.

Definition 1.3.1 For a function f, the **average rate of change** of f on the interval $[a, a+h]$ is given by the value

$$AV_{[a,a+h]} = \frac{f(a+h) - f(a)}{h}.$$

Equivalently, the average rate of change of f on $[a, b]$ is

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}.$$

◊

It is essential to understand how the average rate of change of f on an interval is connected to its graph.

Preview Activity 1.3.1. Suppose that f is the function given by the graph below and that a and $a + h$ are the input values as labeled on the x-axis. Use the graph in Figure 1.3.2 to answer the following questions.

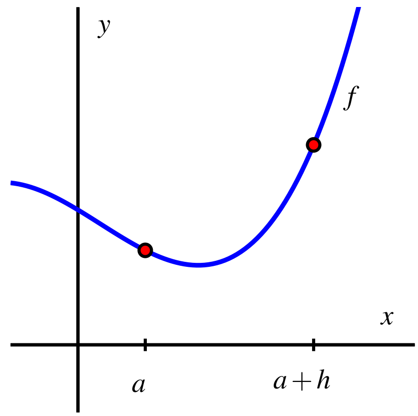


Figure 1.3.2 Plot of $y = f(x)$ for Preview Activity 1.3.1.

(a) Locate and label the points $(a, f(a))$ and $(a + h, f(a + h))$ on the graph.

(b) Construct a right triangle whose hypotenuse is the line segment from $(a, f(a))$ to $(a + h, f(a + h))$. What are the lengths of the respective legs of this triangle?

(c) What is the slope of the line that connects the points $(a, f(a))$ and $(a+h, f(a+h))$?

(d) Write a meaningful sentence that explains how the average rate of change of the function on a given interval and the slope of a related line are connected.

1.3.2 The Derivative of a Function at a Point

Just as we defined instantaneous velocity in terms of average velocity, we now define the instantaneous rate of change of a function at a point in terms of the average rate of change of the function f over related intervals. This instantaneous rate of change of f at a is called "the *derivative* of f at a," and is denoted by $f'(a)$.

Definition 1.3.3 Let f be a function and $x = a$ an input value in the function's domain. We define the **derivative of f with respect to x evaluated at** $x = a$, denoted $f'(a)$, by the formula

$$f'(a) = \lim_{h \to 0} \frac{f(a + h) - f(a)}{h},$$

provided this limit exists. ◊

Aloud, we read the symbol $f'(a)$ as either "f-prime at a" or "the derivative of f evaluated at $x = a$." Much of our work in Chapters 1-3 will be devoted to understanding, computing, applying, and interpreting derivatives. For now, we observe the following important things.

Note 1.3.4

- The derivative of f at the value $x = a$ is defined as the limit of the average rate of change of f on the interval $[a, a+h]$ as $h \to 0$. This limit may not exist, so not every function has a derivative at every point.
- We say that a function is *differentiable* at $x = a$ if it has a derivative at $x = a$.
- The derivative is a generalization of the instantaneous velocity of a position function: if $y = s(t)$ is a position function of a moving body, $s'(a)$ tells us the instantaneous velocity of the body at time $t = a$.
- Because the units on $\frac{f(a+h)-f(a)}{h}$ are "units of $f(x)$ per unit of x," the derivative has these very same units. For instance, if s measures position in feet and t measures time in seconds, the units on $s'(a)$ are feet per second.
- Because the quantity $\frac{f(a+h)-f(a)}{h}$ represents the slope of the line through $(a, f(a))$ and $(a+h, f(a+h))$, when we compute the derivative we are taking the limit of a collection of slopes of lines. Thus, the derivative itself represents the slope of a particularly important line.

We first consider the derivative at a given value as the slope of a certain line.

When we compute an instantaneous rate of change, we allow the interval $[a, a+h]$ to shrink as $h \to 0$. We can think of one endpoint of the interval as "sliding towards" the other. In particular, provided that f has a derivative at $(a, f(a))$, the point $(a+h, f(a+h))$ will approach $(a, f(a))$ as $h \to 0$. Because the process of taking a limit is a dynamic one, it can be helpful to use computing technology to visualize it. One option is an interactive graphic in which the user is able to control the point that is moving. For a helpful collection of examples, consider the work of David Austin[1] of Grand Valley State University, and this particularly relevant example[2]. For interactives that have been built in Geogebra[3], see Marc Renault's library[4] via Shippensburg University, with this example[5] being especially fitting for our work in this section.

Figure 1.3.5 shows a sequence of figures with several different lines through the points $(a, f(a))$ and $(a+h, f(a+h))$, generated by different values of h. These lines (shown in the first three figures in magenta), are often called *secant lines* to the curve $y = f(x)$. A secant line to a curve is simply a line that passes through two points on the curve. For each such line, the slope of the secant line is $m = \frac{f(a+h)-f(a)}{h}$, where the value of h depends on the location of the point we choose. We can see in the diagram how, as $h \to 0$, the secant lines start to approach a single line that passes through the point $(a, f(a))$. If the limit of the slopes of the secant lines exists, we say that the resulting value is the slope of the *tangent line* to the curve. This tangent line (shown in the right-most figure in green) to the graph of

[1] gvsu.edu/s/5r
[2] gvsu.edu/s/5s
[3] You can even consider building your own examples; the fantastic program Geogebra is available for free download and is easy to learn and use.
[4] gvsu.edu/s/5p
[5] gvsu.edu/s/5q

$y = f(x)$ at the point $(a, f(a))$ has slope $m = f'(a)$.

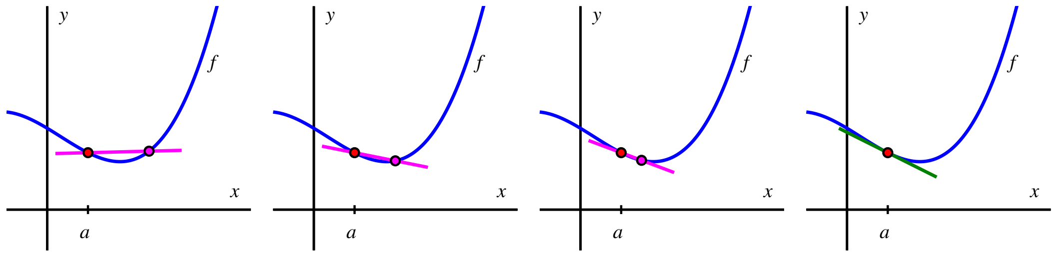


Figure 1.3.5 A sequence of secant lines approaching the tangent line to f at $(a, f(a))$.

If the tangent line at $x = a$ exists, the graph of f looks like a straight line when viewed up close at $(a, f(a))$. In Figure 1.3.6 we combine the four graphs in Figure 1.3.5 into the single one on the left, and zoom in on the box centered at $(a, f(a))$ on the right. Observe how the tangent line sits relative to the curve $y = f(x)$ at $(a, f(a))$ and how closely it resembles the curve near $x = a$.

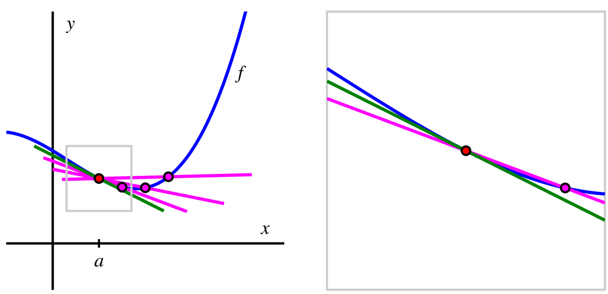


Figure 1.3.6 A sequence of secant lines approaching the tangent line to f at $(a, f(a))$. At right, we zoom in on the point $(a, f(a))$. The slope of the tangent line (in green) to f at $(a, f(a))$ is given by $f'(a)$.

Note 1.3.7 The instantaneous rate of change of f with respect to x at $x = a$, $f'(a)$, also measures the slope of the tangent line to the curve $y = f(x)$ at $(a, f(a))$.

The following example demonstrates several key ideas involving the derivative of a function.

Example 1.3.8 Using the limit definition of the derivative. For the function $f(x) = x - x^2$, use the limit definition of the derivative to compute $f'(2)$. In addition, discuss the meaning of this value and draw a labeled graph that supports your explanation.

Solution. From the limit definition, we know that

$$f'(2) = \lim_{h \to 0} \frac{f(2+h) - f(2)}{h}.$$

Now we use the rule for f, and observe that $f(2) = 2 - 2^2 = -2$ and $f(2+h) = (2+h) - (2+h)^2$. Substituting these values into the limit definition, we have that

$$f'(2) = \lim_{h \to 0} \frac{(2+h) - (2+h)^2 - (-2)}{h}.$$

In order to let $h \to 0$, we must simplify the quotient. Expanding and distributing in the numerator,

$$f'(2) = \lim_{h \to 0} \frac{2 + h - 4 - 4h - h^2 + 2}{h}.$$

Combining like terms, we have

$$f'(2) = \lim_{h \to 0} \frac{-3h - h^2}{h}.$$

Next, we remove a common factor of h in both the numerator and denominator and find that

$$f'(2) = \lim_{h \to 0} (-3 - h).$$

Finally, we are able to take the limit as $h \to 0$, and thus conclude that $f'(2) = -3$. We note that $f'(2)$ is the instantaneous rate of change of f at the point $(2, -2)$. It is also the slope of the tangent line to the graph of $y = x - x^2$ at the point $(2, -2)$. Figure 1.3.9 shows both the function and the line through $(2, -2)$ with slope $m = f'(2) = -3$.

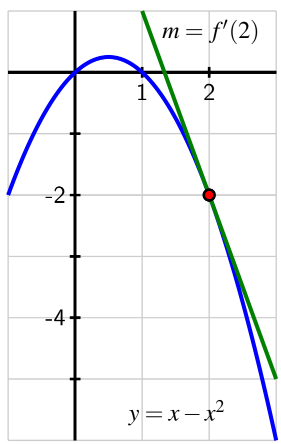


Figure 1.3.9 The tangent line to $y = x - x^2$ at the point $(2, -2)$.

◊

The following activities will help you explore a variety of key ideas related to derivatives.

Activity 1.3.2. Consider the function f whose formula is $f(x) = 3 - 2x$.

(a) What familiar type of function is f? What can you say about the slope of f at every value of x?

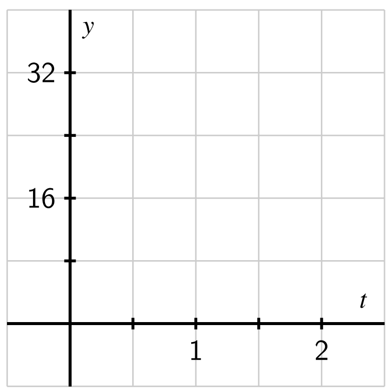
(b) Compute the average rate of change of f on the intervals $[1, 4]$, $[3, 7]$, and $[5, 5 + h]$; simplify each result as much as possible. What do you notice about these quantities?

(c) Use the limit definition of the derivative to compute the exact instantaneous rate of change of f with respect to x at the value $a = 1$. That is, compute $f'(1)$ using the limit definition. Show your work. Is your result surprising?

(d) Without doing any additional computations, what are the values of $f'(2)$, $f'(\pi)$, and $f'(-\sqrt{2})$? Why?

Activity 1.3.3. A water balloon is tossed vertically in the air from a window. The balloon's height in feet at time t in seconds after being launched is given by $s(t) = -16t^2 + 16t + 32$. Use this function to respond to each of the following questions.

(a) Sketch an accurate, labeled graph of s on the axes provided. You should be able to do this without using computing technology.



(b) Compute the average rate of change of s on the time interval $[1, 2]$. Include units on your answer and write one sentence to explain the meaning of the value you found.

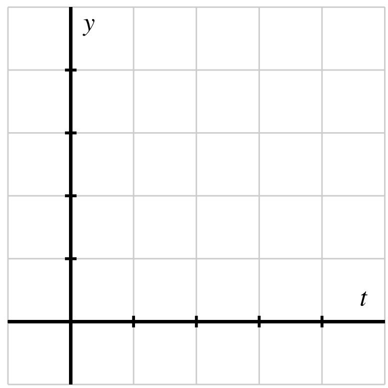
(c) Use the limit definition to compute the instantaneous rate of change of s with respect to time, t, at the instant $a = 1$. Show your work using proper notation, include units on your answer, and write one sentence to explain the meaning of the value you found.

(d) On your graph in (a), sketch two lines: one whose slope represents the average rate of change of s on $[1, 2]$, the other whose slope represents the instantaneous rate of change of s at the instant $a = 1$. Label each line clearly.

(e) For what values of a do you expect $s'(a)$ to be positive? Why? Answer the same questions when "positive" is replaced by "negative" and "zero."

Activity 1.3.4. A rapidly growing city in Arizona has its population P at time t, where t is the number of decades after the year 2010, modeled by the formula $P(t) = 25000e^{t/5}$. Use this function to respond to the following questions.

(a) Sketch an accurate graph of P for $t = 0$ to $t = 5$ on the axes provided. Label the scale on the axes carefully.



(b) Compute the average rate of change of P between 2030 and 2050. Include units on your answer and write one sentence to explain the meaning (in everyday language) of the value you found.

(c) Use the limit definition to write an expression for the instantaneous rate of change of P with respect to time, t, at the instant $a = 2$. Explain why this limit is difficult to evaluate exactly.

(d) Estimate the limit in (c) for the instantaneous rate of change of P at the instant $a = 2$ by using several small h values. Once you have determined an accurate estimate of $P'(2)$, include units on your answer, and write one sentence (using everyday language) to explain the meaning of the value you found.

(e) On your graph, sketch two lines: one whose slope represents the average rate of change of P on $[2, 4]$, the other whose slope represents the instantaneous rate of change of P at the instant $a = 2$.

(f) In a carefully-worded sentence, describe the behavior of $P'(a)$ as a increases in value. What does this reflect about the behavior of the given function P?

1.3.3 Summary

- The average rate of change of a function f on the interval $[a, b]$ is $AV_{[a,b]} = \frac{f(b)-f(a)}{b-a}$. The units on the average rate of change are "units of $f(x)$ per unit of x", and the numerical value of the average rate of change represents the slope of the secant line between the points $(a, f(a))$ and $(b, f(b))$ on the graph of $y = f(x)$. If we view the interval as being

$[a, a+h]$ instead of $[a, b]$, the meaning is still the same, but the average rate of change is now computed by $AV_{[a,b]} = \frac{f(a+h)-f(a)}{h}$.

- The instantaneous rate of change with respect to x of a function f at a value $x = a$ is denoted $f'(a)$ (read "the derivative of f evaluated at a" or "f-prime at a") and is defined by the formula

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h},$$

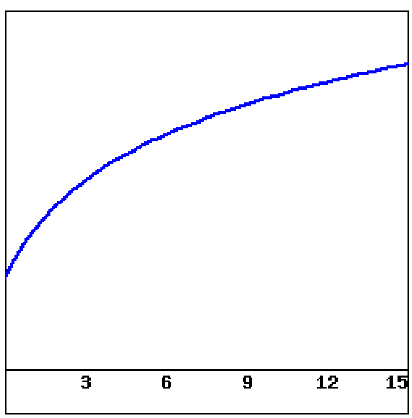
provided the limit exists. Note particularly that the instantaneous rate of change at $x = a$ is the limit of the average rate of change on $[a, a+h]$ as $h \to 0$, and that its units are also "units of $f(x)$ per unit of x".

- Provided the derivative $f'(a)$ exists, its value tells us the instantaneous rate of change of f with respect to x at $x = a$, which geometrically is the slope of the tangent line to the curve $y = f(x)$ at the point $(a, f(a))$. We even say that $f'(a)$ is the "slope of the curve" at the point $(a, f(a))$.

- Limits allow us to move from the rate of change over an interval to the rate of change at a single point.

1.3.4 Exercises

1. Consider the graph of the function $f(x)$ shown below.



Using this graph, for each of the following pairs of numbers decide which is larger. *Be sure that you can explain your answer.*

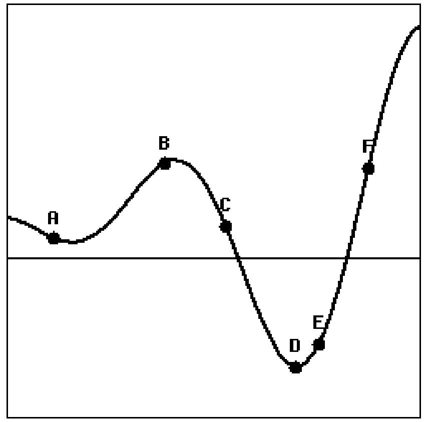
A. $f(9)$ (□ ? □ < □ = □ >) $f(12)$

B. $f(9) - f(6)$ (□ ? □ < □ = □ >) $f(6) - f(3)$

C. $\frac{f(6)-f(3)}{6-3}$ (□ ? □ < □ = □ >) $\frac{f(9)-f(3)}{9-3}$

D. $f'(3)$ (□ ? □ < □ = □ >) $f'(12)$

2. Match the points labeled on the curve below with the given slopes in the following table.



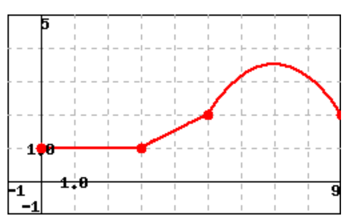
slope	-2
label	(□ ? □ A □ B □ C □ D □ E □ F)
slope	-1/2
label	(□ ? □ A □ B □ C □ D □ E □ F)
slope	0
label	(□ ? □ A □ B □ C □ D □ E □ F)
slope	1/2
label	(□ ? □ A □ B □ C □ D □ E □ F)
slope	2
label	(□ ? □ A □ B □ C □ D □ E □ F)
slope	3
label	(□ ? □ A □ B □ C □ D □ E □ F)

3. If $f(x) = \frac{4}{x^2}$, find $f'(3)$, using the definition of derivative.

$f'(3)$ is the limit as $h \to$ ____________________ of the expression

The value of this limit is ____________________

4. Let $f(x)$ be the function whose graph is shown below.



Determine $f'(a)$ for $a = 1, 2, 4, 7$.

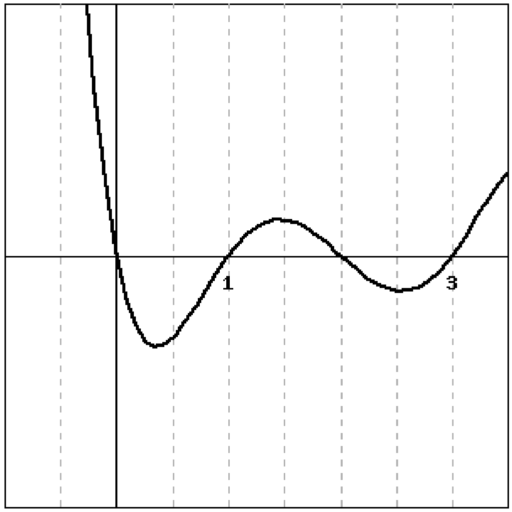
$f'(1) =$ ____________________

$f'(2) =$ ____________________

$f'(4) =$ ____________________

$f'(7) =$ ____________________

5. Consider the function $y = f(x)$ graphed below.



Give the x-coordinate of a point where:

A. the derivative of the function is positive: $x =$ _____

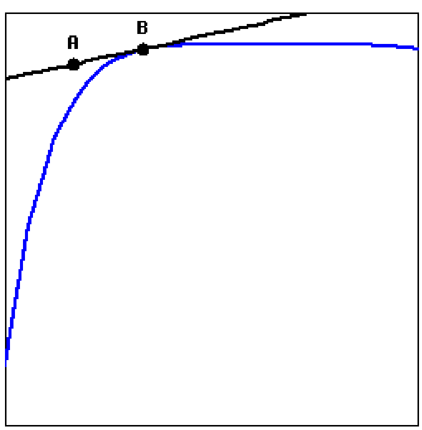
B. the value of the function is positive: $x =$ _____

C. the derivative of the function is largest: $x =$ _____

D. the derivative of the function is zero: $x =$ _____

E. the derivative of the function is approximately the same as the derivative at $x = 2.25$ (be sure that you give a point that is distinct from $x = 2.25$!): $x =$ _____

6. The figure below shows a function $g(x)$ and its tangent line at the point $B = (5.4, 2)$. If the point A on the tangent line is $(5.36, 1.96)$, fill in the blanks below to complete the statements about the function g at the point B.



$g($ _____ $) =$ ________

$g'($ _____ $) =$ ________

7. Let $f(x) = x^2 - 5x$.

(A) Find the slope of the secant line joining $(2, f(2))$ and $(7, f(7))$.

Slope of secant line = ______________________

(B) Find the slope of the secant line joining $(5, f(5))$ and $(5 + h, f(5 + h))$.

Slope of secant line = ______________________

(C) Find the slope of the tangent line at $(5, f(5))$.

Slope of tangent line = ______________________

(D) Find the equation of the tangent line at $(5, f(5))$.

$y =$ ______________________

8. Find $f'(a)$ for

$$f(x) = 1 + 12x - 12x^2.$$

$f'(a) =$ ______________________

9. Find $f'(a)$ for

$$f(x) = \frac{3}{\sqrt{7 - 5x}}.$$

$f'(a) =$ ______________________

10. Consider the graph of $y = f(x)$ provided in Figure 1.3.10.

a. On the graph of $y = f(x)$, sketch and label the following quantities:

- the secant line to $y = f(x)$ on the interval $[-3, -1]$ and the secant line to $y = f(x)$ on the interval $[0, 2]$.
- the tangent line to $y = f(x)$ at $x = -3$ and the tangent line to $y = f(x)$ at $x = 0$.

b. What is the approximate value of the average rate of change of f on $[-3, -1]$? On $[0, 2]$? How are these values related to your work in (a)?

c. What is the approximate value of the instantaneous rate of change of f at $x = -3$? At $x = 0$? How are these values related to your work in (a)?

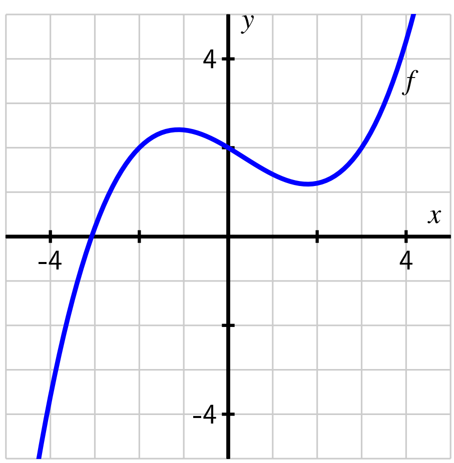


Figure 1.3.10 Plot of $y = f(x)$.

11. For each of the following prompts, sketch a graph on the provided axes in Figure 1.3.11 of a function that has the stated properties.

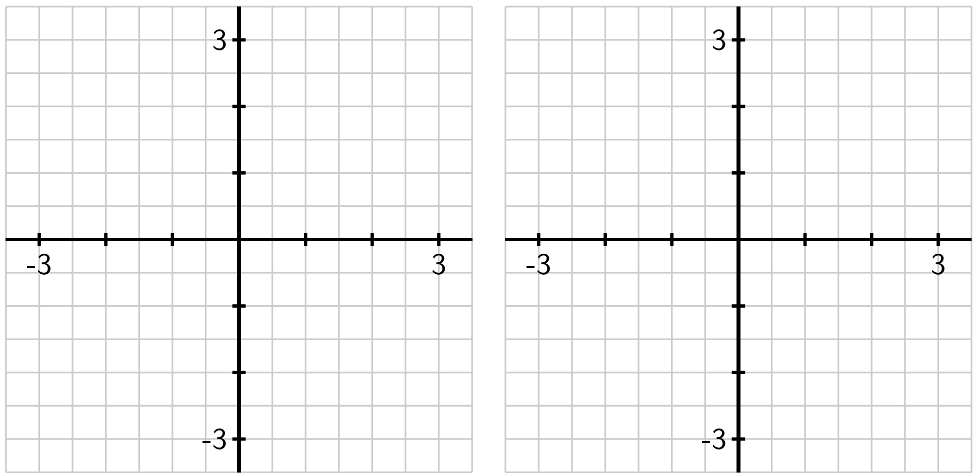


Figure 1.3.11 Axes for plotting $y = f(x)$ in (a) and $y = g(x)$ in (b).

a. $y = f(x)$ such that

- the average rate of change of f on $[-3, 0]$ is -2 and the average rate of change of f on $[1, 3]$ is 0.5, and
- the instantaneous rate of change of f at $x = -1$ is -1 and the instantaneous rate of change of f at $x = 2$ is 1.

b. $y = g(x)$ such that

- $\frac{g(3)-g(-2)}{5} = 0$ and $\frac{g(1)-g(-1)}{2} = -1$, and
- $g'(2) = 1$ and $g'(-1) = 0$

12. Suppose that the population, P, of China (in billions) can be approximated by the function $P(t) = 1.15(1.014)^t$ where t is the number of years since the start of 1993.

a. According to the model, what was the total change in the population of China between January 1, 1993 and January 1, 2000? What will be the average rate of change of the population over this time period? Is this average rate of change greater or less than the instantaneous rate of change of the population on January 1, 2000? Explain and justify, being sure to include proper units on all your answers.

b. According to the model, what is the average rate of change of the population of China in the ten-year period starting on January 1, 2012?

c. Write an expression involving limits that, if evaluated, would give the exact instantaneous rate of change of the population on today's date. Then estimate the value of this limit (discuss how you chose to do so) and explain the meaning (including units) of the value you have found.

d. Find an equation for the tangent line to the function $y = P(t)$ at the point where the t-value is given by today's date.

13. The goal of this problem is to compute the value of the derivative at a point for several different functions, where for each one we do so in three different ways, and then to compare the results to see that each produces the same value.

For each of the following functions, use the limit definition of the derivative to compute the value of $f'(a)$ using three different approaches: strive to use the algebraic approach first (to compute the limit exactly), then test your result using numerical evidence (with small values of h), and finally plot the graph of $y = f(x)$ near $(a, f(a))$ along with the appropriate tangent line to estimate the value of $f'(a)$ visually. Compare your findings among all three approaches; if you are unable to complete the algebraic approach, still work numerically and graphically.

a. $f(x) = x^2 - 3x$, $a = 2$

b. $f(x) = \frac{1}{x}$, $a = 1$

c. $f(x) = \sqrt{x}$, $a = 1$

d. $f(x) = 2 - |x - 1|$, $a = 1$

e. $f(x) = \sin(x)$, $a = \frac{\pi}{2}$

1.4 The derivative function

Motivating Questions

- How does the limit definition of the derivative of a function f lead to an entirely new (but related) function f'?
- What is the difference between writing $f'(a)$ and $f'(x)$?
- How is the graph of the derivative function $f'(x)$ related to the graph of $f(x)$?
- What are some examples of functions f for which f' is not defined at one or more points?

1.4.1 Introduction

We now know that the instantaneous rate of change of a function $f(x)$ at $x = a$, or equivalently the slope of the tangent line to the graph of $y = f(x)$ at $x = a$, is given by the value $f'(a)$. In all of our examples so far, we have identified a particular value of a as our point of interest: $a = 1$, $a = 3$, etc. But it is not hard to imagine that we will often be interested in the derivative value for more than just one a-value, and possibly for many of them. In this section, we explore how we can move from computing the derivative at a single point to computing a formula for $f'(a)$ at any point a. Indeed, the process of "taking the derivative" generates a new function, denoted by $f'(x)$, derived from the original function $f(x)$.

Preview Activity 1.4.1. Consider the function $f(x) = 4x - x^2$.

(a) Use the limit definition to compute each of the following derivative values: $f'(0)$, $f'(1)$, $f'(2)$, and $f'(3)$.

(b) Observe that the work to find $f'(a)$ is the same, regardless of the value of a. Based on your work in (a), what do you conjecture is the value of $f'(4)$? How about $f'(5)$? (***Note***: you should *not* use the limit definition of the derivative to find either value.)

(c) Conjecture a formula for $f'(a)$ that depends only on the value a. That is, in the same way that we have a formula for $f(x)$ (recall $f(x) = 4x - x^2$), see if you can use your work above to guess a formula for $f'(a)$ in terms of a.

1.4.2 How the derivative is itself a function

In your work in Preview Activity 1.4.1 with $f(x) = 4x - x^2$, you may have found several patterns. One comes from observing that $f'(0) = 4$, $f'(1) = 2$, $f'(2) = 0$, and $f'(3) = -2$. That sequence of values leads us naturally to conjecture that $f'(4) = -4$ and $f'(5) = -6$. We also observe that the particular value of a has very little effect on the process of computing the value of the derivative through the limit definition. To see this more clearly, we compute $f'(a)$, where a represents a number to be named later. Following the now standard process of using the limit definition of the derivative,

$$\begin{aligned} f'(a) &= \lim_{h\to 0} \frac{f(a+h)-f(a)}{h} = \lim_{h\to 0} \frac{4(a+h)-(a+h)^2-(4a-a^2)}{h} \\ &= \lim_{h\to 0} \frac{4a+4h-a^2-2ha-h^2-4a+a^2}{h} = \lim_{h\to 0} \frac{4h-2ha-h^2}{h} \\ &= \lim_{h\to 0} \frac{h(4-2a-h)}{h} = \lim_{h\to 0}(4-2a-h). \end{aligned}$$

Here we observe that neither 4 nor $2a$ depend on the value of h, so as $h \to 0$, $(4-2a-h) \to (4-2a)$. Thus, $f'(a) = 4-2a$.

This result is consistent with the specific values we found above: e.g., $f'(3) = 4-2(3) = -2$. And indeed, our work confirms that the value of a has almost no bearing on the process of computing the derivative. We note further that the letter being used is immaterial: whether we call it a, x, or anything else, the derivative at a given value is simply given by "4 minus 2 times the value." We choose to use x for consistency with the original function given by $y = f(x)$, as well as for the purpose of graphing the derivative function. For the function $f(x) = 4x - x^2$, it follows that $f'(x) = 4-2x$.

Because the value of the derivative function is linked to the graph of the original function, it makes sense to look at both of these functions plotted on the same domain.

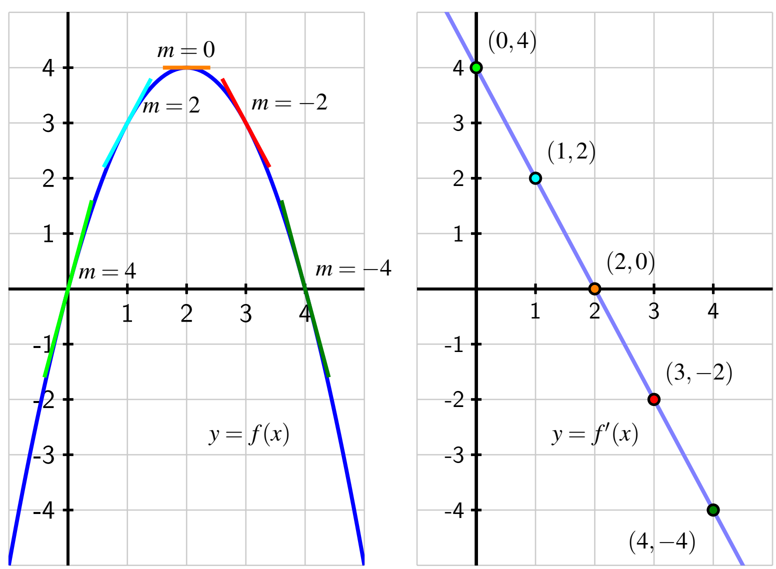


Figure 1.4.1 The graphs of $f(x) = 4x - x^2$ (at left) and $f'(x) = 4-2x$ (at right). Slopes on the graph of f correspond to heights on the graph of f'.

In Figure 1.4.1, on the left we show a plot of $f(x) = 4x - x^2$ together with a selection of tangent lines at the points we've considered above. On the right, we show a plot of $f'(x) = 4-2x$

with emphasis on the heights of the derivative graph at the same selection of points. Notice the connection between colors in the left and right graphs: the green tangent line on the original graph is tied to the green point on the right graph in the following way: *the slope of the tangent line* at a point on the lefthand graph is the same as the *height* at the corresponding point on the righthand graph. That is, at each respective value of x, the slope of the tangent line to the original function is the same as the height of the derivative function. Do note, however, that the units on the vertical axes are different: in the left graph, the vertical units are simply the output units of f. On the righthand graph of $y = f'(x)$, the units on the vertical axis are units of f per unit of x.

An excellent way to explore how the graph of $f(x)$ generates the graph of $f'(x)$ is through an interactive graphic. See, for instance, gvsu.edu/s/5C or gvsu.edu/s/5D, via the sites of David Austin[1] and Marc Renault[2].

In Section 1.3 when we first defined the derivative, we wrote the definition in terms of a value a to find $f'(a)$. As we have seen above, the letter a is merely a placeholder, and it often makes more sense to use x instead. For the record, here we restate the definition of the derivative.

Definition 1.4.2 Let f be a function and x an input value in the function's domain. We define the **derivative of** f, a new function called f', by the formula $f'(x) = \lim_{h \to 0} \frac{f(x+h)-f(x)}{h}$, provided this limit exists. ◊

We now have two different natural questions we can ask about the derivative function:

1. given a graph of $y = f(x)$, how does this graph lead to the graph of the derivative function $y = f'(x)$? and

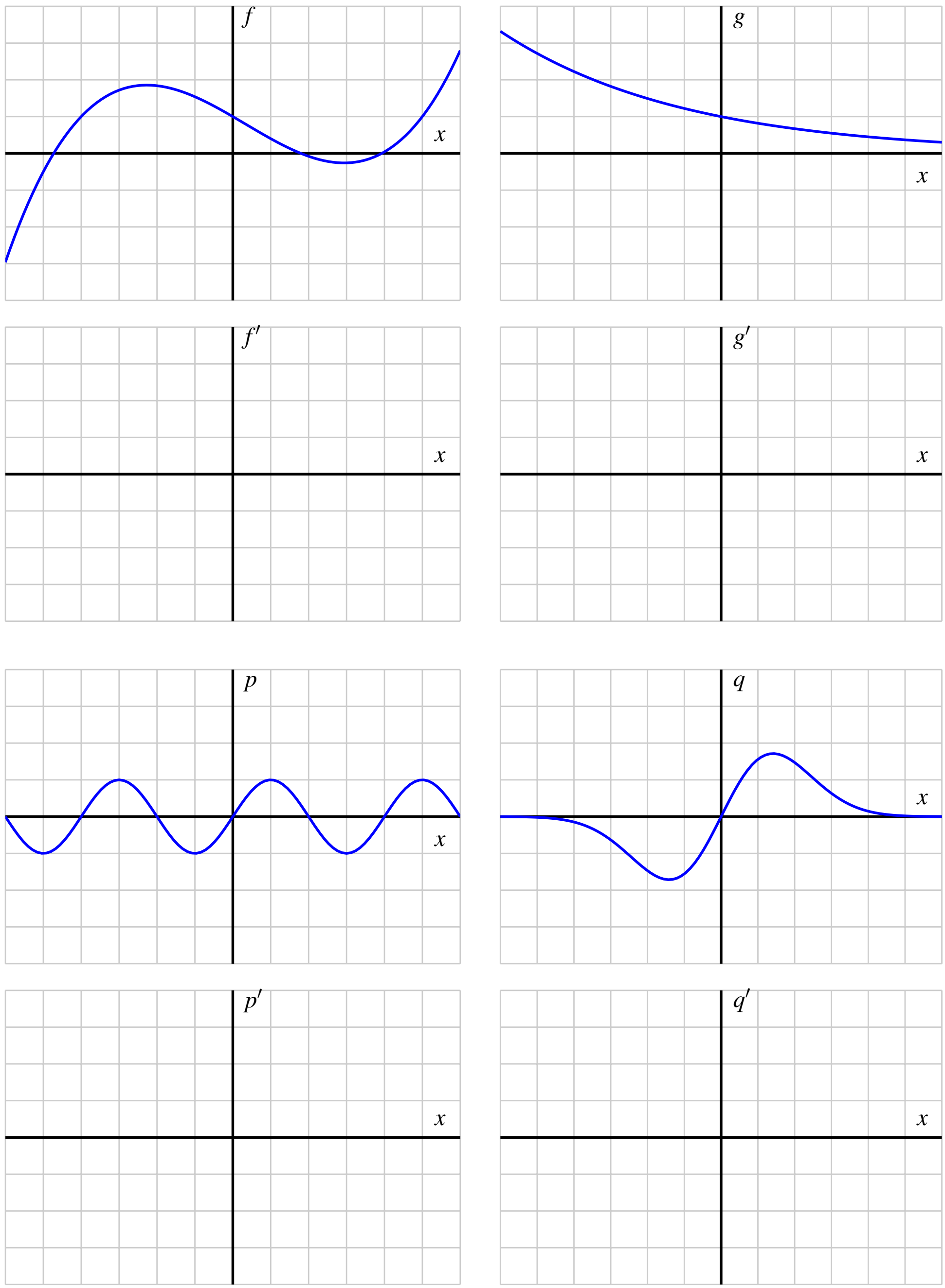
2. given a formula for $y = f(x)$, how does the limit definition of derivative generate a formula for $y = f'(x)$?

Both of these issues are explored in the following activities.

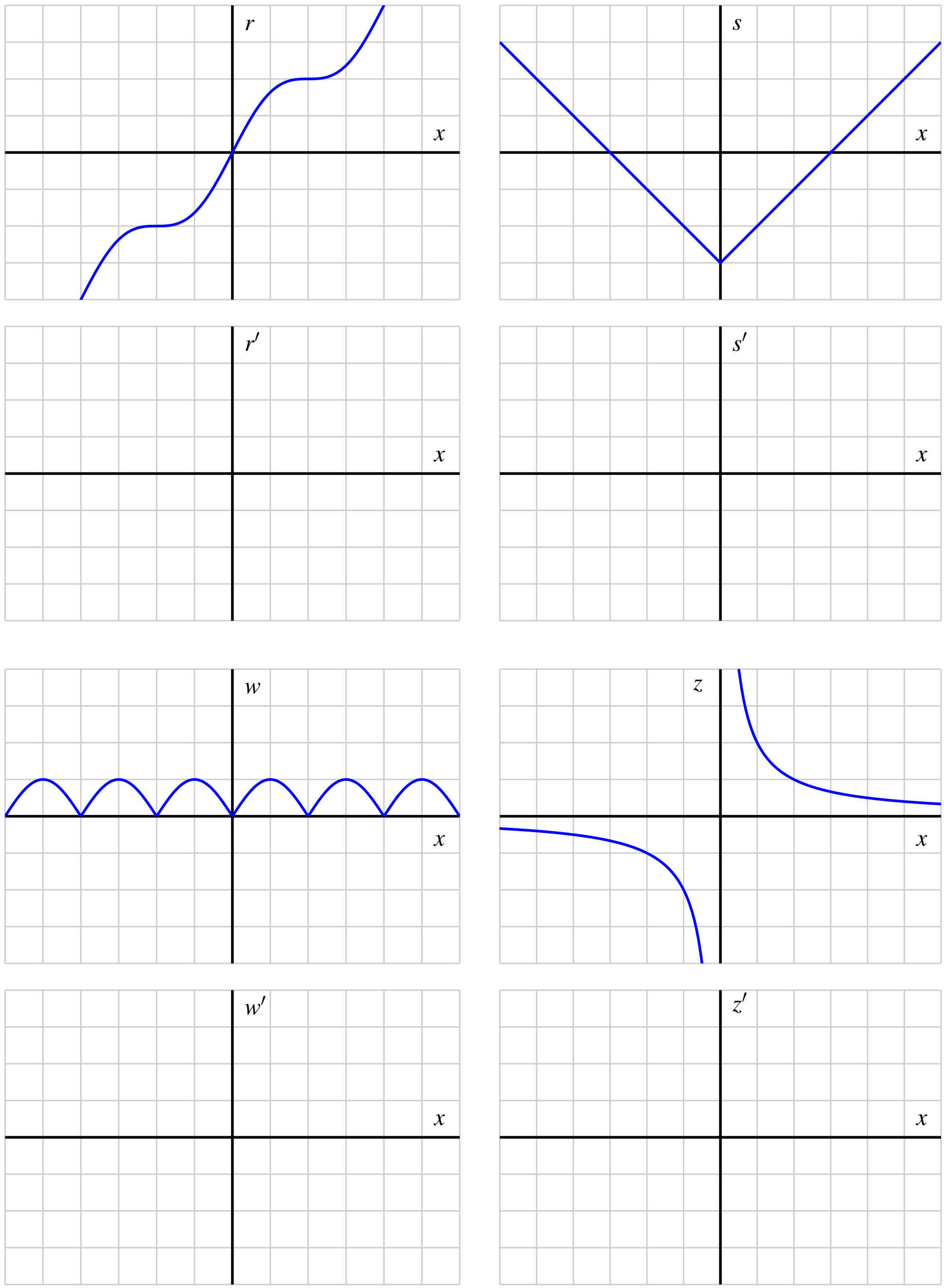
Activity 1.4.2. For each given graph of $y = f(x)$, sketch an approximate graph of its derivative function, $y = f'(x)$, on the axes immediately below. The scale of the grid for the graph of f is 1×1; assume the horizontal scale of the grid for the graph of f' is identical to that for f. If necessary, adjust and label the vertical scale on the axes for f'.

When you are finished with all 8 graphs, write several sentences that describe your overall process for sketching the graph of the derivative function, given the graph the original function. What are the values of the derivative function that you tend to identify first? What do you do thereafter? How do key traits of the graph of the derivative function exemplify properties of the graph of the original function?

[1] gvsu.edu/s/5r
[2] gvsu.edu/s/5p



f
x
g
x
f'
x
g'
x
p
x
q
x
p'
x
q'
x



For a dynamic investigation that allows you to experiment with graphing f' when given the graph of f, see this interactive by Marc Renault[3].

Now, recall the opening example of this section: we began with the function $y = f(x) = 4x -$

[3] gvsu.edu/s/8y

x^2 and used the limit definition of the derivative to show that $f'(a) = 4 - 2a$, or equivalently that $f'(x) = 4-2x$. We subsequently graphed the functions f and f' as shown in Figure 1.4.1. Following Activity 1.4.2, we now understand that we could have constructed a fairly accurate graph of $f'(x)$ *without* knowing a formula for either f or f'. At the same time, it is useful to know a formula for the derivative function whenever it is possible to find one.

In the next activity, we further explore the more algebraic approach to finding $f'(x)$: given a formula for $y = f(x)$, the limit definition of the derivative will be used to develop a formula for $f'(x)$.

Activity 1.4.3. For each of the listed functions, determine a formula for the derivative function. For the first two, determine the formula for the derivative by thinking about the nature of the given function and its slope at various points; do not use the limit definition. For the latter four, use the limit definition. Pay careful attention to the function names and independent variables. It is important to be comfortable with using letters other than f and x. For example, given a function $p(z)$, we call its derivative $p'(z)$.

(a) $f(x) = 1$

(b) $g(t) = t$

(c) $p(z) = z^2$

(d) $q(s) = s^3$

(e) $F(t) = \dfrac{1}{t}$

(f) $G(y) = \sqrt{y}$

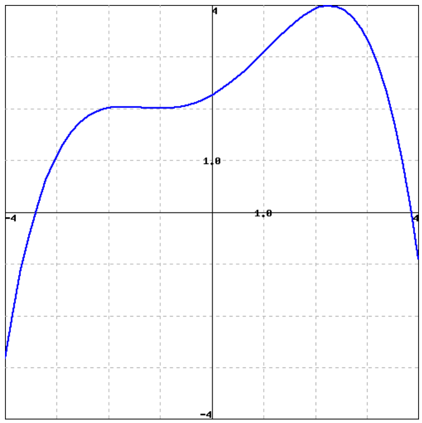
1.4.3 Summary

- The limit definition of the derivative, $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$, produces a value for each x at which the derivative is defined, and this leads to a new function $y = f'(x)$. It is especially important to note that taking the derivative is a process that starts with a given function (f) and produces a new, related function (f').
- There is essentially no difference between writing $f'(a)$ (as we did regularly in Section 1.3) and writing $f'(x)$. In either case, the variable is just a placeholder that is used to define the rule for the derivative function.
- Given the graph of a function $y = f(x)$, we can sketch an approximate graph of its derivative $y = f'(x)$ by observing that *heights* on the derivative's graph correspond to *slopes* on the original function's graph.
- In Activity 1.4.2, we encountered some functions that had sharp corners on their graphs, such as the shifted absolute value function. At such points, the derivative fails to exist, and we say that f is not differentiable there. For now, it suffices to understand this as

a consequence of the jump that must occur in the derivative function at a sharp corner on the graph of the original function.

1.4.4 Exercises

1. Consider the function $f(x)$ shown in the graph below.

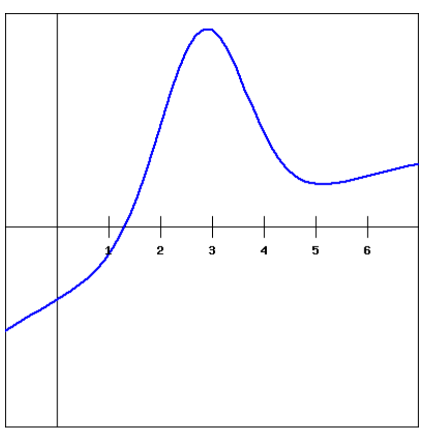


(Note that you can click on the graph to get a larger version of it, and that it may be useful to print that larger version to be able to work with it by hand.)

Carefully sketch the derivative function of the given function (you will want to estimate values on the derivative function at different x values as you do this). Use your derivative function graph to estimate the following values on the derivative function.

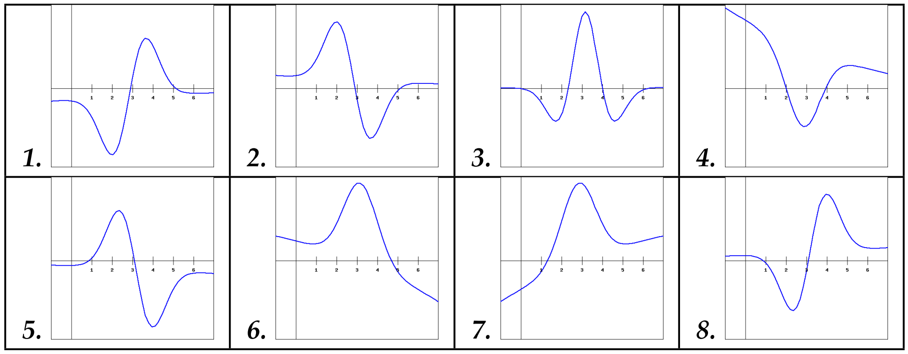
at x =	-3	-1	1	3
the derivative is	______	______	______	______

2. For the function $f(x)$ shown in the graph below, sketch a graph of the derivative. You will then be picking which of the following is the correct derivative graph, but should be sure to first sketch the derivative yourself.

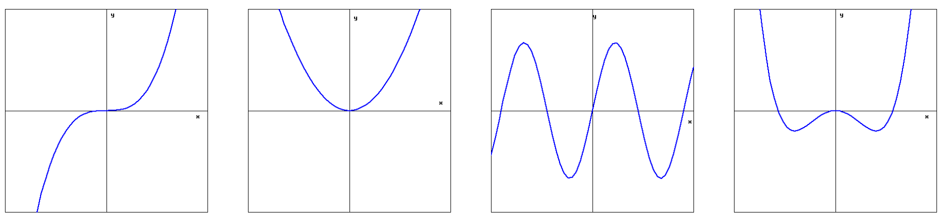


Which of the following graphs is the derivative of $f(x)$? (□ ? □ 1 □ 2 □ 3 □ 4 □ 5 □ 6 □ 7 □ 8)

(Click on a graph to enlarge it.)



3. (Click on a graph to enlarge it.)

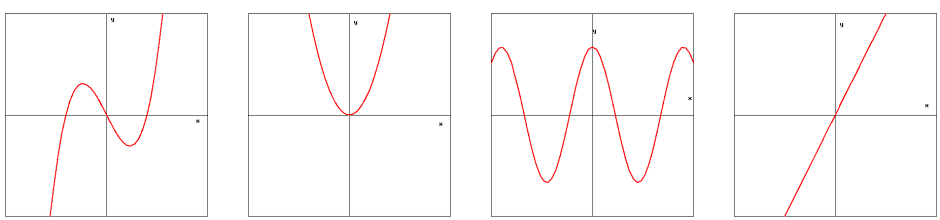


1 2 3 4

The graphs of 4 functions are shown above, and the graphs of the derivatives of those functions are shown below.

Match the number of each graph of $f(x)$ above with the letter of the graph of its derivative below.

1. ______ 2. ______ 3. ______ 4. ______



A B C D

(Click on a graph to enlarge it.)

4. On a piece of paper, draw the graph of a continuous function $y = f(x)$ that satisfies the following three conditions.

$$f'(x) < 0 \quad \text{for} \quad x < -1,$$

$$f'(x) > 0 \quad \text{for} \quad -1 < x < 1,$$

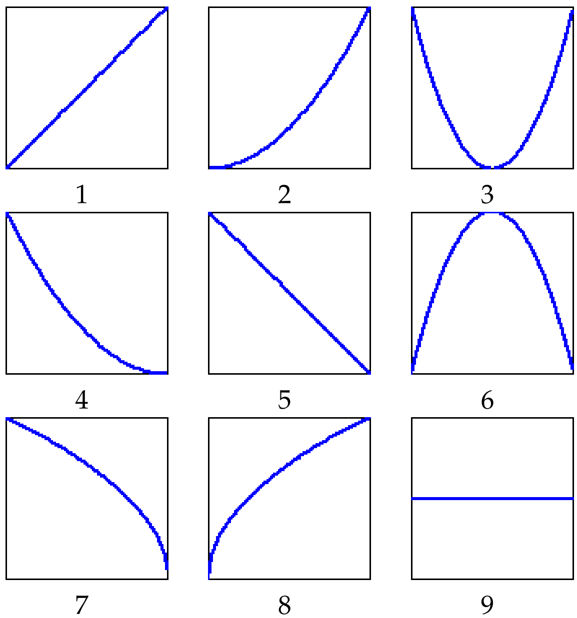
$$f'(x) < 0 \quad \text{for} \quad 1 < x$$

Approximate your function by picking a segment from the following for each of the sections of your graph, first for $x < -1$, then for $-1 < x < 1$, and then for $1 < x$. (You should, of course, imagine sliding the pieces vertically up or down to make the function you create be continuous.)

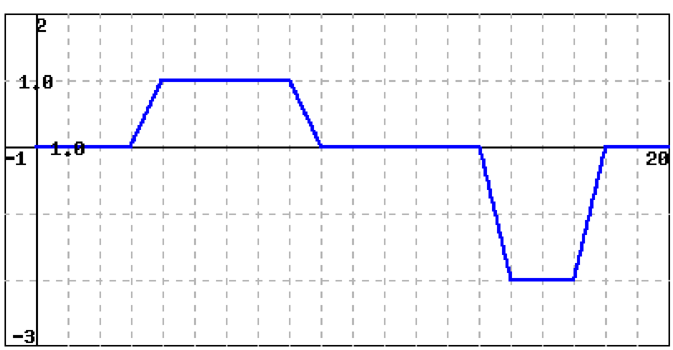
For $x < -1$, use segment _____

For $-1 < x < 1$, use segment _____

For $1 < x$, use segment _____



5. A child fills a pail by using a water hose. After finishing, the child plays in a sandbox for a while before tipping the pail over to empty it. If $V(t)$ gives the volume of the water in the pail at time t, then the figure below shows $V'(t)$ as a function of t.



At what time does the child:

A. Begin to fill the pail? $t =$ _____

B. Finish filling the pail? $t =$ _____

C. Tip the pail over? $t =$ _____

(What would the graph of $V'(t)$ look like if the child filled the pail by using a play shovel to repeatedly scoop water from a larger bucket and dump it in the pail instead of using a hose?)

6. Let $f(x)$ be the function $5x^2 - 6x + 10$. Then the quotient

$\frac{f(9+h)-f(9)}{h}$ can be simplified to $ah + b$ for:

$a =$ __________

and

$b =$ __________

7. Suppose that

$$f(x+h) - f(x) = -1hx^2 - 7hx - 5h^2x + 2h^2 - 4h^3.$$

Find $f'(x)$.

$f'(x) =$ __________________________

8. Find a formula for the derivative of the function $g(x) = 4x^2 - 6$ using difference quotients:

$g'(x) = \lim_{h \to 0} [($ ____________________________________ $)/h]$

$=$ ____________________________________.

(In the first answer blank, fill in the numerator of the difference quotient you use to evaluate the derivative. In the second, fill out the derivative you obtain after completing the limit calculation.)

9. Let $f(x) = -2 + 2\sqrt{x}$. Then the expression

$$\frac{f(x+h) - f(x)}{h}$$

can be written in the form

$$\frac{A}{\left(\sqrt{Bx + Ch}\right) + \left(\sqrt{x}\right)},$$

where A, B, and C are constants. (Note: It's possible for one or more of these constants to be 0.) Find the constants.

$A =$ __________

$B =$ __________

$C =$ __________

Use your answer from above to find $\lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$.

$\lim_{h \to 0} \frac{f(x+h) - f(x)}{h} =$ ____________________

Finally, find each of the following:

$f'(1) =$ __________

$f'(2) =$ __________

$f'(3) =$ __________

10. Let f be a function with the following properties: f is differentiable at every value of x (that is, f has a derivative at every point), $f(-2) = 1$, and $f'(-2) = -2$, $f'(-1) = -1$, $f'(0) = 0$, $f'(1) = 1$, and $f'(2) = 2$.

a. On the axes provided at left in Figure 1.4.3, sketch a possible graph of $y = f(x)$. Explain why your graph meets the stated criteria.

b. Conjecture a formula for the function $y = f(x)$. Use the limit definition of the derivative to determine the corresponding formula for $y = f'(x)$. Discuss both graphical and algebraic evidence for whether or not your conjecture is correct.

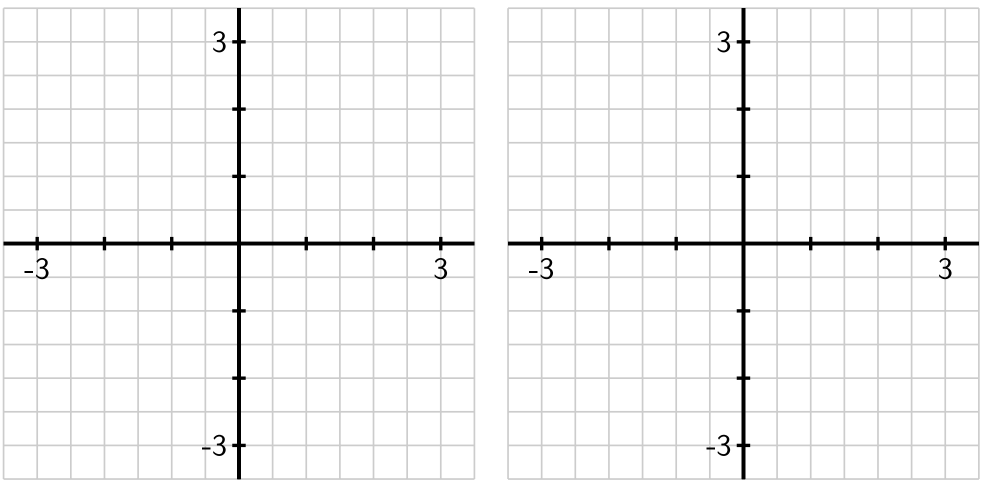


Figure 1.4.3 Axes for plotting $y = f(x)$ in (a) and $y = f'(x)$ in (b).

11. Consider the function $g(x) = x^2 - x + 3$.

a. Use the limit definition of the derivative to determine a formula for $g'(x)$.

b. Use a graphing utility to plot both $y = g(x)$ and your result for $y = g'(x)$; does your formula for $g'(x)$ generate the graph you expected?

c. Use the limit definition of the derivative to find a formula for $p'(x)$ where $p(x) = 5x^2 - 4x + 12$.

d. Compare and contrast the formulas for $g'(x)$ and $p'(x)$ you have found. How do the constants 5, 4, 12, and 3 affect the results?

12. For each graph that provides an original function $y = f(x)$ in Figure 1.4.4, your task is to sketch an approximate graph of its derivative function, $y = f'(x)$, on the axes immediately below. View the scale of the grid for the graph of f as being 1×1, and assume the horizontal scale of the grid for the graph of f' is identical to that for f. If you need to adjust the vertical scale on the axes for the graph of f', you should label that accordingly.

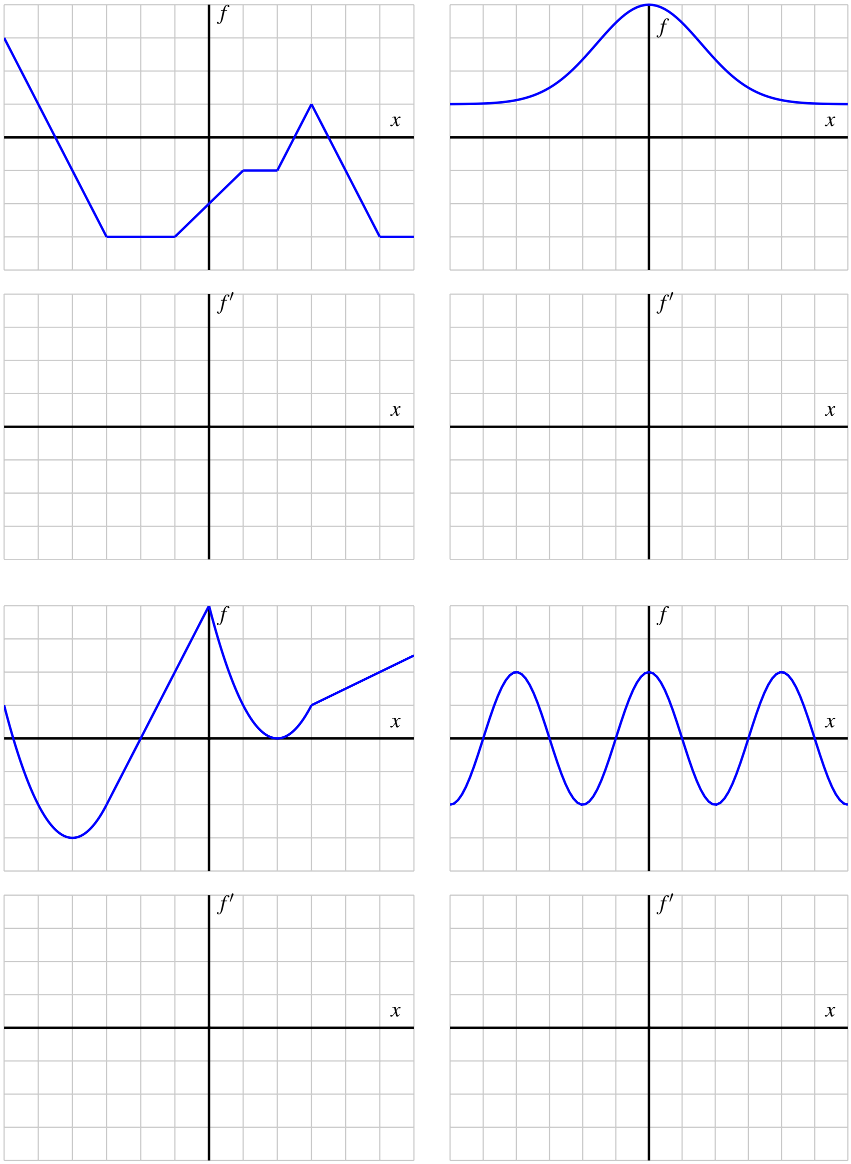


Figure 1.4.4 Graphs of $y = f(x)$ and grids for plotting the corresponding graph of $y = f'(x)$.

13. Let g be a continuous function (that is, one with no jumps or holes in the graph) and suppose that a graph of $y = g'(x)$ is given by the graph on the right in Figure 1.4.5.

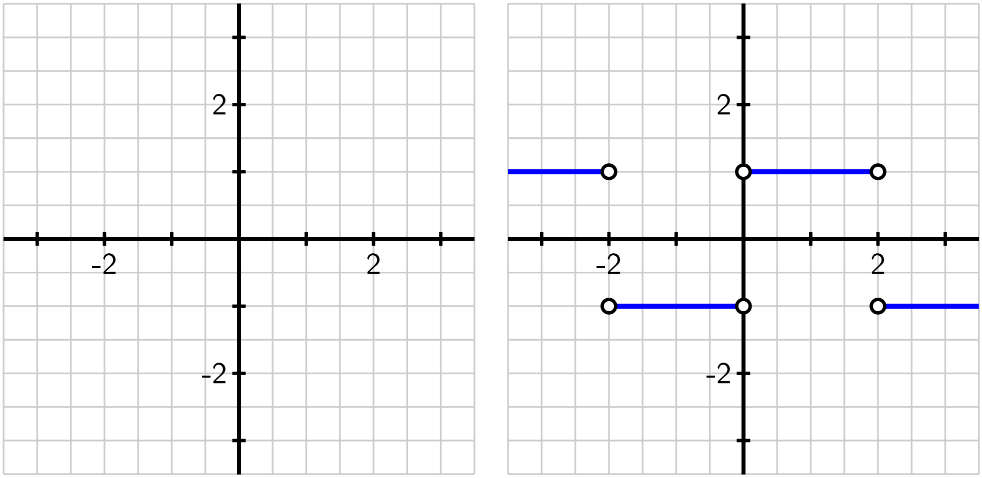


Figure 1.4.5 Axes for plotting $y = g(x)$ and, at right, the graph of $y = g'(x)$.

a. Observe that for every value of x that satisfies $0 < x < 2$, the value of $g'(x)$ is constant. What does this tell you about the behavior of the graph of $y = g(x)$ on this interval?

b. On what intervals other than $0 < x < 2$ do you expect $y = g(x)$ to be a linear function? Why?

c. At which values of x is $g'(x)$ not defined? What behavior does this lead you to expect to see in the graph of $y = g(x)$?

d. Suppose that $g(0) = 1$. On the axes provided at left in Figure 1.4.5, sketch an accurate graph of $y = g(x)$.

1.5 Interpreting, estimating, and using the derivative

Motivating Questions

- In contexts other than the position of a moving object, what does the derivative of a function measure?
- What are the units on the derivative function f', and how are they related to the units of the original function f?
- What is a central difference, and how can one be used to estimate the value of the derivative at a point from given function data?
- Given the value of the derivative of a function at a point, what can we infer about how the value of the function changes nearby?

1.5.1 Introduction

A powerful feature of mathematics is that it can be studied both as an abstract discipline and as an applied one. For instance, calculus can be developed almost entirely as an abstract collection of ideas that focus on properties of functions. At the same time, if we consider functions that represent meaningful processes, calculus can describe our experience of physical reality. We have already learned that for the position function $y = s(t)$ of a ball being tossed straight up in the air, the derivative of the position function, $v(t) = s'(t)$, gives the ball's velocity at time t.

In this section, we investigate several functions with contextual physical meaning and consider how the units on the independent variable, dependent variable, and the derivative function add to our understanding.

Preview Activity 1.5.1. Suppose that you are driving a car on the highway using cruise control so that you drive at a constant speed. The time, T (measured in hours), it takes to drive 50 miles depends on the constant rate, s (measured in miles per hour), at which you are driving. Since

$$\text{distance} = \text{rate} \cdot \text{travel time},$$

it follows that

$$\text{travel time} = \frac{\text{distance}}{\text{rate}},$$

and thus we know that $T(s) = \frac{50}{s}$.

For instance, observe that $T(75) = \frac{50}{75} = \frac{2}{3}$, so it takes $\frac{2}{3}$ of an hour to drive 50 miles at 75 miles per hour, and the point $(75, \frac{2}{3})$ lies on the graph of $T(s)$.

(a) Plot the graph of $T(s) = \frac{50}{s}$ on the axes provided in Figure 1.5.1. Include the point $(75, \frac{2}{3})$, and plot and label 2 additional points of your choice. Write a sentence to explain why the shape of the graph makes sense.

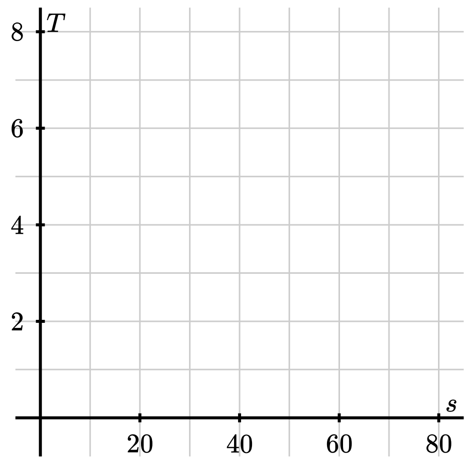


Figure 1.5.1 Axes for plotting $T(s)$.

(b) What is the value of $T(60)$? Include units on your answer. Write a sentence to explain the meaning of this value in context.

(c) Recall the average rate of change of a function f on an interval $[a, b]$, given by $AV_{[a,b]} = \frac{f(b)-f(a)}{b-a}$. Determine the average rate of change of T on the interval $[60, 65]$, $AV_{[60,65]}$. Include units on your answer.

(d) Is the sign of $AV_{[60,65]}$ positive or negative? Why does this sign make sense in light of the graph of $T(s)$? Explain your thinking.

(e) Write a sentence in everyday language that explains the meaning of the value of $AV_{[60,65]}$. Include units in your sentence.

1.5.2 Units of the derivative function

We know that the derivative of the function f at a fixed value x is given by

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h},$$

and that this value has several different interpretations. For instance, if we set $x = a$, $f'(a)$ is the slope of the tangent line to the graph of $y = f(x)$ at the point $(a, f(a))$.

We sometimes write $\frac{df}{dx}$ or $\frac{dy}{dx}$ instead of $f'(x)$, and these alternate notations emphasize the meaning of the derivative as *the instantaneous rate of change of f with respect to x*. To understand the units on the derivative function, we use the fact that the instantaneous rate of change is the limit of the average rate of change. Recall that the average rate of change of f on the interval $[a, a+h]$ is

$$AV_{[a,a+h]} = \frac{f(a+h) - f(a)}{(a+h) - a},$$

which is the quotient

$$\frac{\text{change in } f}{\text{change in } x}.$$

It follows that the units on the average rate of change are:

units of f per unit of x.

Note that since $(a+h) - a = h$, we usually write

$$AV_{[a,a+h]} = \frac{f(a+h) - f(a)}{h}.$$

Thus, when we take the limit of the average rate of change as h goes to zero, the derivative

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

has the same units: units of f per unit of x. Said slightly differently, the units on the derivative function are "units of output per unit of input" for the variables and context of the original function, f.

Example 1.5.2 Population growth. Suppose that the function $y = P(t)$ measures the population of a city (in thousands) at the start of year t (where $t = 0$ corresponds to the year 2020). What does

$$P'(2) = 21.37$$

mean in context?

Solution. The value $P'(2) = 21.37$ tells us that the instantaneous rate of change of the city's population with respect to time at the start of 2022 is 21.37 thousand people per year, since P is measured in thousands of people and t is measured in years. In addition, because the derivative value tells us how fast the population is changing, we can also infer about how much the population will change: we expect that in the year from 2022 to 2023, about 21,370 people will be added to the city's population. ◊

Example 1.5.3 Time as a function of speed. Consider the function $T(s) = \frac{50}{s}$ from Preview Activity 1.5.1, whose output tells us the amount of time (in hours) it takes to drive 50 miles at s miles per hour. What is the meaning of

$$T'(60) = -0.0139?$$

Solution. The value $T'(60) = -0.0139$ means that the instantaneous rate of change of travel time, T, when $s = 60$ is -0.0139 hours per (mile per hour). This also tells us about how much the function value will change if we change the speed at which we drive: if we are driving at a speed of 60 miles per hour and we increase our speed by one mile per hour, we expect the time it takes us to drive 50 miles to drop by about 0.0139 hours. Similarly, if we increase our speed by two miles per hour, we'd expect the time it takes us to drive 50 miles to drop by about $2 \cdot 0.0139 = 0.0278$ hours. ◊

Note 1.5.4 It is usually best *not* to simplify complicated units on a rate of change, but rather to leave their form as "(units of output) per (unit of input)".

The units on $T'(60)$ in Example 1.5.3 are "hours per (mile per hour)", since the number of hours it takes to drive 50 miles depends on the speed (in miles per hour) at which we travel. If we tried to simplify those units, we lose the meaning inherent to the context of the value of $T'(60)$.

Activity 1.5.2. In each of the following contexts, explain the meaning of the given derivative values in two ways that are similar to the sentence structures and discussions of units in the two examples preceding this activity in the text.

First, write a sentence that explicitly describes what we know about the instantaneous rate of change of the given function at a particular instant, with units. Then, write a second sentence that explains the amount of change we expect in the function value if the input variable changes by one unit at the known instant.

Remember not to try to simplify the units on the derivative value, but rather keep the units in the form "units of output per unit of input".

(a) Suppose that $V(m)$ measures the value of a car (in dollars) after the car has been driven m miles. Explain the meaning of the statement

$$V'(10250) = -0.89$$

by writing two sentences that address the instantaneous rate of change and the predicted change in the function, respectively.

Explicitly, your first sentence might have structure like

> $V'(10250) = -0.89$ means that the instantaneous rate of change of ____ when the car has been driven ________ miles is -0.89____________,

where the final blank should be completed using the units on the derivative. Your second sentence might have form

> When the car has been driven ________ miles, if the car is driven one more mile, we expect that the value of the car will ________ by about 0.89__________.

(b) Suppose that $W(h)$ measures the amount of water (in liters) in a tank that is filled with water that is h meters deep. Explain the meaning of the statement

$$W'(0.75) = 3.43$$

by writing two sentences in the structure described above.

(c) Suppose that $S(t)$ measures the temperature of a can of soda (in degrees Celsius) in a refrigerator at time t in minutes. Explain the meaning of the statement

$$S'(20) = -0.527$$

by writing two sentences in the structure described above.

(d) Suppose that $C(s)$ measures the rate at which a person burns calories (in calories per hour) when riding a bike at a speed of s kilometers per hour. Explain the meaning of the statement
$$C'(19) = 52.1$$
by writing two sentences in the structure described above.

1.5.3 Toward more accurate derivative estimates

Recall that to estimate the value of $f'(a)$ at a given value of $x = a$, we can compute a *difference quotient* $\frac{f(a+h)-f(a)}{h}$ with a relatively small value of h. We usually use both positive and negative values of h in order to account for the behavior of the function on both sides of the point of interest. In the setting where we have limited data, we introduce the notion of a *central difference* for estimating the value of $f'(a)$.

Example 1.5.5 Suppose that $y = f(x)$ is a function for which three values are known: $f(1) = 2.5$, $f(2) = 3.25$, and $f(3) = 3.625$. Estimate $f'(2)$.

Solution. We know that $f'(2) = \lim_{h\to 0} \frac{f(2+h)-f(2)}{h}$. But since we don't have a graph or a formula for the function, we can neither sketch a tangent line nor evaluate the limit algebraically, nor can we use smaller and smaller values of h to estimate the limit.

Instead, we have just two choices: using $h = -1$ or $h = 1$, depending on which point we pair with $(2, 3.25)$.

So, one estimate is
$$f'(2) \approx \frac{f(1) - f(2)}{1 - 2} = \frac{2.5 - 3.25}{-1} = 0.75.$$

The other is
$$f'(2) \approx \frac{f(3) - f(2)}{3 - 2} = \frac{3.625 - 3.25}{1} = 0.375.$$

Because the first approximation looks backward from the point $(2, 3.25)$ and the second approximation looks forward, it makes sense to average these two estimates in order to account for behavior on both sides of $x = 2$. Doing so, we find that
$$f'(2) \approx \frac{0.75 + 0.375}{2} = 0.5625.$$

◊

When we have two function values for f at equally spaced locations on opposite sides of $x = a$, the approach of averaging the two estimates — as we did in Example 1.5.5 — results in the best possible estimate we can find for $f'(a)$ using the two known function values.

To see why, let's think about how we look both "backward" and "forward" from $x = a$, like we did for the specific data in Example 1.5.5. Using a *backward difference*, BD, one estimate for $f'(a)$ is

$$BD = \frac{f(a) - f(a-h)}{a - (a-h)} = \frac{f(a) - f(a-h)}{h}.$$

When we instead use a *forward difference*, FD, we now get the estimate

$$FD = \frac{f(a+h) - f(a)}{(a+h) - a} = \frac{f(a+h) - f(a)}{h}.$$

Because these two estimates take into account the function's behavior on both sides of $x = a$, it makes sense to average them to include the most possible information for an even better estimate for the value of $f'(a)$. Doing so, we find that

$$\begin{aligned}
\frac{BD + FD}{2} &= \frac{1}{2}(BD + FD) \\
&= \frac{1}{2}\left(\frac{f(a) - f(a-h)}{h} + \frac{f(a+h) - f(a)}{h}\right) \\
&= \frac{1}{2}\left(\frac{f(a) - f(a-h) + f(a+h) - f(a)}{h}\right) \\
&= \frac{f(a+h) - f(a-h)}{2h}
\end{aligned}$$

We naturally call the resulting estimate a *central difference*.

Definition 1.5.6 For a given value of h, the **central difference approximation** of $f'(a)$ is given by

$$f'(a) \approx \frac{f(a+h) - f(a-h)}{2h}.$$

◊

Observe that the central difference is also the average rate of change of f on $[a-h, a+h]$,

$$AV_{[a-h,a+h]} = \frac{f(a+h) - f(a-h)}{2h},$$

and thus we can even interpret the central difference as a slope of a certain secant line.

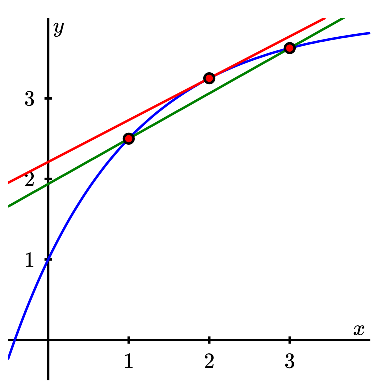


Figure 1.5.7 Plot of a function f that satisfies the data in Example 1.5.5.

In Figure 1.5.7, we see a function that passes through the three data points given in Example 1.5.5: $(1, 2.5)$, $(2, 3.25)$, and $(3, 3.265)$. The top line in the figure is the tangent line to the curve at $(2, f(2))$; for the function shown, the slope of that tangent line turns out to be $f'(2) = 0.5199$. We found in Example 1.5.5 that the central difference approximation is 0.5625, which is the slope of the secant line through the points $(1, f(1))$ and $(3, f(3))$ in Figure 1.5.7.

In general, the central difference

$$AV_{[a-h,a+h]} = \frac{f(a+h) - f(a-h)}{2h}$$

is the slope of the secant line between the points $(a - h, f(a - h))$ and $(a + h, f(a + h))$.

Activity 1.5.3. A potato is placed in an oven whose temperature is 350 degrees Fahrenheit, and the potato's temperature F (in degrees Fahrenheit) is recorded in the following table. Time t is measured in minutes.[1]

Table 1.5.8 Potato temperature data in degrees Fahrenheit.

t	0	10	20	30	40	50	60
$F(t)$	65	94.7	141.7	167.6	182.4	197.9	209.3

(a) Use a central difference to estimate the instantaneous rate of change of the potato's temperature at $t = 20$. Include units on your answer.

(b) Use a central difference to estimate the instantaneous rate of change of the potato's temperature at $t = 40$. Include units on your answer.

(c) Without doing any calculation, which do you expect to be greater: $F'(50)$ or $F'(60)$? Why?

(d) Suppose we know that $F(46) = 192.5$ and $F'(46) = 1.39$. What are the respective units on these two quantities? What do you expect the temperature of the potato to be when $t = 47$? when $t = 48$? Why?

(e) Write a couple of careful sentences that describe the behavior of the temperature of the potato on the time interval $[10, 40]$, as well as the behavior of the instantaneous rate of change of the potato's temperature on the same time interval.

In Section 1.4, we learned how use to the graph of a given function f to plot the graph of its derivative, f'. It is important to remember that when we do so, the scale (often) and the units (always) on the vertical axis have to change to represent f'. For example, suppose that $P(t) = 212 - 148e^{-0.05t}$ tells us the temperature in degrees Fahrenheit of a potato in an oven at time t in minutes. In Figure 1.5.9, we sketch the graph of P, and in Figure 1.5.10 the graph of P'. Notice that the vertical scales are different in size and different in units, as the units of P are °F, while those of P' are °F/min.

[1]Thanks to Nick Owad of Hood College for conducting experiments with actual potatoes in his oven in order to generate the data for this activity.

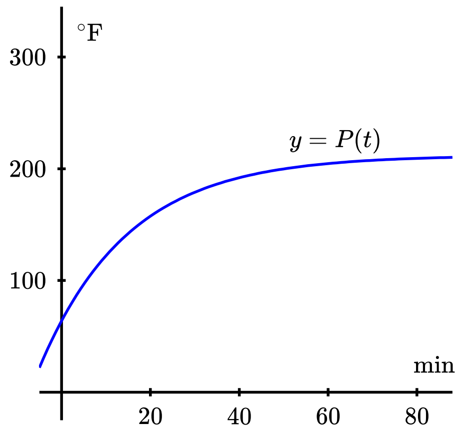


Figure 1.5.9 Plot of the temperature function, $P(t) = 212 - 148e^{-0.05t}$.

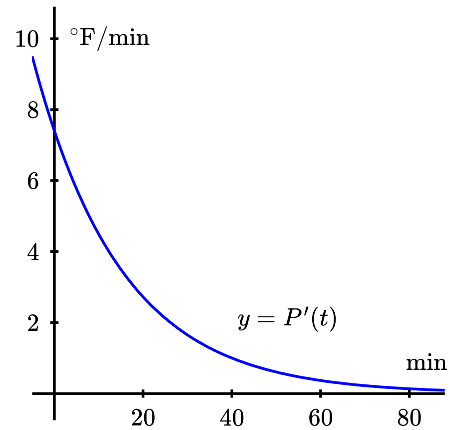


Figure 1.5.10 Plot of the temperature function's derivative, $P'(t)$.

Activity 1.5.4. Researchers at a major car company have found a function that relates gasoline consumption to speed for a particular model of car. In particular, they have determined that the consumption C, in *liters per kilometer*, at a given speed s, is given by a function $C = f(s)$, where s is the car's speed in *kilometers per hour*.

(a) Data provided by the car company tells us that $f(80) = 0.015$, $f(90) = 0.02$, and $f(100) = 0.027$. Use this information to estimate the instantaneous rate of change of fuel consumption with respect to speed at $s = 90$. Be as accurate as possible, use proper notation, and include units on your answer.

(b) By writing a complete sentence, interpret the meaning (in the context of fuel consumption) of "$f(80) = 0.015$."

(c) Write at least one complete sentence that interprets the meaning of the value of $f'(90)$ that you estimated in (a).

1.5.4 Summary

- The derivative of a given function $y = f(x)$ measures the instantaneous rate of change of the output variable with respect to the input variable. It also predicts about how much change we can expect when the input variable changes a small amount.
- The units on the derivative function $y = f'(x)$ are units of f per unit of x. Again, this measures how fast the output of the function f changes when the input variable changes.
- The central difference approximation to the value of the first derivative is given by

$$f'(a) \approx \frac{f(a+h) - f(a-h)}{2h}.$$

This quantity measures the slope of the secant line to $y = f(x)$ through the points $(a - h, f(a-h))$ and $(a+h, f(a+h))$. The central difference generates a good approximation of the derivative's value at $x = a$.

1.5.5 Exercises

1. The temperature, H, in degrees Celsius, of a cup of coffee placed on the kitchen counter is given by $H = f(t)$, where t is in minutes since the coffee was put on the counter.

(a) Is $f'(t)$ positive or negative? (□ ? □ positive □ negative)

(Be sure that you are able to give a reason for your answer.)

(b) What are the units of $f'(30)$? (□ ? □ degC □ min □ degC/min □ min/degC □ hr □ degC/hr □ hr/degC)

Suppose that $|f'(30)| = 0.9$ and $f(30) = 51$. Fill in the blanks and select the appropriate units and terms to complete the following statement about the temperature of the coffee in this case.

At ____ minutes after the coffee was put on the counter, its (□ ? □ derivative □ temperature □ change in temperature) is __________ (□ ? □ degC □ min □ degC/min □ min/degC □ hr □ degC/hr □ hr/degC) and will (□ ? □ increase □ decrease) by about _________ (□ ? □ degC □ min □ degC/min □ min/degC □ hr □ degC/hr □ hr/degC) in the next 75 seconds.

2. The cost, C (in dollars) to produce g gallons of ice cream can be expressed as $C = f(g)$.

(a) In the expression $f(275) = 350$,

what are the units of 275? (□ ? □ dollars □ gallons □ dollars*gallons □ dollars/gallon □ gallons/dollar)

what are the units of 350? (□ ? □ dollars □ gallons □ dollars*gallons □ dollars/gallon □ gallons/dollar)

(b) In the expression $f'(275) = 1.8$,

what are the units of 275? (□ ? □ dollars □ gallons □ dollars*gallons □ dollars/gallon □ gallons/dollar)

what are the units of 1.8? (□ ? □ dollars □ gallons □ dollars*gallons □ dollars/gallon □ gallons/dollar)

(Be sure that you can carefully put into words the meanings of each of these statement in terms of ice cream and money.)

3. A laboratory study investigating the relationship between diet and weight in adult humans found that the weight of a subject, W, in pounds, was a function, $W = f(c)$, of the average number of Calories, c, consumed by the subject in a day.

(a) In the statement $f(1900) = 150$

what are the units of 1900? (□ ? □ lb □ cal □ day □ lb/cal □ cal/lb □ cal/day □ lb/day □ day/lb □ day/cal) what are the units of 150? (□ ? □ lb □ cal □ day □ lb/cal □ cal/lb □ cal/day □ lb/day □ day/lb □ day/cal)

(Think about what this statement means in terms of the weight of the subject and the number of calories that the subject consumes.)

(b) In the statement $f'(2000) = 0$,

what are the units of 2000? (□ ? □ lb □ cal □ day □ lb/cal □ cal/lb □ cal/day □ lb/day □ day/lb □ day/cal)

what are the units of 0? (□ ? □ lb □ cal □ day □ lb/cal □ cal/lb □ cal/day □ lb/day □ day/lb □ day/cal)

(Think about what this statement means in terms of the weight of the subject and the number of calories that the subject consumes.)

(c) In the statement $f^{-1}(156) = 2200$,

what are the units of 156? (□ ? □ lb □ cal □ day □ lb/cal □ cal/lb □ cal/day □ lb/day □ day/lb □ day/cal)

what are the units of 2200? (□ ? □ lb □ cal □ day □ lb/cal □ cal/lb □ cal/day □ lb/day □ day/lb □ day/cal)

(Think about what this statement means in terms of the weight of the subject and the number of calories that the subject consumes.)

(d) What are the units of $f'(c) = dW/dc$? (□ ? □ lb □ cal □ day □ lb/cal □ cal/lb □ cal/day □ lb/day □ day/lb □ day/cal)

(e) Suppose that Sam reads about f' in this study and draws the following conclusion: If Sam increases his average calorie intake from 2600 to 2620 calories per day, then his weight will increase by approximately 0.4 pounds.

Fill in the blanks below so that the equation supports his conclusion.

$f'\Big(\underline{\qquad\qquad\qquad\qquad}\Big) = \underline{\qquad\qquad\qquad\qquad}$

4. Let $f(t)$ be the number of centimeters of rainfall that has fallen since midnight, where t is the time in hours. Match the following statements to their interpretations, given below.

(a) $f(5) = 3.1$: (□ ? □ 1 □ 2 □ 3 □ 4 □ 5 □ 6 □ 7 □ 8 □ 9 □ 10 □ 11 □ 12 □ 13 □ 14 □ 15 □ 16)

(b) $f^{-1}(6) = 11$: (□ ? □ 1 □ 2 □ 3 □ 4 □ 5 □ 6 □ 7 □ 8 □ 9 □ 10 □ 11 □ 12 □ 13 □ 14 □ 15 □ 16)

(c) $f'(5) = 0.1$: (□ ? □ 1 □ 2 □ 3 □ 4 □ 5 □ 6 □ 7 □ 8 □ 9 □ 10 □ 11 □ 12 □ 13 □ 14 □ 15 □ 16)

(d) $(f^{-1})'(6) = 1$: (□ ? □ 1 □ 2 □ 3 □ 4 □ 5 □ 6 □ 7 □ 8 □ 9 □ 10 □ 11 □ 12 □ 13 □ 14 □ 15 □ 16)

Interpretations:

5. Suppose that f is given for x in the interval $[0, 12]$ by

$x =$	0	2	4	6	8	10	12
$f(x) =$	-17	-19	-23	-24	-23	-20	-18

A. Estimate $f'(2)$ using the values of f in the table.

$f'(2) \approx$ ________

B. For what values of x does $f'(x)$ appear to be positive?

(Give your answer as an interval or a list of intervals, e.g., ***(-infinity,8]*** *or* ***(1,5),(7,10)*** *.)*

C. For what values of x does $f'(x)$ appear to be negative?

(Give your answer as an interval or a list of intervals, e.g., ***(-infinity,8]*** *or* ***(1,5),(7,10)*** *.)*

6. A cup of coffee has its temperature F (in degrees Fahrenheit) at time t given by the function $F(t) = 75 + 110e^{-0.05t}$, where time is measured in minutes.

a. Use a central difference with $h = 0.01$ to estimate the value of $F'(10)$.

b. What are the units on the value of $F'(10)$ that you computed in (a)? What is the practical meaning of the value of $F'(10)$?

c. Which do you expect to be greater: $F'(10)$ or $F'(20)$? Why?

d. Write a sentence that describes the behavior of the function $y = F'(t)$ on the time interval $0 \le t \le 30$. How do you think its graph will look? Why?

7. The temperature change T (in Fahrenheit degrees), in a patient, that is generated by a dose q (in milliliters), of a drug, is given by the function $T = f(q)$.

a. What does it mean to say $f(50) = 0.75$? Write a complete sentence to explain, using correct units.

b. A person's sensitivity, s, to the drug is defined by the function $s(q) = f'(q)$. What are the units of sensitivity?

c. Suppose that $f'(50) = -0.02$. Write a complete sentence to explain the meaning of this value. Include in your response the information given in (a).

8. The velocity of a ball that has been tossed vertically in the air is given by $v(t) = 16 - 32t$, where v is measured in feet per second, and t is measured in seconds. The ball is in the air from $t = 0$ until $t = 2$.

a. When is the ball's velocity greatest?

b. Determine the value of $v'(1)$. Justify your thinking.

c. What are the units on the value of $v'(1)$? What does this value and the corresponding units tell you about the behavior of the ball at time $t = 1$?

d. What is the physical meaning of the function $v'(t)$?

9. The value, V, of a particular automobile (in dollars) depends on the number of miles, m, the car has been driven, according to the function $V = h(m)$.

a. Suppose that $h(40000) = 15500$ and $h(55000) = 13200$. What is the average rate of change of h on the interval $[40000, 55000]$, and what are the units on this value?

b. In addition to the information given in (a), say that $h(70000) = 11100$. Determine the best possible estimate of $h'(55000)$ and write one sentence to explain the meaning of your result, including units on your answer.

c. Which value do you expect to be greater: $h'(30000)$ or $h'(80000)$? Why?

d. Write a sentence to describe the long-term behavior of the function $V = h(m)$, plus another sentence to describe the long-term behavior of $h'(m)$. Provide your discussion in practical terms regarding the value of the car and the rate at which that value is changing.

1.6 The second derivative

Motivating Questions

- How does the derivative of a function tell us whether the function is increasing or decreasing on an interval?
- What can we learn by taking the derivative of the derivative (the *second* derivative) of a function f?
- What does it mean to say that a function is concave up or concave down? How are these characteristics connected to certain properties of the derivative of the function?
- What are the units of the second derivative? How do they help us understand the rate of change of the rate of change?

1.6.1 Introduction

Given a differentiable function $y = f(x)$, we know that its derivative, $y = f'(x)$, is a related function whose output at $x = a$ tells us the slope of the tangent line to $y = f(x)$ at the point $(a, f(a))$. That is, heights on the derivative graph tell us the values of slopes on the original function's graph.

At a point where $f'(x)$ is positive, the slope of the tangent line to f is positive. Therefore, on an interval where $f'(x)$ is positive, the function f is increasing (or rising). Similarly, if $f'(x)$ is negative on an interval, the graph of f is decreasing (or falling).

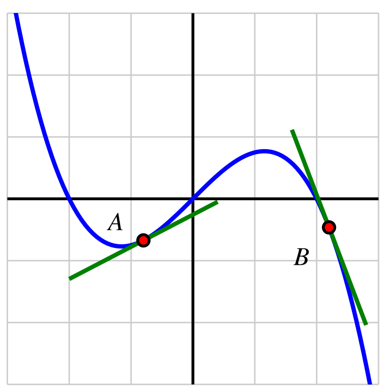


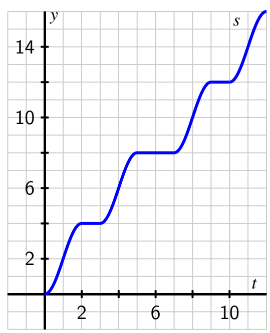
Figure 1.6.1 Two tangent lines on a graph.

The derivative of f tells us not only *whether* the function f is increasing or decreasing on an interval, but also *how* the function f is increasing or decreasing. Look at the two tangent lines shown in Figure 1.6.1. We see that near point A the value of $f'(x)$ is positive and relatively close to zero, so near that point the graph is rising slowly. By contrast, near point B, the derivative is negative and relatively large in absolute value, so f is decreasing rapidly near B.

Besides asking whether the value of the derivative function is positive or negative and whether it is large or small, we can also ask "how is the derivative changing?"

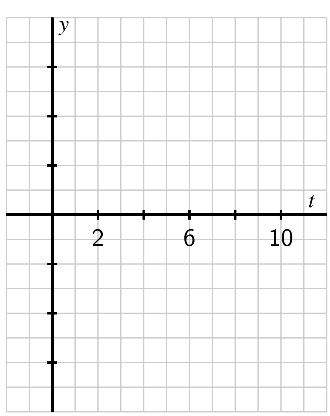
Because the derivative, $y = f'(x)$, is itself a function, we can consider taking its derivative — the derivative of the derivative — and ask "what does the derivative of the derivative tell us about how the original function behaves?" We start with an investigation of a moving object.

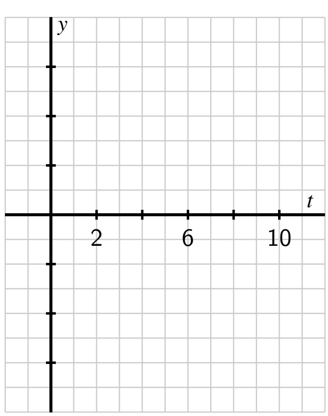
Preview Activity 1.6.1. The position of a car driving along a straight road at time t in minutes is given by the function $y = s(t)$ whose graph is provided. The car's position function has units measured in thousands of feet. For instance, the point $(2, 4)$ on the graph indicates that after 2 minutes, the car has traveled 4000 feet.



(a) In everyday language, describe the behavior of the car over the provided time interval. In particular, you should carefully discuss what is happening on each of the time intervals $[0, 1]$, $[1, 2]$, $[2, 3]$, $[3, 4]$, and $[4, 5]$, plus provide commentary overall on what the car is doing on the interval $[0, 12]$.

(b) On the lefthand axes provided, sketch a careful, accurate graph of $y = s'(t)$.





(c) What is the meaning of the function $y = s'(t)$ in the context of the given problem? What can we say about the car's behavior when $s'(t)$ is positive? when $s'(t)$ is zero? when $s'(t)$ is negative?

(d) Rename the function you graphed in (b) to be called $y = v(t)$. Describe the behavior of v in words, using phrases like "v is increasing on the interval ..." and "v is constant on the interval ...."

(e) Sketch a graph of the function $y = v'(t)$ on the righthand axes provide in (b). Write at least one sentence to explain how the behavior of $v'(t)$ is connected to the graph of $y = v(t)$.

1.6.2 Increasing or decreasing

So far, we have used the words *increasing* and *decreasing* intuitively to describe a function's graph. Here we define these terms more formally.

Definition 1.6.2 Given a function $f(x)$ defined on the interval (a, b), we say that f **is increasing on** (a, b) provided that for all x_1, x_2 in the interval (a, b), if $x_1 < x_2$, then $f(x_1) < f(x_2)$. Similarly, we say that f **is decreasing on** (a, b) provided that for all x_1, x_2 in the interval (a, b), if $x_1 < x_2$, then $f(x_1) > f(x_2)$. ◊

Simply put, an increasing function is one that is rising as we move from left to right along the graph, and a decreasing function is one that falls as the value of the input increases. If the function has a derivative, the sign of the derivative tells us whether the function is increasing or decreasing.

Let f be a function that is differentiable on an interval (a, b). It is possible to show that if $f'(x) > 0$ for every x such that $a < x < b$, then f is increasing on (a, b); similarly, if $f'(x) < 0$ on (a, b), then f is decreasing on (a, b).

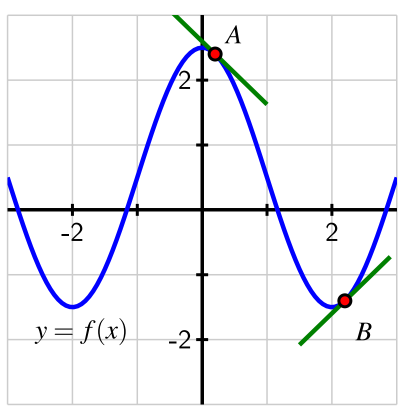


Figure 1.6.3 A function that is decreasing on the intervals $-3 < x < -2$ and $0 < x < 2$ and increasing on $-2 < x < 0$ and $2 < x < 3$.

For example, the function pictured in Figure 1.6.3 is increasing on the entire interval $-2 < x < 0$, and decreasing on the interval $0 < x < 2$. Note that the value $x = 0$ is not included in either interval since at this location, the function is changing from increasing to decreasing.

1.6.3 The Second Derivative

We are now accustomed to investigating the behavior of a function by examining its derivative. The derivative of a function f is a new function given by the rule

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}.$$

Because f' is itself a function, it is perfectly feasible for us to consider the derivative of the derivative, which is the new function $y = [f'(x)]'$. We call this resulting function *the second derivative* of $y = f(x)$, and denote the second derivative by $y = f''(x)$. Consequently, we will sometimes call f' "the first derivative" of f, rather than simply "the derivative" of f.

Definition 1.6.4 Let f be a function and x an input value in the function's domain. We define the **second derivative of** f, a new function called $f''(x)$, by the formula

$$f''(x) = \lim_{h\to 0} \frac{f'(x+h) - f'(x)}{h},$$

provided this limit exists. That is, $f''(x)$ is given by limit definition of the derivative of the first derivative. ◊

The meaning of the derivative function still holds, so when we compute $y = f''(x)$, this new function measures slopes of tangent lines to the curve $y = f'(x)$, as well as the instantaneous rate of change of $y = f'(x)$. In other words, just as the first derivative measures the rate at which the original function changes, the second derivative measures the rate at which the first derivative changes. The second derivative will help us understand how the rate of change of the original function is itself changing.

1.6.4 Concavity

In addition to asking *whether* a function is increasing or decreasing, it is also natural to inquire *how* a function is increasing or decreasing. There are three basic behaviors that an increasing function can demonstrate on an interval, as pictured in Figure 1.6.5: the function can increase more and more rapidly, it can increase at the same rate, or it can increase in a way that is slowing down. Fundamentally, we are beginning to think about how a particular curve bends, with the natural comparison being made to lines, which don't bend at all. More than this, we want to understand how the bend in a function's graph is tied to behavior characterized by the first derivative of the function.

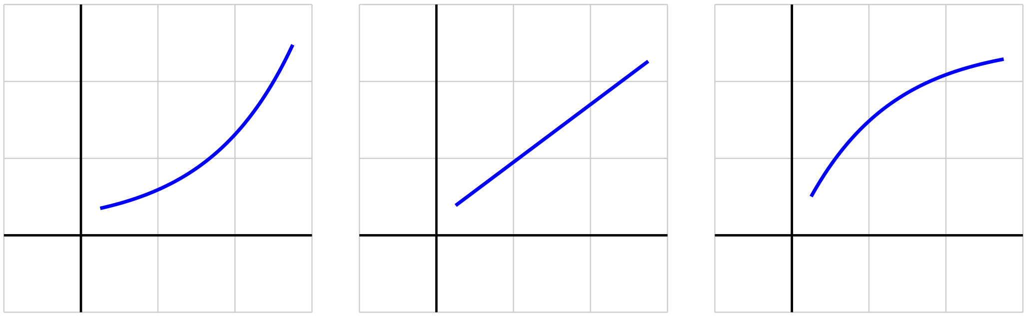


Figure 1.6.5 Three functions that are all increasing, but doing so in different ways.

On the leftmost curve in Figure 1.6.5, imagine drawing a sequence of tangent lines to the curve. As we move from left to right, the slopes of those tangent lines will increase. Therefore, the rate of change of the pictured function is increasing, and this explains why we sometimes say this function is *increasing at an increasing rate*. For the rightmost graph in Figure 1.6.5, observe that as x increases, the function increases, but the slopes of the tangent lines decrease. This function is *increasing at a decreasing rate*.

Similar options hold for how a function can decrease. Here we must be extra careful with our language, because decreasing functions involve negative slopes. Negative numbers present an interesting tension between common language and mathematical language. For example, it can be tempting to say that "-100 is bigger than -2." But we must remember that "greater than" describes how numbers lie on a number line: $x > y$ provided that x lies to the right of y. So of course, -100 is less than -2. Informally, it can be helpful to say that "-100 is more negative than -2." When a function's values are negative, and those values get more negative as the input increases, the function must be decreasing. The situation gets a bit more complicated when we think about how the slope of the tangent line to a decreasing function changes as we move from left to right.

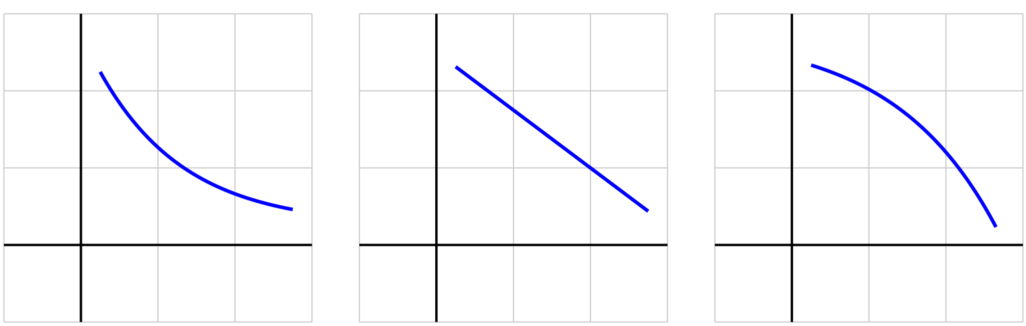


Figure 1.6.6 From left to right, three functions that are all decreasing, but doing so in different ways.

The three graphs shown in Figure 1.6.6 show three possibilities for how a function can decrease in a consistent way. The middle graph depicts a function decreasing at a constant rate. If we imagine a sequence of tangent lines on the leftmost graph, the slopes of the tangent lines get less and less negative as we move from left to right. That means that the values of the first derivative of the leftmost function, while all negative, are increasing. In the rightmost curve in Figure 1.6.6, observe that as x increases, the function value decreases, and the slopes of the tangent lines decrease (because they become more and more negative). That means that the values of the first derivative of the rightmost function are all negative and decreasing.

In calculus, we describe these possible behaviors through the language of *concavity*. When a curve opens upward on a given interval, such as the parabola $y = x^2$ or the natural exponential function $y = e^x$, we say that the curve is *concave up* on that interval. Likewise, when a curve opens down, such as the parabola $y = -x^2$ or the opposite of the exponential function $y = -e^x$, we say that the function is *concave down*. Concavity is linked to both the first and second derivatives of the function.

In Figure 1.6.7, we see two functions and a sequence of tangent lines to each. On the lefthand plot, where the function is concave up, observe that the tangent lines always lie below the curve itself, and the slopes of the tangent lines are increasing as we move from left to right. In other words, the function f is concave up on the interval shown because its derivative, f', is increasing on that interval. Similarly, on the righthand plot in Figure 1.6.7, where the function shown is concave down, we see that the tangent lines always lie above the curve, and the slopes of the tangent lines are decreasing as we move from left to right. The fact that its derivative, f', is decreasing makes f concave down on the interval.

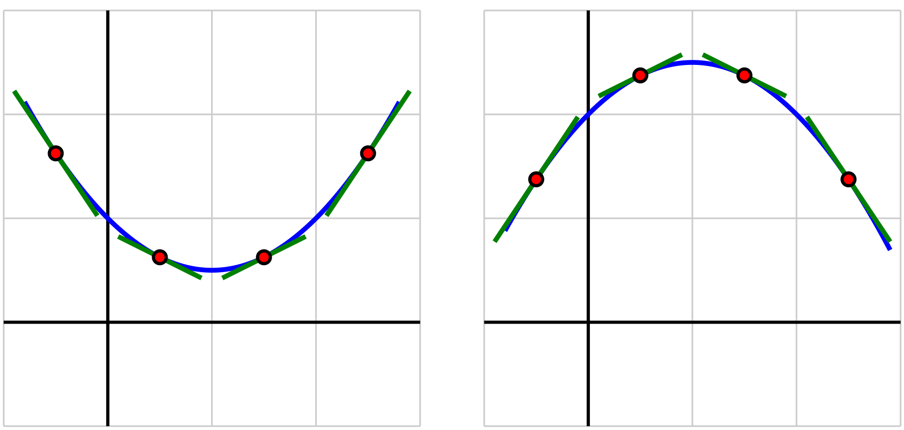


Figure 1.6.7 At left, a function that is concave up; at right, one that is concave down.

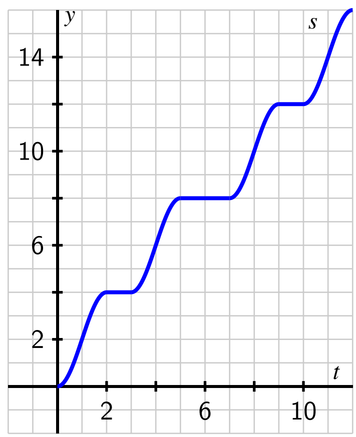
We state these most recent observations formally as the definitions of the terms *concave up* and *concave down*.

Definition 1.6.8 Let f be a differentiable function on an interval (a, b). Then f is **concave up** on (a, b) if and only if f' is increasing on (a, b); f is **concave down** on (a, b) if and only if f' is decreasing on (a, b). ◊

Returning to Figure 1.6.6 with the three decreasing functions, we can now say that the leftmost function is "decreasing and concave up", while the function on the right is "decreasing and concave down". Applying the language of concavity to the options for increasing functions shown in Figure 1.6.5, the leftmost function is "increasing and concave up", while the function on the right is "increasing and concave down".

In Activity 1.6.2, we revisit Preview Activity 1.6.1 to use these latest ideas involving the second derivative and the concavity of a function.

Activity 1.6.2. The position of a car driving along a straight road at time t in minutes is given by the function $y = s(t)$ that is pictured in the following graph. The car's position function has units measured in thousands of feet. Remember that you worked with this function and sketched graphs of $y = v(t) = s'(t)$ and $y = v'(t)$ in Preview Activity 1.6.1.



(a) On what intervals is the position function $y = s(t)$ increasing? decreasing? Why?

(b) On which intervals is the velocity function $y = v(t) = s'(t)$ increasing? decreasing? neither? Why?

(c) *Acceleration* is defined to be the instantaneous rate of change of velocity, as the acceleration of an object measures the rate at which the velocity of the object is changing. Say that the car's acceleration function is named $a(t)$. How is $a(t)$ computed from $v(t)$? How is $a(t)$ computed from $s(t)$? Explain.

(d) What can you say about s'' whenever s' is increasing? Why?

(e) Using only the words *increasing*, *decreasing*, *constant*, *concave up*, *concave down*, and *linear*, complete the following sentences. For the position function s with velocity v and acceleration a,

- on an interval where v is positive, s is __________.
- on an interval where v is negative, s is __________.
- on an interval where v is zero, s is __________.
- on an interval where a is positive, v is __________.
- on an interval where a is negative, v is __________.

- on an interval where a is zero, v is __________.
- on an interval where a is positive, s is __________.
- on an interval where a is negative, s is __________.
- on an interval where a is zero, s is __________.

Exploring the context of position, velocity, and acceleration is a good way to understand how a function, its first derivative, and its second derivative are related to one another. In Activity 1.6.2, we can replace s, v, and a with an arbitrary function f and its derivatives f' and f'', and essentially all the same observations hold. In particular, note that the following are equivalent: on an interval where the graph of f is concave up, f' is increasing and f'' is positive; and where the graph of f is concave down, f' is decreasing and f'' is negative.

Activity 1.6.3. A potato is placed in an oven whose temperature is 350 degrees Fahrenheit, and the potato's temperature F (in degrees Fahrenheit) is recorded in Table 1.6.9. Time t is measured in minutes.[1]

Table 1.6.9 Select values of $F(t)$.

t	0	10	20	30	40	50	60
$F(t)$	65	94.7	141.7	167.6	182.4	197.9	209.3

Table 1.6.10 Select approximate values of $F'(t)$.

t	0	10	20	30	40	50	60
$F'(t)$	NA	3.805	3.645	2.065	1.515	1.35	NA

In Activity 1.5.3, we used this data to compute approximations to $F'(20)$ and $F'(40)$ using central differences. Those values are provided in Table 1.6.10, along with several others computed in the same way.

(a) What are the units on $F'(t)$? What is the precise meaning of the value $F'(20) = 3.645$?

(b) Use a central difference to estimate the value of $F''(30)$.

(c) What is the meaning of the value of $F''(30)$ that you have computed in (b) in terms of the potato's temperature? Write several careful sentences that describe the overall behavior of the potato's temperature at this point in time. In particular, you should cite the values of $F(30)$, $F'(30)$, and $F''(30)$, each with appropriate units. Be sure to explicitly discuss what you expect to happen in the minute that transpires from $t = 30$ to $t = 31$.

(d) On the interval from $t = 10$ to $t = 40$, is the potato's temperature increasing at an increasing rate, increasing at a constant rate, or increasing at a decreasing rate? Why?

---

[1]Thanks to Nick Owad of Hood College for conducting experiments with actual potatoes in his oven in order to generate the data for this activity.

In Subsection 1.5.2, we learned that for a given function $f(x)$, the units on the first derivative function, $f'(x)$ are "units of f per unit of x". Because $f''(x) = [f'(x)]'$, it follows that the units on $f''(x)$ are "units of f' per unit of x". In other words, the units on $f''(x)$ are:

(units of f per unit of x) per unit of x.

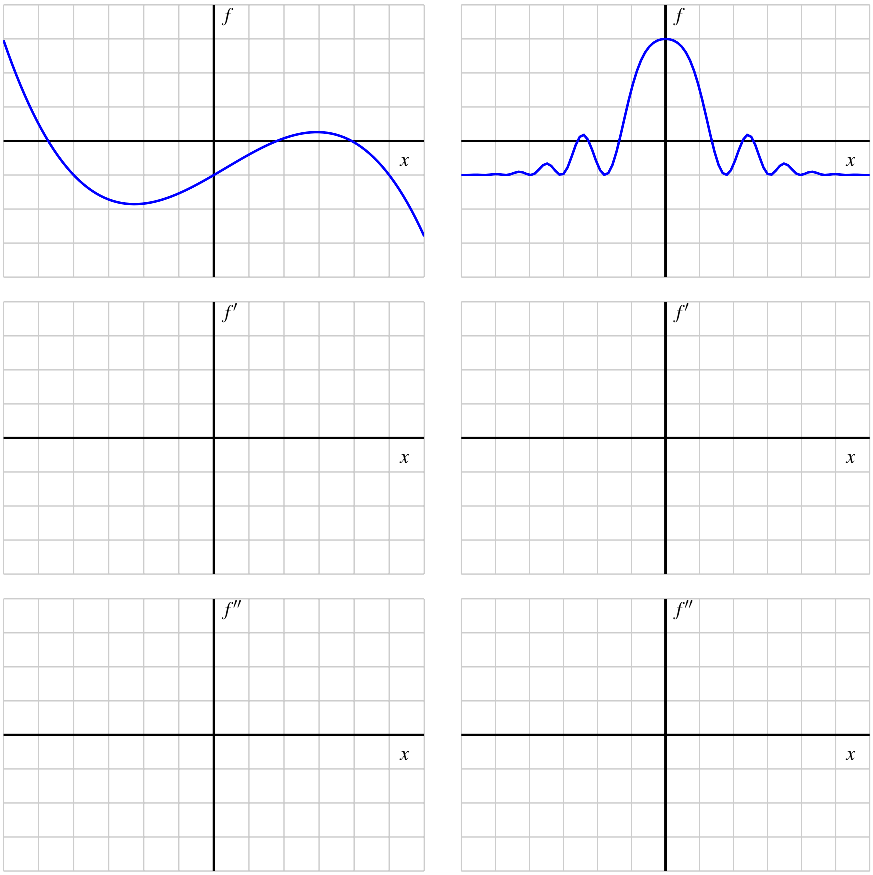
As we observed in Note 1.5.4, it's best *not* to simplify the units. For example, in Activity 1.6.2, where $s(t)$ represents the position of a car driving on a straight road, with t measured in minutes and $s(t)$ in thousands of feet, the units on $s'(t)$ are thousands of feet per minute, which represents the car's instantaneous velocity at time t. In addition, the units on $s''(t)$ are thousands of feet per minute per minute, which represents the car's acceleration at time t. Notice that a "square minute" is not a meaningful quantity when thinking about motion, so it's much better to use "per minute per minute". A similar observation holds from Activity 1.6.3, where $F(t)$ represents the temperature of a potato in degrees Fahrenheit at time t in minutes; in that setting, the units on $F''(t)$ are "degrees Fahrenheit per minute per minute", as the second derivative measures the instantaneous rate of change of the first derivative.

Activity 1.6.4. This activity builds on our experience and understanding of how to sketch the graph of f' given the graph of f.

In the provided figure, we given the respective graphs of two different functions f, sketch the corresponding graph of f' on the first axes below, and then sketch f'' on the second set of axes. In addition, for each, write several careful sentences in the spirit of those in Activity 1.6.2 that connect the behaviors of f, f', and f''. For instance, write something such as

> f' is __________ on the interval __________, which is connected to the fact that f is __________ on the same interval __________, and f'' is __________ on the interval.

but of course with the blanks filled in. Throughout, view the scale of the grid for the graph of f as being 1×1, and assume the horizontal scale of the grid for the graph of f' is identical to that for f. If you need to adjust the vertical scale on the axes for the graph of f' or f'', you should label that accordingly.



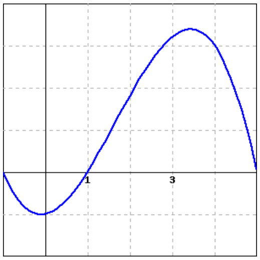
1.6.5 Summary

- A differentiable function f is increasing on an interval whenever its first derivative is positive, and decreasing whenever its first derivative is negative.
- By taking the derivative of the derivative of a function f, we arrive at the second derivative, f''. The second derivative measures the instantaneous rate of change of the first derivative. The sign of the second derivative tells us whether the slope of the tangent line to f is increasing or decreasing.
- A differentiable function is concave up whenever its first derivative is increasing (or equivalently whenever its second derivative is positive), and concave down whenever its first derivative is decreasing (or equivalently whenever its second derivative is negative). When a differentiable function is concave up on an interval, its tangent line always lies below the curve; when a differentiable function is concave down on an interval, its tangent line always lies above the curve. Examples of functions that are everywhere concave up are $y = x^2$ and $y = e^x$; examples of functions that are everywhere concave down are $y = -x^2$ and $y = -e^x$.

- The units on the second derivative are "units of output per unit of input per unit of input." Is is best not to try to simplify these units, as the second derivative's units and value tell us how the value of the derivative function is changing in response to changes in the input. In other words, the second derivative tells us the rate of change of the rate of change of the original function.

1.6.6 Exercises

1. Consider the function $f(x)$ graphed below.



For this function, are the following nonzero quantities positive or negative?

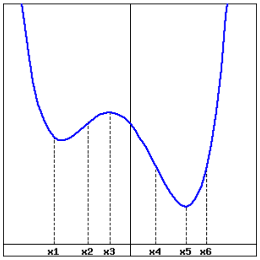
$f(0.5)$ is (□ ? □ positive □ negative)

$f'(0.5)$ is (□ ? □ positive □ negative)

$f''(0.5)$ is (□ ? □ positive □ negative)

(Because this is a multiple choice problem, it will not show which parts of the problem are correct or incorrect when you submit it.)

2. The graph of f' (***not*** f) is given below.



(Note that this is a graph of f', not a graph of f.)

At which of the marked values of x is

A. $f(x)$ greatest? $x =$ ________

B. $f(x)$ least? $x =$ ________

C. $f'(x)$ greatest? $x =$ ________

D. $f'(x)$ least? $x =$ ________

E. $f''(x)$ greatest? $x =$ ________

F. $f''(x)$ least? $x =$ ________

3. The position of a particle moving along the x-axis is given by $s(t) = 5t^2 + 6$. Use difference quotients to find the velocity $v(t)$ and acceleration $a(t)$, filling in the following expressions as you do so:

$v(t) = \lim_{h \to 0}$ [________________________________ / h] = ______________

$a(t) = \lim_{h \to 0}$ [________________________________ / h] = ______________

4. Let $P(t)$ represent the price of a share of stock of a corporation at time t. What does each of the following statements tell us about the signs of the first and second derivatives of $P(t)$?

(a) The price of the stock is rising slower and slower.

The first derivative of $P(t)$ is (□ ? □ positive □ zero □ negative)

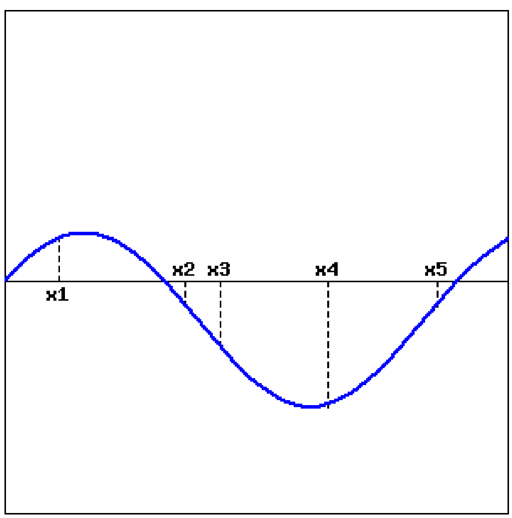
The second derivative of $P(t)$ is (□ ? □ positive □ zero □ negative)

(b) The price of the stock is close to maxing out.

The first derivative of $P(t)$ is (□ ? □ positive □ zero □ negative)

The second derivative of $P(t)$ is (□ ? □ positive □ zero □ negative)

5. Given the graph of $y = f(x)$ below, at which of the marked x-values can the following statements be true?



(For each question, enter your answer as a comma-separated list, e.g., ***x1,x3,x5****. Enter* ***none*** *if no points satisfy the given condition)*

A. $f(x) > 0$ at $x =$ ______________

B. $f'(x) > 0$ at $x =$ ______________

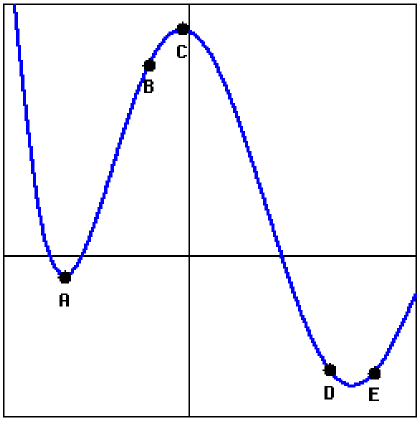
C. $f(x)$ is increasing at $x =$ ______________

D. $f'(x)$ is increasing at $x =$ ______________

E. The slope of $f(x)$ is negative at $x =$ ______________

F. The slope of $f'(x)$ is negative at $x =$ ____________

6. At exactly two of the labeled points in the figure below, which shows a function f, the derivative f' is zero; the second derivative f'' is not zero at any of the labeled points. Select the correct signs for each of f, f' and f'' at each marked point.



Point	A	B	C
f	(□ ? □ + □ 0 □ −)	(□ ? □ + □ 0 □ −)	(□ ? □ + □ 0 □ −)
f'	(□ ? □ + □ 0 □ −)	(□ ? □ + □ 0 □ −)	(□ ? □ + □ 0 □ −)
f''	(□ ? □ + □ 0 □ −)	(□ ? □ + □ 0 □ −)	(□ ? □ + □ 0 □ −)

Point	D	E
f	(□ ? □ + □ 0 □ −)	(□ ? □ + □ 0 □ −)
f'	(□ ? □ + □ 0 □ −)	(□ ? □ + □ 0 □ −)
f''	(□ ? □ + □ 0 □ −)	(□ ? □ + □ 0 □ −)

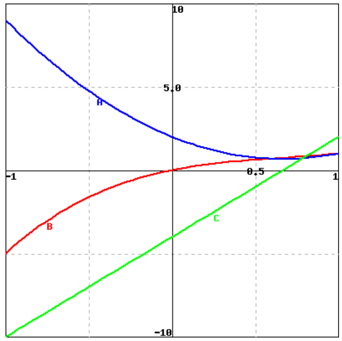
7. Suppose that an accelerating car goes from 0 mph to 68.2 mph in five seconds. Its velocity is given in the following table, converted from miles per hour to feet per second, so that all time measurements are in seconds. (Note: 1 mph is 22/15 ft/sec.) Find the average acceleration of the car over each of the first two seconds.

t (s)	0	1	2	3	4	5
$v(t)$ (ft/s)	0.00	34.09	59.09	77.27	90.91	100.00

average acceleration over the first second = ____________________ (□ ? □ ft □ s □ ft/s □ s/ft □ ft/s^2 □ s^2/ft □ ft^2/s □ s/ft^2 □ mph)

average acceleration over the second second = ____________________ (□ ? □ ft □ s □ ft/s □ s/ft □ ft/s^2 □ s^2/ft □ ft^2/s □ s/ft^2 □ mph)

8.



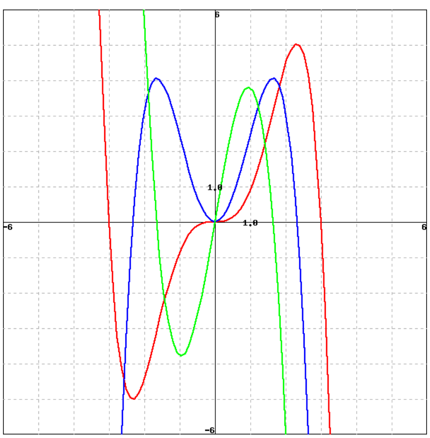
Identify the graphs A (blue), B(red) and C (green) as the graphs of a function and its derivatives:

____ is the graph of the function

____ is the graph of the function's first derivative

____ is the graph of the function's second derivative

9. The figure below shows three graphs: A (in blue), B (in red), and C (in green). Identify each curve as f, f', and f''.



1. f ___ 2. f' ___ 3. f'' ___

Note: *You can click on the graph to enlarge the images.*

10. Suppose that $y = f(x)$ is a twice-differentiable function such that f'' is continuous for which the following information is known: $f(2) = -3$, $f'(2) = 1.5$, $f''(2) = -0.25$.

a. Is f increasing or decreasing near $x = 2$? Is f concave up or concave down near $x = 2$?

b. Do you expect $f(2.1)$ to be greater than -3, equal to -3, or less than -3? Why?

c. Do you expect $f'(2.1)$ to be greater than 1.5, equal to 1.5, or less than 1.5? Why?

d. Sketch a graph of $y = f(x)$ near $(2, f(2))$ and include a graph of the tangent line.

11. For a certain function $y = g(x)$, its derivative is given by the function pictured in Figure 1.6.11.

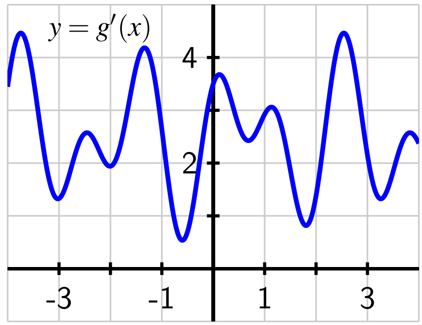


Figure 1.6.11 The graph of $y = g'(x)$.

a. What is the approximate slope of the tangent line to $y = g(x)$ at the point $(2, g(2))$?

b. How many real number solutions can there be to the equation $g(x) = 0$? Justify your conclusion fully and carefully by explaining what you know about how the graph of g must behave based on the given graph of g'.

c. On the interval $-3 < x < 3$, how many times does the concavity of g change? Why?

d. Use the provided graph to estimate the value of $g''(2)$.

12. A bungee jumper's height h (in feet) at time t (in seconds) is given in part by the table:

t	0.0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0
$h(t)$	200	184.2	159.9	131.9	104.7	81.8	65.5	56.8	55.5	60.4	69.8

t	5.5	6.0	6.5	7.0	7.5	8.0	8.5	9.0	9.5	10.0
$h(t)$	81.6	93.7	104.4	112.6	117.7	119.4	118.2	114.8	110.0	104.7

a. Use the given data to estimate $h'(4.5)$, $h'(5)$, and $h'(5.5)$. At which of these times is the bungee jumper rising most rapidly?

b. Use the given data and your work in (a) to estimate $h''(5)$.

c. What physical property of the bungee jumper does the value of $h''(5)$ measure? What are its units?

d. Based on the data, on what approximate time intervals is the function $y = h(t)$ concave down? What is happening to the velocity of the bungee jumper on these time intervals?

13. For each prompt that follows, sketch a possible graph of a function on the interval $-3 < x < 3$ that satisfies the stated properties.

a. $y = f(x)$ such that f is increasing on $-3 < x < 3$, concave up on $-3 < x < 0$, and concave down on $0 < x < 3$.

b. $y = g(x)$ such that g is increasing on $-3 < x < 3$, concave down on $-3 < x < 0$, and concave up on $0 < x < 3$.

c. $y = h(x)$ such that h is decreasing on $-3 < x < 3$, concave up on $-3 < x < -1$, neither concave up nor concave down on $-1 < x < 1$, and concave down on $1 < x < 3$.

d. $y = p(x)$ such that p is decreasing and concave down on $-3 < x < 0$ and is increasing and concave down on $0 < x < 3$.

1.7 Limits, continuity, and differentiability

Motivating Questions

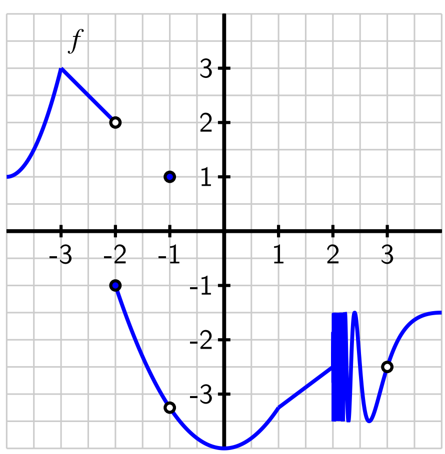
- What does it mean graphically to say that f has limit L as $x \to a$? How is this connected to having a left-hand limit at $x = a$ and having a right-hand limit at $x = a$?
- What does it mean to say that a function f is continuous at $x = a$? What role do limits play in determining whether or not a function is continuous at a point?
- What does it mean graphically to say that a function f is differentiable at $x = a$? How is this connected to the function being locally linear?
- How are the characteristics of a function having a limit, being continuous, and being differentiable at a given point related to one another?

1.7.1 Introduction

In Section 1.2, we learned how limits can be used to study the trending behavior of a function near a fixed input value. In this section, we aim to quantify how the function acts and how its values change near a particular point. If the function has a limit L at $x = a$, we will consider how the value of the function $f(a)$ is related to $\lim_{x \to a} f(x)$, and whether or not the function has a derivative $f'(a)$ at $x = a$.

Preview Activity 1.7.1. A function f is given by the graph in the following figure. Use the graph to answer each of the following questions.

Note: to the right of $x = 2$, the graph of f is exhibiting infinite oscillatory behavior similar to the function $\sin(\frac{\pi}{x})$. Assume that $f(2) = -2.5$.



(a) For each of the values $a = -3, -2, -1, 0, 1, 2, 3$, determine whether or not

$\lim_{x \to a} f(x)$ exists. If the function has a limit L at a given x-value, state the value of the limit using the notation $\lim_{x \to a} f(x) = L$. If the function does not have a limit at a given point, write a sentence to explain why.

(b) For each of the values of a from part (a) where f has a limit, determine the value of $f(a)$ at each such point, or explain why $f(a)$ is undefined. In addition, for each such a value, does $f(a)$ have the same value as $\lim_{x \to a} f(x)$?

(c) For each of the values $a = -3, -2, -1, 0, 1, 2, 3$, determine whether or not $f'(a)$ exists. In particular, based on the given graph, ask yourself if it is reasonable to say that f has a tangent line at $(a, f(a))$ for each of the given a-values. If so, visually estimate the slope of the tangent line to find the value of $f'(a)$.

1.7.2 Having a limit at a point

In Section 1.2, we learned that f has limit L as x approaches a provided that we can make the value of $f(x)$ as close to L as we like by taking x sufficiently close (but not equal to) a. If so, we write $\lim_{x \to a} f(x) = L$.

Essentially there are two behaviors that a function can exhibit near a point where it fails to have a limit. In Figure 1.7.2, at left we see a function f whose graph shows a jump at $a = 1$. If we let x approach 1 from the left side, the value of f approaches 2, but if we let x approach 1 from the right, the value of f tends to 3. Because the value of f does not approach a single number as x gets arbitrarily close to 1 from both sides, we know that f does not have a limit at $a = 1$.

For such cases, we introduce the notion of *left* and *right* (or *one-sided*) limits.

Definition 1.7.1 We say that f *has limit L_1 as x approaches a from the left* and write

$$\lim_{x \to a^-} f(x) = L_1$$

provided that we can make the value of $f(x)$ as close to L_1 as we like by taking x sufficiently close to a while always having $x < a$. We call L_1 the left-hand limit of f as x approaches a. Similarly, we say f *has limit L_1 as x approaches a from the right* and write

$$\lim_{x \to a^+} f(x) = L_2$$

provided that we can make the value of $f(x)$ as close to L_2 as we like by taking x sufficiently close to a while always having $x > a$. We call L_1 the right-hand limit of f as x approaches a.

◊

In the graph of the function f in Figure 1.7.2, we see that

$$\lim_{x \to 1^-} f(x) = 2 \text{ and } \lim_{x \to 1^+} f(x) = 3.$$

Precisely because the left and right limits are not equal, the overall limit of f as $x \to 1$ fails to exist.

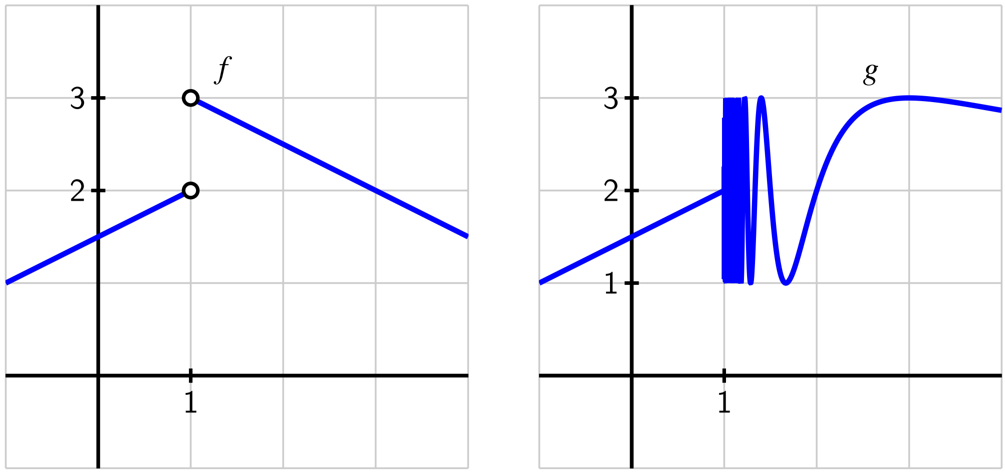


Figure 1.7.2 Functions f and g that each fail to have a limit at $a = 1$.

For the function g pictured at right in Figure 1.7.2, the function fails to have a limit at $a = 1$ for a different reason. While the function does not have a jump in its graph at $a = 1$, it is still not the case that g approaches a single value as x approaches 1. In particular, due to the infinitely oscillating behavior of g to the right of $a = 1$, we say that the right-hand limit of g as $x \to 1^+$ does not exist, and thus $\lim_{x\to 1} g(x)$ does not exist.

To summarize, if either a left- or right-hand limit fails to exist or if the left- and right-hand limits are not equal to each other, the overall limit does not exist.

When a function f has a limit.

A function f has limit L as $x \to a$ if and only if

$$\lim_{x\to a^-} f(x) = L = \lim_{x\to a^+} f(x).$$

That is, a function has a limit at $x = a$ if and only if both the left- and right-hand limits at $x = a$ exist and have the same value.

The function f given in the figure in Preview Activity 1.7.1 fails to have a limit at only two values: at $a = -2$ (where the left- and right-hand limits are 2 and -1, respectively) and at $x = 2$, where $\lim_{x\to 2^+} f(x)$ does not exist. Note well that even at values such as $a = -1$ and $a = 0$ where there are holes in the graph, the limit still exists.

Activity 1.7.2. Consider a function that is piecewise-defined according to the formula

$$f(x) = \begin{cases} 3(x+2)+2 & \text{for } -3 < x < -2 \\ \frac{2}{3}(x+2)+1 & \text{for } -2 \le x < -1 \\ \frac{2}{3}(x+2)+1 & \text{for } -1 < x < 1 \\ 2 & \text{for } x = 1 \\ 4-x & \text{for } x > 1 \end{cases}$$

Use the given formula to answer the following questions.

(a) For each of the values $a = -2, -1, 0, 1, 2$, compute $f(a)$.

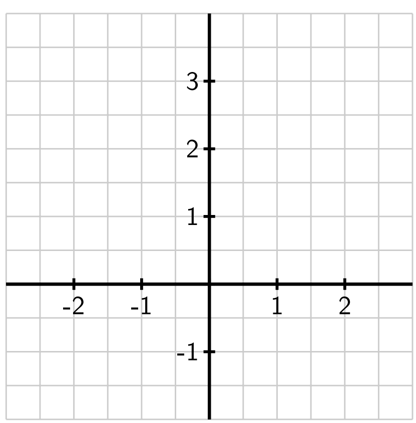
(b) For each of the values $a = -2, -1, 0, 1, 2$, determine $\lim_{x \to a^-} f(x)$ and $\lim_{x \to a^+} f(x)$.

(c) For each of the values $a = -2, -1, 0, 1, 2$, determine $\lim_{x \to a} f(x)$. If the limit fails to exist, explain why by discussing the left- and right-hand limits at the relevant a-value.

(d) For which values of a is the following statement true?

$$\lim_{x \to a} f(x) \neq f(a)$$

(e) On the axes provided, sketch an accurate, labeled graph of $y = f(x)$. Be sure to carefully use open circles (○) and filled circles (●) to represent key points on the graph, as dictated by the piecewise formula.



1.7.3 Being continuous at a point

Intuitively, a function is continuous if we can draw its graph without ever lifting our pencil from the page. Alternatively, we might say that the graph of a continuous function has no jumps or holes in it. In Figure 1.7.3 we consider three functions that have a limit at $a = 1$, and use them to make the idea of continuity more precise.

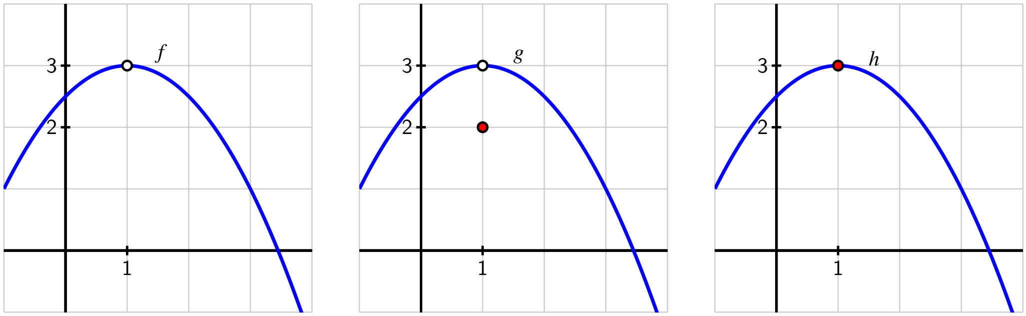


Figure 1.7.3 Functions f, g, and h that demonstrate subtly different behaviors at $a = 1$.

First consider the function in the left-most graph. Note that $f(1)$ is not defined, which leads to the resulting hole in the graph of f at $a = 1$. We will naturally say that f is *not continuous at* $a = 1$. For the function g, we observe that while $\lim_{x\to 1} g(x) = 3$, the value of $g(1) = 2$, and thus the limit does not equal the function value. Here, too, we will say that g is *not continuous*, even though the function is defined at $a = 1$. Finally, the function h appears to be the most well-behaved of all three, since at $a = 1$ its limit and its function value agree. That is,

$$\lim_{x\to 1} h(x) = 3 = h(1).$$

With no hole or jump in the graph of h at $a = 1$, we say that h is *continuous* there. More formally, we make the following definition.

Definition 1.7.4 A function f is **continuous at** $x = a$ provided that

a. f has a limit as $x \to a$,

b. f is defined at $x = a$, and

c. $\lim_{x\to a} f(x) = f(a)$.

◊

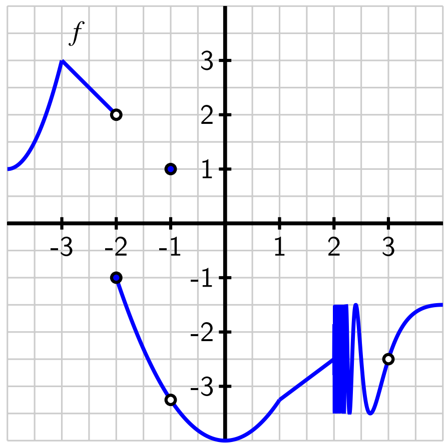
Conditions (a) and (b) are technically contained implicitly in (c), but we state them explicitly to emphasize their individual importance. The definition says that a function is continuous at $x = a$ provided that its limit as $x \to a$ exists and equals its function value at $x = a$. If a function is continuous at every point in an interval $[a, b]$, we say the function is "continuous on $[a, b]$." If a function is continuous at every point in its domain, we simply say the function is "continuous." Thus, continuous functions are particularly nice: to evaluate the limit of a continuous function at a point, all we need to do is evaluate the function.

For example, consider $p(x) = x^2 - 2x + 3$. It can be proved that every polynomial is a continuous function at every real number, and thus if we would like to know $\lim_{x\to 2} p(x)$, we simply compute

$$\lim_{x\to 2}(x^2 - 2x + 3) = 2^2 - 2 \cdot 2 + 3 = 3.$$

This approach of substituting an input value to evaluate a limit works whenever we know that the function being considered is continuous. Besides polynomial functions, all exponential functions and the sine and cosine functions are continuous at every point, as are many other familiar functions and combinations thereof.

Activity 1.7.3. This activity builds on your work in Preview Activity 1.7.1, using the same function f as given by the graph that is repeated in the following figure. Assume that $f(2) = -2.5$.



(a) At which values of a does $\lim_{x\to a} f(x)$ not exist?

(b) At which values of a is $f(a)$ not defined?

(c) At which values of a does f have a limit, but $\lim_{x\to a} f(x) \neq f(a)$?

(d) State all values of a for which f is not continuous at $x = a$.

(e) Which condition is stronger, and hence implies the other: f has a limit at $x = a$ or f is continuous at $x = a$? Explain, and hence complete the following sentence: "If f__________ at $x = a$, then f__________ at $x = a$," where you complete the blanks with *has a limit* and *is continuous*, using each phrase once.

1.7.4 Being differentiable at a point

We recall that a function f is said to be differentiable at $x = a$ if $f'(a)$ exists. Moreover, for $f'(a)$ to exist, we know that the function $y = f(x)$ must have a tangent line at the point $(a, f(a))$, since $f'(a)$ is precisely the slope of this line. In order to even ask if f has a tangent line at $(a, f(a))$, it is necessary that f be continuous at $x = a$: if f fails to have a limit at $x = a$, if $f(a)$ is not defined, or if $f(a)$ does not equal the value of $\lim_{x\to a} f(x)$, then it doesn't make sense to talk about a tangent line to the curve at this point.

Indeed, it can be proved formally that if a function f is differentiable at $x = a$, then it must be continuous at $x = a$. So, if f is not continuous at $x = a$, then it is automatically the case that f is not differentiable there. For example, in Figure 1.7.3, both f and g fail to be differentiable at $x = 1$ because neither function is continuous at $x = 1$. But can a function fail to be differentiable at a point where the function is continuous?

In Figure 1.7.5, the function has a sharp corner at a point. For the pictured function f, we observe that f is clearly continuous at $a = 1$, since $\lim_{x\to 1} f(x) = 1 = f(1)$.

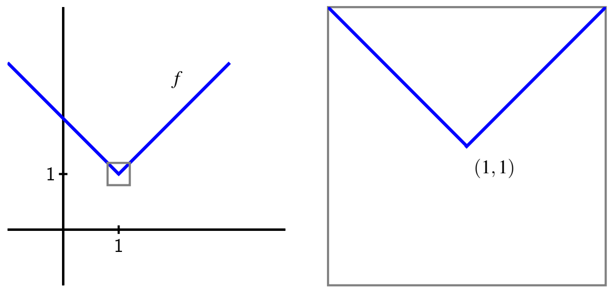


Figure 1.7.5 A function f that is continuous at $a = 1$ but not differentiable at $a = 1$; at right, we zoom in on the point $(1, 1)$ in a magnified version of the box in the left-hand plot.

But the function f in Figure 1.7.5 is not differentiable at $a = 1$ because $f'(1)$ fails to exist. One way to see this is to observe that $f'(x) = -1$ for every value of x that is less than 1, while $f'(x) = +1$ for every value of x that is greater than 1. That makes it seem that either $+1$ or -1 would be equally good candidates for the value of the derivative at $x = 1$. Alternately, we could use the limit definition of the derivative to attempt to compute $f'(1)$, and discover that the derivative does not exist. Finally, we can see visually that the function f in Figure 1.7.5 does not have a tangent line. When we zoom in on $(1, 1)$ on the graph of f, no matter how closely we examine the function, it will always look like a "V", and never like a single line, which tells us there is no possibility for a tangent line there.

If a function does have a tangent line at a given point, when we zoom in on the point of tangency, the function and the tangent line should appear essentially indistinguishable[1]. Conversely, if we zoom in on a point and the function looks like a single straight line, then the function should have a tangent line there, and thus be differentiable. Hence, a function that is differentiable at $x = a$ will, up close, look more and more like its tangent line at $(a, f(a))$. Therefore, we say that a function that is differentiable at $x = a$ is *locally linear*.

To summarize the preceding discussion of differentiability and continuity, we make several important observations.

When a function is differentiable and/or continuous.

- If f is differentiable at $x = a$, then f is continuous at $x = a$. Equivalently, if f fails to be continuous at $x = a$, then f will not be differentiable at $x = a$.
- A function can be continuous at a point, but not be differentiable there. In particular, a function f is not differentiable at $x = a$ if the graph has a sharp

[1]See, for instance, this interactive graphic, where zooming in shows the increasing similarity between the tangent line and the curve.

corner (or *cusp*) at the point $(a, f(a))$.

- If f is differentiable at $x = a$, then f is locally linear at $x = a$. That is, when a function is differentiable, it looks linear when viewed up close because it resembles its tangent line there.

Activity 1.7.4. In this activity, we explore two different functions and classify the points at which each is not differentiable.

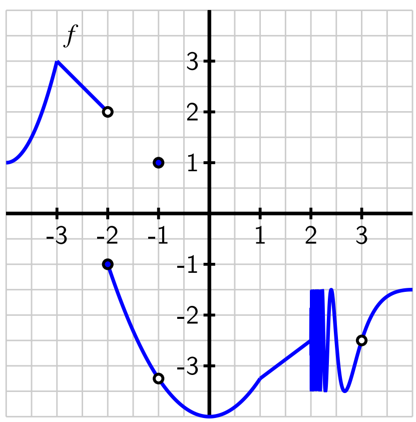
For (a)-(c), let g be the function given by the rule $g(x) = |x|$. Note that for part (d) we use the function f given by the graph in (d).

(a) Reasoning graphically (that is, by discussing the graph of $g(x) = |x|$), explain why g is differentiable at every point x such that $x \neq 0$.

(b) Use the limit definition of the derivative to show that $g'(0) = \lim_{h\to 0} \frac{|h|}{h}$.

(c) Explain why $g'(0)$ fails to exist by using small positive and negative values of h.

(d) Let f be the function that we have previously explored in Preview Activity 1.7.1, whose graph is given again in the following figure.



State all values of a for which f is not differentiable at $x = a$. For each, provide a reason for your conclusion.

(e) True or false: if a function p is differentiable at $x = b$, then $\lim_{x\to b} p(x)$ must exist. Why?

1.7.5 Summary

- A function f has limit L as $x \to a$ if and only if f has a left-hand limit at $x = a$, f has a right-hand limit at $x = a$, and the left- and right-hand limits are equal. Visually, this means that there can be a hole in the graph at $x = a$, but the function must approach the same single value from either side of $x = a$.

- A function f is continuous at $x = a$ whenever $f(a)$ is defined, f has a limit as $x \to a$, and the value of the limit and the value of the function agree. This guarantees that there is not a hole or jump in the graph of f at $x = a$.

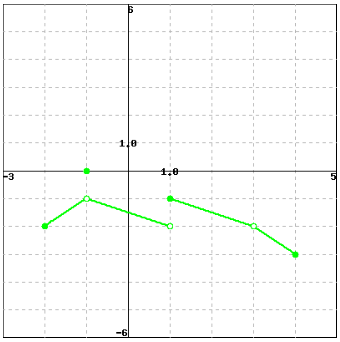
- A function f is differentiable at $x = a$ whenever $f'(a)$ exists, which means that f has a tangent line at $(a, f(a))$ and thus f is locally linear at $x = a$. Informally, this means that the function looks like a line when viewed up close at $(a, f(a))$ and that there is not a corner point or cusp at $(a, f(a))$.

- Of the three conditions discussed in this section (having a limit at $x = a$, being continuous at $x = a$, and being differentiable at $x = a$), the strongest condition is being differentiable, and the next strongest is being continuous. In particular, if f is differentiable at $x = a$, then f is also continuous at $x = a$, and if f is continuous at $x = a$, then f has a limit at $x = a$.

1.7.6 Exercises

1. Let F be the function below.

If you are having a hard time seeing the picture clearly, click on the picture. It will expand to a larger picture on its own page so that you can inspect it more clearly.



Evaluate each of the following expressions.

Note: Enter 'DNE' if the limit does not exist or is not defined.

a) $\lim_{x \to -1^-} F(x) =$ ________

$\lim_{x \to -1^+} F(x) =$ ________

$\lim_{x \to -1} F(x) =$ ________

$F(-1) =$ ________

b) $\lim_{x \to 1^-} F(x) =$ ________

$\lim_{x \to 1^+} F(x) =$ ________

$\lim_{x\to 1} F(x) =$ ________

$F(1) =$ ________

c) $\lim_{x\to 3^-} F(x) =$ ________

$\lim_{x\to 3^+} F(x) =$ ________

$\lim_{x\to 3} F(x) =$ ________

$F(3) =$ ________

2. For the function

$$f(x) = \begin{cases} 3x-3, & 0 \le x < 4 \\ 3, & x = 4 \\ x^2 - 8x + 25, & 4 < x \end{cases}$$

use algebra to find each of the following limits:

$\lim_{x\to 4^+} f(x) =$ ______________

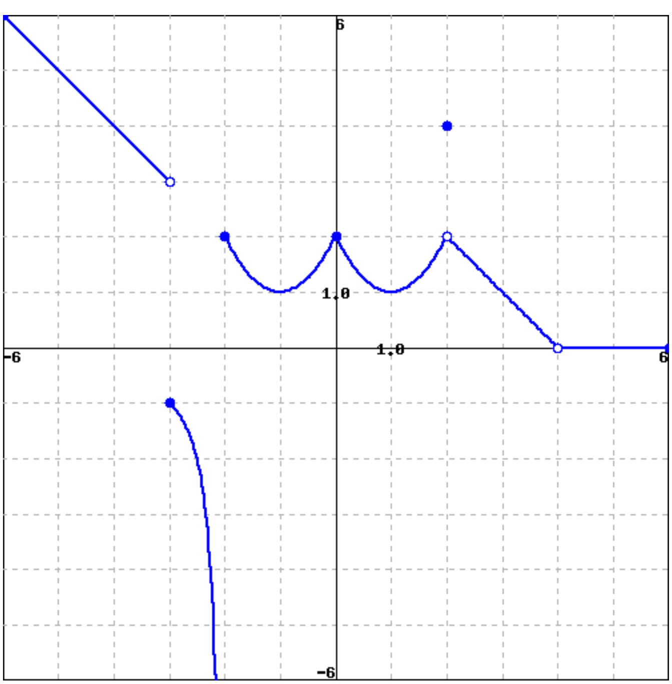
$\lim_{x\to 4^-} f(x) =$ ______________

$\lim_{x\to 4} f(x) =$ ______________

(For each, enter ***DNE*** *if the limit does not exist.)*

Sketch a graph of $f(x)$ to confirm your answers.

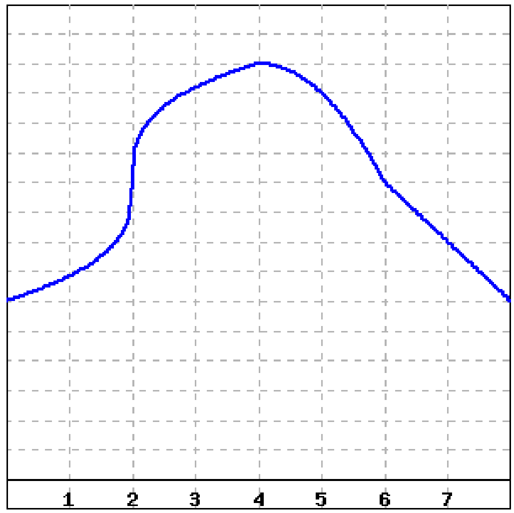
3. Use the given graph of the function to find the x-values for which f is discontinuous.



Answer (separate by commas): $x =$ ________________________________

Note: *You can click on the graph to enlarge the image.*

4. Consider the function graphed below.



At what x-values does the function appear to not be continuous? $x =$ __________

At what x-values does the function appear to not be differentiable? $x =$ __________

(Enter ***none*** *if there are no x-values that apply; enter x-values as a comma-separated list, e.g.,* ***1,3,5****.)*

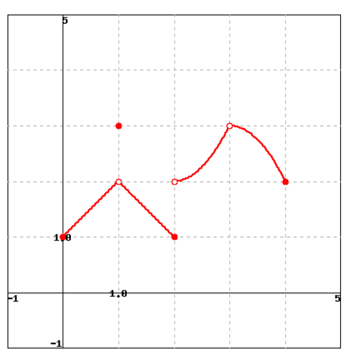
5. If possible, choose k so that the following function is continuous on any interval:

$$f(x) = \begin{cases} \frac{7x^5-35x^4}{x-5} & x \neq 5 \\ k & x = 5. \end{cases}$$

$k =$ ______________

(If no k will make the function continuous, enter ***none****)*

6. Let F be the function below



(Click on graph to enlarge)

7. Consider the graph of the function $y = p(x)$ that is provided in Figure 1.7.6. Assume that each portion of the graph of p is a straight line, as pictured.

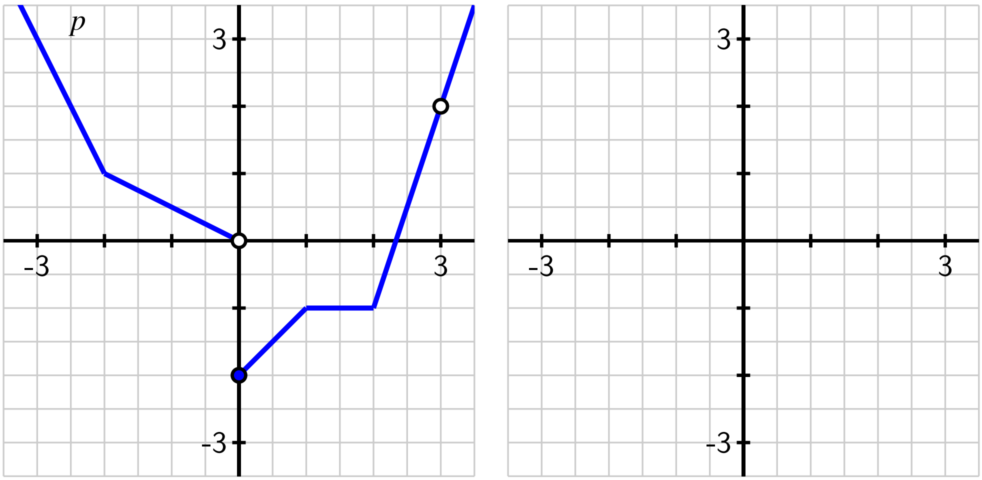


Figure 1.7.6 At left, the piecewise linear function $y = p(x)$. At right, axes for plotting $y = p'(x)$.

a. State all values of a for which $\lim_{x\to a} p(x)$ does not exist.

b. State all values of a for which p is not continuous at a.

c. State all values of a for which p is not differentiable at $x = a$.

d. On the axes provided in Figure 1.7.6, sketch an accurate graph of $y = p'(x)$.

8. For each of the following prompts, give an example of a function that satisfies the stated criteria. A formula or a graph, with reasoning, is sufficient for each. If no such example is possible, explain why.

a. A function f that is continuous at $a = 2$ but not differentiable at $a = 2$.

b. A function g that is differentiable at $a = 3$ but does not have a limit at $a = 3$.

c. A function h that has a limit at $a = -2$, is defined at $a = -2$, but is not continuous at $a = -2$.

d. A function p that satisfies all of the following:

- $p(-1) = 3$ and $\lim_{x\to -1} p(x) = 2$
- $p(0) = 1$ and $p'(0) = 0$
- $\lim_{x\to 1} p(x) = p(1)$ and $p'(1)$ does not exist

9. Let $h(x)$ be a function whose derivative $y = h'(x)$ is given by the graph on the right in Figure 1.7.7.

a. Based on the graph of $y = h'(x)$, what can you say about the behavior of the function $y = h(x)$?

b. At which values of x is $y = h'(x)$ not defined? What behavior does this lead you to expect to see in the graph of $y = h(x)$?

c. Is it possible for $y = h(x)$ to have points where h is not continuous? Explain your answer.

d. On the axes provided at left, sketch at least two distinct graphs that are possible functions $y = h(x)$ that each have a derivative $y = h'(x)$ that matches the provided graph at right. Explain why there are multiple possibilities for $y = h(x)$.

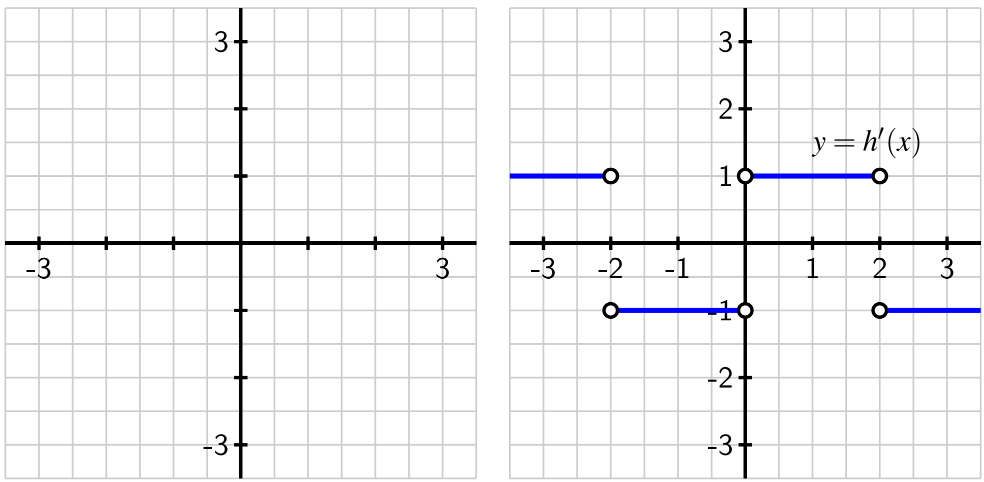


Figure 1.7.7 Axes for plotting $y = h(x)$ and, at right, the graph of $y = h'(x)$.

10. Consider the function $g(x) = \sqrt{|x|}$.

a. Use a graph to explain visually why g is not differentiable at $x = 0$.

b. Use the limit definition of the derivative to show that

$$g'(0) = \lim_{h \to 0} \frac{\sqrt{|h|}}{h}.$$

c. Investigate the value of $g'(0)$ by estimating the limit in (b) using small positive and negative values of h. For instance, you might compute $\frac{\sqrt{|-0.01|}}{0.01}$. Be sure to use several different values of h (both positive and negative), including ones closer to 0 than 0.01. What do your results tell you about $g'(0)$?

d. Use your graph in (a) to sketch an approximate graph of $y = g'(x)$.

1.8 The tangent line approximation

Motivating Questions

- What is the formula for the general tangent line approximation to a differentiable function $y = f(x)$ at the point $(a, f(a))$?
- What is the principle of local linearity and what is the local linearization of a differentiable function f at a point $(a, f(a))$?
- How does knowing just the tangent line approximation tell us information about the behavior of the original function itself near the point of approximation? How does knowing the second derivative's value at this point provide us additional knowledge of the original function's behavior?

1.8.1 Introduction

Among all functions, linear functions are simplest. One of the powerful consequences of a function $y = f(x)$ being differentiable at a point $(a, f(a))$ is that, up close, the function $y = f(x)$ is locally linear and looks like its tangent line at that point. In certain circumstances, this allows us to approximate the original function f with a simpler function L that is linear: this can be advantageous when we have limited information about f or when f is computationally or algebraically complicated. We will explore all of these situations in what follows.

It is essential to recall that when f is differentiable at $x = a$, the value of $f'(a)$ provides the slope of the tangent line to $y = f(x)$ at the point $(a, f(a))$. If we know both a point on the line and the slope of the line we can find the equation of the tangent line and write the equation in point-slope form[1].

Preview Activity 1.8.1. Consider the function $y = g(x) = -x^2 + 3x + 2$.

(a) Use the limit definition of the derivative to compute a formula for $y = g'(x)$.

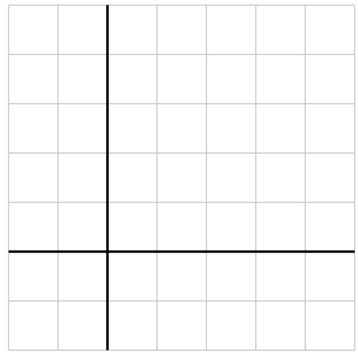
(b) Determine the slope of the tangent line to $y = g(x)$ at the value $x = 2$.

(c) Compute $g(2)$.

(d) Find an equation for the tangent line to $y = g(x)$ at the point $(2, g(2))$. Write your result in point-slope form.

(e) On the axes provided below, sketch an accurate, labeled graph of $y = g(x)$ along with its tangent line at the point $(2, g(2))$.

[1] Recall that a line with slope m that passes through (x_0, y_0) has equation $y - y_0 = m(x - x_0)$, and this is the *point-slope form* of the equation.



1.8.2 The tangent line

Given a function f that is differentiable at $x = a$, we know that we can determine the slope of the tangent line to $y = f(x)$ at $(a, f(a))$ by computing $f'(a)$. The equation of the resulting tangent line is given in point-slope form by

$$y - f(a) = f'(a)(x - a) \text{ or } y = f'(a)(x - a) + f(a).$$

Note well: there is a major difference between $f(a)$ and $f(x)$ in this context. The former is a constant that results from using the given fixed value of a, while the latter is the general expression for the rule that defines the function. The same is true for $f'(a)$ and $f'(x)$: we must carefully distinguish between these expressions. Each time we find the tangent line, we need to evaluate the function and its derivative at a fixed a-value.

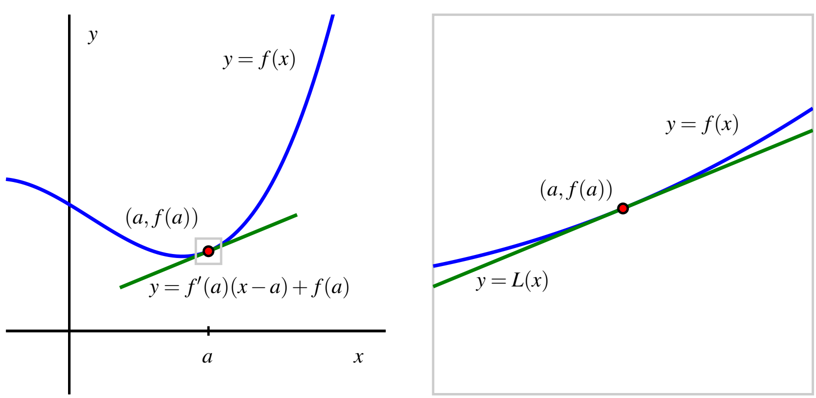


Figure 1.8.1 A function $y = f(x)$ and its tangent line at the point $(a, f(a))$: at left, from a distance, and at right, up close. At right, we label the tangent line function by $y = L(x)$ and observe that for x near a, $f(x) \approx L(x)$.

In Figure 1.8.1, we see the graph of a function f and its tangent line at the point $(a, f(a))$. Notice how when we zoom in we see the local linearity of f more clearly highlighted. The function and its tangent line are nearly indistinguishable up close. Local linearity can also be seen dynamically in this interactive graphic[2].

1.8.3 The local linearization

A slight change in perspective and notation will enable us to be more precise in discussing how the tangent line approximates f near $x = a$. By solving for y, we can write the equation for the tangent line as

$$y = f'(a)(x - a) + f(a)$$

This line is itself a function of x. Replacing the variable y with the expression $L(x)$, we call

$$L(x) = f'(a)(x - a) + f(a)$$

the *local linearization of* f at the point $(a, f(a))$. In this notation, $L(x)$ is nothing more than a new name for the tangent line. As we saw above, for x close to a, $f(x) \approx L(x)$.

Example 1.8.2 Suppose that a function $y = f(x)$ has its tangent line approximation given by $L(x) = 3 - 2(x - 1)$ at the point $(1, 3)$, but we do not know anything else about the function f. How can we estimate the value of $f(x)$ for x near 1?

Solution. To estimate a value of $f(x)$ for x near 1, such as $f(1.2)$, we can use the fact that $f(1.2) \approx L(1.2)$ and hence

$$f(1.2) \approx L(1.2) = 3 - 2(1.2 - 1) = 3 - 2(0.2) = 2.6.$$

◊

We emphasize that $y = L(x)$ is simply a new name for the tangent line function. Using this new notation and our observation that $L(x) \approx f(x)$ for x near a, it follows that we can write

$$f(x) \approx f(a) + f'(a)(x - a) \text{ for } x \text{ near } a.$$

Activity 1.8.2. Suppose it is known that for a given differentiable function $y = g(x)$, its local linearization at the point where $a = -1$ is given by $L(x) = -2 + 3(x + 1)$.

(a) Compute the values of $L(-1)$ and $L'(-1)$.

(b) What must be the values of $g(-1)$ and $g'(-1)$? Why?

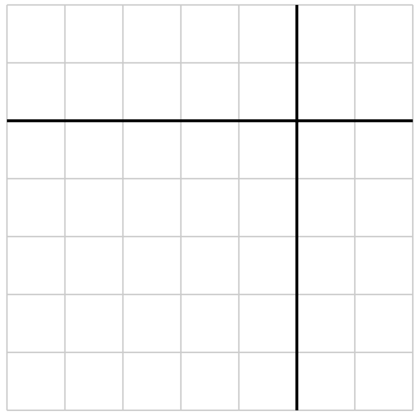
(c) Do you expect the value of $g(-1.03)$ to be greater than or less than the value of $g(-1)$? Why?

(d) Use the local linearization to estimate the value of $g(-1.03)$.

(e) Suppose that you also know that $g''(-1) = 2$. What does this tell you about the graph of $y = g(x)$ at $a = -1$?

[2] gvsu.edu/s/6J

(f) For x near -1, sketch the graph of the local linearization $y = L(x)$ as well as a possible graph of $y = g(x)$ on the axes provided.



From Activity 1.8.2, we see that the local linearization $y = L(x)$ is a linear function that shares two important values with the function $y = f(x)$ that it is derived from. In particular,

- because $L(x) = f(a) + f'(a)(x - a)$, it follows that $L(a) = f(a)$; and
- because L is a linear function, its derivative is its slope.

Hence, $L'(x) = f'(a)$ for every value of x, and specifically $L'(a) = f'(a)$. Therefore, we see that L is a linear function that has both the same value and the same slope as the function f at the point $(a, f(a))$.

Thus, if we know the linear approximation $y = L(x)$ for a function, we know the original function's value and its slope at the point of tangency. What remains unknown, however, is the shape of the function f at the point of tangency. There are essentially four possibilities, as shown in Figure 1.8.3.

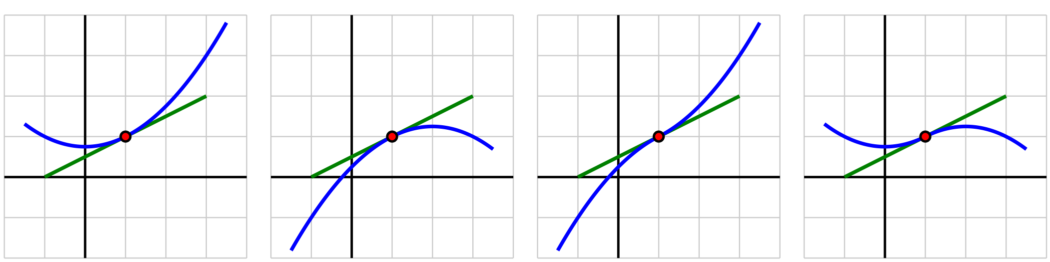


Figure 1.8.3 Four possible graphs for a nonlinear differentiable function and how it can be situated relative to its tangent line at a point.

These possible shapes result from the fact that there are three options for the value of the second derivative: either $f''(a) < 0$, $f''(a) = 0$, or $f''(a) > 0$.

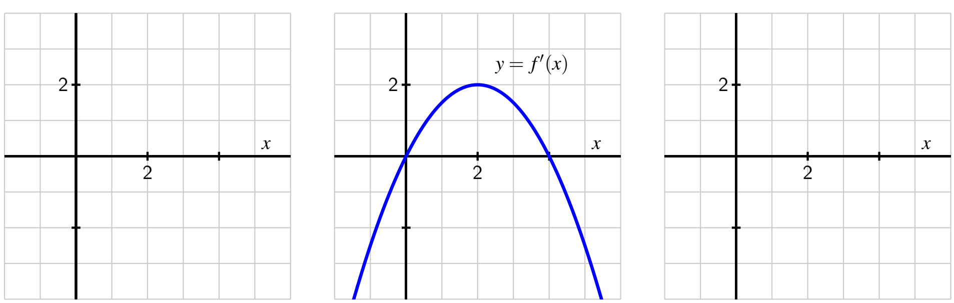
- If $f''(a) > 0$, then we know the graph of f is concave up, and we see the first possibility on the left, where the tangent line lies entirely below the curve.

- If $f''(a) < 0$, then f is concave down and the tangent line lies above the curve, as shown in the second figure.
- If $f''(a) = 0$ and f'' changes sign at $x = a$, the concavity of the graph will change, and we will see either the third or fourth figure.[3]
- A fifth option (which is not very interesting) can occur if the function f itself is linear, so that $f(x) = L(x)$ for all values of x.

The plots in Figure 1.8.3 highlight yet another important thing that we can learn from the concavity of the graph near the point of tangency: whether the tangent line lies above or below the curve itself. This is key because it tells us whether or not the tangent line approximation's values will be too large or too small in comparison to the true value of f. For instance, in the first situation in the leftmost plot in Figure 1.8.3 where $f''(a) > 0$, because the tangent line falls below the curve, we know that $L(x) \leq f(x)$ for all values of x near a.

Activity 1.8.3. This activity concerns a function $f(x)$ about which the following information is known:

- f is a differentiable function defined at every real number x
- $f(2) = -1$
- $y = f'(x)$ has its graph given in the following figure.



Your overall task is to determine as much information as possible about f (especially near the value $a = 2$) by responding to the questions below.

(a) Find a formula for the tangent line approximation, $L(x)$, to f at the point $(2, -1)$.

(b) Use the tangent line approximation to estimate the value of $f(2.07)$. Show your work carefully and clearly.

(c) Sketch a graph of $y = f''(x)$ on the righthand grid in the provided figure; label it appropriately. Write a sentence to explain why your graph looks the way it does.

[3]It is possible that $f''(a) = 0$ but f'' does not change sign at $x = a$, in which case the graph will look like one of the first two options.

(d) Is the slope of the tangent line to $y = f(x)$ increasing, decreasing, or neither when $x = 2$? Explain.

(e) Sketch a possible graph of $y = f(x)$ near $x = 2$ on the lefthand grid in the provided figure. Include a sketch of $y = L(x)$ (found in part (a)). Explain how you know the graph of $y = f(x)$ looks like you have drawn it.

(f) Does your estimate in (b) over- or under-estimate the true value of $f(2.07)$? Why?

The idea that a differentiable function looks linear and can be well-approximated by a linear function is an important one that finds wide application in calculus. For example, by approximating a function with its local linearization, it is possible to develop an effective algorithm to estimate the zeroes of a function. Local linearity also helps us to make further sense of certain challenging limits. For instance, we have seen that the limit

$$\lim_{x \to 0} \frac{\sin(x)}{x}$$

is indeterminate, because both its numerator and denominator tend to 0. While there is no algebra that we can do to simplify $\frac{\sin(x)}{x}$, it is straightforward to show that the linearization of $f(x) = \sin(x)$ at the point $(0, 0)$ is given by $L(x) = x$. Hence, for values of x near 0, $\sin(x) \approx x$, and therefore

$$\frac{\sin(x)}{x} \approx \frac{x}{x} = 1,$$

which makes plausible the fact that

$$\lim_{x \to 0} \frac{\sin(x)}{x} = 1.$$

Another important question — one that we study in depth in second semester calculus — regards how accurate the tangent line approximation remains as we move away from the point of tangency. We explore some related ideas in the next activity.

Activity 1.8.4. Consider $f(x) = 0.1x^2 - 1.1x + 3.4$ and $g(x) = -0.5x^2 + 2.5x - 2$.

(a) Compute $f(3)$, $f'(3)$, $g(3)$, and $g'(3)$. What do you observe?

(b) Determine $L(x)$, the local linear approximation to $f(x)$ at $x = 3$. Explain why this function is also the local linear approximation to $g(x)$ at $x = 3$.

(c) Complete the following table of values for $f(x)$, $g(x)$, and $L(x)$:

Table 1.8.4 Values of $f(x)$, $g(x)$, and $L(x)$.

x	2.9	2.99	2.999	3	3.001	3.01	3.1
$f(x)$							
$g(x)$							
$L(x)$							

What do you notice about how $L(x)$ approximates $f(x)$ and $g(x)$ near $x = 3$?

(d) On the axes provided in Figure 1.8.5, plot $f(x)$, $g(x)$, and $L(x)$ near $x = 3$. In addition, compute $f''(3)$, $g''(3)$, and $L''(3)$. How do the second derivative values and the graph explain why $L(x)$ is a better approximation for one of the functions than it is for the other?

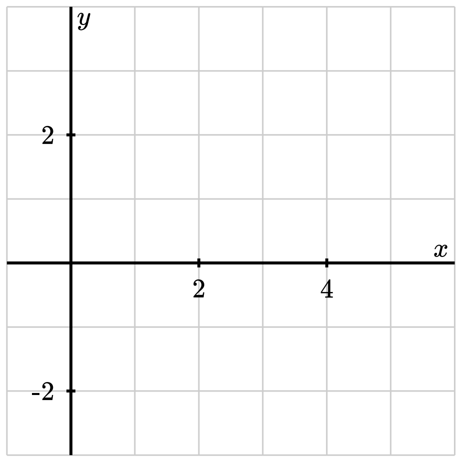


Figure 1.8.5 Axes for plotting $f(x)$, $g(x)$, and $L(x)$ near $x = 3$.

1.8.4 Summary

- The tangent line to a differentiable function $y = f(x)$ at the point $(a, f(a))$ is given in point-slope form by the equation

$$y - f(a) = f'(a)(x - a).$$

- The principle of local linearity tells us that if we zoom in on a point where a function $y = f(x)$ is differentiable, the function will be indistinguishable from its tangent line. That is, a differentiable function looks linear when viewed up close. We rename the tangent line to be the function $y = L(x)$, where $L(x) = f(a) + f'(a)(x - a)$. Thus, $f(x) \approx L(x)$ for all x near $x = a$.

- If we know the tangent line approximation $L(x) = f(a) + f'(a)(x - a)$ to a function $y = f(x)$, then because $L(a) = f(a)$ and $L'(a) = f'(a)$, we also know the values of both the function and its derivative at the point where $x = a$. In other words, the linear approximation tells us the height and slope of the original function. If, in addition, we know the value of $f''(a)$, we then know whether the tangent line lies above or below the graph of $y = f(x)$, depending on the concavity of f.

1.8.5 Exercises

1. Suppose that $f(x)$ is a function with $f(155) = 28$ and $f'(155) = 8$. Estimate $f(152)$.

$f(152) \approx$ ________________

2. Use linear approximation, i.e. the tangent line, to approximate $\frac{1}{0.103}$ as follows: Let $f(x) = \frac{1}{x}$ and find the equation of the tangent line to $f(x)$ at a "nice" point near 0.103. Then use this to approximate $\frac{1}{0.103}$.

$\frac{1}{0.103} \approx$ ________________

3. Use linear approximation, i.e. the tangent line, to approximate 2.9^2 as follows:

Let $f(x) = x^2$. The equation of the tangent line to $f(x)$ at $x = 3$ can be written in the form $y = mx + b$ where:

$m =$ ________________ and

$b =$ ________________

Using this, we find our approximation for 2.9^2 is ________________

4. Use linear approximation to approximate $\sqrt{25.1}$ as follows.

Let $f(x) = \sqrt{x}$. The equation of the tangent line to $f(x)$ at $x = 25$ can be written in the form $y = mx + b$. Compute m and b.

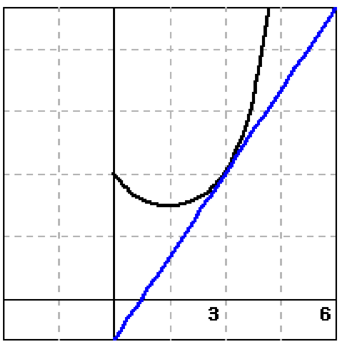
$m =$ ________________

$b =$ ________________

Using this find the approximation for $\sqrt{25.1}$.

Answer: ________________

5. The figure below shows $f(x)$ and its local linearization at $x = a$, $y = 4x - 3$. (The local linearization is shown in blue.)



What is the value of a?

$a =$ ________________

What is the value of $f(a)$?

$f(a) =$ ______________

Use the linearization to approximate the value of $f(3.1)$.

$f(3.1) =$ ______________

Is the approximation an under- or overestimate?

______________ (Enter ***under*** or ***over***.)

6. Given that $f(0.5) = 7.3$ and $f(0.7) = 6.9$, approximate $f'(0.5)$.

$f'(0.5) \approx$ ______

7. A certain function $y = p(x)$ has its local linearization at $a = 3$ given by $L(x) = -2x + 5$.

a. What are the values of $p(3)$ and $p'(3)$? Why?

b. Estimate the value of $p(2.79)$.

c. Suppose that $p''(3) = 0$ and you know that $p''(x) < 0$ for $x < 3$. Is your estimate in (b) too large or too small?

d. Suppose that $p''(x) > 0$ for $x > 3$. Use this fact and the additional information above to sketch an accurate graph of $y = p(x)$ near $x = 3$. Include a sketch of $y = L(x)$ in your work.

8. A potato is placed in an oven, and the potato's temperature F (in degrees Fahrenheit) at various points in time is taken and recorded in the following table. Time t is measured in minutes.

Table 1.8.6 Temperature data for the potato.

t	0	10	20	30	40	50	60
$F(t)$	65	94.7	141.7	167.6	182.4	197.9	209.3

a. Use a central difference to estimate $F'(30)$. Use this estimate as needed in subsequent questions.

b. Find the local linearization $y = L(t)$ to the function $y = F(t)$ at the point where $a = 30$.

c. Determine an estimate for $F(33)$ by employing the local linearization.

d. Do you think your estimate in (c) is too large or too small? Why?

9. An object moving along a straight line path has a differentiable position function $y = s(t)$; $s(t)$ measures the object's position relative to the origin at time t. It is known that at time $t = 9$ seconds, the object's position is $s(9) = 4$ feet (i.e., 4 feet to the right of the origin). Furthermore, the object's instantaneous velocity at $t = 9$ is -1.2 feet per second, and its acceleration at the same instant is 0.08 feet per second per second.

a. Use local linearity to estimate the position of the object at $t = 9.34$.

b. Is your estimate likely too large or too small? Why?

c. In everyday language, describe the behavior of the moving object at $t = 9$. Is it moving toward the origin or away from it? Is its velocity increasing or decreasing?

10. For a certain function f, its derivative is known to be $f'(x) = (x-1)e^{-x^2}$. Note that you do not know a formula for $y = f(x)$.

a. At what x-value(s) is $f'(x) = 0$? Justify your answer algebraically, but include a graph of f' to support your conclusion.

b. Reasoning graphically, for what intervals of x-values is $f''(x) > 0$? What does this tell you about the behavior of the original function f? Explain.

c. Assuming that $f(2) = -3$, estimate the value of $f(1.88)$ by finding and using the tangent line approximation to f at $x = 2$. Is your estimate larger or smaller than the true value of $f(1.88)$? Justify your answer.

CHAPTER 2

Computing Derivatives

2.1 Elementary derivative rules

Motivating Questions

- What are alternate notations for the derivative?
- How can we use the algebraic structure of a function $f(x)$ to compute a formula for $f'(x)$?
- What is the derivative of a power function of the form $f(x) = x^n$? What is the derivative of an exponential function of form $f(x) = a^x$?
- If we know the derivative of $y = f(x)$, what is the derivative of $y = kf(x)$, where k is a constant?
- If we know the derivatives of $y = f(x)$ and $y = g(x)$, how do we compute the derivative of $y = f(x) + g(x)$?

2.1.1 Introduction

In Chapter 1, we developed the concept of the derivative of a function. We now know that the derivative f' of a function f measures the instantaneous rate of change of f with respect to x. The derivative also tells us the slope of the tangent line to $y = f(x)$ at any given value of x. So far, we have focused on interpreting the derivative graphically or, in the context of a physical setting, as a meaningful rate of change. To calculate the value of the derivative at a specific point, we have relied on the limit definition of the derivative,

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

In this chapter, we investigate how the limit definition of the derivative leads to interesting patterns and rules that enable us to find a formula for $f'(x)$ quickly, *without* using the limit definition directly. For example, we would like to apply shortcuts to differentiate a function such as $g(x) = 4x^7 - \sin(x) + 3e^x$ simply by observation.

Preview Activity 2.1.1. Functions of the form $f(x) = x^n$, where $n = 1, 2, 3, \ldots$, are often called **power functions**. The first two questions below revisit work we did earlier in Chapter 1, and the following questions extend those ideas to higher powers of x.

(a) Use the limit definition of the derivative to find $f'(x)$ for $f(x) = x^2$.

(b) Use the limit definition of the derivative to find $f'(x)$ for $f(x) = x^3$.

(c) Use the limit definition of the derivative to find $f'(x)$ for $f(x) = x^4$.
(***Hint***: $(a+b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$. Apply this rule to $(x+h)^4$ within the limit definition.)

(d) Based on your work in (a), (b), and (c), what do you conjecture is the derivative of $f(x) = x^5$? Of $f(x) = x^{13}$?

(e) Conjecture a formula for the derivative of $f(x) = x^n$ that holds for any positive integer n. That is, given $f(x) = x^n$ where n is a positive integer, what do you think is the formula for $f'(x)$?

2.1.2 Some Key Notation

In addition to our usual f' notation, there are other ways to denote the derivative of a function, as well as the instruction to take the derivative. If we are thinking about the relationship between y and x, we sometimes denote the derivative of y with respect to x by the symbol

$$\frac{dy}{dx}$$

which we read "dee-y dee-x." For example, if $y = x^2$, we'll write that the derivative is $\frac{dy}{dx} = 2x$. This notation comes from the fact that the derivative is related to the slope of a line, and slope is measured by $\frac{\Delta y}{\Delta x}$. Note that while we read $\frac{\Delta y}{\Delta x}$ as "change in y over change in x," we view $\frac{dy}{dx}$ as a single symbol, not a quotient of two quantities.

We use a variant of this notation as the instruction to take the derivative. In particular,

$$\frac{d}{dx}[\square]$$

means "take the derivative of the quantity in $\square$ with respect to x." For example, we may write $\frac{d}{dx}[x^2] = 2x$.

It is important to note that the independent variable can be different from x. If we have $f(z) = z^2$, we then write $f'(z) = 2z$. Similarly, if $y = t^2$, we say $\frac{dy}{dt} = 2t$. And it is also true that $\frac{d}{dq}[q^2] = 2q$. This notation may also be used for second derivatives: $f''(z) = \frac{d}{dz}\left[\frac{df}{dz}\right] = \frac{d^2f}{dz^2}$.

In what follows, we'll build a repertoire of functions for which we can quickly compute the derivative.

2.1.3 Constant, Power, and Exponential Functions

So far, we know the derivative formula for two important classes of functions: constant functions and power functions. If $f(x) = c$ is a constant function, its graph is a horizontal line with slope zero at every point. Thus, $\frac{d}{dx}[c] = 0$. We summarize this with the following rule.

The derivative of a constant function.

For any real number c, if $f(x) = c$, then $f'(x) = 0$.

Example 2.1.1 If $f(x) = 7$, then $f'(x) = 0$. Similarly, $\frac{d}{dx}[\sqrt{3}] = 0$. ◊

In your work in Preview Activity 2.1.1, you conjectured that for any positive integer n, if $f(x) = x^n$, then $f'(x) = nx^{n-1}$. This rule can be formally proved for any positive integer n, and even for any nonzero real number (positive or negative).

The derivative of a power function.

For any nonzero real number n, if $f(x) = x^n$, then $f'(x) = nx^{n-1}$.

Example 2.1.2 Using the rule for power functions, we can compute the following derivatives. If $g(z) = z^{-3}$, then $g'(z) = -3z^{-4}$. Similarly, if $h(t) = t^{7/5}$, then $\frac{dh}{dt} = \frac{7}{5}t^{2/5}$, and $\frac{d}{dq}[q^\pi] = \pi q^{\pi-1}$. ◊

It will be helpful to have a derivative formula for one more type of basic function. For now, we simply state this rule without explanation or justification; we explore why this rule is true in one of the exercises and we will encounter graphical reasoning for why the rule is plausible in Preview Activity 2.2.1.

The derivative of an exponential function.

For any positive real number a, if $f(x) = a^x$, then $f'(x) = a^x \ln(a)$.

Example 2.1.3 If $f(x) = 2^x$, then $f'(x) = 2^x \ln(2)$. Similarly, for $p(t) = 10^t$, $p'(t) = 10^t \ln(10)$. It is especially important to note that when $a = e$, where e is the base of the natural logarithm function, we have that

$$\frac{d}{dx}[e^x] = e^x \ln(e) = e^x$$

since $\ln(e) = 1$. This is an extremely important property of the function e^x: its derivative function is itself! ◊

Note carefully the distinction between power functions and exponential functions: in power functions, the variable is in the base, as in x^2, while in exponential functions, the variable is in the power, as in 2^x. As we can see from the rules, this makes a big difference in the form of the derivative.

Activity 2.1.2. Use the rules for constant, power, and exponential functions to determine the derivative of each of the following functions. For each, state your answer using full and proper notation, labeling the derivative with its name. For example, if you are given a function $h(z)$, you should use notation such as "$h'(z) =$" or "$\frac{dh}{dz} =$" as part of your response.

(a) $f(t) = \pi$

(b) $g(z) = 7^z$

(c) $h(w) = w^{3/4}$

(d) $p(x) = 3^{1/2}$

(e) $r(t) = (\sqrt{2})^t$

(f) $s(q) = q^{-1}$

(g) $m(t) = \frac{1}{t^3}$

2.1.4 Constant Multiples and Sums of Functions

Next we will learn how to compute the derivative of a function constructed as an algebraic combination of basic functions. For instance, we'd like to be able to take the derivative of a polynomial function such as

$$p(t) = 3t^5 - 7t^4 + t^2 - 9,$$

which is a sum of constant multiples of powers of t. To that end, we develop two new rules: the Constant Multiple Rule and the Sum Rule.

How is the derivative of $y = kf(x)$ related to the derivative of $y = f(x)$? Recall that when we multiply a function by a constant k, we vertically stretch the graph by a factor of $|k|$ (and reflect the graph across $y = 0$ if $k < 0$). This vertical stretch affects the slope of the graph, so the slope of the function $y = kf(x)$ is k times as steep as the slope of $y = f(x)$. Thus, when we multiply a function by a factor of k, we change the value of its derivative by a factor of k as well.[1]

The Constant Multiple Rule.

For any real number k, if $f(x)$ is a differentiable function with derivative $f'(x)$, then $\frac{d}{dx}[kf(x)] = kf'(x)$.

In words, this rule says that "the derivative of a constant times a function is the constant times the derivative of the function."

Example 2.1.4 If $g(t) = 3 \cdot 5^t$, we have $g'(t) = 3 \cdot 5^t \ln(5)$. Similarly, $\frac{d}{dz}[5z^{-2}] = 5(-2z^{-3})$. ◊

[1]The Constant Multiple Rule can be formally proved as a consequence of properties of limits, using the limit definition of the derivative.

Next we examine a sum of two functions. If we have $y = f(x)$ and $y = g(x)$, we can compute a new function $y = (f + g)(x)$ by adding the outputs of the two functions: $(f + g)(x) = f(x) + g(x)$. Not only is the value of the new function the sum of the values of the two known functions, but the slope of the new function is the sum of the slopes of the known functions. Therefore[2], we arrive at the following Sum Rule for derivatives:

The Sum Rule.

If $f(x)$ and $g(x)$ are differentiable functions with derivatives $f'(x)$ and $g'(x)$ respectively, then $\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$.

In words, the Sum Rule tells us that "the derivative of a sum is the sum of the derivatives." It also tells us that a sum of two differentiable functions is also differentiable. Furthermore, because we can view the difference function $y = (f-g)(x) = f(x)-g(x)$ as $y = f(x)+(-1\cdot g(x))$, the Sum Rule and Constant Multiple Rules together tell us that $\frac{d}{dx}[f(x) + (-1 \cdot g(x))] = f'(x) - g'(x)$, or that "the derivative of a difference is the difference of the derivatives." We can now compute derivatives of sums and differences of elementary functions.

Example 2.1.5 Using the sum rule, $\frac{d}{dw}(2^w + w^2) = 2^w \ln(2) + 2w$. Using both the sum and constant multiple rules, if $h(q) = 3q^6 - 4q^{-3}$, then $h'(q) = 3(6q^5) - 4(-3q^{-4}) = 18q^5 + 12q^{-4}$. ◊

Activity 2.1.3. Use only the rules for constant, power, and exponential functions, together with the Constant Multiple and Sum Rules, to compute the derivative of each function below with respect to the given independent variable. Note well that we do not yet know any rules for how to differentiate the product or quotient of functions. This means that you may have to do some algebra first on certain functions below before you can actually use existing rules to compute the desired derivative formula. For each function whose derivative you find, label the derivative with its name using proper notation such as $f'(x)$, $h'(z)$, dr/dt, etc.

(a) $f(x) = x^{5/3} - x^4 + 2^x$

(b) $g(x) = 14e^x + 3x^5 - x$

(c) $h(z) = \sqrt{z} + \frac{1}{z^4} + 5^z$

(d) $r(t) = \sqrt{53}\,t^7 - \pi e^t + e^4$

(e) $s(y) = (y^2 + 1)(y^2 - 1)$

(f) $q(x) = \frac{x^3-x+2}{x}$

(g) $p(a) = 3a^4 - 2a^3 + 7a^2 - a + 12$

In the same way that we have shortcut rules to help us find derivatives, we introduce some language that is simpler and shorter. Often, rather than say "take the derivative of f," we'll

[2]Like the Constant Multiple Rule, the Sum Rule can be formally proved as a consequence of properties of limits, using the limit definition of the derivative.

instead say simply "differentiate f." Similarly, if the derivative exists at a point, we say "f is differentiable at that point," or that f can be differentiated.

As we work with the algebraic structure of functions, it is important to develop a big picture view of what we are doing. Here, we make several general observations based on the rules we have so far.

- The derivative of any polynomial function will be another polynomial function, and that the degree of the derivative is one less than the degree of the original function. For instance, if $p(t) = 7t^5 - 4t^3 + 8t$, p is a degree 5 polynomial, and its derivative, $p'(t) = 35t^4 - 12t^2 + 8$, is a degree 4 polynomial.
- The derivative of any exponential function is another exponential function: for example, if $g(z) = 7 \cdot 2^z$, then $g'(z) = 7 \cdot 2^z \ln(2)$, which is also exponential.
- We should not lose sight of the fact that all of the meaning of the derivative that we developed in Chapter 1 still holds. The derivative measures the instantaneous rate of change of the original function, as well as the slope of the tangent line at any selected point on its graph.

Activity 2.1.4. Each of the following questions asks you to use derivatives to answer key questions about functions. Be sure to think carefully about each question and to use proper notation in your responses.

(a) Find the slope of the tangent line to $h(z) = \sqrt{z} + \frac{1}{z}$ at the point where $z = 4$.

(b) A population of cells is growing in such a way that its total number in millions is given by the function $P(t) = 2(1.37)^t + 32$, where t is measured in days.

i. Determine the instantaneous rate at which the population is growing on day 4, and include units on your answer.

ii. Is the population growing at an increasing rate or growing at a decreasing rate on day 4? Explain.

(c) Find an equation for the tangent line to the curve $p(a) = 3a^4 - 2a^3 + 7a^2 - a + 12$ at the point where $a = -1$.

(d) What is the difference between being asked to find the *slope* of the tangent line (asked in (a)) and the *equation* of the tangent line (asked in (c))?

2.1.5 Summary

- Given a differentiable function $y = f(x)$, we can express the derivative of f in several different notations: $f'(x)$, $\frac{df}{dx}$, $\frac{dy}{dx}$, and $\frac{d}{dx}[f(x)]$.
- The limit definition of the derivative leads to patterns among certain families of functions that enable us to compute derivative formulas without resorting directly to the

limit definition. For example, if f is a power function of the form $f(x) = x^n$, then $f'(x) = nx^{n-1}$ for any real number n other than 0.

- We have stated a rule for derivatives of exponential functions in the same spirit as the rule for power functions: for any positive real number a, if $f(x) = a^x$, then $f'(x) = a^x \ln(a)$.

- If we are given a constant multiple of a function whose derivative we know, or a sum of functions whose derivatives we know, the Constant Multiple and Sum Rules make it straightforward to compute the derivative of the overall function. More formally, if $f(x)$ and $g(x)$ are differentiable with derivatives $f'(x)$ and $g'(x)$ and a and b are constants, then
$$\frac{d}{dx}\left[af(x) + bg(x)\right] = af'(x) + bg'(x).$$

2.1.6 Exercises

1. Find the derivative of

$y = \sqrt[7]{x}$.

$\frac{dy}{dx} =$ ____________________

2. If $f(x) = x^3 + 3x^2 - 189x + 18$, find analytically all values of x for which $f'(x) = 0$. (Enter your answer as a comma separated list of numbers, e.g., -1,0,2)

$x =$ ____________________

3. Find the derivative of $f(t) = \dfrac{t^7 + t^8 - 1}{t^9}$.

$f'(t) =$ ____________________

4. Find an equation for the line tangent to the graph of f at $(2, 35)$, where f is given by $f(x) = 5x^3 - 5x^2 + 15$.

$y =$ ____________________

5. Certain pieces of antique furniture increased very rapidly in price in the 1970s and 1980s. For example, the value of a particular rocking chair is well approximated by

$$V = 55(1.55)^t,$$

where V is in dollars and t is the number of years since 1975. Find the rate, in dollars per year, at which the price is increasing.

rate = ____________________ dollars/yr

6. Find the derivative of $f(x) = 4e^x + x^4$.

$f'(x) =$ ____________________

7. Find the derivative of the function $y(x) = c^x + x^c$. Assume that c is a positive constant.

$y'(x) =$ ____________________

8. Let f and g be differentiable functions for which the following information is known: $f(2) = 5$, $g(2) = -3$, $f'(2) = -1/2$, $g'(2) = 2$.

a. Let h be the new function defined by the rule $h(x) = 3f(x) - 4g(x)$. Determine $h(2)$ and $h'(2)$.

b. Find an equation for the tangent line to $y = h(x)$ at the point $(2, h(2))$.

c. Let p be the function defined by the rule $p(x) = -2f(x) + \frac{1}{2}g(x)$. Is p increasing, decreasing, or neither at $a = 2$? Why?

d. Estimate the value of $p(2.03)$ by using the local linearization of p at the point $(2, p(2))$.

9. Let functions p and q be the piecewise linear functions given by their respective graphs in Figure 2.1.6. Use the graphs to answer the following questions.

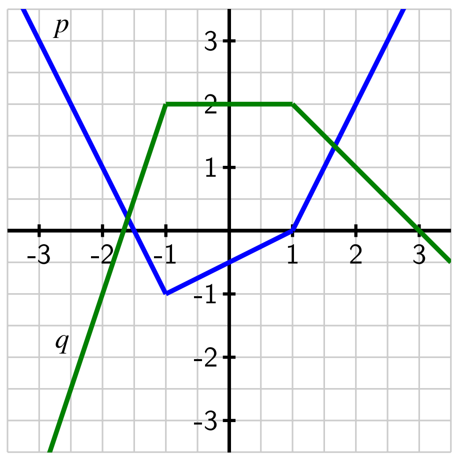


Figure 2.1.6 The graphs of p (in blue) and q (in green).

a. At what values of x is p not differentiable? At what values of x is q not differentiable? Why?

b. Let $r(x) = p(x) + 2q(x)$. At what values of x is r not differentiable? Why?

c. Determine $r'(-2)$ and $r'(0)$.

d. Find an equation for the tangent line to $y = r(x)$ at the point $(2, r(2))$.

10. Consider the functions $r(t) = t^t$ and $s(t) = \arccos(t)$, for which you are given the facts that $r'(t) = t^t(\ln(t) + 1)$ and $s'(t) = -\frac{1}{\sqrt{1-t^2}}$. Do not be concerned with where these derivative formulas come from. We restrict our interest in both functions to the domain $0 < t < 1$.

a. Let $w(t) = 3t^t - 2\arccos(t)$. Determine $w'(t)$.

b. Find an equation for the tangent line to $y = w(t)$ at the point $(\frac{1}{2}, w(\frac{1}{2}))$.

c. Let $v(t) = t^t + \arccos(t)$. Is v increasing or decreasing at the instant $t = \frac{1}{2}$? Why?

11. Let $f(x) = a^x$. The goal of this problem is to explore how the value of a affects the derivative of $f(x)$, without assuming we know the rule for $\frac{d}{dx}[a^x]$ that we have stated and used in earlier work in this section.

a. Use the limit definition of the derivative to show that

$$f'(x) = \lim_{h \to 0} \frac{a^x \cdot a^h - a^x}{h}.$$

b. Explain why it is also true that

$$f'(x) = a^x \cdot \lim_{h \to 0} \frac{a^h - 1}{h}.$$

c. Use computing technology and small values of h to estimate the value of

$$L = \lim_{h \to 0} \frac{a^h - 1}{h}$$

when $a = 2$. Do likewise when $a = 3$.

d. Note that it would be ideal if the value of the limit L was 1, for then f would be a particularly special function: its derivative would be simply a^x, which would mean that its derivative is itself. By experimenting with different values of a between 2 and 3, try to find a value for a for which

$$L = \lim_{h \to 0} \frac{a^h - 1}{h} = 1.$$

e. Compute $\ln(2)$ and $\ln(3)$. What does your work in (b) and (c) suggest is true about $\frac{d}{dx}[2^x]$ and $\frac{d}{dx}[3^x]$?

f. How do your investigations in (d) lead to a particularly important fact about the function $f(x) = e^x$?

2.2 The sine and cosine functions

Motivating Questions

- What is a graphical justification for why $\frac{d}{dx}[a^x] = a^x \ln(a)$?
- What do the graphs of $y = \sin(x)$ and $y = \cos(x)$ suggest as formulas for their respective derivatives?
- Once we know the derivatives of $\sin(x)$ and $\cos(x)$, how do previous derivative rules work when these functions are involved?

2.2.1 Introduction

Throughout Chapter 2, we will develop shortcut derivative rules to help us bypass the limit definition and quickly compute $f'(x)$ from a formula for $f(x)$. In Section 2.1, we stated the rule for power functions,

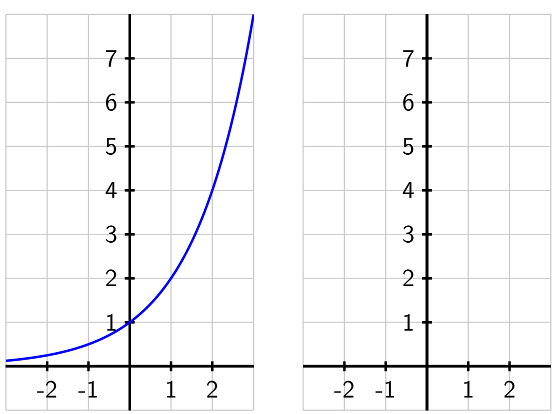
$$\text{if } f(x) = x^n, \text{ then } f'(x) = nx^{n-1},$$

and the rule for exponential functions,

$$\text{if } a \text{ is a positive real number and } f(x) = a^x, \text{ then } f'(x) = a^x \ln(a).$$

Later in this section, we will use a graphical argument to conjecture derivative formulas for the sine and cosine functions. In Preview Activity 2.2.1 we use a graphical approach to understand why the rule for exponential functions is plausible.

Preview Activity 2.2.1. Consider the function $g(x) = 2^x$, which is graphed in the provided figure.



(a) At each of $x = -2, -1, 0, 1, 2$, use a straightedge to sketch an accurate tangent line to $y = g(x)$.

(b) Use the provided grid to estimate the slope of the tangent line you drew at each point in (a).

(c) Use the limit definition of the derivative to estimate $g'(0)$ by using small values of h, and compare the result to your visual estimate for the slope of the tangent line to $y = g(x)$ at $x = 0$ in (b).

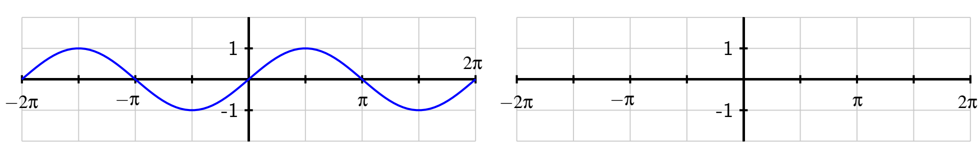
(d) Based on your work in (a), (b), and (c), sketch an accurate graph of $y = g'(x)$ on the axes adjacent to the graph of $y = g(x)$.

(e) Write at least one sentence that explains why it is reasonable to think that $g'(x) = cg(x)$, where c is a constant. In addition, calculate $\ln(2)$, and then discuss how this value, combined with your work above, reasonably suggests that $g'(x) = 2^x \ln(2)$.

2.2.2 The sine and cosine functions

The sine and cosine functions are among the most important functions in all of mathematics. Sometimes called the *circular* functions due to their definition on the unit circle, these periodic functions play a key role in modeling repeating phenomena such as tidal elevations, the behavior of an oscillating mass attached to a spring, or the location of a point on a bicycle tire. Like polynomial and exponential functions, the sine and cosine functions are considered basic functions, ones that are often used in building more complicated functions. As such, we would like to know formulas for $\frac{d}{dx}[\sin(x)]$ and $\frac{d}{dx}[\cos(x)]$, and the next two activities lead us to that end.

Activity 2.2.2. Consider the function $f(x) = \sin(x)$, whose graph is provided in the lefthand side of the given figure. Note carefully that the grid in the diagram does not have boxes that are 1×1, but rather approximately 1.57×1, as the horizontal scale of the grid is $\pi/2$ units per box.



(a) At each of $x = -2\pi, -\frac{3\pi}{2}, -\pi, -\frac{\pi}{2}, 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$, use a straightedge to sketch an accurate tangent line to $y = f(x)$.

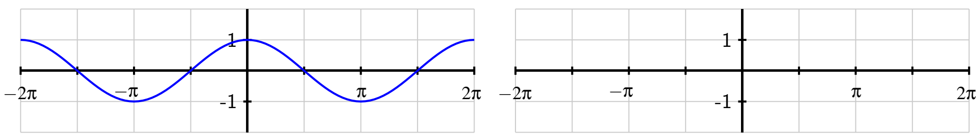
(b) Use the provided grid to estimate the slope of the tangent line you drew at each point. Pay careful attention to the scale of the grid.

(c) Use the limit definition of the derivative to estimate $f'(0)$ by using small values of h, and compare the result to your visual estimate for the slope of the tangent line to $y = f(x)$ at $x = 0$ in (b). Using periodicity, what does this result suggest about $f'(2\pi)$? about $f'(-2\pi)$?

(d) Based on your work in (a), (b), and (c), sketch an accurate graph of $y = f'(x)$ on the axes adjacent to the graph of $y = f(x)$.

(e) What familiar function do you think is the derivative of $f(x) = \sin(x)$?

Activity 2.2.3. Consider the function $g(x) = \cos(x)$, whose graph is provided on the lefthand axes in the figure given below. Note carefully that the grid in the diagram does not have boxes that are 1×1, but rather approximately 1.57×1, as the horizontal scale of the grid is $\pi/2$ units per box.



(a) At each of $x = -2\pi, -\frac{3\pi}{2}, -\pi, -\frac{\pi}{2}, 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$, use a straightedge to sketch an accurate tangent line to $y = g(x)$.

(b) Use the provided grid to estimate the slope of the tangent line you drew at each point. Again, note the scale of the axes and grid.

(c) Use the limit definition of the derivative to estimate $g'(\frac{\pi}{2})$ by using small values of h, and compare the result to your visual estimate for the slope of the tangent line to $y = g(x)$ at $x = \frac{\pi}{2}$ in (b). Using periodicity, what does this result suggest about $g'(-\frac{3\pi}{2})$? can symmetry on the graph help you estimate other slopes easily?

(d) Based on your work in (a), (b), and (c), sketch an accurate graph of $y = g'(x)$ on the axes adjacent to the graph of $y = g(x)$.

(e) What familiar function do you think is the derivative of $g(x) = \cos(x)$?

The results of the two preceding activities suggest that the sine and cosine functions not only have beautiful connections such as the identities $\sin^2(x)+\cos^2(x) = 1$ and $\cos(x-\frac{\pi}{2}) = \sin(x)$, but that they are even further linked through calculus, as the derivative of each involves the other. We have now added the sine and cosine functions to our library of basic functions whose derivatives we know. The constant multiple and sum rules still hold, of course, as well as all of the inherent meaning of the derivative.

The following rules summarize the results of the activities[1].

[1]These two rules may be formally proved using the limit definition of the derivative and the expansion identities for $\sin(x + h)$ and $\cos(x + h)$.

The derivatives of the sine and cosine functions.

For all real numbers x,

$$\frac{d}{dx}[\sin(x)] = \cos(x) \text{ and } \frac{d}{dx}[\cos(x)] = -\sin(x).$$

Activity 2.2.4. Respond to each of the following prompts. Where a derivative is requested, be sure to label the derivative function with its name using proper notation.

(a) Determine the derivative of $h(t) = 3\cos(t) - 4\sin(t)$.

(b) Find the exact slope of the tangent line to $y = f(x) = 2x + \frac{\sin(x)}{2}$ at the point where $x = \frac{\pi}{6}$.

(c) Find the equation of the tangent line to $y = g(x) = x^2 + 2\cos(x)$ at the point where $x = \frac{\pi}{2}$.

(d) Determine the derivative of $p(z) = z^4 + 4^z + 4\cos(z) - \sin(\frac{\pi}{2})$.

(e) The function $P(t) = 24 + 8\sin(t)$ represents a population of a particular kind of animal that lives on a small island, where P is measured in hundreds and t is measured in decades since January 1, 2010. What is the instantaneous rate of change of P on January 1, 2030? What are the units of this quantity? Write a sentence in everyday language that explains how the population is behaving at this point in time.

2.2.3 Summary

- For an exponential function $f(x) = a^x$ $(a > 1)$, the graph of $f'(x)$ appears to be a scaled version of the original function. In particular, careful analysis of the graph of $f(x) = 2^x$, suggests that $\frac{d}{dx}[2^x] = 2^x \ln(2)$, which is a special case of the rule we stated in Section 2.1: $\frac{d}{dx}[a^x] = a^x \ln(a)$.
- By carefully analyzing the graphs of $y = \sin(x)$ and $y = \cos(x)$, and by using the limit definition of the derivative at select points, we found that $\frac{d}{dx}[\sin(x)] = \cos(x)$ and $\frac{d}{dx}[\cos(x)] = -\sin(x)$.
- We note that all previously encountered derivative rules still hold, but now may also be applied to functions involving the sine and cosine. All of the established meaning of the derivative applies to these trigonometric functions as well.

2.2.4 Exercises

1. Let

$$f(x) = 6\sin x + 8\cos x$$

$f'(x) =$ ________________________________

$f'(\frac{4\pi}{3}) =$ ________________________________

2. In this problem, please evaluate the trig functions without a calculator and do not use a decimal point in your answer.

An equation of the tangent line to the curve $y = \sin(x)$ at $x = \pi$ is

$y =$ _______ $+$ _______ $\cdot(x - \pi)$.

An equation of the tangent line to the curve $y = \cos(x)$ at $x = \pi/3$ is

$y =$ _______ $+$ _______ $\cdot(x - \pi/3)$.

3. For what values of x in $[0, 2\pi]$ does the graph of $f(x) = x + 2\sin x$ have a horizontal tangent?

List the values of x below. Separate multiple values with commas.

$x =$____________________________________

4. Suppose that $V(t) = 24 \cdot 1.07^t + 6\sin(t)$ represents the value of a person's investment portfolio in thousands of dollars in year t, where $t = 0$ corresponds to January 1, 2010.

a. At what instantaneous rate is the portfolio's value changing on January 1, 2012? Include units on your answer.

b. Determine the value of $V''(2)$. What are the units on this quantity and what does it tell you about how the portfolio's value is changing?

c. On the interval $0 \le t \le 20$, graph the function $V(t) = 24 \cdot 1.07^t + 6\sin(t)$ and describe its behavior in the context of the problem. Then, compare the graphs of the functions $A(t) = 24 \cdot 1.07^t$ and $V(t) = 24 \cdot 1.07^t + 6\sin(t)$, as well as the graphs of their derivatives $A'(t)$ and $V'(t)$. What is the impact of the term $6\sin(t)$ on the behavior of the function $V(t)$?

5. Let $f(x) = 3\cos(x) - 2\sin(x) + 6$.

a. Determine the exact slope of the tangent line to $y = f(x)$ at the point where $a = \frac{\pi}{4}$.

b. Determine the tangent line approximation to $y = f(x)$ at the point where $a = \pi$.

c. At the point where $a = \frac{\pi}{2}$, is f increasing, decreasing, or neither?

d. At the point where $a = \frac{3\pi}{2}$, does the tangent line to $y = f(x)$ lie above the curve, below the curve, or neither? How can you answer this question without even graphing the function or the tangent line?

6. In this exercise, we explore how the limit definition of the derivative more formally shows that $\frac{d}{dx}[\sin(x)] = \cos(x)$. Letting $f(x) = \sin(x)$, note that the limit definition of the derivative tells us that

$$f'(x) = \lim_{h \to 0} \frac{\sin(x+h) - \sin(x)}{h}.$$

a. Recall the trigonometric identity for the sine of a sum of angles α and β: $\sin(\alpha + \beta) = \sin(\alpha)\cos(\beta) + \cos(\alpha)\sin(\beta)$. Use this identity and some algebra to show that

$$f'(x) = \lim_{h \to 0} \frac{\sin(x)(\cos(h) - 1) + \cos(x)\sin(h)}{h}.$$

b. Next, note that as h changes, x remains constant. Explain why it therefore makes sense to say that

$$f'(x) = \sin(x) \cdot \lim_{h \to 0} \frac{\cos(h) - 1}{h} + \cos(x) \cdot \lim_{h \to 0} \frac{\sin(h)}{h}.$$

c. Finally, use small values of h to estimate the values of the two limits in (c):

$$\lim_{h \to 0} \frac{\cos(h) - 1}{h} \quad \text{and} \quad \lim_{h \to 0} \frac{\sin(h)}{h}.$$

d. What do your results in (b) and (c) thus tell you about $f'(x)$?

e. By emulating the steps taken above, use the limit definition of the derivative to argue convincingly that $\frac{d}{dx}[\cos(x)] = -\sin(x)$.

2.3 The product and quotient rules

Motivating Questions

- How does the algebraic structure of a function guide us in computing its derivative using shortcut rules?
- How do we compute the derivative of a product of two basic functions in terms of the derivatives of the basic functions?
- How do we compute the derivative of a quotient of two basic functions in terms of the derivatives of the basic functions?
- How do the product and quotient rules combine with the sum and constant multiple rules to expand the library of functions we can differentiate quickly?

2.3.1 Introduction

So far, we can differentiate power functions (x^n), exponential functions (a^x), and the two fundamental trigonometric functions ($\sin(x)$ and $\cos(x)$). With the sum rule and constant multiple rules, we can also compute the derivative of combined functions.

Example 2.3.1 Find the derivative of

$$f(x) = 7x^{11} - 4 \cdot 9^x + \pi \sin(x) - \sqrt{3}\cos(x).$$

Because f is a sum of basic functions, we can now quickly say that $f'(x) = 77x^{10} - 4 \cdot 9^x \ln(9) + \pi \cos(x) + \sqrt{3}\sin(x)$. ◊

What about a product or quotient of two basic functions, such as

$$p(z) = z^3 \cos(z),$$

or

$$q(t) = \frac{\sin(t)}{2^t}?$$

While the derivative of a sum is the sum of the derivatives, it turns out that the rules for computing derivatives of products and quotients are more complicated.

Preview Activity 2.3.1. Let f and g be the functions defined by $f(t) = 2t^2$ and $g(t) = t^3 + 4t$.

(a) Determine $f'(t)$ and $g'(t)$.

(b) Let $p(t) = 2t^2(t^3 + 4t)$ and observe that $p(t) = f(t) \cdot g(t)$. Rewrite the formula for p by distributing the $2t^2$ term. Then, compute $p'(t)$ using the sum and constant multiple rules.

(c) True or false: $p'(t) = f'(t) \cdot g'(t)$. Why?

(d) Let $q(t) = \frac{t^3+4t}{2t^2}$ and observe that $q(t) = \frac{g(t)}{f(t)}$. Rewrite the formula for q by dividing each term in the numerator by the denominator and simplify to write q as a sum of constant multiples of powers of t. Then, compute $q'(t)$ using the sum and constant multiple rules.

(e) True or false: $q'(t) = \frac{g'(t)}{f'(t)}$. Why?

2.3.2 The product rule

As part (b) of Preview Activity 2.3.1 shows, it is not true in general that the derivative of a product of two functions is the product of the derivatives of those functions. To see why this is the case, we consider an example involving meaningful functions.

Say that an investor is regularly purchasing stock in a particular company. Let $N(t)$ represent the number of shares owned on day t, where $t = 0$ represents the first day on which shares were purchased. Let $S(t)$ give the value of one share of the stock on day t; note that the units on $S(t)$ are dollars per share. To compute the total value of the stock on day t, we take the product

$$V(t) = N(t) \text{ shares} \cdot S(t) \text{ dollars per share.}$$

Observe that over time, both the number of shares and the value of a given share will vary. The derivative $N'(t)$ measures the rate at which the number of shares is changing, while $S'(t)$ measures the rate at which the value per share is changing. How do these respective rates of change affect the rate of change of the total value function?

To help us understand the relationship among changes in N, S, and V, let's consider some specific data.

- Suppose that on day 100, the investor owns 520 shares of stock and the stock's current value is \$27.50 per share. This tells us that $N(100) = 520$ and $S(100) = 27.50$.
- On day 100, the investor purchases an additional 12 shares (so the number of shares held is rising at a rate of 12 shares per day).
- On that same day the price of the stock is rising at a rate of 0.75 dollars per share per day.

In calculus notation, the latter two facts tell us that $N'(100) = 12$ (shares per day) and $S'(100) = 0.75$ (dollars per share per day). At what rate is the value of the investor's total holdings changing on day 100?

Observe that the increase in total value comes from two sources: the growing number of shares, and the rising value of each share. If only the number of shares is increasing (and the value of each share is constant), the rate at which total value would rise is the product of the current value of the shares and the rate at which the number of shares is changing. That is, the rate at which total value would change is given by

$$S(100) \cdot N'(100) = 27.50 \, \frac{\text{dollars}}{\text{share}} \cdot 12 \, \frac{\text{shares}}{\text{day}} = 330 \, \frac{\text{dollars}}{\text{day}}.$$

Note particularly how the units make sense and show the rate at which the total value V is changing, measured in dollars per day.

If instead the number of shares is constant, but the value of each share is rising, the rate at which the total value would rise is the product of the number of shares and the rate of change of share value. The total value is rising at a rate of

$$N(100) \cdot S'(100) = 520 \, \text{shares} \cdot 0.75 \, \frac{\text{dollars per share}}{\text{day}} = 390 \, \frac{\text{dollars}}{\text{day}}.$$

Of course, when both the number of shares and the value of each share are changing, we have to include both of these sources. In that case the rate at which the total value is rising is

$$V'(100) = S(100) \cdot N'(100) + N(100) \cdot S'(100) = 330 + 390 = 720 \, \frac{\text{dollars}}{\text{day}}.$$

We expect the total value of the investor's holdings to rise by about \$720 on the 100th day.[1]

Next, we expand our perspective from the specific example above to the more general and abstract setting of a product P of two differentiable functions, f and g. If $P(x) = f(x) \cdot g(x)$, our work above suggests that $P'(x) = f(x)g'(x) + g(x)f'(x)$. Indeed, a formal proof using the limit definition of the derivative can be given to show that the following rule, called the *product rule*, holds in general.

The Product Rule.

If f and g are differentiable functions, then their product $P(x) = f(x) \cdot g(x)$ is also a differentiable function, and

$$P'(x) = f(x)g'(x) + g(x)f'(x).$$

In light of the earlier example involving shares of stock, the product rule also makes sense intuitively: the rate of change of P should take into account both how fast f and g are changing, as well as how large f and g are at the point of interest. In words the product rule says: if P is the product of two functions f (the first function) and g (the second), then "the derivative of P is the first times the derivative of the second, plus the second times the derivative of the first." It is often a helpful mental exercise to say this phrasing aloud when executing the product rule.

[1] While this example highlights why the product rule is true, there are some subtle issues to recognize. For one, if the stock's value really does rise exactly \$0.75 on day 100, and the number of shares really rises by 12 on day 100, then we'd expect that $V(101) = N(101) \cdot S(101) = 532 \cdot 28.25 = 15029$. If, as noted above, we expect the total value to rise by \$720, then with $V(100) = N(100) \cdot S(100) = 520 \cdot 27.50 = 14300$, then it seems we should find that $V(101) = V(100) + 720 = 15020$. Why do the two results differ by 9? One way to understand why this difference occurs is to recognize that $N'(100) = 12$ represents an *instantaneous* rate of change, while our (informal) discussion has also thought of this number as the total change in the number of shares over the course of a single day. The formal proof of the product rule reconciles this issue by taking the limit as the change in the input tends to zero.

Example 2.3.2 Use the product rule to differentiate $P(z) = z^3 \cdot \cos(z)$.

Solution. In $P(z) = z^3 \cdot \cos(z)$, the first function is z^3 and the second function is $\cos(z)$. By the product rule, P' will be given by the first, z^3, times the derivative of the second, $-\sin(z)$, plus the second, $\cos(z)$, times the derivative of the first, $3z^2$. That is,

$$P'(z) = z^3(-\sin(z)) + \cos(z)3z^2 = -z^3\sin(z) + 3z^2\cos(z).$$

◊

Activity 2.3.2. Use the product rule to respond to each of the prompts below. Throughout, be sure to carefully label any derivative you find by name. It is not necessary to algebraically simplify any of the derivatives you compute.

(a) Let $m(w) = 3w^{17}4^w$. Find $m'(w)$.

(b) Let $h(t) = (\sin(t) + \cos(t))t^4$. Find $h'(t)$.

(c) Determine the slope of the tangent line to the curve $y = f(x)$ at the point $(1, f(1))$ if f is given by the rule $f(x) = e^x \sin(x)$.

(d) Find the tangent line approximation $L(x)$ to the function $y = g(x)$ at the point $(-1, g(-1))$ if g is given by the rule $g(x) = (x^2 + x)2^x$.

2.3.3 The quotient rule

Because quotients and products are closely linked, we can use the product rule to understand how to take the derivative of a quotient. Let $Q(x)$ be defined by $Q(x) = f(x)/g(x)$, where f and g are both differentiable functions. It turns out that Q is differentiable everywhere that $g(x) \neq 0$. We would like a formula for Q' in terms of f, g, f', and g'. Multiplying both sides of the formula $Q = f/g$ by g, we observe that

$$f(x) = Q(x) \cdot g(x).$$

Applying the product rule to differentiate f, it follows

$$f'(x) = Q(x)g'(x) + g(x)Q'(x).$$

Since we want to know a formula for Q', we solve this most recent equation for $Q'(x)$ and find

$$Q'(x)g(x) = f'(x) - Q(x)g'(x).$$

Dividing both sides by $g(x)$, we have

$$Q'(x) = \frac{f'(x) - Q(x)g'(x)}{g(x)}.$$

Finally, we recall that $Q(x) = \frac{f(x)}{g(x)}$. Substituting this expression in the preceding equation and simplifying, it follows that

$$\begin{aligned} Q'(x) &= \frac{f'(x) - \frac{f(x)}{g(x)}g'(x)}{g(x)} \\ &= \frac{f'(x) - \frac{f(x)}{g(x)}g'(x)}{g(x)} \cdot \frac{g(x)}{g(x)} \\ &= \frac{g(x)f'(x) - f(x)g'(x)}{g(x)^2}. \end{aligned}$$

The preceding argument results in the *quotient rule*.

The Quotient Rule.

If f and g are differentiable functions, then their quotient $Q(x) = \frac{f(x)}{g(x)}$ is also a differentiable function for all x where $g(x) \neq 0$ and

$$Q'(x) = \frac{g(x)f'(x) - f(x)g'(x)}{g(x)^2}.$$

As with the product rule, it can be helpful to think of the quotient rule verbally. If a function Q is the quotient of a top function f and a bottom function g, then Q' is given by "the bottom times the derivative of the top, minus the top times the derivative of the bottom, all over the bottom squared."

Example 2.3.3 Use the quotient rule to differentiate $Q(t) = \dfrac{\sin(t)}{2^t}$.

Solution. Since $Q(t) = \frac{\sin(t)}{2^t}$, we call $\sin(t)$ the top function and 2^t the bottom function. By the quotient rule, Q' is given by the bottom, 2^t, times the derivative of the top, $\cos(t)$, minus the top, $\sin(t)$, times the derivative of the bottom, $2^t \ln(2)$, all over the bottom squared, $(2^t)^2$. That is,

$$Q'(t) = \frac{2^t \cos(t) - \sin(t) 2^t \ln(2)}{(2^t)^2}.$$

In this particular example, it is possible to simplify $Q'(t)$ by removing a factor of 2^t from both the numerator and denominator, so that

$$Q'(t) = \frac{\cos(t) - \sin(t)\ln(2)}{2^t}.$$

◊

In general, we must be careful in doing any such simplification, as we don't want to execute the quotient rule correctly but then make an algebra error.

Activity 2.3.3. Use the quotient rule to respond to each of the prompts below. Throughout, be sure to carefully label any derivative you find by name. That is, if you're given a formula for $f(x)$, clearly label the formula you find for $f'(x)$. It is not necessary to algebraically simplify any of the derivatives you compute.

(a) Let $r(z) = \frac{3^z}{z^4+1}$. Find $r'(z)$.

(b) Let $v(t) = \frac{\sin(t)}{\cos(t)+t^2}$. Find $v'(t)$.

(c) Determine the slope of the tangent line to the curve $R(x) = \dfrac{x^2 - 2x - 8}{x^2 - 9}$ at the point where $x = 0$.

(d) When a camera flashes, the intensity I of light seen by the eye is given by the function

$$I(t) = \frac{100t}{e^t},$$

where I is measured in candles and t is measured in milliseconds. Compute $I'(0.5)$, $I'(2)$, and $I'(5)$; include appropriate units on each value; and discuss the meaning of each.

2.3.4 Combining rules

In order to apply the derivative shortcut rules correctly we must recognize the fundamental structure of a function.

Example 2.3.4 Determine the derivative of the function

$$f(x) = x\sin(x) + \frac{x^2}{\cos(x)+2}.$$

Clearly state which derivative rules you use and how they were applied.

Solution. To differentiate any complicated function, our first task is to recognize the structure of the function. This function f is a sum of two slightly less complicated functions, so we can apply the sum rule[2] to get

$$\begin{aligned} f'(x) &= \frac{d}{dx}\left[x\sin(x) + \frac{x^2}{\cos(x)+2}\right] \\ &= \frac{d}{dx}\left[x\sin(x)\right] + \frac{d}{dx}\left[\frac{x^2}{\cos(x)+2}\right] \end{aligned}$$

Now, the left-hand term above is a product, so the product rule is needed there, while the right-hand term is a quotient, so the quotient rule is required. Applying these rules respectively, we find that

$$f'(x) = (x\cos(x) + \sin(x)) + \frac{(\cos(x)+2)2x - x^2(-\sin(x))}{(\cos(x)+2)^2}$$

$$= x\cos(x) + \sin(x) + \frac{2x\cos(x) + 4x + x^2\sin(x)}{(\cos(x)+2)^2}.$$

◊

Example 2.3.5 Determine the derivative of the function

$$s(y) = \frac{y \cdot 7^y}{y^2+1}.$$

Clearly state which rules you used and how they were applied.

Solution. The function s is a quotient of two simpler functions, so the quotient rule will be needed. To begin, we set up the quotient rule and use the notation $\frac{d}{dy}$ to indicate the derivatives of the numerator and denominator. Thus,

$$s'(y) = \frac{(y^2+1)\cdot\frac{d}{dy}\left[y\cdot 7^y\right] - y\cdot 7^y \cdot \frac{d}{dy}\left[y^2+1\right]}{(y^2+1)^2}.$$

Now, there remain two derivatives to determine. The first one, $\frac{d}{dy}\left[y\cdot 7^y\right]$ calls for use of the product rule, while the second, $\frac{d}{dy}\left[y^2+1\right]$ needs only the sum rule. Applying these rules, we now have

$$s'(y) = \frac{(y^2+1)[y\cdot 7^y\ln(7) + 7^y\cdot 1] - y\cdot 7^y[2y]}{(y^2+1)^2}.$$

While some minor simplification is possible, we are content to leave $s'(y)$ in its current form.

◊

Success in applying derivative rules begins with recognizing the structure of the function, followed by the careful and diligent application of the relevant derivative rules. The best way to become proficient at this process is to do a large number of examples.

Activity 2.3.4. Use relevant derivative rules to respond to each of the prompts below. Throughout, be sure to use proper notation and carefully label any derivative you find by name.

(a) Let $f(r) = (5r^3 + \sin(r))(4^r - 2\cos(r))$. Find $f'(r)$.

(b) Let $p(t) = \dfrac{\cos(t)}{t^6\cdot 6^t}$. Find $p'(t)$.

(c) Let $g(z) = 3z^7e^z - 2z^2\sin(z) + \frac{z}{z^2+1}$. Find $g'(z)$.

(d) A moving particle has its position in feet at time t in seconds given by the function $s(t) = \frac{3\cos(t)-\sin(t)}{e^t}$. Find the particle's instantaneous velocity at the moment $t = 1$.

[2]When taking a derivative that involves the use of multiple derivative rules, it is often helpful to use the notation $\frac{d}{dx}[\;]$ to wait to apply subsequent rules. This is demonstrated both in this example and the one that follows.

(e) Suppose that $f(x)$ and $g(x)$ are differentiable functions and it is known that $f(3) = -2$, $f'(3) = 7$, $g(3) = 4$, and $g'(3) = -1$. If $p(x) = f(x) \cdot g(x)$ and $q(x) = \dfrac{f(x)}{g(x)}$, calculate $p'(3)$ and $q'(3)$.

As the algebraic complexity of the functions we are able to differentiate continues to increase, it is important to remember that all of the derivative's meaning continues to hold. Regardless of the structure of the function f, the value of $f'(a)$ tells us the instantaneous rate of change of f with respect to x at the moment $x = a$, as well as the slope of the tangent line to $y = f(x)$ at the point $(a, f(a))$.

2.3.5 Summary

- If a function is a sum, product, or quotient of simpler functions, then we can use the sum, product, or quotient rules to differentiate it in terms of the simpler functions and their derivatives.
- The product rule tells us that if P is a product of differentiable functions f and g according to the rule $P(x) = f(x)g(x)$, then
$$P'(x) = f(x)g'(x) + g(x)f'(x).$$
- The quotient rule tells us that if Q is a quotient of differentiable functions f and g according to the rule $Q(x) = \frac{f(x)}{g(x)}$, then
$$Q'(x) = \frac{g(x)f'(x) - f(x)g'(x)}{g(x)^2}.$$
- Along with the constant multiple and sum rules, the product and quotient rules enable us to compute the derivative of any function that consists of sums, constant multiples, products, and quotients of basic functions. For instance, if F has the form
$$F(x) = \frac{2a(x) - 5b(x)}{c(x) \cdot d(x)},$$
then F is a quotient, in which the numerator is a sum of constant multiples and the denominator is a product. This, the derivative of F can be found by applying the quotient rule and then using the sum and constant multiple rules to differentiate the numerator and the product rule to differentiate the denominator.

2.3.6 Exercises

1. Let
$$y = 7x^4\left(x^6 - 6\right)$$
Calculate the derivative using the product rule.

$f(x) =$ ________________ and $g(x) =$ ________________

which means that

$f'(x) =$ ________________ and $g'(x) =$ ________________

And thus $y' =$ ________________ + ________________

(It doesn't matter which part of the sum you enter first or second.)

Try checking your answer by multiplying y out and then using power rule.

2. Let

$$y = \sqrt[5]{x}(x^2 + x^8)$$

Calculate the derivative using the product rule.

$f(x) =$ ________________ and $g(x) =$ ________________

which means that

$f'(x) =$ ________________ and $g'(x) =$ ________________

And thus $y' =$ ________________ + ________________

(It doesn't matter which part of the sum you enter first or second.)

Try checking your answer by multiplying y out and then using power rule.

3. Calculate the derivative of the function below using the quotient rule.

$$y = \frac{9x}{8x + 5}$$

$f(x) =$ ________________ and $g(x) =$ ________________

which means that

$f'(x) =$ ________________ and $g'(x) =$ ________________

And thus

$$\left[\frac{9x}{8x + 5}\right]' = \frac{p(x)}{q(x)}$$

where $p(x) =$ ____________________________ and $q(x) =$ ____________________________

4. Calculate the derivative of the function below using the quotient rule.

$$y = \frac{x^2}{\sqrt{x} + 6x^8}$$

$f(x) =$ ________________ and $g(x) =$ ________________

which means that

$f'(x) =$ ________________ and $g'(x) =$ ________________

And thus

$$\left[\frac{x^2}{\sqrt{x}+6x^8}\right]' = \frac{p(x)}{q(x)}$$

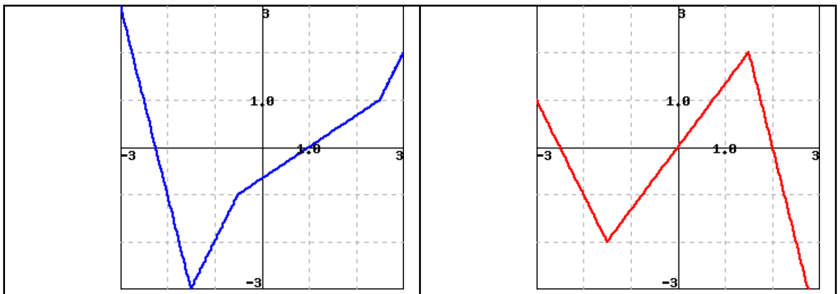
where $p(x) =$ ______________________ and $q(x) =$______________________

5. Let $F(1) = 1, F'(1) = 1, H(1) = 4, H'(1) = 4.$

A. If $G(z) = F(z) \cdot H(z)$, then $G'(1) =$ ______________________

B. If $G(w) = F(w)/H(w)$, then $G'(1) =$ ______________________

6. Let $h(x) = f(x) \cdot g(x)$, and $k(x) = f(x)/g(x)$. Use the figures below to find the ***exact*** values of the indicated derivatives.



$f(x)$	$g(x)$

A. $h'(-1) =$______________________

B. $k'(1) =$______________________

(Enter ***dne*** *for any answer where the derivative does not exist.)*

7. Find the derivative of $s(q) = 8\cos q \sin q$.

$s'(q) =$ ______________________

8. Find an equation for the line tangent to the graph of

$$f(x) = -5xe^x$$

at the point $(a, f(a))$ for $a = 2$.

$y =$ ______________________

9. Find the derivative of the function $f(x)$, below. It may be to your advantage to simplify first.

$f(x) = (x^7 - \sqrt[7]{x})8^x$

$f'(x) =$ ______________________

10. Given that

$f(2) = 4, \; f'(2) = -2,$

$g(2) = -2,\ g'(2) = 3,$

calculate the following:

$(f+g)'(2) =$ ___________

$(f-g)'(2) =$ ___________

$(f\cdot g)'(2) =$ ___________

$(f/g)'(2) =$ ___________

11. Let f and g be differentiable functions for which the following information is known: $f(2) = 5$, $g(2) = -3$, $f'(2) = -1/2$, $g'(2) = 2$.

a. Let h be the new function defined by the rule $h(x) = g(x)\cdot f(x)$. Determine $h(2)$ and $h'(2)$.

b. Find an equation for the tangent line to $y = h(x)$ at the point $(2, h(2))$ (where h is the function defined in (a)).

c. Let r be the function defined by the rule $r(x) = \frac{g(x)}{f(x)}$. Is r increasing, decreasing, or neither at $a = 2$? Why?

d. Estimate the value of $r(2.06)$ (where r is the function defined in (c)) by using the local linearization of r at the point $(2, r(2))$.

12. Let functions p and q be the piecewise linear functions given by their respective graphs in Figure 2.3.6. Use the graphs to answer the following questions.

a. Let $r(x) = p(x)\cdot q(x)$. Determine $r'(-2)$ and $r'(0)$.

b. Are there values of x for which $r'(x)$ does not exist? If so, which values, and why?

c. Find an equation for the tangent line to $y = r(x)$ at the point $(2, r(2))$.

d. Let $z(x) = \frac{q(x)}{p(x)}$. Determine $z'(0)$ and $z'(2)$.

e. Are there values of x for which $z'(x)$ does not exist? If so, which values, and why?

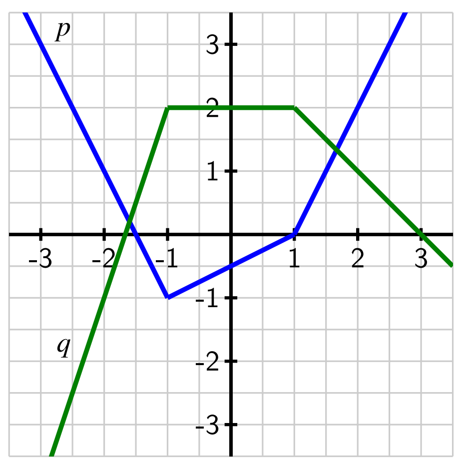


Figure 2.3.6 The graphs of p (in blue) and q (in green).

13. Consider the functions $r(t) = t^t$ and $s(t) = \arccos(t)$, for which you are given the facts that $r'(t) = t^t(\ln(t) + 1)$ and $s'(t) = -\frac{1}{\sqrt{1-t^2}}$. Do not be concerned with where these derivative formulas come from. We restrict our interest in both functions to the domain $0 < t < 1$.

a. Let $w(t) = t^t \arccos(t)$. Determine $w'(t)$.

b. Find an equation for the tangent line to $y = w(t)$ at the point $(\frac{1}{2}, w(\frac{1}{2}))$.

c. Let $v(t) = \frac{t^t}{\arccos(t)}$. Is v increasing or decreasing at the instant $t = \frac{1}{2}$? Why?

14. A farmer with large land holdings has historically grown a wide variety of crops. With the price of ethanol fuel rising, he decides that it would be prudent to devote more and more of his acreage to producing corn. As he grows more and more corn, he learns efficiencies that increase his yield per acre. In the present year, he used 7000 acres of his land to grow corn, and that land had an average yield of 170 bushels per acre. At the current time, he plans to increase his number of acres devoted to growing corn at a rate of 600 acres/year, and he expects that right now his average yield is increasing at a rate of 8 bushels per acre per year. Use this information to answer the following questions.

a. Say that the present year is $t = 0$, that $A(t)$ denotes the number of acres the farmer devotes to growing corn in year t, $Y(t)$ represents the average yield in year t (measured in bushels per acre), and $C(t)$ is the total number of bushels of corn the farmer produces. What is the formula for $C(t)$ in terms of $A(t)$ and $Y(t)$? Why?

b. What is the value of $C(0)$? What does it measure?

c. Write an expression for $C'(t)$ in terms of $A(t)$, $A'(t)$, $Y(t)$, and $Y'(t)$. Explain your thinking.

d. What is the value of $C'(0)$? What does it measure?

e. Based on the given information and your work above, estimate the value of $C(1)$.

15. Let $f(v)$ be the gas consumption (in liters/km) of a car going at velocity v (in km/hour). In other words, $f(v)$ tells you how many liters of gas the car uses to go one kilometer if it is traveling at v kilometers per hour. In addition, suppose that $f(80) = 0.05$ and $f'(80) = 0.0004$.

a. Let $g(v)$ be the distance the same car goes on one liter of gas at velocity v. What is the relationship between $f(v)$ and $g(v)$? Hence find $g(80)$ and $g'(80)$.

b. Let $h(v)$ be the gas consumption in liters per hour of a car going at velocity v. In other words, $h(v)$ tells you how many liters of gas the car uses in one hour if it is going at velocity v. What is the algebraic relationship between $h(v)$ and $f(v)$? Hence find $h(80)$ and $h'(80)$.

c. How would you explain the practical meaning of these function and derivative values to a driver who knows no calculus? Include units on each of the function and derivative values you discuss in your response.

2.4 Derivatives of other trigonometric functions

Motivating Questions

- What are the derivatives of the tangent, cotangent, secant, and cosecant functions?
- How do the derivatives of $\tan(x)$, $\cot(x)$, $\sec(x)$, and $\csc(x)$ combine with other derivative rules we have developed to expand the library of functions we can quickly differentiate?

2.4.1 Introduction

One of the powerful themes in trigonometry comes from a very simple idea: locating a point on the unit circle.

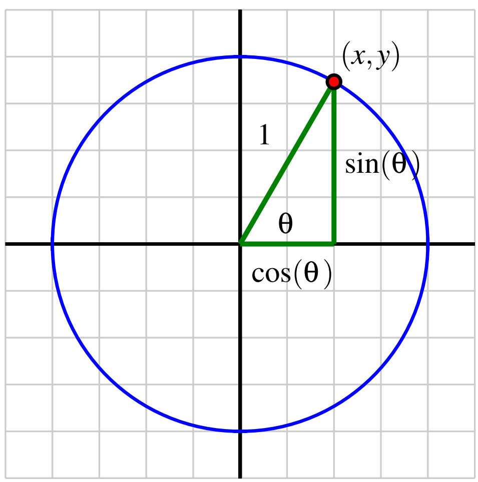


Figure 2.4.1 The unit circle and the definition of the sine and cosine functions.

Because each angle θ in standard position corresponds to one and only one point (x, y) on the unit circle, the x- and y-coordinates of this point are each functions of θ. In fact, this is the very definition of $\cos(\theta)$ and $\sin(\theta)$: $\cos(\theta)$ is the x-coordinate of the point on the unit circle corresponding to the angle θ, and $\sin(\theta)$ is the y-coordinate. From this simple definition, all of trigonometry is founded. For instance, the Fundamental Trigonometric Identity,

$$\sin^2(\theta) + \cos^2(\theta) = 1,$$

is a restatement of the Pythagorean Theorem, applied to the right triangle shown in Figure 2.4.1.

There are four other trigonometric functions, each defined in terms of the sine and/or cosine functions.

- The tangent function is defined by $\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)}$;

- the cotangent function is its reciprocal: $\cot(\theta) = \frac{\cos(\theta)}{\sin(\theta)}$.
- The secant function is the reciprocal of the cosine function, $\sec(\theta) = \frac{1}{\cos(\theta)}$;
- and the cosecant function is the reciprocal of the sine function, $\csc(\theta) = \frac{1}{\sin(\theta)}$.

These six trigonometric functions together offer us a wide range of flexibility in problems involving right triangles.

Because we know the derivatives of the sine and cosine function, we can now develop shortcut differentiation rules for the tangent, cotangent, secant, and cosecant functions. In this section's preview activity, we work through the steps to find the derivative of $y = \tan(x)$.

Preview Activity 2.4.1. Consider the function $f(x) = \tan(x)$, and remember that $\tan(x) = \frac{\sin(x)}{\cos(x)}$.

(a) What is the domain of f?

(b) Use the quotient rule to show that one expression for $f'(x)$ is

$$f'(x) = \frac{\cos(x)\cos(x) + \sin(x)\sin(x)}{\cos^2(x)}.$$

(c) What is the Fundamental Trigonometric Identity? How can this identity be used to find a simpler form for $f'(x)$?

(d) Recall that $\sec(x) = \frac{1}{\cos(x)}$. How can we express $f'(x)$ in terms of the secant function?

(e) For what values of x is $f'(x)$ defined? How does this set compare to the domain of f?

2.4.2 Derivatives of the cotangent, secant, and cosecant functions

In Preview Activity 2.4.1, we found that the derivative of the tangent function can be expressed in several ways, but most simply in terms of the secant function. Next, we develop the derivative of the cotangent function.

Let $g(x) = \cot(x)$. To find $g'(x)$, we observe that $g(x) = \frac{\cos(x)}{\sin(x)}$ and apply the quotient rule. Hence

$$\begin{aligned} g'(x) &= \frac{\sin(x)(-\sin(x)) - \cos(x)\cos(x)}{\sin^2(x)} \\ &= -\frac{\sin^2(x) + \cos^2(x)}{\sin^2(x)} \end{aligned}$$

By the Fundamental Trigonometric Identity, we see that $g'(x) = -\frac{1}{\sin^2(x)}$, and recalling that $\csc(x) = \frac{1}{\sin(x)}$, it follows that we can express g' by the rule

$$g'(x) = -\csc^2(x).$$

Note that neither g nor g' is defined when $\sin(x) = 0$, which occurs at every integer multiple of π. Hence we have the following rule.

The derivative of the cotangent function.

For all real numbers x such that $x \neq k\pi$, where $k = 0, \pm 1, \pm 2, \ldots$,

$$\frac{d}{dx}[\cot(x)] = -\csc^2(x).$$

Notice that the derivative of the cotangent function is very similar to the derivative of the tangent function we discovered in Preview Activity 2.4.1.

The derivative of the tangent function.

For all real numbers x such that $x \neq \frac{(2k+1)\pi}{2}$, where $k = \pm 1, \pm 2, \ldots$,

$$\frac{d}{dx}[\tan(x)] = \sec^2(x).$$

In the next two activities, we develop the rules for differentiating the secant and cosecant functions.

Activity 2.4.2. Let $h(x) = \sec(x)$ and recall that $\sec(x) = \frac{1}{\cos(x)}$.

(a) What is the domain of h?

(b) Use the quotient rule to develop a formula for $h'(x)$ that is expressed completely in terms of $\sin(x)$ and $\cos(x)$.

(c) How can you use other relationships among trigonometric functions to write $h'(x)$ only in terms of $\tan(x)$ and $\sec(x)$?

(d) What is the domain of h'? How does this compare to the domain of h?

Activity 2.4.3. Let $p(x) = \csc(x)$ and recall that $\csc(x) = \frac{1}{\sin(x)}$.

(a) What is the domain of p?

(b) Use the quotient rule to develop a formula for $p'(x)$ that is expressed completely in terms of $\sin(x)$ and $\cos(x)$.

(c) How can you use other relationships among trigonometric functions to write $p'(x)$ only in terms of $\cot(x)$ and $\csc(x)$?

(d) What is the domain of p'? How does this compare to the domain of p?

Using the quotient rule we have determined the derivatives of the tangent, cotangent, secant, and cosecant functions, expanding our overall library of functions we can differentiate. Observe that just as the derivative of any polynomial function is a polynomial, and the derivative of any exponential function is another exponential function, so it is that the derivative of any basic trigonometric function is another function that consists of basic trigonometric functions. This makes sense because all trigonometric functions are periodic, and hence their derivatives will be periodic, too.

The derivative retains all of its fundamental meaning as an instantaneous rate of change and as the slope of the tangent line to the function under consideration.

Activity 2.4.4. Respond to each of the following prompts. Where a derivative is requested, be sure to label the derivative function with its name using proper notation.

(a) Let $f(x) = 5\sec(x) - 2\csc(x)$. Find the slope of the tangent line to f at the point where $x = \frac{\pi}{3}$.

(b) Let $p(z) = z^2 \sec(z) - z\cot(z)$. Find the instantaneous rate of change of p at the point where $z = \frac{\pi}{4}$.

(c) Let $h(t) = \dfrac{\tan(t)}{t^2 + 1} - 2e^t \cos(t)$. Find $h'(t)$.

(d) Let $g(r) = \dfrac{r\sec(r)}{5^r}$. Find $g'(r)$.

(e) When a mass hangs from a spring and is set in motion, the object's position oscillates in a way that the size of the oscillations decrease. This is usually called a *damped oscillation*. Suppose that for a particular object, its displacement from equilibrium (where the object sits at rest) is modeled by the function

$$s(t) = \frac{15\sin(t)}{e^t}.$$

Assume that s is measured in inches and t in seconds. Sketch a graph of this function for $t \geq 0$ to see how it represents the situation described. Then compute ds/dt, state the units on this function, and explain what it tells you about the object's motion. Finally, compute and interpret $s'(2)$.

2.4.3 Summary

- The derivatives of the other four trigonometric functions are

$$\frac{d}{dx}[\tan(x)] = \sec^2(x), \quad \frac{d}{dx}[\cot(x)] = -\csc^2(x),$$

$$\frac{d}{dx}[\sec(x)] = \sec(x)\tan(x), \text{ and } \frac{d}{dx}[\csc(x)] = -\csc(x)\cot(x).$$

Each derivative exists and is defined on the same domain as the original function. For example, both the tangent function and its derivative are defined for all real numbers x such that $x \neq \frac{k\pi}{2}$, where $k = \pm 1, \pm 2, \ldots$.

- The four rules for the derivatives of the tangent, cotangent, secant, and cosecant can be used along with the rules for power functions, exponential functions, and the sine and cosine, as well as the sum, constant multiple, product, and quotient rules, to quickly differentiate a wide range of different functions.

2.4.4 Exercises

1. Find $f'(x)$ if $f(x) = 9\csc(x) - \cot(x)$.

Answer: ____________________

2. Find the equation of the tangent line to the curve $y = 6\sec(x) - 12\cos(x)$ at the point $(\pi/3, 6)$. Write your answer in the form $y = mx + b$ where m is the slope and b is the y-intercept. ____

3. Find the derivative of $h(t) = t\cos t + \sin t$

$h'(t) =$ ____

4. Differentiate $f(t) = \dfrac{\tan t}{e^t}$.

$f'(t) =$ ______________________________

5. Differentiate $y = x^3 \sin x \tan x$.

Answer: $y' =$ ______________________________

6. If

$$f(x) = \frac{3x^2 \tan x}{\sec x},$$

find $f'(x)$.

Find $f'(4)$.

7. Find $\frac{d^2y}{dx^2}$ for

$$y = \sec(x)$$

$\frac{d^2y}{dx^2} =$ ____

8. An object moving vertically has its height at time t (measured in feet, with time in seconds) given by the function $h(t) = 3 + \frac{2\cos(t)}{1.2^t}$.

a. What is the object's instantaneous velocity when $t = 2$?

b. What is the object's acceleration at the instant $t = 2$?

c. Describe in everyday language the behavior of the object at the instant $t = 2$.

9. Let $f(x) = \sin(x)\cot(x)$.

a. Use the product rule to find $f'(x)$.

b. True or false: for all real numbers x, $f(x) = \cos(x)$.

c. Explain why the function that you found in (a) is almost the opposite of the sine function, but not quite. (Hint: convert all of the trigonometric functions in (a) to sines and cosines, and work to simplify. Think carefully about the domain of f and the domain of f'.)

10. Let $p(z)$ be given by the rule

$$p(z) = \frac{z\tan(z)}{z^2\sec(z) + 1} + 3e^z + 1.$$

a. Determine $p'(z)$.

b. Find an equation for the tangent line to p at the point where $z = 0$.

c. At $z = 0$, is p increasing, decreasing, or neither? Why?

2.5 The chain rule

Motivating Questions

- What is a composite function and how do we recognize its structure algebraically?
- Given a composite function $C(x) = f(g(x))$ that is built from differentiable functions f and g, how do we compute $C'(x)$ in terms of f, g, f', and g'? What is the statement of the Chain Rule?

2.5.1 Introduction

In addition to learning how to differentiate a variety of basic functions, we have also been developing our ability to use rules to differentiate certain algebraic combinations of them.

Example 2.5.1 State the rule(s) required to find the derivative of each of the following combinations of $f(x) = \sin(x)$ and $g(x) = x^2$:

$$s(x) = 3x^2 - 5\sin(x),$$

$$p(x) = x^2 \sin(x), \text{and}$$

$$q(x) = \frac{\sin(x)}{x^2}.$$

Solution. Finding s' uses the sum and constant multiple rules, because $s(x) = 3g(x) - 5f(x)$. Determining p' requires the product rule, because $p(x) = g(x) \cdot f(x)$. To calculate q' we use the quotient rule, because $q(x) = \frac{f(x)}{g(x)}$. ◊

There is one more natural way to combine basic functions algebraically, and that is by *composing* them. For instance, let's consider the function

$$C(x) = \sin(x^2),$$

and observe that any input x passes through a *chain* of functions. In the process that defines the function $C(x)$, x is first squared, and then the sine of the result is taken. Using an arrow diagram,

$$x \longrightarrow x^2 \longrightarrow \sin(x^2).$$

In terms of the elementary functions f and g, we observe that x is the input for the function g, and the result is used as the input for f. We write

$$C(x) = f(g(x)) = \sin(x^2)$$

and say that C is the *composition* of f and g. We will refer to g, the function that is first applied to x, as the *inner* function, while f, the function that is applied to the result, is the *outer* function.

Given a composite function $C(x) = f(g(x))$ that is built from differentiable functions f and g, how do we compute $C'(x)$ in terms of f, g, f', and g'? In the same way that the rate of change of a product of two functions, $p(x) = f(x) \cdot g(x)$, depends on the behavior of both f and g, it makes sense intuitively that the rate of change of a composite function $C(x) = f(g(x))$ will also depend on some combination of f and g and their derivatives. The rule that describes how to compute C' in terms of f and g and their derivatives is called the *chain rule*.

But before we can learn what the chain rule says and why it works, we first need to be comfortable decomposing composite functions so that we can correctly identify the inner and outer functions, as we did in the example above with $C(x) = \sin(x^2)$.

Preview Activity 2.5.1. For each function given below, identify its fundamental algebraic structure.

In particular, is the given function a sum, product, quotient, or composition of basic functions? If the function is a composition of basic functions, state a formula for the inner function g and the outer function f so that the overall composite function can be written in the form $f(g(x))$. If the function is a sum, product, or quotient of basic functions, use the appropriate rule to determine its derivative.

(a) $h(x) = \tan(2^x)$

(b) $p(x) = 2^x \tan(x)$

(c) $r(x) = (\tan(x))^2$

(d) $m(x) = e^{\tan(x)}$

(e) $w(x) = \sqrt{x} + \tan(x)$

(f) $z(x) = \sqrt{\tan(x)}$

2.5.2 The chain rule

Often a composite function cannot be written in an alternate algebraic form. For instance, the function $C(x) = \sin(x^2)$ cannot be expanded or otherwise rewritten, so it presents no alternate approaches to taking the derivative. But some composite functions can be expanded or simplified, and these provide a way to explore how the chain rule works.

Example 2.5.2 Let $f(x) = -4x + 7$ and $g(x) = 3x - 5$. Determine a formula for $C(x) = f(g(x))$ and compute $C'(x)$. How is C' related to f and g and their derivatives?

Solution. By the rules given for f and g,

$$\begin{aligned} C(x) &= f(g(x)) \\ &= f(3x-5) \\ &= -4(3x-5)+7 \\ &= -12x+20+7 \end{aligned}$$

$$= -12x + 27.$$

Thus, $C'(x) = -12$. Noting that $f'(x) = -4$ and $g'(x) = 3$, we observe that C' appears to be the product of f' and g'. ◊

It may seem that Example 2.5.2 is too elementary to illustrate how to differentiate a composite function. Linear functions are the simplest of all functions, and composing linear functions yields another linear function. While this example does not illustrate the full complexity of a composition of nonlinear functions, at the same time we remember that any differentiable function is *locally* linear, and thus any function with a derivative behaves like a line when viewed up close. The fact that the derivatives of the linear functions f and g are multiplied to find the derivative of their composition turns out to be a key insight.

We now consider a composition involving a nonlinear function.

Example 2.5.3 Let $C(x) = \sin(2x)$. Use the double angle identity to rewrite C as a product of basic functions, and use the product rule to find C'. Rewrite C' in the simplest form possible.

Solution. Using the double angle identity for the sine function, we write

$$C(x) = \sin(2x) = 2\sin(x)\cos(x).$$

Applying the product rule and simplifying, we find

$$C'(x) = 2\sin(x)(-\sin(x)) + \cos(x)(2\cos(x)) = 2(\cos^2(x) - \sin^2(x)).$$

Next, we recall that a double angle identity for the cosine tells us

$$\cos(2x) = \cos^2(x) - \sin^2(x).$$

Substituting this result into our expression for $C'(x)$, we now have that

$$C'(x) = 2\cos(2x).$$

◊

In Example 2.5.3, if we let $g(x) = 2x$ and $f(x) = \sin(x)$, we observe that $C(x) = f(g(x))$. Now, $g'(x) = 2$ and $f'(x) = \cos(x)$, so we can view the structure of $C'(x)$ as

$$C'(x) = 2\cos(2x) = g'(x)f'(g(x)).$$

In this example, as in the example involving linear functions, we see that the derivative of the composite function $C(x) = f(g(x))$ is found by multiplying the derivatives of f and g, but with f' evaluated at $g(x)$.

It makes sense intuitively that these two quantities are involved in the rate of change of a composite function: if we ask how fast C is changing at a given x value, it clearly matters how fast g is changing at x, as well as how fast f is changing at the value of $g(x)$. It turns out that this structure holds for all differentiable functions[1] as is stated in the Chain Rule.

[1]Like other differentiation rules, the Chain Rule can be proved formally using the limit definition of the deriva-

The Chain Rule.

If g is differentiable at x and f is differentiable at $g(x)$, then the composite function C defined by $C(x) = f(g(x))$ is differentiable at x and

$$C'(x) = f'(g(x))g'(x).$$

As with the product and quotient rules, it is often helpful to think verbally about what the chain rule says: "If C is a composite function defined by an outer function f and an inner function g, then C' is given by the derivative of the outer function evaluated at the inner function, times the derivative of the inner function."

It is helpful to identify clearly the inner function g and outer function f, compute their derivatives individually, and then put all of the pieces together by the chain rule.

Example 2.5.4 Determine the derivative of the function

$$r(x) = (\tan(x))^2.$$

Solution. The function r is composite, with inner function $g(x) = \tan(x)$ and outer function $f(x) = x^2$. Organizing the key information involving f, g, and their derivatives, we have

$$\begin{array}{ll} f(x) = x^2 & g(x) = \tan(x) \\ f'(x) = 2x & g'(x) = \sec^2(x) \\ f'(g(x)) = 2\tan(x) & \end{array}$$

Applying the chain rule, we find that

$$r'(x) = f'(g(x))g'(x) = 2\tan(x)\sec^2(x).$$

◊

As a side note, we remark that $r(x)$ is usually written as $\tan^2(x)$. This is common notation for powers of trigonometric functions: $\cos^4(x)$, $\sin^5(x)$, and $\sec^2(x)$ are all composite functions, with the outer function a power function and the inner function a trigonometric one.

Activity 2.5.2. For each function given below, identify an inner function g and outer function f to write the function in the form $f(g(x))$. Determine $f'(x)$, $g'(x)$, and $f'(g(x))$, and then apply the chain rule to determine the derivative of the given function.

(a) $h(x) = \cos(x^4)$

(b) $p(x) = \sqrt{\tan(x)}$

(c) $s(x) = 2^{\sin(x)}$

(d) $z(x) = \cot^5(x)$

---

tive.

(e) $m(x) = (\sec(x) + e^x)^9$

2.5.3 Using multiple rules simultaneously

The chain rule now joins the sum, constant multiple, product, and quotient rules in our collection of techniques for finding the derivative of a function through understanding its algebraic structure and the basic functions that constitute it. It takes practice to get comfortable applying multiple rules to differentiate a single function, but using proper notation and taking a few extra steps will help.

Example 2.5.5 Find a formula for the derivative of $h(t) = 3^{t^2+2t}\sec^4(t)$.

Solution. We first observe that h is the product of two functions: $h(t) = a(t)\cdot b(t)$, where $a(t) = 3^{t^2+2t}$ and $b(t) = \sec^4(t)$. We will need to use the product rule to differentiate h. And because a and b are composite functions, we will need the chain rule. We therefore begin by computing $a'(t)$ and $b'(t)$.

Writing $a(t) = f(g(t)) = 3^{t^2+2t}$, and finding the derivatives of f and g, we have

$$\begin{aligned} f(t) &= 3^t & g(t) &= t^2 + 2t \\ f'(t) &= 3^t\ln(3) & g'(t) &= 2t + 2 \\ f'(g(t)) &= 3^{t^2+2t}\ln(3) \end{aligned}$$

Thus, by the chain rule, it follows that $a'(t) = f'(g(t))g'(t) = 3^{t^2+2t}\ln(3)(2t+2)$.

Turning next to b, we write $b(t) = r(s(t)) = \sec^4(t)$ and find the derivatives of r and s.

$$\begin{aligned} r(t) &= t^4 & s(t) &= \sec(t) \\ r'(t) &= 4t^3 & s'(t) &= \sec(t)\tan(t) \\ r'(s(t)) &= 4\sec^3(t) \end{aligned}$$

By the chain rule,

$$b'(t) = r'(s(t))s'(t) = 4\sec^3(t)\sec(t)\tan(t) = 4\sec^4(t)\tan(t).$$

Now we are finally ready to compute the derivative of the function h. Recalling that $h(t) = 3^{t^2+2t}\sec^4(t)$, by the product rule we have

$$h'(t) = 3^{t^2+2t}\frac{d}{dt}[\sec^4(t)] + \sec^4(t)\frac{d}{dt}[3^{t^2+2t}].$$

From our work above with a and b, we know the derivatives of 3^{t^2+2t} and $\sec^4(t)$, and therefore

$$h'(t) = 3^{t^2+2t}4\sec^4(t)\tan(t) + \sec^4(t)3^{t^2+2t}\ln(3)(2t+2).$$

◊

Activity 2.5.3. For each of the following functions, find the function's derivative. State the rule(s) you use, label relevant derivatives appropriately, and be sure to clearly identify your overall answer.

(a) $p(r) = 4\sqrt{r^6 + 2e^r}$

(b) $m(v) = \sin(v^2)\cos(v^3)$

(c) $h(y) = \frac{\cos(10y)}{e^{4y}+1}$

(d) $s(z) = 2^{z^2 \sec(z)}$

(e) $c(x) = \sin(e^{x^2})$

The chain rule now adds substantially to our ability to compute derivatives. Whether we are finding the equation of the tangent line to a curve, the instantaneous velocity of a moving particle, or the instantaneous rate of change of a certain quantity, if the function under consideration is a composition, the chain rule is often an essential tool.

Activity 2.5.4. Use known derivative rules, including the chain rule, as needed to respond to each of the following prompts.

(a) Find an equation for the tangent line to the curve $y = \sqrt{e^x + 3}$ at the point where $x = 0$.

(b) If $s(t) = \dfrac{1}{(t^2+1)^3}$ represents the position function of a particle moving horizontally along an axis at time t (where s is measured in inches and t in seconds), find the particle's instantaneous velocity at $t = 1$. Is the particle moving to the left or right at that instant?

(c) At sea level, air pressure is 30 inches of mercury. At an altitude of h feet above sea level, the air pressure, P, in inches of mercury, is given by the function $P = 30e^{-0.0000323h}$. Compute dP/dh and explain what this derivative function tells you about air pressure, including a discussion of the units on dP/dh. In addition, determine how fast the air pressure is changing for a pilot of a small plane passing through an altitude of 1000 feet.

(d) Suppose that $f(x)$ and $g(x)$ are differentiable functions and that the following information about them is known:

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
-1	2	-5	-3	4
2	-3	4	-1	2

If $C(x)$ is a function given by the formula $f(g(x))$, determine $C'(2)$. In addition, if $D(x)$ is the function $f(f(x))$, find $D'(-1)$.

2.5.4 The composite version of basic function rules

As we gain more experience with differention, we will become more comfortable in simply writing down the derivative without taking multiple steps. This is particularly simple when the inner function is linear, since the derivative of a linear function is a constant.

Example 2.5.6 For each of the following composite functions whose inside function is linear, find the overall function's derivative using the chain rule: $f(x) = (5x+7)^{10}$, $g(x) = \tan(17x)$, and $h(x) = e^{-3x}$.

Solution. For each of the three given functions, the derivative of the inner function is constant. By the chain rule, we see

$$\frac{d}{dx}\left[(5x+7)^{10}\right] = 10(5x+7)^9 \cdot 5,$$

$$\frac{d}{dx}\left[\tan(17x)\right] = 17\sec^2(17x), \text{ and}$$

$$\frac{d}{dx}\left[e^{-3x}\right] = -3e^{-3x}.$$

◊

More generally, we can think about how each basic function rule has a corresponding chain rule version. The next example demonstrates this for two familiar functions.

Example 2.5.7 Develop a chain rule version of the two basic derivative rules that state $\frac{d}{dx}[\sin(x)] = \cos(x)$ and $\frac{d}{dx}[a^x] = a^x \ln(a)$.

Solution. To determine

$$\frac{d}{dx}[\sin(u(x))],$$

where u is a differentiable function of x, we use the chain rule with the sine function as the outer function. Applying the chain rule, we find that

$$\frac{d}{dx}[\sin(u(x))] = \cos(u(x)) \cdot u'(x).$$

This rule is analogous to the basic derivative rule that $\frac{d}{dx}[\sin(x)] = \cos(x)$.

Similarly, since $\frac{d}{dx}[a^x] = a^x \ln(a)$, it follows by the chain rule that

$$\frac{d}{dx}[a^{u(x)}] = a^{u(x)} \ln(a) \cdot u'(x).$$

This rule is analogous to the basic derivative rule that $\frac{d}{dx}[a^x] = a^x \ln(a)$. ◊

An excellent exercise for getting comfortable with the derivative rules is to complete Example 2.5.7 for every basic function. That is, write down a list of all the basic functions whose derivatives you know, and list their corresponding derivatives. Then, corresponding to each basic rule, write a composite function with the inner function being an unknown function $u(x)$ and the outer function being a basic function. Finally, write the chain rule for the composite function, such as $\frac{d}{dx}[\sin(u(x))] = \cos(u(x)) \cdot u'(x)$.

2.5.5 Summary

- A composite function is one where the input variable x first passes through one function, and then the resulting output passes through another. For example, the function $h(x) = 2^{\sin(x)}$ is composite since $x \longrightarrow \sin(x) \longrightarrow 2^{\sin(x)}$.
- Given a composite function $C(x) = f(g(x))$ where f and g are differentiable functions, the chain rule tells us that
$$C'(x) = f'(g(x))g'(x).$$

2.5.6 Exercises

1. A boat at anchor is bobbing up and down in the sea. The vertical distance, y, in feet, between the sea floor and the boat is given as a function of time, t, in minutes, by $y = 26 + \sin(2\pi t)$ ft.

Find the vertical velocity, v, of the boat at time t.

$v =$ ______________________________ ft/min

2. Find the derivative of $g(q) = \sin(\cos q)$

$g'(q) =$ ______________________________

3. Given $F(4) = 1, F'(4) = 7, F(5) = 3, F'(5) = 4$ and $G(1) = 7, G'(1) = 3, G(4) = 5, G'(4) = 4$, find each of the following. (Enter ***dne*** for any derivative that cannot be computed from this information alone.)

A. $H(4)$ if $H(x) = F(G(x))$ ____________________

B. $H'(4)$ if $H(x) = F(G(x))$ ____________________

C. $H(4)$ if $H(x) = G(F(x))$ ____________________

D. $H'(4)$ if $H(x) = G(F(x))$ ____________________

E. $H'(4)$ if $H(x) = F(x)/G(x)$ ____________________

4. Find the derivative of

$f(y) = e^{e^{(y^7)}}$

$f'(y) =$ ______________________________

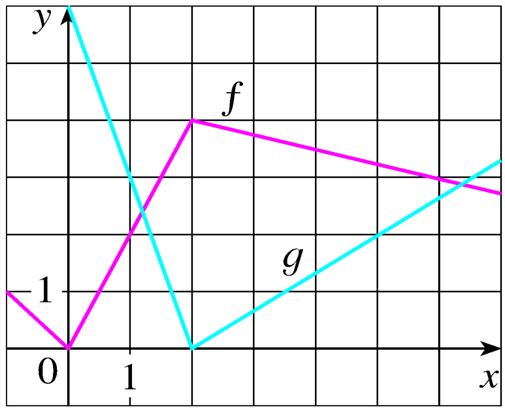
5. If f and g are the functions whose graphs are shown, let $u(x) = f(g(x)), v(x) = g(f(x))$, and $w(x) = g(g(x))$.

Find each of the following derivatives, if they exist. If it does not exist, enter "dne" (without the quotes) below.

(a) $u'(1) =$ __________

(b) $v'(1) =$ __________

(c) $w'(1) =$ __________



6. Let $F(x) = f(x^4)$ and $G(x) = (f(x))^4$. You also know that $a^3 = 14$, $f(a) = 3$, $f'(a) = 8$, $f'(a^4) = 11$.

Find $F'(a) =$ __________ and $G'(a) =$ __________.

7. A table of values for f, g, f', and g' is given below.

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	1	2	1	2
2	2	2	2	2
3	1	1	2	3

(A) If $h(x) = f(g(x))$, then $h'(3) =$ ____________________

(B) If $H(x) = g(f(x))$, then $H'(2) =$ ____________________

8. Suppose that

$$f(x) = (4x - 4)^3.$$

(A) Find an equation for the tangent line to the graph of f at $x = 1$.

Tangent line: $y =$ ________________________

(B) Find the values of x where the tangent line is horizontal. If there are no such values, enter -1000.

Values of $x =$ ________________________

9. Evaluate

$\frac{d}{dx} e^{5x^2+7x} =$ ________________

10. Suppose that

$$y = (5x^2 + 4x + 4)^{1/3}.$$

Find $\frac{dy}{dx}$.

$\frac{dy}{dx} =$ ______________________

11. Consider the basic functions $f(x) = x^3$ and $g(x) = \sin(x)$.

a. Let $h(x) = f(g(x))$. Find the exact instantaneous rate of change of h at the point where $x = \frac{\pi}{4}$.

b. Which function is changing most rapidly at $x = 0.25$: $h(x) = f(g(x))$ or $r(x) = g(f(x))$? Why?

c. Let $h(x) = f(g(x))$ and $r(x) = g(f(x))$. Which of these functions has a derivative that is periodic? Why?

12. Let $u(x)$ be a differentiable function. For each of the following functions, determine the derivative. Each response will involve u and/or u'.

a. $p(x) = e^{u(x)}$

b. $q(x) = u(e^x)$

c. $r(x) = \cot(u(x))$

d. $s(x) = u(\cot(x))$

e. $a(x) = u(x^4)$

f. $b(x) = u^4(x)$

13. Let functions p and q be the piecewise linear functions given by their respective graphs in Figure 2.5.8. Use the graphs to answer the following questions.

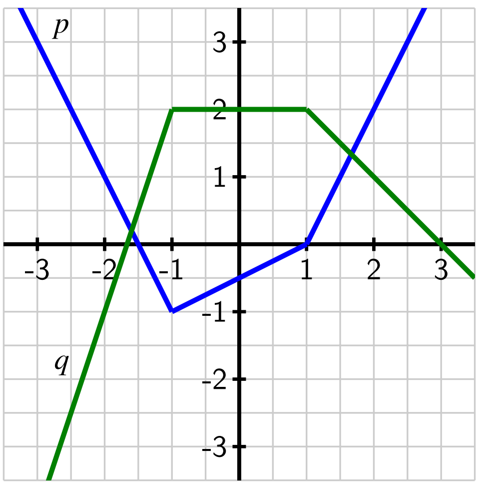


Figure 2.5.8 The graphs of p (in blue) and q (in green).

a. Let $C(x) = p(q(x))$. Determine $C'(0)$ and $C'(3)$.

b. Let $Y(x) = q(q(x))$ and $Z(x) = q(p(x))$. Determine $Y'(-2)$ and $Z'(0)$.

14. If a spherical tank of radius 4 feet has h feet of water present in the tank, then the volume of water in the tank is given by the formula

$$V = \frac{\pi}{3}h^2(12 - h).$$

a. At what instantaneous rate is the volume of water in the tank changing with respect to the *height* of the water at the instant $h = 1$? What are the units on this quantity?

b. Now suppose that the height of water in the tank is being regulated by an inflow and outflow (e.g., a faucet and a drain) so that the height of the water at time t is given by the rule $h(t) = \sin(\pi t) + 1$, where t is measured in hours (and h is still measured in feet). At what rate is the height of the water changing with respect to time at the instant $t = 2$?

c. Continuing under the assumptions in (b), at what instantaneous rate is the volume of water in the tank changing with respect to *time* at the instant $t = 2$?

d. What are the main differences between the rates found in (a) and (c)? Include a discussion of the relevant units.

2.6 Derivatives of inverse functions

Motivating Questions

- What is the derivative of the natural logarithm function?
- What are the derivatives of the inverse trigonometric functions $\arcsin(x)$ and $\arctan(x)$?
- If g is the inverse of a differentiable function f, how is g' computed in terms of f, f', and g?

2.6.1 Introduction

Much of mathematics relies on the notion of function. Indeed, throughout our study of calculus, we are investigating the behavior of functions, with particular emphasis on how fast the output of the function changes in response to changes in the input. Because each function represents a process, a natural question to ask is whether or not the particular process can be reversed. That is, if we know the output that results from the function, can we determine the input that led to it? And if we know how fast a particular process is changing, can we determine how fast the inverse process is changing?

One of the most important functions in all of mathematics is the natural exponential function $f(x) = e^x$, along with its inverse, the natural logarithm $g(x) = \ln(x)$. One of our goals in this section is to learn how to differentiate the logarithm function. First, we review some of the basic concepts surrounding functions and their inverses.

Preview Activity 2.6.1. The equation $y = \frac{5}{9}(x - 32)$ relates a temperature given in x degrees Fahrenheit to the corresponding temperature y measured in degrees Celsius.

(a) Solve the equation $y = \frac{5}{9}(x - 32)$ for x to write x (Fahrenheit temperature) in terms of y (Celsius temperature).

(b) Let $C(x) = \frac{5}{9}(x - 32)$ be the function that takes a Fahrenheit temperature as input and produces the Celsius temperature as output. In addition, let $F(y)$ be the function that converts a temperature given in y degrees Celsius to the temperature $F(y)$ measured in degrees Fahrenheit. Use your work in (a) to write a formula for $F(y)$.

(c) Next consider the new function defined by $p(x) = F(C(x))$. Use the formulas for F and C to determine an expression for $p(x)$ and simplify this expression as much as possible. What do you observe?

(d) Now, let $r(y) = C(F(y))$. Use the formulas for F and C to determine an expression for $r(y)$ and simplify this expression as much as possible. What do you observe?

(e) What is the value of $C'(x)$? of $F'(y)$? How do these values appear to be related?

2.6.2 Basic facts about inverse functions

A *function* $f : A \to B$ is a rule that associates each element in the set A to one and only one element in the set B. We call A the *domain* of f and B the *codomain* of f. If there exists a function $g : B \to A$ such that $g(f(a)) = a$ for every possible choice of a in the set A and $f(g(b)) = b$ for every b in the set B, then we say that g is the *inverse* of f.

We often use the notation f^{-1} (read "f-inverse") to denote the inverse of f. The inverse function undoes the work of f. Indeed, if $y = f(x)$, then

$$f^{-1}(y) = f^{-1}(f(x)) = x.$$

Thus, the equations $y = f(x)$ and $x = f^{-1}(y)$ say the same thing. The only difference between the two equations is one of perspective — one is solved for x, while the other is solved for y.

Here we briefly remind ourselves of some key facts about inverse functions.

Note 2.6.1 For a function $f : A \to B$,

- f has an inverse if and only if f is one-to-one [1] and onto [2];
- provided f^{-1} exists, the domain of f^{-1} is the codomain of f, and the codomain of f^{-1} is the domain of f;
- $f^{-1}(f(x)) = x$ for every x in the domain of f and $f(f^{-1}(y)) = y$ for every y in the codomain of f;
- $y = f(x)$ if and only if $x = f^{-1}(y)$.

The last fact reveals a special relationship between the graphs of f and f^{-1}. If a point (x, y) that lies on the graph of $y = f(x)$, then it is also true that $x = f^{-1}(y)$, which means that the point (y, x) lies on the graph of f^{-1} (provided we view f^{-1} as a function of the same independent variable, x). This shows us that the graphs of f and f^{-1} are the reflections of each other across the line $y = x$, because this reflection is precisely the geometric action that swaps the coordinates in an ordered pair. In Figure 2.6.2, we see this illustrated by the function $y = f(x) = 2^x$ and its inverse, with the points $(-1, \frac{1}{2})$ and $(\frac{1}{2}, -1)$ highlighting the reflection of the curves across $y = x$.

To close our review of important facts about inverses, we recall that the natural exponential function $y = f(x) = e^x$ has an inverse function, namely the natural logarithm, $x = f^{-1}(y) = \ln(y)$. Thus, writing $y = e^x$ is interchangeable with $x = \ln(y)$, plus $\ln(e^x) = x$ for every real number x and $e^{\ln(y)} = y$ for every positive real number y.

[1] A function f is *one-to-one* provided that no two distinct inputs lead to the same output. A one-to-one function is sometimes called an *injection*.

[2] A function f is *onto* provided that every possible element of the codomain can be realized as an output of the function for some choice of input from the domain. An onto function is sometimes called a *surjection*.

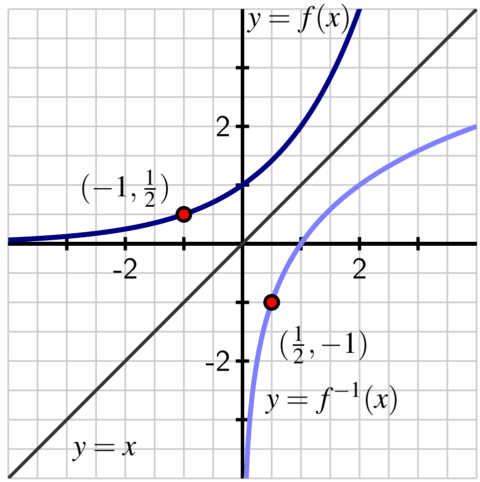


Figure 2.6.2 A graph of a function $y = f(x)$ along with its inverse, $y = f^{-1}(x)$.

2.6.3 The derivative of the natural logarithm function

In what follows, we find a formula for the derivative of $g(x) = \ln(x)$. To do so, we take advantage of the fact that we know the derivative of the natural exponential function, the inverse of g. In particular, we know that writing $g(x) = \ln(x)$ is equivalent to writing $e^{g(x)} = x$. Now we differentiate both sides of this equation and observe that

$$\frac{d}{dx}\left[e^{g(x)}\right] = \frac{d}{dx}[x].$$

The righthand side is simply 1; by applying the chain rule to the left side, we find that

$$e^{g(x)}g'(x) = 1.$$

Next we solve for $g'(x)$, to get

$$g'(x) = \frac{1}{e^{g(x)}}.$$

Finally, we recall that $g(x) = \ln(x)$, so $e^{g(x)} = e^{\ln(x)} = x$, and thus

$$g'(x) = \frac{1}{x}.$$

The derivative of the natural logarithm.

For all positive real numbers x, $\frac{d}{dx}[\ln(x)] = \frac{1}{x}$.

This rule for the natural logarithm function now joins our list of basic derivative rules. Note that this rule applies only to positive values of x, as these are the only values for which $\ln(x)$ is defined.

Also notice that for the first time in our work, differentiating a basic function of a particular type has led to a function of a very different nature: the derivative of the natural logarithm is not another logarithm, nor even an exponential function, but rather a rational one.

Derivatives of logarithms may now be computed in concert with all of the rules known to date. For instance, if $f(t) = \ln(t^2 + 1)$, then by the chain rule, $f'(t) = \frac{1}{t^2+1} \cdot 2t$.

There are interesting connections between the graphs of $f(x) = e^x$ and $f^{-1}(x) = \ln(x)$.

In Figure 2.6.3, we are reminded that since the natural exponential function has the property that its derivative is itself, the slope of the tangent to $y = e^x$ is equal to the height of the curve at that point. For instance, at the point $A = (\ln(0.5), 0.5)$, the slope of the tangent line is $m_A = 0.5$, and at $B = (\ln(5), 5)$, the tangent line's slope is $m_B = 5$. At the corresponding points A' and B' on the graph of the natural logarithm function (which come from reflecting A and B across the line $y = x$), we know that the slope of the tangent line is the reciprocal of the x-coordinate of the point (since $\frac{d}{dx}[\ln(x)] = \frac{1}{x}$). Thus, at $A' = (0.5, \ln(0.5))$, we have $m_{A'} = \frac{1}{0.5} = 2$, and at $B' = (5, \ln(5))$, $m_{B'} = \frac{1}{5}$.

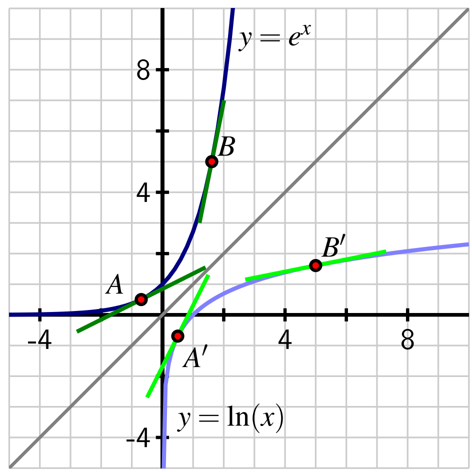


Figure 2.6.3 A graph of the function $y = e^x$ along with its inverse, $y = \ln(x)$, where both functions are viewed using the input variable x.

In particular, we observe that $m_{A'} = \frac{1}{m_A}$ and $m_{B'} = \frac{1}{m_B}$. This is not a coincidence, but in fact holds for any function $y = f(x)$ that has an inverse, due to the reflection across $y = x$. Reflection changes the roles of x and y, thus reversing the rise and run, so the slope of the inverse function at the reflected point is the reciprocal of the slope of the original function.

Activity 2.6.2. For each function given below, find its derivative.

(a) $h(x) = x^2 \ln(x)$

(b) $p(t) = \frac{\ln(t)}{e^t+1}$

(c) $s(y) = \ln(\cos(y) + 2)$

(d) $z(x) = \tan(\ln(x))$

(e) $m(z) = \ln(\ln(z))$

2.6.4 Inverse trigonometric functions and their derivatives

Trigonometric functions are periodic, so they fail to be one-to-one, and thus do not have inverse functions. However, we can restrict the domain of each trigonometric function so that it is one-to-one on that domain.

For instance, consider the sine function on the domain $[-\frac{\pi}{2}, \frac{\pi}{2}]$. Because no output of the sine function is repeated on this interval, the function is one-to-one and thus has an inverse. Thus, the function $f(x) = \sin(x)$ with $[-\frac{\pi}{2}, \frac{\pi}{2}]$ and codomain $[-1, 1]$ has an inverse function f^{-1} such that

$$f^{-1} : [-1, 1] \to [-\frac{\pi}{2}, \frac{\pi}{2}],$$

as we see in Figure 2.6.4. We call f^{-1} the *arcsine* (or inverse sine) function and write $f^{-1}(y) = \arcsin(y)$. "The arcsine of y" means "the angle whose sine is y." It is especially important to remember that

$$y = \sin(x) \text{ and } x = \arcsin(y)$$

say the same thing.

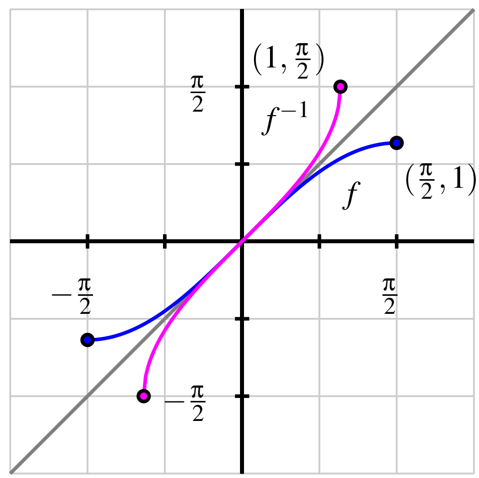


Figure 2.6.4 A graph of $f(x) = \sin(x)$ (in blue), restricted to the domain $[-\frac{\pi}{2}, \frac{\pi}{2}]$, along with its inverse, $f^{-1}(x) = \arcsin(x)$ (in magenta).

For example, $\arcsin(\frac{1}{2}) = \frac{\pi}{6}$ means that $\frac{\pi}{6}$ is the angle whose sine is $\frac{1}{2}$, which is equivalent to writing $\sin(\frac{\pi}{6}) = \frac{1}{2}$.

Next, we determine the derivative of the arcsine function. Letting $h(x) = \arcsin(x)$, our goal is to find $h'(x)$. Since $h(x)$ is the angle whose sine is x, it is equivalent to write

$$\sin(h(x)) = x.$$

Differentiating both sides of the previous equation, we have

$$\frac{d}{dx}[\sin(h(x))] = \frac{d}{dx}[x].$$

The righthand side is simply 1, and by applying the chain rule applied to the left side,

$$\cos(h(x))h'(x) = 1.$$

Solving for $h'(x)$, it follows that

$$h'(x) = \frac{1}{\cos(h(x))}. \tag{2.6.1}$$

Finally, we recall that $h(x) = \arcsin(x)$, so the denominator of $h'(x)$ is the function $\cos(\arcsin(x))$, or in other words, "the cosine of the angle whose sine is x." A bit of right triangle trigonometry allows us to simplify this expression considerably.

Let's say that $\theta = \arcsin(x)$, so that θ is the angle whose sine is x. We can picture θ as an angle in a right triangle with hypotenuse 1 and a vertical leg of length x, as shown in Figure 2.6.5. The horizontal leg must be $\sqrt{1-x^2}$, by the Pythagorean Theorem. Now, because $\theta = \arcsin(x)$, it follows that

$$\begin{aligned}\cos(\arcsin(x)) &= \cos(\theta)\\ &= \sqrt{1-x^2}\end{aligned}$$

with the last equality holding from the relationships in the right triangle in Figure 2.6.5.

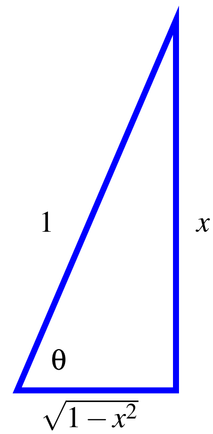


Figure 2.6.5 The right triangle that corresponds to $\theta = \arcsin(x)$.

Substituting this most recent expression for $\cos(\arcsin(x))$ into Equation (2.6.1), we have now shown that

$$h'(x) = \frac{1}{\cos(\arcsin(x))} = \frac{1}{\sqrt{1-x^2}}.$$

The derivative of the arcsine function.

For all real numbers x such that $-1 < x < 1$,

$$\frac{d}{dx}[\arcsin(x)] = \frac{1}{\sqrt{1-x^2}}.$$

Activity 2.6.3. The following prompts in this activity will lead you to develop the derivative of the inverse tangent function.

(a) Let $r(x) = \arctan(x)$. Use the relationship between the arctangent and tangent functions to rewrite this equation using only the tangent function.

(b) Differentiate both sides of the equation you found in (a). Solve the resulting equation for $r'(x)$, writing $r'(x)$ as simply as possible in terms of a trigonometric function evaluated at $r(x)$.

(c) Recall that $r(x) = \arctan(x)$. Update your expression for $r'(x)$ so that it only involves trigonometric functions and the independent variable x.

(d) Introduce a right triangle with angle θ so that $\theta = \arctan(x)$. What are the three sides of the triangle?

(e) In terms of only x and 1, what is the value of $\cos(\arctan(x))$?

(f) Use the results of your work above to find an expression involving only 1 and x for $r'(x)$.

While derivatives for other inverse trigonometric functions can be established similarly, for now we limit ourselves to the arcsine and arctangent functions.

Activity 2.6.4. Determine the derivative of each of the following functions.

(a) $f(x) = x^3 \arctan(x) + e^x \ln(x)$

(b) $p(t) = 2^{t \arcsin(t)}$

(c) $h(z) = (\arcsin(5z) + \arctan(4 - z))^{27}$

(d) $s(y) = \cot(\arctan(y))$

(e) $m(v) = \ln(\sin^2(v) + 1)$

(f) $g(w) = \arctan\left(\dfrac{\ln(w)}{1 + w^2}\right)$

2.6.5 The link between the derivative of a function and the derivative of its inverse

In Figure 2.6.3, we saw an interesting relationship between the slopes of tangent lines to the natural exponential and natural logarithm functions at points reflected across the line $y = x$. In particular, we observed that at the point $(\ln(2), 2)$ on the graph of $f(x) = e^x$, the slope of the tangent line is $f'(\ln(2)) = 2$, while at the corresponding point $(2, \ln(2))$ on the graph of $f^{-1}(x) = \ln(x)$, the slope of the tangent line is $(f^{-1})'(2) = \frac{1}{2}$, which is the reciprocal of $f'(\ln(2))$.

That the two corresponding tangent lines have reciprocal slopes is not a coincidence. If f and g are differentiable inverse functions, then $y = f(x)$ if and only if $x = g(y)$, so $f(g(x)) = x$ for every x in the domain of f^{-1}. Differentiating both sides of this equation, we have

$$\frac{d}{dx}[f(g(x))] = \frac{d}{dx}[x],$$

and by the chain rule,

$$f'(g(x))g'(x) = 1.$$

Solving for $g'(x)$, we have $g'(x) = \frac{1}{f'(g(x))}$. Here we see that the slope of the tangent line to the inverse function g at the point $(x, g(x))$ is precisely the reciprocal of the slope of the tangent line to the original function f at the point $(g(x), f(g(x))) = (g(x), x)$.

To see this more clearly, consider the graph of the function $y = f(x)$ shown in Figure 2.6.6, along with its inverse $y = g(x)$. Given a point (a, b) that lies on the graph of f, we know that (b, a) lies on the graph of g; because $f(a) = b$ and $g(b) = a$.

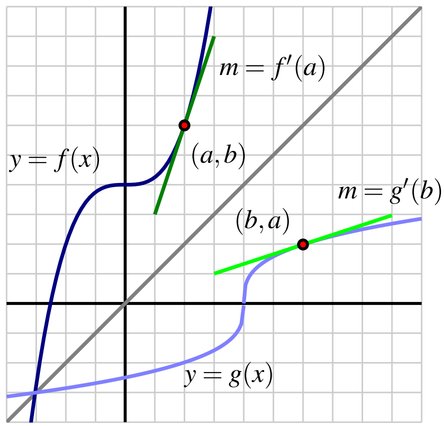


Figure 2.6.6 A graph of function $y = f(x)$ along with its inverse, $y = g(x) = f^{-1}(x)$. Observe that the slopes of the two tangent lines are reciprocals of one another.

Now, applying the rule that $g'(x) = 1/f'(g(x))$ to the value $x = b$, we have

$$g'(b) = \frac{1}{f'(g(b))} = \frac{1}{f'(a)},$$

which is precisely what we see in the figure: the slope of the tangent line to g at (b, a) is the reciprocal of the slope of the tangent line to f at (a, b), since these two lines are reflections of one another across the line $y = x$.

The derivative of the inverse function.

Suppose that f is a differentiable function with inverse g and that (a, b) is a point that lies on the graph of f at which $f'(a) \neq 0$. Then

$$g'(b) = \frac{1}{f'(a)}.$$

More generally, for any x in the domain of g', we have $g'(x) = 1/f'(g(x))$.

The rules we derived for $\ln(x)$, $\arcsin(x)$, and $\arctan(x)$ are all examples of this general property of the derivative of an inverse function. For instance, with $g(x) = \ln(x)$ and $f(x) = e^x$, it follows that

$$g'(x) = \frac{1}{f'(g(x))} = \frac{1}{e^{\ln(x)}} = \frac{1}{x}.$$

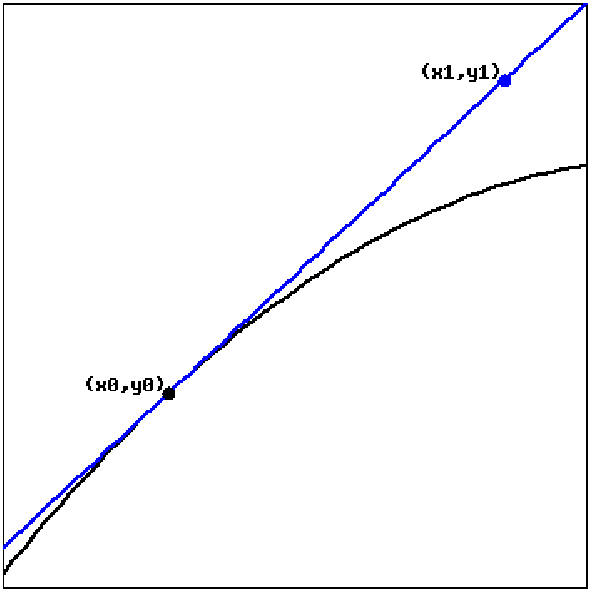
2.6.6 Summary

- For all positive real numbers x, $\frac{d}{dx}[\ln(x)] = \frac{1}{x}$.

- For all real numbers x such that $-1 < x < 1$, $\frac{d}{dx}[\arcsin(x)] = \frac{1}{\sqrt{1-x^2}}$. In addition, for all real numbers x, $\frac{d}{dx}[\arctan(x)] = \frac{1}{1+x^2}$.
- If g is the inverse of a differentiable function f, then for any point x in the domain of g', $g'(x) = \frac{1}{f'(g(x))}$.

2.6.7 Exercises

1. Let $(x_0, y_0) = (3, 3)$ and $(x_1, y_1) = (3.4, 3.2)$. Use the following graph of the function f to find the indicated derivatives.



If $h(x) = (f(x))^3$, then

$h'(3) =$ ______________________________

If $g(x) = f^{-1}(x)$, then

$g'(3) =$ ______________________________

2. Find the derivative of the function $f(t)$, below.

$f(t) = \ln(t^8 + 8)$

$f'(t) =$ ____

3. Let

$$f(x) = 8\sin^{-1}\left(x^3\right)$$

$f'(x) =$ ____

NOTE: The webwork system will accept $\arcsin(x)$ or $\sin^{-1}(x)$ as the inverse of $\sin(x)$.

4. If $f(x) = 2x^4 \arctan(2x^3)$, find $f'(x)$.

$f'(x) =$ ______

5. For each of the given functions $f(x)$, find the derivative $\left(f^{-1}\right)'(c)$ at the given point c, first finding $a = f^{-1}(c)$.

a) $f(x) = 3x + 9x^{13}$; $c = -12$

$a =$ ______

$\left(f^{-1}\right)'(c) =$ ______

b) $f(x) = x^2 - 9x + 26$ on the interval $[4.5, \infty)$; $c = 8$

$a =$ ______

$\left(f^{-1}\right)'(c) =$ ______

6. Given that $f(x) = 2x + \cos(x)$ is one-to-one, use the formula

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

to find $(f^{-1})'(1)$.

$(f^{-1})'(1) =$ ______

7. Consider the graph of $y = f(x)$ provided in Figure 2.6.7 and use it to answer the following questions.

a. Use the provided graph to estimate the value of $f'(1)$.

b. Sketch an approximate graph of $y = f^{-1}(x)$. Label at least three distinct points on the graph that correspond to three points on the graph of f.

c. Based on your work in (a), what is the value of $(f^{-1})'(-1)$? Why?

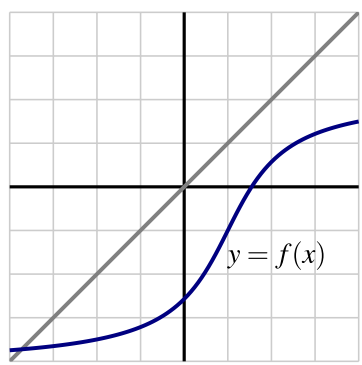


Figure 2.6.7 A function $y = f(x)$

8. Determine the derivative of each of the following functions. Use proper notation and clearly identify the derivative rules you use.

a. $f(x) = \ln(2\arctan(x) + 3\arcsin(x) + 5)$

b. $r(z) = \arctan(\ln(\arcsin(z)))$

c. $q(t) = \arctan^2(3t)\arcsin^4(7t)$

d. $g(v) = \ln\left(\dfrac{\arctan(v)}{\arcsin(v) + v^2}\right)$

9. Let $f(x) = \frac{1}{4}x^3 + 4$.

a. Sketch a graph of $y = f(x)$ and explain why f is an invertible function.

b. Let g be the inverse of f and determine a formula for g.

c. Compute $f'(x)$, $g'(x)$, $f'(2)$, and $g'(6)$. What is the special relationship between $f'(2)$ and $g'(6)$? Why?

10. Let $h(x) = x + \sin(x)$.

a. Sketch a graph of $y = h(x)$ and explain why h must be invertible.

b. Explain why it does not appear to be algebraically possible to determine a formula for h^{-1}.

c. Observe that the point $(\frac{\pi}{2}, \frac{\pi}{2} + 1)$ lies on the graph of $y = h(x)$. Determine the value of $(h^{-1})'(\frac{\pi}{2} + 1)$.

2.7 Derivatives of functions given implicitly

Motivating Questions

- What does it mean to say that a curve is an implicit function of x, rather than an explicit function of x?
- How does implicit differentiation enable us to find a formula for $\frac{dy}{dx}$ when y is an implicit function of x?
- In the context of an implicit curve, how can we use $\frac{dy}{dx}$ to answer important questions about the tangent line to the curve?

2.7.1 Introduction

In all of our studies with derivatives so far, we have worked with functions whose formula is given explicitly in terms of x. But there are many interesting curves whose equations involving x and y are impossible to solve for y in terms of x.

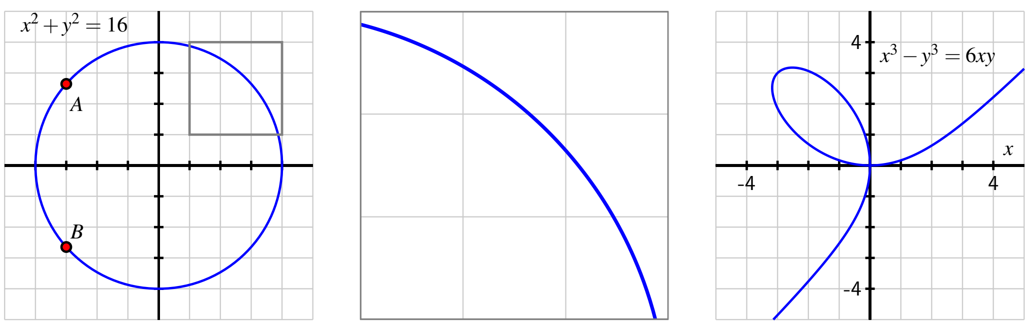


Figure 2.7.1 At left, the circle given by $x^2 + y^2 = 16$. In the middle, the portion of the circle $x^2 + y^2 = 16$ that has been highlighted in the box at left. And at right, the curve given by $x^3 - y^3 = 6xy$.

Perhaps the simplest and most natural of all such curves are circles. Because of the circle's symmetry, for each x value strictly between the endpoints of the horizontal diameter, there are two corresponding y-values. For instance, in Figure 2.7.1, we have labeled $A = (-3, \sqrt{7})$ and $B = (-3, -\sqrt{7})$, and these points demonstrate that the circle fails the vertical line test. Hence, it is impossible to represent the circle through a single function of the form $y = f(x)$. But portions of the circle can be represented explicitly as a function of x, such as the highlighted arc that is magnified in the center of Figure 2.7.1. Moreover, it is evident that the circle is locally linear, so we ought to be able to find a tangent line to the curve at every point. Thus, it makes sense to wonder if we can compute $\frac{dy}{dx}$ at any point on the circle, even though we cannot write y explicitly as a function of x.

We say that the equation $x^2 + y^2 = 16$ defines y *implicitly* as a function of x. The graph of the equation can be broken into pieces where each piece can be defined by an explicit function of x. For the circle, we could choose to take the top half as one explicit function of x, namely $y = \sqrt{16 - x^2}$ and the bottom half as the explicit function $y = -\sqrt{16 - x^2}$. The equation for the circle defines an *implicit function* of x.

The righthand curve in Figure 2.7.1 is called the *folium of Descartes* and is just one of many fascinating possibilities for implicitly given curves.

How can we find an equation for $\frac{dy}{dx}$ without an explicit formula for y in terms of x? To begin answering this question, the following preview activity reminds us of some ways we can compute derivatives of functions in settings where the function's formula is not known.

Preview Activity 2.7.1. Let f be a differentiable function of x (whose formula is not known) and recall that $\frac{d}{dx}[f(x)]$ and $f'(x)$ are interchangeable notations. Determine each of the following derivatives of combinations of explicit functions of x, the unknown function f, and an arbitrary constant c.

(a) $\frac{d}{dx}\left[x^2 + f(x)\right]$

(b) $\frac{d}{dx}\left[x^2 f(x)\right]$

(c) $\frac{d}{dx}\left[c + x + f(x)^2\right]$

(d) $\frac{d}{dx}\left[f(x^2)\right]$

(e) $\frac{d}{dx}\left[x f(x) + f(cx) + c f(x)\right]$

2.7.2 Implicit Differentiation

We begin our exploration of implicit differentiation with the example of the circle given by $x^2 + y^2 = 16$. How can we find a formula for $\frac{dy}{dx}$?

By viewing y as an *implicit* function of x, we think of y as some function whose formula $f(x)$ is unknown, but which we can differentiate. Just as y represents an unknown formula, so too its derivative with respect to x, $\frac{dy}{dx}$, will be (at least temporarily) unknown.

So we view y as an unknown differentiable function of x and differentiate both sides of the equation with respect to x.

$$\frac{d}{dx}\left[x^2 + y^2\right] = \frac{d}{dx}[16].$$

On the right, the derivative of the constant 16 is 0, and on the left we can apply the sum rule, so it follows that

$$\frac{d}{dx}\left[x^2\right] + \frac{d}{dx}\left[y^2\right] = 0.$$

Note carefully the different roles being played by x and y. Because x is the independent variable, $\frac{d}{dx}\left[x^2\right] = 2x$. But y is the dependent variable and y is an implicit function of x.

Recall Preview Activity 2.7.1, where we computed $\frac{d}{dx}[f(x)^2]$. Computing $\frac{d}{dx}[y^2]$ is the same, and requires the chain rule, by which we find that $\frac{d}{dx}[y^2] = 2y^1\frac{dy}{dx}$. We now have that

$$2x + 2y\frac{dy}{dx} = 0.$$

We solve this equation for $\frac{dy}{dx}$ by subtracting $2x$ from both sides and dividing by $2y$.

$$\frac{dy}{dx} = -\frac{2x}{2y} = -\frac{x}{y}.$$

Let's think further about the result that $\frac{dy}{dx} = -\frac{x}{y}$. First, notice that this expression for the derivative involves both x and y. This makes sense because there are two corresponding points on the circle for each value of x between -4 and 4, and the slope of the tangent line is different at each of these points. Second, this formula is entirely consistent with our understanding of circles. The slope of the radius from the origin to the point (a, b) is $m_r = \frac{b}{a}$. The tangent line to the circle at (a, b) is perpendicular to the radius, and thus has slope $m_t = -\frac{a}{b}$, as shown in Figure 2.7.2. In particular, the slope of the tangent line is zero at $(0, 4)$ and $(0, -4)$, and is undefined at $(-4, 0)$ and $(4, 0)$.

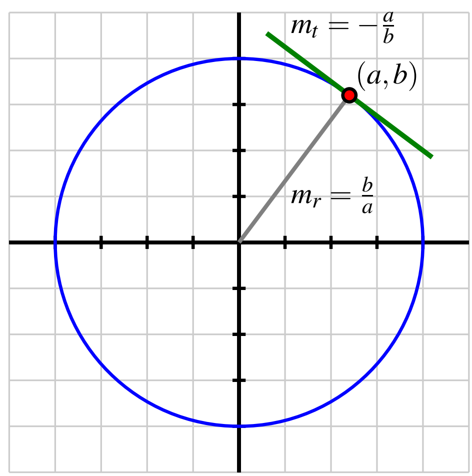


Figure 2.7.2 The circle given by $x^2+y^2 = 16$ with point (a, b) on the circle and the tangent line at that point, with labeled slopes of the radial line, m_r, and tangent line, m_t.

All of these observations about the circle are consistent with the formula $\frac{dy}{dx} = -\frac{x}{y}$.

Example 2.7.3 For the curve given implicitly by $x^3+y^2-2xy = 2$, find the slope of the tangent line at $(-1, 1)$.

Solution. We begin by differentiating the curve's equation implicitly. Taking the derivative of each side with respect to x,

$$\frac{d}{dx}\left[x^3 + y^2 - 2xy\right] = \frac{d}{dx}\left[2\right],$$

by the sum rule and the fact that the derivative of a constant is zero, we have

$$\frac{d}{dx}[x^3] + \frac{d}{dx}[y^2] - \frac{d}{dx}[2xy] = 0.$$

For the three derivatives we now must execute, the first uses the simple power rule, the second requires the chain rule (since y is an implicit function of x), and the third necessitates the product rule (again since y is a function of x). Applying these rules, we now find that

$$3x^2 + 2y\frac{dy}{dx} - [2x\frac{dy}{dx} + 2y] = 0.$$

We want to solve this equation for $\frac{dy}{dx}$. To do so, we first collect all of the terms involving $\frac{dy}{dx}$ on one side of the equation.

$$2y\frac{dy}{dx} - 2x\frac{dy}{dx} = 2y - 3x^2.$$

Then we factor the left side to isolate $\frac{dy}{dx}$.

$$\frac{dy}{dx}(2y - 2x) = 2y - 3x^2.$$

Finally, we divide both sides by $(2y - 2x)$ and conclude that

$$\frac{dy}{dx} = \frac{2y - 3x^2}{2y - 2x}.$$

Note that the expression for $\frac{dy}{dx}$ depends on both x and y. To find the slope of the tangent line at $(-1, 1)$, we substitute the coordinates into the formula for $\frac{dy}{dx}$, using the notation

$$\left.\frac{dy}{dx}\right|_{(-1,1)} = \frac{2(1) - 3(-1)^2}{2(1) - 2(-1)} = -\frac{1}{4}.$$

If we plot[1] the implicit curve and its tangent line, as shown in Figure 2.7.4, we see that the slope we have found matches the graph and confirms the local linearity of this curve at $(-1, 1)$.

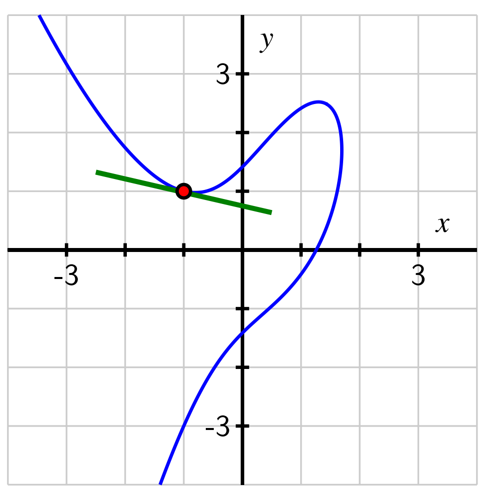


Figure 2.7.4 The curve $x^3 + y^2 - 2xy = 2$ and its tangent line at $(-1, 1)$.

◊

Example 2.7.3 shows that it is possible when differentiating implicitly to have multiple terms involving $\frac{dy}{dx}$. We use addition and subtraction to collect all terms involving $\frac{dy}{dx}$ on one side of the equation, then factor to get a single term of $\frac{dy}{dx}$. Finally, we divide to solve for $\frac{dy}{dx}$.

We use the notation

$$\left.\frac{dy}{dx}\right|_{(a,b)}$$

to denote the evaluation of $\frac{dy}{dx}$ at the point (a,b). This is analogous to writing $f'(a)$ when f' depends on a single variable.

There is a big difference between writing $\frac{d}{dx}$ and $\frac{dy}{dx}$. For example,

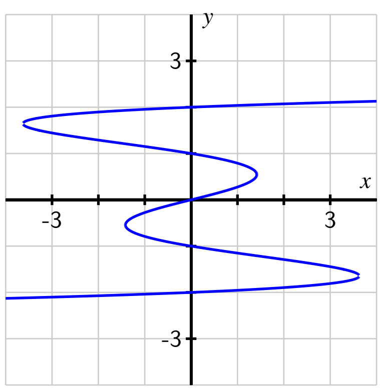
$$\frac{d}{dx}[x^2+y^2]$$

gives an instruction to take the derivative with respect to x of the quantity x^2+y^2, presumably where y is a function of x. On the other hand,

$$\frac{dy}{dx}(x^2+y^2)$$

means the product of the derivative of y with respect to x with the quantity x^2+y^2. Understanding this notational subtlety is essential.

Activity 2.7.2. Consider the curve defined by the equation $x = y^5 - 5y^3 + 4y$, whose graph is pictured in the following figure



(a) Explain why it is not possible to express y as an explicit function of x.

(b) Use implicit differentiation to find a formula for dy/dx.

(c) Use your result from part (b) to find an equation of the line tangent to the graph of $x = y^5 - 5y^3 + 4y$ at the point $(0,1)$.

[1]In *Desmos*, entering the equation $x^3 + y^2 - 2xy = 2$ will automatically graph the corresponding curve.

(d) Use your result from part (b) to determine all of the points at which the graph of $x = y^5 - 5y^3 + 4y$ has a vertical tangent line.

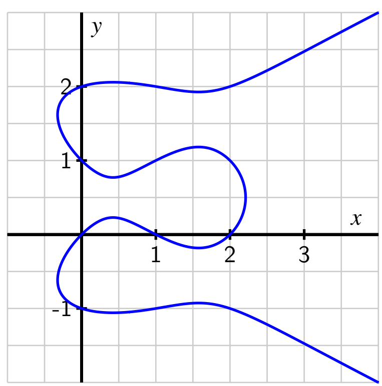
It is natural to ask where the tangent line to a curve is vertical or horizontal. The slope of a horizontal tangent line must be zero, while the slope of a vertical tangent line is undefined. Often the formula for $\frac{dy}{dx}$ is expressed as a quotient of functions of x and y, say

$$\frac{dy}{dx} = \frac{p(x,y)}{q(x,y)}.$$

The tangent line is horizontal precisely when the numerator is zero and the denominator is nonzero, making the slope of the tangent line zero. If we can solve the equation $p(x,y) = 0$ for either x and y in terms of the other, we can substitute that expression into the original equation for the curve. This gives an equation in a single variable, and if we can solve that equation we can find the point(s) on the curve where $p(x,y) = 0$. At those points, the tangent line is horizontal.

Similarly, the tangent line is vertical whenever $q(x,y) = 0$ and $p(x,y) \neq 0$, making the slope undefined.

Activity 2.7.3. Consider the curve defined by the equation $y(y^2 - 1)(y - 2) = x(x - 1)(x - 2)$, whose graph is pictured in the following figure.



Through implicit differentiation, it can be shown that

$$\frac{dy}{dx} = \frac{(x-1)(x-2) + x(x-2) + x(x-1)}{(y^2-1)(y-2) + 2y^2(y-2) + y(y^2-1)}.$$

Use this fact to answer each of the following questions.

(a) Determine all points (x,y) at which the tangent line to the curve is horizontal. (Use technology appropriately to find the needed zeros of the relevant polynomial function.)

(b) Determine all points (x,y) at which the tangent line is vertical. (Use technology appropriately to find the needed zeros of the relevant polynomial function.)

(c) Find the equation of the tangent line to the curve at one of the points where $x = 1$.

Activity 2.7.4. For each of the following curves, use implicit differentiation to find dy/dx and determine the equation of the tangent line at the given point.

(a) $x^3 - y^3 = 6xy,\ (-3, 3)$

(b) $\sin(y) + y = x^3 + x,\ (0, 0)$

(c) $3xe^{-xy} = y^2,\ (0.619061, 1)$

2.7.3 Summary

- In an equation involving x and y where portions of the graph can be defined by explicit functions of x, we say that y is an implicit function of x. A good example of such a curve is the unit circle.
- We use implicit differentiation to differentiate an implicitly defined function. We differentiate both sides of the equation with respect to x, treating y as a function of x by applying the chain rule. If possible, we subsequently solve for $\frac{dy}{dx}$ using algebra.
- While $\frac{dy}{dx}$ may now involve both the variables x and y, $\frac{dy}{dx}$ still gives the slope of the tangent line to the curve. It may be used to decide where the tangent line is horizontal ($\frac{dy}{dx} = 0$) or vertical ($\frac{dy}{dx}$ is undefined), or to find the equation of the tangent line at a particular point on the curve.

2.7.4 Exercises

1. Find dy/dx in terms of x and y if $x^3y - x - 2y - 4 = 0$.

$\frac{dy}{dx} =$ ____________________

2. Find the slope of the tangent to the curve $x^4 + 3xy + y^2 = 29$ at $(1, 4)$.

The slope is _____.

*(Enter **undef** if the slope is not defined at this point.)*

3. Find dy/dx by implicit differentiation.

$$\sqrt{x+y} = 7 + x^2y^2$$

$dy/dx =$ ____

4. Find dy/dx by implicit differentiation.

$$e^{x^2 y} = x + y$$

$dy/dx =$ ______

5. Find $\dfrac{dy}{dx}$ in terms of x and y if $x \ln y + y^7 = 7 \ln x$.

$\dfrac{dy}{dx} =$ ______

6. Find dy/dx in terms of x and y if $\arcsin(x^7 y) = xy^7$.

$\frac{dy}{dx} =$ ______

7. Find the slope of the tangent line to the ellipse $\dfrac{x^2}{4} + \dfrac{y^2}{36} = 1$ at the point (x, y).

slope = ______

Are there any points where the slope is not defined? (Enter them as comma-separated ordered-pairs, e.g., (1,3), (-2,5). Enter ***none*** if there are no such points.)

slope is undefined at ______

8. Find the slope of the tangent to the curve $y^2 = \dfrac{x^3}{xy + 6}$ at $(2, 1)$

$\frac{dy}{dx} =$ ______

*(Enter **undef** if the slope is not defined at this point.)*

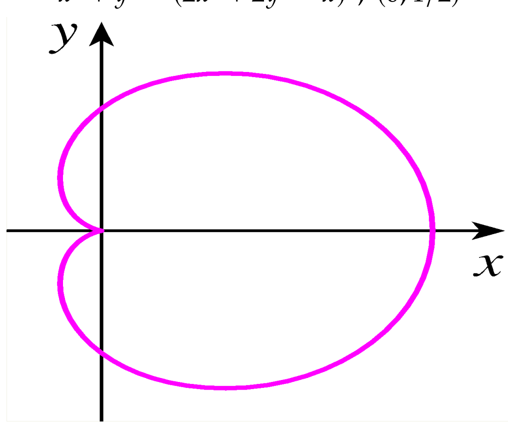
9. If

$$g(x) + x\sin(g(x)) = x^2 - 9 \qquad \text{and} \qquad g(3) = 0,$$

then $g'(3) =$ ______

10. Use implicit differentiation to find an equation of the tangent line to the curve at the given point.

$$x^2 + y^2 = (2x^2 + 2y^2 - x)^2, \ (0, 1/2)$$



(cardioid)

$y =$ ____________________

11. Use implicit differentiation to find an equation of the tangent line to curve called the ***devil's curve***,defined by $y^2(y^2 - 4) = x^2(x^2 - 7)$, at the point $(0, 2)$.

An equation of the tangent line to the devil's curve at the given point is __________.

(Be sure to enter the entire equation of the line, including the "$y =$".

12. Consider the curve given by the equation $2y^3 + y^2 - y^5 = x^4 - 2x^3 + x^2$. Find all points at which the tangent line to the curve is horizontal or vertical. Be sure to use a graphing utility to plot this implicit curve and to visually check the results of algebraic reasoning that you use to determine where the tangent lines are horizontal and vertical.

13. For the curve given by the equation $\sin(x + y) + \cos(x - y) = 1$, find the equation of the tangent line to the curve at the point $(\frac{\pi}{2}, \frac{\pi}{2})$.

14. Implicit differentiation enables us a different perspective from which to see why the rule $\frac{d}{dx}[a^x] = a^x \ln(a)$ holds, if we assume that $\frac{d}{dx}[\ln(x)] = \frac{1}{x}$. This exercise leads you through the key steps to do so.

a. Let $y = a^x$. Rewrite this equation using the natural logarithm function to write x in terms of y (and the constant a).

b. Differentiate both sides of the equation you found in (a) with respect to x, keeping in mind that y is implicitly a function of x.

c. Solve the equation you found in (b) for $\frac{dy}{dx}$, and then use the definition of y to write $\frac{dy}{dx}$ solely in terms of x. What have you found?

CHAPTER 3

Using Derivatives

3.1 Related rates

Motivating Questions

- If two quantities that are related, such as the radius and volume of a spherical balloon, are both changing as implicit functions of time, how are their rates of change related? That is, how does the relationship between the values of the quantities affect the relationship between their respective derivatives with respect to time?

3.1.1 Introduction

In most of our applications of the derivative so far, we have been interested in the instantaneous rate at which one variable, say y, changes with respect to another, say x, leading us to compute and interpret $\frac{dy}{dx}$. We next consider situations where several variable quantities are related, but where each quantity is implicitly a function of time, which will be represented by the variable t. Through understanding how the quantities themselves are related, we will be able to determine how their respective rates of change with respect to time are related.

For example, suppose that air is being pumped into a spherical balloon so that its volume increases at a constant rate of 20 cubic inches per second. Since the balloon's volume and radius are related, by knowing how fast the volume is changing, we ought to be able to discover how fast the radius is changing. We are interested in questions such as: can we determine how fast the radius of the balloon is increasing at the moment the balloon's diameter is 12 inches?

Preview Activity 3.1.1. A spherical balloon is being inflated at a constant rate of 20 cubic inches per second. How fast is the radius of the balloon changing at the instant the balloon's diameter is 12 inches? Is the radius changing more rapidly when $d = 12$ or when $d = 16$? Why?

(a) Draw at least three different spheres with different radii. As the volume changes, what other aspects of the balloon also change?

(b) Recall that the volume of a sphere of radius r is $V = \frac{4}{3}\pi r^3$. Note well that in the setting of this problem, *both* V and r are changing as time t changes, and thus both V and r may be viewed as implicit functions of t, with respective derivatives $\frac{dV}{dt}$ and $\frac{dr}{dt}$. Differentiate both sides of the equation $V = \frac{4}{3}\pi r^3$ with respect to t (using the chain rule on the right) to find a formula for $\frac{dV}{dt}$ that depends on both r and $\frac{dr}{dt}$.

(c) At this point in our work, by differentiating with respect to t we have "related the rates" of change of V and r. Recall that we are given in the problem that the balloon is being inflated at a constant *rate* of 20 cubic inches per second. Is this rate the value of $\frac{dr}{dt}$ or $\frac{dV}{dt}$? Why?

(d) From part (c), we know the value of $\frac{dV}{dt}$ at every value of t. Next, observe that when the diameter of the balloon is 12, we know the value of the radius. In the equation $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$, substitute these values for the relevant quantities and solve for the remaining unknown quantity, which is $\frac{dr}{dt}\big|_{d=12}$. How fast is the radius changing at the instant $d = 12$? What are the units on this quantity?

(e) How is the situation different when $d = 16$? When is the radius changing more rapidly, when $d = 12$ or when $d = 16$?

3.1.2 Related Rates Problems

In problems where two or more quantities can be related to one another, and all of the variables involved are implicitly functions of time, t, we are often interested in how their rates are related; we call these *related rates* problems. Once we have an equation establishing the relationship among the variables, we differentiate implicitly with respect to time to find connections among the rates of change.

Example 3.1.1 Sand is being dumped by a conveyor belt onto a pile so that the sand forms a right circular cone, as pictured in Figure 3.1.2. How are the instantaneous rates of change of the sand's volume, height, and radius related to one another?

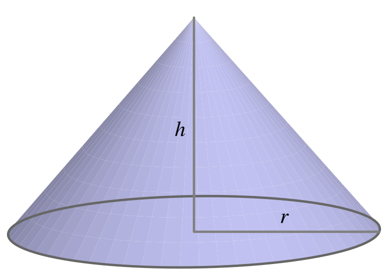


Figure 3.1.2 A conical pile of sand.

Solution. As sand falls from the conveyor belt, several features of the sand pile will change: the volume of the pile will grow, the height will increase, and the radius will get bigger, too. All of these quantities are related to one another, and the rate at which each is changing is related to the rate at which sand falls from the conveyor.

We begin by identifying which variables are changing and how they are related. In this situation, we observe that the radius and height of the pile are related to its volume by the standard equation for the volume of a cone,

$$V = \frac{1}{3}\pi r^2 h.$$

Viewing each of V, r, and h as functions of t, we differentiate implicitly to arrive at an equation that relates their respective rates of change. Taking the derivative of each side of the equation with respect to t, we find

$$\frac{d}{dt}[V] = \frac{d}{dt}\left[\frac{1}{3}\pi r^2 h\right].$$

On the left, $\frac{d}{dt}[V]$ is simply $\frac{dV}{dt}$. On the right, the situation is more complicated, as both r and h are implicit functions of t. Hence we need the product and chain rules. We find that

$$\begin{aligned}\frac{dV}{dt} &= \frac{d}{dt}\left[\frac{1}{3}\pi r^2 h\right]\\ &= \frac{1}{3}\pi r^2 \frac{d}{dt}[h] + \frac{1}{3}\pi h \frac{d}{dt}[r^2]\\ &= \frac{1}{3}\pi r^2 \frac{dh}{dt} + \frac{1}{3}\pi h 2r \frac{dr}{dt}\end{aligned}$$

(Note particularly how we are using ideas from Section 2.7 on implicit differentiation. There we found that when y is an implicit function of x, $\frac{d}{dx}[y^2] = 2y\frac{dy}{dx}$. The same ideas are applied here when we compute $\frac{d}{dt}[r^2] = 2r\frac{dr}{dt}$.)

The equation

$$\frac{dV}{dt} = \frac{1}{3}\pi r^2 \frac{dh}{dt} + \frac{2}{3}\pi r h \frac{dr}{dt},$$

relates the rates of change of V, h, and r. ◊

If we are given sufficient additional information, we may then find the value of one or more of these rates of change at a specific point in time.

Example 3.1.3 In the setting of Example 3.1.1, suppose we also know the following: (a) sand falls from the conveyor in such a way that the height of the pile is always half the radius, and (b) sand falls from the conveyor belt at a constant rate of 10 cubic feet per minute. How fast is the height of the sandpile changing at the moment the radius is 4 feet?

Solution. The information that the height is always half the radius tells us that for all values of t, $h = \frac{1}{2}r$. Differentiating with respect to t, it follows that $\frac{dh}{dt} = \frac{1}{2}\frac{dr}{dt}$. These relationships enable us to relate $\frac{dV}{dt}$ to just one of r or h. Substituting the expressions involving r and $\frac{dr}{dt}$

for h and $\frac{dh}{dt}$, we now have that

$$\frac{dV}{dt} = \frac{1}{3}\pi r^2 \cdot \frac{1}{2}\frac{dr}{dt} + \frac{2}{3}\pi r \cdot \frac{1}{2}r \cdot \frac{dr}{dt}. \tag{3.1.1}$$

Since sand falls from the conveyor at the constant rate of 10 cubic feet per minute, the value of $\frac{dV}{dt}$, the rate at which the volume of the sand pile changes, is $\frac{dV}{dt} = 10$ ft^3/min. We are interested in how fast the height of the pile is changing at the instant when $r = 4$, so we substitute $r = 4$ and $\frac{dV}{dt} = 10$ into Equation (3.1.1), to find

$$10 = \frac{1}{3}\pi 4^2 \cdot \frac{1}{2}\left.\frac{dr}{dt}\right|_{r=4} + \frac{2}{3}\pi 4 \cdot \frac{1}{2}4 \cdot \left.\frac{dr}{dt}\right|_{r=4} = \frac{8}{3}\pi \left.\frac{dr}{dt}\right|_{r=4} + \frac{16}{3}\pi \left.\frac{dr}{dt}\right|_{r=4}.$$

Only the value of $\left.\frac{dr}{dt}\right|_{r=4}$ remains unknown. We combine like terms on the right side of the equation above to get $10 = 8\pi \left.\frac{dr}{dt}\right|_{r=4}$, and solve for $\left.\frac{dr}{dt}\right|_{r=4}$ to find

$$\left.\frac{dr}{dt}\right|_{r=4} = \frac{10}{8\pi} \approx 0.39789$$

feet per minute. Because we were interested in how fast the height of the pile was changing at this instant, we want to know $\frac{dh}{dt}$ when $r = 4$. Since $\frac{dh}{dt} = \frac{1}{2}\frac{dr}{dt}$ for all values of t, it follows

$$\left.\frac{dh}{dt}\right|_{r=4} = \frac{5}{8\pi} \approx 0.19894 \text{ ft/min.}$$

◊

Observe the difference between the notations $\frac{dr}{dt}$ and $\left.\frac{dr}{dt}\right|_{r=4}$. The former represents the rate of change of r with respect to t at an arbitrary value of t, while the latter is the rate of change of r with respect to t at a particular moment, the moment when $r = 4$.

Had we known that $h = \frac{1}{2}r$ at the beginning of Example 3.1.1, we could have immediately simplified our work by writing V solely in terms of r to have

$$V = \frac{1}{3}\pi r^2 \left(\frac{1}{2}r\right) = \frac{1}{6}\pi r^3.$$

From this last equation, differentiating with respect to t implies

$$\frac{dV}{dt} = \frac{1}{2}\pi r^2 \frac{dr}{dt},$$

from which the same conclusions can be made.

Our work with the sandpile problem above is similar in many ways to our approach in Preview Activity 3.1.1, and these steps are typical of most related rates problems. In certain ways, they also resemble work we do in applied optimization problems, and here we summarize the main approach for consideration in subsequent problems.

Note 3.1.4

- Draw two or three possible figures that clearly represent the situation. Identify the quantities in the problem that are changing and choose clearly defined variable names for them. Identify the quantities or relationships that are not changing.
- Determine all rates of change that are known or given and identify the rate(s) of change to be found.
- Find an equation that relates the variables whose rates of change are known to those variables whose rates of change are to be found.
- Differentiate implicitly with respect to t to relate the rates of change of the involved quantities.
- Evaluate the derivatives and variables at the information relevant to the instant at which a certain rate of change is sought. Use proper notation to identify when a derivative is being evaluated at a particular instant, such as $\left.\frac{dr}{dt}\right|_{r=4}$.

When identifying variables and drawing pictures, it is important to think about the dynamic ways in which the quantities change. Usually a sequence of several pictures is helpful; for some pictures that can be easily modified as interactive built in Geogebra, see the following links,[1] which represent

- how a circular oil slick's area grows[2] as its radius increases;
- how the location of the base of a ladder and its height along a wall change[3] as the ladder slides;
- how the water level changes[4] in a conical tank as it fills with water at a constant rate (compare the setting in Activity 3.1.2);
- how a skateboarder's shadow changes[5] as he moves past a lamppost.

Drawing well-labeled diagrams (usually several of them) and envisioning how different parts of the figure change is a key part of understanding related rates problems and being successful at solving them.

Activity 3.1.2. A water tank has the shape of an inverted circular cone (point down) with a base of radius 6 feet and a depth of 8 feet. Suppose that water is being pumped into the tank at a constant instantaneous rate of 4 cubic feet per minute.

(a) Draw 2-3 pictures of the conical tank that show sketches of the water level at different points in time when the tank is not yet full. Introduce variables that measure the radius of the water's surface and the water's depth in the tank, and label them on your figure.

(b) Say that r is the radius and h the depth of the water at a given time, t. What

[1]We again refer to the work of Prof. Marc Renault, found at gvsu.edu/s/5p.
[2]gvsu.edu/s/9n
[3]gvsu.edu/s/9o
[4]gvsu.edu/s/9p
[5]gvsu.edu/s/9q

equation relates the radius and height of the water, and why?

(c) Determine an equation that relates the volume of water in the tank at time t to the depth h of the water at that time.

(d) Through differentiation, find an equation that relates the instantaneous rate of change of water volume with respect to time to the instantaneous rate of change of water depth at time t.

(e) Find the instantaneous rate at which the water level is rising when the water in the tank is 3 feet deep.

(f) When is the water rising most rapidly: at $h = 3$, $h = 4$, or $h = 5$? Why?

Recognizing which geometric relationships are relevant in a given problem often simplifies the process of differentiation when we relate the rates. For instance, although the problem in Activity 3.1.2 is about a conical tank, the most important fact is that there are two similar right triangles involved. In another setting, we might use the Pythagorean Theorem to relate the legs of the triangle. But in the conical tank, the fact that the water fills the tank in such a way that that the ratio of radius to depth is constant turns out to be the important relationship. In other situations where a changing angle is involved, trigonometric functions may provide the means to find relationships among various parts of the triangle.

Activity 3.1.3. A television camera is positioned 4000 feet from the base of a rocket launching pad. The angle of elevation of the camera has to change at the correct rate in order to keep the rocket in sight. In addition, the auto-focus of the camera has to take into account the increasing distance between the camera and the rocket. We assume that the rocket rises vertically. (A similar situation is discussed and pictured dynamically in this interactive graphic[6]. Exploring the interactive will be helpful to you in answering the questions that follow.)

(a) Draw several figures that show the rocket at different points in time. What quantities are changing? What quantities are constant? Introduce appropriate variables to represent the quantities that are changing.

(b) Find an equation that relates the camera's angle of elevation to the height of the rocket, and then find an equation that relates the instantaneous rate of change of the camera's elevation angle to the instantaneous rate of change of the rocket's height (where all rates of change are with respect to time).

(c) Find an equation that relates the distance from the camera to the rocket to the rocket's height, as well as an equation that relates the instantaneous rate of change of distance from the camera to the rocket to the instantaneous rate of change of the rocket's height (where all rates of change are with respect to time).

(d) Suppose that the rocket's speed is 600 ft/sec at the instant it has risen 3000 feet. How fast is the distance from the television camera to the rocket changing at

[6] webspace.ship.edu/msrenault/GeoGebraCalculus/derivative_app_rr_conical_tank.html

that moment? If the camera is following the rocket, how fast is the camera's angle of elevation changing at that same moment?

(e) If from an elevation of 3000 feet onward the rocket continues to rise at 600 feet/sec, will the rate of change of distance with respect to time be greater when the elevation is 4000 feet than it was at 3000 feet, or less? Why?

In addition to finding instantaneous rates of change at specific instants, we can often make more general observations about how particular rates themselves will change over time. For instance, when a conical tank is filling with water at a constant rate, it seems obvious that the depth of the water should increase more slowly over time. Note how carefully we must phrase the relationship: we mean to say that while the depth, h, of the water is increasing, its rate of change, $\frac{dh}{dt}$, is decreasing (both as a function of t and as a function of h). We make this observation by solving the equation that relates the various rates for one particular rate, without substituting any particular values for known variables or rates. For instance, in the conical tank problem in Activity 3.1.2, we established that

$$\frac{dV}{dt} = \frac{1}{16}\pi h^2 \frac{dh}{dt},$$

and hence

$$\frac{dh}{dt} = \frac{16}{\pi h^2} \frac{dV}{dt}.$$

Provided that $\frac{dV}{dt}$ is constant, it is apparent that as h gets larger, $\frac{dh}{dt}$ will get smaller but remain positive. Hence, the depth of the water is increasing at a decreasing rate.

Activity 3.1.4. As pictured in the interactive graphic[7], a skateboarder who is 6 feet tall rides under a 15 foot tall lamppost at a constant rate of 3 feet per second. We are interested in understanding how fast his shadow is changing at various points in time.

(a) Draw several rights triangle that represents snapshots in time of the skateboarder, lamppost, and his shadow. Let x denote the horizontal distance from the base of the lamppost to the skateboarder and s represent the length of his shadow. Label these quantities, as well as the skateboarder's height and the lamppost's height on the diagram.

(b) Observe that the skateboarder and the lamppost represent parallel line segments in the diagram, and thus similar triangles are present. Use similar triangles to establish an equation that relates x and s.

(c) Use your work in (b) to find an equation that relates $\frac{dx}{dt}$ and $\frac{ds}{dt}$.

(d) At what rate is the length of the skateboarder's shadow increasing at the instant the skateboarder is 8 feet from the lamppost?

[7] gvsu.edu/s/9t

(e) As the skateboarder's distance from the lamppost increases, is his shadow's length increasing at an increasing rate, increasing at a decreasing rate, or increasing at a constant rate?

(f) Which is moving more rapidly: the skateboarder or the tip of his shadow? Explain, and justify your answer.

In the first three activities of this section, we provided guided instruction to build a solution in a step by step way. For the closing activity and the following exercises, most of the detailed work is left to the reader.

Activity 3.1.5. A baseball diamond is 90′ square. A batter hits a ball along the third base line and runs to first base. At what rate is the distance between the ball and first base changing when the ball is halfway to third base, if at that instant the ball is traveling 100 feet/sec? At what rate is the distance between the ball and the runner changing at the same instant, if at the same instant the runner is 1/8 of the way to first base running at 30 feet/sec?

3.1.3 Summary

- When two or more related quantities are changing as implicit functions of time, their rates of change can be related by implicitly differentiating the equation that relates the quantities themselves. For instance, if the sides of a right triangle are all changing as functions of time, say having lengths x, y, and z, then these quantities are related by the Pythagorean Theorem: $x^2 + y^2 = z^2$. It follows by implicitly differentiating with respect to t that their rates are related by the equation

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 2z\frac{dz}{dt},$$

  so that if we know the values of x, y, and z at a particular time, as well as two of the three rates, we can deduce the value of the third.

3.1.4 Exercises

1. If $L = \sqrt{x^2 + y^2}$, $\dfrac{dx}{dt} = -2$, and $\dfrac{dy}{dt} = 2$, find $\dfrac{dL}{dt}$ when $x = 9$ and $y = 5$.

$\dfrac{dL}{dt} =$ ____________________

2. Suppose that water is pouring into a swimming pool in the shape of a right circular cylinder at a constant rate of 8 cubic feet per minute. If the pool has radius 7 feet and height 6 feet, what is the rate of change of the height of the water in the pool when the depth of the water in the pool is 4 feet?

[8] gvsu.edu/s/9q

______________________________ ft/min

3. You say goodbye to your friend at the intersection of two perpendicular roads. At time $t = 0$ you drive off North at a (constant) speed v and your friend drives West at a (constant) speed w. You badly want to know: how fast is the distance between you and your friend increasing at time t?

Enter here the derivative of the distance from your friend with respect to t:

Being scientifically minded you ask yourself how does the speed of separation change with time. In other words, what is the second derivative of the distance between you and your friend?

4. Oil spilled from a ruptured tanker spreads in a circle whose area increases at a constant rate of 9 mi^2/hr. How rapidly is radius of the spill increasing when the area is 5 mi^2?

The radius is increasing at ______________________________ mi/hr.

5. A 12 foot ladder is leaning against a wall.

If the top slips down the wall at a rate of 3 ft/s, how fast will the foot be moving away from the wall when the top is 7 feet above the ground?

The foot will be moving at _________________________ ft/s.

6. A voltage V across a resistance R generates a current $I = V/R$. If a constant voltage of 5 volts is put across a resistance that is increasing at a rate of 0.4 ohms per second when the resistance is 8 ohms, at what rate is the current changing?

rate = ___________________________________

7. When air expands adiabatically (without gaining or losing heat), its pressure P and volume V are related by the equation $PV^{1.4} = C$ where C is a constant. Suppose that at a certain instant the volume is 410 cubic centimeters and the pressure is 77 kPa and is decreasing at a rate of 9 kPa/minute. At what rate in cubic centimeters per minute is the volume increasing at this instant?

Answer: ______________________________

Note: *Pa* stands for Pascal. One *Pa* is equivalent to one Newton/m^2. *kPa* is a kiloPascal or 1000 Pascals.

8. Water is leaking out of an inverted conical tank at a rate of 9500.0 cm^3/min at the same time that water is being pumped into the tank at a constant rate. The tank has height 13.0 m and the the diameter at the top is 6.0 m. If the water level is rising at a rate of 16.0 cm/min when the height of the water is 3.5 m, find the rate at which water is being pumped into the tank in cubic centimeters per minute.

Answer: ______________________________ cm^3/min

9. A sailboat is sitting at rest near its dock. A rope attached to the bow of the boat is drawn in over a pulley that stands on a post on the end of the dock that is 5 feet higher than the bow. If the rope is being pulled in at a rate of 2 feet per second, how fast is the boat approaching the dock when the length of rope from bow to pulley is 13 feet?

10. A swimming pool is 60 feet long and 25 feet wide. Its depth varies uniformly from 3 feet at the shallow end to 15 feet at the deep end, as shown in the Figure 3.1.5.

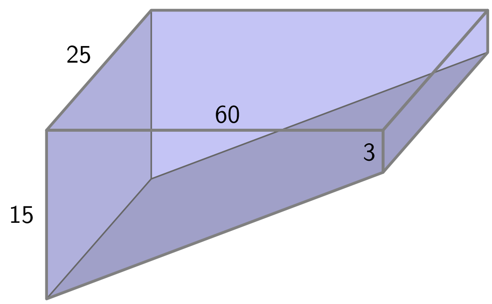


Figure 3.1.5 The swimming pool.

Suppose the pool has been emptied and is now being filled with water at a rate of 800 cubic feet per minute. At what rate is the depth of water (measured at the deepest point of the pool) increasing when it is 5 feet deep at that end? Over time, describe how the depth of the water will increase: at an increasing rate, at a decreasing rate, or at a constant rate. Explain.

11. A baseball diamond is a square with sides 90 feet long. Suppose a baseball player is advancing from second to third base at the rate of 24 feet per second, and an umpire is standing on home plate. Let θ be the angle between the third baseline and the line of sight from the umpire to the runner. How fast is θ changing when the runner is 30 feet from third base?

12. Sand is being dumped off a conveyor belt onto a pile in such a way that the pile forms in the shape of a cone whose radius is always equal to its height. Assuming that the sand is being dumped at a rate of 10 cubic feet per minute, how fast is the height of the pile changing when there are 1000 cubic feet on the pile?

3.2 Using derivatives to evaluate limits

Motivating Questions

- How can derivatives be used to help us evaluate indeterminate limits of the form $\frac{0}{0}$?
- What does it mean to say that $\lim_{x\to\infty} f(x) = L$ and $\lim_{x\to a} f(x) = \infty$?
- How can derivatives assist us in evaluating indeterminate limits of the form $\frac{\infty}{\infty}$?

3.2.1 Introduction

Because differential calculus is based on the definition of the derivative, and the definition of the derivative involves a limit, there is a sense in which all of calculus rests on limits. In addition, the limit involved in the definition of the derivative always generates the indeterminate form $\frac{0}{0}$. If f is a differentiable function, then in the definition

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h},$$

not only does $h \to 0$ in the denominator, but also $(f(x+h) - f(x)) \to 0$ in the numerator, since f is continuous. Remember, saying that a limit has an indeterminate form only means that we don't yet know its value and have more work to do: indeed, limits of the form $\frac{0}{0}$ can take on any value, as is evidenced by evaluating $f'(x)$ for varying values of x for a function such as $f(x) = x^2$.

We have learned many techniques for evaluating the limits that result from the derivative definition, including a large number of shortcut rules. In this section, we turn the situation upside-down: instead of using limits to evaluate derivatives, we explore how to use derivatives to evaluate certain limits.

Preview Activity 3.2.1. Let h be the function given by $h(x) = \frac{x^5+x-2}{x^2-1}$.

(a) What is the domain of h?

(b) Explain why $\lim_{x\to 1} \dfrac{x^5+x-2}{x^2-1}$ results in an indeterminate form.

(c) Next we will investigate the behavior of both the numerator and denominator of h near the point where $x = 1$. Let $f(x) = x^5 + x - 2$ and $g(x) = x^2 - 1$. Find the local linearizations of f and g at $a = 1$, and call these functions $L_f(x)$ and $L_g(x)$, respectively.

(d) Explain why $h(x) \approx \frac{L_f(x)}{L_g(x)}$ for x near $a = 1$.

(e) Using your work from (c) and (d), evaluate

$$\lim_{x \to 1} \frac{L_f(x)}{L_g(x)}.$$

What do you think your result tells us about $\lim_{x \to 1} h(x)$?

(f) Investigate the function $h(x)$ graphically and numerically near $x = 1$. What do you think is the value of $\lim_{x \to 1} h(x)$?

3.2.2 Using derivatives to evaluate indeterminate limits of the form $\frac{0}{0}$.

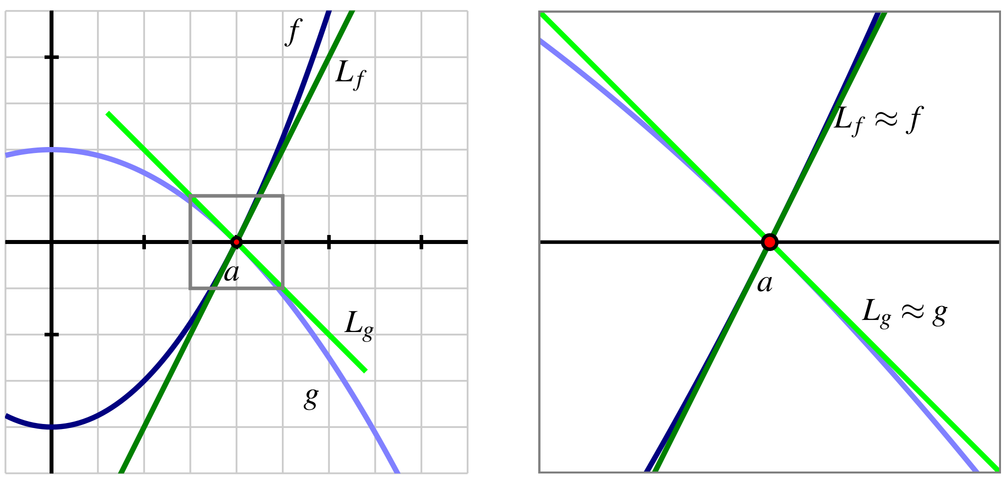


Figure 3.2.1 At left, the graphs of f and g near the value a, along with their tangent line approximations L_f and L_g at $x = a$. At right, zooming in on the point a and the four graphs.

The idea demonstrated in Preview Activity 3.2.1 — that we can evaluate an indeterminate limit of the form $\frac{0}{0}$ by replacing each of the numerator and denominator with their local linearizations at the point of interest — can be generalized in a way that enables us to evaluate a wide range of limits. We have a function $h(x)$ that can be written as a quotient $h(x) = \frac{f(x)}{g(x)}$, where f and g are both differentiable at $x = a$ and for which $f(a) = g(a) = 0$. We would like to evaluate the indeterminate limit given by $\lim_{x \to a} h(x)$. Figure 3.2.1 illustrates the situation. We see that both f and g have an x-intercept at $x = a$. Their respective tangent line approximations L_f and L_g at $x = a$ are also shown in the figure. We can take advantage of the fact that a function and its tangent line approximation become indistinguishable as $x \to a$.

First, let's recall that $L_f(x) = f'(a)(x - a) + f(a)$ and $L_g(x) = g'(a)(x - a) + g(a)$. Because x is getting arbitrarily close to a when we take the limit, we can replace f with L_f and replace g

with L_g, and thus we observe that

$$\begin{aligned}\lim_{x \to a} \frac{f(x)}{g(x)} &= \lim_{x \to a} \frac{L_f(x)}{L_g(x)} \\ &= \lim_{x \to a} \frac{f'(a)(x-a) + f(a)}{g'(a)(x-a) + g(a)}.\end{aligned}$$

Next, we remember that both $f(a) = 0$ and $g(a) = 0$, which is precisely what makes the original limit indeterminate. Substituting these values for $f(a)$ and $g(a)$ in the limit above, we now have

$$\begin{aligned}\lim_{x \to a} \frac{f(x)}{g(x)} &= \lim_{x \to a} \frac{f'(a)(x-a)}{g'(a)(x-a)} \\ &= \lim_{x \to a} \frac{f'(a)}{g'(a)},\end{aligned}$$

where the latter equality holds because $\frac{x-a}{x-a} = 1$ when x is approaching (but not equal to) a. Finally, we note that $\frac{f'(a)}{g'(a)}$ is constant with respect to x, and thus

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{f'(a)}{g'(a)}.$$

This result holds as long as $g'(a)$ is not equal to zero. The formal name of the result is L'Hôpital's Rule.

L'Hôpital's Rule.

Let f and g be differentiable on an open interval that includes $x = a$, and suppose that $f(a) = g(a) = 0$ and that $g'(a) \neq 0$. Then

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{f'(a)}{g'(a)}.$$

In practice, we typically work with a slightly more general version of L'Hôpital's Rule, which states that (under the identical assumptions as the boxed rule above plus the extra assumption that g' is continuous at $x = a$)

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)},$$

provided the righthand limit exists. This form reflects the basic idea of L'Hôpital's Rule: if $\frac{f(x)}{g(x)}$ produces an indeterminate limit of form $\frac{0}{0}$ as $x \to a$, that limit is equivalent to the limit of the quotient of the two functions' derivatives, $\frac{f'(x)}{g'(x)}$.

For example, if we consider the limit from Preview Activity 3.2.1,

$$\lim_{x \to 1} \frac{x^5 + x - 2}{x^2 - 1},$$

by L'Hôpital's Rule we have that

$$\lim_{x\to 1} \frac{x^5+x-2}{x^2-1} = \lim_{x\to 1} \frac{5x^4+1}{2x} = \frac{6}{2} = 3.$$

By replacing the numerator and denominator with their respective derivatives, we often replace an indeterminate limit with one whose value we can easily determine.

Activity 3.2.2. Evaluate each of the following limits. If you use L'Hôpital's Rule, indicate where it was used, and be certain the limit is indeterminate before you apply it.

(a) $\lim_{x\to 0} \frac{\ln(1+x)}{x}$

(b) $\lim_{x\to \pi} \frac{\cos(x)}{x}$

(c) $\lim_{x\to 1} \frac{2\ln(x)}{1-e^{x-1}}$

(d) $\lim_{x\to 0} \frac{\sin(x)-x}{\cos(2x)-1}$

While L'Hôpital's Rule can be applied in an entirely algebraic way, it is important to remember that the justification of the rule is graphical: the main idea is that the slopes of the tangent lines to f and g at $x=a$ determine the value of the limit of $\frac{f(x)}{g(x)}$ as $x \to a$.

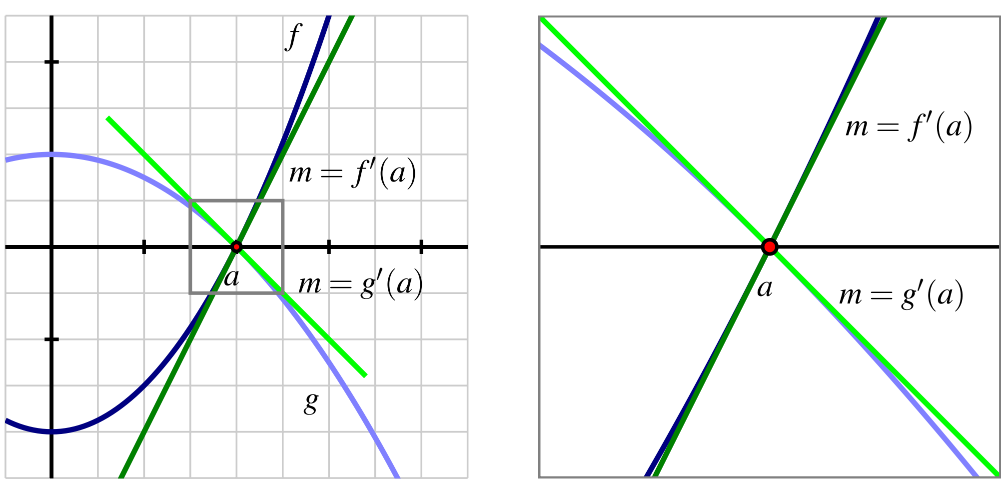


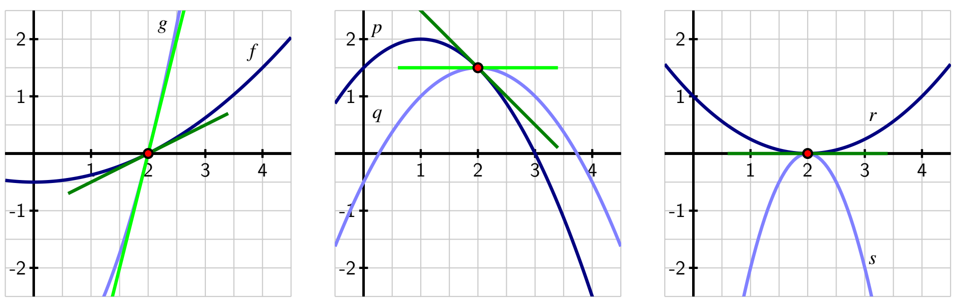
Figure 3.2.2 Two functions f and g that satisfy L'Hôpital's Rule.

We see this in Figure 3.2.2, where we can see from the grid that $f'(a) = 2$ and $g'(a) = -1$, hence by L'Hôpital's Rule,

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{f'(a)}{g'(a)} = \frac{2}{-1} = -2.$$

It's not the fact that f and g both approach zero that matters most, but rather the *rate* at which each approaches zero that determines the value of the limit. This is a good way to remember what L'Hôpital's Rule says: if $f(a) = g(a) = 0$, the the limit of $\frac{f(x)}{g(x)}$ as $x \longrightarrow a$ is given by the ratio of the slopes of f and g at $x = a$.

Activity 3.2.3. In this activity, we reason graphically from the following figure to evaluate limits of ratios of functions about which some information is known.



(a) Use the left-hand graph to determine the values of $f(2)$, $f'(2)$, $g(2)$, and $g'(2)$. Then, evaluate $\lim_{x \to 2} \frac{f(x)}{g(x)}$.

(b) Use the middle graph to find $p(2)$, $p'(2)$, $q(2)$, and $q'(2)$. Then, determine the value of $\lim_{x \to 2} \frac{p(x)}{q(x)}$.

(c) Assume that r and s are functions whose for which $r''(2) \neq 0$ and $s''(2) \neq 0$ Use the right-hand graph to compute $r(2)$, $r'(2)$, $s(2)$, $s'(2)$. Explain why you cannot determine the exact value of $\lim_{x \to 2} \frac{r(x)}{s(x)}$ without further information being provided, but that you can determine the sign of $\lim_{x \to 2} \frac{r(x)}{s(x)}$. In addition, state what the sign of the limit will be, with justification.

3.2.3 Limits involving ∞

The concept of infinity, denoted ∞, arises naturally in calculus, as it does in much of mathematics. It is important to note from the outset that ∞ is a concept, but not a number itself. Indeed, the notion of ∞ naturally invokes the idea of limits. Consider, for example, the function $f(x) = \frac{1}{x}$, whose graph is pictured in Figure 3.2.3.

We note that $x = 0$ is not in the domain of f, so we may naturally wonder what happens as $x \longrightarrow 0$. As $x \longrightarrow 0^+$, we observe that $f(x)$ *increases without bound*. That is, we can make the value of $f(x)$ as large as we like by taking x closer and closer (but not equal) to 0, while keeping $x > 0$. This is a good way to think about what infinity represents: a quantity is tending to infinity if there is no single number that the quantity is always less than.

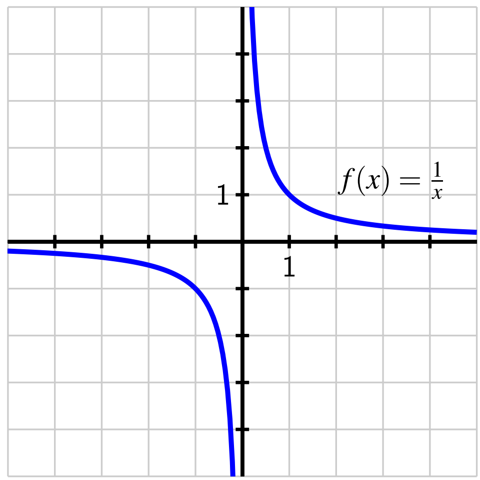


Figure 3.2.3 The graph of $f(x) = \frac{1}{x}$.

Recall that the statement $\lim_{x\to a} f(x) = L$, means that we can make $f(x)$ as close to L as we'd like by taking x sufficiently close (but not equal) to a. We now expand this notation and language to include the possibility that either L or a can be ∞. For instance, for $f(x) = \frac{1}{x}$, we now write

$$\lim_{x\to 0^+} \frac{1}{x} = \infty,$$

by which we mean that we can make $\frac{1}{x}$ as large as we like by taking x sufficiently close (but not equal) to 0. In a similar way, we write

$$\lim_{x\to\infty} \frac{1}{x} = 0,$$

since we can make $\frac{1}{x}$ as close to 0 as we'd like by taking x sufficiently large (i.e., by letting x increase without bound).

In general, the notation $\lim_{x\to a} f(x) = \infty$ means that we can make $f(x)$ as large as we like by taking x sufficiently close (but not equal) to a, and the notation $\lim_{x\to\infty} f(x) = L$ means that we can make $f(x)$ as close to L as we like by taking x sufficiently large. This notation also applies to left- and right-hand limits, and to limits involving $-\infty$. For example, returning to Figure 3.2.3 and $f(x) = \frac{1}{x}$, we can say that

$$\lim_{x\to 0^-} \frac{1}{x} = -\infty \quad \text{and} \quad \lim_{x\to -\infty} \frac{1}{x} = 0.$$

Finally, we write

$$\lim_{x\to\infty} f(x) = \infty$$

if we can make the value of $f(x)$ as large as we'd like by taking x sufficiently large. For example,

$$\lim_{x\to\infty} x^2 = \infty.$$

Limits involving infinity identify *vertical* and *horizontal asymptotes* of a function. If $\lim_{x\to a} f(x) = \infty$, then $x = a$ is a vertical asymptote of f, while if $\lim_{x\to\infty} f(x) = L$, then $y = L$ is a horizontal asymptote of f. Similar statements can be made using $-\infty$, and with left- and right-hand limits as $x \to a^-$ or $x \to a^+$.

In precalculus classes, it is common to study the *end behavior* of certain families of functions, by which we mean the behavior of a function as $x \to \infty$ and as $x \to -\infty$. Here we briefly examine some familiar functions and note the values of several limits involving ∞.

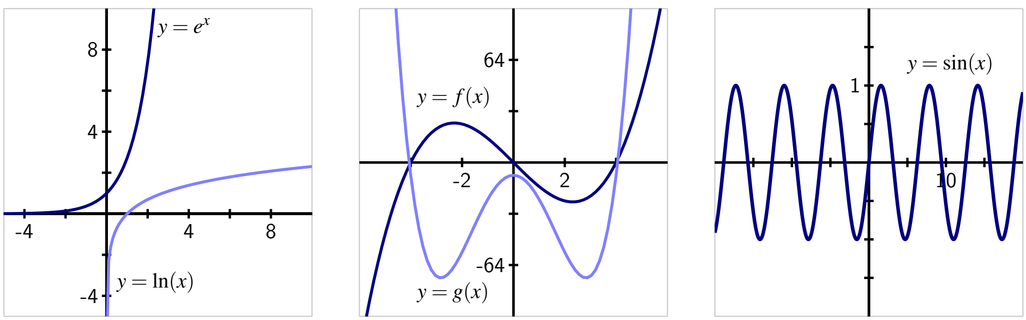


Figure 3.2.4 Graphs of some familiar functions whose end behavior as $x \to \pm\infty$ is known. In the middle graph, $f(x) = x^3 - 16x$ and $g(x) = x^4 - 16x^2 - 8$.

For the natural exponential function e^x, we note that $\lim_{x\to\infty} e^x = \infty$ and $\lim_{x\to-\infty} e^x = 0$. For the exponential decay function e^{-x}, these limits are reversed, with $\lim_{x\to\infty} e^{-x} = 0$ and $\lim_{x\to-\infty} e^{-x} = \infty$. Turning to the natural logarithm function, we have $\lim_{x\to 0^+} \ln(x) = -\infty$ and $\lim_{x\to\infty} \ln(x) = \infty$. While both e^x and $\ln(x)$ grow without bound as $x \to \infty$, the exponential function does so much more quickly than the logarithm function does. We'll soon use limits to quantify what we mean by "quickly."

For polynomial functions of the form

$$p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0,$$

the end behavior depends on the sign of a_n and whether the highest power n is even or odd. If n is even and a_n is positive, then $\lim_{x\to\infty} p(x) = \infty$ and $\lim_{x\to-\infty} p(x) = \infty$, as in the plot of g in Figure 3.2.4. If instead a_n is negative, then $\lim_{x\to\infty} p(x) = -\infty$ and $\lim_{x\to-\infty} p(x) = -\infty$. In the situation where n is odd, then either $\lim_{x\to\infty} p(x) = \infty$ and $\lim_{x\to-\infty} p(x) = -\infty$ (which occurs when a_n is positive, as in the graph of f in Figure 3.2.4), or $\lim_{x\to\infty} p(x) = -\infty$ and $\lim_{x\to-\infty} p(x) = \infty$ (when a_n is negative).

A function can fail to have a limit as $x \to \infty$. For example, consider the plot of the sine function at right in Figure 3.2.4. Because the function continues oscillating between -1 and 1 as $x \to \infty$, we say that $\lim_{x\to\infty} \sin(x)$ does not exist.

Finally, it is straightforward to analyze the behavior of any rational function as $x \to \infty$.

Example 3.2.5 Determine the limit of the function

$$q(x) = \frac{3x^2 - 4x + 5}{7x^2 + 9x - 10}$$

as $x \to \infty$.

Solution. Note that both $(3x^2 - 4x + 5) \to \infty$ as $x \to \infty$ and $(7x^2 + 9x - 10) \to \infty$ as $x \to \infty$. Here we say that $\lim_{x\to\infty} q(x)$ has indeterminate form $\frac{\infty}{\infty}$. We can determine the value of this limit through a standard algebraic approach. Multiplying the numerator and denominator each by $\frac{1}{x^2}$ (we choose this quantity since x^2 is the highest power present in the numerator or denominator), we find that

$$\begin{aligned}\lim_{x\to\infty} q(x) &= \lim_{x\to\infty} \frac{3x^2 - 4x + 5}{7x^2 + 9x - 10} \cdot \frac{\frac{1}{x^2}}{\frac{1}{x^2}} \\ &= \lim_{x\to\infty} \frac{3 - 4\frac{1}{x} + 5\frac{1}{x^2}}{7 + 9\frac{1}{x} - 10\frac{1}{x^2}} = \frac{3}{7}\end{aligned}$$

since $\frac{1}{x^2} \to 0$ and $\frac{1}{x} \to 0$ as $x \to \infty$. This shows that the rational function q has a horizontal asymptote at $y = \frac{3}{7}$. A similar approach can be used to determine the limit of any rational function as $x \to \infty$. ◊

But how should we handle a limit such as

$$\lim_{x\to\infty} \frac{x^2}{e^x}?$$

Here, both $x^2 \to \infty$ and $e^x \to \infty$. Like the situation we considered in Example 3.2.5, $\lim_{x\to\infty} \frac{3x^2-4x+5}{7x^2+9x-10}$, here we have both $x^2 \to \infty$ and $e^x \to \infty$, so

$$\lim_{x\to\infty} \frac{x^2}{e^x}$$

is also indeterminate with form $\frac{\infty}{\infty}$. With the presence of e^x, there is not an algebraic approach that enables us to find the limit's value. Fortunately, it turns out that L'Hôpital's Rule can be extended[1] to cases involving infinity.

L'Hôpital's Rule (∞).

Suppose that f and g are both differentiable on an open interval that contains a, both approach zero or both approach $\pm\infty$ as $x \to a$, and $g'(x) \neq 0$ on an open interval that contains a (where a is allowed to be ∞), then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)},$$

provided the righthand limit exists.

[1] The mathematical argument for why this is possible is beyond the scope of this text. One reason to think that an infinite version of L'Hôpital's Rule is plausible is that if we have $\frac{f(x)}{g(x)} \to \frac{\infty}{\infty}$, it's equivalent to consider $\frac{1/g(x)}{1/f(x)}$ which has the indeterminate form $\frac{0}{0}$.

To evaluate $\lim_{x\to\infty} \frac{x^2}{e^x}$, we can now apply the infinite version of L'Hôpital's Rule, since both $x^2 \to \infty$ and $e^x \to \infty$. Doing so, it follows that

$$\lim_{x\to\infty} \frac{x^2}{e^x} = \lim_{x\to\infty} \frac{2x}{e^x}.$$

This updated limit is still indeterminate and of the form $\frac{\infty}{\infty}$, but it is simpler since $2x$ has replaced x^2. Hence, we can apply L'Hôpital's Rule again, and find that

$$\lim_{x\to\infty} \frac{x^2}{e^x} = \lim_{x\to\infty} \frac{2x}{e^x} = \lim_{x\to\infty} \frac{2}{e^x}.$$

Now, since 2 is constant and $e^x \to \infty$ as $x \to \infty$, it follows that $\frac{2}{e^x} \to 0$ as $x \to \infty$, which shows that

$$\lim_{x\to\infty} \frac{x^2}{e^x} = 0.$$

Activity 3.2.4. Evaluate each of the following limits. If you use L'Hôpital's Rule, indicate where it was used, and be certain its hypotheses are met before you apply it.

(a) $\displaystyle\lim_{x\to\infty} \frac{x}{\ln(x)}$

(b) $\displaystyle\lim_{x\to\infty} \frac{e^x + x}{2e^x + x^2}$

(c) $\displaystyle\lim_{x\to 0^+} \frac{\ln(x)}{\frac{1}{x}}$

(d) $\displaystyle\lim_{x\to \frac{\pi}{2}^-} \frac{\tan(x)}{x - \frac{\pi}{2}}$

(e) $\displaystyle\lim_{x\to\infty} xe^{-x}$

To evaluate the limit of a quotient of two functions $\frac{f(x)}{g(x)}$ that results in an indeterminate form of $\frac{\infty}{\infty}$, in essence we are asking which function is growing faster without bound. We say that the function g *dominates* the function f as $x \to \infty$ provided that

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = 0,$$

whereas f dominates g provided that $\lim_{x\to\infty} \frac{f(x)}{g(x)} = \infty$. Finally, if the value of $\lim_{x\to\infty} \frac{f(x)}{g(x)}$ is finite and nonzero, we say that f and g *grow at the same rate*. For example, we saw that

$$\lim_{x\to\infty} \frac{x^2}{e^x} = 0,$$

so e^x dominates x^2 (i.e., e^x grows faster than x^2 as $x \to \infty$), while

$$\lim_{x\to\infty} \frac{3x^2 - 4x + 5}{7x^2 + 9x - 10} = \frac{3}{7},$$

so $f(x) = 3x^2 - 4x + 5$ and $g(x) = 7x^2 + 9x - 10$ grow at the same rate.

3.2.4 Summary

- Derivatives can be used to help us evaluate indeterminate limits of the form $\frac{0}{0}$ through L'Hôpital's Rule, by replacing the functions in the numerator and denominator with their tangent line approximations. In particular, if $f(a) = g(a) = 0$ and f and g are differentiable on an open interval containing a, L'Hôpital's Rule tells us that

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}.$$

- When we write $x \to \infty$, this means that x is increasing without bound. Thus, writing $\lim_{x\to\infty} f(x) = L$ means that we can make $f(x)$ as close to L as we like by choosing x to be sufficiently large. Similarly, $\lim_{x\to a} f(x) = \infty$ means that we can make $f(x)$ as large as we like by choosing x sufficiently close to a.

- A version of L'Hôpital's Rule also helps us evaluate indeterminate limits of the form $\frac{\infty}{\infty}$. If f and g are differentiable and both approach zero or both approach $\pm\infty$ as $x \to a$ (where a is allowed to be ∞), then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}.$$

3.2.5 Exercises

1. Based on your knowledge of the behavior of the numerator and denominator, predict the value of the limit $\displaystyle\lim_{x\to 0^+} \frac{\sin x}{x^{1/4}}$

the limit is ____________________

Find the limit using l'Hopital's rule:

$\displaystyle\lim_{x\to 0^+} \frac{\sin x}{x^{1/4}} = \lim_{x\to 0^+}$ [______________ / ______________] = ______________

2. Evaluate the following limit using L'Hospital's rule where appropriate.

$$\lim_{x\to 0} \frac{\sin(10x)}{\tan(3x)}$$

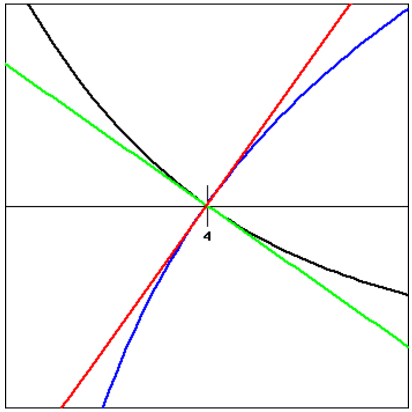
Answer: ____________________

3. Find the limit. Use l'Hospital's Rule where appropriate.

$$\lim_{x\to -\infty} x^3 e^x$$

Limit: ________________

4. The functions f and g and their tangent lines at $(4, 0)$ are shown in the figure below.

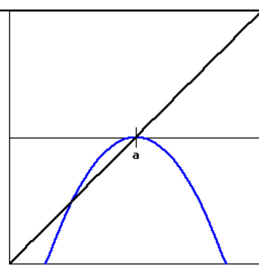


f is shown in blue, g in black, and the tangent line to f is $y = 1.8(x-4)$, and is graphed in red, and the tangent line to g is $y = -0.9(x-4)$, and is graphed in green.

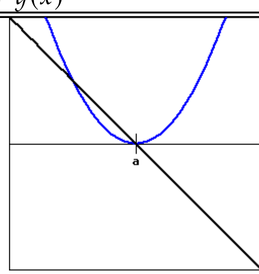
Find the limit

$\lim\limits_{x\to 4} \dfrac{f(x)}{g(x)} =$ ______

5. For the figures below, determine the nature of $\lim\limits_{x\to a} \dfrac{f(x)}{g(x)}$, if $f(x)$ is shown as the blue curve and $g(x)$ as the black curve.



$\lim\limits_{x\to a} \dfrac{f(x)}{g(x)} =$ (□ ? □ pos □ neg □ zero □ und)



$\lim\limits_{x\to a} \dfrac{f(x)}{g(x)} =$ (□ ? □ pos □ neg □ zero □ und)

6. Find the limit. Use l'Hospital's Rule where appropriate.

$$\lim_{x\to\infty} \frac{6(\ln x)^2}{x}$$

Limit: ____________________

7. If f' is continuous, $f(2) = 0$, and $f'(2) = 12$, evaluate

$$\lim_{x\to 0} \frac{f(2+3x) + f(2+5x)}{x}.$$

Limit: ____________________

8. Evaluate the limit

$$\lim_{x\to\infty} \frac{\sqrt{4+2x^2}}{(2+6x)}$$

Enter ***I*** for ∞, ***-I*** for $-\infty$, and ***DNE*** if the limit does not exist.

Limit = _______________

9. A tank contains 5000 L of pure water. Brine that contains 30 g of salt per liter of water is pumped into the tank at a rate of 25 L/min. The concentration of salt after t minutes can be shown to be given by:

$$C(t) = \frac{30t}{200+t}$$

What happens to the concentration as $t \to \infty$?

$\lim_{t\to\infty} C(t) =$ __________ g/L

10. Evaluate the limit

$$\lim_{x\to\infty} \left(\sqrt{x^2+6} - \sqrt{x^2-5}\right)$$

Answer: _______________

11. Let f and g be differentiable functions about which the following information is known: $f(3) = g(3) = 0$, $f'(3) = g'(3) = 0$, $f''(3) = -2$, and $g''(3) = 1$. Let a new function h be given by the rule $h(x) = \frac{f(x)}{g(x)}$. On the same set of axes, sketch possible graphs of f and g near $x = 3$, and use the provided information to determine the value of

$$\lim_{x\to 3} h(x).$$

Provide explanation to support your conclusion.

12. Find all vertical and horizontal asymptotes of the function

$$R(x) = \frac{3(x-a)(x-b)}{5(x-a)(x-c)},$$

where a, b, and c are distinct, arbitrary constants. In addition, state all values of x for which R is not continuous. Sketch a possible graph of R, clearly labeling the values of a, b, and c.

13. Consider the function $g(x) = x^{2x}$, which is defined for all $x > 0$. Observe that $\lim_{x\to 0^+} g(x)$ is indeterminate due to its form of 0^0. (Think about how we know that $0^k = 0$ for all $k > 0$, while $b^0 = 1$ for all $b \neq 0$, but that neither rule can apply to 0^0.)

a. Let $h(x) = \ln(g(x))$. Explain why $h(x) = 2x\ln(x)$.

b. Next, explain why it is equivalent to write $h(x) = \frac{2\ln(x)}{\frac{1}{x}}$.

c. Use L'Hôpital's Rule and your work in (b) to compute $\lim_{x\to 0^+} h(x)$.

d. Based on the value of $\lim_{x\to 0^+} h(x)$, determine $\lim_{x\to 0^+} g(x)$.

14. Recall we say that function g **dominates** function f provided that $\lim_{x\to\infty} f(x) = \infty$, $\lim_{x\to\infty} g(x) = \infty$, and $\lim_{x\to\infty} \frac{f(x)}{g(x)} = 0$.

a. Which function dominates the other: $\ln(x)$ or $\sqrt{x}$?

b. Which function dominates the other: $\ln(x)$ or $\sqrt[n]{x}$? (n can be any positive integer)

c. Explain why e^x will dominate any polynomial function.

d. Explain why x^n will dominate $\ln(x)$ for any positive integer n.

e. Give any example of two nonlinear functions such that neither dominates the other.

3.3 Using derivatives to identify extreme values

Motivating Questions

- What are the critical numbers of a function f and how are they connected to identifying the most extreme values the function achieves?
- How does the first derivative of a function reveal important information about the behavior of the function, including the function's extreme values?
- How can the second derivative of a function be used to help identify extreme values of the function?

3.3.1 Introduction

In many different settings, we are interested in knowing where a function achieves its least and greatest values. These can be important in applications — say to identify a point at which maximum profit or minimum cost occurs — or in theory to characterize the behavior of a function or a family of related functions.

Consider the simple and familiar example of a parabolic function such as $s(t) = -16t^2 + 24t + 32$ (shown at left in Figure 3.3.1) that represents the height of an object tossed vertically: its maximum value occurs at the vertex of the parabola and represents the greatest height the object reaches. This maximum value is an especially important point on the graph, the point at which the curve changes from increasing to decreasing.

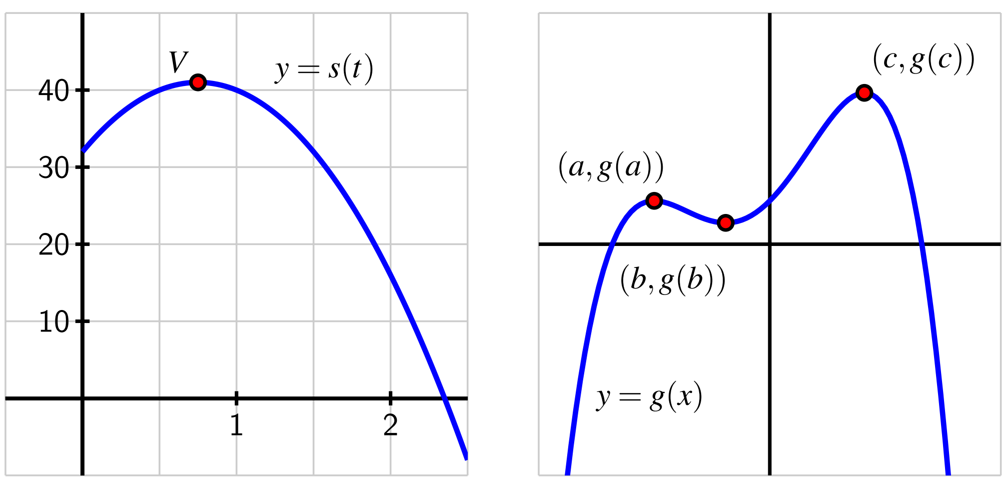


Figure 3.3.1 At left, $s(t) = -16t^2 + 24t + 32$ whose vertex is $(\frac{3}{4}, 41)$; at right, a function g that demonstrates several high and low points.

Definition 3.3.2 Given a function f, we say that $f(c)$ is a **global** or **absolute maximum** of f provided that $f(c) \geq f(x)$ for all x in the domain of f, and similarly we call $f(c)$ a **global** or **absolute minimum** of f whenever $f(c) \leq f(x)$ for all x in the domain of f. ◊

For instance, on the right in Figure 3.3.1, g has a global maximum of $g(c)$, but g does not appear to have a global minimum, as the graph of g seems to decrease indefinitely. Note that the point $(c, g(c))$ marks a fundamental change in the behavior of g, where g changes from increasing to decreasing; similar things happen at both $(a, g(a))$ and $(b, g(b))$, although these points are not global minima or maxima.

Definition 3.3.3 We say that $f(c)$ is a **local maximum** or **relative maximum** of f provided that $f(c) \geq f(x)$ for all x near c, and $f(c)$ is called a **local** or **relative minimum** of f whenever $f(c) \leq f(x)$ for all x near c. ◊

For example, on the right in Figure 3.3.1, g has a relative minimum of $g(b)$ at the point $(b, g(b))$ and a relative maximum of $g(a)$ at $(a, g(a))$. We have already identified the global maximum of g as $g(c)$; it can also be considered a relative maximum. Any maximum or minimum may also be called an *extreme value* of f.

We would like to use calculus ideas to identify and classify key function behavior, including the location of relative extremes. Of course, if we are given a graph of a function, it is often straightforward to locate these important behaviors visually.

Preview Activity 3.3.1. Consider the function h given by the graph in Figure 3.3.4. Use the graph to answer each of the following questions.

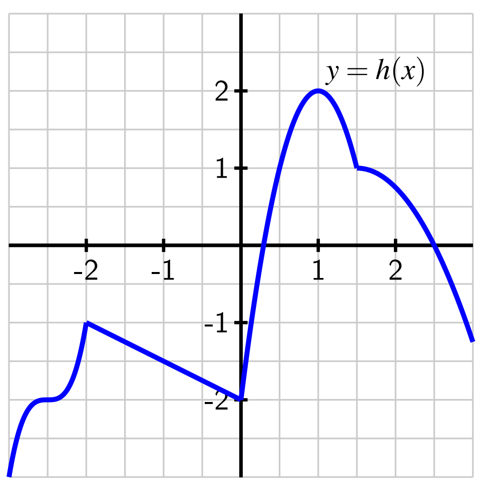


Figure 3.3.4 The graph of a function h on the interval $[-3, 3]$.

(a) Identify all of the values of c such that $-3 < c < 3$ for which $h(c)$ is a local maximum of h.

(b) Identify all of the values of c such that $-3 < c < 3$ for which $h(c)$ is a local minimum of h.

(c) Does h have a global maximum on the interval $[-3, 3]$? If so, what is the value of this global maximum?

(d) Does h have a global minimum on the interval $[-3, 3]$? If so, what is its value?

(e) Identify all values of c for which $h'(c) = 0$.

(f) Identify all values of c for which $h'(c)$ does not exist.

(g) True or false: every relative maximum and minimum of h on the interval $-3 < x < 3$ occurs at a point c such that where $h'(c)$ is either zero or does not exist. Explain your conclusion.

(h) True or false: at every point where $h'(c)$ is zero or does not exist on the interval $-3 < x < 3$, h has a relative maximum or minimum. Explain your conclusion.

3.3.2 Critical numbers and the first derivative test

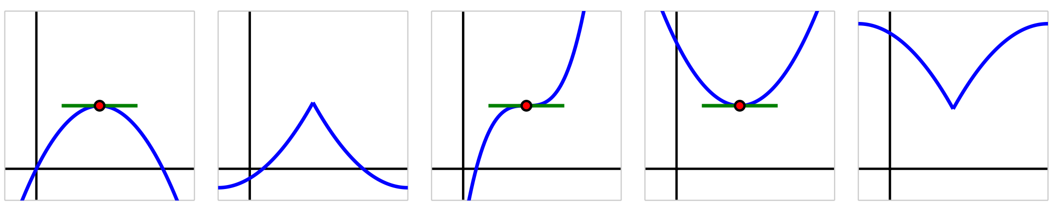


Figure 3.3.5 Five different possible behaviors of a function near one of its critical numbers.

As seen in the first two functions on the left in Figure 3.3.5, when a continuous function defined on (a, b) changes from being always increasing on interval (a, c) to being always decreasing on interval (c, b) (where $a < c < b$), the function has a relative maximum at c. Similarly, when a continuous function defined on (a, b) changes from being always decreasing on interval (a, c) to being always increasing on interval (c, b) (as in the two rightmost functions in the figure), the function has a relative minimum at c. The middle option in Figure 3.3.5 demonstrates that it's possible for a function to have neither a maximum nor minimum at a critical number, as it can be the case that the function doesn't change from increasing to decreasing or vice versa.

Because the sign of the derivative changes at every location c where a continuous function changes from increasing to decreasing or from decreasing to increasing, there are only two possible ways for these changes in behavior to occur: either $f'(c) = 0$ or $f'(c)$ is undefined. Because these values of c are so important, we call them *critical numbers*.

Definition 3.3.6 We say that a function f has a **critical number** at $x = c$ provided that c is in the domain of f, and either $f'(c) = 0$ or $f'(c)$ is undefined. ◊

Critical numbers are the only possible locations where the function f may have relative extremes. Again, note that not every critical number produces a maximum or minimum; in the middle graph of Figure 3.3.5, the function pictured there has a horizontal tangent line at the noted point, but the function is increasing before and increasing after, so the critical number does not yield a maximum or minimum.

When c is a critical number, we say that $(c, f(c))$ is a *critical point* of the function, or that $f(c)$ is a *critical value*. The *first derivative test* summarizes how sign changes in the first derivative (which can only occur at critical numbers) indicate the presence of a local maximum or minimum for a given function.

The First Derivative Test.

Let p be a critical number of a continuous function f that is differentiable near p (except possibly at $x = p$). If f' changes sign from positive to negative at p, then f has a relative maximum at p. If f' changes sign from negative to positive at p, then f has a relative minimum at p.

In Example 3.3.7, we show how to apply the First Derivative Test to determine whether relative maxima or minima occur at various critical numbers and introduce the idea of a **sign chart** to visualize important function and derivative behavior.

Example 3.3.7 Let f be a function whose derivative is given by the formula $f'(x) = e^{-2x}(3 - x)(x + 1)^2$. Determine all critical numbers of f and decide whether a relative maximum, relative minimum, or neither occurs at each.

Solution. Since we already have $f'(x)$ written in factored form, it is straightforward to find the critical numbers of f. Because $f'(x)$ is defined for all values of x, we need only determine where $f'(x) = 0$. From the equation

$$e^{-2x}(3 - x)(x + 1)^2 = 0$$

and the zero product property, it follows that $x = 3$ and $x = -1$ are critical numbers of f. (There is no value of x that makes $e^{-2x} = 0$.)

Next, to apply the first derivative test, we'd like to know the sign of $f'(x)$ at inputs near the critical numbers. Because the critical numbers are the only locations at which f' can change sign, it follows that the sign of the derivative is the same on each of the intervals created by the critical numbers: for instance, the sign of f' must be the same for every $x < -1$. We create a first derivative sign chart to summarize the sign of f' on the relevant intervals, along with the corresponding behavior of f.

To produce the first derivative sign chart in Figure 3.3.8 we identify the sign of each factor of $f'(x)$ at one selected point in each interval. For instance, for $x < -1$, we can determine the sign of e^{-2x}, $(3 - x)$, and $(x + 1)^2$ at the value $x = -2$. We note that both e^{-2x} and $(x + 1)^2$ are positive regardless of the value of x, while $(3 - x)$ is also positive at $x = -2$. Hence, each of the three terms in f' is positive, which we indicate by writing "+ + +." Taking the product of three positive terms results in a positive value for f', which we denote by the "+" in the interval to the left of $x = -1$. And, since f' is positive on that interval, we know that f is increasing, so we write "INC" to represent the behavior of f. In a similar way, we find that f' is positive and f is increasing on $-1 < x < 3$, and f' is negative and f is decreasing for $x > 3$.

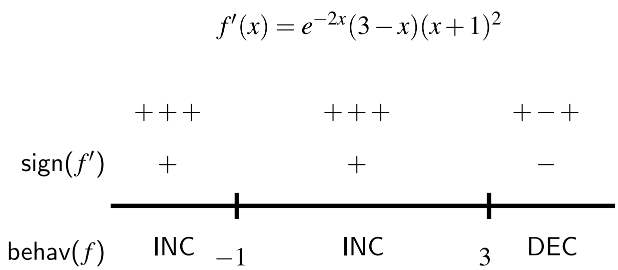


Figure 3.3.8 The first derivative sign chart for a function f whose derivative is given by the formula $f'(x) = e^{-2x}(3-x)(x+1)^2$.

Now we look for critical numbers at which f' changes sign. In this example, f' changes sign only at $x = 3$, from positive to negative, so f has a relative maximum at $x = 3$. Although f has a critical number at $x = -1$, since f is increasing both before and after $x = -1$, f has neither a minimum nor a maximum at $x = -1$. ◊

Activity 3.3.2. Suppose that $g(x)$ is a function whose first derivative is

$$g'(x) = \frac{(x+4)(x-2)}{x^2+1}.$$

(a) Determine, with justification, all critical numbers of g.

(b) By developing a carefully labeled first derivative sign chart, decide whether g has as a local maximum, local minimum, or neither at each critical number.

(c) Does g have a global maximum? global minimum? Justify your claims.

(d) Sketch a possible graph of $y = g(x)$. Clearly label any local or global extrema on the graph.

3.3.3 The second derivative test

Recall that the second derivative of a function tells us several important things about the behavior of the function itself. For instance, if f'' is positive on an interval, then we know that f' is increasing on that interval and, consequently, that f is concave up, so throughout that interval the tangent line to $y = f(x)$ lies below the curve at every point. At a point where $f'(p) = 0$, the sign of the second derivative determines whether f has a local minimum or local maximum at the critical number p.

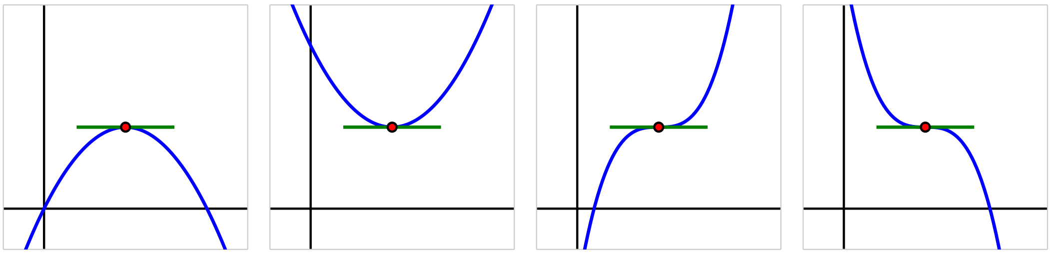


Figure 3.3.9 Four possible graphs of a function f with a horizontal tangent line at a critical point.

In Figure 3.3.9, we see the four possibilities for a function f that has a critical number p at which $f'(p) = 0$, provided $f''(p)$ is not zero on an interval including p (except possibly at p). On either side of the critical number, f'' can be either positive or negative, and hence f can be either concave up or concave down. In the first two graphs, f does not change concavity at p, and in those situations, f has either a local minimum or local maximum. In particular, if $f'(p) = 0$ and $f''(p) < 0$, then f is concave down at p with a horizontal tangent line, so f has a local maximum there. This fact, along with the corresponding statement for when $f''(p)$ is positive, is the substance of the *second derivative test*.

Second Derivative Test.

If p is a critical number of a continuous function f such that $f'(p) = 0$ and $f''(p) \neq 0$, then f has a relative maximum at p if and only if $f''(p) < 0$, and f has a relative minimum at p if and only if $f''(p) > 0$.

In the event that $f''(p) = 0$, the second derivative test is inconclusive. That is, the test doesn't provide us any information. This is because if $f''(p) = 0$, it is possible that f has a local minimum, local maximum, or neither.[1]

Just as a first derivative sign chart reveals all of the increasing and decreasing behavior of a function, we can construct a second derivative sign chart that demonstrates all of the important information involving concavity.

Example 3.3.10 Let $f(x)$ be a function whose first derivative is $f'(x) = 3x^4 - 9x^2$. Construct both first and second derivative sign charts for f, fully discuss where f is increasing and decreasing and concave up and concave down, identify all relative extreme values, and sketch a possible graph of f.

Solution. Since we know $f'(x) = 3x^4 - 9x^2$, we can find the critical numbers of f by solving $3x^4 - 9x^2 = 0$. Factoring, we observe that

$$0 = 3x^2(x^2 - 3) = 3x^2(x + \sqrt{3})(x - \sqrt{3}),$$

so that $x = 0, \pm\sqrt{3}$ are the three critical numbers of f. The first derivative sign chart for f is given in Figure 3.3.11.

[1]Consider the functions $f(x) = x^4$, $g(x) = -x^4$, and $h(x) = x^3$ at the critical point $p = 0$.

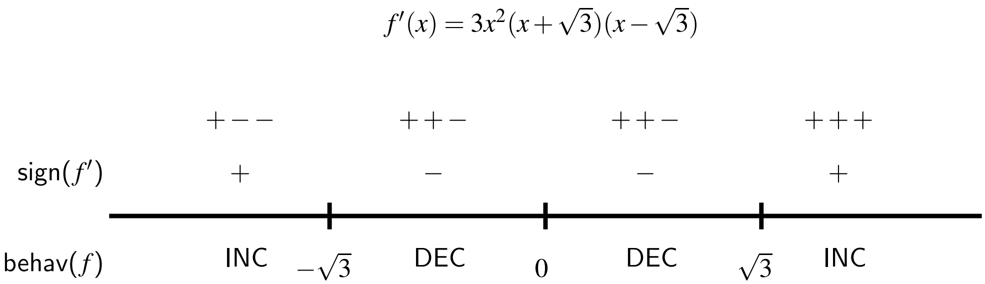


Figure 3.3.11 The first derivative sign chart for f when $f'(x) = 3x^4 - 9x^2 = 3x^2(x^2 - 3)$.

We see that f is increasing on the intervals $(-\infty, -\sqrt{3})$ and $(\sqrt{3}, \infty)$, and f is decreasing on $(-\sqrt{3}, 0)$ and $(0, \sqrt{3})$. By the first derivative test, this information tells us that f has a local maximum at $x = -\sqrt{3}$ and a local minimum at $x = \sqrt{3}$. Although f also has a critical number at $x = 0$, neither a maximum nor minimum occurs there since f' does not change sign at $x = 0$.

Next, we move on to investigate concavity. Differentiating $f'(x) = 3x^4 - 9x^2$, we see that $f''(x) = 12x^3 - 18x$. Since we are interested in knowing the intervals on which f'' is positive and negative, we first find where $f''(x) = 0$. Observe that

$$0 = 12x^3 - 18x = 12x\left(x^2 - \frac{3}{2}\right) = 12x\left(x + \sqrt{\frac{3}{2}}\right)\left(x - \sqrt{\frac{3}{2}}\right).$$

This equation has solutions $x = 0, \pm\sqrt{\frac{3}{2}}$. Building a sign chart for f'' in the exact same way we do for f', we see the result shown in Figure 3.3.12.

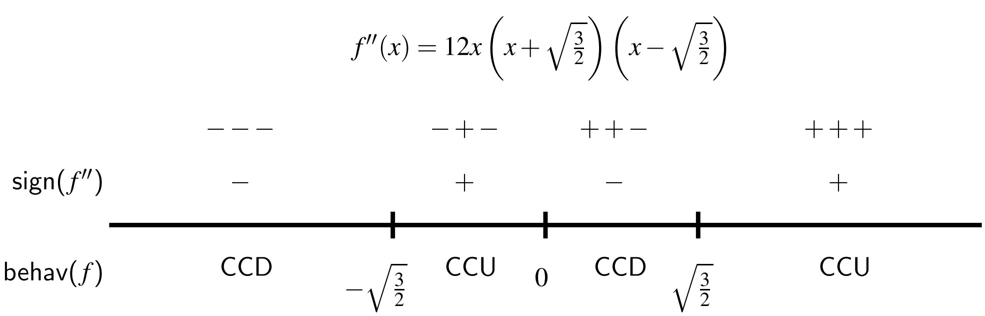


Figure 3.3.12 The second derivative sign chart for f when $f''(x) = 12x^3 - 18x = 12x^2\left(x^2 - \frac{3}{2}\right)$.

Therefore, f is concave down on the intervals $(-\infty, -\sqrt{\frac{3}{2}})$ and $(0, \sqrt{\frac{3}{2}})$, and concave up on $(-\sqrt{\frac{3}{2}}, 0)$ and $(\sqrt{\frac{3}{2}}, \infty)$.

Putting all of this information together, we now see a complete and accurate possible graph of f in Figure 3.3.13.

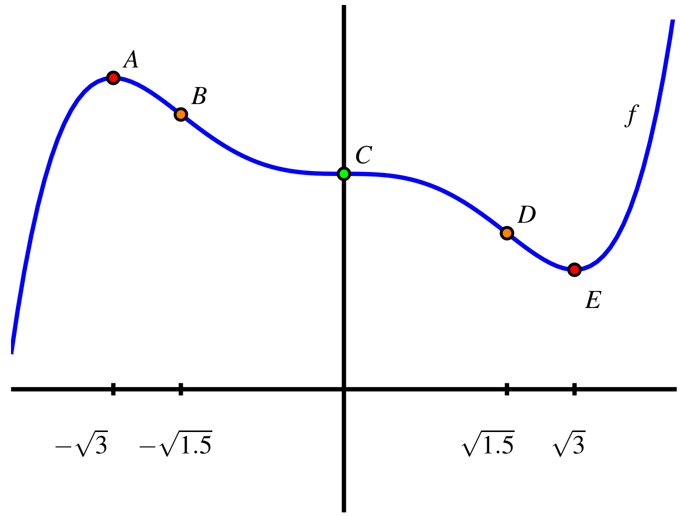


Figure 3.3.13 A possible graph of the function f in Example 3.3.10.

The point $A = (-\sqrt{3}, f(-\sqrt{3}))$ is a local maximum, because f is increasing prior to A and decreasing after; similarly, the point $E = (\sqrt{3}, f(\sqrt{3})$ is a local minimum. Note, too, that f is concave down at A and concave up at B, which is consistent both with our second derivative sign chart and the second derivative test. At points B and D, concavity changes, as we saw in the results of the second derivative sign chart in Figure 3.3.12. Finally, at point C, f has a critical point with a horizontal tangent line, but neither a maximum nor a minimum occurs there, since f is decreasing both before and after C. It is also the case that concavity changes at C.

While we completely understand where f is increasing and decreasing, where f is concave up and concave down, and where f has relative extremes, we do not know any specific information about the y-coordinates of points on the curve. For instance, while we know that f has a local maximum at $x = -\sqrt{3}$, we don't know the value of that maximum because we do not know $f(-\sqrt{3})$. Any vertical translation of our sketch of f in Figure 3.3.13 would satisfy the given criteria for f. ◊

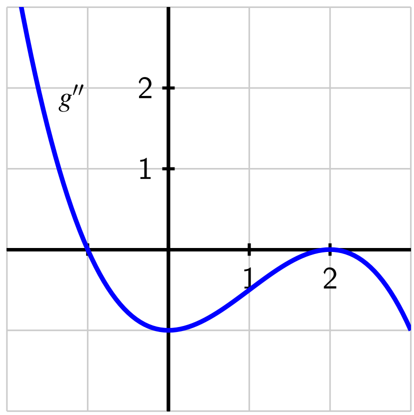
Points B, C, and D in Figure 3.3.13 are locations at which the concavity of f changes. We give a special name to any such point.

Definition 3.3.14 If p is a value in the domain of a continuous function f at which f changes concavity, then we say that $(p, f(p))$ is an **inflection point** (or **point of inflection**) of f. ◊

Just as we look for locations where f changes from increasing to decreasing at points where $f'(p) = 0$ or $f'(p)$ is undefined, so too we find where $f''(p) = 0$ or $f''(p)$ is undefined to see if there are points of inflection at these locations.

At this point in our study, it is important to remind ourselves of the big picture that derivatives help to paint: the sign of the first derivative f' tells us *whether* the function f is increasing or decreasing, while the sign of the second derivative f'' tells us *how* the function f is increasing or decreasing.

Activity 3.3.3. Suppose that g is a function whose second derivative, g'', is given by the graph in the following figure.



(a) Identify all x-values where $g''(x) = 0$ or $g''(x)$ is undefined. Then, construct a second derivative sign chart for g and hence state the intervals on which g is concave up, as well as the intervals on which g is concave down.

(b) State the x-coordinates of all points of inflection of g.

(c) Suppose you are given that $g'(-1.67857351) = 0$. Is there is a local maximum, local minimum, or neither (for the function g) at this critical number of g, or is it impossible to say? Why?

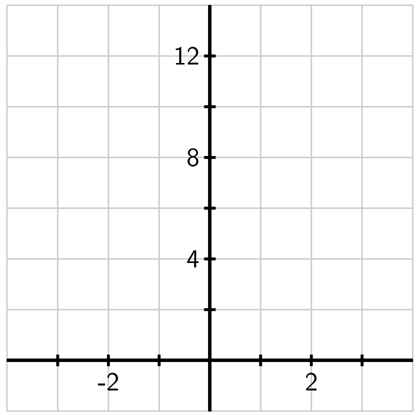
(d) Assuming that $g''(x)$ is a polynomial (and that all important behavior of g'' is seen in the graph above), what degree polynomial do you think $g(x)$ is? Why?

As we will see in more detail in the following section, derivatives also help us to understand families of functions that differ only by changing one or more parameters. For instance, we might be interested in understanding the behavior of all functions of the form $f(x) = a(x-h)^2 + k$ where a, h, and k are parameters. Each parameter has considerable impact on how the graph appears.

In the final activity in this section, we explore the impact a parameter has on the concavity of an interesting function.

Activity 3.3.4. Consider the family of functions given by $h(x) = x^2 + \cos(kx)$, where k is an arbitrary positive real number.

(a) Use a graphing utility to sketch the graph of h for several different k-values, including $k = 1, 3, 5, 10$. Plot $h(x) = x^2 + \cos(3x)$ on the axes provided. What is the smallest value of k at which you think you can see (just by looking at the graph) at least one inflection point on the graph of h?



(b) Explain why the graph of h has no inflection points if $k \leq \sqrt{2}$, but infinitely many inflection points if $k > \sqrt{2}$.

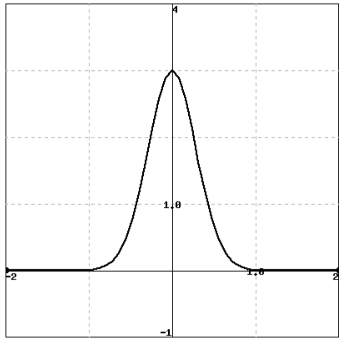
(c) Explain why, no matter the value of k, h can only have finitely many critical numbers.

3.3.4 Summary

- The critical numbers of a continuous function f are the values of p for which $f'(p) = 0$ or $f'(p)$ does not exist. These values are important because they identify horizontal tangent lines or corner points on the graph, which are the only possible locations at which a local maximum or local minimum can occur.
- Given a differentiable function f, whenever f' is positive, f is increasing; whenever f' is negative, f is decreasing. The first derivative test tells us that at any point where f changes from increasing to decreasing, f has a local maximum, while conversely at any point where f changes from decreasing to increasing f has a local minimum.
- Given a twice differentiable function f, if we have a horizontal tangent line at $x = p$ and $f''(p)$ is nonzero, the sign of f'' tells us the concavity of f and hence whether f has a maximum or minimum at $x = p$. In particular, if $f'(p) = 0$ and $f''(p) < 0$, then f is concave down at p and f has a local maximum there, while if $f'(p) = 0$ and $f''(p) > 0$, then f has a local minimum at p. If $f'(p) = 0$ and $f''(p) = 0$, then the second derivative does not tell us whether f has a local extreme at p or not.

3.3.5 Exercises

1. Use a graph below of $f(x) = 3e^{-6x^2}$ to estimate the x-values of any critical points and inflection points of $f(x)$.



critical points (enter as a comma-separated list): $x =$ ____________________

inflection points (enter as a comma-separated list): $x =$ ____________________

Next, use derivatives to find the x-values of any critical points and inflection points exactly.

critical points (enter as a comma-separated list): $x =$ ____________________

inflection points (enter as a comma-separated list): $x =$ ____________________

2. For the function $f(x) = -7x + 5\sin(x)$, find all intervals where the function is increasing.

f is increasing on ____________________

(Give your answer as an interval or a list of intervals, e.g., ***(-infinity,8]*** *or* ***(1,5),(7,10)*** *. Enter* ***none*** *if there are no such intervals.)*

Similarly, find all intervals where the function is decreasing:

f is decreasing on ____________________

(Give your answer as an interval or a list of intervals, e.g., ***(-infinity,8]*** *or* ***(1,5),(7,10)*** *. Enter* ***none*** *if there are no such intervals.)*

Finally, find all critical points in the graph of $f(x)$ (enter x values as a comma-separated list, or ***none*** if there are no critical points): ____________________

3. The following table gives values of the differentiable function $y = f(x)$.

x	0	1	2	3	4	5	6	7	8	9	10
y	-1	3	2	-2	-1	1	-2	-4	-5	-7	-9

Estimate the x-values of critical points of $f(x)$ on the interval $0 < x < 10$. Classify each critical point as a local maximum, local minimum, or neither.

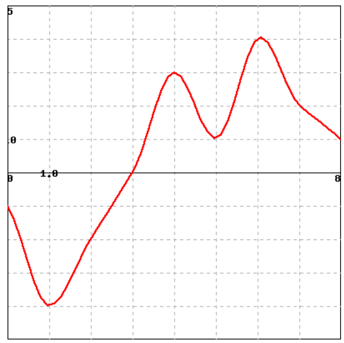
(Enter your critical points as comma-separated xvalue,classification pairs. For example, if you found the critical points $x = -2$ and $x = 3$, and that the first was a local minimum and the second neither a minimum nor a maximum, you should enter ***(-2,min), (3,neither)****. Enter* ***none*** *if there are no critical points.)*

critical points and classifications: ____________________

Now assume that the table gives values of the continuous function $y = f'(x)$ (instead of $f(x)$). Estimate and classify critical points of the function $f(x)$.

critical points and classifications: ______________________________

4. Below is the graph of ***the derivative*** $f'(x)$ of a function defined on the interval (0,8). You can click on the graph to see a larger version in a separate window.



Refer to the graph to answer each of the following questions. For parts (A) and (B), use interval notation to report your answer. (If needed, you use ***U*** for the union symbol.)

(A) For what values of x in (0,8) is $f(x)$ increasing? (If the function is not increasing anywhere, enter ***None***.)

Answer: ______________________

(B) For what values of x in (0,8) is $f(x)$ concave down? (If the function is not concave down anywhere, enter ***None***.)

Answer: ______________________

(C) Find all values of x in (0,8) is where $f(x)$ has a local minimum, and list them (separated by commas) in the box below. (If there are no local minima, enter ***None***.)

Local Minima: ______________________

(D) Find all values of x in (0,8) is where $f(x)$ has an inflection point, and list them (separated by commas) in the box below. (If there are no inflection points, enter ***None***.)

Inflection Points: ______________________

5. Let $f(x) = 2x^3 - 24x + 2$

Input the interval(s) on which f is increasing.

Input the interval(s) on which f is decreasing.

Find the point(s) at which f achieves a local maximum.

Find the point(s) at which f achieves a local minimum.

Find the intervals on which f is concave up.

Find the intervals on which f is concave down.

Find all inflection points.

6. Let $f(x) = 2x^4 - 4x^2 + 6$

Input the interval(s) on which f is increasing.

Input the interval(s) on which f is decreasing.

Find the point(s) at which f achieves a local maximum.

Find the point(s) at which f achieves a local minimum.

Find the interval(s) on which f is concave up.

Find the interval(s) on which f is concave down.

Find all inflection points.

7. Let

$$f(x) = -3x^2 + 8x + 9$$

Use the second derivative test to find any local maxima and minima.

First, find the critical point. Enter the x-value below. __________

Then find the value of the second derivative at that x-values. __________

If the function has a local maximum at that value, enter MAX. If it has a local minimum, enter MIN. __________

8. Find and classify the critical points of $f(x) = 5x^6(5-x)^5$ as local maxima and minima.

Critical points: $x =$ ________________________

Classifications: ________________________

*(Enter your critical points and classifications as comma-separated lists, and enter the types in the same order as your critical points. Note that you must enter something in both blanks for either to be evaluated. For the types, enter **min**, **max**, or **neither**.*

9. This problem concerns a function about which the following information is known:

- f is a differentiable function defined at every real number x
- $f(0) = -1/2$
- $y = f'(x)$ has its graph given at center in Figure 3.3.15

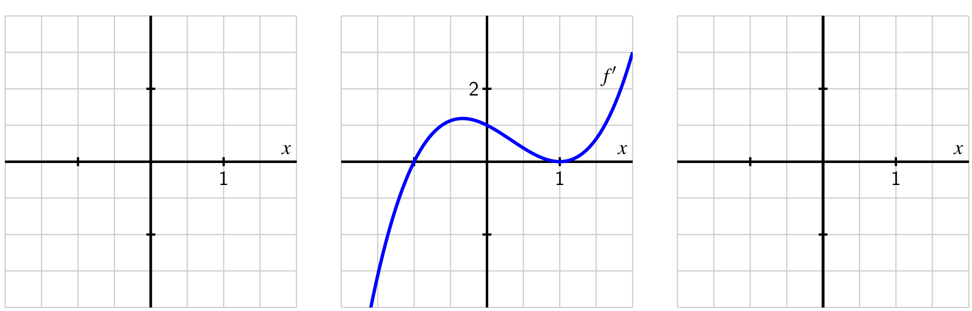


Figure 3.3.15 At center, a graph of $y = f'(x)$; at left, axes for plotting $y = f(x)$; at right, axes for plotting $y = f''(x)$.

a. Construct a first derivative sign chart for f. Clearly identify all critical numbers of f, where f is increasing and decreasing, and where f has local extrema.

b. On the right-hand axes, sketch an approximate graph of $y = f''(x)$.

c. Construct a second derivative sign chart for f. Clearly identify where f is concave up and concave down, as well as all inflection points.

d. On the left-hand axes, sketch a possible graph of $y = f(x)$.

10. Suppose that g is a differentiable function and $g'(2) = 0$. In addition, suppose that on $1 < x < 2$ and $2 < x < 3$ it is known that $g'(x)$ is positive.

a. Does g have a local maximum, local minimum, or neither at $x = 2$? Why?

b. Suppose that $g''(x)$ exists for every x such that $1 < x < 3$. Reasoning graphically, describe the behavior of $g''(x)$ for x-values near 2.

c. Besides being a critical number of g, what is special about the value $x = 2$ in terms of the behavior of the graph of g?

11. Suppose that h is a differentiable function whose first derivative is given by the graph in Figure 3.3.16.

a. How many real number solutions can the equation $h(x) = 0$ have? Why?

b. If $h(x) = 0$ has two distinct real solutions, what can you say about the signs of the two solutions? Why?

c. Assume that $\lim_{x\to\infty} h'(x) = 3$, as appears to be indicated in Figure 3.3.16. How will the graph of $y = h(x)$ appear as $x \to \infty$? Why?

d. Describe the concavity of $y = h(x)$ as fully as you can from the provided information.

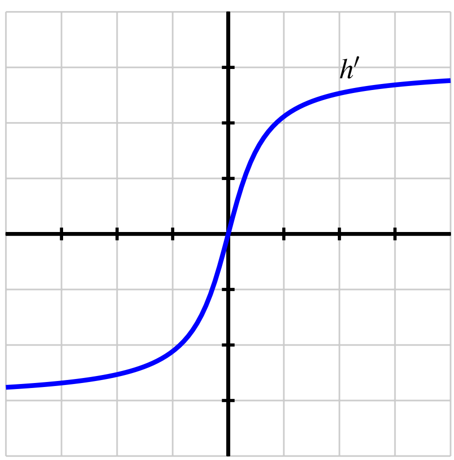


Figure 3.3.16 The graph of $y = h'(x)$.

12. Suppose that $g(x)$ is a function continuous for every value of $x \neq 2$ whose first derivative is $g'(x) = \frac{(x+4)(x-1)^2}{x-2}$. Further, assume that it is known that g has a vertical asymptote at $x = 2$.

a. Determine all critical numbers of g.

b. By developing a carefully labeled first derivative sign chart, decide whether g has as a local maximum, local minimum, or neither at each critical number. ***Note***: observe that $g'(x)$ can change sign at the vertical asymptote of $g(x)$.

c. Does g have a global maximum? global minimum? Justify your claims.

d. What is the value of $\lim_{x\to\infty} g'(x)$? What does the value of this limit tell you about the long-term behavior of g?

e. Sketch a possible graph of $y = g(x)$.

13. Let p be a function whose second derivative is $p''(x) = (x+1)(x-2)e^{-x}$.

a. Construct a second derivative sign chart for p and determine all inflection points of p.

b. Suppose you also know that $x = \frac{\sqrt{5}-1}{2}$ is a critical number of p. Does p have a local minimum, local maximum, or neither at $x = \frac{\sqrt{5}-1}{2}$? Why?

c. If the point $(2, \frac{12}{e^2})$ lies on the graph of $y = p(x)$ and $p'(2) = -\frac{5}{e^2}$, find the equation of the tangent line to $y = p(x)$ at the point where $x = 2$. Does the tangent line lie above the curve, below the curve, or neither at this value? Why?

3.4 Using derivatives to describe families of functions

Motivating Questions

- Given a family of functions that depends on one or more parameters, how does the shape of the graph of a typical function in the family depend on the value of the parameters?
- How can we construct first and second derivative sign charts of functions that depend on one or more parameters while allowing those parameters to remain arbitrary constants?

3.4.1 Introduction

Mathematicians are often interested in making general observations to describe patterns that hold in a large number of related situations. Think about the Pythagorean Theorem: it doesn't tell us something about a single right triangle, but rather a fact about *every* right triangle. In the next part of our studies, we use calculus to make general observations about families of functions that depend on one or more parameters. People who use applied mathematics, such as engineers and economists, often encounter the same types of functions where only small changes to certain constants occur. These constants are called *parameters*.

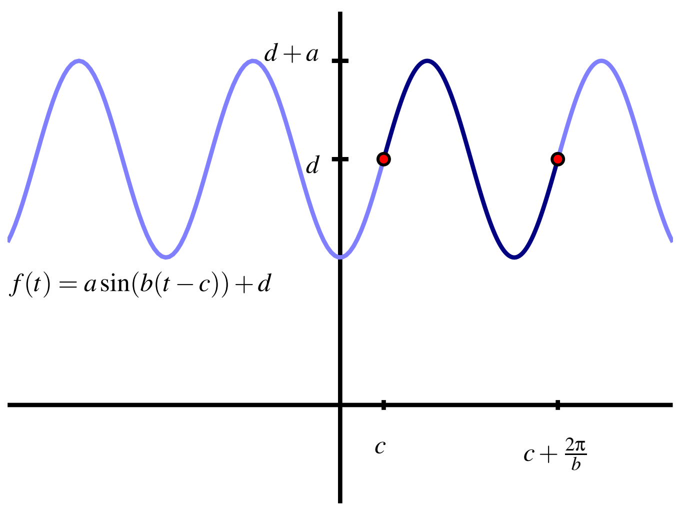


Figure 3.4.1 The graph of $f(t) = a\sin(b(t-c)) + d$ based on parameters a, b, c, and d.

You are already familiar with certain families of functions. For example,

$$f(t) = a\sin(b(t-c)) + d$$

is a stretched and shifted version of the sine function with amplitude a, period $\frac{2\pi}{b}$, phase shift c, and vertical shift d. We know that a affects the size of the oscillation, b the rapidity of oscillation, and c where the oscillation starts, as shown in Figure 3.4.1, while d affects the vertical positioning of the graph.

As another example, every function of the form $y = mx + b$ is a line with slope m and y-intercept $(0, b)$. The value of m affects the line's steepness, and the value of b situates the line vertically on the coordinate axes. These two parameters describe all possible non-vertical lines.

For other less familiar families of functions, we can use calculus to discover where key behavior occurs: where members of the family are increasing or decreasing, concave up or concave down, where relative extremes occur, and more, all in terms of the parameters involved. To get started, we revisit a common collection of functions to see how calculus confirms things we already know.

Preview Activity 3.4.1. Let a, h, and k be arbitrary real numbers with $a \neq 0$, and let f be the function given by the rule $f(x) = a(x-h)^2 + k$.

(a) What familiar type of function is f? What information do you know about f just by looking at its form? (Think about the roles of a, h, and k.)

(b) Next we use some calculus to develop familiar ideas from a different perspective. To start, treat a, h, and k as constants and compute $f'(x)$.

(c) Find all critical numbers of f. (These will depend on at least one of a, h, and k.)

(d) Assume that $a < 0$. Construct a first derivative sign chart for f.

(e) Based on the information you've found above, classify the critical values of f as maxima or minima.

3.4.2 Describing families of functions in terms of parameters

Our goal is to describe the key characteristics of the overall behavior of each member of a family of functions in terms of its parameters. By finding the first and second derivatives and constructing sign charts (each of which may depend on one or more of the parameters), we can often make broad conclusions about how each member of the family will appear.

Example 3.4.2 Consider the two-parameter family of functions given by $g(x) = axe^{-bx}$, where a and b are positive real numbers. Fully describe the behavior of a typical member of the family in terms of a and b, including the location of all critical numbers, where g is increasing, decreasing, concave up, and concave down, and the long term behavior of g.

Solution. We begin by computing $g'(x)$. By the product rule,

$$g'(x) = ax\frac{d}{dx}\left[e^{-bx}\right] + e^{-bx}\frac{d}{dx}[ax].$$

By applying the chain rule and constant multiple rule, we find that

$$g'(x) = axe^{-bx}(-b) + e^{-bx}(a).$$

To find the critical numbers of g, we solve the equation $g'(x) = 0$. By factoring $g'(x)$, we find

$$0 = ae^{-bx}(-bx + 1).$$

Since we are given that $a \neq 0$ and we know that $e^{-bx} \neq 0$ for all values of x, the only way this equation can hold is when $-bx + 1 = 0$. Solving for x, we find $x = \frac{1}{b}$, and this is therefore the only critical number of g.

We construct the first derivative sign chart for g that is shown in Figure 3.4.3.

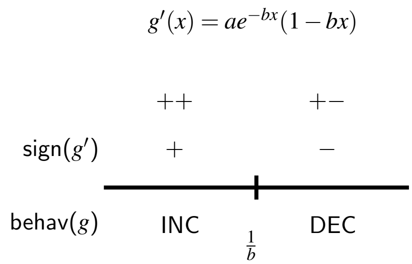


Figure 3.4.3 The first derivative sign chart for $g(x) = axe^{-bx}$.

Because the factor ae^{-bx} is always positive, the sign of g' depends on the linear factor $(1-bx)$, which is positive for $x < \frac{1}{b}$ and negative for $x > \frac{1}{b}$. Hence we can conclude that g is always increasing for $x < \frac{1}{b}$ and decreasing for $x > \frac{1}{b}$, and also that g has a global maximum at $(\frac{1}{b}, g(\frac{1}{b}))$ and no local minimum.

We turn next to analyzing the concavity of g. With $g'(x) = -abxe^{-bx} + ae^{-bx}$, we differentiate to find that

$$g''(x) = -abxe^{-bx}(-b) + e^{-bx}(-ab) + ae^{-bx}(-b).$$

Combining like terms and factoring, we now have

$$g''(x) = ab^2xe^{-bx} - 2abe^{-bx} = abe^{-bx}(bx - 2).$$

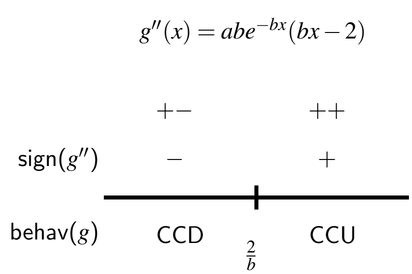


Figure 3.4.4 The second derivative sign chart for $g(x) = axe^{-bx}$.

We observe that abe^{-bx} is always positive, and thus the sign of g'' depends on the sign of $(bx - 2)$, which is zero when $x = \frac{2}{b}$. Since b is positive, the value of $(bx - 2)$ is negative for

$x < \frac{2}{b}$ and positive for $x > \frac{2}{b}$. The sign chart for g'' is shown in Figure 3.4.4. Thus, g is concave down for all $x < \frac{2}{b}$ and concave up for all $x > \frac{2}{b}$.

Finally, we analyze the long term behavior of g by considering two limits. First, we note that

$$\lim_{x\to\infty} g(x) = \lim_{x\to\infty} axe^{-bx} = \lim_{x\to\infty} \frac{ax}{e^{bx}}.$$

This limit has indeterminate form $\frac{\infty}{\infty}$, so we apply L'Hôpital's Rule and find that $\lim_{x\to\infty} g(x) = 0$. In the other direction,

$$\lim_{x\to-\infty} g(x) = \lim_{x\to-\infty} axe^{-bx} = -\infty,$$

because $ax \to -\infty$ and $e^{-bx} \to \infty$ as $x \to -\infty$. Hence, as we move left on its graph, g decreases without bound, while as we move to the right, $g(x) \to 0$.

All of this information now helps us produce the graph of a typical member of this family of functions without using a graphing utility (and without choosing particular values for a and b), as shown in Figure 3.4.5.

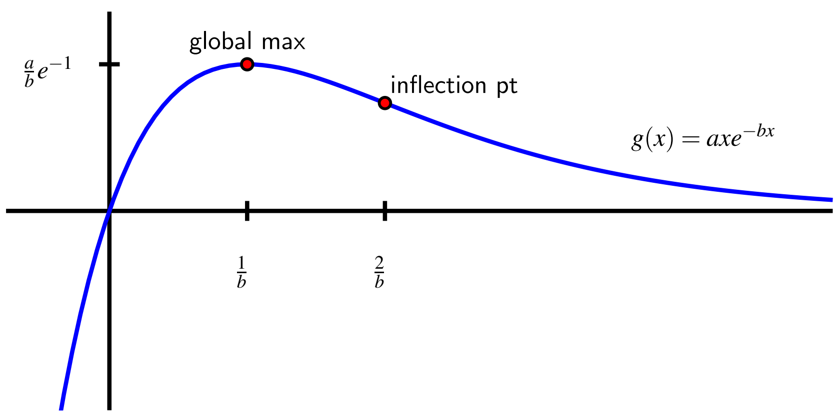


Figure 3.4.5 The graph of $g(x) = axe^{-bx}$.

Note that the value of b controls the horizontal location of the global maximum and the inflection point, as neither depends on a. The value of a affects the vertical stretch of the graph. For example, the global maximum occurs at the point $(\frac{1}{b}, g(\frac{1}{b})) = (\frac{1}{b}, \frac{a}{b}e^{-1})$, so the larger the value of a, the greater the value of the global maximum. ◊

The work we've completed in Example 3.4.2 can often be replicated for other families of functions that depend on parameters. Normally we are most interested in determining all critical numbers, a first derivative sign chart, a second derivative sign chart, and the limit of the function as $x \to \infty$. Throughout, we prefer to work with the parameters as arbitrary constants. In addition, we can experiment with some particular values of the parameters present to reduce the algebraic complexity of our work. The following activities offer several key examples where we see that the values of the parameters substantially affect the behavior of individual functions within a given family.

Activity 3.4.2. Consider the family of functions defined by $p(x) = x^3 - ax$, where $a \neq 0$ is an arbitrary constant.

(a) Find $p'(x)$ and determine the critical numbers of p. How many critical numbers does p have?

(b) Construct a first derivative sign chart for p. What can you say about the overall behavior of p if the constant a is positive? Why? What if the constant a is negative? In each case, describe the relative extremes of p.

(c) Find $p''(x)$ and construct a second derivative sign chart for p. What does this tell you about the concavity of p? What role does a play in determining the concavity of p?

(d) Without using a graphing utility, sketch and label typical graphs of $p(x)$ for the cases where $a > 0$ and $a < 0$. Label all inflection points and local extrema.

Activity 3.4.3. Consider the two-parameter family of functions of the form $h(x) = a(1 - e^{-bx})$, where a and b are positive real numbers.

(a) Find the first derivative and the critical numbers of h. Use these to construct a first derivative sign chart and determine for which values of x the function h is increasing and decreasing.

(b) Find the second derivative and build a second derivative sign chart. For which values of x is a function in this family concave up? concave down?

(c) What is the value of $\lim_{x\to\infty} a(1 - e^{-bx})$? $\lim_{x\to-\infty} a(1 - e^{-bx})$?

(d) How does changing the value of b affect the shape of the curve?

(e) Without using a graphing utility, sketch the graph of a typical member of this family. Write several sentences to describe the overall behavior of a typical function h and how this behavior depends on a and b.

Activity 3.4.4. Let $L(t) = \frac{A}{1+ce^{-kt}}$, where A, c, and k are all positive real numbers.

(a) Observe that we can equivalently write $L(t) = A(1 + ce^{-kt})^{-1}$. Find $L'(t)$ and explain why L has no critical numbers. Is L always increasing or always decreasing? Why?

(b) It turns out that

$$L''(t) = Ack^2e^{-kt}\frac{ce^{-kt} - 1}{(1 + ce^{-kt})^3}.$$

Given this fact, find all values of t such that $L''(t) = 0$ and hence construct a second derivative sign chart. For which values of t is a function in this family concave up? concave down?

(c) What is the value of $\lim_{t\to\infty} \frac{A}{1+ce^{-kt}}$? $\lim_{t\to-\infty} \frac{A}{1+ce^{-kt}}$?

(d) Find the value of $L(x)$ at the inflection point found in (b).

(e) Without using a graphing utility, sketch the graph of a typical member of this family. Write several sentences to describe the overall behavior of a typical function L and how this behavior depends on A, c, and k number.

3.4.3 Summary

- Given a family of functions that depends on one or more parameters, by investigating how critical numbers and locations where the second derivative is zero depend on the values of these parameters, we can often accurately describe the shape of the function in terms of the parameters.
- In particular, just as we can create first and second derivative sign charts for a single function, we can often do so for entire families of functions where critical numbers and possible inflection points depend on arbitrary constants. These sign charts then reveal where members of the family are increasing or decreasing, concave up or concave down, and help us to identify relative extremes and inflection points.

3.4.4 Exercises

1. A function with parameters a and b is given. Describe the critical points and possible points of inflection of f in terms of a and b. Assume $a, b > 0$.

$$f(x) = \frac{a}{x^2 + b^2}$$

Critical point: ____________________

Possible points of inflection: (two answers)

____________________ (smaller value)

____________________ (larger value)

2. A function with parameters a and b is given. Describe the critical points and possible points of inflection of f in terms of a and b.

$$f(x) = (x - a)(x - b)$$

Critical point: ____________________

Possible point of inflection: ____________________ (enter ***DNE*** if none exist)

3. Let $p(x) = x^2(a - x)$, where a is constant and $a > 0$.

Find the local maxima and minima of p.

*(Enter your maxima and minima as comma-separated xvalue,classification pairs. For example, if you found that $x = -2$ was a local minimum and $x = 3$ was a local maximum, you should enter **(-2,min), (3,max)**. If there were no maximum, you **must** drop the parentheses and enter **-2,min**.)*

maxima and minima: ______________________

What effect does increasing the value of a have on the x-position of the maximum(s) you found? (Enter ***left***, ***none*** or ***right*** if it moves left, has no effect, or moves right.)

____

What effect does increasing the value of a have on the x-position of the minimum(s) you found? (Enter ***left***, ***none*** or ***right*** if it moves left, has no effect, or moves right.)

____

What effect does increasing the value of a have on the y-coordinate of the maximum(s) you found? (Enter ***up***, ***none*** or ***down*** if it moves up, has no effect, or moves down.)

____

What effect does increasing the value of a have on the y-coordinate of the minimum(s) you found? (Enter ***up***, ***none*** or ***down*** if it moves up, has no effect, or moves down.)

____

4. Consider $f(x) = ax + x\ln(bx)$ for $a > 0$, $b > 1$, and $x > 1$.

Find $f'(x)$: $f'(x) =$ ______________________

Based on your expression for $f'(x)$, is $f(x)$ increasing or decreasing? (Enter ***increasing*** or ***decreasing***.) ________

(Be sure that you can see why this is true for all values $x > 1$.)

Find $f''(x)$: $f''(x) =$ ______________________

Based on your expression for $f''(x)$, is $f(x)$ concave up or concave down? (Enter ***up*** or ***down***.) ________

(Be sure that you can see why this is true for all values $x > 1$.)

5. Find constants a and b in the function $f(x) = axe^{bx}$ such that $f(\frac{1}{9}) = 1$ and the function has a local maximum at $x = \frac{1}{9}$.

$a =$ ______________

$b =$ ______________

6. Consider the one-parameter family of functions given by $p(x) = x^3 - ax^2$, where $a > 0$.

a. Sketch a plot of a typical member of the family, using the fact that each is a cubic polynomial with a repeated zero at $x = 0$ and another zero at $x = a$.

b. Find all critical numbers of p.

c. Compute p'' and find all values for which $p''(x) = 0$. Hence construct a second derivative sign chart for p.

d. Describe how the location of the critical numbers and the inflection point of p

change as a changes. That is, if the value of a is increased, what happens to the critical numbers and inflection point?

7. Let $q(x) = \frac{e^{-x}}{x-c}$ be a one-parameter family of functions where $c > 0$.

a. Explain why q has a vertical asymptote at $x = c$.

b. Determine $\lim_{x\to\infty} q(x)$ and $\lim_{x\to-\infty} q(x)$.

c. Compute $q'(x)$ and find all critical numbers of q.

d. Construct a first derivative sign chart for q and determine whether each critical number leads to a local minimum, local maximum, or neither for the function q.

e. Sketch a typical member of this family of functions with important behaviors clearly labeled.

8. Let $E(x) = e^{-\frac{(x-m)^2}{2s^2}}$, where m is any real number and s is a positive real number.

a. Compute $E'(x)$ and hence find all critical numbers of E.

b. Construct a first derivative sign chart for E and classify each critical number of the function as a local minimum, local maximum, or neither.

c. It can be shown that $E''(x)$ is given by the formula

$$E''(x) = e^{-\frac{(x-m)^2}{2s^2}} \left(\frac{(x-m)^2 - s^2}{s^4} \right).$$

Find all values of x for which $E''(x) = 0$.

d. Determine $\lim_{x\to\infty} E(x)$ and $\lim_{x\to-\infty} E(x)$.

e. Construct a labeled graph of a typical function E that clearly shows how important points on the graph of $y = E(x)$ depend on m and s.

3.5 Global optimization

Motivating Questions

- What are the differences between finding relative extreme values and global extreme values of a function?
- How is the process of finding the global maximum or minimum of a function over the function's entire domain different from determining the global maximum or minimum on a restricted domain?
- For a function that is guaranteed to have both a global maximum and global minimum on a closed, bounded interval, what are the possible points at which these extreme values occur?

3.5.1 Introduction

We have seen that we can use the first derivative of a function to determine where the function is increasing or decreasing, and the second derivative to know where the function is concave up or concave down. This information helps us determine the overall shape and behavior of the graph, as well as whether the function has relative extrema.

Recall the difference between a relative maximum and a global maximum: there is a *relative* maximum of f at $x = p$ if $f(p) \geq f(x)$ for all x *near* p, while there is a *global* maximum at p if $f(p) \geq f(x)$ for *all* x in the domain of f. For instance, in Figure 3.5.1, f has a global maximum at $x = c$ and a relative maximum at $x = a$, since $f(c)$ is greater than $f(x)$ for every value of x, while $f(a)$ is only greater than the value of $f(x)$ for x near a. Since the function appears to decrease without bound, f has no global minimum, though clearly f has a relative minimum at $x = b$.

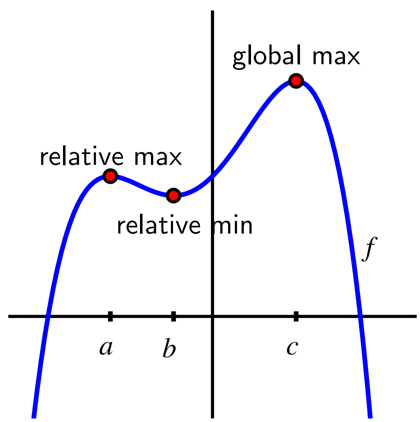


Figure 3.5.1 A function f with a global maximum, but no global minimum.

Our emphasis in this section is on finding the global extreme values of a function (if they exist), either over its entire domain or on some restricted portion.

Preview Activity 3.5.1. Let $f(x) = 2 + \frac{3}{1+(x+1)^2}$.

(a) Find $f'(x)$ and use it to determine all of the critical numbers of f.

(b) Construct a first derivative sign chart for f and thus determine all intervals on which f is increasing or decreasing.

(c) Does f have a global maximum? If so, explain why, plus state the location and value of the global maximum. If not, explain why not.

(d) Determine $\lim_{x \to \infty} f(x)$ and $\lim_{x \to -\infty} f(x)$.

(e) Explain why $f(x) > 2$ for every value of x.

(f) Does f have a global minimum? If so, explain why, plus state the location and value of the global minimum. If not, explain why not.

3.5.2 Global Optimization

In Figure 3.5.1 and Preview Activity 3.5.1, we were interested in finding the global minimum and global maximum for f on its entire domain. At other times, we might focus on some restriction of the domain.

For example, rather than considering $f(x) = 2 + \frac{3}{1+(x+1)^2}$ for every value of x, perhaps instead we are only interested in those x for which $0 \le x \le 4$, and we would like to know which values of x in the interval $[0, 4]$ produce the largest possible and smallest possible values of f. We are accustomed to critical numbers playing a key role in determining the location of extreme values of a function; now, by restricting the domain to an interval, it makes sense that the endpoints of the interval will also be important to consider, as we see in the following activity. When limiting ourselves to a particular interval, we will often refer to the *absolute* maximum or minimum value, rather than the *global* maximum or minimum.

Activity 3.5.2. Let $g(x) = \frac{1}{3}x^3 - 2x + 2$.

(a) Find all critical numbers of g that lie in the interval $-2 \le x \le 3$.

(b) Use a graphing utility to construct the graph of g on the interval $-2 \le x \le 3$.

(c) From the graph, determine the x-values at which the absolute minimum and absolute maximum of g occur on the interval $[-2, 3]$.

(d) How do your answers change if we instead consider the interval $-2 \le x \le 2$?

(e) What if we instead consider the interval $-2 \le x \le 1$?

In Activity 3.5.2, we saw how the absolute maximum and absolute minimum of a function on a closed, bounded interval $[a, b]$, depend not only on the critical numbers of the function, but also on the values of a and b. These observations demonstrate several important facts that hold more generally. First, we state an important result called the Extreme Value Theorem.

The Extreme Value Theorem.

If f is a continuous function on a closed interval $[a, b]$, then f attains both an absolute minimum and absolute maximum on $[a, b]$.

The Extreme Value Theorem tells us that on any closed interval $[a, b]$, a continuous function has to achieve both an absolute minimum and an absolute maximum. The theorem does not tell us where these extreme values occur, but rather only that they must exist. As we saw in Activity 3.5.2, the only possible locations for relative extremes are at the endpoints of the interval or at a critical number.

Note 3.5.2 Thus, we have the following approach to finding the absolute maximum and minimum of a continuous function f on the interval $[a, b]$:

- find all critical numbers of f that lie in the interval;
- evaluate the function f at each critical number in the interval and at each endpoint of the interval;
- from among those function values, the smallest is the absolute minimum of f on the interval, while the largest is the absolute maximum.

Activity 3.5.3. Find the *exact* absolute maximum and minimum of each function on the stated interval.

(a) $h(x) = xe^{-x}$, $[0, 3]$

(b) $p(t) = \sin(t) + \cos(t)$, $[-\frac{\pi}{2}, \frac{\pi}{2}]$

(c) $q(x) = \frac{x^2}{x-2}$, $[3, 7]$

(d) $f(x) = 4 - e^{-(x-2)^2}$, $(-\infty, \infty)$

(e) $h(x) = xe^{-ax}$, $[0, \frac{2}{a}]$ $(a > 0)$

(f) $f(x) = b - e^{-(x-a)^2}$, $(-\infty, \infty)$, $a, b > 0$

The interval we choose has nearly the same influence on extreme values as the function under consideration. Consider, for instance, the function pictured in Figure 3.5.3.

In sequence, from left to right in the figure, the interval under consideration is changed from $[-2, 3]$ to $[-2, 2]$ to $[-2, 1]$.

- On the interval $[-2, 3]$, there are two critical numbers, with the absolute minimum at one critical number and the absolute maximum at the right endpoint.
- On the interval $[-2, 2]$, both critical numbers are in the interval, with the absolute minimum and maximum at the two critical numbers.
- On the interval $[-2, 1]$, just one critical number lies in the interval, with the absolute maximum at one critical number and the absolute minimum at one endpoint.

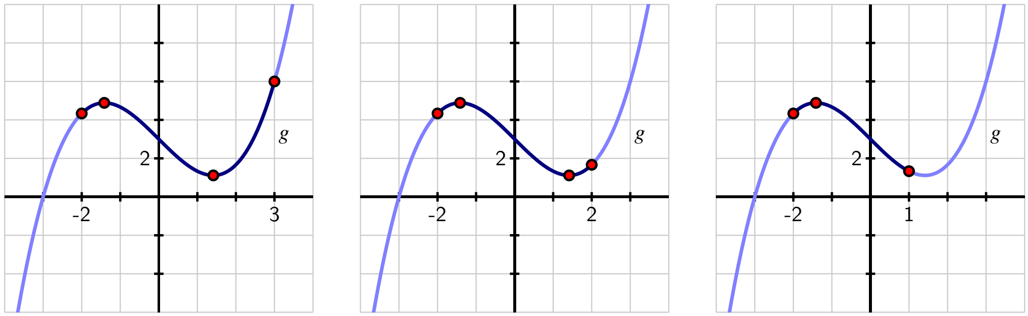


Figure 3.5.3 A function g considered on three different intervals.

For optimizing on a closed, bounded interval, it's important to remember to consider only the critical numbers that lie within the interval.

3.5.3 Moving toward applications

We conclude this section with an example of an applied optimization problem. It highlights the role that a closed, bounded domain can play in finding absolute extrema.

Example 3.5.4 A 20 cm piece of wire is cut into two pieces. One piece is used to form a square and the other to form an equilateral triangle. How should the wire be cut to maximize the total area enclosed by the square and triangle? to minimize the area?

Solution. We begin by sketching a picture that illustrates the situation. The variable in the problem is where we decide to cut the wire. We thus label the cut point at a distance x from one end of the wire, and note that the remaining portion of the wire then has length $20 - x$

As shown in Figure 3.5.5, we see that the x cm of wire that is used to form the equilateral triangle with three sides of length $\frac{x}{3}$. For the remaining $20 - x$ cm of wire, the square that results will have each side of length $\frac{20-x}{4}$.

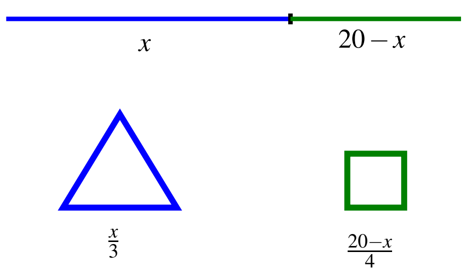


Figure 3.5.5 A 20 cm piece of wire cut into two pieces, one of which forms an equilateral triangle, the other which yields a square.

At this point, we note that there are obvious restrictions on x: in particular, $0 \le x \le 20$. In the extreme cases, all of the wire is being used to make just one figure. For instance, if $x = 0$, then all 20 cm of wire are used to make a square that is 5×5.

Now, our overall goal is to find the minimum and maximum areas that can be enclosed. Because the height of an equilateral triangle is $\sqrt{3}$ times half the length of the base, the area of the triangle is

$$A_{\Delta} = \frac{1}{2}bh = \frac{1}{2} \cdot \frac{x}{3} \cdot \frac{x\sqrt{3}}{6}.$$

The area of the square is $A_{\square} = s^2 = \left(\frac{20-x}{4}\right)^2$. Therefore, the total area function is

$$A(x) = \frac{\sqrt{3}x^2}{36} + \left(\frac{20-x}{4}\right)^2.$$

Remember that we are considering this function only on the restricted domain $[0, 20]$.

Differentiating $A(x)$, we have

$$A'(x) = \frac{\sqrt{3}x}{18} + 2\left(\frac{20-x}{4}\right)\left(-\frac{1}{4}\right) = \frac{\sqrt{3}}{18}x + \frac{1}{8}x - \frac{5}{2}.$$

When we set $A'(x) = 0$, we find that $x = \frac{180}{4\sqrt{3}+9} \approx 11.3007$ is the only critical number of A in the interval $[0, 20]$.

Evaluating A at the critical number and endpoints, we see that

- $A\left(\frac{180}{4\sqrt{3}+9}\right) = \frac{\sqrt{3}(\frac{180}{4\sqrt{3}+9})^2}{4} + \left(\frac{20 - \frac{180}{4\sqrt{3}+9}}{4}\right)^2 \approx 10.8741$
- $A(0) = 25$
- $A(20) = \frac{\sqrt{3}}{36}(400) = \frac{100}{9}\sqrt{3} \approx 19.2450$

Thus, the absolute minimum occurs when $x \approx 11.3007$ and results in the minimum area of approximately 10.8741 square centimeters. The absolute maximum occurs when we invest all of the wire in the square (and none in the triangle), resulting in 25 square centimeters of area. These results are confirmed by a plot of $y = A(x)$ on the interval $[0, 20]$, as shown in Figure 3.5.6.

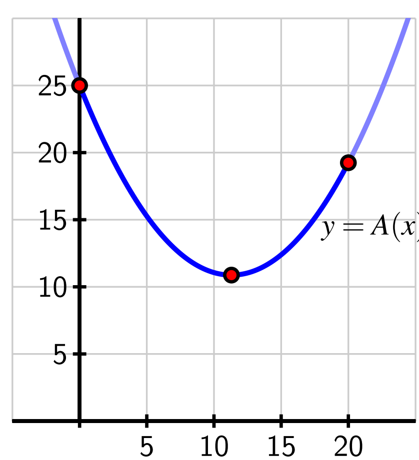


Figure 3.5.6 A plot of the area function from Example 3.5.4.

◊

Activity 3.5.4. A piece of cardboard that is 10×15 (each measured in inches) is being made into a box without a top. To do so, squares are cut from each corner of the box and the remaining sides are folded up. If the box needs to be at least 1 inch deep and no more than 3 inches deep, what is the maximum possible volume of the box? what is the minimum volume? Justify your answers using calculus.

(a) Draw a labeled diagram that shows the given information. What variable should we introduce to represent the choice we make in creating the box? Label the diagram appropriately with the variable, and write a sentence to state what the variable represents.

(b) Determine a formula for the function V (that depends on the variable in (a)) that tells us the volume of the box.

(c) What is the domain of the function V? That is, what values of x make sense for input? Are there additional restrictions provided in the problem?

(d) Determine all critical numbers of the function V.

(e) Evaluate V at each of the endpoints of the domain and at any critical numbers that lie in the domain.

(f) What is the maximum possible volume of the box? the minimum?

Example 3.5.4 and Activity 3.5.4 illustrate standard steps that we undertake in almost every applied optimization problem: we draw a picture to demonstrate the situation, introduce one or more variables to represent quantities that are changing, find a function that models the quantity to be optimized, and then decide on an appropriate domain for that function. Once that is done, we are in the familiar situation of finding the absolute minimum and maximum of a function over a particular domain, so we apply the calculus ideas that we have been studying to this point in Chapter 3.

3.5.4 Summary

- To find relative extreme values of a function, we use a first derivative sign chart and classify all of the function's critical numbers. If instead we are interested in absolute extreme values, we first decide whether we are considering the entire domain of the function or a particular interval.
- In the case of finding global extremes over the function's entire domain, we again use a first or second derivative sign chart. If we are working to find absolute extremes on a restricted interval, then we first identify all critical numbers of the function that lie in the interval.
- For a continuous function on a closed, bounded interval, the only possible points at which absolute extreme values occur are the critical numbers and the endpoints. Thus, we simply evaluate the function at each endpoint and each critical number in the interval, and compare the results to decide which is largest (the absolute maximum) and which is smallest (the absolute minimum).

3.5.5 Exercises

1. Find the exact global maximum and minimum values of the function $g(x) = 2x - x^2 - 7$ if its domain is all real numbers.

global maximum at x = ____________________

global minimum at x = ____________________

*(Enter **none** if there is no global maximum or global minimum for this function.)*

2. Find the exact global maximum and minimum values of the function $g(t) = 3te^{-t}$ if $t > 0$.

global maximum at t = ____________________

global minimum at t = ____________________

*(Enter **none** if there is no global maximum or global minimum for this function.)*

3. For some positive constant C, a patient's temperature change, T, due to a dose, D, of a drug is given by $T = \left(\frac{C}{2} - \frac{D}{3}\right) D^2$.

What dosage maximizes the temperature change?

D = ____________________

The sensitivity of the body to the drug is defined as dT/dD. What dosage maximizes sensitivity?

D = ____________________

4. Find the x-coordinate of the absolute maximum and absolute minimum for the function

$$f(x) = 6x + \frac{11}{x}, \qquad x > 0.$$

Enter *None* for any absolute extrema that does not exist.

x-coordinate of absolute maximum = __________

x-coordinate of absolute minimum = __________

5. Consider the function

$$f(x) = 2x^3 + 18x^2 - 42x + 5 \quad \text{with} \quad -7 \le x \le 2$$

This function has an absolute minimum at the point ____________________

and an absolute maximum at the point ____________________

Note: both parts of this answer should be entered as an ordered pair, including the parentheses, such as (5, 11).

6. Consider the function $f(x) = x - 4\ln(x)$, $\frac{1}{4} \le x \le 10$.

The absolute maximum value is ____________________

and this occurs at x equals ____________________

The absolute minimum value is ____________________

and this occurs at x equals ____________________

7. Based on the given information about each function, decide whether the function has global maximum, a global minimum, neither, both, or that it is not possible to say without more information. Assume that each function is twice differentiable and defined for all real numbers, unless noted otherwise. In each case, write one sentence to explain your conclusion.

a. f is a function such that $f''(x) < 0$ for every x.

b. g is a function with two critical numbers a and b (where $a < b$), and $g'(x) < 0$ for $x < a$, $g'(x) < 0$ for $a < x < b$, and $g'(x) > 0$ for $x > b$.

c. h is a function with two critical numbers a and b (where $a < b$), and $h'(x) < 0$ for $x < a$, $h'(x) > 0$ for $a < x < b$, and $h'(x) < 0$ for $x > b$. In addition, $\lim_{x\to\infty} h(x) = 0$ and $\lim_{x\to-\infty} h(x) = 0$.

d. p is a function differentiable everywhere except at $x = a$ and $p''(x) > 0$ for $x < a$ and $p''(x) < 0$ for $x > a$.

8. For each family of functions that depends on one or more parameters, determine the function's absolute maximum and absolute minimum on the given interval.

a. $p(x) = x^3 - a^2 x$, $[0, a]$ $(a > 0)$

b. $r(x) = axe^{-bx}$, $[\frac{1}{2b}, \frac{2}{b}]$ $(a > 0, b > 1)$

c. $w(x) = a(1 - e^{-bx})$, $[b, 3b]$ $(a, b > 0)$

d. $s(x) = \sin(kx)$, $\left[\frac{\pi}{3k}, \frac{5\pi}{6k}\right]$ $(k > 0)$

9. For each of the functions described below (each continuous on $[a, b]$), state the location of the function's absolute maximum and absolute minimum on the interval $[a, b]$, or say there is not enough information provided to make a conclusion. Assume that any critical numbers mentioned in the problem statement represent all of the critical numbers the function has in $[a, b]$. In each case, write one sentence to explain your answer.

a. $f'(x) \leq 0$ for all x in $[a, b]$

b. g has a critical number at c such that $a < c < b$ and $g'(x) > 0$ for $x < c$ and $g'(x) < 0$ for $x > c$

c. $h(a) = h(b)$ and $h''(x) < 0$ for all x in $[a, b]$

d. $p(a) > 0$, $p(b) < 0$, and for the critical number c such that $a < c < b$, $p'(x) < 0$ for $x < c$ and $p'(x) > 0$ for $x > c$

10. Let $s(t) = 3\sin(2(t - \frac{\pi}{6})) + 5$. Find the exact absolute maximum and minimum of s on the provided intervals by testing the endpoints and finding and evaluating all relevant critical numbers of s.

a. $[\frac{\pi}{6}, \frac{7\pi}{6}]$

b. $[0, \frac{\pi}{2}]$

c. $[0, 2\pi]$

d. $[\frac{\pi}{3}, \frac{5\pi}{6}]$

3.6 Applied optimization

Motivating Questions

- In a setting where a situation is described for which optimal parameters are sought, how do we develop a function that models the situation and use calculus to find the desired maximum or minimum?

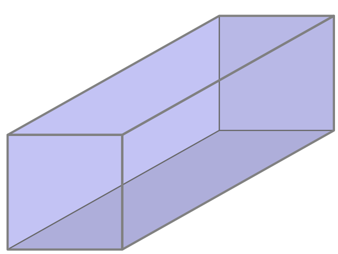
3.6.1 Introduction

Near the conclusion of Section 3.5, we considered two optimization problems where determining the function to be optimized was part of the problem. In Example 3.5.4, we sought to use a single piece of wire to build an equilateral triangle and square in order to maximize the total combined area enclosed. In the subsequent Activity 3.5.4, we investigated how the volume of a box constructed from a piece of cardboard by removing squares from each corner and folding up the sides depends on the size of the squares removed.

In neither of these problems was a function to optimize explicitly provided. Rather, we first tried to understand the problem by drawing a figure and introducing variables, and then sought to develop a formula for a function that modeled the quantity to be optimized. Once the function was established, we then considered what domain was appropriate. At that point, we were finally ready to apply the ideas of calculus to determine the absolute minimum or maximum.

Throughout what follows in the current section, the primary emphasis is on the reader solving problems. Initially, some substantial guidance is provided, with the problems progressing to require greater independence as we move along.

Preview Activity 3.6.1. According to U.S. postal regulations, the girth plus the length of a parcel sent by mail may not exceed 108 inches, where by "girth" we mean the perimeter of the smallest end. What is the largest possible volume of a rectangular parcel with a square end that can be sent by mail? What are the dimensions of the package of largest volume?



(a) Let x represent the length of one side of the square end and y the length of the longer side. Label these quantities appropriately on the image shown in the provided figure.

(b) What is the quantity to be optimized in this situation? Find a formula for this quantity in terms of x and y.

(c) The problem statement tells us that the parcel's girth plus length may not exceed 108 inches. In order to maximize volume, we assume that we will actually need the girth plus length to equal 108 inches. What equation does this produce involving x and y?

(d) Solve the equation you found in (c) for one of x or y (whichever is easier).

(e) Now use your work in (b) and (d) to determine a formula for the volume of the parcel so that this formula is a function of a single variable.

(f) Over what domain should we consider this function? Note that both x and y must be positive; how does the constraint that girth plus length is 108 inches produce intervals of possible values for x and y?

(g) Find the absolute maximum of the volume of the parcel on the domain you established in (f) and hence also determine the dimensions of the box of greatest volume. Justify that you've found the maximum using calculus.

3.6.2 More applied optimization problems

Many of the steps in Preview Activity 3.6.1 are ones that we will execute in any applied optimization problem. We briefly summarize those here to provide an overview of our approach in subsequent questions.

Note 3.6.1

- Draw a picture and introduce variables. It is essential to first understand what quantities are allowed to vary in the problem and then to represent those values with variables. Constructing a figure with the variables labeled is almost always an essential first step. Sometimes drawing several diagrams can be especially helpful to get a sense of the situation. A nice example of this can be seen in this interactive graphic[1], where the choice of where to bend a piece of wire into the shape of a rectangle determines both the rectangle's shape and area.

- Identify the quantity to be optimized as well as any key relationships among the variable quantities. Essentially this step involves writing equations that involve the variables that have been introduced: one to represent the quantity whose minimum or maximum is sought, and possibly others that show how multiple variables in the problem may be interrelated.

- Determine a function of a single variable that models the quantity to be optimized; this may involve using other relationships among variables to eliminate one or more variables in the function formula. For example, in Preview Activity 3.6.1, we initially found that $V = x^2y$, but then the additional relationship that $4x + y = 108$ (girth plus length equals 108 inches) allows us to relate x and y and thus observe equivalently that

$y = 108 - 4x$. Substituting for y in the volume equation yields $V(x) = x^2(108 - 4x)$, and thus we have written the volume as a function of the single variable x. The equation $4x + y = 108$ that shows a key relationship that always holds between the variables is often called a **constraint equation**.

- Decide the domain on which to consider the function being optimized. Often the physical constraints of the problem will limit the possible values that the independent variable can take on. Thinking back to the diagram describing the overall situation and any relationships among variables in the problem often helps identify the smallest and largest values of the input variable.
- Use calculus to identify the absolute maximum and/or minimum of the quantity being optimized. This always involves finding the critical numbers of the function first. Then, depending on the domain, we either construct a first derivative sign chart (for an open or unbounded interval) or evaluate the function at the endpoints and critical numbers (for a closed, bounded interval), using ideas we've studied so far in Chapter 3.
- Finally, we make certain we have answered the question: does the question seek the absolute maximum of a quantity, or the values of the variables that produce the maximum? That is, finding the absolute maximum volume of a parcel is different from finding the dimensions of the parcel that produce the maximum.

Activity 3.6.2. A soup can in the shape of a right circular cylinder is to be made from two materials. The material for the side of the can costs \$0.015 per square inch and the material for the lids costs \$0.027 per square inch. Suppose that we desire to construct a can that has a volume of 16 cubic inches. What dimensions minimize the cost of the can?

(a) Draw a picture of the can and label its dimensions with appropriate variables.

(b) Use your variables to determine expressions for the volume, surface area, and cost of the can.

(c) Determine the total cost function as a function of a single variable. What is the domain on which you should consider this function?

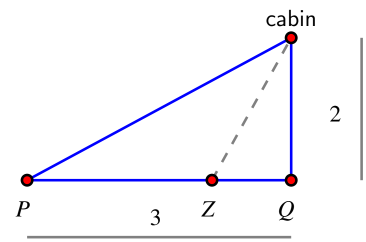
(d) Find the absolute minimum cost and the dimensions that produce this value.

Familiarity with common geometric formulas is particularly helpful in problems such as the one in Activity 3.6.2. Sometimes those involve perimeter, area, volume, or surface area. At other times, the constraints of a problem introduce right triangles (where the Pythagorean Theorem applies) or other functions whose formulas provide relationships among the variables.

Activity 3.6.3. A hiker starting at a point P on a straight road walks east towards point Q, which is on the road and 3 kilometers from point P. Two kilometers due north of point Q is a cabin. The hiker will walk down the road for a while, at a pace

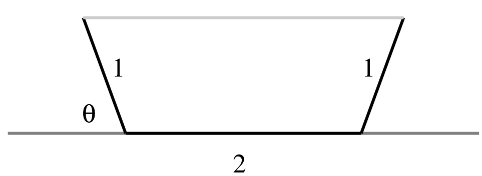
[1] gvsu.edu/s/99

of 8 kilometers per hour. At some point Z between P and Q, the hiker leaves the road and makes a straight line towards the cabin through the woods, hiking at a pace of 3 kph, as pictured in the provided figure. In order to minimize the time to go from P to Z to the cabin, where should the hiker turn into the forest?



In more geometric problems, we often use curves or functions to provide natural constraints. For instance, we could investigate which isosceles triangle that circumscribes a unit circle has the smallest area, which you can explore for yourself in this interactive graphic[2]. Or similarly, for a region bounded by a parabola, we might seek the rectangle of largest area that fits beneath the curve, as shown in this interactive graphic[3]. The final activity in the section is similar to the latter situation.

Activity 3.6.4. A trough is being constructed by bending a 4×24 (measured in feet) rectangular piece of sheet metal. Two symmetric folds 2 feet apart will be made parallel to the longest side of the rectangle so that the trough has cross-sections in the shape of a trapezoid, as pictured in the provided figure. At what angle should the folds be made to produce the trough of maximum volume?



Activity 3.6.5. Consider the region in the x-y plane that is bounded by the x-axis and the function $f(x) = 25 - x^2$. Construct a rectangle whose base lies on the x-axis and is centered at the origin, and whose sides extend vertically until they intersect the curve $y = 25 - x^2$. Which such rectangle has the maximum possible area? Which such rectangle has the greatest perimeter?

[2] gvsu.edu/s/9b
[3] gvsu.edu/s/9c

(Challenge: answer the same questions in terms of positive parameters a and b for the function $f(x) = b - ax^2$.)

3.6.3 Summary

- While there is no single algorithm that works in every situation where optimization is used, in most of the problems we consider, the following steps are helpful: draw a picture and introduce variables; identify the quantity to be optimized and find relationships among the variables; determine a function of a single variable that models the quantity to be optimized; decide the domain on which to consider the function being optimized; use calculus to identify the absolute maximum and/or minimum of the quantity being optimized.

3.6.4 Exercises

1. An open box is to be made out of a 11-inch by 16-inch piece of cardboard by cutting out squares of equal size from the four corners and bending up the sides. If the box needs to be at least 1 inch deep, and no more than 3 inches deep, find the absolute maximum and absolute minimum volumes.

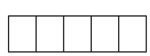
Maximum volume: __________ Minimum volume: __________

2. Find two numbers differing by 48 whose product is as small as possible.

Enter your two numbers as a comma separated list, e.g. 2, 3.

The two numbers are ____________________.

3. An ostrich farmer wants to enclose a rectangular area and then divide it into five pens with fencing parallel to one side of the rectangle (see the figure below). There are 430 feet of fencing available to complete the job. What is the largest possible total area of the five pens?

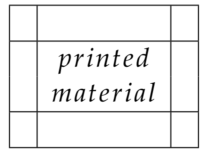


Largest area = ____________________ ft^2

4. A fence is to be built to enclose a rectangular area of 200 square feet. The fence along three sides is to be made of material that costs 4 dollars per foot, and the material for the fourth side costs 14 dollars per foot. Find the dimensions of the enclosure that is most economical to construct.

Dimensions: __________ x __________

5. The top and bottom margins of a poster are 2 cm and the side margins are each 8 cm. If the area of printed material on the poster is fixed at 390 square centimeters, find the dimensions of the poster with the smallest area.



Width = ______________________ cm

Height = ______________________ cm

6. If 1100 square centimeters of material is available to make a box with a square base and an open top, find the largest possible volume of the box.

Volume = __________________________ cm^3

7. A piece of wire 25 m long is cut into two pieces. One piece is bent into a square and the other is bent into a circle.

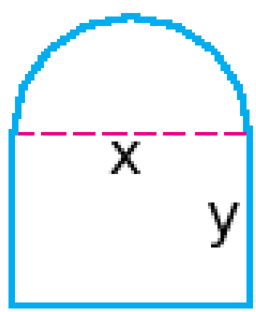
(a) How much of the wire should go to the square to maximize the total area enclosed by both figures?

__________________ m

(b) How much of the wire should go to the square to minimize the total area enclosed by both figures?

__________________ m

8. Suppose that 393 ft of fencing are used to enclose a corral in the shape of a rectangle with a semicircle whose diameter is a side of the rectangle as the following figure:



Find the dimensions of the corral with maximum area.

x = __________________ ft.

y = __________________ ft.

9. A rectangular box with a square bottom and closed top is to be made from two materials. The material for the sides costs \$1.50 per square foot and the material for the top and bottom costs \$3.00 per square foot. If you are willing to spend \$15 on the box, what is the largest volume it can contain? Justify your answer completely using calculus.

10. A farmer wants to start raising cows, horses, goats, and sheep, and desires to have a rectangular pasture for the animals to graze in. However, no two different kinds of animals can graze together. In order to minimize the amount of fencing she will need, she has decided to enclose a large rectangular area and then divide it into four equally sized pens by adding three segments of fence inside the large rectangle that are parallel to two existing sides. She has decided to purchase 7500 ft of fencing. What is the maximum possible area that each of the four pens will enclose?

11. Two vertical poles of heights 60 ft and 80 ft stand on level ground, with their bases 100 ft apart. A cable that is stretched from the top of one pole to some point on the ground between the poles, and then to the top of the other pole. What is the minimum possible length of cable required? Justify your answer completely using calculus.

12. A company is designing propane tanks that are cylindrical with hemispherical ends. Assume that the company wants tanks that will hold 1000 cubic feet of gas, and that the ends are more expensive to make, costing \$5 per square foot, while the cylindrical barrel between the ends costs \$2 per square foot. Use calculus to determine the minimum cost to construct such a tank.

CHAPTER 4

The Definite Integral

4.1 Determining distance traveled from velocity

Motivating Questions

- If we know the velocity of a moving body at every point in a given interval, can we determine the distance the object has traveled on the time interval?
- How is the problem of finding distance traveled related to finding the area under a certain curve?
- What does it mean to antidifferentiate a function and why is this process relevant to finding distance traveled?
- If velocity is negative, how does this impact the problem of finding distance traveled?

4.1.1 Introduction

In the first section of the text, we considered a moving object with known position at time t, namely, a tennis ball tossed into the air with height s (in feet) at time t (in seconds) given by $s(t) = 64 - 16(t-1)^2$. We investigated the average velocity of the ball on an interval $[a, b]$, computed by the difference quotient $\frac{s(b)-s(a)}{b-a}$. We found that we could determine the instantaneous velocity of the ball at time t by taking the derivative of the position function,

$$s'(t) = \lim_{h \to 0} \frac{s(t+h) - s(t)}{h}.$$

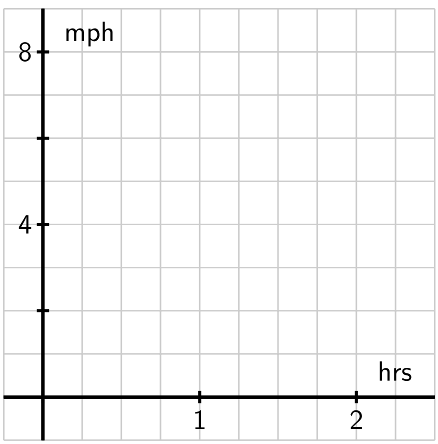
Thus, if its position function is differentiable, we can find the velocity of a moving object at any point in time.

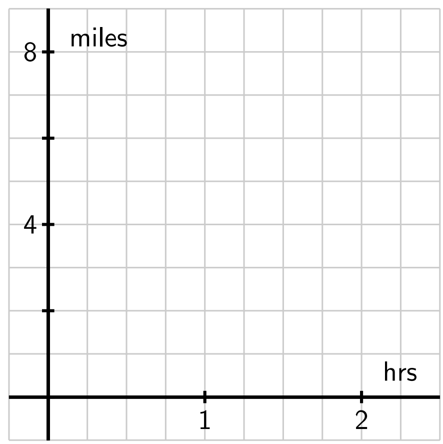
From this study of position and velocity we have learned a great deal. We can use the derivative to find a function's instantaneous rate of change at any point in the domain, to find where the function is increasing or decreasing, where it is concave up or concave down, and to locate relative extremes. The vast majority of the problems and applications we have considered have involved the situation where a particular function is known and we seek information that relies on knowing the function's instantaneous rate of change. For all these tasks, we proceed from a function f to its derivative, f', and use the meaning of the derivative to help us answer important questions.

We have also encountered the reverse situation, where we know the derivative of a function, f', and try to deduce information about f. We will focus our attention in Chapter 4 on this problem: if we know the instantaneous rate of change of a function, can we find the function itself? We start with a more specific question: if we know the instantaneous velocity of an object moving along a straight line path, can we find its corresponding position function?

Preview Activity 4.1.1. Suppose that a person is taking a walk along a long straight path and walks at a constant rate of 3 miles per hour.

(a) On the left-hand axes provided in the following figure, sketch a labeled graph of the velocity function $v(t) = 3$. Note that while the scale on the two sets of axes is the same, the units on the right-hand axes differ from those on the left. The right-hand axes will be used in question (d).





(b) How far did the person travel during the two hours? How is this distance related to the area of a certain region under the graph of $y = v(t)$?

(c) Find an algebraic formula, $s(t)$, for the position of the person at time t, assuming that $s(0) = 0$. Explain your thinking.

(d) On the right-hand axes provided in the provided figure, sketch a labeled graph of the position function $y = s(t)$.

(e) For what values of t is the position function s increasing? Explain why this is the case using relevant information about the velocity function v.

4.1.2 Area under the graph of the velocity function

In Preview Activity 4.1.1, we learned that when the velocity of a moving object's velocity is constant (and positive), the area under the velocity curve over an interval of time tells us the distance the object traveled.

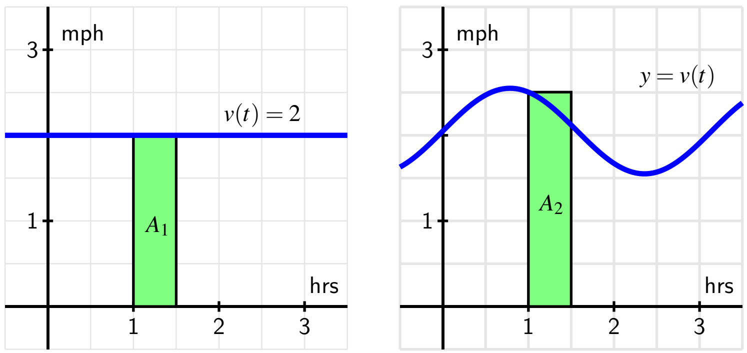


Figure 4.1.1 At left, a constant velocity function; at right, a non-constant velocity function.

The left-hand graph of Figure 4.1.1 shows the velocity of an object moving at 2 miles per hour over the time interval $[1, 1.5]$. The area A_1 of the shaded region under $y = v(t)$ on $[1, 1.5]$ is

$$A_1 = 2\,\frac{\text{miles}}{\text{hour}} \cdot \frac{1}{2}\,\text{hours} = 1\,\text{mile}.$$

This result is simply the fact that distance equals rate times time, provided the rate is constant. Thus, if $v(t)$ is constant on the interval $[a, b]$, the distance traveled on $[a, b]$ is equal to the area A given by

$$A = v(a)(b - a) = v(a)\Delta t,$$

where Δt is the change in t over the interval. (Since the velocity is constant, we can use any value of $v(t)$ on the interval $[a, b]$, we simply chose $v(a)$, the value at the interval's left endpoint.) For several examples where the velocity function is piecewise constant, see this interactive[1].

The situation is more complicated when the velocity function is not constant. But on relatively small intervals where $v(t)$ does not vary much, we can use the area principle to estimate the distance traveled. The graph at right in Figure 4.1.1 shows a non-constant velocity function. On the interval $[1, 1.5]$, the velocity varies from $v(1) = 2.5$ down to $v(1.5) \approx 2.1$. One estimate for the distance traveled is the area of the pictured rectangle,

$$A_2 = v(1)\Delta t = 2.5\,\frac{\text{miles}}{\text{hour}} \cdot \frac{1}{2}\,\text{hours} = 1.25\,\text{miles}.$$

Note that because v is decreasing on $[1, 1.5]$, $A_2 = 1.25$ is an over-estimate of the actual distance traveled.

[1] `gvsu.edu/s/9T`

To estimate the area under this non-constant velocity function on a wider interval, say $[0, 3]$, one rectangle will not give a good approximation. Instead, we could use the six rectangles pictured in Figure 4.1.2, find the area of each rectangle, and add up the total. Obviously there are choices to make and issues to understand: How many rectangles should we use? Where should we evaluate the function to decide the rectangle's height? What happens if the velocity is sometimes negative? Can we find the exact area under any non-constant curve?

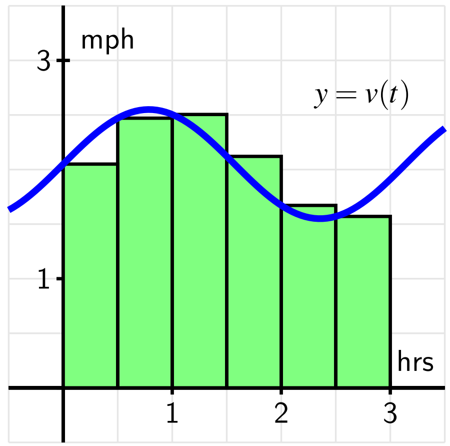
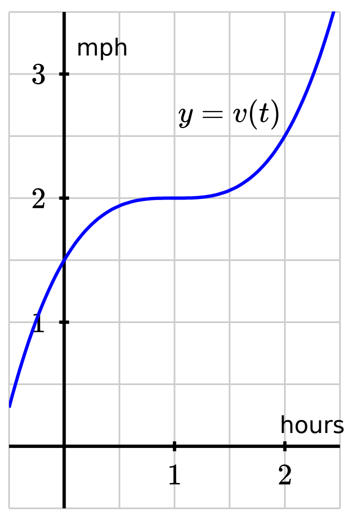


Figure 4.1.2 Estimating the area under $y = v(t)$ on $[0, 3]$.

We will study these questions and more in what follows; for now it suffices to observe that the simple idea of the area of a rectangle gives us a powerful tool for estimating distance traveled from a velocity function, as well as for estimating the area under an arbitrary curve. To explore the use of multiple rectangles to approximate area under a non-constant velocity function, see this interactive graphic[2].

Activity 4.1.2. Suppose that a person is walking in such a way that their velocity varies slightly according to the information given in the following table and graph.

t	$v(t)$
0.00	1.500
0.25	1.789
0.50	1.938
0.75	1.992
1.00	2.000
1.25	2.008
1.50	2.063
1.75	2.211
2.00	2.500



(a) Using the grid, graph, and given data appropriately, estimate the distance traveled by the walker during the two hour interval from $t = 0$ to $t = 2$. You should

[2] gvsu.edu/s/9U

use time intervals of width $\Delta t = 0.5$, choosing a way to use the function consistently to determine the height of each rectangle in order to approximate distance traveled.

(b) How could you get a better approximation of the distance traveled on $[0, 2]$? Explain, and then find this new estimate.

(c) Now suppose that you know that v is given by $v(t) = 0.5t^3 - 1.5t^2 + 1.5t + 1.5$. Remember that v is the derivative of the walker's position function, s. Find a formula for s so that $s' = v$.

(d) Based on your work in (c), what is the value of $s(2) - s(0)$? What is the meaning of this quantity?

4.1.3 Two approaches: area and antidifferentiation

When the velocity of a moving object is positive, the object's position is always increasing. (We will soon consider situations where velocity is negative; for now, we focus on the situation where velocity is always positive.) We have established that whenever v is constant on an interval, the exact distance traveled is the area under the velocity curve. When v is not constant, we can estimate the total distance traveled by finding the areas of rectangles that approximate the area under the velocity curve.

Thus, we see that finding the area between a curve and the horizontal axis is an important exercise: besides being an interesting geometric question, if the curve gives the velocity of a moving object, the area under the curve tells us the exact distance traveled on an interval. We can estimate this area if we have a graph or a table of values for the velocity function.

In Activity 4.1.2, we encountered an alternate approach to finding the distance traveled. If $y = v(t)$ is a formula for the instantaneous velocity of a moving object, then v must be the derivative of the object's position function, s. If we can find a formula for $s(t)$ from the formula for $v(t)$, we will know the position of the object at time t, and the change in position over a particular time interval tells us the distance traveled on that interval.

For a simple example, consider the situation from Preview Activity 4.1.1, where a person is walking along a straight line with velocity function $v(t) = 3$ mph.

On the left-hand graph of the velocity function in Figure 4.1.3, we see the relationship between area and distance traveled,

$$A = 3\,\frac{\text{miles}}{\text{hour}} \cdot 1.25\,\text{hours} = 3.75\,\text{miles}.$$

In addition, we observe[3] that if $s(t) = 3t$, then $s'(t) = 3$, so $s(t) = 3t$ is the position function whose derivative is the given velocity function, $v(t) = 3$. The respective locations of the person at times $t = 0.25$ and $t = 1.5$ are $s(1.5) = 4.5$ and $s(0.25) = 0.75$, and therefore

$$s(1.5) - s(0.25) = 4.5 - 0.75 = 3.75 \text{ miles}.$$

[3]Here we are making the implicit assumption that $s(0) = 0$; we will discuss different possibilities for values of $s(0)$ in subsequent study.

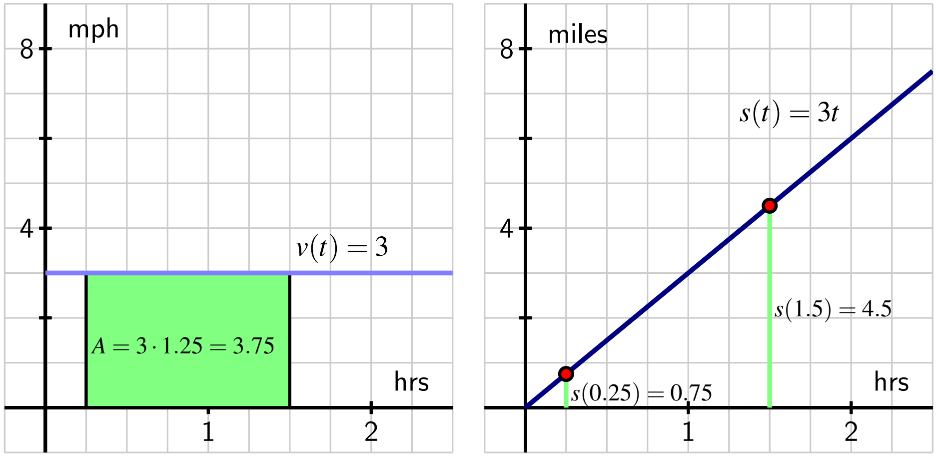


Figure 4.1.3 The velocity function $v(t) = 3$ and corresponding position function $s(t) = 3t$.

The quantity $s(1.5) - s(0.25)$ is the person's change in position on $[0.25, 1.5]$, as well as the distance traveled since their velocity is always positive. In this example there are profound ideas and connections that we will study throughout Chapter 4.

For now, observe that if we know a formula for a velocity function v, it can be very helpful to find a function s that satisfies $s' = v$. We say that s is an *antiderivative* of v. More generally, we have the following formal definition.

Definition 4.1.4 If g and G are functions such that $G' = g$, we say that G is an **antiderivative** of g. ◊

For example, if $g(x) = 3x^2 + 2x$, $G(x) = x^3 + x^2$ is an antiderivative of g, because $G'(x) = g(x)$. Note that we say "an" antiderivative of g rather than "the" antiderivative of g, because $H(x) = x^3+x^2+5$ is also a function whose derivative is g, and thus H is another antiderivative of g.

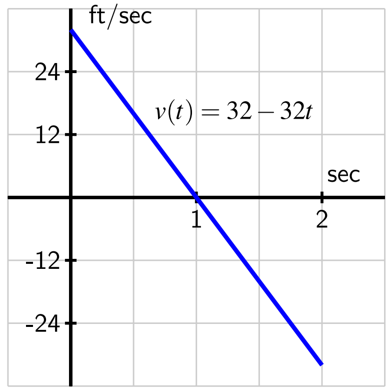
Activity 4.1.3. A ball is tossed vertically in such a way that its velocity function is given by $v(t) = 32 - 32t$, where t is measured in seconds and v in feet per second. Assume that this function is valid for $0 \le t \le 2$.

(a) For what values of t is the velocity of the ball positive? What does this tell you about the motion of the ball on this interval of time values?

(b) Find an antiderivative, s, of v that satisfies $s(0) = 0$.

(c) Compute the value of $s(1) - s(\frac{1}{2})$. What is the meaning of the value you find?

(d) Using the graph of $y = v(t)$ provided in the following figure, find the exact area of the region between the velocity curve and the t-axis between $t = \frac{1}{2}$ and $t = 1$. What is the meaning of the value you find?



(e) Answer the same questions as in (c) and (d) but instead using the interval $[0,1]$.

(f) What is the value of $s(2) - s(0)$? What does this result tell you about the flight of the ball? How is this value connected to the provided graph of $y = v(t)$? Explain.

4.1.4 When velocity is negative

The assumption that the velocity is positive on a given interval guarantees that the movement of an object is always in a single direction, and hence ensures that its change in position is the same as the distance it travels. As we saw in Activity 4.1.3, there are natural settings in which an object's velocity is negative, and we would like to understand this scenario as well.

Consider a simple example where a person goes for a walk on the beach along a stretch of very straight shoreline that runs east-west. We assume that her initial position is $s(0) = 0$, and that her position function increases as she moves east from her starting location. For instance, $s = 1$ mile represents one mile east of the start location, while $s = -1$ tells us she is one mile west of where she began walking on the beach.

Now suppose she walks in the following manner. From the outset at $t = 0$, she walks due east at a constant rate of 3 mph for 1.5 hours. After 1.5 hours, she stops abruptly and begins walking due west at a constant rate of 4 mph and does so for 0.5 hours. Then, after another abrupt stop and start, she resumes walking at a constant rate of 3 mph to the east for one more hour. What is the total distance she traveled on the time interval from $t = 0$ to $t = 3$? What is the total change in her position over that time?

These questions are possible to answer without calculus because the velocity is constant on each interval. From $t = 0$ to $t = 1.5$, she traveled

$$D_{[0,1.5]} = 3 \text{ miles per hour} \cdot 1.5 \text{ hours} = 4.5 \text{ miles}.$$

On $t = 1.5$ to $t = 2$, the distance traveled is

$$D_{[1.5,2]} = 4 \text{ miles per hour} \cdot 0.5 \text{ hours} = 2 \text{ miles}.$$

Finally, in the last hour she walked

$$D_{[2,3]} = 3 \text{ miles per hour} \cdot 1 \text{ hour} = 3 \text{ miles},$$

so the total distance she traveled is

$$D = D_{[0,1.5]} + D_{[1.5,2]} + D_{[2,3]} = 4.5 + 2 + 3 = 9.5 \text{ miles}.$$

Since the velocity for $1.5 < t < 2$ is $v = -4$, indicating motion in the westward direction, the person first walked 4.5 miles east, then 2 miles west, followed by 3 more miles east. Thus, the total change in her position is

$$\text{change in position} = 4.5 - 2 + 3 = 5.5 \text{ miles}.$$

It's also valuable to think about our conclusions from a graphical perspective.

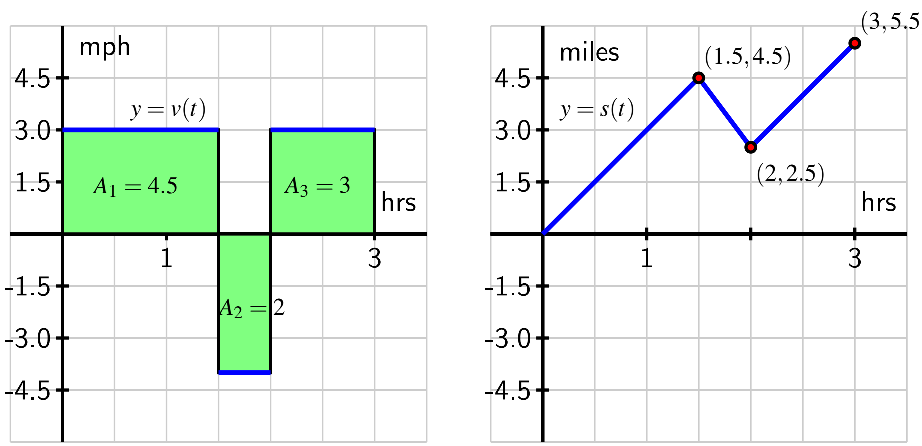


Figure 4.1.5 At left, the velocity function of the person walking; at right, the corresponding position function.

In Figure 4.1.5, we see how the distances we computed can be viewed as areas: $A_1 = 4.5$ comes from multiplying rate times time $(3 \cdot 1.5)$, as do A_2 and A_3. But while A_2 is an area (and is therefore positive), because the velocity function is negative for $1.5 < t < 2$, this area has a negative sign associated with it. The negative area distinguishes between distance traveled and change in position.

The distance traveled is the sum of the areas,

$$D = A_1 + A_2 + A_3 = 4.5 + 2 + 3 = 9.5 \text{ miles}.$$

But the change in position has to account for travel in the negative direction. An area above the t-axis is considered positive because it represents distance traveled in the positive direction, while one below the t-axis is viewed as negative because it represents travel in theneg-ative direction. Thus, the change in the person's position is

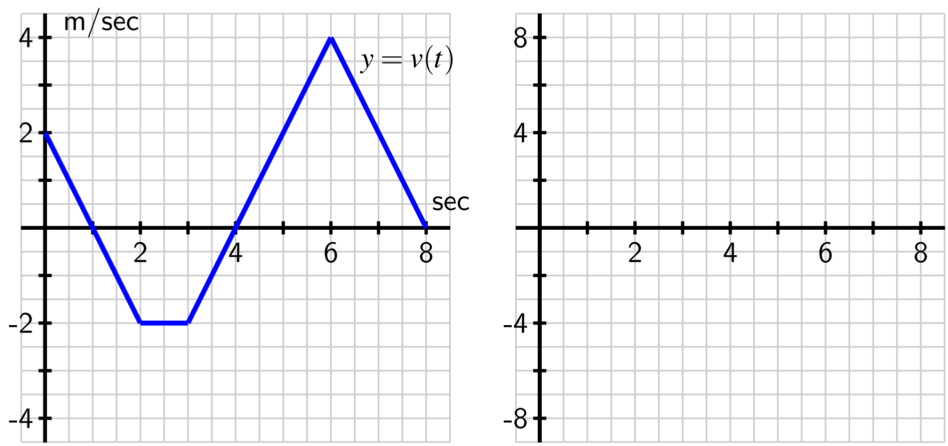
$$s(3) - s(0) = (+4.5) + (-2) + (+3) = 5.5 \text{ miles}.$$

In other words, the person walks 4.5 miles in the positive direction, followed by two miles in the negative direction, and then 3 more miles in the positive direction.

Negative velocity is also seen in the graph of the position function $y = s(t)$. Its slope is negative (specifically, -4) on the interval $1.5 < t < 2$ because the velocity is -4 on that interval. The negative slope shows the position function is decreasing because the person is walking east, rather than west.

To summarize, we see that if velocity is sometimes negative, a moving object's change in position is different from its distance traveled. If we compute separately the distance traveled on each interval where velocity is positive or negative, we can calculate either the total distance traveled or the total change in position. We assign a negative value to distances traveled in the negative direction when we calculate change in position, but a positive value when we calculate the total distance traveled.

Activity 4.1.4. Suppose that an object moving along a straight line path has its velocity v (in meters per second) at time t (in seconds) given by the piecewise linear function whose graph is pictured at left in the following figure. We view movement to the right as being in the positive direction (with positive velocity), while movement to the left is in the negative direction.



Suppose further that the object's initial position at time $t = 0$ is $s(0) = 1$.

(a) Determine the total distance traveled and the total change in position on the time interval $0 \le t \le 2$. What is the object's position at $t = 2$?

(b) On what time intervals is the moving object's position function increasing? Why? On what intervals is the object's position decreasing? Why?

(c) What is the object's position at $t = 8$? How many total meters has it traveled to get to this point (including distance in both directions)? Is this different from the object's total change in position on $t = 0$ to $t = 8$?

(d) Find the exact position of the object at $t = 1, 2, 3, \ldots, 8$ and use this data to sketch an accurate graph of $y = s(t)$ on the axes provided at right in the figure. How can you use the provided information about $y = v(t)$ to determine the concavity of s on each relevant interval?

4.1.5 Summary

- If we know the velocity of a moving body at every point in a given interval and the velocity is positive throughout, we can estimate the object's distance traveled and in some circumstances determine this value exactly.
- In particular, when velocity is positive on an interval, we can find the total distance traveled by finding the area under the velocity curve and above the t-axis on the given time interval. We may only be able to estimate this area, depending on the shape of the velocity curve.
- An antiderivative of a function f is a new function F whose derivative is f. That is, F is an antiderivative of f provided that $F' = f$. In the context of velocity and position, if we know a velocity function v, an antiderivative of v is a position function s that satisfies $s' = v$. If v is positive on a given interval, say $[a, b]$, then the change in position, $s(b) - s(a)$, measures the distance the moving object traveled on $[a, b]$.
- If its velocity is sometimes negative, a moving object is sometimes traveling in the opposite direction or backtracking. To determine distance traveled, we have to compute the distance separately on intervals where velocity is positive or negative, and account for the change in position on each such interval.

4.1.6 Exercises

1. A car comes to a stop six seconds after the driver applies the brakes. While the brakes are on, the following velocities are recorded:

Time since brakes applied (sec)	0	2	4	6
Velocity (ft/s)	92	47	17	0

Give lower and upper estimates (using all of the available data) for the distance the car traveled after the brakes were applied.

lower: ______________ ft

upper: ______________ ft

On a sketch of velocity against time, show the lower and upper estimates you found above..

2. The velocity of a car is $f(t) = 12t$ meters/second. Use a graph of $f(t)$ to find the exact distance traveled by the car, in meters, from $t = 0$ to $t = 10$ seconds.

distance = ________________________________ m

3. The velocity of a particle moving along the x-axis is given by $f(t) = 15 - 5t$ cm/sec. Use a graph of $f(t)$ to find the exact change in position of the particle from time $t = 0$ to $t = 4$ seconds.

change in position = ______________________ cm

4. The velocity function is $v(t) = t^2 - 6t + 8$ for a particle moving along a line. Find the displacement (net distance covered) of the particle during the time interval $[-2, 6]$.

displacement = ______________

5. A car traveling at 39 ft/sec decelerates at a constant 3 feet per second squared. How many feet does the car travel before coming to a complete stop?

______________ ft

6. A stone is thrown straight up from the edge of a roof, 650 feet above the ground, at a speed of 20 feet per second.

A. Remembering that the acceleration due to gravity is -32 feet per second squared, how high is the stone 5 seconds later? ______________________ ft.

B. At what time does the stone hit the ground? ______________________ s.

C. What is the velocity of the stone when it hits the ground? ______________________ ft/s.

7. Along the eastern shore of Lake Michigan from Lake Macatawa (near Holland) to Grand Haven, there is a bike path that runs almost directly north-south. For the purposes of this problem, assume the road is completely straight, and that the function $s(t)$ tracks the position of the biker along this path in miles north of Pigeon Lake, which lies roughly halfway between the ends of the bike path.

Suppose that the biker's velocity function is given by the graph in Figure 4.1.6 on the time interval $0 \leq t \leq 4$ (where t is measured in hours), and that $s(0) = 1$.

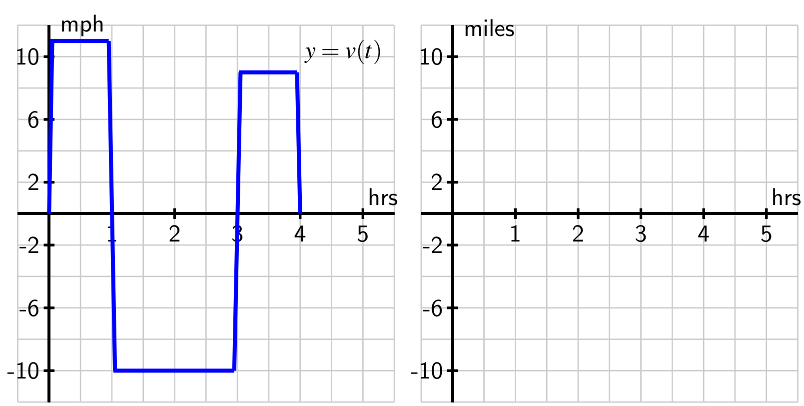


Figure 4.1.6 The graph of the biker's velocity, $y = v(t)$, at left. At right, axes to plot an approximate sketch of $y = s(t)$.

a. Approximately how far north of Pigeon Lake was the cyclist when she was the greatest distance away from Pigeon Lake? At what time did this occur?

b. What is the cyclist's total change in position on the time interval $0 \le t \le 2$? At $t = 2$, was she north or south of Pigeon Lake?

c. What is the total distance the biker traveled on $0 \le t \le 4$? At the end of the ride, how close was she to the point at which she started?

d. Sketch an approximate graph of $y = s(t)$, the position function of the cyclist, on the interval $0 \le t \le 4$. Label at least four important points on the graph of s.

8. A toy rocket is launched vertically from the ground on a day with no wind. The rocket's vertical velocity at time t (in seconds) is given by $v(t) = 500 - 32t$ feet/sec.

a. At what time after the rocket is launched does the rocket's velocity equal zero? Call this time value a. What happens to the rocket at $t = a$?

b. Find the value of the total area enclosed by $y = v(t)$ and the t-axis on the interval $0 \le t \le a$. What does this area represent in terms of the physical setting of the problem?

c. Find an antiderivative s of the function v. That is, find a function s such that $s'(t) = v(t)$.

d. Compute the value of $s(a) - s(0)$. What does this number represent in terms of the physical setting of the problem?

e. Compute $s(5) - s(1)$. What does this number tell you about the rocket's flight?

9. An object moving along a horizontal axis has its instantaneous velocity at time t in seconds given by the function v pictured in Figure 4.1.7, where v is measured in feet/sec. Assume that the curves that make up the parts of the graph of $y = v(t)$ are either portions of straight lines or portions of circles.

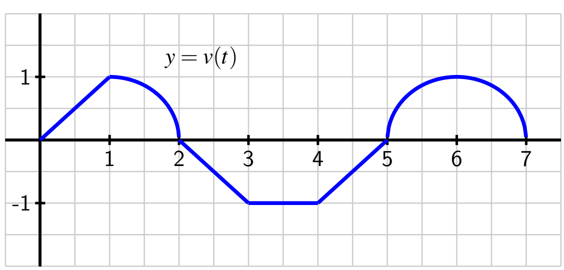


Figure 4.1.7 The graph of $y = v(t)$, the velocity function of a moving object.

a. Determine the exact total distance the object traveled on $0 \le t \le 2$.

b. What is the value and meaning of $s(5) - s(2)$, where $y = s(t)$ is the position function of the moving object?

c. On which time interval did the object travel the greatest distance: $[0, 2]$, $[2, 4]$, or $[5, 7]$?

d. On which time interval(s) is the position function s increasing? At which point(s) does s achieve a relative maximum?

10. Filters at a water treatment plant become dirtier over time and thus become less effective; they are replaced every 30 days. During one 30-day period, the rate at which pollution passes through the filters into a nearby lake (in units of particulate matter per day) is measured every 6 days and is given in the following table. The time t is measured in days since the filters were replaced.

Table 4.1.8 Pollution data for the water filters.

Day, t	0	6	12	18	24	30
Rate of pollution in units per day, $p(t)$	7	8	10	13	18	35

a. Plot the given data on a set of axes with time on the horizontal axis and the rate of pollution on the vertical axis.

b. Explain why the amount of pollution that entered the lake during this 30-day period would be given exactly by the area bounded by $y = p(t)$ and the t-axis on the time interval $[0, 30]$.

c. Estimate the total amount of pollution entering the lake during this 30-day period. Carefully explain how you determined your estimate.

4.2 Riemann sums

Motivating Questions

- How can we use a Riemann sum to estimate the area between a given curve and the horizontal axis over a particular interval?
- What are the differences among left, right, middle, and random Riemann sums?
- How can we write Riemann sums in an abbreviated form?

4.2.1 Introduction

In Section 4.1, we learned that if an object moves with positive velocity v, the area between $y = v(t)$ and the t-axis over a given time interval tells us the distance traveled by the object over that time period. If $v(t)$ is sometimes negative and we view the area of any region below the t-axis as having an associated negative sign, then the sum of these signed areas tells us the moving object's change in position over a given time interval.

For instance, for the velocity function given in Figure 4.2.1, if the areas of shaded regions are A_1, A_2, and A_3 as labeled, then the total distance D traveled by the moving object on $[a, b]$ is

$$D = A_1 + A_2 + A_3,$$

while the total change in the object's position on $[a, b]$ is

$$s(b) - s(a) = A_1 - A_2 + A_3.$$

Because the motion is in the negative direction on the interval where $v(t) < 0$, we subtract A_2 to determine the object's total change in position.

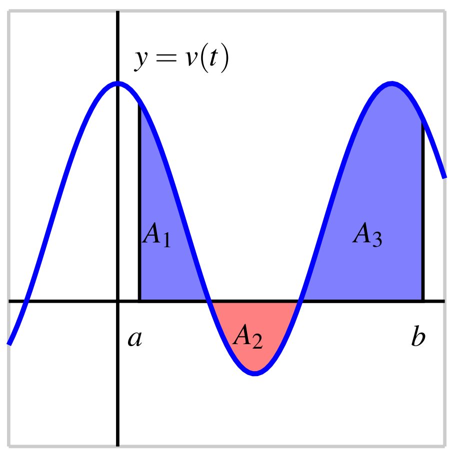
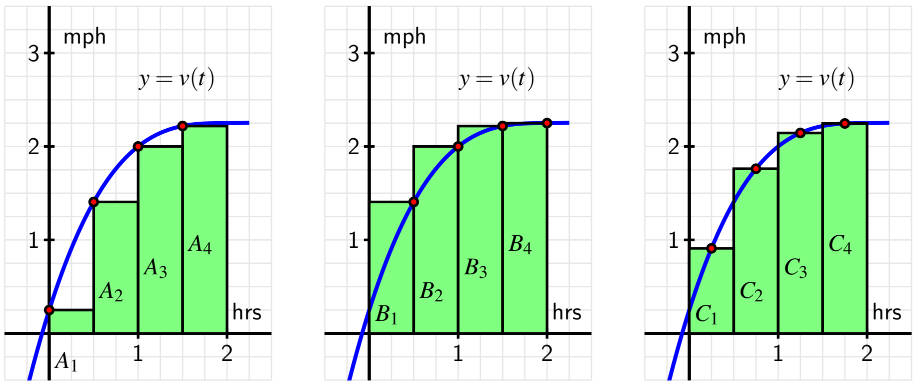


Figure 4.2.1 A velocity function that is sometimes negative.

Of course, finding D and $s(b) - s(a)$ for the graph in Figure 4.2.1 presumes that we can actually find the areas A_1, A_2, and A_3. So far, we have worked with velocity functions that were either constant or linear so that we could find the exact area bounded by the velocity function and the horizontal axis through a combination of rectangles and triangles. But when the curve bounds a region that is not a familiar geometric shape, we cannot find its area exactly without some key ideas in calculus. Indeed, this is one of our biggest goals in Chapter 4: to learn how to find the exact area bounded between a curve and the horizontal axis for as many different types of functions as possible.

In Activity 4.1.2, we approximated the area under a nonlinear velocity function using rectangles. In the following preview activity, we consider three different options for the heights of the rectangles we will use.

Preview Activity 4.2.1. A person walking along a straight path has her velocity in miles per hour at time t given by the function $v(t) = 0.25t^3 - 1.5t^2 + 3t + 0.25$, for times in the interval $0 \le t \le 2$. The graph of this function is also given in each of the three diagrams in the following figure.



Note that in each diagram, we use four rectangles to estimate the area under $y = v(t)$ on the interval $[0, 2]$, but the method by which the four rectangles' respective heights are decided varies among the three individual graphs.

(a) How are the heights of rectangles in the left-most diagram being chosen? Explain, and hence determine the value of

$$S = A_1 + A_2 + A_3 + A_4$$

by evaluating the function $y = v(t)$ at appropriately chosen values and observing the width of each rectangle. Note, for example, that

$$A_3 = v(1) \cdot \frac{1}{2} = 2 \cdot \frac{1}{2} = 1.$$

(b) Explain how the heights of rectangles are being chosen in the middle diagram and find the value of

$$T = B_1 + B_2 + B_3 + B_4.$$

(c) Likewise, determine the pattern of how heights of rectangles are chosen in the right-most diagram and determine

$$U = C_1 + C_2 + C_3 + C_4.$$

(d) Of the estimates S, T, and U, which do you think is the best approximation of D, the total distance the person traveled on $[0, 2]$? Why?

4.2.2 Sigma Notation

We have used sums of areas of rectangles to approximate the area under a curve. Intuitively, we expect that using a larger number of thinner rectangles will provide a better estimate for the area. Consequently, we anticipate dealing with sums of a large number of terms. To do so, we introduce *sigma notation*, named for the Greek letter Σ, which is the capital letter S in the Greek alphabet.

For example, say we are interested in the sum

$$1 + 2 + 3 + \cdots + 100,$$

the sum of the first 100 natural numbers. In sigma notation we write

$$\sum_{k=1}^{100} k = 1 + 2 + 3 + \cdots + 100.$$

We read the symbol $\sum_{k=1}^{100} k$ as "the sum from k equals 1 to 100 of k." The variable k is called the index of summation, and any letter can be used for this variable. The pattern in the terms of the sum is denoted by a function of the index; for example,

$$\sum_{k=1}^{10}(k^2 + 2k) = (1^2 + 2 \cdot 1) + (2^2 + 2 \cdot 2) + (3^2 + 2 \cdot 3) + \cdots + (10^2 + 2 \cdot 10),$$

and more generally,

$$\sum_{k=1}^{n} f(k) = f(1) + f(2) + \cdots + f(n).$$

Sigma notation allows us to easily change the function being used to describe the terms in the sum, and to adjust the number of terms in the sum simply by changing the value of n. We test our understanding of this new notation in the following activity.

Activity 4.2.2. For each sum written in sigma notation, write the sum long-hand and evaluate the sum to find its value. For each sum written in expanded form, write the sum in sigma notation.

(a) $\displaystyle\sum_{k=1}^{5}(k^2 + 2)$

(b) $\displaystyle\sum_{i=3}^{6}(2i - 1)$

(c) $3 + 7 + 11 + 15 + \cdots + 27$

(d) $4 + 8 + 16 + 32 + \cdots + 256$

(e) $\displaystyle\sum_{i=1}^{6} \frac{1}{2^i}$

4.2.3 Riemann Sums

When a moving body has a positive velocity function $y = v(t)$ on a given interval $[a, b]$, the area under the curve over the interval gives the total distance the body travels on $[a, b]$. We are also interested in finding the exact area bounded by $y = f(x)$ on an interval $[a, b]$, regardless of the meaning or context of the function f. For now, we continue to focus on finding an accurate estimate of this area by using a sum of the areas of rectangles. Unless otherwise indicated, we assume that f is continuous and non-negative on $[a, b]$.

The first choice we make in such an approximation is the number of rectangles.

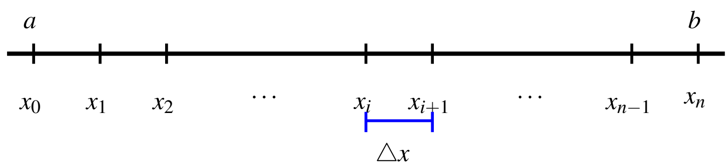


Figure 4.2.2 Subdividing the interval $[a, b]$ into n subintervals of equal length Δx.

If we desire n rectangles of equal width to subdivide the interval $[a, b]$, then each rectangle must have width $\Delta x = \frac{b-a}{n}$. We let $x_0 = a$, $x_n = b$, and define $x_i = a + i\Delta x$, so that $x_1 = x_0 + \Delta x$, $x_2 = x_0 + 2\Delta x$, and so on, as pictured in Figure 4.2.2.

We use each subinterval $[x_i, x_{i+1}]$ as the base of a rectangle, and next choose the height of the rectangle on that subinterval. There are three standard choices: we can use the left endpoint of each subinterval, the right endpoint of each subinterval, or the midpoint of each. These are precisely the options encountered in Preview Activity 4.2.1 and seen in its figure. We next explore how these choices can be described in sigma notation.

Consider an arbitrary positive function f on $[a, b]$ with the interval subdivided as shown in Figure 4.2.2, and choose to use left endpoints. Then on each interval $[x_i, x_{i+1}]$, the area of the rectangle formed is given by

$$A_{i+1} = f(x_i) \cdot \Delta x,$$

as seen in Figure 4.2.3.

If we let L_n denote the sum of the areas of these rectangles, we see that

$$\begin{aligned} L_n &= A_1 + A_2 + \cdots + A_{i+1} + \cdots + A_n \\ &= f(x_0) \cdot \Delta x + f(x_1) \cdot \Delta x + \cdots + f(x_i) \cdot \Delta x + \cdots + f(x_{n-1}) \cdot \Delta x. \end{aligned}$$

In the more compact sigma notation, we have

$$L_n = \sum_{i=0}^{n-1} f(x_i) \Delta x.$$

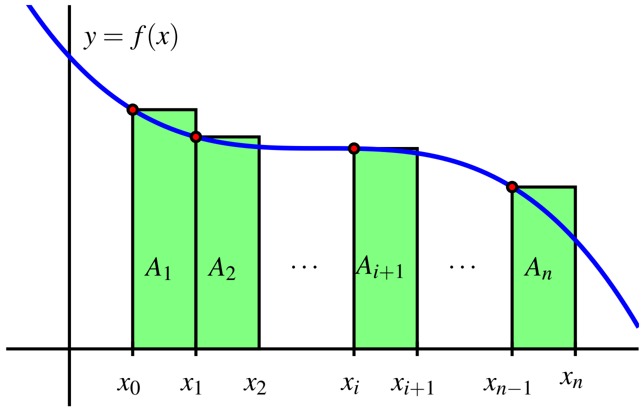


Figure 4.2.3 Subdividing the interval $[a, b]$ into n subintervals of equal length Δx and approximating the area under $y = f(x)$ over $[a, b]$ using left rectangles.

Note that since the index of summation begins at 0 and ends at $n - 1$, there are indeed n terms in this sum. We call L_n the *left Riemann sum* for the function f on the interval $[a, b]$.

To see how the Riemann sums for right endpoints and midpoints are constructed, we consider Figure 4.2.4.

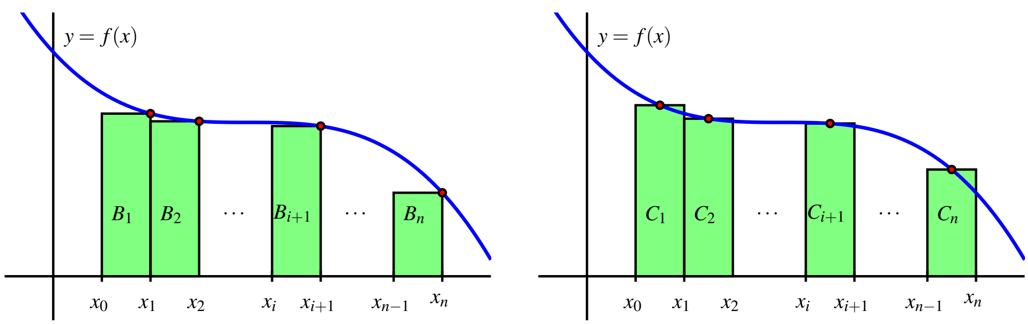


Figure 4.2.4 Riemann sums using right endpoints and midpoints.

For the sum with right endpoints, we see that the area of the rectangle on an arbitrary interval $[x_i, x_{i+1}]$ is given by $B_{i+1} = f(x_{i+1}) \cdot \Delta x$, and that the sum of all such areas of rectangles is given by

$$\begin{aligned} R_n &= B_1 + B_2 + \cdots + B_{i+1} + \cdots + B_n \\ &= f(x_1) \cdot \Delta x + f(x_2) \cdot \Delta x + \cdots + f(x_{i+1}) \cdot \Delta x + \cdots + f(x_n) \cdot \Delta x \\ &= \sum_{i=1}^{n} f(x_i)\Delta x. \end{aligned}$$

We call R_n the *right Riemann sum* for the function f on the interval $[a, b]$.

For the sum that uses midpoints, we introduce the notation

$$\overline{x}_{i+1} = \frac{x_i + x_{i+1}}{2}$$

so that $\overline{x}_{i+1}$ is the midpoint of the interval $[x_i, x_{i+1}]$. For instance, for the rectangle with area C_1 in Figure 4.2.4, we now have

$$C_1 = f(\overline{x}_1) \cdot \Delta x.$$

Hence, the sum of all the areas of rectangles that use midpoints is

$$\begin{aligned} M_n &= C_1 + C_2 + \cdots + C_{i+1} + \cdots + C_n \\ &= f(\overline{x}_1) \cdot \Delta x + f(\overline{x}_2) \cdot \Delta x + \cdots + f(\overline{x}_{i+1}) \cdot \Delta x + \cdots + f(\overline{x}_n) \cdot \Delta x \\ &= \sum_{i=1}^{n} f(\overline{x}_i)\Delta x, \end{aligned}$$

and we say that M_n is the *middle Riemann sum* for f on $[a, b]$.

Thus, we have two variables to explore: the number of rectangles and the height of each rectangle. We can explore these choices dynamically, and this interactive[1] is a particularly useful one. There we see the image shown in Figure 4.2.5, but with the opportunity to adjust the slider bars for the heights and the number of rectangles.

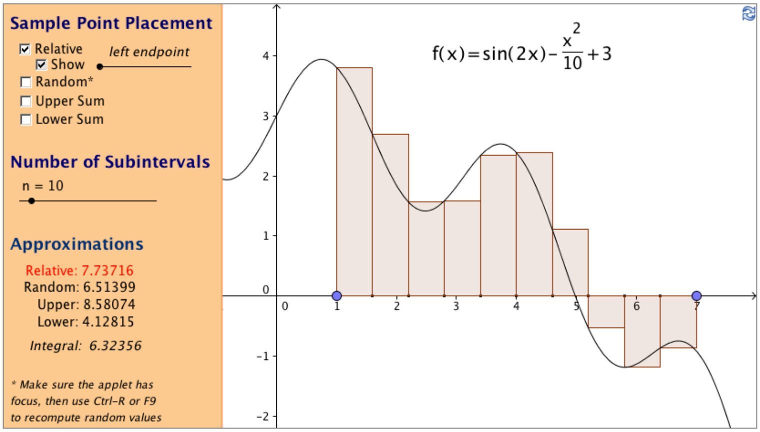


Figure 4.2.5 A snapshot of the interactive graphic found at gvsu.edu/s/a9.

By moving the sliders, we can see how the heights of the rectangles change as we consider left endpoints, midpoints, and right endpoints, as well as the impact that a larger number of narrower rectangles has on the approximation of the exact area bounded by the function and the horizontal axis.

When $f(x) \geq 0$ on $[a, b]$, each of the Riemann sums L_n, R_n, and M_n provides an estimate of the area under the curve $y = f(x)$ over the interval $[a, b]$. We also recall that in the context of a nonnegative velocity function $y = v(t)$, the corresponding Riemann sums approximate the distance traveled on $[a, b]$ by a moving object with velocity function v.

[1] `gvsu.edu/s/a9`

There is a more general way to think of Riemann sums, and that is to allow any choice of where the function is evaluated to determine the rectangle heights. Rather than saying we'll always choose left endpoints, or always choose midpoints, we simply say that a point x_{i+1}^* will be selected at random in the interval $[x_i, x_{i+1}]$ (so that $x_i \le x_{i+1}^* \le x_{i+1}$). The Riemann sum is then given by

$$f(x_1^*) \cdot \Delta x + f(x_2^*) \cdot \Delta x + \cdots + f(x_{i+1}^*) \cdot \Delta x + \cdots + f(x_n^*) \cdot \Delta x = \sum_{i=1}^{n} f(x_i^*)\Delta x.$$

In the interactive graphic noted earlier[2] and referenced in Figure 4.2.5, by unchecking the "relative" box at the top left, and instead checking "random," we can easily explore the effect of using random point locations in subintervals on a Riemann sum. In computational practice, we most often use L_n, R_n, or M_n, while the random Riemann sum is useful in theoretical discussions. In the following activity, we investigate several different Riemann sums for a particular velocity function.

Activity 4.2.3. Suppose that an object moving along a straight line path has its velocity in feet per second at time t in seconds given by $v(t) = \frac{2}{9}(t-3)^2 + 2$.

(a) Carefully sketch the region whose exact area will tell you the value of the distance the object traveled on the time interval $2 \le t \le 5$.

(b) Estimate the distance traveled on $[2, 5]$ by computing L_4, R_4, and M_4.

(c) Does averaging L_4 and R_4 result in the same value as M_4? If not, what do you think the average of L_4 and R_4 measures?

(d) For this question, think about an arbitrary function f, rather than the particular function v given above. If f is positive and increasing on $[a, b]$, will L_n over-estimate or under-estimate the exact area under f on $[a, b]$? Will R_n over- or under-estimate the exact area under f on $[a, b]$? Explain.

4.2.4 When the function is sometimes negative

For a Riemann sum such as

$$L_n = \sum_{i=0}^{n-1} f(x_i)\Delta x,$$

we can of course compute the sum even when f takes on negative values. We know that when f is positive on $[a, b]$, a Riemann sum estimates the area bounded between f and the horizontal axis over the interval.

[2] gvsu.edu/s/a9

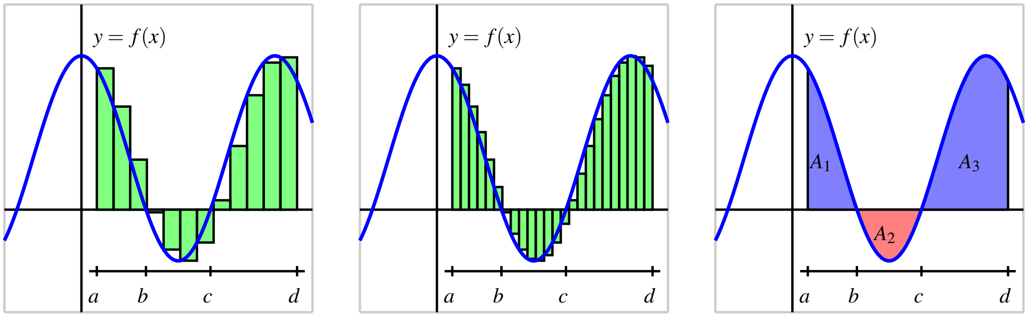


Figure 4.2.6 At left and center, two left Riemann sums for a function f that is sometimes negative; at right, the areas bounded by f on the interval $[a, d]$.

For the function pictured in the first graph of Figure 4.2.6, a left Riemann sum with 12 subintervals over $[a, d]$ is shown. The function is negative on the interval $b \leq x \leq c$, so at the four left endpoints that fall in $[b, c]$, the terms $f(x_i)\Delta x$ are negative. This means that those four terms in the Riemann sum produce an estimate of the *opposite* of the area bounded by $y = f(x)$ and the x-axis on $[b, c]$.

In the middle graph of Figure 4.2.6, we see that increasing the number of rectangles appears to improve the approximation of the area (or the opposite of the area) bounded by the curve.

In general, any Riemann sum of a continuous function f on an interval $[a, b]$ approximates the difference between the area that lies above the horizontal axis on $[a, b]$ and under f and the area that lies below the horizontal axis on $[a, b]$ and above f. In the notation of Figure 4.2.6, we may say that

$$L_{24} \approx A_1 - A_2 + A_3,$$

where L_{24} is the left Riemann sum using 24 subintervals shown in the middle graph. A_1 and A_3 are the areas of the regions where f is positive, and A_2 is the area where f is negative. We will call the quantity $A_1 - A_2 + A_3$ the *net signed area* bounded by f over the interval $[a, d]$, where by the phrase "signed area" we indicate that we are attaching a minus sign to the areas of regions that fall below the horizontal axis.

Finally, we recall that if the function f represents the velocity of a moving object, the sum of the areas bounded by the curve tells us the total distance traveled over the relevant time interval, while the net signed area bounded by the curve computes the object's change in position on the interval.

Activity 4.2.4. Suppose that an object moving along a straight line path has its velocity v (in feet per second) at time t (in seconds) given by

$$v(t) = \frac{1}{2}t^2 - 3t + \frac{7}{2}.$$

(a) Compute M_5, the middle Riemann sum, for v on the time interval $[1, 5]$. Be sure to clearly identify the value of Δt as well as the locations of $t_0, t_1, \cdots, t_5$. In addition, provide a careful sketch of the function and the corresponding rectangles that are being used in the sum.

(b) Building on your work in (a), estimate the total change in position of the object on the interval $[1, 5]$.

(c) Re-interpret your work in (a) and (b) to estimate the total distance traveled by the object on $[1, 5]$.

(d) Use appropriate computing technology such as this applet[3] to compute M_{10} and M_{20}. What exact value do you think the middle sum eventually approaches as n increases without bound? What does that number represent in the physical context of the overall problem?

4.2.5 Summary

- A Riemann sum is simply a sum of products of the form $f(x_i^*)\Delta x$ that estimates the area between a positive function and the horizontal axis over a given interval. If the function is sometimes negative on the interval, the Riemann sum estimates the difference between the areas that lie above the horizontal axis and those that lie below the axis.
- The three most common types of Riemann sums are left, right, and middle sums, but we can also work with a more general Riemann sum. The only difference among these sums is the location of the point at which the function is evaluated to determine the height of the rectangle whose area is being computed. For a left Riemann sum, we evaluate the function at the left endpoint of each subinterval, while for right and middle sums, we use right endpoints and midpoints, respectively.
- The left, right, and middle Riemann sums are denoted L_n, R_n, and M_n, with formulas

$$L_n = f(x_0)\Delta x + f(x_1)\Delta x + \cdots + f(x_{n-1})\Delta x = \sum_{i=0}^{n-1} f(x_i)\Delta x,$$

$$R_n = f(x_1)\Delta x + f(x_2)\Delta x + \cdots + f(x_n)\Delta x = \sum_{i=1}^{n} f(x_i)\Delta x,$$

$$M_n = f(\overline{x}_1)\Delta x + f(\overline{x}_2)\Delta x + \cdots + f(\overline{x}_n)\Delta x = \sum_{i=1}^{n} f(\overline{x}_i)\Delta x,$$

where $x_0 = a$, $x_i = a + i\Delta x$, and $x_n = b$, using $\Delta x = \frac{b-a}{n}$. For the midpoint sum, $\overline{x}_i = (x_{i-1} + x_i)/2$.

[3] gvsu.edu/s/a9

4.2.6 Exercises

1. In this problem, use the general expressions for left and right sums,

$$\text{left-hand sum} = f(t_0)\Delta t + f(t_1)\Delta t + \cdots + f(t_{n-1})\Delta t$$

and

$$\text{right-hand sum} = f(t_1)\Delta t + f(t_2)\Delta t + \cdots + f(t_n)\Delta t,$$

and the following table:

t	0	3	6	9	12
$f(t)$	30	29	27	24	20

A. If we use $n = 4$ subdivisions, fill in the values:

$\Delta t =$ ______

$t_0 =$ ______; $t_1 =$ ______; $t_2 =$ ______; $t_3 =$ ______; $t_4 =$ ______

$f(t_0) =$ ______; $f(t_1) =$ ______; $f(t_2) =$ ______; $f(t_3) =$ ______; $f(t_4) =$ ______

B. Find the left and right sums using $n = 4$

left sum = ________________________________

right sum = ________________________________

C. If we use $n = 2$ subdivisions, fill in the values:

$\Delta t =$ ______

$t_0 =$ ______; $t_1 =$ ______; $t_2 =$ ______

$f(t_0) =$ ______; $f(t_1) =$ ______; $f(t_2) =$ ______

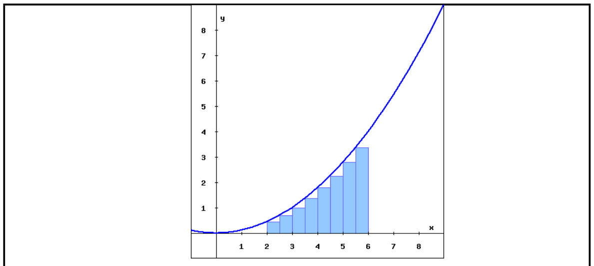
D. Find the left and right sums using $n = 2$

left sum = ________________________________

right sum = ________________________________

2. The rectangles in the graph below illustrate a left endpoint Riemann sum for $f(x) = \frac{x^2}{9}$ on the interval $[2, 6]$.

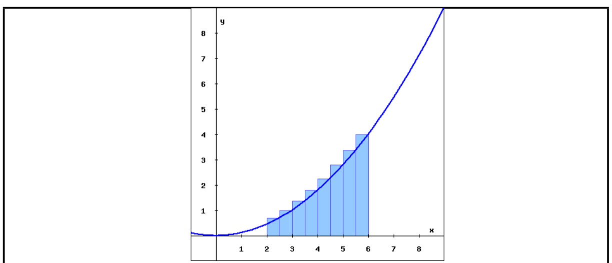
The value of this left endpoint Riemann sum is ______________________________, and this Riemann sum is (☐ [select an answer] ☐ an overestimate of ☐ equal to ☐ an underestimate of ☐ there is ambiguity) the area of the region enclosed by $y = f(x)$, the x-axis, and the vertical lines x = 2 and x = 6.



Left endpoint Riemann sum for $y = \frac{x^2}{9}$ on $[2, 6]$

The rectangles in the graph below illustrate a right endpoint Riemann sum for $f(x) = \dfrac{x^2}{9}$ on the interval $[2, 6]$.

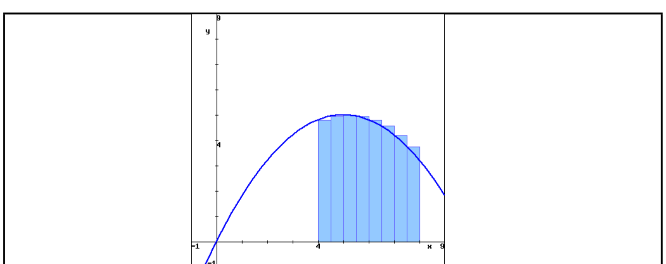
The value of this right endpoint Riemann sum is ______________________________, and this Riemann sum is (☐ [select an answer] ☐ an overestimate of ☐ equal to ☐ an underestimate of ☐ there is ambiguity) the area of the region enclosed by $y = f(x)$, the x-axis, and the vertical lines x = 2 and x = 6.



Right endpoint Riemann sum for $y = \frac{x^2}{9}$ on $[2, 6]$

3. The rectangles in the graph below illustrate a left endpoint Riemann sum for $f(x) = \dfrac{-x^2}{5} + 2x$ on the interval $[4, 8]$.

The value of this left endpoint Riemann sum is ______________________________, and this Riemann sum is (☐ [select an answer] ☐ an overestimate of ☐ equal to ☐ an underestimate of ☐ there is ambiguity) the area of the region enclosed by $y = f(x)$, the x-axis, and the vertical lines x = 4 and x = 8.

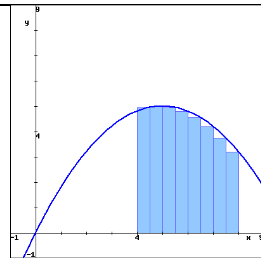


Left endpoint Riemann sum for $y = \frac{-x^2}{5} + 2x$ on $[4, 8]$

The rectangles in the graph below illustrate a right endpoint Riemann sum for $f(x) =$

$\dfrac{-x^2}{5} + 2x$ on the interval $[4, 8]$.

The value of this right endpoint Riemann sum is ________________, and this Riemann sum is (☐ [select an answer] ☐ an overestimate of ☐ equal to ☐ an underestimate of ☐ there is ambiguity) the area of the region enclosed by $y = f(x)$, the x-axis, and the vertical lines x = 4 and x = 8.



Right endpoint Riemann sum for $y = \frac{-x^2}{5} + 2x$ on $[4, 8]$

4. In this problem, use the general expressions for left and right sums,

$$\text{left-hand sum} = f(t_0)\Delta t + f(t_1)\Delta t + \cdots + f(t_{n-1})\Delta t$$

and

$$\text{right-hand sum} = f(t_1)\Delta t + f(t_2)\Delta t + \cdots + f(t_n)\Delta t,$$

and the following table:

t	0	4	8	12	16
$f(t)$	29	27	24	22	20

A. If we use $n = 4$ subdivisions, fill in the values:

$\Delta t =$ ______

$t_0 =$ ______; $t_1 =$ ______; $t_2 =$ ______; $t_3 =$ ______; $t_4 =$ ______

$f(t_0) =$ ______; $f(t_1) =$ ______; $f(t_2) =$ ______; $f(t_3) =$ ______; $f(t_4) =$ ______

B. Find the left and right sums using $n = 4$

left sum = ________________

right sum = ________________

C. If we use $n = 2$ subdivisions, fill in the values:

$\Delta t =$ ______

$t_0 =$ ______; $t_1 =$ ______; $t_2 =$ ______

$f(t_0) =$ ______; $f(t_1) =$ ______; $f(t_2) =$ ______

D. Find the left and right sums using $n = 2$

left sum = ________________

right sum = ________________

5. On a sketch of $y = \ln(x)$, represent the left Riemann sum with $n = 2$ approximating $\int_1^2 \ln(x)\, dx$. Write out the terms of the sum, but do not evaluate it:

Sum = ______________ + ______________

On another sketch, represent the right Riemann sum with $n = 2$ approximating $\int_1^2 \ln(x)\, dx$. Write out the terms of the sum, but do not evaluate it:

Sum = ______________ + ______________

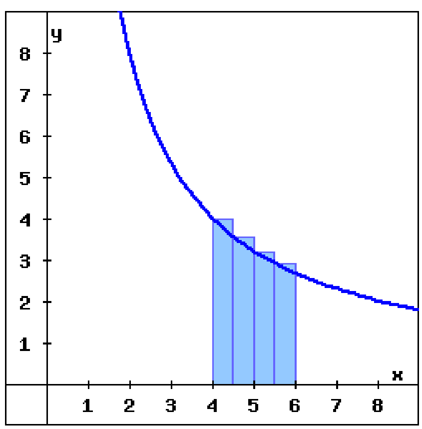
Which sum is an overestimate?

- ◯ the right Riemann sum
- ◯ the left Riemann sum
- ◯ neither sum

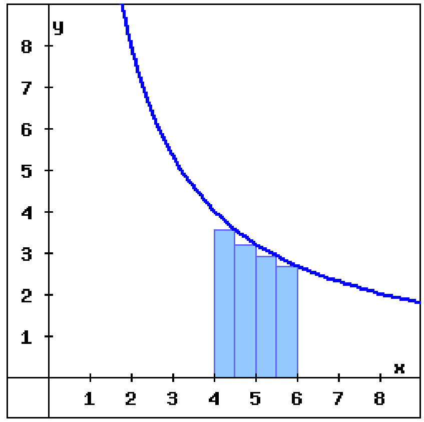
Which sum is an underestimate?

- ◯ the left Riemann sum
- ◯ the right Riemann sum
- ◯ neither sum

6. Suppose $f(x) = \dfrac{16}{x}$



(a) The rectangles in the graph on above illustrate a left endpoint Riemann sum for $f(x)$ on the interval $4 \le x \le 6$. The value of this left endpoint Riemann sum is ____________________, and it is an (☐ overestimate of ☐ equal to ☐ there is ambiguity ☐ underestimate of) the area of the region enclosed by $y = f(x)$, the x-axis, and the vertical lines $x = 4$ and $x = 6$.



(b) The rectangles in the graph above illustrate a right endpoint Riemann sum for $f(x)$ on the interval $4 \leq x \leq 6$. The value of this right endpoint Riemann sum is ____________________, and it is an (☐ there is ambiguity ☐ equal to ☐ overestimate of ☐ underestimate of) the area of the region enclosed by $y = f(x)$, the x-axis, and the vertical lines $x = 4$ and $x = 6$.

7. The rate (in liters per minute) at which water drains from a tank is recorded at half-minute intervals. Use the average of the left- and right-endpoint approximations to estimate the total amount of water drained during the first 3 min.

t min	0	0.5	1	1.5	2	2.5	3
rate in l/min	60	58	56	54	52	50	48

Answer: ________ liters.

8. Consider the function $f(x) = 3x + 4$.

a. Compute M_4 for $y = f(x)$ on the interval $[2, 5]$. Be sure to clearly identify the value of Δx, as well as the locations of $x_0, x_1, \ldots, x_4$. Include a careful sketch of the function and the corresponding rectangles being used in the sum.

b. Use a familiar geometric formula to determine the exact value of the area of the region bounded by $y = f(x)$ and the x-axis on $[2, 5]$.

c. Explain why the values you computed in (a) and (b) turn out to be the same. Will this be true if we use a number different than $n = 4$ and compute M_n? Will L_4 or R_4 have the same value as the exact area of the region found in (b)?

d. Describe the collection of functions g for which it will always be the case that M_n, regardless of the value of n, gives the exact net signed area bounded between the function g and the x-axis on the interval $[a, b]$.

9. Let S be the sum given by

$$S = ((1.4)^2 + 1) \cdot 0.4 + ((1.8)^2 + 1) \cdot 0.4 + ((2.2)^2 + 1) \cdot 0.4 + ((2.6)^2 + 1) \cdot 0.4 + ((3.0)^2 + 1) \cdot 0.4.$$

a. Assume that S is a right Riemann sum. For what function f and what interval

$[a, b]$ is S this function's Riemann sum? Why?

b. How does your answer to (a) change if S is a left Riemann sum? a middle Riemann sum?

c. Suppose that S really is a right Riemann sum. What is geometric quantity does S approximate?

d. Use sigma notation to write a new sum R that is the right Riemann sum for the same function, but that uses twice as many subintervals as S.

10. A car traveling along a straight road is braking and its velocity is measured at several different points in time, as given in the following table.

Table 4.2.7 Data for the braking car.

seconds, t	0	0.3	0.6	0.9	1.2	1.5	1.8
Velocity in ft/sec, $v(t)$	100	88	74	59	40	19	0

a. Plot the given data on a set of axes with time on the horizontal axis and the velocity on the vertical axis.

b. Estimate the total distance traveled during the car the time brakes using a middle Riemann sum with 3 subintervals.

c. Estimate the total distance traveled on $[0, 1.8]$ by computing L_6, R_6, and $\frac{1}{2}(L_6 + R_6)$.

d. Assuming that $v(t)$ is always decreasing on $[0, 1.8]$, what is the maximum possible distance the car traveled before it stopped? Why?

11. The rate at which pollution escapes a scrubbing process at a manufacturing plant increases over time as filters and other technologies become less effective. For this particular example, assume that the rate of pollution (in tons per week) is given by the function r that is pictured in Figure 4.2.8.

a. Use the graph to estimate the value of M_4 on the interval $[0, 4]$.

b. What is the meaning of M_4 in terms of the pollution discharged by the plant?

c. Suppose that $r(t) = 0.5e^{0.5t}$. Use this formula for r to compute L_5 on $[0, 4]$.

d. Determine an upper bound on the total amount of pollution that can escape the plant during the pictured four week time period that is accurate within an error of at most one ton of pollution.

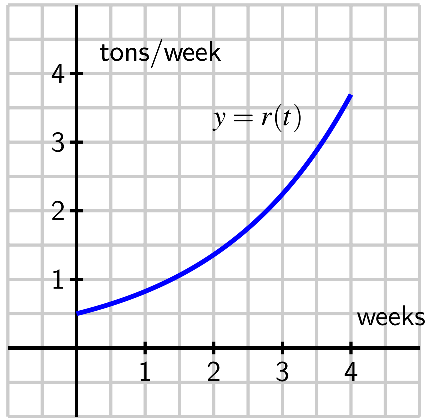


Figure 4.2.8 The rate, $r(t)$, of pollution in tons per week.

4.3 The definite integral

Motivating Questions

- How does increasing the number of subintervals affect the accuracy of the approximation generated by a Riemann sum?
- What is the definition of the definite integral of a function f over the interval $[a, b]$?
- What does the definite integral measure exactly, and what are some of the key properties of the definite integral?

4.3.1 Introduction

In Figure 4.3.1, we see evidence that increasing the number of rectangles in a Riemann sum improves the accuracy of the approximation of the net signed area bounded by the given function.

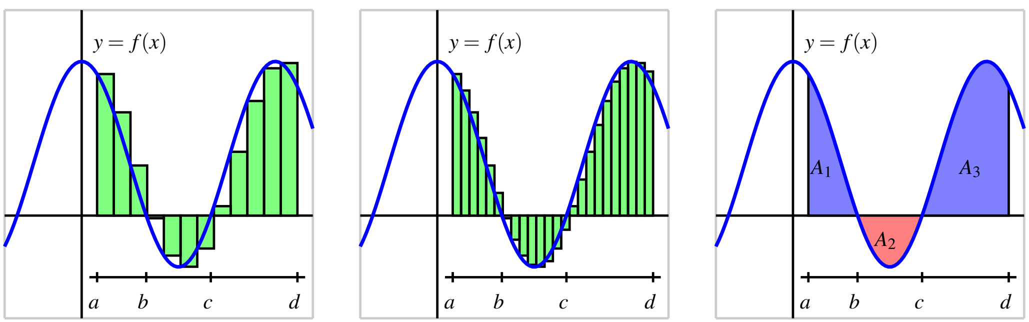
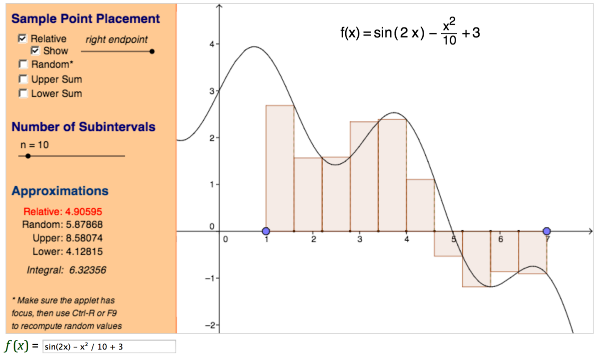


Figure 4.3.1 At left and center, two left Riemann sums for a function f that is sometimes negative; at right, the exact areas bounded by f on the interval $[a, d]$.

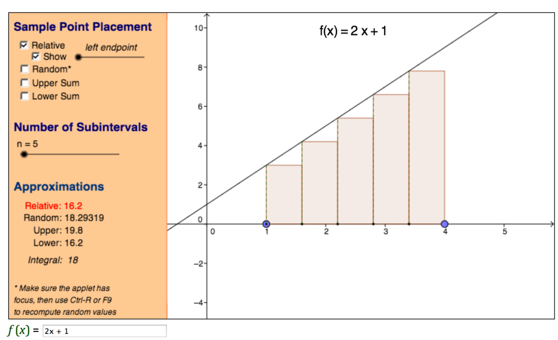
We therefore explore the natural idea of allowing the number of rectangles to increase without bound. In an effort to compute the exact net signed area we also consider the differences among left, right, and middle Riemann sums and the different results they generate as the value of n increases. We begin with functions that are exclusively positive on the interval under consideration.

Preview Activity 4.3.1. Consider this interactive graphic[1], in which you will initially see the situation shown in the following figure. The screenshot shows a right Riemann sum with 10 subintervals for the function $f(x) = \sin(2x) - \frac{x^2}{10} + 3$ on the interval $[1, 7]$. The value of the sum is $R_{10} = 4.90595$.



Note that the value of the chosen Riemann sum is displayed next to the word "relative," and that you can change the type of Riemann sum being computed by dragging the point on the slider bar below the phrase "sample point placement." Explore to see how you can change the window in which the function is viewed, as well as the function itself. You can set the minimum and maximum values of x by clicking and dragging on the blue points that set the endpoints; you can change the function by typing a new formula in the "f(x)" window at the bottom; and you can adjust the overall window by "panning and zooming" by using the Shift key and the scrolling feature of your mouse. More information on how to pan and zoom is available[2].

(a) Update the interactive (and view window, as needed) so that the function being considered is $f(x) = 2x + 1$ on $[1, 4]$. When you do so, you should see the following figure.



For this function on this interval, compute L_n, M_n, R_n for $n = 5$, $n = 25$, and $n = 100$. What appears to be the exact area bounded by $f(x) = 2x + 1$ and the x-axis on $[1, 4]$?

(b) Use basic geometry to determine the exact area bounded by $f(x) = 2x + 1$ and the x-axis on $[1, 4]$.

(c) Based on your work in (a) and (b), what do you observe occurs when we increase the number of subintervals used in the Riemann sum?

(d) Update the interactive to consider the function $f(x) = x^2 + 1$ on the interval $[1, 4]$ (note that you need to enter "x ^ 2 + 1" for the function formula), and hence compute L_n, M_n, R_n for $n = 5$, $n = 25$, and $n = 100$. What do you conjecture is the exact area bounded by $f(x) = x^2 + 1$ and the x-axis on $[1, 4]$?

(e) Why can we not compute the exact value of the area bounded by $f(x) = x^2 + 1$ and the x-axis on $[1, 4]$ using a formula like we did in (b)?

4.3.2 The definition of the definite integral

In Preview Activity 4.3.1, we saw that as the number of rectangles got larger and larger, the values of L_n, M_n, and R_n all grew closer and closer to the same value. It turns out that this occurs for any continuous function on an interval $[a, b]$, and also for a Riemann sum using any point x^*_{i+1} in the interval $[x_i, x_{i+1}]$. Thus, as we let $n \to \infty$, it doesn't really matter where we choose to evaluate the function within a given subinterval, because

$$\lim_{n\to\infty} L_n = \lim_{n\to\infty} R_n = \lim_{n\to\infty} M_n = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x.$$

That these limits always exist (and share the same value) when f is continuous[3] allows us to make the following definition.

Definition 4.3.2 The **definite integral** of a continuous function f on the interval $[a, b]$, denoted $\int_a^b f(x)\,dx$, is the real number given by

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x,$$

where $\Delta x = \frac{b-a}{n}$, $x_i = a + i\Delta x$ (for $i = 0, \ldots, n$), and x_i^* satisfies $x_{i-1} \le x_i^* \le x_i$ (for $i = 1, \ldots, n$). ◊

We call the symbol $\int$ the *integral sign*, the values a and b the *limits of integration*, and the function f the *integrand*. The process of determining the real number $\int_a^b f(x)\,dx$ is called *evaluating the definite integral*. While there are several different interpretations of the definite integral, for now the most important is that $\int_a^b f(x)\,dx$ measures the net signed area bounded by $y = f(x)$ and the x-axis on the interval $[a, b]$.

For example, if f is the function pictured in Figure 4.3.3, and A_1, A_2, and A_3 are the exact areas bounded by f and the x-axis on the respective intervals $[a, b]$, $[b, c]$, and $[c, d]$, then

$$\int_a^b f(x)\,dx = A_1, \quad \int_b^c f(x)\,dx = -A_2,$$

[1] `gvsu.edu/s/a9`

[2] `gvsu.edu/s/Fl`

[3] It turns out that a function need not be continuous in order to have a definite integral. For our purposes, we assume that the functions we consider are continuous on the interval(s) of interest. It is straightforward to see that any function that is piecewise continuous on an interval of interest will also have a well-defined definite integral.

$$\int_c^d f(x)\,dx = A_3,$$

$$\text{and } \int_a^d f(x)\,dx = A_1 - A_2 + A_3.$$

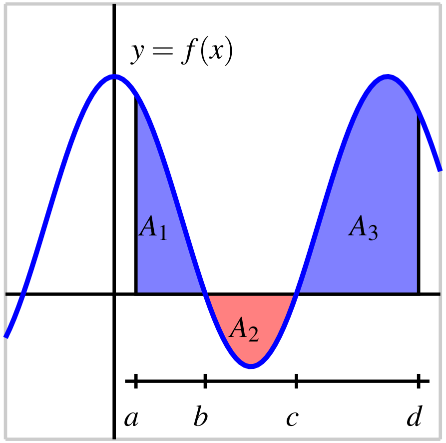


Figure 4.3.3 A continuous function f on the interval $[a, d]$.

We can also use definite integrals to express the change in position and the distance traveled by a moving object. If v is a velocity function on an interval $[a, b]$, then the change in position of the object, $s(b) - s(a)$, is given by

$$s(b) - s(a) = \int_a^b v(t)\,dt.$$

If the velocity function is nonnegative on $[a, b]$, then $\int_a^b v(t)\,dt$ tells us the distance the object traveled. If the velocity is sometimes negative on $[a, b]$, we can use definite integrals to find the areas bounded by the function on each interval where v does not change sign, and the sum of these areas will tell us the distance the object traveled.

To compute the value of a definite integral from the definition, we have to take the limit of a sum. While this is possible to do in select circumstances, it is also tedious and time-consuming, and does not offer much additional insight into the meaning or interpretation of the definite integral. Instead, in Section 4.4, we will learn the Fundamental Theorem of Calculus, which provides a shortcut for evaluating a large class of definite integrals. This will enable us to determine the exact net signed area bounded by a continuous function and the x-axis in many circumstances.

For now, our goal is to understand the meaning and properties of the definite integral, rather than to compute its value. To do this, we will rely on the net signed area interpretation of the definite integral. So we will use as examples curves that produce regions whose areas we can compute exactly through area formulas. We can thus compute the exact value of the corresponding integral.

For instance, if we wish to evaluate the definite integral $\int_1^4 (2x + 1)\,dx$, we observe that the region bounded by this function and the x-axis is the trapezoid shown in Figure 4.3.4. By the formula for the area of a trapezoid, $A = \frac{1}{2}(3+9)\cdot 3 = 18$, so

$$\int_1^4 (2x+1)\,dx = 18.$$

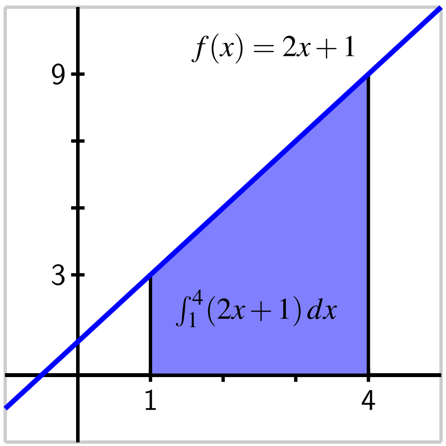


Figure 4.3.4 The area bounded by $f(x) = 2x+1$ and the x-axis on the interval $[1,4]$.

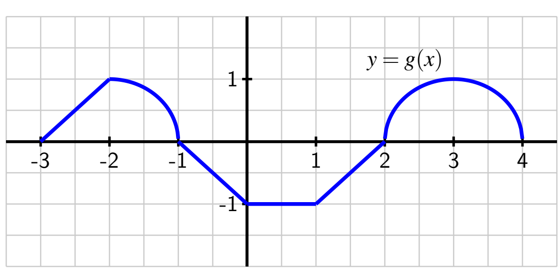
Activity 4.3.2. Use known geometric formulas and the net signed area interpretation of the definite integral to evaluate each of the definite integrals below.

(a) $\int_0^1 3x\,dx$

(b) $\int_{-1}^4 (2-2x)\,dx$

(c) $\int_{-1}^1 \sqrt{1-x^2}\,dx$

(d) $\int_{-3}^4 g(x)\,dx$, where g is the function pictured in the following figure. Assume that each portion of g is either part of a line or part of a circle.



4.3.3 Some properties of the definite integral

Regarding the definite integral of a function f over an interval $[a, b]$ as the net signed area bounded by f and the x-axis, we discover several standard properties of the definite integral. It is helpful to remember that the definite integral is defined in terms of Riemann sums, which consist of the areas of rectangles.

For any real number a and the definite integral $\int_a^a f(x)\,dx$ it is evident that no area is enclosed, because the interval begins and ends with the same point. Hence,

Integrating from $x = a$ to $x = a$.

If f is a continuous function and a is a real number, then $\int_a^a f(x)\,dx = 0$.

Next, we consider the result of subdividing the interval of integration. In Figure 4.3.5, we see that

$$\int_a^b f(x)\,dx = A_1, \quad \int_b^c f(x)\,dx = A_2,$$
$$\text{and} \int_a^c f(x)\,dx = A_1 + A_2,$$

which illustrates the following general rule.

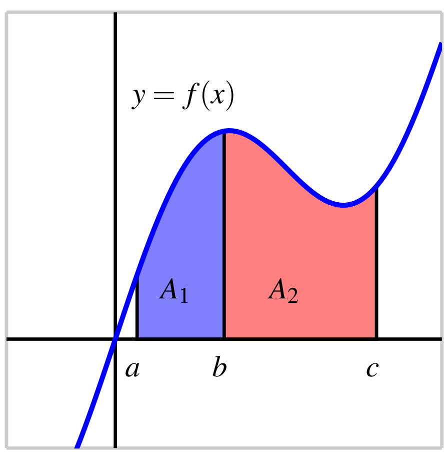


Figure 4.3.5 The area bounded by $y = f(x)$ on the interval $[a, c]$.

Subdividing the interval of integration.

If f is a continuous function and a, b, and c are real numbers, then

$$\int_a^c f(x)\,dx = \int_a^b f(x)\,dx + \int_b^c f(x)\,dx.$$

While this rule is easy to see if $a < b < c$, it in fact holds in general for any values of a, b, and c. Another property of the definite integral states that if we reverse the order of the limits of integration, we change the sign of the integral's value.

Reversing the limits of integration.

If f is a continuous function and a and b are real numbers, then

$$\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx.$$

This result makes sense because if we integrate from a to b, then in the defining Riemann sum we set $\Delta x = \frac{b-a}{n}$, while if we integrate from b to a, we have $\Delta x = \frac{a-b}{n} = -\frac{b-a}{n}$, and this is the only change in the sum used to define the integral.

There are two additional useful properties of the definite integral. When we worked with derivative rules in Chapter 2, we learned the Constant Multiple Rule and the Sum Rule. Recall that the Constant Multiple Rule says that if f is a differentiable function and k is a constant, then

$$\frac{d}{dx}[k f(x)] = k f'(x),$$

and the Sum Rule says that if f and g are differentiable functions, then

$$\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x).$$

These rules are useful because they allow to deal individually with the simplest parts of certain functions by taking advantage of addition and multiplying by a constant. In other words, the process of taking the derivative respects addition and multiplying by constants in the simplest possible way.

It turns out that similar rules hold for the definite integral. First, let's consider the functions pictured in Figure 4.3.6.

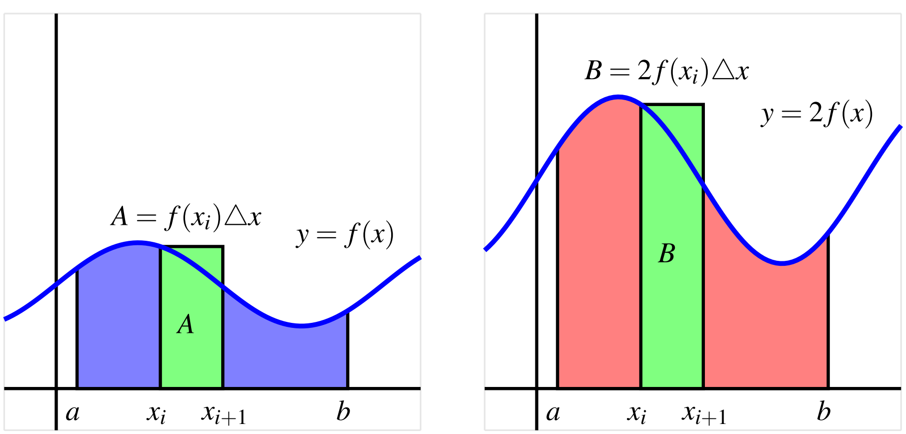


Figure 4.3.6 The areas bounded by $y = f(x)$ and $y = 2f(x)$ on $[a, b]$.

Because multiplying the function by 2 doubles its height at every x-value, we see that the height of each rectangle in a left Riemann sum is doubled, $f(x_i)$ for the original function, versus $2f(x_i)$ in the doubled function. For the areas A and B, it follows $B = 2A$. As this is true regardless of the value of n or the type of sum we use, we see that in the limit, the area of the red region bounded by $y = 2f(x)$ will be twice the area of the blue region bounded by $y = f(x)$. As there is nothing special about the value 2 compared to an arbitrary constant k, the following general principle holds.

The Constant Multiple Rule.

If f is a continuous function and k is any real number, then

$$\int_a^b k \cdot f(x)\,dx = k \int_a^b f(x)\,dx.$$

We see a similar situation with the sum of two functions f and g.

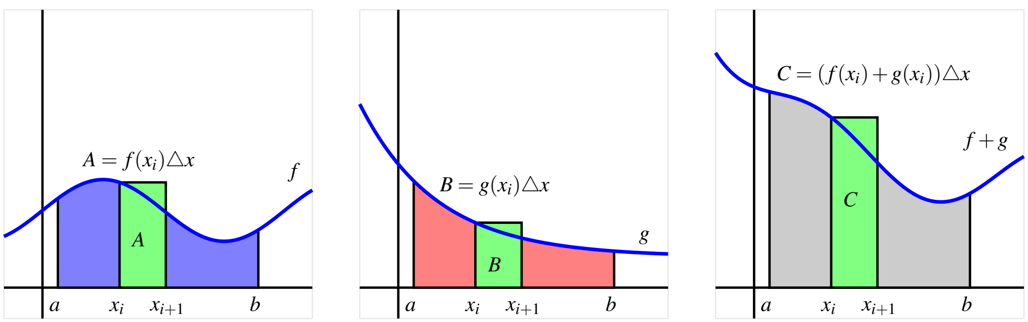


Figure 4.3.7 The areas bounded by $y = f(x)$ and $y = g(x)$ on $[a, b]$, as well as the area bounded by $y = f(x) + g(x)$.

If we take the sum of two functions f and g at every point in the interval, the height of the function $f + g$ is given by $(f + g)(x_i) = f(x_i) + g(x_i)$. Hence, for the pictured rectangles with areas A, B, and C, it follows that $C = A + B$. Because this will occur for every such rectangle, in the limit the area of the gray region will be the sum of the areas of the blue and red regions. In terms of definite integrals, we have the following general rule.

The Sum Rule.

If f and g are continuous functions, then

$$\int_a^b [f(x) + g(x)]\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx.$$

The Constant Multiple and Sum Rules can be combined to say that for any continuous functions f and g and any constants c and k,

$$\int_a^b [cf(x) \pm kg(x)]\,dx = c\int_a^b f(x)\,dx \pm k\int_a^b g(x)\,dx.$$

Activity 4.3.3. Suppose that the following information is known about the functions f, g, x^2, and x^3:

- $\int_0^2 f(x)\,dx = -3$; $\int_2^5 f(x)\,dx = 2$
- $\int_0^2 g(x)\,dx = 4$; $\int_2^5 g(x)\,dx = -1$
- $\int_0^2 x^2\,dx = \frac{8}{3}$; $\int_2^5 x^2\,dx = \frac{117}{3}$
- $\int_0^2 x^3\,dx = 4$; $\int_2^5 x^3\,dx = \frac{609}{4}$

Use the provided information and the rules discussed in the preceding section to evaluate each of the following definite integrals.

(a) $\int_5^2 f(x)\,dx$

(b) $\int_0^5 g(x)\,dx$

(c) $\int_0^5 (f(x) + g(x))\,dx$

(d) $\int_2^5 (3x^2 - 4x^3)\,dx$

(e) $\int_5^0 (2x^3 - 7g(x))\,dx$

4.3.4 How the definite integral is connected to a function's average value

One of the most valuable applications of the definite integral is that it provides a way to discuss the average value of a function, even for a function that takes on infinitely many values. Recall that if we wish to take the average of n numbers $y_1, y_2, \ldots, y_n$, we compute

$$AVG = \frac{y_1 + y_2 + \cdots + y_n}{n}.$$

Since integrals arise from Riemann sums in which we add n values of a function, it should not be surprising that evaluating an integral is similar to averaging the output values of a function. Consider, for instance, the right Riemann sum R_n of a function f, which is given by

$$R_n = f(x_1)\Delta x + f(x_2)\Delta x + \cdots + f(x_n)\Delta x = (f(x_1) + f(x_2) + \cdots + f(x_n))\Delta x.$$

Since $\Delta x = \frac{b-a}{n}$, we can thus write

$$R_n = (f(x_1) + f(x_2) + \cdots + f(x_n)) \cdot \frac{b-a}{n}$$

$$= (b-a)\frac{f(x_1) + f(x_2) + \cdots + f(x_n)}{n}. \tag{4.3.1}$$

We see that the right Riemann sum with n subintervals is just the length of the interval $(b-a)$ times the average of the n function values found at the right endpoints. And just as with our efforts to compute area, the larger the value of n we use, the more accurate our average will be. Indeed, we will define the average value of f on $[a, b]$ to be

$$f_{\text{AVG}[a,b]} = \lim_{n\to\infty} \frac{f(x_1) + f(x_2) + \cdots + f(x_n)}{n}.$$

But we also know that for any continuous function f on $[a, b]$, taking the limit of a Riemann sum leads precisely to the definite integral. That is, $\lim_{n\to\infty} R_n = \int_a^b f(x)\,dx$, and thus taking the limit as $n \to \infty$ in Equation (4.3.1), we have that

$$\int_a^b f(x)\,dx = (b-a)\cdot f_{\text{AVG}[a,b]}. \tag{4.3.2}$$

Solving Equation (4.3.2) for $f_{\text{AVG}[a,b]}$, we have the following general principle.

The average value of a function.

If f is a continuous function on $[a, b]$, then its average value on $[a, b]$ is given by the formula

$$f_{\text{AVG}[a,b]} = \frac{1}{b-a}\cdot \int_a^b f(x)\,dx.$$

Equation (4.3.2) tells us another way to interpret the definite integral: the definite integral of a function f from a to b is the length of the interval $(b-a)$ times the average value of the function on the interval. In addition, when the function f is nonnegative on $[a, b]$, Equation (4.3.2) has a natural visual interpretation.

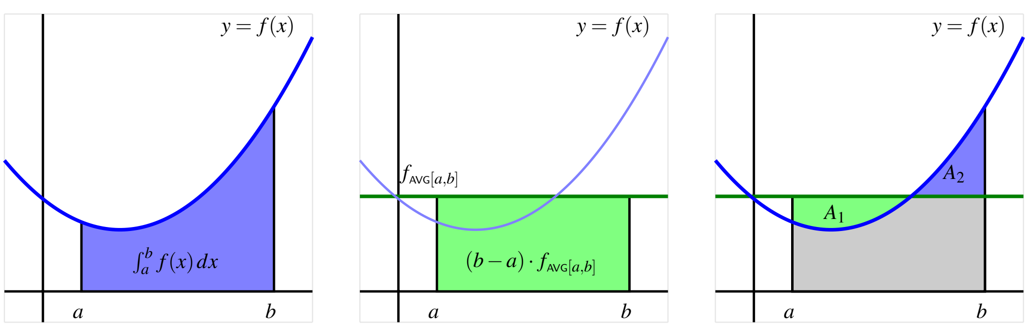


Figure 4.3.8 A function $y = f(x)$, the area it bounds, and its average value on $[a, b]$.

Consider Figure 4.3.8, where we see at left the shaded region whose area is $\int_a^b f(x)\,dx$, at center the shaded rectangle whose dimensions are $(b-a)$ by $f_{\text{AVG}[a,b]}$, and at right these two figures superimposed. Note that in dark green we show the horizontal line $y = f_{\text{AVG}[a,b]}$.

Thus, the area of the green rectangle is given by $(b-a)\cdot f_{\text{AVG}[a,b]}$, which is precisely the value of $\int_a^b f(x)\,dx$. The area of the blue region in the left figure is the same as the area of the green rectangle in the center figure. We can also observe that the areas A_1 and A_2 in the rightmost figure appear to be equal. Thus, knowing the average value of a function enables us to construct a rectangle whose area is the same as the value of the definite integral of the function on the interval. This interactive graphic[4] provides an opportunity to explore how the average value of the function changes as the interval changes, through an image similar to that found in Figure 4.3.8.

Activity 4.3.4. Suppose that $v(t) = \sqrt{4-(t-2)^2}$ tells us the instantaneous velocity of a moving object on the interval $0 \le t \le 4$, where t is measured in minutes and v is measured in meters per minute.

(a) Sketch an accurate graph of $y = v(t)$. What kind of curve is $y = \sqrt{4-(t-2)^2}$?

(b) Evaluate $\int_0^4 v(t)\,dt$ exactly.

(c) In terms of the physical problem of the moving object with velocity $v(t)$, what is the meaning of $\int_0^4 v(t)\,dt$? Include units on your answer.

(d) Determine the exact average value of $v(t)$ on $[0,4]$. Include units on your answer.

(e) Sketch a rectangle whose base is the line segment from $t = 0$ to $t = 4$ on the t-axis such that the rectangle's area is equal to the value of $\int_0^4 v(t)\,dt$. What is the rectangle's exact height?

(f) How can you use the average value you found in (d) to compute the total distance traveled by the moving object over $[0,4]$?

4.3.5 Summary

- Any Riemann sum of a continuous function f on an interval $[a,b]$ provides an estimate of the net signed area bounded by the function and the horizontal axis on the interval. Increasing the number of subintervals in the Riemann sum improves the accuracy of this estimate, and letting the number of subintervals increase without bound results in the values of the corresponding Riemann sums approaching the exact value of the enclosed net signed area.
- When we take the limit of Riemann sums, we arrive at what we call the definite integral of f over the interval $[a,b]$. In particular, the symbol $\int_a^b f(x)\,dx$ denotes the definite integral of f over $[a,b]$, and this quantity is defined by the equation

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x,$$

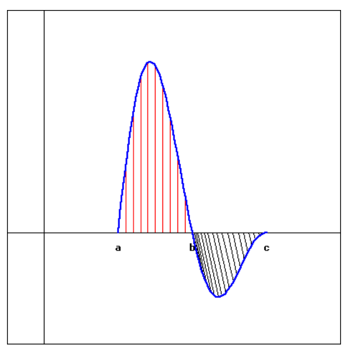
[4]gvsu.edu/s/az

where $\Delta x = \frac{b-a}{n}$, $x_i = a + i\Delta x$ (for $i = 0, \ldots, n$), and x_i^* satisfies $x_{i-1} \leq x_i^* \leq x_i$ (for $i = 1, \ldots, n$).

- The definite integral $\int_a^b f(x)\,dx$ measures the exact net signed area bounded by f and the horizontal axis on $[a, b]$; in addition, the value of the definite integral is related to what we call the average value of the function on $[a, b]$: $f_{\text{AVG}[a,b]} = \frac{1}{b-a} \cdot \int_a^b f(x)\,dx$. In the setting where we consider the integral of a velocity function v, $\int_a^b v(t)\,dt$ measures the exact change in position of the moving object on $[a, b]$; when v is nonnegative, $\int_a^b v(t)\,dt$ is the object's distance traveled on $[a, b]$.
- The definite integral is a sophisticated sum, and thus has some of the same natural properties that finite sums have. Perhaps most important of these is how the definite integral respects sums and constant multiples of functions, which can be summarized by the rule
$$\int_a^b [c f(x) \pm k g(x)]\,dx = c \int_a^b f(x)\,dx \pm k \int_a^b g(x)\,dx$$
where f and g are continuous functions on $[a, b]$ and c and k are arbitrary constants.

4.3.6 Exercises

1. Use the following figure, which shows a graph of $f(x)$ to find each of the indicated integrals.



(Click on the graph for a larger version.)

Note that the first area (with vertical, red shading) is 9 and the second (with oblique, black shading) is 3.

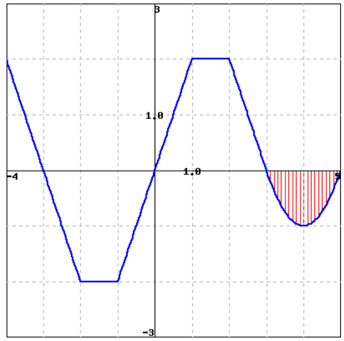
A. $\int_a^b f(x)dx =$ ______________

B. $\int_b^c f(x)dx =$ ______________

C. $\int_a^c f(x)dx =$ ______________

D. $\int_a^c |f(x)|dx =$ ______________

2. Use the graph of $f(x)$ shown below to find the following integrals.



(Click on the graph for a larger version.)

A. $\int_{-3}^{0} f(x)dx =$ ______________

B. If the vertical red shaded area in the graph has area A, estimate: $\int_{-3}^{5} f(x)dx =$ ______________

(Your estimate may be written in terms of A.)

3. Find the average value of $f(x) = 4x + 4$ over $[5, 10]$

average value = ________________________________

4. Coal gas is produced at a gasworks. Pollutants in the gas are removed by scrubbers, which become less and less efficient as time goes on. The following measurements, made at the start of each month, show the rate at which pollutants are escaping (in tons/month) in the gas:

Time (months)	0	1	2	3	4	5	6
Rate	4	5	6	9	13	20	29

A. Make an overestimate and an underestimate of the total quantity of pollutants that escape during the first month.

overestimate = ________________________________ tons

underestimate = ________________________________ tons

B. Make an overestimate and an underestimate of the total quantity of pollutants that escape for the whole six months for which we have data.

overestimate = ________________________________

underestimate = ________________________________

5. The following table gives the the approximate amount of emissions, E, of nitrogen oxides in millions of metric tons per year in the US. Let t be the number of years since 1940 and $E = f(t)$.

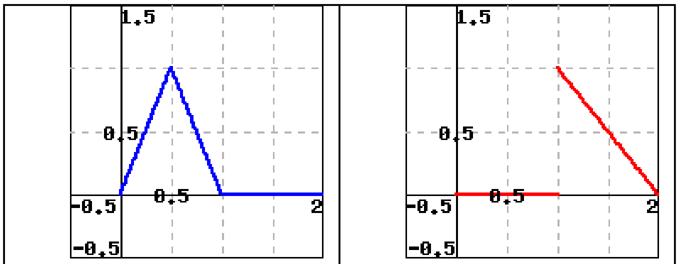
	1940	1950	1960	1970	1980	1990
t	0	10	20	30	40	50
E	6.9	9.6	13.2	18.4	20.8	19.4

Estimate the integral: $\int_0^{50} f(t)dt \approx$ ______________________________

Be sure that you know what the units of your answer are, and what its meaning is!

(Original data from the Statistical Abstract of the US, 1992)

6. The figure below to the left is a graph of $f(x)$, and below to the right is $g(x)$.



$f(x)$	$g(x)$

(a)

What is the average value of $f(x)$ on $0 \leq x \leq 2$?

avg value = ________________

(b)

What is the average value of $g(x)$ on $0 \leq x \leq 2$?

avg value = ________________

(c)

What is the average value of $f(x) \cdot g(x)$ on $0 \leq x \leq 2$?

avg value = ________________

(d)

Is the following statement true?

$$\text{Average}(f) \cdot \text{Average(g)} = \text{Average}(f \cdot g)$$

○ Yes

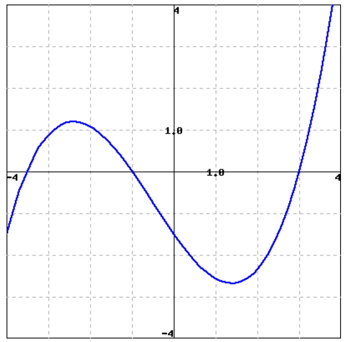
○ No

7. Suppose $\displaystyle\int_4^{8.5} f(x)dx = 10, \int_4^{5.5} f(x)dx = 5, \int_7^{8.5} f(x)dx = 9.$

$\displaystyle\int_{5.5}^{7} f(x)dx =$ __________

$\displaystyle\int_{7}^{5.5} (10f(x) - 5)dx =$ __________

8. Use the figure below, which shows the graph of $y = f(x)$, to answer the following questions.



(Click on the graph to get a larger version.)

A. Estimate the integral: $\int_{-3}^{3} f(x)\,dx \approx$ ______________________________

(You will certainly want to use an enlarged version of the graph to obtain your estimate.)

B. Which of the following average values of f is larger?

◯ Between $x = 0$ and $x = 3$

◯ Between $x = -3$ and $x = 3$

9. The velocity of an object moving along an axis is given by the piecewise linear function v that is pictured in Figure 4.3.9. Assume that the object is moving to the right when its velocity is positive, and moving to the left when its velocity is negative. Assume that the given velocity function is valid for $t = 0$ to $t = 4$.

a. Write an expression involving definite integrals whose value is the total change in position of the object on the interval $[0, 4]$.

b. Use the provided graph of v to determine the value of the total change in position on $[0, 4]$.

c. Write an expression involving definite integrals whose value is the total distance traveled by the object on $[0, 4]$. What is the exact value of the total distance traveled on $[0, 4]$?

d. What is the object's exact average velocity on $[0, 4]$?

e. Find an algebraic formula for the object's position function on $[0, 1.5]$ that satisfies $s(0) = 0$.

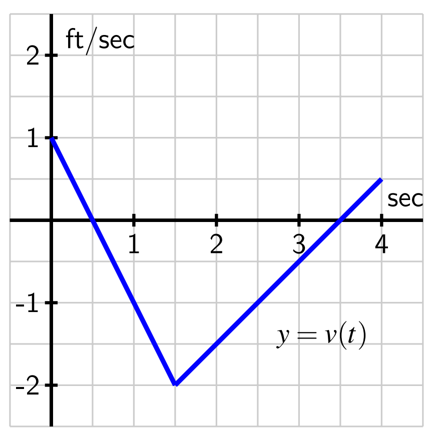


Figure 4.3.9 The velocity function of a moving object.

10. Suppose that the velocity of a moving object is given by $v(t) = t(t-1)(t-3)$, measured in feet per second, and that this function is valid for $0 \le t \le 4$.

a. Write an expression involving definite integrals whose value is the total change in position of the object on the interval $[0, 4]$.

b. Use appropriate technology (such as an applet[5]) to compute Riemann sums to estimate the object's total change in position on $[0, 4]$. Work to ensure that your estimate is accurate to two decimal places, and explain how you know this to be the case.

c. Write an expression involving definite integrals whose value is the total distance traveled by the object on $[0, 4]$.

d. Use appropriate technology to compute Riemann sums to estimate the object's total distance travelled on $[0, 4]$. Work to ensure that your estimate is accurate to two decimal places, and explain how you know this to be the case.

e. What is the object's average velocity on $[0, 4]$, accurate to two decimal places?

11. Consider the graphs of two functions f and g that are provided in Figure 4.3.10. Each piece of f and g is either part of a straight line or part of a circle.

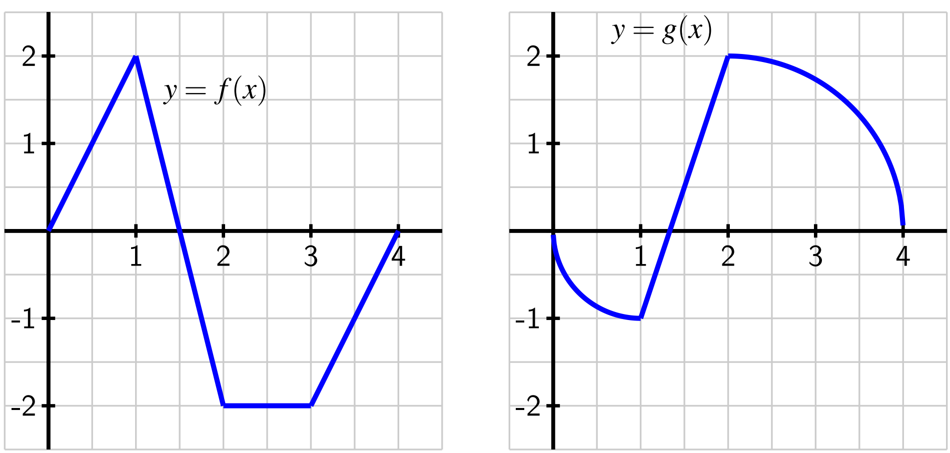


Figure 4.3.10 Two functions f and g.

a. Determine the exact value of $\int_0^1 [f(x) + g(x)]\, dx$.

b. Determine the exact value of $\int_1^4 [2f(x) - 3g(x)]\, dx$.

c. Find the exact average value of $h(x) = g(x) - f(x)$ on $[0, 4]$.

[5] `gvsu.edu/s/a9`

d. For what constant c does the following equation hold?

$$\int_0^4 c\,dx = \int_0^4 [f(x) + g(x)]\,dx$$

12. Let $f(x) = 3 - x^2$ and $g(x) = 2x^2$.

a. On the interval $[-1, 1]$, sketch a labeled graph of $y = f(x)$ and write a definite integral whose value is the exact area bounded by $y = f(x)$ on $[-1, 1]$.

b. On the interval $[-1, 1]$, sketch a labeled graph of $y = g(x)$ and write a definite integral whose value is the exact area bounded by $y = g(x)$ on $[-1, 1]$.

c. Write an expression involving a difference of definite integrals whose value is the exact area that lies between $y = f(x)$ and $y = g(x)$ on $[-1, 1]$.

d. Explain why your expression in (c) has the same value as the single integral $\int_{-1}^{1} [f(x) - g(x)]\,dx$.

e. Explain why, in general, if $p(x) \geq q(x)$ for all x in $[a, b]$, the exact area between $y = p(x)$ and $y = q(x)$ is given by

$$\int_a^b [p(x) - q(x)]\,dx.$$

4.4 The Fundamental Theorem of Calculus

Motivating Questions

- How can we find the exact value of a definite integral without taking the limit of a Riemann sum?
- What is the statement of the Fundamental Theorem of Calculus, and how do antiderivatives of functions play a key role in applying the theorem?
- What is the meaning of the definite integral of a rate of change in contexts other than when the rate of change represents velocity?

4.4.1 Introduction

Much of our work in Chapter 4 has been motivated by the velocity-distance problem: if we know the instantaneous velocity function, $v(t)$, for a moving object on a given time interval $[a, b]$, can we determine the distance it traveled on $[a, b]$? If the velocity function is nonnegative on $[a, b]$, the area bounded by $y = v(t)$ and the t-axis on $[a, b]$ is equal to the distance traveled. This area is also the value of the definite integral $\int_a^b v(t)\,dt$. If the velocity is sometimes negative, the total area bounded by the velocity function still tells us distance traveled, while the net signed area tells us the object's change in position.

For instance, for the velocity function in Figure 4.4.1, the total distance D traveled by the moving object on $[a, b]$ is

$$D = A_1 + A_2 + A_3,$$

and the total change in the object's position is

$$s(b) - s(a) = A_1 - A_2 + A_3.$$

The areas A_1, A_2, and A_3 are each given by definite integrals, which may be computed by limits of Riemann sums (and in special circumstances by geometric formulas).

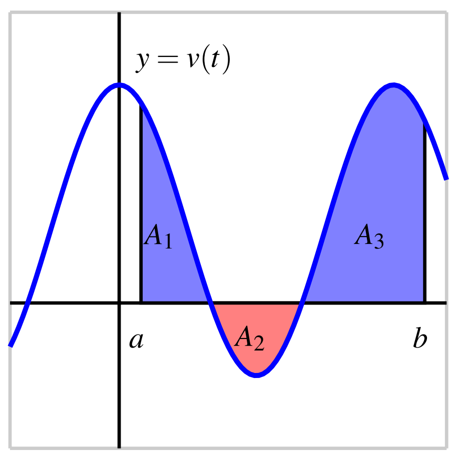


Figure 4.4.1 A velocity function that is sometimes negative.

We now turn our attention to an alternate approach.

Preview Activity 4.4.1. A student with a third floor dormitory window 32 feet off the ground tosses a water balloon straight up in the air with an initial velocity of 16 feet per second. It turns out that the instantaneous velocity of the water balloon is given by $v(t) = -32t + 16$, where v is measured in feet per second and t is measured in seconds.

(a) Let $s(t)$ represent the height of the water balloon above ground at time t, and note that s is an antiderivative of v. That is, v is the derivative of s: $s'(t) = v(t)$. Find a formula for $s(t)$ that satisfies the initial condition that the balloon is tossed from 32 feet above ground. In other words, make your formula for s satisfy $s(0) = 32$.

(b) When does the water balloon reach its maximum height? When does it land?

(c) Compute $s(\frac{1}{2}) - s(0)$, $s(2) - s(\frac{1}{2})$, and $s(2) - s(0)$. What do these represent?

(d) What is the total vertical distance traveled by the water balloon from the time it is tossed until the time it lands?

(e) Sketch a graph of the velocity function $y = v(t)$ on the time interval $[0, 2]$. What is the total net signed area bounded by $y = v(t)$ and the t-axis on $[0, 2]$? Answer this question in two ways: first by using your work above, and then by using a familiar geometric formula to compute areas of certain relevant regions.

4.4.2 The Fundamental Theorem of Calculus

Suppose we know the position function $s(t)$ and the velocity function $v(t)$ of an object moving in a straight line, and for the moment let us assume that $v(t)$ is positive on $[a, b]$.

Then, as shown in Figure 4.4.2, we know two different ways to compute the distance, D, the object travels: one is that $D = s(b) - s(a)$, the object's change in position. The other is the area under the velocity curve, which is given by the definite integral, so $D = \int_a^b v(t)\,dt$. Since both of these expressions tell us the distance traveled, it follows that they are equal, so

$$s(b) - s(a) = \int_a^b v(t)\,dt. \tag{4.4.1}$$

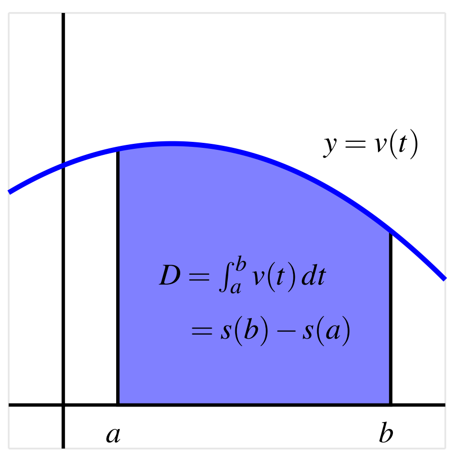


Figure 4.4.2 Finding distance traveled when we know a velocity function v.

Equation (4.4.1) holds even when velocity is sometimes negative, because $s(b) - s(a)$, the object's change in position, is also measured by the net signed area on $[a, b]$, which is given by $\int_a^b v(t)\,dt$.

Perhaps the most powerful fact Equation (4.4.1) reveals is that we can compute the integral's value if we can find a formula for s. Remember, s and v are related by the fact that v is the derivative of s, or equivalently that s is an antiderivative of v.

Example 4.4.3 Determine the exact distance traveled on $[1, 5]$ by an object with velocity function $v(t) = 3t^2 + 40$ feet per second.

Solution. Since the velocity function is positive on the chosen interval, the distance traveled on $[1, 5]$ is given by

$$D = \int_1^5 v(t)\,dt = \int_1^5 (3t^2 + 40)\,dt = s(5) - s(1),$$

where s is an antiderivative of v. Now, the derivative of t^3 is $3t^2$ and the derivative of $40t$ is 40, so it follows that $s(t) = t^3 + 40t$ is an antiderivative of v. Therefore,

$$\begin{aligned} D &= \int_1^5 3t^2 + 40\,dt = s(5) - s(1) \\ &= (5^3 + 40 \cdot 5) - (1^3 + 40 \cdot 1) = 284 \text{ feet.} \end{aligned}$$

◊

Note the key lesson of Example 4.4.3: to find the distance traveled, we need to compute the area under a curve, which is given by the definite integral. But to evaluate the integral, we can find an antiderivative, s, of the velocity function, and then compute the total change in s on the interval. In particular, we can evaluate the integral *without* computing the limit of a Riemann sum.

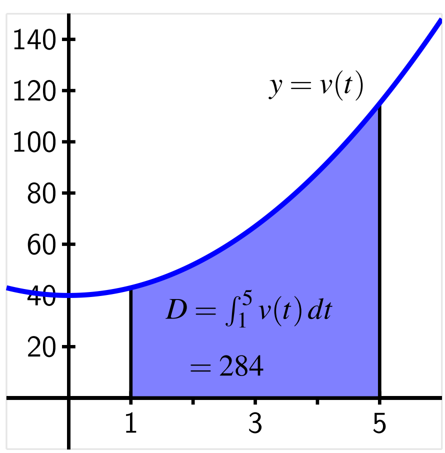


Figure 4.4.4 The exact area of the region enclosed by $v(t) = 3t^2 + 40$ on $[1, 5]$.

It will be convenient to have a shorthand symbol for a function's antiderivative. For a continuous function f, we will often denote an antiderivative of f by F, so that $F'(x) = f(x)$ for all relevant x. Using the notation V in place of s (so that V is an antiderivative of v) in Equation (4.4.1), we can write

$$V(b) - V(a) = \int_a^b v(t)\,dt. \tag{4.4.2}$$

Now, to evaluate the definite integral $\int_a^b f(x)\,dx$ for an arbitrary continuous function f, we could certainly think of f as representing the velocity of some moving object, and x as the variable that represents time. But Equations (4.4.1) and (4.4.2) hold for any continuous velocity function, even when v is sometimes negative. So Equation (4.4.2) offers a shortcut route to evaluating any definite integral, provided that we can find an antiderivative of the integrand. The Fundamental Theorem of Calculus (FTC) summarizes these observations.

The Fundamental Theorem of Calculus (FTC).

If f is a continuous function on $[a, b]$, and F is any antiderivative of f, then $\int_a^b f(x)\,dx = F(b) - F(a)$.

A common alternate notation for $F(b) - F(a)$ is

$$F(b) - F(a) = F(x)\Big|_a^b,$$

where we read the righthand side as "the function F evaluated from a to b." In this notation, the FTC says that

$$\int_a^b f(x)\,dx = F(x)\Big|_a^b.$$

The FTC opens the door to evaluating a wide range of integrals if we can find an antiderivative F for the integrand f. For instance since $\frac{d}{dx}[\frac{1}{3}x^3] = x^2$, the FTC tells us that

$$\begin{aligned}\int_0^1 x^2\,dx &= \frac{1}{3}x^3\Big|_0^1\\ &= \frac{1}{3}(1)^3 - \frac{1}{3}(0)^3\\ &= \frac{1}{3}.\end{aligned}$$

But finding an antiderivative can be far from simple; it is often difficult or even impossible. While we can differentiate just about any function, even some relatively simple functions don't have an elementary antiderivative. A significant portion of integral calculus (which is the main focus of second semester college calculus) is devoted to the problem of finding antiderivatives.

Activity 4.4.2. Use the Fundamental Theorem of Calculus to evaluate each of the following integrals exactly. For each, sketch a graph of the integrand on the relevant interval and write one sentence that explains the meaning of the value of the integral in terms of the (net signed) area bounded by the curve.

(a) $\int_{-1}^{4}(2-2x)\,dx$

(b) $\int_{0}^{\frac{\pi}{2}} \sin(x)\,dx$

(c) $\int_{0}^{1} e^{x}\,dx$

(d) $\int_{-1}^{1} x^{5}\,dx$

(e) $\int_{0}^{2}(3x^{3}-2x^{2}-e^{x})\,dx$

4.4.3 Basic antiderivatives

The general problem of finding an antiderivative is difficult. In part, this is due to the fact that we are trying to undo the process of differentiating, and the undoing is much more difficult than the doing. For example, while it is evident that an antiderivative of $f(x) = \sin(x)$ is $F(x) = -\cos(x)$ and that an antiderivative of $g(x) = x^2$ is $G(x) = \frac{1}{3}x^3$, combinations of f and g can be far more complicated. Consider the functions

$$5\sin(x) - 4x^2,\ x^2\sin(x),\ \frac{\sin(x)}{x^2},\ \text{and}\ \sin(x^2).$$

What is involved in trying to find an antiderivative for each? From our experience with derivative rules, we know that derivatives of sums and constant multiples of basic functions are simple to execute, but derivatives involving products, quotients, and composites of familiar functions are more complicated. Therefore, it stands to reason that antidifferentiating products, quotients, and composites of basic functions may be even more challenging. We defer our study of all but the most elementary antiderivatives to later in the text.

We do note that whenever we know the derivative of a function, we have a *function-derivative pair*, so we also know the antiderivative of a function. For instance, since we know that

$$\frac{d}{dx}[-\cos(x)] = \sin(x),$$

we also know that $F(x) = -\cos(x)$ is an antiderivative of $f(x) = \sin(x)$. F and f together form a function-derivative pair. Clearly, every basic derivative rule leads us to such a pair, and thus to a known antiderivative.

In Activity 4.4.3, we will construct a list of the basic antiderivatives we know at this time. Those rules will help us antidifferentiate sums and constant multiples of basic functions. For example, since $-\cos(x)$ is an antiderivative of $\sin(x)$ and $\frac{1}{3}x^3$ is an antiderivative of x^2, it follows that

$$F(x) = -5\cos(x) - \frac{4}{3}x^3$$

is an antiderivative of $f(x) = 5\sin(x) - 4x^2$, by the sum and constant multiple rules for differentiation.

Finally, before proceeding to build a list of common functions whose antiderivatives we know, we recall that each function has more than one antiderivative. Because the derivative of any constant is zero, we may add a constant of our choice to any antiderivative. For instance, we know that $G(x) = \frac{1}{3}x^3$ is an antiderivative of $g(x) = x^2$. But we could also have chosen $G(x) = \frac{1}{3}x^3 + 7$, since in this case as well, $G'(x) = x^2$. If $g(x) = x^2$, we say that the *general antiderivative* of g is

$$G(x) = \frac{1}{3}x^3 + C,$$

where C represents an arbitrary real number constant. Provided that g is a continuous function, including $+C$ in the formula for its antiderivative G results in the most general possible antiderivative.

Our current interest in antiderivatives is so that we can evaluate definite integrals by the Fundamental Theorem of Calculus. For that task, the constant C is irrelevant, and we usually omit it. To see why, consider the definite integral

$$\int_0^1 x^2\,dx.$$

For the integrand $g(x) = x^2$, suppose we find and use the general antiderivative $G(x) = \frac{1}{3}x^3 + C$. Then, by the FTC,

$$\begin{aligned}
\int_0^1 x^2\,dx &= \frac{1}{3}x^3 + C\Big|_0^1 \\
&= \left(\frac{1}{3}(1)^3 + C\right) - \left(\frac{1}{3}(0)^3 + C\right) \\
&= \frac{1}{3} + C - 0 - C \\
&= \frac{1}{3}.
\end{aligned}$$

Observe that the C-values appear as opposites in the evaluation of the integral and thus do not affect the definite integral's value.

In the following activity, we work to build a list of basic functions whose antiderivatives we already know.

Activity 4.4.3. Use your knowledge of derivatives of basic functions to complete Table 4.4.5 of antiderivatives. For each entry, your task is to find a function F whose derivative is the given function f. When finished, use the FTC and the results in the table to evaluate the three given definite integrals.

Table 4.4.5 Familiar basic functions and their antiderivatives.

given function, $f(x)$	antiderivative, $F(x)$
k, (k is constant)	
x^n, $n \neq -1$	
$\frac{1}{x}$, $x > 0$	
$\sin(x)$	
$\cos(x)$	
$\sec(x)\tan(x)$	
$\csc(x)\cot(x)$	
$\sec^2(x)$	
$\csc^2(x)$	
e^x	
a^x $(a > 1)$	
$\frac{1}{1+x^2}$	
$\frac{1}{\sqrt{1-x^2}}$	

(a) $\displaystyle\int_0^1 \left(x^3 - x - e^x + 2\right)\, dx$

(b) $\displaystyle\int_0^{\pi/3} (2\sin(t) - 4\cos(t) + \sec^2(t) - \pi)\, dt$

(c) $\displaystyle\int_0^1 (\sqrt{x} - x^2)\, dx$

The list of antiderivatives that we developed in Table 4.4.5 is important to know by heart, as anytime we want to evaluate a definite integral using the FTC, we need to find antiderivatives.

There are some subtle issues that arise with some of the functions in Table 4.4.5 due to them having one or more x-values at which they are not defined. To understand the situation better, we focus on the function $f(x) = \frac{1}{x}$ and its antiderivative, $F(x) = \ln(x)$. In the table, we specified that this relationship holds for $x > 0$. We can also observe that if $x < 0$ and $G(x) = \ln(-x)$, then

$$G'(x) = \frac{1}{-x} \cdot (-1) = \frac{1}{x}$$

so $G(x) = \ln(-x)$ is an antiderivative of $\frac{1}{x}$ when $x < 0$. Since $|x| = x$ when $x > 0$ and $|x| = -x$ when $x < 0$, we can say that $F(x) = \ln(|x|)$ is an antiderivative of $f(x) = \frac{1}{x}$ for all $x \neq 0$. This situation is explored in more detail in Exercise 4.4.6.15.

Since $F(x) = \ln(|x|)$ is an antiderivative of $f(x) = \frac{1}{x}$ for all $x \neq 0$, it seems natural to write that the general antiderivative of $f(x) = \frac{1}{x}$ is $F(x) = \ln(|x|) + C$. But doing so is technically incorrect. This is due to the discontinuity of $f(x) = \frac{1}{x}$ at $x = 0$ and the fact that there's one antiderivative, $\ln(-x) + C_1$, for $x < 0$, and a second, $\ln(x) + C_2$, for $x > 0$. While we will still sometimes write "$F(x) = \ln(|x|) + C$" as the general antiderivative, we should be especially careful when the value $x = 0$, where $\frac{1}{x}$ is undefined, is included in the interval of values in which we are interested.

This situation arises for any function that has a discontinuity (e.g., $\frac{1}{x^2}$, $\tan(x)$); some of these issues will be considered further when we study improper integrals in Section 6.5.

4.4.4 The total change theorem

Let us review three interpretations of the definite integral.

- For a moving object with instantaneous velocity $v(t)$, the object's change in position on the time interval $[a, b]$ is given by $\int_a^b v(t)\,dt$, and whenever $v(t) \geq 0$ on $[a, b]$, $\int_a^b v(t)\,dt$ tells us the total distance traveled by the object on $[a, b]$.
- For any continuous function f, its definite integral $\int_a^b f(x)\,dx$ represents the net signed area bounded by $y = f(x)$ and the x-axis on $[a, b]$, where regions that lie below the x-axis have a minus sign associated with their area.
- The value of a definite integral is linked to the average value of a function: for a continuous function f on $[a, b]$, its average value $f_{\mathrm{AVG}[a,b]}$ is given by

$$f_{\mathrm{AVG}[a,b]} = \frac{1}{b-a} \int_a^b f(x)\,dx.$$

The Fundamental Theorem of Calculus now enables us to evaluate exactly (without taking a limit of Riemann sums) any definite integral for which we are able to find an antiderivative of the integrand.

A slight change in perspective allows us to gain even more insight into the meaning of the definite integral. Recall Equation (4.4.2), where we wrote the Fundamental Theorem of Calculus for a velocity function v with antiderivative V as

$$V(b) - V(a) = \int_a^b v(t)\,dt.$$

If we instead replace V with s (which represents position) and replace v with s' (since velocity is the derivative of position), Equation (4.4.2) then reads as

$$s(b) - s(a) = \int_a^b s'(t)\,dt. \tag{4.4.3}$$

In words, this version of the FTC tells us that the total change in an object's position function on a particular interval is given by the definite integral of the position function's derivative over that interval.

Of course, this result is not limited to only the setting of position and velocity. Writing the result in terms of a more general function f, we have the Total Change Theorem.

The Total Change Theorem.

If f is a continuously differentiable function on $[a, b]$ with derivative f', then $f(b) - f(a) = \int_a^b f'(x)\,dx$. That is, the definite integral of the rate of change of a function on $[a, b]$ is the total change of the function itself on $[a, b]$.

The Total Change Theorem tells us more about the relationship between the graph of a function and that of its derivative. Recall that heights on the graph of the derivative function are equal to slopes on the graph of the function itself. If instead we know f' and are seeking information about f, we can say the following:

differences in heights on f correspond to net signed areas bounded by f'.

To see why this is so, let's revisit the function $f(x) = 4x - x^2$ and its derivative $f'(x) = 4 - 2x$ that were the focus in Figure 1.4.1. As one example of a difference in heights on the graph of f, consider the difference $f(1) - f(0)$. This value is 3, because $f(1) = 3$ and $f(0) = 0$, but also because the net signed area bounded by $y = f'(x)$ on $[0, 1]$ is 3. That is,

$$f(1) - f(0) = \int_0^1 f'(x)\,dx.$$

Again, this means that differences in heights on f correspond to net signed areas bounded by f', as shown in Figure 4.4.6.

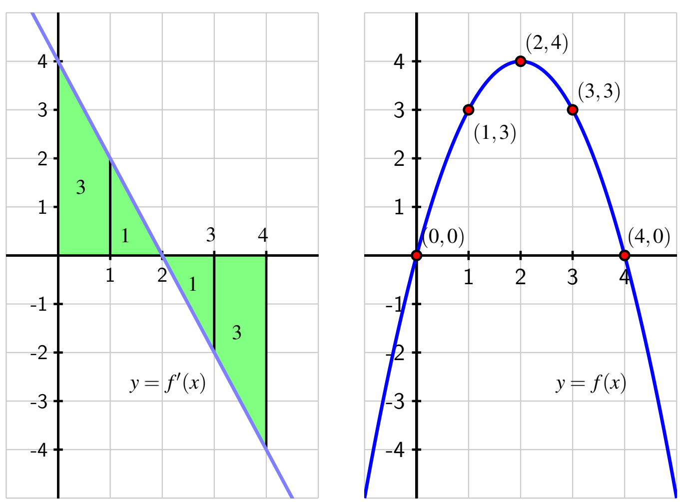


Figure 4.4.6 The graphs of $f'(x) = 4 - 2x$ (at left) and an antiderivative $f(x) = 4x - x^2$ at right. Differences in heights on f correspond to net signed areas bounded by f'.

In addition to this observation about area, the Total Change Theorem enables us to answer questions about a function whose rate of change we know.

Example 4.4.7 Suppose that pollutants are leaking out of an underground storage tank at a rate of $r(t)$ gallons/day, where t is measured in days. It is conjectured that $r(t)$ is given by the formula $r(t) = 0.0069t^3 - 0.125t^2 + 11.079$ over a certain 12-day period. What is the meaning of $\int_4^{10} r(t)\,dt$ and what is its value? What is the average rate at which pollutants are leaving the tank on the time interval $4 \le t \le 10$?

Solution. It's helpful to think about the graph of $y = r(t)$, which is given in Figure 4.4.8.

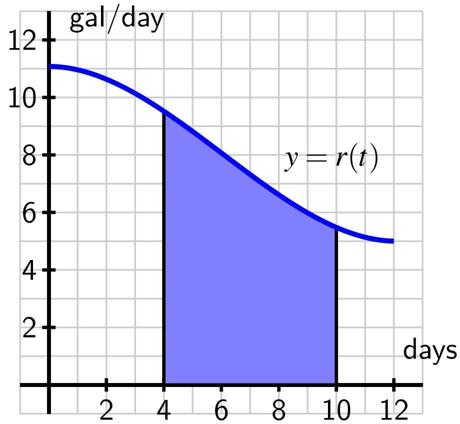


Figure 4.4.8 The rate $r(t)$ of pollution leaking from a tank, measured in gallons per day.

Since $r(t) \ge 0$, the value of $\int_4^{10} r(t)\,dt$ is the area under the curve on the interval $[4, 10]$. A Riemann sum for this area will have rectangles with heights measured in gallons per day and widths measured in days, so the area of each rectangle will have units of

$$\frac{\text{gallons}}{\text{day}} \cdot \text{days} = \text{gallons}.$$

Thus, the definite integral tells us the total number of gallons of pollutant that leak from the tank from day 4 to day 10. The Total Change Theorem tells us the same thing: if we let $R(t)$ denote the total number of gallons of pollutant that have leaked from the tank up to day t, then $R'(t) = r(t)$, and

$$\int_4^{10} r(t)\,dt = R(10) - R(4),$$

the number of gallons that have leaked from day 4 to day 10.

To compute the exact value of the integral, we use the Fundamental Theorem of Calculus. Antidifferentiating $r(t) = 0.0069t^3 - 0.125t^2 + 11.079$, we find that

$$\begin{aligned}\int_4^{10} 0.0069t^3 - 0.125t^2 + 11.079\,dt &= 0.0069 \cdot \frac{1}{4}t^4 - 0.125 \cdot \frac{1}{3}t^3 + 11.079t\Big|_4^{10} \\ &\approx 44.282.\end{aligned}$$

Thus, approximately 44.282 gallons of pollutant leaked over the six day time period.

To find the average rate at which pollutant leaked from the tank over $4 \le t \le 10$, we compute the average value of r on $[4, 10]$. Thus,

$$r_{\text{AVG}[4,10]} = \frac{1}{10-4}\int_4^{10} r(t)\,dt \approx \frac{44.282}{6} = 7.380$$

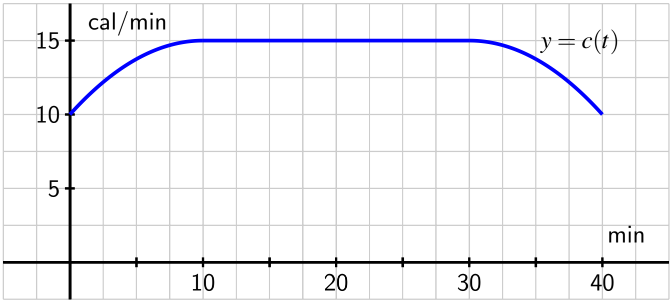
gallons per day. ◊

Activity 4.4.4. During a 40-minute workout, a person riding an exercise machine burns calories at a rate of c calories per minute, where the function $y = c(t)$ is given by the following information. On the interval $0 \le t \le 10$, the formula for c is $c(t) = -0.05t^2 + t + 10$; on the interval $10 \le t \le 30$, $c(t) = 15$; and on $30 \le t \le 40$, its formula is $c(t) = -0.05t^2 + 3t - 30$.

(a) What is the exact total number of calories the person burns during the first 10 minutes of her workout?

(b) Let $C(t)$ be an antiderivative of $c(t)$. What is the meaning of $C(40) - C(0)$ in the context of the person exercising? Include units on your answer.

(c) Determine the exact average rate at which the person burned calories during the 40-minute workout. Sketch a representation of the average rate at which calories were burned on the provided graph of $c(t)$.



(d) At what time(s), if any, is the instantaneous rate at which the person is burning calories equal to the average rate at which she burns calories, on the time interval $0 \le t \le 40$?

4.4.5 Summary

- We can find the exact value of a definite integral without taking the limit of a Riemann sum or using a familiar area formula by finding the antiderivative of the integrand, and hence applying the Fundamental Theorem of Calculus.
- The Fundamental Theorem of Calculus says that if f is a continuous function on $[a, b]$

and F is an antiderivative of f, then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Hence, if we can find an antiderivative for the integrand f, evaluating the definite integral comes from simply computing the change in F on $[a, b]$.

- A slightly different perspective on the FTC allows us to restate it as the Total Change Theorem, which says that

$$\int_a^b f'(x)\,dx = f(b) - f(a),$$

for any continuously differentiable function f. This means that the definite integral of the instantaneous rate of change of a function f on an interval $[a, b]$ is equal to the total change in the function f on $[a, b]$.

4.4.6 Exercises

1. The velocity function is $v(t) = -t^2 + 5t - 6$ for a particle moving along a line. Find the displacement (net distance covered) of the particle during the time interval $[-2, 5]$.

displacement = ____________________

2. Evaluate the definite integral

$$\int_3^9 \frac{2}{\sqrt{x}} dx$$

____

3. Evaluate exactly, using the Fundamental Theorem of Calculus:

$\int_0^c 3e^x dx =$ ___________________________________

4. Evaluate the definite integral

$$\int_0^\pi 4\sin(x) dx$$

____

5. Evaluate the definite integral:

$\int_{-7}^{-1} \left(-5x^{-2} + x^{-1}\right)\,dx =$ ___________________________________

6. Use the Fundamental Theorem to determine the value of b if the area under the graph of $f(x) = 6x$ between $x = 1$ and $x = b$ is equal to 72. Assume $b > 1$.

$b =$ ___________________________________

7. It can be shown that $\int_1^3 e^x\,dx = e^3 - e$. Use this fact and the properties of integrals to evaluate $\int_1^3 -6e^{x+2}\,dx$.

8. Evaluate the definite integral.

$$\int_0^{-3} (x+4)^2 \, dx = \underline{\hspace{10cm}}$$

9. Evaluate the definite integral:

$$\int_3^6 \frac{t^8 - t^4}{t^6} \, dt = \underline{\hspace{9cm}}$$

10. Evaluate the definite integral:

$$\int_0^1 u(\sqrt[2]{u} + \sqrt[7]{u}) \, du = \underline{\hspace{9cm}}$$

11. A function f is given piecewise by the formula

$$f(x) = \begin{cases} -x^2 + 2x + 1, & \text{if } 0 \le x < 2 \\ -x + 3, & \text{if } 2 \le x < 3 \\ x^2 - 8x + 15, & \text{if } 3 \le x \le 5 \end{cases}.$$

a. Determine the exact value of the net signed area enclosed by f and the x-axis on the interval $[2, 5]$.

b. Compute the exact average value of f on $[0, 5]$.

c. Find a formula for a function g on $5 \le x \le 7$ so that if we extend the above definition of f so that $f(x) = g(x)$ if $5 \le x \le 7$, it follows that $\int_0^7 f(x)\, dx = 0$.

12. The instantaneous velocity (in meters per minute) of a moving object is given by the function v as pictured in Figure 4.4.9. Assume that on the interval $0 \le t \le 4$, $v(t)$ is given by $v(t) = -\frac{1}{4}t^3 + \frac{3}{2}t^2 + 1$, and that on every other interval v is piecewise linear, as shown.

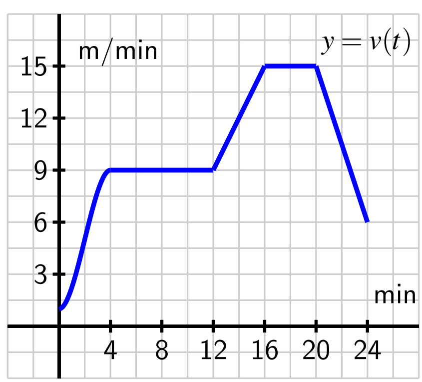


Figure 4.4.9 The velocity function of a moving body.

a. Determine the exact distance traveled by the object on the time interval $0 \le t \le 4$.

b. What is the object's average velocity on $[12, 24]$?

c. At what time is the object's acceleration greatest?

d. Suppose that the velocity of the object is increased by a constant value c for all values of t. What value of c will make the object's total distance traveled on $[12, 24]$ be 210 meters?

13. When an aircraft attempts to climb as rapidly as possible, its climb rate (in feet per minute) decreases as altitude increases, because the air is less dense at higher altitudes. Given below is a table showing performance data for a certain single engine aircraft, giving its climb rate at various altitudes, where $c(h)$ denotes the climb rate of the airplane at an altitude h.

h (feet)	0	1000	2000	3000	4000	5000	6000	7000	8000	9000	10,000
c (ft/min)	925	875	830	780	730	685	635	585	535	490	440

Let a new function called $m(h)$ measure the number of minutes required for a plane at altitude h to climb the next foot of altitude.

a. Determine a similar table of values for $m(h)$ and explain how it is related to the table above. Be sure to explain the units.

b. Give a careful interpretation of a function whose derivative is $m(h)$. Describe what the input is and what the output is. Also, explain in plain English what the function tells us.

c. Determine a definite integral whose value tells us exactly the number of minutes required for the airplane to ascend to 10,000 feet of altitude. Clearly explain why the value of this integral has the required meaning.

d. Use the Riemann sum M_5 to estimate the value of the integral you found in (c). Include units on your result.

14. In Chapter 1, we showed that for an object moving along a straight line with position function $s(t)$, the object's "average velocity on the interval $[a, b]$" is given by

$$AV_{[a,b]} = \frac{s(b) - s(a)}{b - a}.$$

More recently in Chapter 4, we found that for an object moving along a straight line with velocity function $v(t)$, the object's "average value of its velocity function on $[a, b]$" is

$$v_{\text{AVG}[a,b]} = \frac{1}{b-a} \int_a^b v(t)\, dt.$$

Are the "average velocity on the interval $[a, b]$" and the "average value of the velocity function on $[a, b]$" the same thing? Why or why not? Explain.

15. In Table 4.4.5 in Activity 4.4.3, we noted that for $x > 0$, an antiderivative of $f(x) = \frac{1}{x}$ is $F(x) = \ln(x)$. Here we observe that a key difference between $f(x)$ and $F(x)$ is that f is defined for all $x \neq 0$, while F is only defined for $x > 0$, and see how we can actually define an antiderivative of f for all values of x.

a. Suppose that $x < 0$, and let $G(x) = \ln(-x)$. Compute $G'(x)$.

b. Explain why G is an antiderivative of f for $x < 0$.

c. Let $H(x) = \ln(|x|)$, and recall that

$$|x| = \begin{cases} -x, & \text{if } x < 0 \\ x, & \text{if } x \geq 0 \end{cases}.$$

Explain why $H(x) = G(x)$ for $x < 0$ and $H(x) = F(x)$ for $x > 0$.

d. Now discuss why we say that an antiderivative of $\frac{1}{x}$ is $\ln(|x|)$ for all $x \neq 0$.

CHAPTER 5

Evaluating Integrals

5.1 Constructing accurate graphs of antiderivatives

Motivating Questions

- Given the graph of a function's derivative, how can we construct a completely accurate graph of the original function?
- How many antiderivatives does a given function have? What do those antiderivatives all have in common?
- Given a function f, how does the rule $A(x) = \int_0^x f(t)\,dt$ define a new function A?

5.1.1 Introduction

A recurring theme in our discussion of differential calculus has been the question "Given information about the derivative of an unknown function f, how much information can we obtain about f itself?" In Activity 1.8.3, the graph of $y = f'(x)$ was known (along with the value of f at a single point) and we endeavored to sketch a possible graph of f near the known point. In Example 3.3.7, we investigated how the first derivative test enables us to use information about f' to determine where the original function f is increasing and decreasing, as well as where f has relative extreme values. If we know a formula or graph of f', by computing f'' we can find where the original function f is concave up and concave down. Thus, knowing f' and f'' enables us to understand the shape of the graph of f.

We returned to this question in even more detail in Section 4.1. In that setting, we knew the instantaneous velocity of a moving object and worked to determine as much as possible about the object's position function. We found connections between the net signed area under the velocity function and the corresponding change in position of the function, and the Total Change Theorem further illuminated these connections between f' and f, showing that the total change in the value of f over an interval $[a, b]$ is determined by the net signed area bounded by f' and the x-axis on the same interval.

In what follows, we explore the situation where we possess an accurate graph of the derivative function along with a single value of the function f. From that information, we'd like to determine a graph of f that shows where f is increasing, decreasing, concave up, and concave down, and also provides an accurate function value at any point.

Preview Activity 5.1.1. Suppose that the following information is known about a function f: the graph of its derivative, $y = f'(x)$, is given in Figure 5.1.1. Further, assume that f' is piecewise linear (as pictured) and that for $x \le 0$ and $x \ge 6$, $f'(x) = 0$. Finally, it is given that $f(0) = 1$.

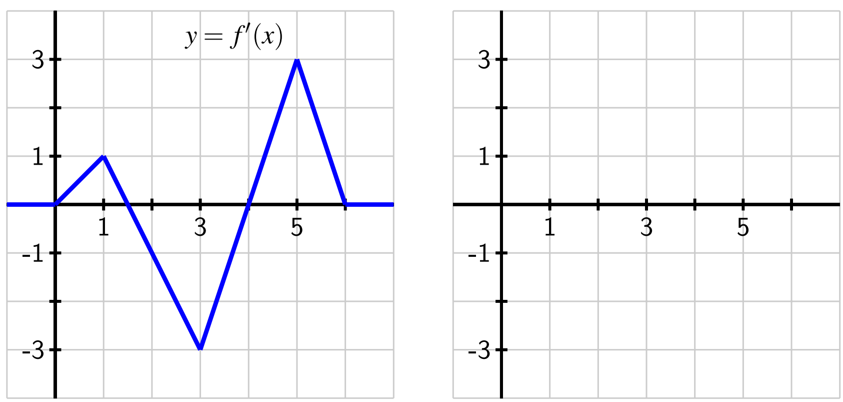


Figure 5.1.1 At left, the graph of $y = f'(x)$; at right, axes for plotting $y = f(x)$.

(a) On what interval(s) is f an increasing function? On what intervals is f decreasing?

(b) On what interval(s) is f concave up? concave down?

(c) At what point(s) does f have a relative minimum? a relative maximum?

(d) Recall that the Total Change Theorem tells us that

$$f(1) - f(0) = \int_0^1 f'(x)\,dx.$$

What is the exact value of $f(1)$?

(e) Use the given information and similar reasoning to that in (d) to determine the exact value of $f(2)$, $f(3)$, $f(4)$, $f(5)$, and $f(6)$.

(f) Based on your responses to all of the preceding questions, sketch a complete and accurate graph of $y = f(x)$ on the axes provided, being sure to indicate the behavior of f for $x < 0$ and $x > 6$.

5.1.2 Constructing the graph of an antiderivative

Preview Activity 5.1.1 demonstrates that when we can find the exact area under the graph of a function on any given interval, it is possible to construct a graph of the function's antiderivative. That is, we can find a function whose derivative is given. We can now determine not only the overall shape of the antiderivative graph, but also the actual *height* of the graph at any point of interest.

This is a consequence of the Fundamental Theorem of Calculus: if we know a function f and the value of the antiderivative F at some starting point a, we can determine the value of $F(b)$ via the definite integral. Since $F(b) - F(a) = \int_a^b f(x)\,dx$, it follows that

$$F(b) = F(a) + \int_a^b f(x)\,dx. \tag{5.1.1}$$

We can also interpret the equation $F(b) - F(a) = \int_a^b f(x)\,dx$ in terms of the graphs of f and F as follows. On an interval $[a, b]$,

> *differences in heights on the graph of the antiderivative given by $F(b) - F(a)$ correspond to the net signed area bounded by the original function on the interval $[a, b]$, which is given by $\int_a^b f(x)\,dx$.*

Activity 5.1.2. Suppose that the function $y = f(x)$ is given by the graph shown in Figure 5.1.2, and that the pieces of f are either portions of lines or portions of circles. In addition, let F be an antiderivative of f and say that $F(0) = -1$. Finally, assume that for $x \le 0$ and $x \ge 7$, $f(x) = 0$.

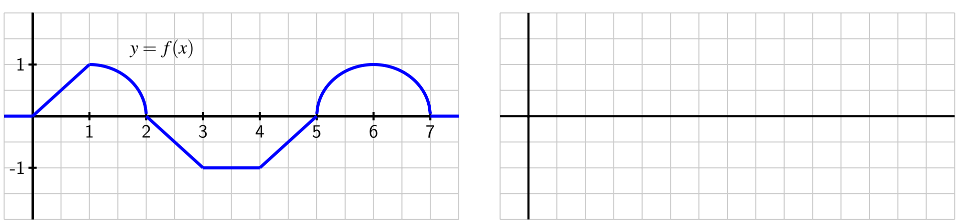


Figure 5.1.2 At left, the graph of $y = f(x)$.

(a) On what interval(s) is F an increasing function? On what intervals is F decreasing?

(b) On what interval(s) is F concave up? concave down? neither?

(c) At what point(s) does F have a relative minimum? a relative maximum?

(d) Use the given information to determine the exact value of $F(x)$ for $x = 1, 2, \ldots, 7$. In addition, what are the values of $F(-1)$ and $F(8)$?

(e) Based on your responses to all of the preceding questions, sketch a complete and accurate graph of $y = F(x)$ on the axes provided, being sure to indicate the behavior of F for $x < 0$ and $x > 7$. Clearly indicate the scale on the vertical and horizontal axes of your graph.

(f) Suppose we change one key piece of information: in particular, say that G is an antiderivative of f and $G(0) = 0$. How (if at all) would your answers to the preceding questions change? Sketch a graph of G on the same axes as the graph of F you constructed in (e).

5.1.3 Multiple antiderivatives of a single function

In the final question of Activity 5.1.2, we encountered a very important idea: a function f has more than one antiderivative. Each antiderivative of f is determined uniquely by its value at a single point. For example, suppose that f is the function given at left in Figure 5.1.3, and suppose further that F is an antiderivative of f that satisfies $F(0) = 1$.

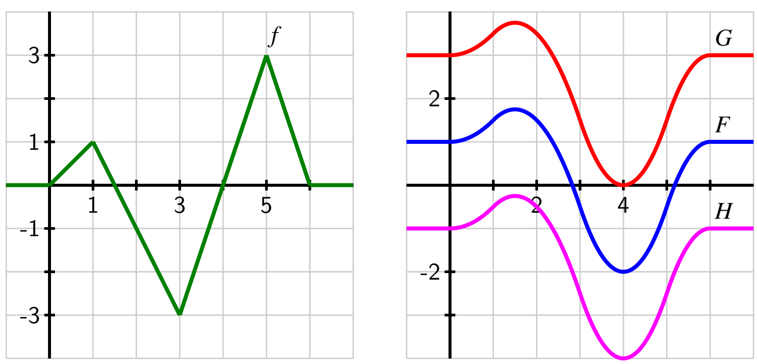


Figure 5.1.3 At left, the graph of $y = f(x)$. At right, three different antiderivatives of f.

Then, using Equation (5.1.1), we can compute

$$\begin{aligned} F(1) &= F(0) + \int_0^1 f(x)\,dx \\ &= 1 + 0.5 \\ &= 1.5. \end{aligned}$$

Similarly, $F(2) = 1.5$, $F(3) = -0.5$, $F(4) = -2$, $F(5) = -0.5$, and $F(6) = 1$. In addition, we can use the fact that $F' = f$ to ascertain where F is increasing and decreasing, concave up and concave down, and has relative extremes and inflection points. We ultimately find that the graph of F is the one given in blue in Figure 5.1.3.

If we want an antiderivative G for which $G(0) = 3$, then G will have the exact same shape as F (since both share the derivative f), but G will be shifted vertically from the graph of F, as pictured in red in Figure 5.1.3. Note that $G(1) - G(0) = \int_0^1 f(x)\,dx = 0.5$, just as $F(1) - F(0) =$

0.5, but since $G(0) = 3$, $G(1) = G(0) + 0.5 = 3.5$, whereas $F(1) = 1.5$. In the same way, if we assigned a different initial value to the antiderivative, say $H(0) = -1$, we would get still another antiderivative, as shown in magenta in Figure 5.1.3.

This example demonstrates an important fact that holds more generally:

Antiderivatives of the same function differ by a constant.

If G and H are both antiderivatives of a function f on an interval (a, b), then the function $G - H$ must be constant on the interval (a, b).

To see why this result holds, observe that if G and H are both antiderivatives of f, then $G' = f$ and $H' = f$. Hence,

$$\frac{d}{dx}[G(x) - H(x)] = G'(x) - H'(x) = f(x) - f(x) = 0.$$

Since the only way a function can have derivative zero is by being a constant function, it follows that the function $G - H$ must be constant.

We now see that if a function has at least one antiderivative, it must have infinitely many: we can add any constant of our choice to the antiderivative and get another antiderivative. For this reason, we sometimes refer to the *general antiderivative* of a function f.

To identify a particular antiderivative of f, we must know a single value of the antiderivative F (this value is often called an *initial condition*). For example, if $f(x) = x^2$, its general antiderivative is $F(x) = \frac{1}{3}x^3 + C$, where we include the "$+C$" to indicate that F includes *all* of the possible antiderivatives of f. If we know that $F(2) = 3$, we substitute 2 for x in $F(x) = \frac{1}{3}x^3 + C$, and find that

$$3 = \frac{1}{3}(2)^3 + C,$$

or $C = 3 - \frac{8}{3} = \frac{1}{3}$. Therefore, the particular antiderivative in this case is $F(x) = \frac{1}{3}x^3 + \frac{1}{3}$.

Activity 5.1.3. For each of the following functions, sketch an accurate graph of the antiderivative that satisfies the given initial condition. In addition, sketch the graph of two additional antiderivatives of the given function, and state the corresponding initial conditions that each of them satisfy. If possible, find an algebraic formula for the antiderivative that satisfies the initial condition.

(a) original function: $g(x) = |x| - 1$; initial condition: $G(-1) = 0$; interval for sketch: $[-2, 2]$

(b) original function: $h(x) = \sin(x)$; initial condition: $H(0) = 1$; interval for sketch: $[0, 4\pi]$

(c) original function: $p(x) = \begin{cases} x^2, & \text{if } 0 < x < 1 \\ -(x-2)^2, & \text{if } 1 < x < 2 \\ 0 & \text{otherwise} \end{cases}$; initial condition: $P(0) = 1$; interval for sketch: $[-1, 3]$

5.1.4 Functions defined by integrals

Equation (5.1.1) allows us to compute the value of the antiderivative F at a point b, provided that we know $F(a)$ and can evaluate the definite integral from a to b of f. That is,

$$F(b) = F(a) + \int_a^b f(x)\,dx.$$

In several situations, we have used this formula to compute $F(b)$ for several different values of b, and then plotted the points $(b, F(b))$ to help us draw an accurate graph of F. This suggests that we may want to think of b, the upper limit of integration, as a variable itself. To that end, we introduce the idea of an *integral function*, a function whose formula involves a definite integral.

Definition 5.1.4 If f is a continuous function, we define the corresponding **integral function** A according to the rule

$$A(x) = \int_a^x f(t)\,dt. \tag{5.1.2}$$

◊

Note that because x is the independent variable in the function A, and determines the end-point of the interval of integration, we need to use a different variable as the variable of integration. A standard choice is t, but any variable other than x is acceptable.

One way to think of the function A is as the "net signed area from a up to x" function, where we consider the region bounded by $y = f(t)$. For example, in Figure 5.1.5, we see a function f pictured at left, and its corresponding area function (choosing $a = 0$), $A(x) = \int_0^x f(t)\,dt$ shown at right.

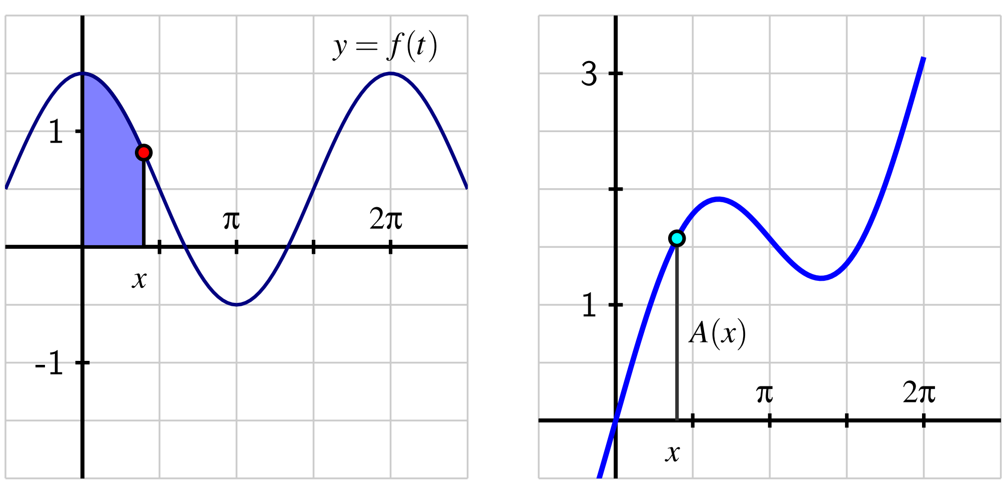


Figure 5.1.5 At left, the graph of the given function f. At right, the area function $A(x) = \int_0^x f(t)\,dt$.

The function A measures the net signed area from $t = 0$ to $t = x$ bounded by the curve $y = f(t)$; this value is then reported as the corresponding height on the graph of $y = A(x)$. This interactive graphic[1] brings the static picture in Figure 5.1.5 to life. There, the user can move the red point on the function f and see how the corresponding height changes at the light blue point on the graph of A.

The choice of a is somewhat arbitrary. In the activity that follows, we explore how the value of a affects the graph of the integral function.

Activity 5.1.4. Suppose that g is given by the graph at left in Figure 5.1.6 and that A is the corresponding integral function defined by $A(x) = \int_1^x g(t)\,dt$.

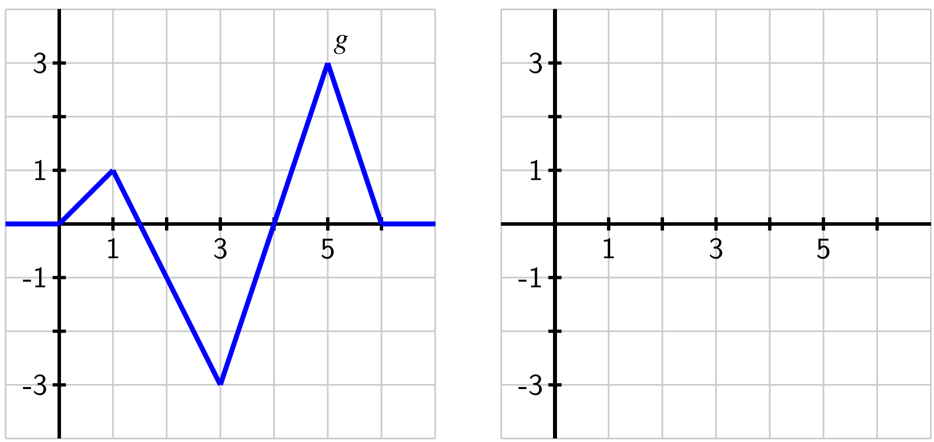


Figure 5.1.6 At left, the graph of $y = g(t)$; at right, axes for plotting $y = A(x)$, where A is defined by the formula $A(x) = \int_1^x g(t)\,dt$.

(a) On what interval(s) is A an increasing function? On what intervals is A decreasing? Why?

(b) On what interval(s) do you think A is concave up? concave down? Why?

(c) At what point(s) does A have a relative minimum? a relative maximum?

(d) Use the given information to determine the exact values of $A(0)$, $A(1)$, $A(2)$, $A(3)$, $A(4)$, $A(5)$, and $A(6)$.

(e) Based on your responses to all of the preceding questions, sketch a complete and accurate graph of $y = A(x)$ on the axes provided, being sure to indicate the behavior of A for $x < 0$ and $x > 6$.

(f) How does the graph of B compare to A if B is instead defined by $B(x) = \int_0^x g(t)\,dt$?

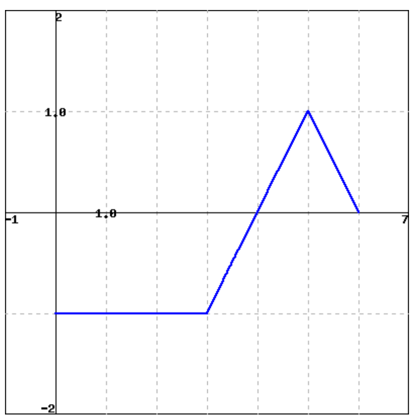
[1] gvsu.edu/s/cz

5.1.5 Summary

- Given the graph of a function f, we can construct the graph of its antiderivative F provided that (a) we know a starting value of F, say $F(a)$, and (b) we can evaluate the integral $\int_a^b f(x)\,dx$ exactly for relevant choices of a and b. For instance, if we wish to know $F(3)$, we can compute $F(3) = F(a) + \int_a^3 f(x)\,dx$. When we combine this information about the function values of F together with our understanding of how the behavior of $F' = f$ affects the overall shape of F, we can develop a completely accurate graph of the antiderivative F.
- Because the derivative of a constant is zero, if F is an antiderivative of f, it follows that $G(x) = F(x) + C$ will also be an antiderivative of f. Moreover, any two antiderivatives of a function f differ precisely by a constant. Thus, any function with at least one antiderivative in fact has infinitely many, and the graphs of any two antiderivatives will differ only by a vertical translation.
- Given a function f, the rule $A(x) = \int_a^x f(t)\,dt$ defines a new function A that measures the net-signed area bounded by f on the interval $[a, x]$. We call the function A the integral function corresponding to f.

5.1.6 Exercises

1. The figure below shows f.

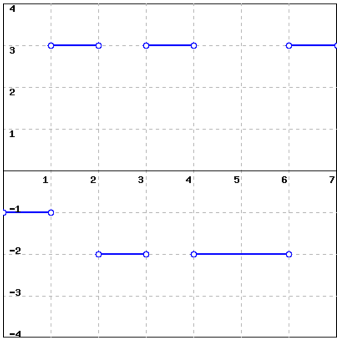


(Click on the graph for a larger version.)

If $F' = f$ and $F(0) = 0$, find $F(b)$ for b =1, 2, 3, 4, 5, 6, and fill these values in the following table.

b	1	2	3	4	5	6
$F(b)$	______	______	______	______	______	______

2. Assume f' is given by the graph below. Suppose f is continuous and that $f(3) = 0$.



(Click on the graph for a larger version.)

Sketch, on a sheet of work paper, an accurate graph of f, and use it to find each of

$f(0) =$ __________

and

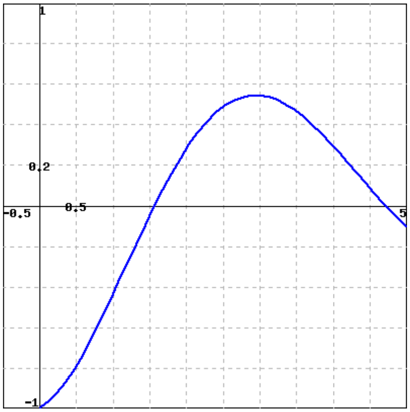
$f(7) =$ __________

Then find the value of the integral:

$\int_0^7 f'(x)\,dx =$ __________

(Note that you can do this in two different ways!)

3. The graph of $f(t)$ is shown below.



(Click on the graph for a larger version.)

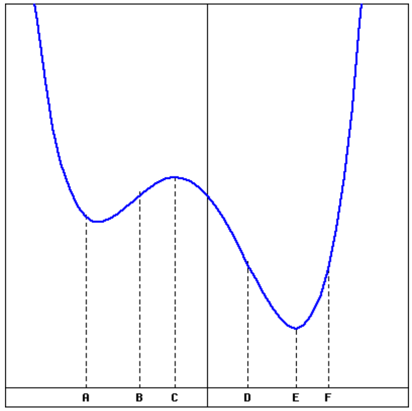
Carefully sketch, on a separate page, a graph of $F(x) = \int_0^x f(t)\,dt$. (You will want to use the larger version of the graph for this.)

Using your graph, at what x-values (give your answer(s) to the nearest 0.25) is $F(x) = -0.5$?

$x =$ ____________________________________

*(Give your answer as a list of x-values, or enter **none** if $F(x)$ is never equal to -0.5.).*

4. The graph of f' (***not*** f) is given below.



(Note that this is a graph of f', not a graph of f. Click on the graph for a larger version.)

At which of the marked values of x is

A. $f(x)$ greatest? $x =$ ________

B. $f(x)$ least? $x =$ ________

C. $f'(x)$ greatest? $x =$ ________

D. $f'(x)$ least? $x =$ ________

E. $f''(x)$ greatest? $x =$ ________

F. $f''(x)$ least? $x =$ ________

5. A moving particle has its velocity given by the quadratic function v pictured in Figure 5.1.7. In addition, it is given that $A_1 = \frac{7}{6}$ and $A_2 = \frac{8}{3}$, as well as that for the corresponding position function s, $s(0) = 0.5$.

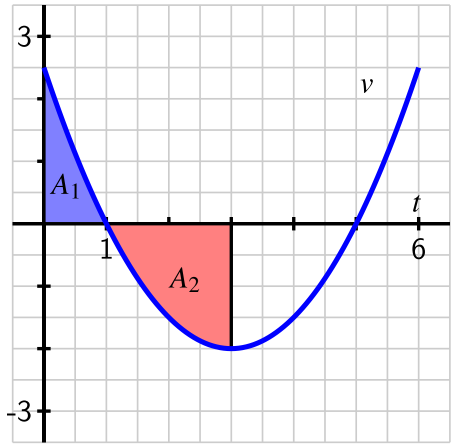
a. Use the given information to determine $s(1)$, $s(3)$, $s(5)$, and $s(6)$. What do these values mean in the context of the moving particle?

b. On what interval(s) is s increasing? That is, when is the particle moving forward? On what interval(s) is s decreasing? That is, when is the particle moving backward?

c. On what interval(s) is s concave up? That is, when is the particle accelerating? On what interval(s) is s concave down? That is, when is the particle decelerating?

d. Sketch an accurate, labeled graph of s on the axes at right in Figure 5.1.7.

e. Note that $v(t) = -2 + \frac{1}{2}(t-3)^2$. Find a formula for s.



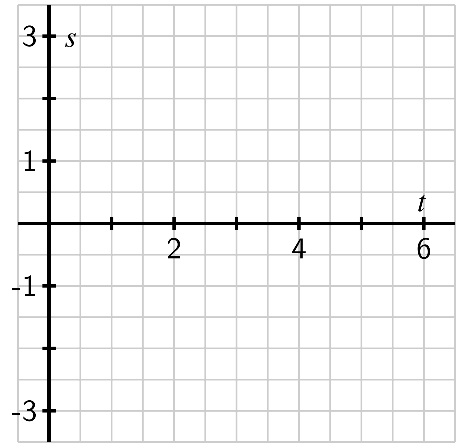
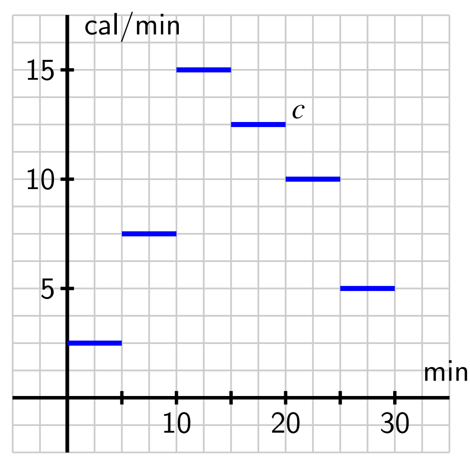


Figure 5.1.7 At left, the given graph of v. At right, axes for plotting s.

6. A person exercising on a treadmill experiences different levels of resistance and thus burns calories at different rates, depending on the treadmill's setting. In a particular workout, the rate at which a person is burning calories is given by the piecewise constant function c pictured in Figure 5.1.8. Note that the units on c are "calories per minute."



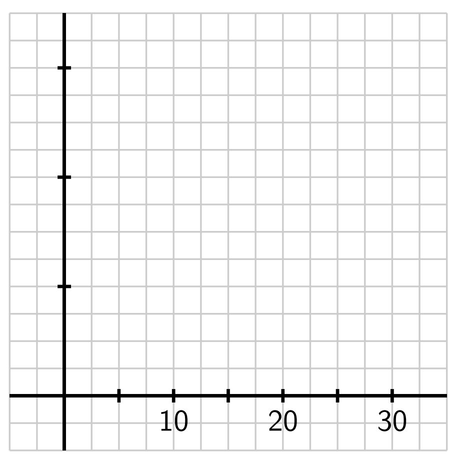


Figure 5.1.8 At left, the given graph of c. At right, axes for plotting C.

a. Let C be an antiderivative of c. What does the function C measure? What are its units?

b. Assume that $C(0) = 0$. Determine the exact value of $C(t)$ at the values $t = 5, 10, 15, 20, 25, 30$.

c. Sketch an accurate graph of C on the axes provided at right in Figure 5.1.8. Be certain to label the scale on the vertical axis.

d. Determine a formula for C that does not involve an integral and is valid for $5 \le t \le 10$.

7. Consider the piecewise linear function f given in Figure 5.1.9. Let the functions A, B, and C be defined by the rules $A(x) = \int_{-1}^{x} f(t)\,dt$, $B(x) = \int_{0}^{x} f(t)\,dt$, and $C(x) = \int_{1}^{x} f(t)\,dt$.

a. For the values $x = -1, 0, 1, \ldots, 6$, make a table that lists corresponding values of $A(x)$, $B(x)$, and $C(x)$.

b. On the axes provided in Figure 5.1.9, sketch the graphs of A, B, and C.

c. How are the graphs of A, B, and C related?

d. How would you best describe the relationship between the function A and the function f?

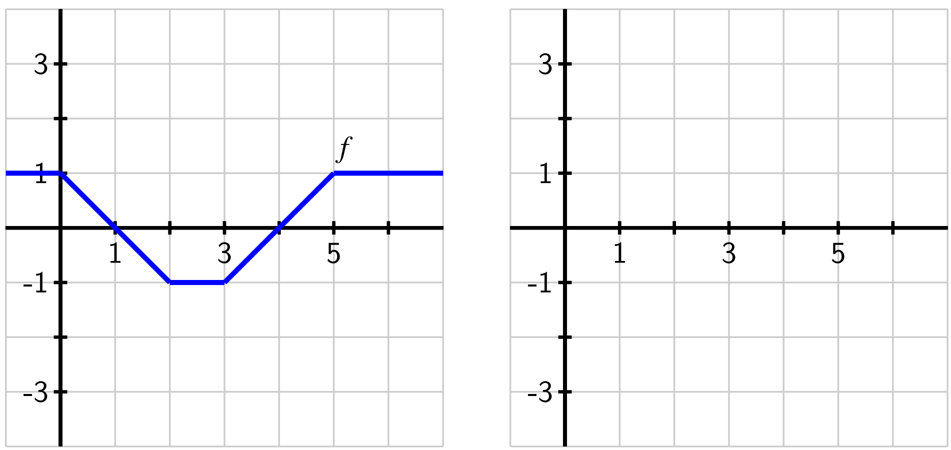


Figure 5.1.9 At left, the given graph of f. At right, axes for plotting A, B, and C.

5.2 The Second Fundamental Theorem of Calculus

Motivating Questions

- How does the integral function $A(x) = \int_1^x f(t)\,dt$ define an antiderivative of f?
- What is the statement of the Second Fundamental Theorem of Calculus?
- How do the First and Second Fundamental Theorems of Calculus enable us to formally see how differentiation and integration are almost inverse processes?

5.2.1 Introduction

In Section 4.4, we learned the Fundamental Theorem of Calculus (FTC), which from here forward will be referred to as the *First* Fundamental Theorem of Calculus, as in this section we develop a corresponding result that follows it. Recall that the First FTC tells us that if f is a continuous function on $[a, b]$ and F is any antiderivative of f (that is, $F' = f$), then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

We have used this result in two settings:

1. If we have a graph of f and we can compute the exact area bounded by f on an interval $[a, b]$, we can compute the change in an antiderivative F over the interval.
2. If we can find an algebraic formula for an antiderivative of f, we can evaluate the integral to find the net signed area bounded by the function on the interval.

For the former, see Preview Activity 5.1.1 or Activity 5.1.2. For the latter, we can easily evaluate exactly integrals such as

$$\int_1^4 x^2\,dx,$$

since we know that the function $F(x) = \frac{1}{3}x^3$ is an antiderivative of $f(x) = x^2$. Thus,

$$\begin{aligned}\int_1^4 x^2\,dx &= \frac{1}{3}x^3\Big|_1^4 \\ &= \frac{1}{3}(4)^3 - \frac{1}{3}(1)^3 \\ &= 21.\end{aligned}$$

Thus, the First FTC can be used in two ways. First, to find the difference $F(b) - F(a)$ for an antiderivative F of the integrand f, even if we may not have a formula for F itself. To do this, we must know the value of the integral $\int_a^b f(x)\,dx$ exactly, perhaps through known geometric formulas for area. In addition, the First FTC provides a way to find the exact value of a definite integral, and hence a certain net signed area exactly, by finding an antiderivative of the integrand and evaluating its total change over the interval. In this case, we need to know a formula for the antiderivative F. Both of these perspectives are reflected in Figure 5.2.1.

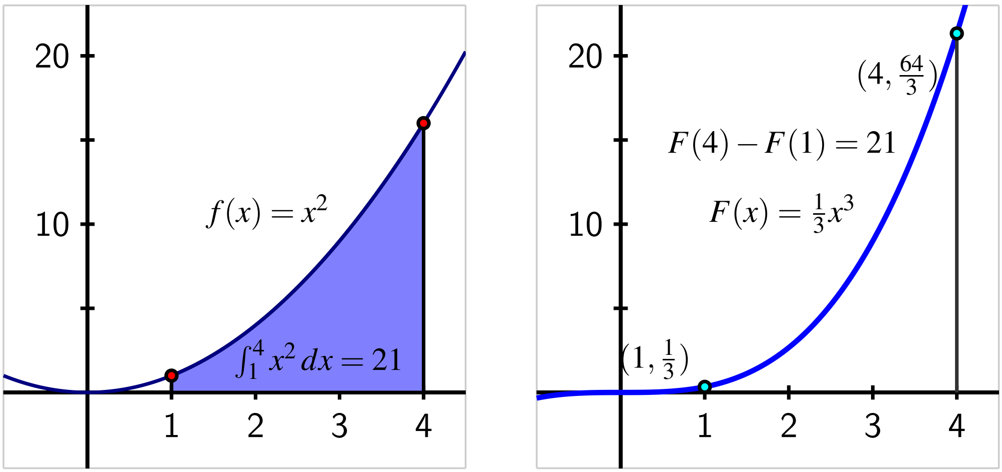


Figure 5.2.1 At left, the graph of $f(x) = x^2$ on the interval $[1, 4]$ and the area it bounds. At right, the antiderivative function $F(x) = \frac{1}{3}x^3$, whose total change on $[1, 4]$ is the value of the definite integral at left.

The value of a definite integral may have additional meaning depending on context: as the change in position when the integrand is a velocity function, the total amount of pollutant leaked from a tank when the integrand is the rate at which pollution is leaking, or other total changes if the integrand is a rate function. Also, the value of the definite integral is connected to the average value of a continuous function on a given interval: $f_{\text{AVG}[a,b]} = \frac{1}{b-a}\int_a^b f(x)\,dx$.

In the last part of Section 5.1, we studied integral functions of the form $A(x) = \int_c^x f(t)\,dt$. Figure 5.1.5 is a particularly important image to keep in mind as we work with integral functions, and the corresponding interactive[1] can help us understand the function A. In what follows, we use the First FTC to gain additional understanding of the function $A(x) = \int_c^x f(t)\,dt$, where the integrand f is given (either through a graph or a formula), and c is a constant.

Preview Activity 5.2.1. Consider the function A defined by the rule

$$A(x) = \int_1^x f(t)\,dt,$$

where $f(t) = 4 - 2t$.

(a) Compute $A(1)$ and $A(2)$ exactly.

[1]gvsu.edu/s/cz

(b) Use the First Fundamental Theorem of Calculus to find a formula for $A(x)$ that does not involve an integral. That is, use the first FTC to evaluate $\int_1^x (4-2t)\,dt$.

(c) Observe that f is a linear function; what kind of function is A?

(d) Using the formula you found in (b) that does not involve an integral, compute $A'(x)$.

(e) While we have defined f by the rule $f(t) = 4-2t$, it is equivalent to say that f is given by the rule $f(x) = 4-2x$. What do you observe about the relationship between A and f?

5.2.2 The Second Fundamental Theorem of Calculus

The result of Preview Activity 5.2.1 is not particular to the function $f(t) = 4-2t$, nor to the choice of "1" as the lower bound in the integral that defines the function A. For instance, if we let $f(t) = \cos(t) - t$ and set $A(x) = \int_2^x f(t)\,dt$, we can determine a formula for A by the First FTC. Specifically,

$$\begin{aligned} A(x) &= \int_2^x (\cos(t) - t)\,dt \\ &= \sin(t) - \frac{1}{2}t^2\bigg|_2^x \\ &= \sin(x) - \frac{1}{2}x^2 - (\sin(2) - 2)\,. \end{aligned}$$

Differentiating $A(x)$, since $(\sin(2)-2)$ is constant, it follows that

$$A'(x) = \cos(x) - x,$$

and thus we see that $A'(x) = f(x)$, so A is an antiderivative of f. And since $A(2) = \int_2^2 f(t)\,dt = 0$, A is the only antiderivative of f for which $A(2) = 0$.

In general, if f is any continuous function, and we define the function A by the rule

$$A(x) = \int_c^x f(t)\,dt,$$

where c is an arbitrary constant, then we can show that A is an antiderivative of f. To see why, let's demonstrate that $A'(x) = f(x)$ by using the limit definition of the derivative. Doing so, we observe that

$$\begin{aligned} A'(x) &= \lim_{h\to 0} \frac{A(x+h) - A(x)}{h} \\ &= \lim_{h\to 0} \frac{\int_c^{x+h} f(t)\,dt - \int_c^x f(t)\,dt}{h} \\ &= \lim_{h\to 0} \frac{\int_x^{x+h} f(t)\,dt}{h}, \end{aligned} \tag{5.2.1}$$

where Equation (5.2.1) follows from the fact that $\int_c^x f(t)\,dt + \int_x^{x+h} f(t)\,dt = \int_c^{x+h} f(t)\,dt$. Now, observe that for small values of h,

$$\int_x^{x+h} f(t)\,dt \approx f(x) \cdot h,$$

by a simple left-hand approximation of the integral. Thus, as we take the limit in Equation (5.2.1), it follows that

$$A'(x) = \lim_{h \to 0} \frac{\int_x^{x+h} f(t)\,dt}{h} = \lim_{h \to 0} \frac{f(x) \cdot h}{h} = f(x).$$

Hence, A is indeed an antiderivative of f. In addition, $A(c) = \int_c^c f(t)\,dt = 0$. The preceding argument demonstrates the truth of the Second Fundamental Theorem of Calculus, which we state as follows.

The Second Fundamental Theorem of Calculus.

If f is a continuous function and c is any constant, then f has a unique antiderivative A that satisfies $A(c) = 0$, and that antiderivative is given by the rule $A(x) = \int_c^x f(t)\,dt$.

Activity 5.2.2. Suppose that f is the function given in Figure 5.2.2 and that f is a piecewise function whose parts are either portions of lines or portions of circles, as pictured.

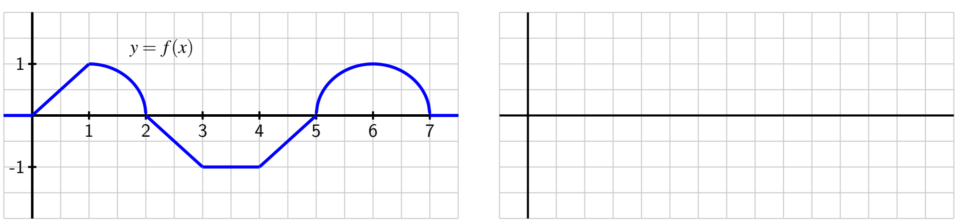


Figure 5.2.2 At left, the graph of $y = f(x)$. At right, axes for sketching $y = A(x)$.

In addition, let A be the function defined by the rule $A(x) = \int_2^x f(t)\,dt$.

(a) What does the Second FTC tell us about the relationship between A and f?

(b) Compute $A(1)$ and $A(3)$ exactly.

(c) Sketch a precise graph of $y = A(x)$ on the axes at right that accurately reflects where A is increasing and decreasing, where A is concave up and concave down, and the exact values of A at $x = 0, 1, \ldots, 7$.

(d) How is A similar to, but different from, the function F that you found in Activity 5.1.2?

(e) With as little additional work as possible, sketch precise graphs of the functions $B(x) = \int_3^x f(t)\,dt$ and $C(x) = \int_1^x f(t)\,dt$. Justify your results with at least one sentence of explanation.

5.2.3 Understanding Integral Functions

The Second FTC provides us with a way to construct an antiderivative of any continuous function. In particular, if we are given a continuous function g and wish to find an antiderivative G, we can now say that

$$G(x) = \int_c^x g(t)\,dt$$

provides the rule for such an antiderivative, and moreover that $G(c) = 0$. Note especially that we know that $G'(x) = g(x)$, or

$$\frac{d}{dx}\left[\int_c^x g(t)\,dt\right] = g(x). \tag{5.2.2}$$

This result is useful for understanding the graph of G.

Example 5.2.3 Investigate the behavior of the integral function

$$E(x) = \int_0^x e^{-t^2}\,dt.$$

Solution. E is closely related to the well known *error function* [2] in probability and statistics. It turns out that the function e^{-t^2} does not have an elementary antiderivative.

While we cannot evaluate E exactly for any value other than $x = 0$, we still can gain a tremendous amount of information about the function E. By applying the rule in Equation (5.2.2) to E, it follows that

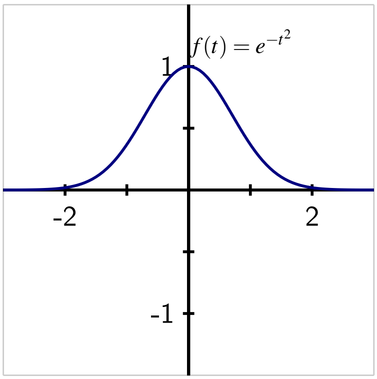
$$E'(x) = \frac{d}{dx}\left[\int_0^x e^{-t^2}\,dt\right] = e^{-x^2},$$

so we know a formula for the derivative of E, and we know that $E(0) = 0$. This information is precisely the type we were given in Activity 3.3.2, where we were given information about the derivative of a function, but lacked a formula for the function itself.

Using the first and second derivatives of E, along with the fact that $E(0) = 0$, we can determine more information about the behavior of E. First, we note that for all real numbers x, $e^{-x^2} > 0$, and thus $E'(x) > 0$ for all x. Thus E is an always increasing function. Further, as $x \to \infty$, $E'(x) = e^{-x^2} \to 0$, so the slope of the function E tends to zero as $x \to \infty$ (and similarly as $x \to -\infty$). Indeed, it turns out that E has horizontal asymptotes as x increases or decreases without bound.

In addition, we can observe that $E''(x) = -2xe^{-x^2}$, and that $E''(0) = 0$, while $E''(x) < 0$ for $x > 0$ and $E''(x) > 0$ for $x < 0$. This information tells us that E is concave up for $x < 0$ and concave down for $x > 0$ with a point of inflection at $x = 0$.

The only thing we lack at this point is a sense of how big E can get as x increases. If we use a midpoint Riemann sum with 10 subintervals to estimate $E(2)$, we see that $E(2) \approx 0.8822$; a similar calculation to estimate $E(3)$ shows little change ($E(3) \approx 0.8862$), so it appears that as x increases without bound, E approaches a value just larger than 0.886, which aligns with the fact that E has horizontal asymptotes. Putting all of this information together (and using the symmetry of $f(t) = e^{-t^2}$), we see the results shown in Figure 5.2.4.



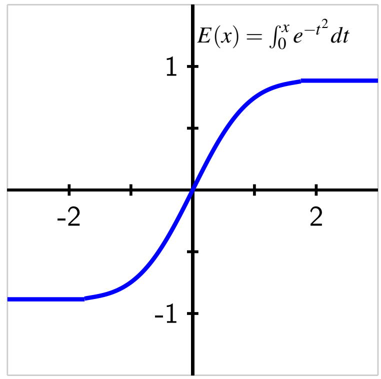


Figure 5.2.4 At left, the graph of $f(t) = e^{-t^2}$. At right, the integral function $E(x) = \int_0^x e^{-t^2}\,dt$, which is the unique antiderivative of f that satisfies $E(0) = 0$.

Because E is the antiderivative of $f(t) = e^{-t^2}$ that satisfies $E(0) = 0$, values on the graph of $y = E(x)$ represent the net signed area of the region bounded by $f(t) = e^{-t^2}$ from 0 up to x. We see that the value of E increases rapidly near zero but then levels off as x increases, since there is less and less additional accumulated area bounded by $f(t) = e^{-t^2}$ as x increases. ◊

Activity 5.2.3. Suppose that $f(t) = \frac{t}{1+t^2}$.

Note that while we know a function whose derivative is $\frac{1}{1+t^2}$, namely $\arctan(t)$, we haven't yet encountered an elementary function whose derivative is $f(t) = \frac{t}{1+t^2}$.

Let $F(x) = \int_0^x f(t)\,dt$. We will explore $F(x)$ from numerical, graphical, and algebraic perspectives.

(a) What is the key relationship between F and f, according to the Second FTC?

(b) Analyze the first derivative of F algebraically to determine the intervals on which F is increasing and decreasing.

(c) Analyze the second derivative of F algebraically to determine the intervals on which F is concave up and concave down. Note that $f'(t)$ can be simplified to be written in the form $f'(t) = \frac{1-t^2}{(1+t^2)^2}$.

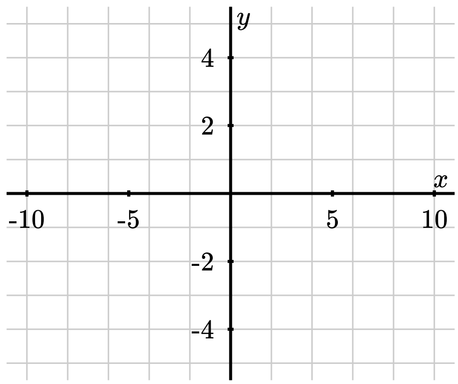
[2]The error function is defined by the rule $\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\,dt$ and has the key property that $0 \le \text{erf}(x) < 1$ for all $x \ge 0$ and moreover that $\lim_{x\to\infty} \text{erf}(x) = 1$.

(d) Use a Riemann sum calculator with midpoints and 10 subintervals to calculate approximate values of $F(x)$ in the table shown below.

Note that $F(0) = \int_0^0 f(t)\,dt = 0$.

x	-10	-5	0	5	10
$F(x)$	________	________	0	________	________

(e) Use your work above to sketch an accurate graph of $y = F(x)$ on the axes provided.



(f) Now consider the function $g(x) = \frac{1}{2}\ln(1 + x^2)$. Calculate $g(0)$ and $g'(x)$. Use appropriate computing technology to plot the graph of $g(x)$. What can you conclude about the functions $F(x)$ and $g(x)$?

5.2.4 Differentiating an Integral Function

We have seen that the Second FTC enables us to construct an antiderivative F for any continuous function f as the integral function $F(x) = \int_c^x f(t)\,dt$. If we have a function of the form $F(x) = \int_c^x f(t)\,dt$, then we know that $F'(x) = \frac{d}{dx}\left[\int_c^x f(t)\,dt\right] = f(x)$. This shows that integral functions, while perhaps having the most complicated formulas of any functions we have encountered, are nonetheless particularly simple to differentiate. For instance, if

$$F(x) = \int_\pi^x \sin(t^2)\,dt,$$

then by the Second FTC, we know immediately that

$$F'(x) = \sin(x^2).$$

In general, we know by the Second FTC that

$$\frac{d}{dx}\left[\int_a^x f(t)\,dt\right] = f(x).$$

This equation says that "the derivative of the integral function whose integrand is f, is f." We see that if we first integrate the function f from $t = a$ to $t = x$, and then differentiate with respect to x, these two processes "undo" each other.

What happens if we differentiate a function $f(t)$ and then integrate the result from $t = a$ to $t = x$? That is, what can we say about the quantity

$$\int_a^x \frac{d}{dt}\left[f(t)\right]\, dt?$$

We note that $f(t)$ is an antiderivative of $\frac{d}{dt}\left[f(t)\right]$ and apply the First FTC. We see that

$$\begin{aligned}\int_a^x \frac{d}{dt}\left[f(t)\right]\, dt &= f(t)\Big|_a^x \\ &= f(x) - f(a).\end{aligned}$$

Thus, we see that if we first differentiate f and then integrate the result from a to x, we return to the function f, minus the constant value $f(a)$. So the two processes almost undo each other, up to the constant $f(a)$.

The observations made in the preceding two paragraphs demonstrate that differentiating and integrating (where we integrate from a constant up to a variable) are almost inverse processes. This should not be surprising: integrating involves antidifferentiating, which reverses the process of differentiating. On the other hand, we see that there is some subtlety involved, because integrating the derivative of a function does not quite produce the function itself. This is because every function has an entire family of antiderivatives, and any two of those antiderivatives differ only by a constant.

Activity 5.2.4. Evaluate each of the following derivatives and definite integrals. Clearly cite whether you use the First or Second FTC in so doing.

(a) $\frac{d}{dx}\left[\int_4^x e^{t^2}\, dt\right]$

(b) $\int_{-2}^x \frac{d}{dt}\left[\frac{t^4}{1+t^4}\right]\, dt$

(c) $\frac{d}{dx}\left[\int_x^1 \cos(t^3)\, dt\right]$

(d) $\int_3^x \frac{d}{dt}\left[\ln(1+t^2)\right]\, dt$

(e) $\frac{d}{dx}\left[\int_4^{x^3} \sin(t^2)\, dt\right]$

5.2.5 Summary

- For a continuous function f, the integral function $A(x) = \int_1^x f(t)\, dt$ defines an antiderivative of f.

- The Second Fundamental Theorem of Calculus is the formal, more general statement of the preceding fact: if f is a continuous function and c is any constant, then $A(x) = \int_c^x f(t)\,dt$ is the unique antiderivative of f that satisfies $A(c) = 0$.
- Together, the First and Second FTC enable us to formally see how differentiation and integration are almost inverse processes through the observations that

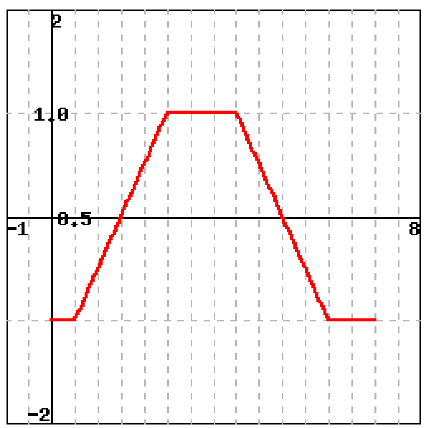
$$\int_c^x \frac{d}{dt}\left[f(t)\right]\,dt = f(x) - f(c)$$

and

$$\frac{d}{dx}\left[\int_c^x f(t)\,dt\right] = f(x).$$

5.2.6 Exercises

1. Let $A(x) = \int_0^x f(t)\,dt$, with $f(x)$ as in figure.



$A(x)$ has a local minimum on $(0, 6)$ at $x =$ ______________

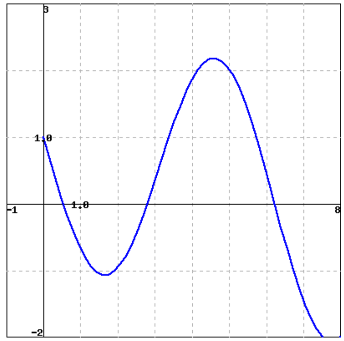
$A(x)$ has a local maximum on $(0, 6)$ at $x =$ ______________

2. Let $f(x) = \displaystyle\int_0^x t^6 dt$. Evaluate the following.

$f'(x) =$ ______________

$f'(-4) =$ ______________

3. Let $g(x) = \int_2^x f(t)\,dt$, where $f(t)$ is given in the figure below.



(Click on the graph for a larger version.)

Find each of the following:

A. $g(2) =$ ________________

B. $g'(5) =$ ________________

C. The interval (with endpoints given to the nearest 0.25) where g is concave up:

interval = ________________

(Give your answer as an interval or a list of intervals, e.g., ***(-infinity,8]*** *or* ***(1,5),(7,10)****, or enter* ***none****for no intervals.)*

D. The value of x where g takes its maximum on the interval $0 \le x \le 8$.

$x =$ ________________

4. Find the interval on which the curve

$$y = \int_1^x \frac{1}{7 + t + 3t^2}\, dt$$

is concave upward.

Interval = ________________

5. The tide removes sand from the beach at a small ocean park at a rate modeled by the function

$$R(t) = 2 + 5\sin\left(\frac{4\pi t}{25}\right)$$

A pumping station adds sand to the beach at rate modeled by the function

$$S(t) = \frac{15t}{1 + 3t}$$

Both $R(t)$ and $S(t)$ are measured in cubic yards of sand per hour, t is measured in hours, and the valid times are $0 \le t \le 6$. At time $t = 0$, the beach holds 2500 cubic yards of sand.

a. What definite integral measures how much sand the tide will remove during the time period $0 \le t \le 6$? Why?

b. Write an expression for $Y(x)$, the total number of cubic yards of sand on the beach at time x. Carefully explain your thinking and reasoning.

c. At what instantaneous rate is the total number of cubic yards of sand on the beach at time $t = 4$ changing?

d. Over the time interval $0 \leq t \leq 6$, at approximately what time t is the amount of sand on the beach least? What is the corresponding approximate minimum value? Explain and justify your answers fully.

6. Let g be the function pictured at left in Figure 5.2.5, and let F be defined by $F(x) = \int_2^x g(t)\,dt$. Assume that the shaded areas have values $A_1 = 4.29$, $A_2 = 12.75$, $A_3 = 0.36$, and $A_4 = 1.79$. Assume further that the portion of A_2 that lies between $x = 0.5$ and $x = 2$ is 6.06.

Sketch a carefully labeled graph of F on the axes provided, and include a written analysis of how you know where F is zero, increasing, decreasing, concave up, and concave down.

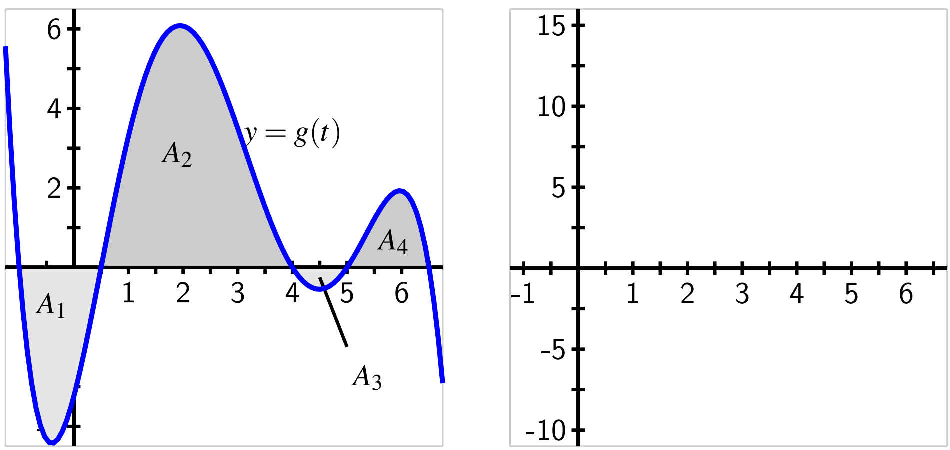


Figure 5.2.5 At left, the graph of g. At right, axes for plotting F.

7. When an aircraft attempts to climb as rapidly as possible, its climb rate (in feet per minute) decreases as altitude increases, because the air is less dense at higher altitudes. Table 5.2.6 provides performance data for a certain single engine aircraft, giving its climb rate at various altitudes, where $c(h)$ denotes the climb rate of the airplane at an altitude h.

Table 5.2.6 Data for the climbing aircraft.

h (feet)	0	1000	2000	3000	4000	5000
c (ft/min)	925	875	830	780	730	685

h (feet)	6000	7000	8000	9000	10,000
c (ft/min)	635	585	535	490	440

Let a new function m, that also depends on h, (say $y = m(h)$) measure the number of minutes required for a plane at altitude h to climb the next foot of altitude.

a. Determine a similar table of values for $m(h)$ and explain how it is related to the table above. Be sure to discuss the units on m.

b. Give a careful interpretation of a function whose derivative is $m(h)$. Describe what the input is and what the output is. Also, explain in plain English what the function tells us.

c. Determine a definite integral whose value tells us exactly the number of minutes required for the airplane to ascend to 10,000 feet of altitude. Clearly explain why the value of this integral has the required meaning.

d. Determine a formula for a function $M(h)$ whose value tells us the exact number of minutes required for the airplane to ascend to h feet of altitude.

e. Estimate the values of $M(6000)$ and $M(10000)$ as accurately as you can. Include units on your results.

5.3 Integration by substitution

Motivating Questions

- How can we begin to find algebraic formulas for antiderivatives of more complicated algebraic functions?
- What is an indefinite integral and how is its notation used in discussing antiderivatives?
- How does the technique of u-substitution work to help us evaluate certain indefinite integrals, and how does this process rely on identifying function-derivative pairs?

5.3.1 Introduction

In Section 4.4, we learned the key role that antiderivatives play in the process of evaluating definite integrals exactly. The Fundamental Theorem of Calculus tells us that if F is any antiderivative of f, then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Furthermore, we realized that each elementary derivative rule developed in Chapter 2 leads to a corresponding elementary antiderivative, as summarized in Table 4.4.5. Thus, if we wish to evaluate an integral such as

$$\int_0^1 \left(x^3 - \sqrt{x} + 5^x\right)\,dx,$$

it is straightforward to do so, since we can easily antidifferentiate $f(x) = x^3 - \sqrt{x} + 5^x$. Because one antiderivative of f is $F(x) = \frac{1}{4}x^4 - \frac{2}{3}x^{3/2} + \frac{1}{\ln(5)}5^x$, the Fundamental Theorem of Calculus tells us that

$$\begin{aligned}\int_0^1 \left(x^3 - \sqrt{x} + 5^x\right)\,dx &= \frac{1}{4}x^4 - \frac{2}{3}x^{3/2} + \frac{1}{\ln(5)}5^x\Big|_0^1 \\ &= \left(\frac{1}{4}(1)^4 - \frac{2}{3}(1)^{3/2} + \frac{1}{\ln(5)}5^1\right) - \left(0 - 0 + \frac{1}{\ln(5)}5^0\right) \\ &= -\frac{5}{12} + \frac{4}{\ln(5)}.\end{aligned}$$

We see that we have a natural interest in being able to find such algebraic antiderivatives. We emphasize *algebraic* antiderivatives, as opposed to any antiderivative, since we know by the Second Fundamental Theorem of Calculus that $G(x) = \int_a^x f(t)\,dt$ is indeed an antiderivative of the given function f, but one that still involves a definite integral. Our goal in this section is to "undo" the process of differentiation to find an algebraic antiderivative for a given function.

Preview Activity 5.3.1. In Section 2.5, we learned the Chain Rule and how it can be applied to find the derivative of a composite function. In particular, if u is a differentiable function of x, and f is a differentiable function of $u(x)$, then

$$\frac{d}{dx}\left[f(u(x))\right] = f'(u(x)) \cdot u'(x).$$

In words, we say that the derivative of a composite function $c(x) = f(u(x))$, where f is considered the "outer" function and u the "inner" function, is "the derivative of the outer function, evaluated at the inner function, times the derivative of the inner function."

(a) For each of the following functions, use the Chain Rule to find the function's derivative. Be sure to label each derivative by name (e.g., the derivative of $g(x)$ should be labeled $g'(x)$).

i. $g(x) = e^{3x}$
ii. $h(x) = \sin(5x + 1)$
iii. $p(x) = \arctan(2x)$
iv. $q(x) = (2 - 7x)^4$
v. $r(x) = 3^{4-11x}$

(b) For each of the following functions, use your work in (a) to help you determine the general antiderivative (that is, include "$+C$") of the function. Label each antiderivative by name (e.g., the antiderivative of m should be called M). In addition, check your work by computing the derivative of each proposed antiderivative.

i. $m(x) = e^{3x}$
ii. $n(x) = \cos(5x + 1)$
iii. $s(x) = \dfrac{1}{1 + 4x^2}$
iv. $v(x) = (2 - 7x)^3$
v. $w(x) = 3^{4-11x}$

(c) Based on your experience in parts (a) and (b), conjecture an antiderivative for each of the following functions. Test your conjectures by computing the derivative of each proposed antiderivative.

i. $a(x) = \cos(\pi x)$
ii. $b(x) = (4x + 7)^{11}$
iii. $c(x) = xe^{x^2}$

5.3.2 Reversing the Chain Rule: First Steps

Whenever f is a familiar function whose antiderivative is known and $u(x)$ is a linear function, it is straightforward to antidifferentiate a function of the form

$$h(x) = f(u(x)).$$

Example 5.3.1 Determine the general antiderivative of

$$h(x) = (5x - 3)^6.$$

Check the result by differentiating.

Solution. For this composite function, the outer function f is $f(u) = u^6$, while the inner function is $u(x) = 5x - 3$. Since the antiderivative of f is $F(u) = \frac{1}{7}u^7 + C$, we see that the antiderivative of h is

$$H(x) = \frac{1}{7}(5x - 3)^7 \cdot \frac{1}{5} + C = \frac{1}{35}(5x - 3)^7 + C.$$

The inclusion of the constant $\frac{1}{5}$ is essential precisely because the derivative of the inner function is $u'(x) = 5$. Indeed, if we now compute $H'(x)$, we find by the Chain Rule (and Constant Multiple Rule) that

$$H'(x) = \frac{1}{35} \cdot 7(5x - 3)^6 \cdot 5 = (5x - 3)^6 = h(x),$$

and thus H is indeed the general antiderivative of h. ◊

Hence, in the special case where the outer function is familiar and the inner function is linear, we can antidifferentiate composite functions according to the following rule.

Antidifferentiating a composite function whose inner function is linear.

If $h(x) = f(ax + b)$ and F is a known algebraic antiderivative of f, then the general antiderivative of h is given by

$$H(x) = \frac{1}{a}F(ax + b) + C.$$

It is useful to have shorthand notation that indicates the instruction to find an antiderivative. Thus, in a similar way to how the notation

$$\frac{d}{dx}\left[f(x)\right]$$

represents the derivative of $f(x)$ with respect to x, we use the notation of the *indefinite integral*,

$$\int f(x)\,dx$$

to represent the general antiderivative of f with respect to x. Returning to the earlier example with $h(x) = (5x - 3)^6$, we can rephrase the relationship between h and its antiderivative H through the notation

$$\int (5x - 3)^6\,dx = \frac{1}{35}(5x - 6)^7 + C.$$

When we find an antiderivative, we will often say that we *evaluate an indefinite integral*. Just as the notation $\frac{d}{dx}[\square]$ means "find the derivative with respect to x of $\square$," the notation $\int \square\,dx$ means "find a function of x whose derivative is $\square$."

Activity 5.3.2. Evaluate each of the following indefinite integrals. Check each antiderivative that you find by differentiating.

(a) $\int \sin(8-3x)\,dx$

(b) $\int \sec^2(4x)\,dx$

(c) $\int \frac{1}{11x-9}\,dx$

(d) $\int \csc(2x+1)\cot(2x+1)\,dx$

(e) $\int \frac{1}{\sqrt{1-16x^2}}\,dx$

(f) $\int 5^{-x}\,dx$

Note: one of the integrals in Activity 5.3.2 has the form $\int \frac{1}{u}\,du$. As we discussed briefly in Section 4.4 immediately following the development of Table 4.4.5 in Activity 4.4.3, there is a subtle issue with writing

$$\int \frac{1}{u}\,du = \ln(|u|) + C$$

due to the fact that $\frac{1}{u}$ has a discontinuity at $u = 0$, which leads to an antiderivative for one piece of the function and another antiderivative for the other. In most applications (such as when we use partial fractions to solve certain differential equations such as the Logistic Equation in Section 7.6), the domain of input values on which we seek the antiderivative leads to a natural resolution of this ambiguity, so we will still write $\int \frac{1}{u}\,du = \ln(|u|) + C$.

5.3.3 Reversing the Chain Rule: u-substitution

A natural question arises from our recent work: what happens when the inner function is not linear? For example, can we find antiderivatives of such functions as

$$g(x) = xe^{x^2} \text{ and } h(x) = e^{x^2}?$$

It is important to remember that differentiation and antidifferentiation are almost inverse processes (that they are not is due to the $+C$ that arises when antidifferentiating). This almost-inverse relationship enables us to take any known derivative rule and rewrite it as a corresponding rule for an indefinite integral. For example, since

$$\frac{d}{dx}\left[x^5\right] = 5x^4,$$

we can equivalently write

$$\int 5x^4\,dx = x^5 + C.$$

Recall that the Chain Rule states that

$$\frac{d}{dx}\left[f(g(x))\right] = f'(g(x)) \cdot g'(x).$$

Restating this relationship in terms of an indefinite integral,

$$\int f'(g(x))g'(x)\,dx = f(g(x)) + C. \tag{5.3.1}$$

Equation (5.3.1) tells us that if we can view a given function as $f'(g(x))g'(x)$ for some appropriate choices of f and g, then we can antidifferentiate the function by reversing the Chain Rule. Note that both $g(x)$ and $g'(x)$ appear in the form of $f'(g(x))g'(x)$; we will sometimes say that we seek to *identify a function-derivative pair* ($g(x)$ and $g'(x)$) when trying to apply the rule in Equation (5.3.1).

If we can identify a function-derivative pair, we will introduce a new variable u to represent the function $g(x)$. With $u = g(x)$, it follows in Leibniz notation that $\frac{du}{dx} = g'(x)$, so that in terms of differentials[1], $du = g'(x)\,dx$. Now converting the indefinite integral to a new one in terms of u, we have

$$\int f'(g(x))g'(x)\,dx = \int f'(u)\,du.$$

Provided that f' is an elementary function whose antiderivative is known, we can easily evaluate the indefinite integral in u, and then go on to determine the desired overall antiderivative of $f'(g(x))g'(x)$. We call this process *u-substitution*, and summarize the rule as follows:

u-substitution.

With the substitution $u = g(x)$,

$$\int f'(g(x))g'(x)\,dx = \int f'(u)\,du = f(u) + C = f(g(x)) + C.$$

To see u-substitution at work, we consider the following example.

Example 5.3.2 Evaluate the indefinite integral

$$\int x^3 \cdot \sin(7x^4 + 3)\,dx$$

and check the result by differentiating.

Solution. We can make two algebraic observations regarding the integrand, $x^3 \cdot \sin(7x^4 + 3)$. First, $\sin(7x^4 + 3)$ is a composite function; as such, we know we'll need a more sophisticated approach to antidifferentiating. Second, x^3 is almost the derivative of $(7x^4 + 3)$; the only issue is a missing constant. Thus, x^3 and $(7x^4 + 3)$ are nearly a function-derivative pair. Furthermore, we know the antiderivative of $f(u) = \sin(u)$. The combination of these observations suggests that we can evaluate the given indefinite integral by reversing the chain rule through u-substitution.

Letting u represent the inner function of the composite function $\sin(7x^4 + 3)$, we have $u = 7x^4 + 3$, and thus $\frac{du}{dx} = 28x^3$. In differential notation, it follows that $du = 28x^3\,dx$, and thus

[1]If we recall from the definition of the derivative that $\frac{du}{dx} \approx \frac{\Delta u}{\Delta x}$ and use the fact that $\frac{du}{dx} = g'(x)$, then we see that $g'(x) \approx \frac{\Delta u}{\Delta x}$. Solving for Δu, $\Delta u \approx g'(x)\Delta x$. It is this last relationship that, when expressed in "differential" notation enables us to write $du = g'(x)\,dx$ in the change of variable formula.

$x^3\,dx = \frac{1}{28}\,du$. The original indefinite integral may be slightly rewritten as

$$\int \sin(7x^4+3)\cdot x^3\,dx,$$

and so by substituting u for $7x^4+3$ and $\frac{1}{28}\,du$ for $x^3\,dx$, it follows that

$$\int \sin(7x^4+3)\cdot x^3\,dx = \int \sin(u)\cdot\frac{1}{28}\,du.$$

Now we may evaluate the easier integral in u, and then replace u by the expression $7x^4+3$. Doing so, we find

$$\begin{aligned}\int \sin(7x^4+3)\cdot x^3\,dx &= \int \sin(u)\cdot\frac{1}{28}\,du\\ &= \frac{1}{28}\int \sin(u)\,du\\ &= \frac{1}{28}(-\cos(u))+C\\ &= -\frac{1}{28}\cos(7x^4+3)+C.\end{aligned}$$

To check our work, we observe by the Chain Rule that

$$\frac{d}{dx}\left[-\frac{1}{28}\cos(7x^4+3)\right] = -\frac{1}{28}\cdot(-1)\sin(7x^4+3)\cdot 28x^3 = \sin(7x^4+3)\cdot x^3,$$

which is indeed the original integrand. ◊

The u-substitution worked because the function multiplying $\sin(7x^4+3)$ was x^3. If instead that function was x^2 or x^4, the substitution process would not have worked. This is one of the primary challenges of antidifferentiation: slight changes in the integrand make tremendous differences. For instance, we can use u-substitution with $u = x^2$ and $du = 2x\,dx$ to find that

$$\begin{aligned}\int xe^{x^2}\,dx &= \int e^u\cdot\frac{1}{2}\,du\\ &= \frac{1}{2}\int e^u\,du\\ &= \frac{1}{2}e^u+C\\ &= \frac{1}{2}e^{x^2}+C.\end{aligned}$$

However, for the similar indefinite integral

$$\int e^{x^2}\,dx,$$

the u-substitution $u = x^2$ is no longer possible because the factor of x is missing. Hence, part of the lesson of u-substitution is just how specialized the process is: it only applies to situations where, up to a missing constant, the integrand is the result of applying the Chain Rule to a different, related function.

Activity 5.3.3. Evaluate each of the following indefinite integrals by using these steps:

- Find two functions within the integrand that form (up to a possible missing constant) a function-derivative pair;
- Make a substitution and convert the integral to one involving u and du;
- Evaluate the new integral in u;
- Convert the resulting function of u back to a function of x by using your earlier substitution;
- Check your work by differentiating the function of x. You should come up with the integrand originally given.

(a) $\int \frac{x^2}{5x^3+1}\, dx$

(b) $\int e^x \sin(e^x)\, dx$

(c) $\int \frac{\cos(\sqrt{x})}{\sqrt{x}}\, dx$

5.3.4 Evaluating Definite Integrals via u-substitution

We have introduced u-substitution as a means to evaluate indefinite integrals of functions that can be written, up to a constant multiple, in the form $f(g(x))g'(x)$. This same technique can be used to evaluate definite integrals involving such functions, though we need to be careful with the corresponding limits of integration. Consider, for instance, the definite integral

$$\int_2^5 xe^{x^2}\, dx.$$

Whenever we write a definite integral, it is implicit that the limits of integration correspond to the variable of integration. To be more explicit, observe that

$$\int_2^5 xe^{x^2}\, dx = \int_{x=2}^{x=5} xe^{x^2}\, dx.$$

When we execute a u-substitution, we change the *variable* of integration; it is essential to note that this also changes the *limits* of integration. For instance, with the substitution $u = x^2$ and $du = 2x\, dx$, it also follows that when $x = 2$, $u = 2^2 = 4$, and when $x = 5$, $u = 5^2 = 25$. Thus, under the change of variables of u-substitution, we now have

$$\int_{x=2}^{x=5} xe^{x^2}\, dx = \int_{u=4}^{u=25} e^u \cdot \frac{1}{2}\, du$$

$$\begin{aligned} &= \frac{1}{2}e^u \Big|_{u=4}^{u=25} \\ &= \frac{1}{2}e^{25} - \frac{1}{2}e^4. \end{aligned}$$

Alternatively, we could consider the related indefinite integral $\int xe^{x^2}\,dx$, find the antiderivative $\frac{1}{2}e^{x^2}$ through u-substitution, and then evaluate the original definite integral. With that method, we'd have

$$\begin{aligned} \int_2^5 xe^{x^2}\,dx &= \frac{1}{2}e^{x^2}\Big|_2^5 \\ &= \frac{1}{2}e^{25} - \frac{1}{2}e^4, \end{aligned}$$

which is, of course, the same result.

Activity 5.3.4. Evaluate each of the following definite integrals exactly through an appropriate u-substitution.

(a) $\int_1^2 \frac{x}{1+4x^2}\,dx$

(b) $\int_0^1 e^{-x}(2e^{-x}+3)^9\,dx$

(c) $\int_{2/\pi}^{4/\pi} \frac{\cos\left(\frac{1}{x}\right)}{x^2}\,dx$

5.3.5 Summary

- To find algebraic formulas for antiderivatives of more complicated algebraic functions, we need to think carefully about how we can reverse known differentiation rules. To that end, it is essential that we understand and recall known derivatives of basic functions, as well as the standard derivative rules.
- The indefinite integral provides notation for antiderivatives. When we write "$\int f(x)\,dx$," we mean "the general antiderivative of f." In particular, if we have functions f and F such that $F' = f$, the following two statements say the exact thing:

$$\frac{d}{dx}[F(x)] = f(x) \text{ and } \int f(x)\,dx = F(x) + C.$$

  That is, f is the derivative of F, and F is an antiderivative of f.
- The technique of u-substitution helps us to evaluate indefinite integrals of the form $\int f(g(x))g'(x)\,dx$ through the substitutions $u = g(x)$ and $du = g'(x)\,dx$, so that

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du.$$

A key part of choosing the expression in x to be represented by u is the identification of a function-derivative pair. To do so, we often look for an "inner" function $g(x)$ that is part of a composite function, while investigating whether $g'(x)$ (or a constant multiple of $g'(x)$) is present as a multiplying factor of the integrand.

5.3.6 Exercises

1. Solve the integral below with u-substitution.

$$\int \frac{-9\sec(-10x)\tan(-10x)}{\sec^2(-10x)}\,dx$$

Let $u =$ ____________________

Then $du =$ ____________________

Then, we substitute in the integral to get $\int$ ____________________, (which has 'u' and 'du' in it, not 'x' or 'dx'.)

Now this is a basic integral. Evaluate it to get all possible antiderivatives in terms of 'u': ____________________

and then the final answer is $\int \frac{-9\sec(-10x)\tan(-10x)}{\sec^2(-10x)}\,dx =$ ______________________________

2. Solve the integral below with u-substitution.

$$\int -4\csc^2(2x)e^{\cot(2x)}\,dx$$

Let $u =$ ____________________

Then $du =$ ____________________

Then, we substitute in the integral to get $\int$ ____________________, (which has 'u' and 'du' in it, not 'x' or 'dx'.)

Now this is a basic integral. Evaluate it to get all possible antiderivatives in terms of 'u': ____________________

and then the final answer is $\int -4\csc^2(2x)e^{\cot(2x)}\,dx =$ ______________________________

3. Find the following integral. Note that you can check your answer by differentiation.

$\int t^2\left(t^3-4\right)^2\,dt =$ ___________________________________

4. Find the the general antiderivative $F(x)$ of the function $f(x)$ given below. Note that you can check your answer by differentiation.

$f(x) = 9x\cos(x^2)$

antiderivative $F(x) =$ ____________________

5. Find the following integral. Note that you can check your answer by differentiation.

$$\int \frac{\ln^4(z)}{z}\,dz = \_\_\_\_\_\_\_\_\_\_\_\_$$

6. Find the following integral. Note that you can check your answer by differentiation.

$$\int \frac{e^{3x}}{3+e^{3x}}\,dx = \_\_\_\_\_\_\_\_\_\_\_\_$$

7. Use the Fundamental Theorem of Calculus to find

$$\int_{3\pi}^{3\pi/2} e^{-\cos(q)} \cdot \sin(q)\; dq = \_\_\_\_\_\_\_\_\_\_\_\_$$

8. Consider the integral $\int 6\,x^3\,(x^4+1)\,dx$. In the following, we will evaluate the integral using two methods.

A. First, rewrite the integral by multiplying out the integrand:

$$\int 6\,x^3\,(x^4+1)\,dx = \int \_\_\_\_\_\_\_\_\_\_\_\_\,dx$$

Then evaluate the resulting integral term-by-term:

$$\int 6\,x^3\,(x^4+1)\,dx = \_\_\_\_\_\_\_\_\_\_\_\_$$

B. Next, rewrite the integral using the substitution $w = x^4+1$:

$$\int 6\,x^3\,(x^4+1)\,dx = \int \_\_\_\_\_\_\_\_\_\_\_\_\,dw$$

Evaluate this integral (and back-substitute for w) to find the value of the original integral:

$$\int 6\,x^3\,(x^4+1)\,dx = \_\_\_\_\_\_\_\_\_\_\_\_$$

C. How are your expressions from parts (A) and (B) different? What is the difference between the two? (Ignore the constant of integration.)

(answer from B)−(answer from A) = ____________________

Are both of the answers correct? *(Be sure you can explain why they are!)*

9. Evaluate the indefinite integral

$$\int -9(1+\tan(t))^5\sec^2(t)\,dt$$

Note: Any arbitrary constants used must be an upper-case "C".

10. Evaluate the integral

$$\int 10\cot(x)\ln(\sin(x))\,dx$$

Note: Use an upper-case "C" for the constant of integration.

11. This problem centers on finding antiderivatives for the basic trigonometric functions other than $\sin(x)$ and $\cos(x)$.

a. Consider the indefinite integral $\int \tan(x)\,dx$. By rewriting the integrand as $\tan(x) = \frac{\sin(x)}{\cos(x)}$ and identifying an appropriate function-derivative pair, make a u-substitution and hence evaluate $\int \tan(x)\,dx$.

b. In a similar way, evaluate $\int \cot(x)\,dx$.

c. Consider the indefinite integral

$$\int \frac{\sec^2(x) + \sec(x)\tan(x)}{\sec(x) + \tan(x)}\,dx.$$

Evaluate this integral using the substitution $u = \sec(x) + \tan(x)$.

d. Simplify the integrand in (c) by factoring the numerator. What is a far simpler way to write the integrand?

e. Combine your work in (c) and (d) to determine $\int \sec(x)\,dx$.

f. Using (c)-(e) as a guide, evaluate $\int \csc(x)\,dx$.

12. Consider the indefinite integral $\int x\sqrt{x-1}\,dx$.

a. At first glance, this integrand may not seem suited to substitution due to the presence of x in separate locations in the integrand. Nonetheless, using the composite function $\sqrt{x-1}$ as a guide, let $u = x - 1$. Determine expressions for both x and dx in terms of u.

b. Convert the given integral in x to a new integral in u.

c. Evaluate the integral in (b) by noting that $\sqrt{u} = u^{1/2}$ and observing that it is now possible to rewrite the integrand in u by expanding through multiplication.

d. Evaluate each of the integrals $\int x^2\sqrt{x-1}\,dx$ and $\int x\sqrt{x^2-1}\,dx$. Write a paragraph to discuss the similarities among the three indefinite integrals in this problem and the role of substitution and algebraic rearrangement in each.

13. Consider the indefinite integral $\int \sin^3(x)\,dx$.

a. Explain why the substitution $u = \sin(x)$ will not work to help evaluate the given integral.

b. Recall the Fundamental Trigonometric Identity, which states that $\sin^2(x) + \cos^2(x) = 1$. By observing that $\sin^3(x) = \sin(x)\cdot\sin^2(x)$, use the Fundamental

Trigonometric Identity to rewrite the integrand as the product of $\sin(x)$ with another function.

c. Explain why the substitution $u = \cos(x)$ now provides a possible way to evaluate the integral in (b).

d. Use your work in (a)-(c) to evaluate the indefinite integral $\int \sin^3(x)\,dx$.

e. Use a similar approach to evaluate $\int \cos^3(x)\,dx$.

14. For the town of Mathland, MI, residential power consumption has shown certain trends over recent years. Based on data reflecting average usage, engineers at the power company have modeled the town's rate of energy consumption by the function

$$r(t) = 4 + \sin(0.263t + 4.7) + \cos(0.526t + 9.4).$$

Here, t measures time in hours after midnight on a typical weekday, and r is the rate of consumption in megawatts at time t. Units are critical throughout this problem; note that the unit *megawatt* is itself a rate, which measures energy consumption per unit time. A *megawatt-hour* is the total amount of energy that is equivalent to a constant stream of 1 megawatt of power being sustained for 1 hour.

a. Sketch a carefully labeled graph of $r(t)$ on the interval [0,24] and explain its meaning. Why is this a reasonable model of power consumption?

b. Without calculating its value, explain the meaning of $\int_0^{24} r(t)\,dt$. Include appropriate units on your answer.

c. Determine the exact amount of energy Mathland consumes in a typical day.

d. What is Mathland's average rate of power consumption in a given 24-hour period? What are the units on this quantity?

5.4 Integration by parts

Motivating Questions

- How do we evaluate indefinite integrals that involve products of basic functions such as $\int x \sin(x)\, dx$ and $\int xe^x\, dx$?
- What is the method of integration by parts and how can we consistently apply it to integrate products of basic functions?
- How does the algebraic structure of functions guide us in identifying u and dv in using integration by parts?

5.4.1 Introduction

In Section 5.3, we learned the technique of u-substitution for evaluating indefinite integrals. For example, the indefinite integral $\int x^3 \sin(x^4)\, dx$ is perfectly suited to u-substitution, because one factor is a composite function and the other factor is the derivative (up to a constant) of the inner function. Recognizing the algebraic structure of a function can help us to find its antiderivative.

Next we consider integrands with a different elementary algebraic structure: a product of basic functions. For instance, suppose we are interested in evaluating the indefinite integral

$$\int x \sin(x)\, dx.$$

The integrand is the product of the basic functions $f(x) = x$ and $g(x) = \sin(x)$. We know that it is relatively complicated to compute the derivative of the product of two functions, so we should expect that antidifferentiating a product should be similarly involved. Intuitively, we expect that evaluating $\int x \sin(x)\, dx$ will involve somehow reversing the Product Rule.

To that end, in Preview Activity 5.4.1 we refresh our understanding of the Product Rule and then investigate some indefinite integrals that involve products of basic functions.

Preview Activity 5.4.1. In Section 2.3, we developed the Product Rule and studied how it is employed to differentiate a product of two functions. In particular, recall that if f and g are differentiable functions of x, then

$$\frac{d}{dx}\left[f(x) \cdot g(x)\right] = f(x) \cdot g'(x) + g(x) \cdot f'(x).$$

(a) For each of the following functions, use the Product Rule to find the function's derivative. Be sure to label each derivative by name (e.g., the derivative of $g(x)$ should be labeled $g'(x)$).

i. $g(x) = x\sin(x)$
ii. $h(x) = xe^x$
iii. $p(x) = x\ln(x)$
iv. $q(x) = x^2\cos(x)$
v. $r(x) = e^x\sin(x)$

(b) Use your work in (a) to help you evaluate the following indefinite integrals. Use differentiation to check your work.

i. $\displaystyle\int xe^x + e^x\,dx$
ii. $\displaystyle\int e^x(\sin(x) + \cos(x))\,dx$
iii. $\displaystyle\int 2x\cos(x) - x^2\sin(x)\,dx$
iv. $\displaystyle\int x\cos(x) + \sin(x)\,dx$
v. $\displaystyle\int 1 + \ln(x)\,dx$

(c) Observe that the examples in (b) work nicely because of the derivatives you were asked to calculate in (a). Each integrand in (b) is precisely the result of differentiating one of the products of basic functions found in (a). To see what happens when an integrand is still a product but not necessarily the result of differentiating an elementary product, we consider how to evaluate

$$\int x\cos(x)\,dx.$$

i. First, observe that

$$\frac{d}{dx}[x\sin(x)] = x\cos(x) + \sin(x).$$

Integrating both sides indefinitely and using the fact that the integral of a sum is the sum of the integrals, we find that

$$\int \left(\frac{d}{dx}[x\sin(x)]\right)dx = \int x\cos(x)\,dx + \int \sin(x)\,dx.$$

In this last equation, evaluate the indefinite integral on the left side as well as the rightmost indefinite integral on the right.

ii. In the most recent equation from (i.), solve the equation for the expression $\int x\cos(x)\,dx$.

iii. For which product of basic functions have you now found the antiderivative?

5.4.2 Reversing the Product Rule: Integration by Parts

Problem (c) in Preview Activity 5.4.1 provides a clue to the general technique known as Integration by Parts, which comes from reversing the Product Rule. Recall that the Product

Rule states that

$$\frac{d}{dx}\left[f(x)g(x)\right] = f(x)g'(x) + g(x)f'(x).$$

Integrating both sides of this equation indefinitely with respect to x, we find

$$\int \frac{d}{dx}\left[f(x)g(x)\right]\,dx = \int f(x)g'(x)\,dx + \int g(x)f'(x)\,dx. \tag{5.4.1}$$

On the left side of Equation (5.4.1), we have the indefinite integral of the derivative of a function. Temporarily omitting the constant that may arise, we have

$$f(x)g(x) = \int f(x)g'(x)\,dx + \int g(x)f'(x)\,dx. \tag{5.4.2}$$

We solve for the first indefinite integral on the left to generate the rule

$$\int f(x)g'(x)\,dx = f(x)g(x) - \int g(x)f'(x)\,dx. \tag{5.4.3}$$

Often we express Equation (5.4.3) in terms of the variables u and v, where $u = f(x)$ and $v = g(x)$. In differential notation, $du = f'(x)\,dx$ and $dv = g'(x)\,dx$, so we can state the rule for Integration by Parts in its most common form as follows:

Integration by parts.

$$\int u\,dv = uv - \int v\,du.$$

To apply integration by parts, we look for a product of basic functions that we can identify as u and dv. If we can antidifferentiate dv to find v, and evaluating $\int v\,du$ is not more difficult than evaluating $\int u\,dv$, then this substitution usually proves to be fruitful. To demonstrate, we consider the following example.

Example 5.4.1 Evaluate the indefinite integral

$$\int x\cos(x)\,dx$$

using integration by parts.

Solution. When we use integration by parts, we have a choice for u and dv. In this problem, we can either let $u = x$ and $dv = \cos(x)\,dx$, or let $u = \cos(x)$ and $dv = x\,dx$. While there is not a universal rule for how to choose u and dv, a good guideline is this: do so in a way that $\int v\,du$ is at least as simple as the original problem $\int u\,dv$.

This leads us to choose[1] $u = x$ and $dv = \cos(x)\,dx$, from which it follows that $du = 1\,dx$ and $v = \sin(x)$. With this substitution, the rule for integration by parts tells us that

$$\int x\cos(x)\,dx = x\sin(x) - \int \sin(x)\cdot 1\,dx.$$

All that remains to do is evaluate the (simpler) integral $\int \sin(x) \cdot 1\,dx$. Doing so, we find

$$\int x\cos(x)\,dx = x\sin(x) - (-\cos(x)) + C = x\sin(x) + \cos(x) + C.$$

Observe that when we get to the final stage of evaluating the last remaining antiderivative, it is at this step that we include the integration constant, $+C$. ◊

The general technique of integration by parts involves trading the problem of integrating the product of two functions for the problem of integrating the product of two related functions. That is, we convert the problem of evaluating $\int u\,dv$ to that of evaluating $\int v\,du$. This clearly shapes our choice of u and v. In Example 5.4.1, the original integral to evaluate was $\int x\cos(x)\,dx$, and through the substitution provided by integration by parts, we were instead able to evaluate $\int \sin(x) \cdot 1\,dx$. Note that the original function x was replaced by its derivative, while $\cos(x)$ was replaced by its antiderivative.

Activity 5.4.2. Evaluate each of the following indefinite integrals. Check each antiderivative that you find by differentiating.

(a) $\int te^{-t}\,dt$

(b) $\int 4x\sin(3x)\,dx$

(c) $\int z\sec^2(z)\,dz$

(d) $\int x\ln(x)\,dx$

5.4.3 Some Subtleties with Integration by Parts

Sometimes integration by parts is not an obvious choice, but the technique is appropriate nonetheless. Integration by parts allows us to replace one function in a product with its derivative while replacing the other with its antiderivative. For instance, consider evaluating

$$\int \arctan(x)\,dx.$$

Initially, this problem seems ill-suited to integration by parts, since there does not appear to be a product of functions present. But if we note that $\arctan(x) = \arctan(x)\cdot 1$, and realize that we know the derivative of $\arctan(x)$ as well as the antiderivative of 1, we see the possibility for the substitution $u = \arctan(x)$ and $dv = 1\,dx$. We explore this substitution further in Activity 5.4.3.

In a related problem, consider $\int t^3\sin(t^2)\,dt$. Observe that there is a composite function present in $\sin(t^2)$, but there is not an obvious function-derivative pair, as we have t^3 (rather

[1]Observe that if we considered the alternate choice, and let $u = \cos(x)$ and $dv = x\,dx$, then $du = -\sin(x)\,dx$ and $v = \frac{1}{2}x^2$, from which we would write $\int x\cos(x)\,dx = \frac{1}{2}x^2\cos(x) - \int \frac{1}{2}x^2(-\sin(x))\,dx$. Thus we have replaced the problem of integrating $x\cos(x)$ with that of integrating $\frac{1}{2}x^2\sin(x)$; the latter is clearly more complicated, which shows that this alternate choice is not as helpful as the first choice.

than simply t) multiplying $\sin(t^2)$. In this problem we use both u-substitution and integration by parts. First we write $t^3 = t \cdot t^2$ and consider the indefinite integral

$$\int t \cdot t^2 \cdot \sin(t^2)\,dt.$$

We let $z = t^2$ so that $dz = 2t\,dt$, and thus $t\,dt = \frac{1}{2}\,dz$. (We are using the variable z to perform a "z-substitution" first so that we may then apply integration by parts.) Under this z-substitution, we now have

$$\int t \cdot t^2 \cdot \sin(t^2)\,dt = \int z \cdot \sin(z) \cdot \frac{1}{2}\,dz.$$

The resulting integral can be evaluated by parts. This, too, is explored further in Activity 5.4.3.

These problems show that we sometimes must think creatively in choosing the variables for substitution in integration by parts, and that we may need to use substitution for an additional change of variables.

Activity 5.4.3. Evaluate each of the following indefinite integrals, using the provided hints.

(a) Evaluate $\int \arctan(x)\,dx$ by using integration by parts with the substitution $u = \arctan(x)$ and $dv = 1\,dx$.

(b) Evaluate $\int \ln(z)\,dz$. Consider a similar substitution to the one in (a).

(c) Use the substitution $z = t^2$ to transform the integral $\int t^3 \sin(t^2)\,dt$ to a new integral in the variable z, and evaluate that new integral by parts.

(d) Evaluate $\int s^5 e^{s^3}\,ds$ using an approach similar to that described in (c).

(e) Evaluate $\int e^{2t} \cos(e^t)\,dt$. You will find it helpful to note that $e^{2t} = e^t \cdot e^t$.

5.4.4 Using Integration by Parts Multiple Times

Integration by parts is well suited to integrating the product of basic functions, allowing us to trade a given integrand for a new one where one function in the product is replaced by its derivative, and the other is replaced by its antiderivative. The goal in this trade of $\int u\,dv$ for $\int v\,du$ is that the new integral be simpler to evaluate than the original one. Sometimes it is necessary to apply integration by parts more than once in order to evaluate a given integral.

Example 5.4.2 Evaluate $\int t^2 e^t\,dt$.

Solution. Let $u = t^2$ and $dv = e^t\,dt$. Then $du = 2t\,dt$ and $v = e^t$, and thus

$$\int t^2 e^t\,dt = t^2 e^t - \int 2te^t\,dt.$$

The integral on the right side is simpler to evaluate than the one on the left, but it still requires integration by parts. Now letting $u = 2t$ and $dv = e^t\,dt$, we have $du = 2\,dt$ and $v = e^t$, so that

$$\int t^2 e^t\,dt = t^2 e^t - \left(2te^t - \int 2e^t\,dt\right).$$

(Note the parentheses, which remind us to distribute the minus sign to the entire value of the integral $\int 2te^t\,dt$.) The final integral on the right is a basic one; evaluating that integral and distributing the minus sign, we find

$$\int t^2 e^t\,dt = t^2 e^t - 2te^t + 2e^t + C.$$

◊

Of course, even more than two applications of integration by parts may be necessary. In the preceding example, if the integrand had been $t^3 e^t$, we would have had to use integration by parts three times.

Next, we consider the slightly different scenario.

Example 5.4.3 Evaluate $\int e^t \cos(t)\,dt$.

Solution. We can choose to let u be either e^t or $\cos(t)$; we pick $u = \cos(t)$, and thus $dv = e^t\,dt$. With $du = -\sin(t)\,dt$ and $v = e^t$, integration by parts tells us that

$$\int e^t \cos(t)\,dt = e^t \cos(t) - \int e^t(-\sin(t))\,dt,$$

or equivalently that

$$\int e^t \cos(t)\,dt = e^t \cos(t) + \int e^t \sin(t)\,dt. \tag{5.4.4}$$

The new integral has the same algebraic structure as the original one. While the overall situation isn't necessarily better than what we started with, it hasn't gotten worse. Thus, we proceed to integrate by parts again. This time we let $u = \sin(t)$ and $dv = e^t\,dt$, so that $du = \cos(t)\,dt$ and $v = e^t$, which implies

$$\int e^t \cos(t)\,dt = e^t \cos(t) + \left(e^t \sin(t) - \int e^t \cos(t)\,dt\right). \tag{5.4.5}$$

We seem to be back where we started, as two applications of integration by parts has led us back to the original problem, $\int e^t \cos(t)\,dt$. But if we look closely at Equation (5.4.5), we see that we can use algebra to solve for the value of the desired integral. Adding $\int e^t \cos(t)\,dt$ to both sides of the equation, we have

$$2\int e^t \cos(t)\,dt = e^t \cos(t) + e^t \sin(t),$$

and therefore

$$\int e^t \cos(t)\,dt = \frac{1}{2}\left(e^t \cos(t) + e^t \sin(t)\right) + C.$$

Note that since we never actually encountered an integral we could evaluate directly, we didn't have the opportunity to add the integration constant C until the final step. ◊

Activity 5.4.4. Evaluate each of the following indefinite integrals.

(a) $\int x^2 \sin(x)\, dx$

(b) $\int t^3 \ln(t)\, dt$

(c) $\int e^z \sin(z)\, dz$

(d) $\int s^2 e^{3s}\, ds$

(e) $\int t \arctan(t)\, dt$ (*Hint:* At a certain point in this problem, it is very helpful to note that $\frac{t^2}{1+t^2} = 1 - \frac{1}{1+t^2}$.)

5.4.5 Evaluating Definite Integrals Using Integration by Parts

We can use the technique of integration by parts to evaluate a definite integral as well.

Example 5.4.4 Evaluate

$$\int_0^{\pi/2} t \sin(t)\, dt.$$

Solution. One option is to find an antiderivative (using indefinite integral notation) and then apply the Fundamental Theorem of Calculus to find that

$$\begin{aligned}
\int_0^{\pi/2} t \sin(t)\, dt &= (-t\cos(t) + \sin(t))\Big|_0^{\pi/2} \\
&= \left(-\frac{\pi}{2}\cos(\frac{\pi}{2}) + \sin(\frac{\pi}{2})\right) - (-0\cos(0) + \sin(0)) \\
&= 1.
\end{aligned}$$

Alternatively, we can apply integration by parts and work with definite integrals throughout. With this method, we must remember to evaluate the product uv over the given limits of integration. Using the substitution $u = t$ and $dv = \sin(t)\, dt$, so that $du = dt$ and $v = -\cos(t)$, we write

$$\begin{aligned}
\int_0^{\pi/2} t \sin(t)\, dt &= -t\cos(t)\Big|_0^{\pi/2} - \int_0^{\pi/2} (-\cos(t))\, dt \\
&= -t\cos(t)\Big|_0^{\pi/2} + \sin(t)\Big|_0^{\pi/2} \\
&= \left(-\frac{\pi}{2}\cos(\frac{\pi}{2}) + \sin(\frac{\pi}{2})\right) - (-0\cos(0) + \sin(0)) \\
&= 1.
\end{aligned}$$

◊

As with any substitution technique, it is important to use notation carefully and completely, and to ensure that the end result makes sense.

5.4.6 When u-substitution and Integration by Parts Fail to Help

Both integration techniques we have discussed apply in relatively limited circumstances. It is not hard to find examples of functions for which neither technique produces an antiderivative; indeed, there are many, many functions that appear elementary but that do not have an elementary algebraic antiderivative. For instance, neither u-substitution nor integration by parts proves fruitful for the indefinite integrals

$$\int e^{x^2}\,dx \text{ and } \int x\tan(x)\,dx.$$

While there are other integration techniques, some of which we will consider briefly, none of them enables us to find an algebraic antiderivative for e^{x^2} or $x\tan(x)$. We do know from the Second Fundamental Theorem of Calculus that we can construct an integral antiderivative for each function; $F(x) = \int_0^x e^{t^2}\,dt$ is an antiderivative of $f(x) = e^{x^2}$, and $G(x) = \int_0^x t\tan(t)\,dt$ is an antiderivative of $g(x) = x\tan(x)$. But finding an elementary algebraic formula that doesn't involve integrals for either F or G turns out not only to be impossible through u-substitution or integration by parts, but indeed impossible altogether. Antidifferentiation is much harder in general than differentiation.

5.4.7 Summary

- Through the method of integration by parts, we can evaluate indefinite integrals that involve products of basic functions such as $\int x\sin(x)\,dx$ and $\int x\ln(x)\,dx$. Using a substitution enables us to trade one of the functions in the product for its derivative, and the other for its antiderivative, in an effort to find a different product of functions that is easier to integrate.
- If the algebraic structure of an integrand is a product of basic functions in the form $\int f(x)g'(x)\,dx$, we can use the substitution $u = f(x)$ and $dv = g'(x)\,dx$ and apply the rule
$$\int u\,dv = uv - \int v\,du$$
to evaluate the original integral $\int f(x)g'(x)\,dx$ by instead evaluating
$$\int v\,du = \int f'(x)g(x)\,dx.$$
- When deciding to integrate by parts, we have to select both u and dv. That selection is guided by the overall principle that the new integral $\int v\,du$ not be more difficult than the original integral $\int u\,dv$. In addition, it is often helpful to recognize if one of the functions present is much easier to differentiate than antidifferentiate (such as $\ln(x)$), in which case that function often is best assigned the variable u. In addition, dv must be a function that we can antidifferentiate.

5.4.8 Exercises

1. Calculate the following integral:

$$\int x\cos(2x)\,dx$$

We let $u =$ __________ and $dv =$ __________

So $du =$ __________ and $v =$ __________

and then use the integration-by-parts formula to find that

$\int x\cos(2x)\,dx =$ _______________ - $\int$ _______________

and finally, we finish evaluating to get that $\int x\cos(2x)dx =$ ______________________________

2. Calculate the following integral:

$$\int x\sec^2(-3x)\ dx$$

We let $u =$ __________ and $dv =$ __________

So $du =$ __________ and $v =$ __________

and then use the integration-by-parts formula to find that

$\int \sec^2(-3x)\,x\,dx =$ _______________ - $\int$ _______________

and finally, we finish evaluating to get that $\int \sec^2(-3x)\,x\,dx =$ ______________________________

3. Calculate the following integral:

$$\int x^{10}\ln(4x)\ dx$$

We let $u =$ __________ and $dv =$ __________

So $du =$ __________ and $v =$ __________

and then use the integration-by-parts formula to find that

$\int x^{10}\ln(4x)\ dx =$ _______________ - $\int$ _______________

and finally, we finish evaluating to get that $\int x^{10}\ln(4x)\ dx =$ ______________________________

4. Evaluate the definite integral.

$\int_0^3 te^{-t}\,dt =$ ______________________

5. Find the integral

$\int y\sqrt[3]{y+2}\,dy =$ ______________________

6. Find the integral

$\int (z+1)\,e^{6z}\,dz =$ ______________________

7. Evaluate the indefinite integral.

$$\int x\arctan(6x)dx$$

Answer: ______________________ $+C$

8. Find the integral

$\int 2t^2\,e^t\,dt =$ ______________________

9. Use integration by parts to evaluate the integral.

$\int (\ln(7x))^2 dx =$ ______________________ $+C$

10. If $g(1) = -5$, $g(5) = -2$, and $\int_1^5 g(x)\,dx = -6$, evaluate the integral $\int_1^5 xg'(x)\,dx$.

Answer: ______________

11. Evaluate the indefinite integral.

$\int x^3\cos\left(x^2\right)\;dx =$ ______________________ $+C$.

Hint: First make a substitution and then use integration by parts to evaluate the integral.

12. For each of the following integrals, indicate whether integration by substitution or integration by parts is more appropriate, or if neither method is appropriate. Do not evaluate the integrals.

1. $\int x\sin x\,dx$

- ○ neither
- ○ integration by parts
- ○ substitution

2. $\int \frac{x^3}{1+x^4}\,dx$

- ○ integration by parts

- ○ neither
- ○ substitution

3. $\int x^3 e^{x^4}\, dx$

- ○ substitution
- ○ integration by parts
- ○ neither

4. $\int x^3 \cos(x^4)\, dx$

- ○ substitution
- ○ neither
- ○ integration by parts

5. $\int \frac{1}{\sqrt{3x+1}}\, dx$

- ○ neither
- ○ integration by parts
- ○ substitution

(Note that because this is multiple choice, you will not be able to see which parts of the problem you got correct.)

13. Let $f(t) = te^{-2t}$ and $F(x) = \int_0^x f(t)\, dt$.

a. Determine $F'(x)$.

b. Use the First FTC to find a formula for F that does not involve an integral.

c. Is F an increasing or decreasing function for $x > 0$? Why?

14. Consider the indefinite integral given by $\int e^{2x} \cos(e^x)\, dx$.

a. Noting that $e^{2x} = e^x \cdot e^x$, use the substitution $z = e^x$ to determine a new, equivalent integral in the variable z.

b. Evaluate the integral you found in (a) using an appropriate technique.

c. How is the problem of evaluating $\int e^{2x} \cos(e^{2x})\, dx$ different from evaluating the integral in (a)? Do so.

d. Evaluate each of the following integrals as well, keeping in mind the approach(es) used earlier in this problem:

- $\displaystyle\int e^{2x} \sin(e^x)\,dx$
- $\displaystyle\int e^{3x} \sin(e^{3x})\,dx$
- $\displaystyle\int xe^{x^2} \cos(e^{x^2}) \sin(e^{x^2})\,dx$

15. For each of the following indefinite integrals, determine whether you would use u-substitution, integration by parts, neither$*$, or both to evaluate the integral. In each case, write one sentence to explain your reasoning, and include a statement of any substitutions used. (That is, if you decide in a problem to let $u = e^{3x}$, you should state that, as well as that $du = 3e^{3x}\,dx$.) Finally, use your chosen approach to evaluate each integral. ($*$ one of the following problems does not have an elementary antiderivative and you are not expected to actually evaluate this integral; this will correspond with a choice of "neither" among those given.)

a. $\displaystyle\int x^2 \cos(x^3)\,dx$

b. $\int x^5 \cos(x^3)\,dx$ (Hint: $x^5 = x^2 \cdot x^3$)

c. $\displaystyle\int x \ln(x^2)\,dx$

d. $\displaystyle\int \sin(x^4)\,dx$

e. $\displaystyle\int x^3 \sin(x^4)\,dx$

f. $\displaystyle\int x^7 \sin(x^4)\,dx$

5.5 Other options for finding algebraic antiderivatives

Motivating Questions

- How does the method of partial fractions enable any rational function to be antidifferentiated?
- What role have integral tables historically played in the study of calculus and how can a table be used to evaluate integrals such as $\int \sqrt{a^2+u^2}\,du$?
- What role can a computer algebra system play in the process of finding antiderivatives?

5.5.1 Introduction

We have learned two antidifferentiation techniques: u-substitution and integration by parts. The former is used to reverse the chain rule, while the latter to reverse the product rule. But we have seen that each works only in specialized circumstances. For example, while $\int xe^{x^2}\,dx$ may be evaluated by u-substitution and $\int xe^x\,dx$ by integration by parts, neither method provides a route to evaluate $\int e^{x^2}\,dx$, and in fact an elementary algebraic antiderivative for e^{x^2} does not exist. No antidifferentiation method will provide us with a simple algebraic formula for a function $F(x)$ that satisfies $F'(x) = e^{x^2}$.

In this section of the text, our main goals are to identify some classes of functions that can be antidifferentiated, and to learn some methods to do so. We should also recognize that there are many functions for which an algebraic formula for an antiderivative does not exist, and appreciate the role that computing technology can play in finding antiderivatives of other complicated functions.

Preview Activity 5.5.1. For each of the indefinite integrals below, the main question is to decide whether the integral can be evaluated using u-substitution, integration by parts, a combination of the two, or neither. For integrals for which your answer is affirmative, state the substitution(s) you would use. It is not necessary to actually evaluate any of the integrals completely, unless the integral can be evaluated immediately using a familiar basic antiderivative.

(a) $\int x^2\sin(x^3)\,dx$, $\int x^2\sin(x)\,dx$, $\int \sin(x^3)\,dx$, $\int x^5\sin(x^3)\,dx$

(b) $\int \frac{1}{1+x^2}\,dx$, $\int \frac{x}{1+x^2}\,dx$, $\int \frac{2x+3}{1+x^2}\,dx$, $\int \frac{e^x}{1+(e^x)^2}\,dx$,

(c) $\int x\ln(x)\,dx$, $\int \frac{\ln(x)}{x}\,dx$, $\int \ln(1+x^2)\,dx$, $\int x\ln(1+x^2)\,dx$,

(d) $\int x\sqrt{1-x^2}\,dx$, $\int \frac{1}{\sqrt{1-x^2}}\,dx$, $\int \frac{x}{\sqrt{1-x^2}}\,dx$, $\int \frac{1}{x\sqrt{1-x^2}}\,dx$,

5.5.2 The Method of Partial Fractions

The method of partial fractions is used to integrate rational functions. It involves reversing the process of finding a common denominator.

Example 5.5.1 Evaluate

$$\int \frac{5x}{x^2 - x - 2}\,dx.$$

Solution. If we factor the denominator, we can see how R might be the sum of two fractions of the form $\frac{A}{x-2} + \frac{B}{x+1}$, so we suppose that

$$\frac{5x}{(x-2)(x+1)} = \frac{A}{x-2} + \frac{B}{x+1}$$

and look for the constants A and B.

Multiplying both sides of this equation by $(x-2)(x+1)$, we find that

$$5x = A(x+1) + B(x-2).$$

Since we want this equation to hold for every value of x, we can use insightful choices of specific x-values to help us find A and B. Taking $x = -1$, we have

$$5(-1) = A(0) + B(-3),$$

so $B = \frac{5}{3}$. Choosing $x = 2$, it follows

$$5(2) = A(3) + B(0),$$

so $A = \frac{10}{3}$. Thus,

$$\int \frac{5x}{x^2 - x - 2}\,dx = \int \frac{10/3}{x-2} + \frac{5/3}{x+1}\,dx.$$

Recalling the fact that $\ln(|u|)$ is an antiderivative of $f(u) = \frac{1}{u}$ for all $u \neq 0$, this integral is straightforward to evaluate, and hence we find that

$$\int \frac{5x}{x^2 - x - 2}\,dx = \frac{10}{3}\ln|x-2| + \frac{5}{3}\ln|x+1| + C.$$

Note: as we discussed briefly in Section 4.4 immediately following the development of Table 4.4.5 in Activity 4.4.3, there is a subtle issue with writing

$$\int \frac{1}{u}\,du = \ln(|u|) + C$$

due to the fact that $\frac{1}{u}$ has a discontinuity at $u = 0$, which leads to an antiderivative for one piece of the function and another antiderivative for the other. In most applications (such as when we use partial fractions to solve certain differential equations such as the Logistic Equation in Section 7.6), the domain of input values on which we seek the antiderivative leads to a natural resolution of this ambiguity, so we will still write $\int \frac{1}{u}\,du = \ln(|u|) + C$. ◊

It turns out that we can use the method of partial fractions, together with u-substitution and other algebraic techniques,[1] to rewrite any rational function $R(x) = \frac{P(x)}{Q(x)}$ where the degree of the polynomial P is less than[2] the degree of Q as a sum of simpler rational functions of one of the following forms:

$$\frac{A}{x-c}, \quad \frac{A}{(x-c)^n}, \quad \frac{Ax+B}{x^2+k}, \text{ or } \frac{Ax+B}{(x^2+k)^n}$$

where A, B, and c are real numbers, and k is a positive real number. Because we can antidifferentiate each of these basic forms, partial fractions enables us to antidifferentiate any rational function.

A computer algebra system such as *Maple*, *Mathematica*, or *WolframAlpha* can be used to find the partial fraction decomposition of any rational function. In *WolframAlpha*, entering

```
partial fraction 5x/(x^2-x-2)
```

results in the output

$$\frac{5x}{x^2-x-2} = \frac{10}{3(x-2)} + \frac{5}{3(x+1)}.$$

We will use technology to generate partial fraction decompositions of rational functions, and then evaluate the integrals using established methods.

Activity 5.5.2. For each of the following problems, evaluate the integral by using the partial fraction decomposition provided.

(a) $\int \frac{1}{x^2-2x-3}\,dx$, given that $\frac{1}{x^2-2x-3} = \frac{1/4}{x-3} - \frac{1/4}{x+1}$

(b) $\int \frac{x^2+1}{x^3-x^2}\,dx$, given that $\frac{x^2+1}{x^3-x^2} = -\frac{1}{x} - \frac{1}{x^2} + \frac{2}{x-1}$

(c) $\int \frac{x-2}{x^4+x^2}\,dx$, given that $\frac{x-2}{x^4+x^2} = \frac{1}{x} - \frac{2}{x^2} + \frac{-x+2}{1+x^2}$

5.5.3 Using an Integral Table

Calculus has a long history, going back to Greek mathematicians in 400-300 BC. Its main foundations were first investigated and understood independently by Isaac Newton and Gottfried Wilhelm Leibniz in the late 1600s, making the modern ideas of calculus well over 300 years old. It is instructive to realize that until the late 1980s, the personal computer did not exist, so calculus (and other mathematics) had to be done by hand for roughly 300 years. In the 21st century, however, computers have revolutionized many aspects of the world we live in, including mathematics. In this section we take a short historical tour to precede

[1]Specifically, if we encounter a quadratic function of the form x^2+bx+c that we cannot factor, we complete the square to rewrite it as $(x-h)^2+k$. Then the substitution $u = x-h$ enables us to antidifferentiate the result.

[2]If the degree of P is greater than or equal to the degree of Q, long division may be used to write R as the sum of a polynomial plus a rational function where the numerator's degree is less than the denominator's.

discussing the role computer algebra systems can play in evaluating indefinite integrals. In particular, we consider a class of integrals involving certain radical expressions.

As seen in the short table of integrals found in Appendix A, there are many forms of integrals that involve $\sqrt{a^2 \pm w^2}$ and $\sqrt{w^2 - a^2}$. These integral rules can be developed using a technique known as *trigonometric substitution* that we choose to omit; instead, we will simply accept the results presented in the table. To see how these rules are used, consider the differences among

$$\int \frac{1}{\sqrt{1-x^2}}\,dx, \quad \int \frac{x}{\sqrt{1-x^2}}\,dx, \quad \text{and} \quad \int \sqrt{1-x^2}\,dx.$$

The first integral is a familiar basic one, and results in $\arcsin(x) + C$. The second integral can be evaluated using a standard u-substitution with $u = 1 - x^2$. The third, however, is not familiar and does not lend itself to u-substitution.

In Appendix A, we find the rule

$$(h) \quad \int \sqrt{a^2 - u^2}\,du = \frac{u}{2}\sqrt{a^2 - u^2} + \frac{a^2}{2}\arcsin\left(\frac{u}{a}\right) + C.$$

Using the substitutions $a = 1$ and $u = x$ (so that $du = dx$), it follows that

$$\int \sqrt{1-x^2}\,dx = \frac{x}{2}\sqrt{1-x^2} - \frac{1}{2}\arcsin(x) + C.$$

Whenever we are applying a rule in the table, we are doing a u-substitution, especially when the substitution is more complicated than setting $u = x$ as in the last example.

Example 5.5.2 Evaluate the integral

$$\int \sqrt{9 + 64x^2}\,dx.$$

Solution. Here, we want to use Rule (c) from the table, but we now set $a = 3$ and $u = 8x$. We also choose the "+" option in the rule. With this substitution, it follows that $du = 8dx$, so $dx = \frac{1}{8}du$. Applying the substitution,

$$\int \sqrt{9 + 64x^2}\,dx = \int \sqrt{9 + u^2} \cdot \frac{1}{8}\,du = \frac{1}{8}\int \sqrt{9 + u^2}\,du.$$

By Rule (c), we now find that

$$\begin{aligned}\int \sqrt{9 + 64x^2}\,dx &= \frac{1}{8}\left(\frac{u}{2}\sqrt{u^2+9} + \frac{9}{2}\ln|u + \sqrt{u^2+9}| + C\right) \\ &= \frac{1}{8}\left(\frac{8x}{2}\sqrt{64x^2+9} + \frac{9}{2}\ln|8x + \sqrt{64x^2+9}| + C\right).\end{aligned}$$

◊

Whenever we use a u-subsitution in conjunction with Appendix A, it's important that we not forget to address any constants that arise and include them in our computations, such as the $\frac{1}{8}$ that appeared in Example 5.5.2.

Activity 5.5.3. For each of the following integrals, evaluate the integral using u-substitution and/or an entry from the table found in Appendix A.

(a) $\int \sqrt{x^2+4}\,dx$

(b) $\int \frac{x}{\sqrt{x^2+4}}\,dx$

(c) $\int \frac{2}{\sqrt{16+25x^2}}\,dx$

(d) $\int \frac{1}{x^2\sqrt{49-36x^2}}\,dx$

5.5.4 Using Computer Algebra Systems

A computer algebra system (CAS) is a computer program that is capable of executing symbolic mathematics. For example, if we ask a CAS to solve the equation $ax^2+bx+c=0$ for the variable x, where a, b, and c are arbitrary constants, the program will return $x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$. Research to develop the first CAS dates to the 1960s, and these programs became publicly available in the early 1990s. Two prominent examples are the programs *Maple* and *Mathematica*, which were among the first computer algebra systems to offer a graphical user interface. Today, *Maple* and *Mathematica* are exceptionally powerful professional software packages that can execute an amazing array of sophisticated mathematical computations. They are also very expensive, as each is a proprietary program. The CAS *SAGE* is an open-source, free alternative to *Maple* and *Mathematica*.

For the purposes of this text, when we need to use a CAS, we are going to turn instead to a similar, but somewhat different computational tool, the web-based "computational knowledge engine" called *WolframAlpha*. There are two features of *WolframAlpha* that make it stand out from the CAS options mentioned above: (1) unlike *Maple* and *Mathematica*, *WolframAlpha* is free (provided we are willing to navigate some pop-up advertising); and (2) unlike any of the three, the syntax in *WolframAlpha* is flexible. Think of *WolframAlpha* as being a little bit like doing a Google search: the program will interpret what is input, and then provide a summary of options.

If we want to have *WolframAlpha* evaluate an integral for us, we can provide it syntax such as

```
integrate x^2 dx
```

to which the program responds with

$$\int x^2\,dx = \frac{x^3}{3} + \text{constant}.$$

While there is much to be enthusiastic about regarding CAS programs such as *WolframAlpha*, there are several things we should be cautious about: (1) a CAS only responds to exactly what is input; (2) a CAS can answer using powerful functions from very advanced mathematics; and (3) there are problems that even a CAS cannot do without additional human insight.

Although (1) likely goes without saying, we have to be careful with our input: if we enter syntax that defines the wrong function, the CAS will work with precisely the function we define. For example, if we are interested in evaluating the integral

$$\int \frac{1}{16-5x^2}\,dx,$$

and we mistakenly enter

```
integrate 1/16 - 5x^2 dx
```

a CAS will (correctly) reply with

$$\frac{1}{16}x - \frac{5}{3}x^3.$$

But if we are sufficiently well-versed in antidifferentiation, we will recognize that this function cannot be the one that we seek: integrating a rational function such as $\frac{1}{16-5x^2}$, we expect the logarithm function to be present in the result.

Regarding (2), even for a relatively simple integral such as $\int \frac{1}{16-5x^2}\,dx$, some CASs will invoke advanced functions rather than simple ones. For instance, if we use *Maple* to execute the command

```
int(1/(16-5*x^2), x);
```

the program responds with

$$\int \frac{1}{16-5x^2}\,dx = \frac{\sqrt{5}}{20}\,\text{arctanh}(\frac{\sqrt{5}}{4}x).$$

While this is correct (save for the missing arbitrary constant, which *Maple* never reports), the inverse hyperbolic tangent function is not a common nor familiar one; a simpler way to express this function can be found by using the partial fractions method, and happens to be the result reported by *WolframAlpha*:

$$\int \frac{1}{16-5x^2}\,dx = \frac{1}{8\sqrt{5}}\Big(\log(4\sqrt{5}+5x) - \log(4\sqrt{5}-5x)\Big) + \text{constant}.$$

Using sophisticated functions from more advanced mathematics is sometimes the way a CAS says to the user "I don't know how to do this problem." For example, if we want to evaluate

$$\int e^{-x^2}\,dx,$$

and we ask *WolframAlpha* to do so, the input

```
integrate exp(-x^2) dx
```

results in the output

$$\int e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2}\operatorname{erf}(x) + \text{constant}.$$

The function "erf(x)" is the *error function*, which is actually defined by an integral:

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}\,dt.$$

So, in producing output involving an integral, the CAS has basically reported back to us the very question we asked.

Finally, as remarked at (3) above, there are times that a CAS will actually fail without some additional human insight. If we consider the integral

$$\int (1+x)e^x\sqrt{1+x^2e^{2x}}\,dx$$

and ask *WolframAlpha* to evaluate

```
int (1+x) * exp(x) * sqrt(1+x^2 * exp(2x)) dx,
```

the program thinks for a moment and then reports

(*no result found in terms of standard mathematical functions*)

But in fact this integral is not that difficult to evaluate. If we let $u = xe^x$, then $du = (1+x)e^x\,dx$, which means that the preceding integral has form

$$\int (1+x)e^x\sqrt{1+x^2e^{2x}}\,dx = \int \sqrt{1+u^2}\,du,$$

which is a straightforward one for any CAS to evaluate.

So, we should proceed with some caution: while any CAS is capable of evaluating a wide range of integrals (both definite and indefinite), there are times when the result can mislead us. We must think carefully about the meaning of the output, whether it is consistent with what we expect, and whether or not it makes sense to proceed.

5.5.5 Summary

- We can antidifferentiate any rational function with the method of partial fractions. Any polynomial function can be factored into a product of linear and irreducible quadratic terms, so any rational function may be written as the sum of a polynomial plus rational terms of the form $\frac{A}{(x-c)^n}$ (where n is a natural number) and $\frac{Bx+C}{x^2+k}$ (where k is a positive real number).

- Until the development of compute algebra systems, integral tables enabled students of calculus to evaluate integrals such as $\int \sqrt{a^2 + u^2}\, du$, where a is a positive real number. A short table of integrals may be found in Appendix A.
- Computer algebra systems can play an important role in finding antiderivatives, though we must be cautious to use correct input, to watch for unusual or unfamiliar advanced functions that the CAS may cite in its result, and to consider the possibility that a CAS may need further assistance or insight from us in order to answer a particular question.

5.5.6 Exercises

1. Calculate the integral below by partial fractions and by using the indicated substitution. Be sure that you can show how the results you obtain are the same.

$$\int \frac{2x}{x^2 - 9}\, dx$$

First, rewrite this with partial fractions:

$\int \frac{2x}{x^2-9}\, dx = \int$ ____________ $dx + \int$ ____________ $dx =$ ____________ $+$ ____________ $+C$.

(Note that you should not include the +C in your entered answer, as it has been provided at the end of the expression.)

Next, use the substitution $w = x^2 - 9$ to find the integral:

$\int \frac{2x}{x^2-9}\, dx = \int$ ____________ $dw =$ ____________ $+C =$ ____________ $+C$.

(For the second answer blank, give your antiderivative in terms of the variable w. Again, note that you should not include the +C in your answer.)

2. Calculate the integral:

$$\int \frac{1}{(x+5)(x+3)}\, dx = \underline{\qquad\qquad\qquad\qquad\qquad\qquad}$$

3. Calculate the integral

$$\int \frac{6x+5}{x^2 - 3x + 2} dx = \underline{\qquad\qquad\qquad\qquad\qquad\qquad}$$

4. Consider the following indefinite integral.

$$\int \frac{5x^3 + 8x^2 + 48x + 80}{x^4 + 16x^2}\, dx$$

The integrand has partial fractions decomposition:

$$\frac{a}{x^2} + \frac{b}{x} + \frac{cx + d}{x^2 + 16}$$

where

$a =$ ____________

$b =$ ____________________

$c =$ ____________________

$d =$ ____________________

Now integrate term by term to evaluate the integral.

Answer: ____ $+C$

5. The form of the partial fraction decomposition of a rational function is given below.

$$\frac{3x - 4x^2 - 27}{(x+3)(x^2+9)} = \frac{A}{x+3} + \frac{Bx+C}{x^2+9}$$

$A =$ __________ $B =$ __________ $C =$ __________

Now evaluate the indefinite integral.

$\displaystyle\int \frac{3x - 4x^2 - 27}{(x+3)(x^2+9)}\,dx =$ ____

6. For each of the following integrals involving rational functions, (1) use a CAS to find the partial fraction decomposition of the integrand; (2) evaluate the integral of the resulting function without the assistance of technology; (3) use a CAS to evaluate the original integral to test and compare your result in (2).

a. $\displaystyle\int \frac{x^3+x+1}{x^4-1}\,dx$

b. $\displaystyle\int \frac{x^5+x^2+3}{x^3-6x^2+11x-6}\,dx$

c. $\displaystyle\int \frac{x^2-x-1}{(x-3)^3}\,dx$

7. For each of the following integrals involving radical functions, (1) use an appropriate u-substitution along with Appendix A to evaluate the integral without the assistance of technology, and (2) use a CAS to evaluate the original integral to test and compare your result in (1).

a. $\displaystyle\int \frac{1}{x\sqrt{9x^2+25}}\,dx$

b. $\displaystyle\int x\sqrt{1+x^4}\,dx$

c. $\displaystyle\int e^x\sqrt{4+e^{2x}}\,dx$

d. $\displaystyle\int \frac{\tan(x)}{\sqrt{9-\cos^2(x)}}\,dx$

8. Consider the indefinite integral given by

$$\int \frac{\sqrt{x+\sqrt{1+x^2}}}{x}\,dx.$$

a. Explain why u-substitution does not offer a way to simplify this integral by discussing at least two different options you might try for u.

b. Explain why integration by parts does not seem to be a reasonable way to proceed, either, by considering one option for u and dv.

c. Is there any line in the integral table in Appendix A that is helpful for this integral?

d. Evaluate the given integral using *WolframAlpha*. What do you observe?

5.6 Numerical integration

Motivating Questions

- How do we accurately evaluate a definite integral such as $\int_0^1 e^{-x^2}\,dx$ when we cannot use the First Fundamental Theorem of Calculus because the integrand lacks an elementary algebraic antiderivative? Are there ways to generate accurate estimates without using extremely large values of n in Riemann sums?
- What is the Trapezoid Rule, and how is it related to left, right, and middle Riemann sums?
- How are the errors in the Trapezoid Rule and Midpoint Rule related, and how can they be used to develop an even more accurate rule?

5.6.1 Introduction

When we first explored finding the net signed area bounded by a curve, we developed the concept of a Riemann sum as a helpful estimation tool and a key step in the definition of the definite integral. Recall that the left, right, and middle Riemann sums of a function f on an interval $[a, b]$ are given by

$$L_n = f(x_0)\Delta x + f(x_1)\Delta x + \cdots + f(x_{n-1})\Delta x = \sum_{i=0}^{n-1} f(x_i)\Delta x, \tag{5.6.1}$$

$$R_n = f(x_1)\Delta x + f(x_2)\Delta x + \cdots + f(x_n)\Delta x = \sum_{i=1}^{n} f(x_i)\Delta x, \tag{5.6.2}$$

$$M_n = f(\overline{x}_1)\Delta x + f(\overline{x}_2)\Delta x + \cdots + f(\overline{x}_n)\Delta x = \sum_{i=1}^{n} f(\overline{x}_i)\Delta x, \tag{5.6.3}$$

where $x_0 = a$, $x_i = a + i\Delta x$, $x_n = b$, and $\Delta x = \frac{b-a}{n}$. For the middle sum, we defined $\overline{x}_i = (x_{i-1} + x_i)/2$.

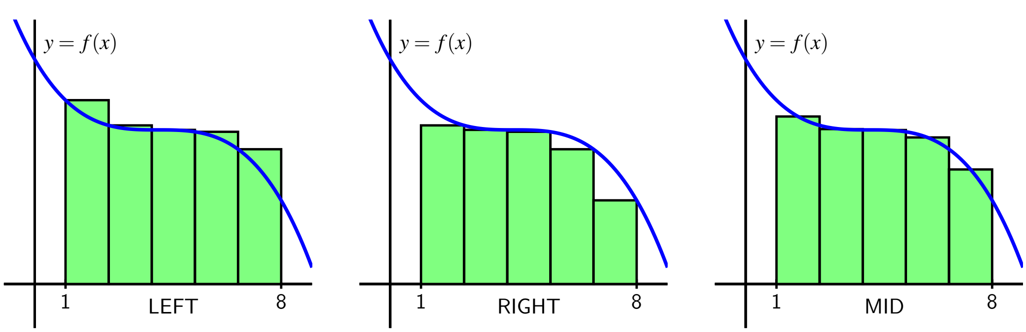


Figure 5.6.1 Left, right, and middle Riemann sums for $y = f(x)$ on $[1, 8]$ with 5 subintervals.

A Riemann sum is a sum of (possibly signed) areas of rectangles. The value of n determines the number of rectangles, and our choice of left endpoints, right endpoints, or midpoints determines the heights of the rectangles. We can see the similarities and differences among these three options in Figure 5.6.1, where we consider the function $f(x) = \frac{1}{20}(x-4)^3 + 7$ on the interval $[1, 8]$, and use 5 rectangles for each of the Riemann sums.

While it is a good exercise to compute a few Riemann sums by hand, just to ensure that we understand how they work and how varying the function, the number of subintervals, and the choice of endpoints or midpoints affects the result, using computing technology is the best way to determine L_n, R_n, and M_n. Any computer algebra system will offer this capability; as we saw in Preview Activity 4.3.1, a straightforward option that is freely available online is this interactive[1]. Note that we can adjust the formula for $f(x)$, the window of x- and y-values of interest, the number of subintervals, and the type of Riemann sum. (See Preview Activity 4.3.1 for any needed reminders on how the interactive works.)

In this section we explore several different alternatives for estimating definite integrals. Our main goal is to develop formulas to estimate definite integrals accurately without using a large numbers of rectangles.

Preview Activity 5.6.1. As we begin to investigate ways to approximate definite integrals, it will be insightful to compare results to integrals whose exact values we know. To that end, the following sequence of questions centers on $\int_0^3 x^2\,dx$.

(a) Use the interactive[2] with the function $f(x) = x^2$ on the window of x values from 0 to 3 to compute L_3, the left Riemann sum with three subintervals.

(b) Likewise, use the interactive to compute R_3 and M_3, the right and middle Riemann sums with three subintervals, respectively.

(c) Use the Fundamental Theorem of Calculus to compute the exact value of $I = \int_0^3 x^2\,dx$.

(d) We define the *error* that results from an approximation of a definite integral to be the approximation's value minus the integral's exact value. What is the error that results from using L_3? From R_3? From M_3?

(e) In what follows in this section, we will learn a new approach to estimating the value of a definite integral known as the Trapezoid Rule. The basic idea is to use trapezoids, rather than rectangles, to estimate the area under a curve. What is the formula for the area of a trapezoid with bases of length b_1 and b_2 and height h?

(f) Working by hand, estimate the area under $f(x) = x^2$ on $[0, 3]$ using three subintervals and three corresponding trapezoids. What is the error in this approximation? How does it compare to the errors you calculated in (d)?

[1] gvsu.edu/s/a9
[2] gvsu.edu/s/a9

5.6.2 The Trapezoid Rule

So far, we have used the simplest possible quadrilaterals (that is, rectangles) to estimate areas. It is natural, however, to wonder if other familiar shapes might serve us even better.

An alternative to L_n, R_n, and M_n is called the *Trapezoid Rule*. Rather than using a rectangle to estimate the (signed) area bounded by $y = f(x)$ on a small interval, we use a trapezoid. For example, in Figure 5.6.2, we estimate the area under the curve using three subintervals and the trapezoids that result from connecting the corresponding points on the curve with straight lines.

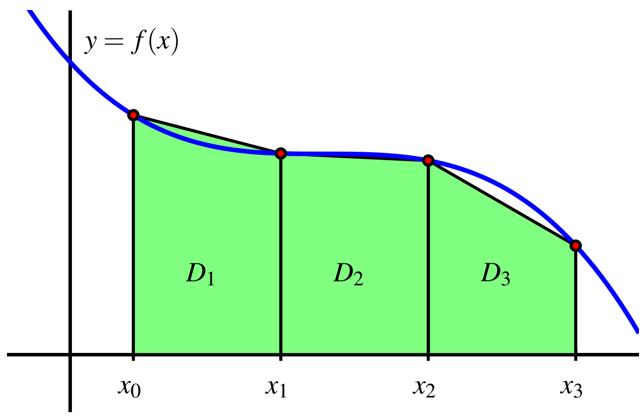


Figure 5.6.2 Estimating $\int_a^b f(x)\,dx$ using three subintervals and trapezoids, rather than rectangles, where $a = x_0$ and $b = x_3$.

The biggest difference between the Trapezoid Rule and a Riemann sum is that on each subinterval, the Trapezoid Rule uses two function values, rather than one, to estimate the (signed) area bounded by the curve. For instance, to compute D_1, the area of the trapezoid on $[x_0, x_1]$, we observe that the left base has length $f(x_0)$, while the right base has length $f(x_1)$. The height of the trapezoid is $x_1 - x_0 = \Delta x = \frac{b-a}{3}$. The area of a trapezoid is the average of the bases times the height, so we have

$$D_1 = \frac{1}{2}(f(x_0) + f(x_1)) \cdot \Delta x.$$

Using similar computations for D_2 and D_3, we find that T_3, the trapezoidal approximation to $\int_a^b f(x)\,dx$ is given by

$$\begin{aligned} T_3 &= D_1 + D_2 + D_3 \\ &= \frac{1}{2}(f(x_0) + f(x_1)) \cdot \Delta x + \frac{1}{2}(f(x_1) + f(x_2)) \cdot \Delta x + \frac{1}{2}(f(x_2) + f(x_3)) \cdot \Delta x. \end{aligned}$$

Because both left and right endpoints are being used, we recognize within the trapezoidal approximation the use of both left and right Riemann sums. Rearranging the expression for

T_3 by removing factors of $\frac{1}{2}$ and Δx, grouping the left endpoint and right endpoint evaluations of f, we see that

$$T_3 = \frac{1}{2}\left[f(x_0) + f(x_1) + f(x_2)\right]\Delta x + \frac{1}{2}\left[f(x_1) + f(x_2) + f(x_3)\right]\Delta x. \tag{5.6.4}$$

We now observe that two familiar sums have arisen. The left Riemann sum L_3 is $L_3 = f(x_0)\Delta x + f(x_1)\Delta x + f(x_2)\Delta x$, and the right Riemann sum is $R_3 = f(x_1)\Delta x + f(x_2)\Delta x + f(x_3)\Delta x$. Substituting L_3 and R_3 for the corresponding expressions in Equation (5.6.4), it follows that $T_3 = \frac{1}{2}[L_3 + R_3]$. We have thus seen a very important result: using trapezoids to estimate the (signed) area bounded by a curve is the same as averaging the estimates generated by using left and right endpoints.

The Trapezoid Rule.

The trapezoidal approximation, T_n, of the definite integral $\int_a^b f(x)\,dx$ using n subintervals is given by the rule

$$\begin{aligned} T_n &= \left[\frac{f(x_0)+f(x_1)}{2} + \frac{f(x_1)+f(x_2)}{2} + \cdots + \frac{f(x_{n-1})+f(x_n)}{2}\right]\Delta x. \\ &= \sum_{i=0}^{n-1} \frac{1}{2}(f(x_i) + f(x_{i+1}))\Delta x. \end{aligned}$$

Moreover, $T_n = \frac{1}{2}[L_n + R_n]$.

Activity 5.6.2. In this activity, we explore the relationships among the errors generated by left, right, midpoint, and trapezoid approximations to the definite integral $\int_1^2 \frac{1}{x^2}\,dx$.

(a) Use the First FTC to evaluate $\int_1^2 \frac{1}{x^2}\,dx$ exactly.

(b) Use appropriate computing technology to compute the following approximations for $\int_1^2 \frac{1}{x^2}\,dx$: T_4, M_4, T_8, and M_8.

(c) Let the *error* that results from an approximation be the approximation's value minus the exact value of the definite integral. For instance, if we let $E_{T,4}$ represent the error that results from using the trapezoid rule with 4 subintervals to estimate the integral, we have

$$E_{T,4} = T_4 - \int_1^2 \frac{1}{x^2}\,dx.$$

Similarly, we compute the error of the midpoint rule approximation with 8 subintervals by the formula

$$E_{M,8} = M_8 - \int_1^2 \frac{1}{x^2}\,dx.$$

Based on your work in (a) and (b) above, compute $E_{T,4}$, $E_{T,8}$, $E_{M,4}$, $E_{M,8}$.

(d) Which rule consistently over-estimates the exact value of the definite integral? Which rule consistently under-estimates the definite integral?

(e) What behavior(s) of the function $f(x) = \frac{1}{x^2}$ lead to your observations in (d)?

5.6.3 Comparing the Midpoint and Trapezoid Rules

We know from the definition of the definite integral that if we let n be large enough, we can make any of the approximations L_n, R_n, and M_n as close as we'd like (in theory) to the exact value of $\int_a^b f(x)\,dx$. Thus, it may be natural to wonder why we ever use any rule other than L_n or R_n (with a sufficiently large n value) to estimate a definite integral. One of the primary reasons is that as $n \to \infty$, $\Delta x = \frac{b-a}{n} \to 0$, and thus in a Riemann sum calculation with a large n value, we end up multiplying by a number that is very close to zero. Doing so often generates roundoff error, because representing numbers close to zero accurately is a persistent challenge for computers.

Hence, we explore ways to estimate definite integrals to high levels of precision, but without using extremely large values of n. Paying close attention to patterns in errors, such as those observed in Activity 5.6.2, is one way to begin to see some alternate approaches.

To begin, we compare the errors in the Midpoint and Trapezoid rules. First, consider a function that is concave up on a given interval, and picture approximating the area bounded on that interval by both the Midpoint and Trapezoid rules using a single subinterval, as shown in Figure 5.6.3.

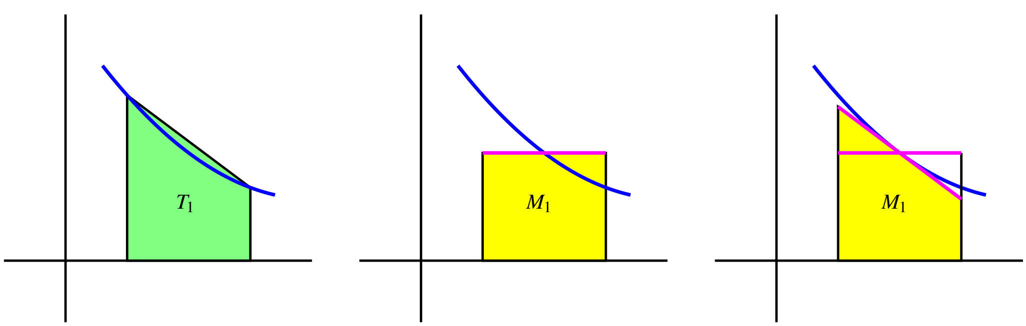


Figure 5.6.3 Estimating $\int_a^b f(x)\,dx$ using a single subinterval: at left, the trapezoid rule; in the middle, the midpoint rule; at right, a modified way to think about the midpoint rule.

As seen at left in Figure 5.6.3, it is evident that whenever the function is concave up on an interval, the Trapezoid Rule with one subinterval, T_1, will overestimate the exact value of the definite integral on that interval. From a careful analysis of the line that bounds the top of the rectangle for the Midpoint Rule (shown in magenta), we see that if we rotate this line segment until it is tangent to the curve at the midpoint of the interval (as shown at right in Figure 5.6.3), the resulting trapezoid has the same area as M_1, and this value is less than the exact value of the definite integral. Thus, when the function is concave up on the interval, M_1 underestimates the integral's true value.

These observations extend easily to the situation where the function's concavity remains consistent but we use larger values of n in the Midpoint and Trapezoid Rules. Hence, whenever f is concave up on $[a, b]$, T_n will overestimate the value of $\int_a^b f(x)\,dx$, while M_n will underestimate $\int_a^b f(x)\,dx$. The reverse observations are true in the situation where f is concave down.

Next, we compare the size of the errors between M_n and T_n. Again, we focus on M_1 and T_1 on an interval where the concavity of f is consistent. In Figure 5.6.4, where the error of the Trapezoid Rule is shaded in red, while the error of the Midpoint Rule is shaded lighter red, it is visually apparent that the error in the Trapezoid Rule is more significant.

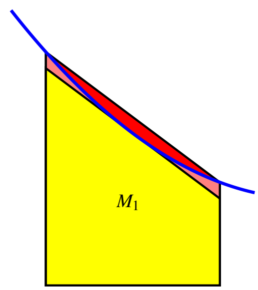


Figure 5.6.4 Comparing the error in estimating $\int_a^b f(x)\,dx$ using a single subinterval: in red, the error from the Trapezoid rule; in light red, the error from the Midpoint rule.

To see how much more significant the error from the Trapezoid Rule is, let's consider two examples and some particular computations. If we let $f(x) = 1 - x^2$ and consider $\int_0^1 f(x)\,dx$, we know by the First FTC that the exact value of the integral is

$$\int_0^1 (1 - x^2)\,dx = x - \frac{x^3}{3}\bigg|_0^1 = \frac{2}{3}.$$

Using appropriate technology to compute M_4, M_8, T_4, and T_8, as well as the corresponding errors $E_{M,4}$, $E_{M,8}$, $E_{T,4}$, and $E_{T,8}$, as we did in Activity 5.6.2, we find the results summarized in Table 5.6.5. We also include the approximations and their errors for the example $\int_1^2 \frac{1}{x^2}\,dx$ from Activity 5.6.2.

For a given function f and interval $[a, b]$, $E_{T,4} = T_4 - \int_a^b f(x)\,dx$ calculates the difference between the approximation generated by the Trapezoid Rule with $n = 4$ and the exact value of the definite integral. If we look at not only $E_{T,4}$, but also the other errors generated by using T_n and M_n with $n = 4$ and $n = 8$ in the two examples noted in Table 5.6.5, we see an evident pattern. Not only is the sign of the error (which measures whether the rule generates an over- or under-estimate) tied to the rule used and the function's concavity, but the magnitude of the errors generated by T_n and M_n seems closely connected. In particular, the errors generated by the Midpoint Rule seem to be about half the size (in absolute value) of those generated by the Trapezoid Rule.

Table 5.6.5 Calculations of T_4, M_4, T_8, and M_8, along with corresponding errors, for the definite integrals $\int_0^1 (1-x^2)\,dx$ and $\int_1^2 \frac{1}{x^2}\,dx$.

Rule	$\int_0^1 (1-x^2)\,dx = 0.\overline{6}$	error	$\int_1^2 \frac{1}{x^2}\,dx = 0.5$	error
T_4	0.65625	−0.0104166667	0.5089937642	0.0089937642
M_4	0.671875	0.0052083333	0.4955479365	−0.0044520635
T_8	0.6640625	−0.0026041667	0.5022708502	0.0022708502
M_8	0.66796875	0.0013020833	0.4988674899	−0.0011325101

That is, we can observe in both examples that $E_{M,4} \approx -\frac{1}{2}E_{T,4}$ and $E_{M,8} \approx -\frac{1}{2}E_{T,8}$. This property of the Midpoint and Trapezoid Rules turns out to hold in general: for a function of consistent concavity, the error in the Midpoint Rule has the opposite sign and approximately half the magnitude of the error of the Trapezoid Rule. Written symbolically,

$$E_{M,n} \approx -\frac{1}{2}E_{T,n}.$$

This important relationship suggests a way to combine the Midpoint and Trapezoid Rules to create an even more accurate approximation to a definite integral.

5.6.4 Simpson's Rule

When we first developed the Trapezoid Rule, we observed that it is an average of the Left and Right Riemann sums:

$$T_n = \frac{1}{2}(L_n + R_n).$$

If a function is always increasing or always decreasing on the interval $[a,b]$, one of L_n and R_n will over-estimate the true value of $\int_a^b f(x)\,dx$, while the other will under-estimate the integral. Thus, the errors that result from L_n and R_n will have opposite signs; so averaging L_n and R_n eliminates a considerable amount of the error present in the respective approximations. In a similar way, it makes sense to think about averaging M_n and T_n in order to generate a still more accurate approximation.

We've just observed that M_n is typically about twice as accurate as T_n. So we use the weighted average

$$S_{2n} = \frac{2M_n + T_n}{3}. \tag{5.6.5}$$

The rule for S_{2n} giving by Equation (5.6.5) is usually known as *Simpson's Rule*.[3] Note that we use "S_{2n}" rather that "S_n" since the n points the Midpoint Rule uses are different from the n points the Trapezoid Rule uses, and thus Simpson's Rule is using $2n$ points at which to evaluate the function. We build upon the results in Table 5.6.5 to see the approximations generated by Simpson's Rule. In particular, in Table 5.6.6, we include all of the results in Table 5.6.5, but include additional results for $S_8 = \frac{2M_4+T_4}{3}$ and $S_{16} = \frac{2M_8+T_8}{3}$.

[3]Thomas Simpson was an 18th century mathematician; his idea was to extend the Trapezoid rule, but rather than using straight lines to build trapezoids, to use quadratic functions to build regions whose area was bounded by parabolas (whose areas he could find exactly). Simpson's Rule is often developed from the more sophisticated perspective of using interpolation by quadratic functions.

Table 5.6.6 Table 5.6.5 updated to include S_8, S_{16}, and the corresponding errors.

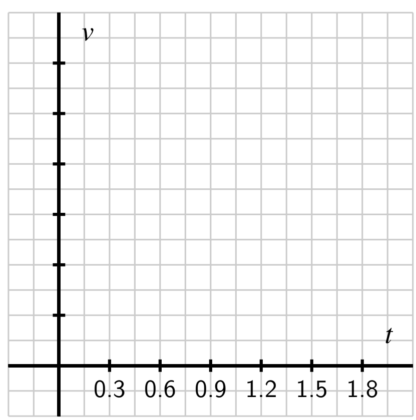
Rule	$\int_0^1 (1-x^2)\,dx = 0.\overline{6}$	error	$\int_1^2 \frac{1}{x^2}\,dx = 0.5$	error
T_4	0.65625	−0.0104166667	0.5089937642	0.0089937642
M_4	0.671875	0.0052083333	0.4955479365	−0.0044520635
S_8	0.6666666667	0	0.5000298792	0.0000298792
T_8	0.6640625	−0.0026041667	0.5022708502	0.0022708502
M_8	0.66796875	0.0013020833	0.4988674899	−0.0011325101
S_{16}	0.6666666667	0	0.5000019434	0.0000019434

The results seen in Table 5.6.6 are striking. If we consider the S_{16} approximation of $\int_1^2 \frac{1}{x^2}\,dx$, the error is only $E_{S,16} = 0.0000019434$. By contrast, $L_8 = 0.5491458502$, so the error of that estimate is $E_{L,8} = 0.0491458502$. Moreover, we observe that generating the approximations for Simpson's Rule is almost no additional work: once we have L_n, R_n, and M_n for a given value of n, it is a simple exercise to generate T_n, and from there to calculate S_{2n}. Finally, note that the error in the Simpson's Rule approximations of $\int_0^1 (1-x^2)\,dx$ is zero![4]

These rules are not only useful for approximating definite integrals such as $\int_0^1 e^{-x^2}\,dx$, for which we cannot find an elementary antiderivative of e^{-x^2}, but also for approximating definite integrals when we are given a function through a table of data.

Activity 5.6.3. A car traveling along a straight road is braking and its velocity is measured at several different points in time, as given in the following table. Assume that v is continuous, always decreasing, and always concave down, as is suggested by the data.

seconds, t	Velocity in ft/sec, $v(t)$
0	100
0.3	99
0.6	96
0.9	90
1.2	80
1.5	50
1.8	0



(a) Plot the given data on the set of axes provided in the figure, with time on the horizontal axis and the velocity on the vertical axis.

[4]Similar to how the Midpoint and Trapezoid approximations are exact for linear functions, Simpson's Rule approximations are exact for quadratic and cubic functions. See additional discussion on this issue later in the section and in the exercises.

(b) What definite integral will give you the exact distance the car traveled on $[0, 1.8]$?

(c) Estimate the total distance traveled on $[0, 1.8]$ by computing L_3, R_3, and T_3. Which of these under-estimates the true distance traveled?

(d) Estimate the total distance traveled on $[0, 1.8]$ by computing M_3. Is this an over- or under-estimate? Why?

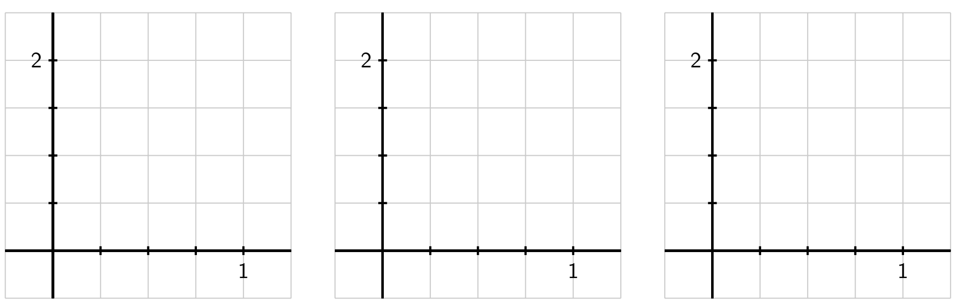
(e) Using your results from (c) and (d), improve your estimate further by using Simpson's Rule.

(f) What is your best estimate of the average velocity of the car on $[0, 1.8]$? Why? What are the units on this quantity?

5.6.5 Overall observations regarding L_n, R_n, T_n, M_n, and S_{2n}.

As we conclude our discussion of numerical approximation of definite integrals, it is important to summarize general trends in how the various rules over- or under-estimate the true value of a definite integral, and by how much. To revisit some past observations and see some new ones, we consider the following activity.

Activity 5.6.4. Consider the functions $f(x) = 2 - x^2$, $g(x) = 2 - x^3$, and $h(x) = 2 - x^4$, all on the interval $[0, 1]$. For each of the questions that require a numerical answer in what follows, write your answer exactly in fraction form.



(a) On the three sets of axes in the provided figure, sketch a graph of each function on the interval $[0, 1]$, and compute L_1 and R_1 for each. What do you observe?

(b) Compute M_1 for each function to approximate $\int_0^1 f(x)\,dx$, $\int_0^1 g(x)\,dx$, and $\int_0^1 h(x)\,dx$, respectively.

(c) Compute T_1 for each of the three functions, and hence compute S_2 for each of the three functions.

(d) Evaluate each of the integrals $\int_0^1 f(x)\,dx$, $\int_0^1 g(x)\,dx$, and $\int_0^1 h(x)\,dx$ exactly using the First FTC.

(e) For each of the three functions f, g, and h, compare the results of L_1, R_1, M_1, T_1, and S_2 to the true value of the corresponding definite integral. What patterns do you observe?

The results seen in Activity 5.6.4 generalize nicely. For instance, if f is decreasing on $[a, b]$, L_n will over-estimate the exact value of $\int_a^b f(x)\,dx$, and if f is concave down on $[a, b]$, M_n will over-estimate the exact value of the integral. An excellent exercise is to write a collection of scenarios of possible function behavior, and then categorize whether each of L_n, R_n, T_n, and M_n is an over- or under-estimate.

Finally, we make two important notes about Simpson's Rule. When T. Simpson first developed this rule, his idea was to replace the function f on a given interval with a quadratic function that shared three values with the function f. In so doing, he guaranteed that this new approximation rule would be exact for the definite integral of any quadratic polynomial. In one of the pleasant surprises of numerical analysis, it turns out that even though it was designed to be exact for quadratic polynomials, Simpson's Rule is exact for any cubic polynomial: that is, if we are interested in an integral such as $\int_2^5 (5x^3 - 2x^2 + 7x - 4)\,dx$, S_{2n} will always be exact, regardless of the value of n. This is just one more piece of evidence that shows how effective Simpson's Rule is as an approximation tool for estimating definite integrals.[5]

5.6.6 Summary

- For a definite integral such as $\int_0^1 e^{-x^2}\,dx$ when we cannot use the First Fundamental Theorem of Calculus because the integrand lacks an elementary algebraic antiderivative, we can estimate the integral's value by using a sequence of Riemann sum approximations. Typically, we start by computing L_n, R_n, and M_n for one or more chosen values of n.
- The Trapezoid Rule, which estimates $\int_a^b f(x)\,dx$ by using trapezoids, rather than rectangles, can also be viewed as the average of Left and Right Riemann sums. That is, $T_n = \frac{1}{2}(L_n + R_n)$.
- The Midpoint Rule is typically twice as accurate as the Trapezoid Rule, and the signs of the respective errors of these rules are opposites. Hence, by taking the weighted average $S_{2n} = \frac{2M_n + T_n}{3}$, we can build a much more accurate approximation to $\int_a^b f(x)\,dx$ by using approximations we have already computed. The rule for S_{2n} is known as Simpson's Rule, which can also be developed by approximating a given continuous function with pieces of quadratic polynomials.

[5]One reason that Simpson's Rule is so effective is that S_{2n} benefits from using $2n + 1$ points of data. Because it combines M_n, which uses n midpoints, and T_n, which uses the $n + 1$ endpoints of the chosen subintervals, S_{2n} takes advantage of the maximum amount of information we have when we know function values at the endpoints and midpoints of n subintervals.

5.6.7 Exercises

1. Calculate the integral approximations T_6 , M_6 and S_{12} for

$\int_2^6 x^4\,dx.$

$T_6 =$ ____________________

$M_6 =$ ____________________

$S_{12} =$ ____________________

2. Estimate $\int_0^4 x^2\,dx$ using SIMP(4).

$\int_0^4 x^2\,dx \approx$ ____________________________________

3. ***Note: for this problem, because later answers depend on earlier ones, you must enter answers for all answer blanks for the problem to be correctly graded. If you would like to get feedback before you completed all computations, enter a "1" for each answer you did not yet compute and then submit the problem. (But note that this will, obviously, result in a problem submission.)***

(a) What is the exact value of $\int_0^3 e^x\,dx$?

$\int_0^3 e^x\,dx =$ __________

(b)

Find LEFT(2), RIGHT(2), TRAP(2), MID(2), and SIMP(4); compute the error for each.

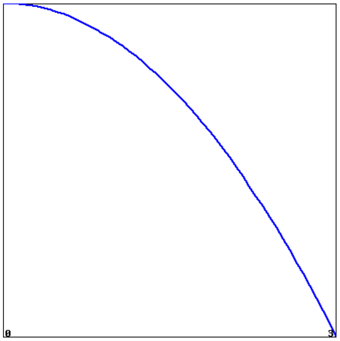
	LEFT(2)	RIGHT(2)	TRAP(2)	MID(2)	SIMP(4)
value	__________	__________	__________	__________	__________
error	__________	__________	__________	__________	__________

(c)

Repeat part (b) with $n = 4$ (instead of $n = 2$).

	LEFT(4)	RIGHT(4)	TRAP(4)	MID(4)	SIMP(8)
value	__________	__________	__________	__________	__________
error	__________	__________	__________	__________	__________

4. Using the figure showing $f(x)$ below, order the following approximations to the integral $\int_0^3 f(x)\,dx$ and its exact value from smallest to largest.



(Click on the graph for a larger version.)

Enter each of "LEFT(n)", "RIGHT(n)", "TRAP(n)", "MID(n)" and "Exact" in one of the following answer blanks to indicate the correct ordering:

__________ < __________ < __________ < __________ < __________

5. Consider the definite integral $\int_0^1 x\tan(x)\,dx$.

a. Explain why this integral cannot be evaluated exactly by using either u-substitution or by integrating by parts.

b. Using appropriate subintervals, compute L_4, R_4, M_4, T_4, and S_8.

c. Which of the approximations in (b) is an over-estimate to the true value of $\int_0^1 x\tan(x)\,dx$? Which is an under-estimate? How do you know?

6. For an unknown function $f(x)$, the following information is known.

- f is continuous on $[3, 6]$;
- f is either always increasing or always decreasing on $[3, 6]$;
- f has the same concavity throughout the interval $[3, 6]$;
- As approximations to $\int_3^6 f(x)\,dx$, $L_4 = 7.23$, $R_4 = 6.75$, and $M_4 = 7.05$.

a. Is f increasing or decreasing on $[3, 6]$? What data tells you?

b. Is f concave up or concave down on $[3, 6]$? Why?

c. Determine the best possible estimate you can for $\int_3^6 f(x)\,dx$, based on the given information.

7. The rate at which water flows through Table Rock Dam on the White River in Branson, MO, is measured in cubic feet per second (CFS). As engineers open the floodgates, flow rates are recorded according to the following chart.

Table 5.6.7 Water flow data.

seconds, t	0	10	20	30	40	50	60
flow in CFS, $r(t)$	2000	2100	2400	3000	3900	5100	6500

a. What definite integral measures the total volume of water to flow through the dam in the 60 second time period provided by the table above?

b. Use the given data to calculate M_n for the largest possible value of n to approximate the integral you stated in (a). Do you think M_n over- or under-estimates the exact value of the integral? Why?

c. Approximate the integral stated in (a) by calculating S_n for the largest possible value of n, based on the given data.

d. Compute $\frac{1}{60}S_n$ and $\frac{2000+2100+2400+3000+3900+5100+6500}{7}$. What quantity do both of these values estimate? Which is a more accurate approximation?

CHAPTER 6

Using Definite Integrals

6.1 Using definite integrals to find area and length

Motivating Questions

- How can we use definite integrals to measure the area between two curves?
- How do we decide whether to integrate with respect to x or with respect to y when we try to find the area of a region?
- How can a definite integral be used to measure the length of a curve?

6.1.1 Introduction

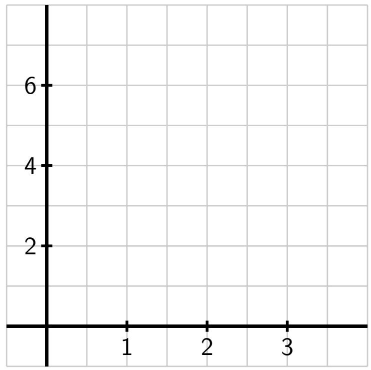
Early on in our work with the definite integral, we learned that for an object moving along an axis, the area under a non-negative velocity function v between a and b tells us the distance the object traveled on that time interval, and that area is given precisely by the definite integral $\int_a^b v(t)\,dt$. In general, for any nonnegative function f on an interval $[a, b]$, $\int_a^b f(x)\,dx$ measures the area bounded by the curve and the x-axis between $x = a$ and $x = b$.

Next, we will explore how definite integrals can be used to represent other physically important properties. In Preview Activity 6.1.1, we investigate how a single definite integral may be used to represent the area between two curves.

Preview Activity 6.1.1. Consider the functions given by $f(x) = 5 - (x-1)^2$ and $g(x) = 4 - x$.

(a) Use algebra to find the points where the graphs of f and g intersect.

(b) Sketch an accurate graph of f and g on the axes provided, labeling the curves by name and the intersection points with ordered pairs.



(c) Find and evaluate exactly an integral expression that represents the area between $y = f(x)$ and the x-axis on the interval between the intersection points of f and g.

(d) Find and evaluate exactly an integral expression that represents the area between $y = g(x)$ and the x-axis on the interval between the intersection points of f and g.

(e) What is the exact area between f and g between their intersection points? Why?

6.1.2 The Area Between Two Curves

In Preview Activity 6.1.1, we saw a natural way to think about the area between two curves: it is the area beneath the upper curve minus the area below the lower curve.

Example 6.1.1 Find the area bounded between the graphs of $f(x) = (x-1)^2 + 1$ and $g(x) = x + 2$.

Solution. We can find the intersection points of the graphs of f and g algebraically by solving the system of equations given by $y = x + 2$ and $y = (x-1)^2 + 1$: substituting $x + 2$ for y in the second equation yields $x + 2 = (x-1)^2 + 1$, so $x + 2 = x^2 - 2x + 1 + 1$, and thus

$$x^2 - 3x = x(x-3) = 0,$$

from which it follows that $x = 0$ or $x = 3$. Using $y = x+2$, we find the corresponding y-values of the intersection points. In Figure 6.1.2, we confirm this result and show the y-coordinates of the points, $(0,2)$ and $(3,5)$. Furthermore, the figure shows how we can think of the area beneath f and the area beneath g separately, and use the difference of these quantities to find the difference between the curves.

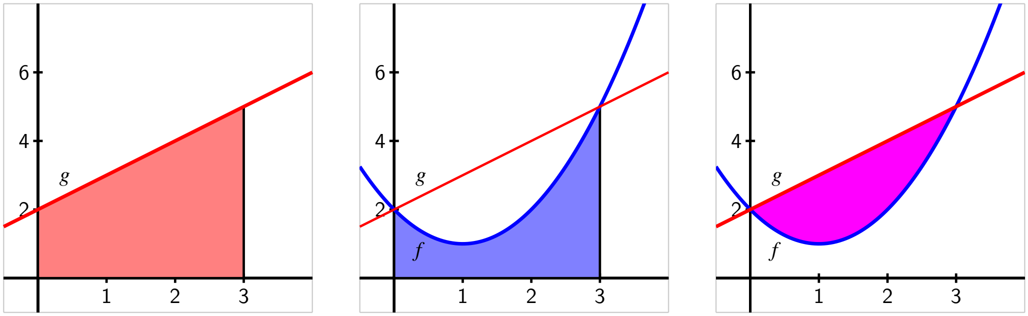


Figure 6.1.2 The areas bounded by the functions $f(x) = (x-1)^2 + 1$ and $g(x) = x + 2$ on the interval $[0,3]$.

Now, using definite integrals to find the values of the respective areas beneath the curves, on the interval $[0,3]$, the area beneath g is

$$\int_0^3 (x+2)\,dx = \frac{21}{2},$$

while the area under f on the same interval is

$$\int_0^3 [(x-1)^2 + 1]\,dx = 6.$$

Thus, the area between the curves is

$$A = \int_0^3 (x+2)\,dx - \int_0^3 [(x-1)^2+1]\,dx = \frac{21}{2} - 6 = \frac{9}{2}. \tag{6.1.1}$$

◊

We can also think of the area in this slightly different way: if we slice up the region between two curves into thin vertical rectangles (in the same spirit as we originally sliced the region between a single curve and the x-axis in Section 4.2), we see (as shown in Figure 6.1.3) that the height of a typical rectangle is given by the difference between the two functions, $g(x) - f(x)$, and its width is Δx. Thus the area of the rectangle is

$$A_{\text{rect}} = (g(x) - f(x))\Delta x.$$

The area between the two curves on $[0,3]$ is thus approximated by the Riemann sum

$$A \approx \sum_{i=1}^{n} (g(x_i) - f(x_i))\Delta x,$$

and as we let $n \to \infty$, it follows that the area is given by the single definite integral

$$A = \int_0^3 (g(x) - f(x))\,dx. \tag{6.1.2}$$

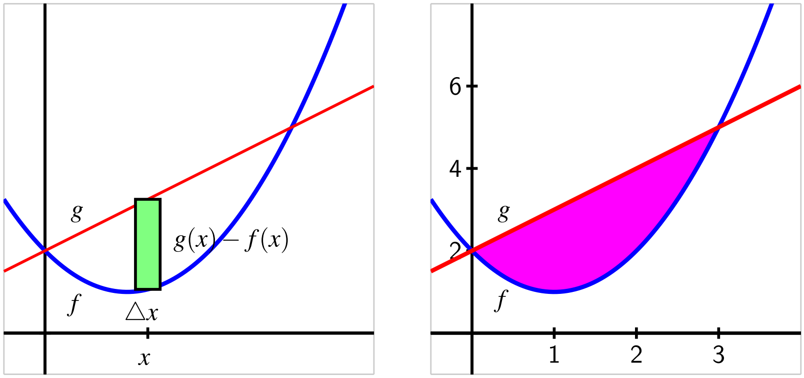


Figure 6.1.3 The area bounded by the functions $f(x) = (x-1)^2 + 1$ and $g(x) = x + 2$ on the interval $[0, 3]$.

In many applications of the definite integral, we will find it helpful to think of a "representative slice" and use the definite integral to add up these slices. Here, the integral sums the areas of thin rectangles.

Finally, it doesn't matter whether we think of the area between two curves as the difference between the area bounded by the individual curves (as in (6.1.1)) or as the limit of a Riemann sum of the areas of thin rectangles between the curves (as in (6.1.2)). These two results are the same, since the difference of two integrals is the integral of the difference:

$$\int_0^3 g(x)\,dx - \int_0^3 f(x)\,dx = \int_0^3 (g(x) - f(x))\,dx.$$

Our work so far in this section illustrates the following general principle.

The area between two curves.

If two curves $y = g(x)$ and $y = f(x)$ intersect at $(a, g(a))$ and $(b, g(b))$, and for all x such that $a \le x \le b$, $g(x) \ge f(x)$, then the area between the curves is

$$A = \int_a^b (g(x) - f(x))\,dx.$$

Activity 6.1.2. In each of the following situations, our goal is to determine the area of the region described. For each region, (i) determine the intersection points of the curves, (ii) sketch the region whose area is being found, (iii) draw and label a representative slice, and (iv) state the area of the representative slice. Then, state a definite integral whose value is the exact area of the region, and evaluate the integral to find the numeric value of the region's area.

(a) The finite region bounded by $y = \sqrt{x}$ and $y = \frac{1}{4}x$.

(b) The finite region bounded by $y = 12 - 2x^2$ and $y = x^2 - 8$.

(c) The area bounded by the y-axis, $f(x) = \cos(x)$, and $g(x) = \sin(x)$, where we consider the region formed by the first positive value of x for which f and g intersect.

(d) The finite regions between the curves $y = x^3 - x$ and $y = x^2$.

6.1.3 Finding Area with Horizontal Slices

At times, the shape of a region may dictate that we use horizontal rectangular slices, instead of vertical ones.

Example 6.1.4 Find the area of the region bounded by the parabola $x = y^2 - 1$ and the line $y = x - 1$, shown at left in Figure 6.1.5.

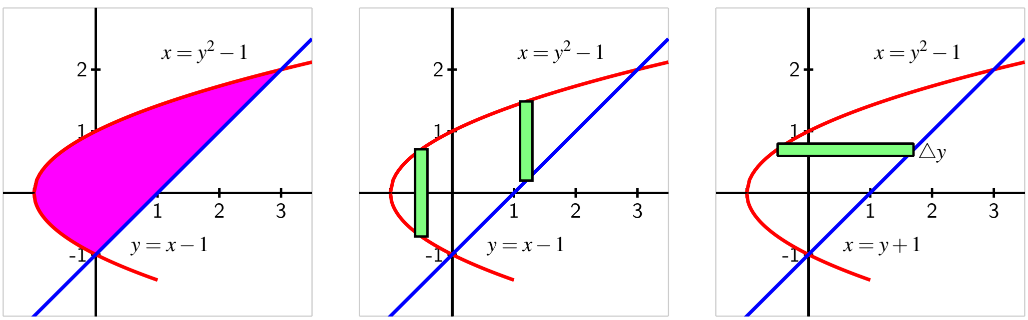


Figure 6.1.5 The area bounded by the functions $x = y^2 - 1$ and $y = x - 1$ (at left), with the region sliced vertically (center) and horizontally (at right).

Solution. By solving the second equation for x and writing $x = y + 1$, we find that $y + 1 = y^2 - 1$. Hence the curves intersect where $y^2 - y - 2 = 0$. Thus, we find $y = -1$ or $y = 2$, so the intersection points of the two curves are $(0, -1)$ and $(3, 2)$.

If we attempt to use vertical rectangles to slice up the area (as in the center graph of Figure 6.1.5), we see that from $x = -1$ to $x = 0$ the curves that bound the top and bottom of the rectangle are one and the same. This suggests, as shown in the rightmost graph in the figure, that we try using horizontal rectangles.

Note that the width of a horizontal rectangle depends on y. Between $y = -1$ and $y = 2$, the right end of a representative rectangle is determined by the line $x = y + 1$, and the left end is determined by the parabola, $x = y^2 - 1$. The thickness of the rectangle is Δy.

Therefore, the area of the rectangle is

$$A_{\text{rect}} = [(y + 1) - (y^2 - 1)]\Delta y,$$

and the area between the two curves on the y-interval $[-1, 2]$ is approximated by the Riemann sum

$$A \approx \sum_{i=1}^{n} [(y_i + 1) - (y_i^2 - 1)]\Delta y.$$

Taking the limit of the Riemann sum, it follows that the area of the region is

$$A = \int_{y=-1}^{y=2} [(y+1) - (y^2 - 1)]\, dy. \tag{6.1.3}$$

We emphasize that we are integrating with respect to y; this is because we chose to use horizontal rectangles whose widths depend on y and whose thickness is denoted Δy. It is a straightforward exercise to evaluate the integral in Equation (6.1.3) and find that $A = \frac{9}{2}$. ◊

Just as with the use of vertical rectangles of thickness Δx, we have a general principle for finding the area between two curves, which we state as follows.

Using horizontal slices to find the area between two curves.

If two curves $x = g(y)$ and $x = f(y)$ intersect at $(g(c), c)$ and $(g(d), d)$, and for all y such that $c \le y \le d$, $g(y) \ge f(y)$, then the area between the curves is

$$A = \int_{y=c}^{y=d} (g(y) - f(y))\, dy.$$

Activity 6.1.3. In each of the following situations, our goal is to determine the area of the region described. For each region, (i) determine the intersection points of the curves, (ii) sketch the region whose area is being found, (iii) draw and label a representative slice, and (iv) state the area of the representative slice. Then, state a definite integral whose value is the exact area of the region, and evaluate the integral to find the numeric value of the region's area. *Note well:* At the step where you draw a representative slice, you need to make a choice about whether to slice vertically or horizontally.

(a) The finite region bounded by $x = y^2$ and $x = 6 - 2y^2$.

(b) The finite region bounded by $x = 1 - y^2$ and $x = 2 - 2y^2$.

(c) The area bounded by the x-axis, $y = x^2$, and $y = 2 - x$.

(d) The finite regions between the curves $x = y^2 - 2y$ and $y = x$.

6.1.4 Finding the length of a curve

We can also use the definite integral to find the length of a portion of a curve. We use the same fundamental principle: we slice the curve up into small pieces whose lengths we can easily approximate. Specifically, we subdivide the curve into small approximating line segments, as shown at left in Figure 6.1.6.

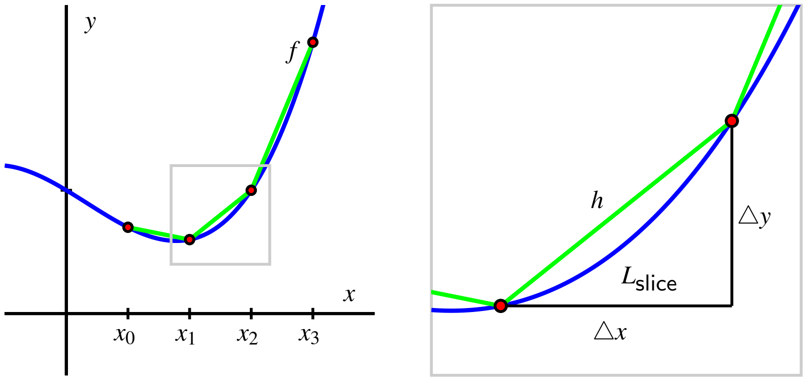


Figure 6.1.6 At left, a continuous function $y = f(x)$ whose length we seek on the interval $a = x_0$ to $b = x_3$. At right, a close up view of a portion of the curve.

We estimate the length L_{slice} of each portion of the curve on a small interval of length Δx. We use the right triangle with legs parallel to the coordinate axes and hypotenuse connecting the endpoints of the slice, as seen at right in Figure 6.1.6. The length, h, of the hypotenuse approximates the length, L_{slice}, of the curve between the two selected points. Thus,

$$L_{\text{slice}} \approx h = \sqrt{(\Delta x)^2 + (\Delta y)^2}.$$

Next we use algebra to rearrange the expression for the length of the hypotenuse into a form that we can integrate. By removing a factor of $(\Delta x)^2$, we find

$$\begin{aligned} L_{\text{slice}} &\approx \sqrt{(\Delta x)^2 + (\Delta y)^2} \\ &= \sqrt{(\Delta x)^2 \left(1 + \frac{(\Delta y)^2}{(\Delta x)^2}\right)} \\ &= \sqrt{1 + \frac{(\Delta y)^2}{(\Delta x)^2}} \cdot \Delta x. \end{aligned}$$

Then, as $n \to \infty$ and $\Delta x \to 0$, we have that $\frac{\Delta y}{\Delta x} \to \frac{dy}{dx} = f'(x)$. Thus, we can say that

$$L_{\text{slice}} \approx \sqrt{1 + f'(x)^2}\Delta x.$$

Taking a Riemann sum of all of these slices and letting $n \to \infty$, the total length of the curve is therefore

$$L = \int_a^b \sqrt{1 + f'(x)^2}\, dx.$$

We summarize this result more formally as follows.

The arc length of a curve.

Given a differentiable function f on an interval $[a,b]$, the total arc length, L, along the curve $y = f(x)$ from $x = a$ to $x = b$ is given by

$$L = \int_a^b \sqrt{1 + f'(x)^2}\, dx.$$

Activity 6.1.4. Each of the following questions involves the arc length along a curve.

(a) Use the definition and appropriate computational technology to determine the arc length along $y = x^2$ from $x = -1$ to $x = 1$.

(b) Find the arc length of $y = \sqrt{4 - x^2}$ on the interval $-2 \le x \le 2$. Find this value in two different ways: (a) by using a definite integral, and (b) by using a familiar property of the curve.

(c) Determine the arc length of $y = xe^{3x}$ on the interval $[0, 1]$.

(d) Will the integrals that arise calculating arc length typically be ones that we can evaluate exactly using the First FTC, or ones that we need to approximate? Why?

(e) A moving particle is traveling along the curve given by $y = f(x) = 0.1x^2 + 1$, and does so at a constant rate of 7 cm/sec, where both x and y are measured in cm (that is, the curve $y = f(x)$ is the path along which the object actually travels; the curve is not a "position function"). Find the position of the particle when $t = 4$ sec, assuming that when $t = 0$, the particle's location is $(0, f(0))$.

6.1.5 Summary

- To find the area between two curves, we think about slicing the region into thin rectangles. If, for instance, the area of a typical rectangle on the interval $x = a$ to $x = b$ is given by $A_{\text{rect}} = (g(x) - f(x))\Delta x$, then the exact area of the region is given by the definite integral

$$A = \int_a^b (g(x) - f(x))\, dx.$$

- The shape of the region usually dictates whether we should use vertical rectangles of thickness Δx or horizontal rectangles of thickness Δy. We want the height of the rectangle given by the difference between two curves: if those curves are best thought of as functions of y, we use horizontal rectangles, whereas if those curves are best viewed as functions of x, we use vertical rectangles.

- The arc length, L, along the curve $y = f(x)$ from $x = a$ to $x = b$ is given by

$$L = \int_a^b \sqrt{1 + f'(x)^2}\, dx.$$

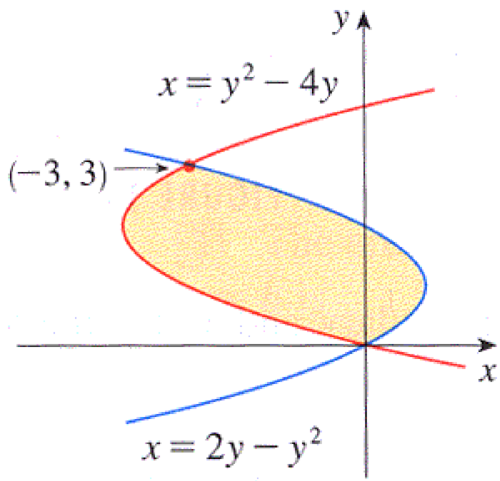
6.1.6 Exercises

1. Find the area enclosed between $f(x) = 0.7x^2 + 5$ and $g(x) = x$ from $x = -3$ to $x = 3$.

2. Find the area of the region between $y = x^{1/2}$ and $y = x^{1/4}$ for $0 \le x \le 1$.

area = ____________________

3. Find the area of the shaded region below.

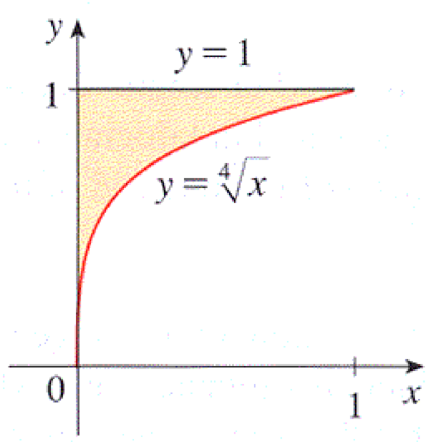


Area = ____________________

4. Find the area between $y = 7\sin x$ and $y = 10\cos x$ over the interval $[0, \pi]$. Sketch the curves if necessary.

A =____________

5. The boundaries of the shaded region are the y-axis, the line $y = 1$, and the curve $y = \sqrt[4]{x}$.



This area can be computed in two different ways using integrals.

First, it can be computed as an integral with respect to x,

$$\int_a^b [f(x) - g(x)]\,dx$$

where $a =$ ____________________, $b =$ ____________________, and

$f(x) - g(x) =$ ____

Alternatively, this area can be computed as an integral with respect to y,

$$\int_c^d h(y)\,dy$$

where $c =$ ____________________, $d =$ ____________________ and

$h(y) =$ ____

Either way we find that the area is ____________________.

6. Find the area of the region between the curves $4x + y^2 = 12$ and $x = y$.

These two curves intersect when $y =$ ___ and $y =$ ___.

This is better to integrate with respect to y, since we know the starting value and ending value of y, and the region can be described with one integral,

$$\int_c^d \left[h(y) - k(y)\right]\,dy$$

where $c =$ ____________________, $d =$ ____________________ and

$h(y) - k(y) =$ ____

Then we find that the area is ____________________.

7. Consider the curve defined by the equation $y = 2x^2 + 3x$. Set up an integral that represents the length of curve from the point $(-3, 9)$ to the point $(3, 27)$.

arclength = $\int_c^d f(x)\,dx$,

where $c =$ _____ $d =$ _____ $f(x) =$ __________

8. Find the arc length of the graph of the function $f(x) = 2\sqrt{x^3}$ from $x = 4$ to $x = 7$.

arc length = ___________________________________

9. Approximate the arc length of the curve $y = \frac{1}{4}x^4$ over the interval $[1, 2]$ using the trapezoidal rule and 6 intervals (i.e., T_6).

$T_6 =$ ____________________

10. Find the exact area of each described region.

a. The finite region between the curves $x = y(y-2)$ and $x = -(y-1)(y-3)$.

b. The region between the sine and cosine functions on the interval $[\frac{\pi}{4}, \frac{3\pi}{4}]$.

c. The finite region between $x = y^2 - y - 2$ and $y = 2x - 1$.

d. The finite region between $y = mx$ and $y = x^2 - 1$, where m is a positive constant.

11. Let $f(x) = 1 - x^2$ and $g(x) = ax^2 - a$, where a is an unknown positive real number. For what value(s) of a is the area between the curves f and g equal to 2?

12. Let $f(x) = 2 - x^2$. Recall that the average value of any continuous function f on an interval $[a, b]$ is given by $\frac{1}{b-a}\int_a^b f(x)\,dx$.

a. Find the average value of $f(x) = 2 - x^2$ on the interval $[0, \sqrt{2}]$. Call this value r.

b. Sketch a graph of $y = f(x)$ and $y = r$. Find their intersection point(s).

c. Show that on the interval $[0, \sqrt{2}]$, the amount of area that lies below $y = f(x)$ and above $y = r$ is equal to the amount of area that lies below $y = r$ and above $y = f(x)$.

d. Will the result of (c) be true for any continuous function and its average value on any interval? Why?

6.2 Using definite integrals to find volume

Motivating Questions

- How can we use a definite integral to find the volume of a three-dimensional solid of revolution that results from revolving a two-dimensional region about a particular axis?
- In what circumstances do we integrate with respect to y instead of integrating with respect to x?
- What adjustments do we need to make if we revolve about a line other than the x- or y-axis?

6.2.1 Introduction

Just as we can use definite integrals to add the areas of rectangular slices to find the exact area that lies between two curves, we can also use integrals to find the volume of regions whose cross-sections have a particular shape.

In particular, we can determine the volume of solids whose cross-sections are all thin cylinders (or washers) by adding up the volumes of these individual slices. We first consider a familiar shape in Preview Activity 6.2.1: a circular cone.

Preview Activity 6.2.1. Consider a circular cone of radius 3 and height 5, which we view horizontally as pictured in Figure 6.2.1. Our goal in this activity is to use a definite integral to determine the volume of the cone.

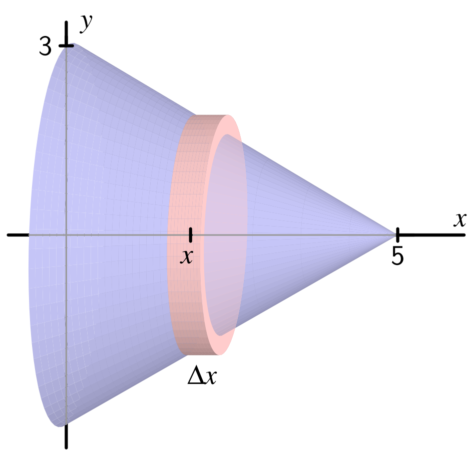


Figure 6.2.1 The circular cone described in Preview Activity 6.2.1

(a) Find a formula for the linear function $y = f(x)$ that is pictured in Figure 6.2.1.

(b) For the representative slice of thickness Δx that is located horizontally at a location x (somewhere between $x = 0$ and $x = 5$), what is the radius of the representative slice? Note that the radius depends on the value of x.

(c) What is the volume of the representative slice you found in (b)?

(d) What definite integral will sum the volumes of the thin slices across the full horizontal span of the cone? What is the exact value of this definite integral?

(e) Compare the result of your work in (d) to the volume of the cone that comes from using the formula $V_{\text{cone}} = \frac{1}{3}\pi r^2 h$.

6.2.2 The Volume of a Solid of Revolution

A solid of revolution is a three-dimensional solid that can be generated by revolving one or more curves around a fixed axis. For example, the circular cone in Preview Activity 6.2.1 is the solid of revolution generated by revolving the portion of the line $y = 3 - \frac{3}{5}x$ from $x = 0$ to $x = 5$ about the x-axis. Notice that if we slice a solid of revolution perpendicular to the axis of revolution, the resulting cross-section is a circle.

We first consider solids whose slices are thin cylinders. Recall that the volume of a cylinder is given by $V = \pi r^2 h$.

Example 6.2.2 Find the volume of the solid of revolution generated when the region R bounded by $y = 4 - x^2$ and the x-axis is revolved about the x-axis.

Solution. First, we observe that $y = 4 - x^2$ intersects the x-axis at the points $(-2, 0)$ and $(2, 0)$. When we revolve the region R about the x-axis, we get the three-dimensional solid pictured in Figure 6.2.3.

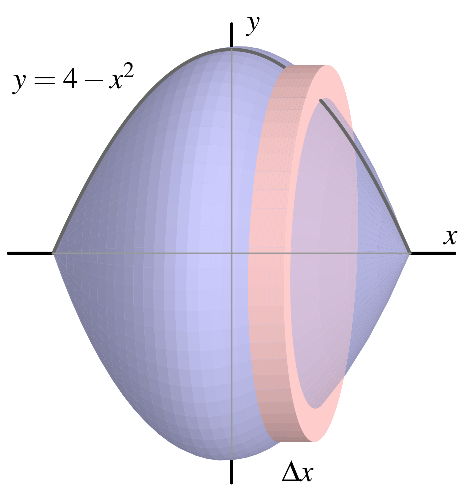


Figure 6.2.3 The solid of revolution in Example 6.2.2.

We slice the solid into vertical slices of thickness Δx between $x = -2$ and $x = 2$. A representative slice is a cylinder of height Δx and radius $4 - x^2$. Hence, the volume of the slice is

$$V_{\text{slice}} = \pi(4 - x^2)^2 \Delta x.$$

Using a definite integral to sum the volumes of the representative slices, it follows that

$$V = \int_{-2}^{2} \pi(4 - x^2)^2 \, dx.$$

It is straightforward to evaluate the integral and find that the volume is $V = \frac{512}{15}\pi$. ◊

For a solid such as the one in Example 6.2.2, where each slice is a cylindrical disk, we first find the volume of a typical slice (noting particularly how this volume depends on x), and then integrate over the range of x-values that bound the solid. Often, we will be content with simply finding the integral that represents the volume; if we desire a numeric value for the integral, we typically use a calculator or computer algebra system to find that value.

This method for finding the volume of a solid of revolution is often called the *disk method*.

The Disk Method.

If $y = r(x)$ is a nonnegative continuous function on $[a, b]$, then the volume of the solid of revolution generated by revolving the curve about the x-axis over this interval is given by

$$V = \int_{a}^{b} \pi r(x)^2 \, dx.$$

A different type of solid can emerge when two curves are involved, as we see in the following example.

Example 6.2.4 Find the volume of the solid of revolution generated when the finite region R that lies between $y = 4 - x^2$ and $y = x + 2$ is revolved about the x-axis.

Solution. First, we must determine where the curves $y = 4 - x^2$ and $y = x + 2$ intersect. Substituting the expression for y from the second equation into the first equation, we find that $x + 2 = 4 - x^2$. Rearranging, it follows that

$$x^2 + x - 2 = 0,$$

and the solutions to this equation are $x = -2$ and $x = 1$. The curves therefore cross at $(-2, 0)$ and $(1, 3)$.

When we revolve the region R about the x-axis, we get the three-dimensional solid pictured at left in Figure 6.2.5. Immediately we see a major difference between the solid in this example and the one in Example 6.2.2: here, the three-dimensional solid of revolution isn't "solid" because it has open space in its center along the axis of revolution.

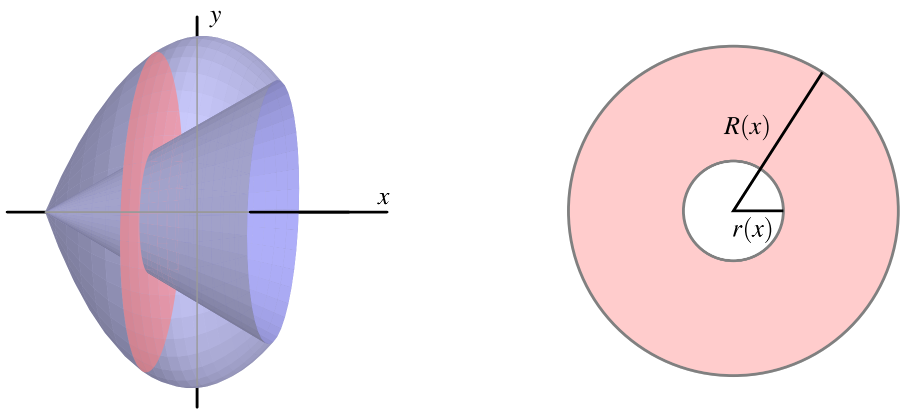


Figure 6.2.5 At left, the solid of revolution in Example 6.2.4. At right, a typical slice with inner radius $r(x)$ and outer radius $R(x)$.

If we slice the solid perpendicular to the axis of revolution, we observe that the resulting slice is not a solid disk, but rather a *washer*, as pictured at right in Figure 6.2.5. At a given location x between $x = -2$ and $x = 1$, the small radius $r(x)$ of the inner circle is determined by the curve $y = x + 2$, so $r(x) = x + 2$. Similarly, the big radius $R(x)$ comes from the function $y = 4 - x^2$, and thus $R(x) = 4 - x^2$.

To find the volume of a representative slice, we compute the volume of the outer disk and subtract the volume of the inner disk. Since

$$\pi R(x)^2 \Delta x - \pi r(x)^2 \Delta x = \pi[R(x)^2 - r(x)^2]\Delta x,$$

it follows that the volume of a typical slice is

$$V_{\text{slice}} = \pi[(4 - x^2)^2 - (x + 2)^2]\Delta x.$$

Using a definite integral to sum the volumes of the respective slices across the integral, we find that

$$V = \int_{-2}^{1} \pi[(4 - x^2)^2 - (x + 2)^2]\, dx.$$

Evaluating the integral, we find that the volume of the solid of revolution is $V = \frac{108}{5}\pi$. ◊

This method for finding the volume of a solid of revolution generated by two curves is often called the *washer method*.

The Washer Method.

If $y = R(x)$ and $y = r(x)$ are nonnegative continuous functions on $[a, b]$ that satisfy $R(x) \geq r(x)$ for all x in $[a, b]$, then the volume of the solid of revolution generated by revolving the region between them about the x-axis over this interval is given by

$$V = \int_a^b \pi[R(x)^2 - r(x)^2]\, dx.$$

Activity 6.2.2. In each of the following questions, draw a careful, labeled sketch of the region described, as well as the resulting solid that results from revolving the region about the stated axis. In addition, draw a representative slice and state the volume of that slice, along with a definite integral whose value is the volume of the entire solid. It is not necessary to evaluate the integrals you find.

(a) The region S bounded by the x-axis, the curve $y = \sqrt{x}$, and the line $x = 4$; revolve S about the x-axis.

(b) The region S bounded by the y-axis, the curve $y = \sqrt{x}$, and the line $y = 2$; revolve S about the x-axis.

(c) The finite region S bounded by the curves $y = \sqrt{x}$ and $y = x^3$; revolve S about the x-axis.

(d) The finite region S bounded by the curves $y = 2x^2 + 1$ and $y = x^2 + 4$; revolve S about the x-axis.

(e) The region S bounded by the y-axis, the curve $y = \sqrt{x}$, and the line $y = 2$; revolve S about the y-axis. How is this problem different from the one posed in part (b)?

6.2.3 Revolving about the y-axis

When we revolve a given region about the y-axis, the representative slices now have thickness Δy, which means that we must integrate with respect to y.

Example 6.2.6 Find the volume of the solid of revolution generated when the finite region R that lies between $y = \sqrt{x}$ and $y = x^4$ is revolved about the y-axis.

Solution. These two curves intersect at $x = 0$ and $x = 1$, hence at the points $(0, 0)$ and $(1, 1)$. When we revolve the region R about the y-axis, we get the three-dimensional solid pictured at left in Figure 6.2.7.

Note that the slices are cylindrical washers only if taken perpendicular to the y-axis. We slice the solid horizontally, starting at $y = 0$ and proceeding up to $y = 1$. The thickness of a representative slice is Δy, so we must express the integrand in terms of y. The inner radius is determined by the curve $y = \sqrt{x}$, so we solve for x and get $x = y^2 = r(y)$. In the same way, we solve the curve $y = x^4$ (which governs the outer radius) for x in terms of y, and hence $x = \sqrt[4]{y}$. Therefore, the volume of a typical slice is

$$V_{\text{slice}} = \pi[R(y)^2 - r(y)^2] = \pi[(\sqrt[4]{y})^2 - (y^2)^2]\Delta y.$$

We use a definite integral to sum the volumes of all the slices from $y = 0$ to $y = 1$. The total volume is

$$V = \int_{y=0}^{y=1} \pi \left[(\sqrt[4]{y})^2 - (y^2)^2\right] \, dy.$$

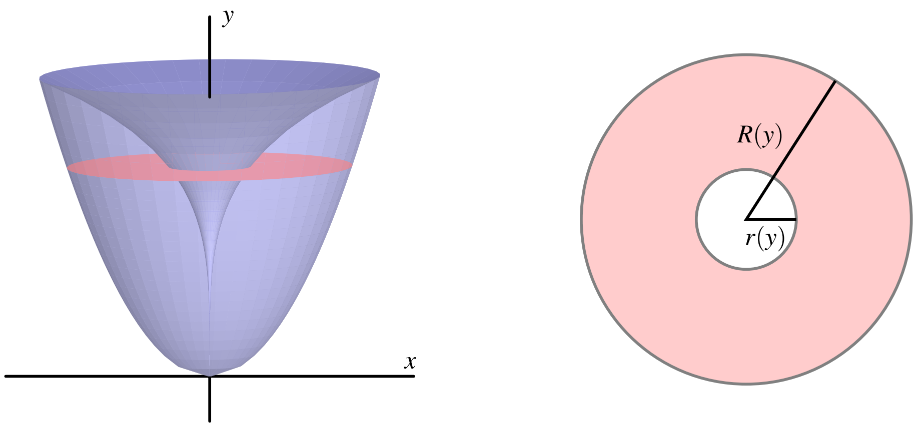


Figure 6.2.7 At left, the solid of revolution in Example 6.2.6. At right, a typical slice with inner radius $r(y)$ and outer radius $R(y)$.

It is straightforward to evaluate the integral and find that $V = \frac{7}{15}\pi$. ◊

Activity 6.2.3. In each of the following questions, draw a careful, labeled sketch of the region described, as well as the resulting solid that results from revolving the region about the stated axis. In addition, draw a representative slice and state the volume of that slice, along with a definite integral whose value is the volume of the entire solid. It is not necessary to evaluate the integrals you find.

(a) The region S bounded by the y-axis, the curve $y = \sqrt{x}$, and the line $y = 2$; revolve S about the y-axis.

(b) The region S bounded by the x-axis, the curve $y = \sqrt{x}$, and the line $x = 4$; revolve S about the y-axis.

(c) The finite region S in the first quadrant bounded by the curves $y = 2x$ and $y = x^3$; revolve S about the x-axis.

(d) The finite region S in the first quadrant bounded by the curves $y = 2x$ and $y = x^3$; revolve S about the y-axis.

(e) The finite region S bounded by the curves $x = (y-1)^2$ and $y = x - 1$; revolve S about the y-axis

6.2.4 Revolving about horizontal and vertical lines other than the coordinate axes

It is possible to revolve a region around any horizontal or vertical line. Doing so adjusts the radii of the cylinders or washers involved by a constant value. A careful, well-labeled plot of the solid of revolution will usually reveal how the different axis of revolution affects the definite integral.

Example 6.2.8 Find the volume of the solid of revolution generated when the finite region S that lies between $y = x^2$ and $y = x$ is revolved about the line $y = -1$.

Solution. Graphing the region between the two curves in the first quadrant between their points of intersection ($(0,0)$ and $(1,1)$) and then revolving the region about the line $y = -1$, we see the solid shown in Figure 6.2.9. Each slice of the solid perpendicular to the axis of revolution is a washer, and the radii of each washer are governed by the curves $y = x^2$ and $y = x$. But we also see that there is one added change: the axis of revolution adds a fixed length to each radius. The inner radius of a typical slice, $r(x)$, is given by $r(x) = x^2 + 1$, while the outer radius is $R(x) = x + 1$.

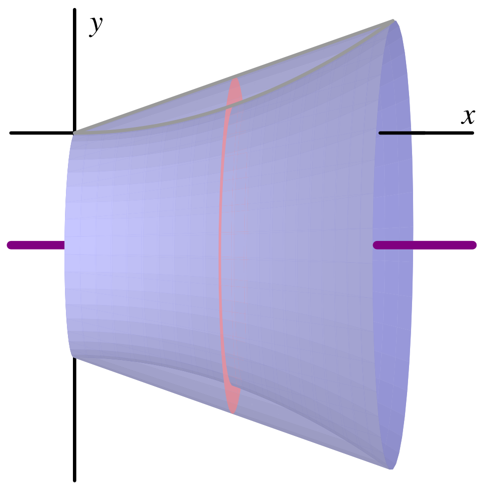


Figure 6.2.9 The solid of revolution described in Example 6.2.8.

Therefore, the volume of a typical slice is

$$V_{\text{slice}} = \pi[R(x)^2 - r(x)^2]\Delta x = \pi\left[(x+1)^2 - (x^2+1)^2\right]\Delta x.$$

Finally, we integrate to find the total volume, and

$$V = \int_0^1 \pi\left[(x+1)^2 - (x^2+1)^2\right]\, dx = \frac{7}{15}\pi.$$

◊

Activity 6.2.4. In each of the following questions, draw a careful, labeled sketch of the region described, as well as the resulting solid that results from revolving the region about the stated axis. In addition, draw a representative slice and state the volume of that slice, along with a definite integral whose value is the volume of the entire solid. It is not necessary to evaluate the integrals you find. For each prompt, use the finite region S in the first quadrant bounded by the curves $y = 2x$ and $y = x^3$.

(a) Revolve S about the line $y = -2$.

(b) Revolve S about the line $y = 4$.

(c) Revolve S about the line $x = -1$.

(d) Revolve S about the line $x = 5$.

6.2.5 Summary

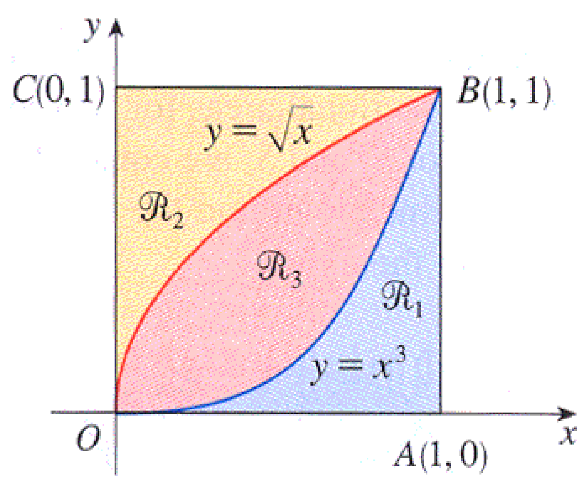
- We can use a definite integral to find the volume of a three-dimensional solid of revolution that results from revolving a two-dimensional region about a particular axis by taking slices perpendicular to the axis of revolution which will then be circular disks or washers.
- If we revolve about a vertical line and slice perpendicular to that line, then our slices are horizontal and of thickness Δy. This leads us to integrate with respect to y, as opposed to with respect to x when we slice a solid vertically.
- If we revolve about a line other than the x- or y-axis, we need to carefully account for the shift that occurs in the radius of a typical slice. Normally, this shift involves taking a sum or difference of the function along with the constant connected to the equation for the horizontal or vertical line; a well-labeled diagram is usually the best way to decide the new expression for the radius.

6.2.6 Exercises

1. The region bounded by $y = e^x, y = 0, x = -2, x = -1$ is rotated around the x-axis. Find the volume.

volume = ______________________________

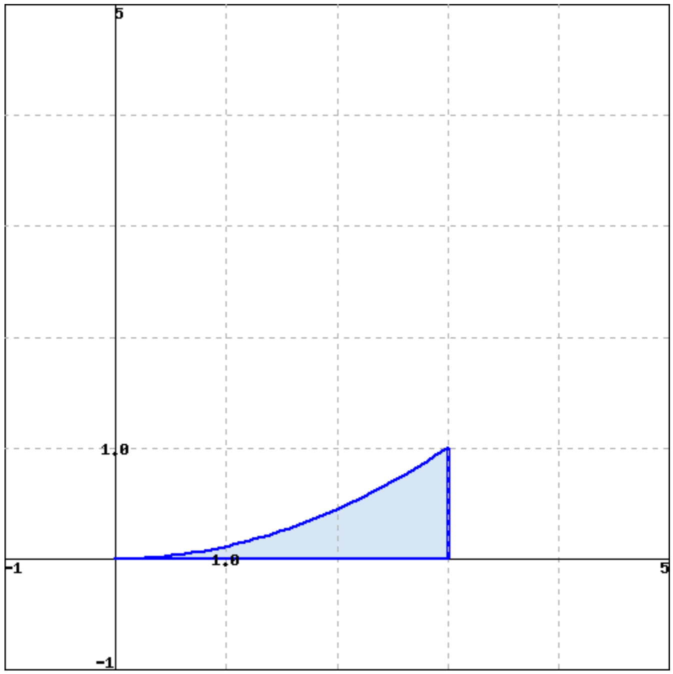
2.



Referring to the figure above, find the volume generated by rotating the region $\mathcal{R}_2$ about the line OA.

Volume = ____________________

3. Find the volume of the solid obtained by rotating the region bounded by $y = \frac{1}{9}x^2$, $x = 3$, and $y = 0$ about the y-axis. Below is a graph of the bounded region.



Volume = ____________________

Note: *You can click on the graph to enlarge the image.*

4. Find the volume of the solid obtained by rotating the region bounded by $y = x$ and $y = \sqrt{x}$ about the line $x = 4$.

Volume = ____________________

5. Which of the following integrals represents the volume of the solid obtained by rotating the region bounded by the curves $y = (x-2)^4$ and $8x - y = 16$ about the line $x = 10$?

○ $\pi \int_0^{16} \left\{ \left[10 - \left(\frac{1}{8}y + 2 \right)^2 \right] - \left[10 - \left(2 + \sqrt[4]{y} \right)^2 \right] \right\} dy$

○ $\pi \int_2^{4} \left\{ \left[10 - \left(\frac{1}{8}y + 2 \right) \right]^2 - \left[10 - \left(2 + \sqrt[4]{y} \right) \right]^2 \right\} dy$

○ $\pi \int_2^{4} \left\{ \left[10 - \left(\frac{1}{8}y + 2 \right)^2 \right] - \left[10 - \left(2 + \sqrt[4]{y} \right)^2 \right] \right\} dy$

○ $\pi \int_0^{16} \left\{ \left[10 - \left(\frac{1}{8}y + 2 \right) \right] - \left[10 - (2 + \sqrt[4]{y}) \right] \right\}^2 dy$

○ $\pi \int_0^{16} \left\{ \left[10 - \left(\frac{1}{8}y + 2 \right) \right]^2 - \left[10 - (2 + \sqrt[4]{y}) \right]^2 \right\} dy$

○ $\pi \int_2^{4} \left\{ \left[10 - \left(\frac{1}{8}y + 2 \right) \right] - \left[10 - (2 + \sqrt[4]{y}) \right] \right\}^2 dy$

6. Consider the curve $f(x) = 3\cos(\frac{x^3}{4})$ and the portion of its graph that lies in the first quadrant between the y-axis and the first positive value of x for which $f(x) = 0$. Let R denote the region bounded by this portion of f, the x-axis, and the y-axis.

a. Set up a definite integral whose value is the exact arc length of f that lies along the upper boundary of R. Use technology appropriately to evaluate the integral you find.

b. Set up a definite integral whose value is the exact area of R. Use technology appropriately to evaluate the integral you find.

c. Suppose that the region R is revolved around the x-axis. Set up a definite integral whose value is the exact volume of the solid of revolution that is generated. Use technology appropriately to evaluate the integral you find.

d. Suppose instead that R is revolved around the y-axis. If possible, set up an integral expression whose value is the exact volume of the solid of revolution and evaluate the integral using appropriate technology. If not possible, explain why.

7. Consider the curves given by $y = \sin(x)$ and $y = \cos(x)$. For each of the following problems, you should include a sketch of the region/solid being considered, as well as a labeled representative slice.

a. Sketch the region R bounded by the y-axis and the curves $y = \sin(x)$ and $y = \cos(x)$ up to the first positive value of x at which they intersect. What is the exact intersection point of the curves?

b. Set up a definite integral whose value is the exact area of R.

c. Set up a definite integral whose value is the exact volume of the solid of revolution generated by revolving R about the x-axis.

d. Set up a definite integral whose value is the exact volume of the solid of revolution generated by revolving R about the y-axis.

e. Set up a definite integral whose value is the exact volume of the solid of revolution generated by revolving R about the line $y = 2$.

f. Set up a definite integral whose value is the exact volume of the solid of revolution generated by revolving R about the line $x = -1$.

8. Consider the finite region R that is bounded by the curves $y = 1 + \frac{1}{2}(x-2)^2$, $y = \frac{1}{2}x^2$, and $x = 0$.

a. Determine a definite integral whose value is the area of the region enclosed by the two curves.

b. Find an expression involving one or more definite integrals whose value is the volume of the solid of revolution generated by revolving the region R about the line $y = -1$.

c. Determine an expression involving one or more definite integrals whose value is the volume of the solid of revolution generated by revolving the region R about the y-axis.

d. Find an expression involving one or more definite integrals whose value is the perimeter of the region R.

6.3 Density, mass, and center of mass

Motivating Questions

- How are mass, density, and volume related?
- How is the mass of an object with varying density computed?
- What is the center of mass of an object, and how are definite integrals used to compute it?

6.3.1 Introduction

Studying the units on the integrand and variable of integration helps us understand the meaning of a definite integral. For instance, if $v(t)$ is the velocity of an object moving along an axis, measured in feet per second, and t measures time in seconds, then both the definite integral and its Riemann sum approximation,

$$\int_a^b v(t)\,dt \approx \sum_{i=1}^{n} v(t_i)\Delta t,$$

have units given by the product of the units of $v(t)$ and t:

$$(\text{feet/sec}) \cdot (\text{sec}) = \text{feet}.$$

Thus, $\int_a^b v(t)\,dt$ measures the total change in position of the moving object in feet.

Unit analysis will be particularly helpful to us in what follows.

Preview Activity 6.3.1. In each of the following scenarios, we consider the distribution of a quantity along an axis.

(a) Suppose that the function $c(x) = 200 + 100e^{-0.1x}$ models the density of traffic on a straight road, measured in cars per mile, where x is number of miles east of a major interchange, and consider the definite integral $\int_0^2 (200 + 100e^{-0.1x})\,dx$.

i. What are the units on the product $c(x) \cdot \Delta x$?

ii. What are the units on the definite integral and its Riemann sum approximation given by

$$\int_0^2 c(x)\,dx \approx \sum_{i=1}^{n} c(x_i)\Delta x?$$

iii. Evaluate the definite integral $\int_0^2 c(x)\,dx = \int_0^2 \left(200 + 100e^{-0.1x}\right)\,dx$ and write one sentence to explain the meaning of the value you find.

(b) On a 6 foot long shelf filled with books, the function B models the distribution of the weight of the books, in pounds per inch, where x is the number of inches from the left end of the bookshelf. Let $B(x)$ be given by the rule $B(x) = 0.5 + \frac{1}{(x+1)^2}$.

i. What are the units on the product $B(x) \cdot \Delta x$?

ii. What are the units on the definite integral and its Riemann sum approximation given by

$$\int_{12}^{36} B(x)\,dx \approx \sum_{i=1}^{n} B(x_i)\Delta x?$$

iii. Evaluate the definite integral $\int_0^{72} B(x)\,dx = \int_0^{72} \left(0.5 + \frac{1}{(x+1)^2}\right) dx$ and write one sentence to explain the meaning of the value you find.

6.3.2 Density

The *mass* of a quantity, typically measured in metric units such as grams or kilograms, is a measure of the amount of the quantity. In a corresponding way, the *density* of an object measures the distribution of mass per unit volume. For instance, if a brick has mass 3 kg and volume 0.002 m^3, then the density of the brick is

$$\frac{3\text{kg}}{0.002\text{m}^3} = 1500\frac{\text{kg}}{\text{m}^3}.$$

As another example, the mass density of water is 1000 kg/m^3. Each of these relationships demonstrate the following general principle.

Density, mass, and volume.

For an object of constant density d, with mass m and volume V,

$$d = \frac{m}{V}, \text{ or } m = d \cdot V.$$

But what happens when the density is not constant?

The formula $m = d \cdot V$ is reminiscent of two other equations that we have used in our work: for a body moving in a fixed direction, distance = rate · time, and, for a rectangle, its area is given by $A = l \cdot w$. These formulas hold when the principal quantities involved, such as the rate the body moves and the height of the rectangle, are *constant*. When these quantities are not constant, we have turned to the definite integral for assistance. By working with small slices on which the quantity of interest (such as velocity) is approximately constant, we can use a definite integral to add up the values on the pieces.

For example, if we have a nonnegative velocity function that is not constant, over a short time interval Δt we know that the distance traveled is approximately $v(t)\Delta t$, since $v(t)$ is almost constant on a small interval. Similarly, if we are thinking about the area under a nonnegative function f whose value is changing, on a short interval Δx the area under the curve is approximately the area of the rectangle whose height is $f(x)$ and whose width is Δx: $f(x)\Delta x$. Both of these principles are represented visually in Figure 6.3.1.

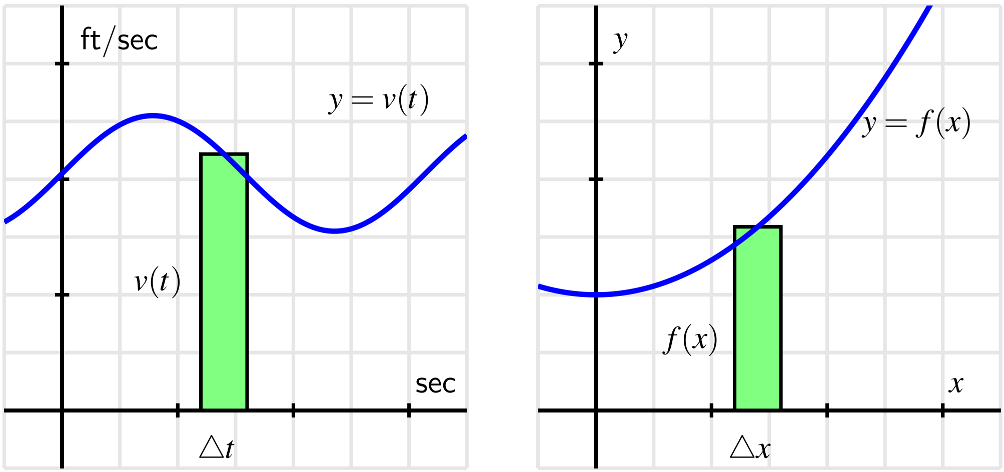


Figure 6.3.1 At left, estimating a small amount of distance traveled, $v(t)\Delta t$, and at right, a small amount of area under the curve, $f(x)\Delta x$.

In a similar way, if the density of some object is not constant, we can use a definite integral to compute the overall mass of the object. We will focus on problems where the density varies in only one dimension, say along a single axis.

Let's consider a thin bar of length b whose left end is at the origin, where $x = 0$, and assume that the bar has constant cross-sectional area of 1 cm^2. We let $\rho(x)$ represent the mass density function of the bar, measured in grams per cubic centimeter. That is, given a location x, $\rho(x)$ tells us approximately how much mass will be found in a one-centimeter wide slice of the bar at x.

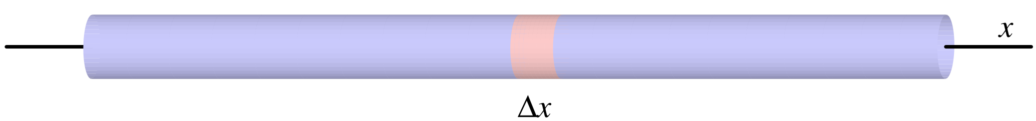


Figure 6.3.2 A thin bar of constant cross-sectional area 1 cm^2 with density function $\rho(x)$ g/cm^3.

The volume of a thin slice of the bar of width Δx, as pictured in Figure 6.3.2, is the cross-sectional area times Δx. Since the cross-sections each have constant area 1 cm^2, it follows that the volume of the slice is $1\Delta x$ cm^3. And because mass is the product of density and volume, we see that the mass of this slice is approximately

$$\text{mass}_{\text{slice}} \approx \rho(x)\,\frac{\text{g}}{\text{cm}^3} \cdot 1\Delta x\ \text{cm}^3 = \rho(x)\cdot\Delta x\ \text{g}.$$

The corresponding Riemann sum (and the integral that it approximates),

$$\sum_{i=1}^{n} \rho(x_i)\Delta x \approx \int_0^b \rho(x)\,dx,$$

therefore measure the mass of the bar between 0 and b. (The Riemann sum is an approximation, while the integral will be the exact mass.)

For objects whose cross-sectional area is constant and whose mass is distributed relative to horizontal location, x, it makes sense to think of the density function $\rho(x)$ with units "mass per unit length," such as g/cm. Thus, when we compute $\rho(x) \cdot \Delta x$ on a small slice Δx, the resulting units are g/cm · cm = g, which thus measures the mass of the slice. The general principle follows.

The mass of a thin rod.

For an object of constant cross-sectional area whose mass is distributed along a single axis according to the function $\rho(x)$ (whose units are units of mass per unit of length), the total mass, M, of the object between $x = a$ and $x = b$ is given by

$$M = \int_a^b \rho(x)\,dx.$$

Activity 6.3.2. Consider the following situations in which mass is distributed in a non-constant manner.

(a) Suppose that a thin rod with constant cross-sectional area of 1 cm^2 has its mass distributed according to the density function $\rho(x) = 2e^{-0.2x}$, where x is the distance in cm from the left end of the rod, and the units on $\rho(x)$ are g/cm. If the rod is 10 cm long, determine the exact mass of the rod.

(b) Consider the cone that has a base of radius 4 m and a height of 5 m. Picture the cone lying horizontally with the center of its base at the origin and think of the cone as a solid of revolution.

i. Write and evaluate a definite integral whose value is the volume of the cone.

ii. Next, suppose that the cone has uniform density of 800 kg/m^3. What is the mass of the solid cone?

iii. Now suppose that the cone's density is not uniform, but rather that the cone is most dense at its base. In particular, assume that the density of the cone is uniform across cross sections parallel to its base, but that in each such cross section that is a distance x units from the origin, the density of the cross section is given by the function $\rho(x) = 400 + \frac{200}{1+x^2}$, measured in kg/m^3. Determine and evaluate a definite integral whose value is the mass of this cone of non-uniform density. Do so by first thinking about the mass of a given slice of the cone x units away from the base; remember that in such a slice, the density will be *essentially constant*.

(c) Let a thin rod of constant cross-sectional area 1 cm^2 and length 12 cm have its mass be distributed according to the density function $\rho(x) = \frac{1}{25}(x-15)^2$, measured in g/cm. Find the exact location z at which to cut the bar so that the two pieces will each have identical mass.

6.3.3 Weighted Averages

The concept of an average is a natural one, and one that we have used repeatedly as part of our understanding of the meaning of the definite integral. If we have n values $a_1, a_2, \ldots, a_n$, we know that their average is given by

$$\frac{a_1 + a_2 + \cdots + a_n}{n},$$

and for a quantity being measured by a function f on an interval $[a,b]$, the average value of the quantity on $[a,b]$ is

$$\frac{1}{b-a}\int_a^b f(x)\,dx.$$

As we continue to think about problems involving the distribution of mass, it is natural to consider the idea of a *weighted* average, where certain quantities involved are counted more in the average.

A common use of weighted averages is in the computation of a student's GPA, where grades are weighted according to credit hours. Let's consider the scenario in Table 6.3.3.

class	grade	grade points	credits
chemistry	B+	3.3	5
calculus	A-	3.7	4
history	B-	2.7	3
psychology	B-	2.7	3

Table 6.3.3 A college student's semester grades.

If all of the classes were of the same weight (i.e., the same number of credits), the student's GPA would simply be calculated by taking the average

$$\frac{3.3 + 3.7 + 2.7 + 2.7}{4} = 3.1.$$

But since the chemistry and calculus courses have higher weights (of 5 and 4 credits respectively), we actually compute the GPA according to the weighted average

$$\frac{3.3 \cdot 5 + 3.7 \cdot 4 + 2.7 \cdot 3 + 2.7 \cdot 3}{5 + 4 + 3 + 3} = 3.1\overline{6}.$$

The weighted average reflects the fact that chemistry and calculus, as courses with higher credits, have a greater impact on the students' grade point average. Note particularly that in the weighted average, each grade gets multiplied by its weight, and we divide by the sum of the weights.

In the following activity, we explore further how weighted averages can be used to find the balancing point of a physical system.

Activity 6.3.3. For quantities of equal weight, such as two children on a teeter-totter, the balancing point is found by taking the average of their locations. When the weights of the quantities differ, we use a weighted average of their respective locations to find the balancing point.

(a) Suppose that a shelf is 6 feet long, with its left end situated at $x = 0$. If one book of weight 1 lb is placed at $x_1 = 0$, and another book of weight 1 lb is placed at $x_2 = 6$, what is the location of $\overline{x}$, the point at which the shelf would (theoretically) balance on a fulcrum?

(b) Now, say that we place four books on the shelf, each weighing 1 lb: at $x_1 = 0$, at $x_2 = 2$, at $x_3 = 4$, and at $x_4 = 6$. Find $\overline{x}$, the balancing point of the shelf.

(c) How does $\overline{x}$ change if we change the location of the third book? Say the locations of the 1-lb books are $x_1 = 0$, $x_2 = 2$, $x_3 = 3$, and $x_4 = 6$.

(d) Next, suppose that we place four books on the shelf, but of varying weights: at $x_1 = 0$ a 2-lb book, at $x_2 = 2$ a 3-lb book, at $x_3 = 4$ a 1-lb book, and at $x_4 = 6$ a 1-lb book. Use a weighted average of the locations to find $\overline{x}$, the balancing point of the shelf. How does the balancing point in this scenario compare to that found in (b)?

(e) What happens if we change the location of one of the books? Say that we keep everything the same in (d), except that $x_3 = 5$. How does $\overline{x}$ change?

(f) What happens if we change the weight of one of the books? Say that we keep everything the same in (d), except that the book at $x_3 = 4$ now weighs 2 lbs. How does $\overline{x}$ change?

(g) Experiment with a couple of different scenarios of your choosing where you move one of the books to the left, or you decrease the weight of one of the books.

(h) Write a couple of sentences to explain how adjusting the location of one of the books or the weight of one of the books affects the location of the balancing point of the shelf. Think carefully here about how your changes should be considered relative to the location of the balancing point $\overline{x}$ of the current scenario.

6.3.4 Center of Mass

In Activity 6.3.3, we saw that the balancing point of a system of point-masses[1] (such as books on a shelf) is found by taking a weighted average of their respective locations. In the activity, we were computing the *center of mass* of a system of masses distributed along an axis, which

[1] In the activity, we actually used *weight* rather than *mass*. Since weight is proportional to mass, the computations for the balancing point result in the same location regardless of whether we use weight or mass. The gravitational constant is present in both the numerator and denominator of the weighted average.

is the balancing point of the axis on which the masses rest. Here we formally summarize the results of the activity, which hold for any finite collection of point-masses. This forms an important foundation for when mass varies continuously.

Center of Mass (point-masses).

For a collection of n masses $m_1, \ldots, m_n$ that are distributed along a single axis at the locations $x_1, \ldots, x_n$, the center of mass is given by

$$\overline{x} = \frac{x_1 m_1 + x_2 m_2 + \cdots + x_n m_n}{m_1 + m_2 + \cdots + m_n}.$$

Now consider a thin bar over which density is distributed continuously. If the density is constant, it is obvious that the balancing point of the bar is its midpoint. But if density is not constant, we used the idea of a weighted average to find its balancing point. Let's say that the function $\rho(x)$ tells us the density distribution along the bar, measured in g/cm. If we slice the bar into small sections, we can think of the bar as holding a collection of adjacent point-masses. The mass m_i of a slice of thickness Δx at location x_i, is $m_i \approx \rho(x_i)\Delta x$.

If we slice the bar into n pieces, we can approximate its center of mass by

$$\overline{x} \approx \frac{x_1 \cdot \rho(x_1)\Delta x + x_2 \cdot \rho(x_2)\Delta x + \cdots + x_n \cdot \rho(x_n)\Delta x}{\rho(x_1)\Delta x + \rho(x_2)\Delta x + \cdots + \rho(x_n)\Delta x}.$$

Rewriting the sums in sigma notation, we have

$$\overline{x} \approx \frac{\sum_{i=1}^{n} x_i \cdot \rho(x_i)\Delta x}{\sum_{i=1}^{n} \rho(x_i)\Delta x}. \tag{6.3.1}$$

The greater the number of slices, the more accurate our estimate of the balancing point will be. The sums in Equation (6.3.1) can be viewed as Riemann sums, so in the limit as $n \to \infty$, we find that the center of mass is given by the quotient of two integrals.

Center of Mass (continuous mass distribution).

For a thin rod of density $\rho(x)$ distributed along an axis from $x = a$ to $x = b$, the center of mass of the rod is given by

$$\overline{x} = \frac{\int_a^b x\rho(x)\, dx}{\int_a^b \rho(x)\, dx}.$$

Note that the denominator of $\overline{x}$ is the mass of the bar, and that this quotient of integrals is simply the continuous version of the weighted average of locations, x, along the bar.

Activity 6.3.4. Consider a thin bar of length 20 cm whose density is distributed according to the function $\rho(x) = 4 + 0.1x$, where $x = 0$ represents the left end of the bar. Assume that ρ is measured in g/cm and x is measured in cm.

(a) Find the total mass, M, of the bar.

(b) Without doing any calculations, do you expect the center of mass of the bar to be equal to 10, less than 10, or greater than 10? Why?

(c) Compute $\overline{x}$, the exact center of mass of the bar.

(d) What is the average density of the bar?

(e) Now consider a different density function, given by $p(x) = 4e^{0.020732x}$, also for a bar of length 20 cm whose left end is at $x = 0$. Plot both $\rho(x)$ and $p(x)$ on the same axes. Without doing any calculations, which bar do you expect to have the greater center of mass? Why?

(f) Compute the exact center of mass of the bar described in (e) whose density function is $p(x) = 4e^{0.020732x}$. Check the result against the prediction you made in (e).

6.3.5 Summary

- For an object of constant density D, with volume V and mass m, we know that

$$m = D \cdot V.$$

- If an object with constant cross-sectional area (such as a thin bar) has its density distributed along an axis according to the function $\rho(x)$, then we can find the mass of the object between $x = a$ and $x = b$ by

$$m = \int_a^b \rho(x)\,dx.$$

- For a system of point-masses distributed along an axis, say $m_1, \ldots, m_n$ at locations $x_1, \ldots, x_n$, the center of mass, $\overline{x}$, is given by the weighted average

$$\overline{x} = \frac{\sum_{i=1}^n x_i m_i}{\sum_{i=1}^n m_i}.$$

  If instead we have mass continuously distributed along an axis, such as by a density function $\rho(x)$ for a thin bar of constant cross-sectional area, the center of mass of the portion of the bar between $x = a$ and $x = b$ is given by

$$\overline{x} = \frac{\int_a^b x\rho(x)\,dx}{\int_a^b \rho(x)\,dx}.$$

  In each situation, $\overline{x}$ represents the balancing point of the system of masses or of the portion of the bar.

6.3.6 Exercises

1. **Center of mass for a linear density function.** A rod has length 4 meters. At a distance x meters from its left end, the density of the rod is given by $\delta(x) = 5 + 4x$ g/m.

   (a) Complete the Riemann sum for the total mass of the rod (use Dx in place of Δx):

   mass = Σ ________________

   (b) Convert the Riemann sum to an integral and find the exact mass.

   mass = ________________

   (include)

2. **Center of mass for a nonlinear density function.** A rod with uniform density (mass/unit length) $\delta(x) = 2 + \sin(x)$ lies on the x-axis between $x = 0$ and $x = \pi$. Find the mass and center of mass of the rod.

   mass = ______________________________________

   center of mass = ______________________________________

3. **Interpreting the density of cars on a road.** Suppose that the density of cars (in cars per mile) down a 20-mile stretch of the Pennsylvania Turnpike is approximated by $\delta(x) = 250\left(2 + \sin\left(4\sqrt{x + 0.175}\right)\right)$, at a distance x miles from the Breezewood toll plaza. Sketch a graph of this function for $0 \le x \le 20$.

   (a) Complete the Riemann sum that approximates the total number of cars on this 20-mile stretch (use Dx instead of Δx):

   Number = Σ______________________________________

   (b) Find the total number of cars on the 20-mile stretch.

   Number = ___________________________

4. **Center of mass in a point-mass system.** A point mass of 2 grams located 6 centimeters to the left of the origin and a point mass of 3 grams located 7 centimeters to the right of the origin are connected by a thin, light rod. Find the center of mass of the system.

   Center of Mass = __________________________________ (□ ? □ to the left of the origin □ to the right of the origin □ (at the origin))

   (include in your center of mass)

5. Let a thin rod of length a have density distribution function $\rho(x) = 10e^{-0.1x}$, where x is measured in cm and ρ in grams per centimeter.

   a. If the mass of the rod is 30 g, what is the value of a?

   b. For the 30g rod, will the center of mass lie at its midpoint, to the left of the midpoint, or to the right of the midpoint? Why?

   c. For the 30g rod, find the center of mass, and compare your prediction in (b).

   d. At what value of x should the 30g rod be cut in order to form two pieces of equal mass?

6. Consider two thin bars of constant cross-sectional area, each of length 10 cm, with respective mass density functions $\rho(x) = \frac{1}{1+x^2}$ and $p(x) = e^{-0.1x}$.

a. Find the mass of each bar.

b. Find the center of mass of each bar.

c. Now consider a new 10 cm bar whose mass density function is $f(x) = \rho(x) + p(x)$.

(i) Explain how you can easily find the mass of this new bar with little to no additional work.

(ii) Similarly, compute $\int_0^{10} x f(x)\,dx$ as simply as possible, in light of earlier computations.

(iii) True or false: the center of mass of this new bar is the average of the centers of mass of the two earlier bars. Write at least one sentence to say why your conclusion makes sense.

7. Consider the curve given by $y = f(x) = 2xe^{-1.25x} + (30 - x)e^{-0.25(30-x)}$.

a. Plot this curve in the window $x = 0 \ldots 30$, $y = 0 \ldots 3$ (with constrained scaling so the units on the x and y axis are equal), and use it to generate a solid of revolution about the x-axis. Explain why this curve could generate a reasonable model of a baseball bat.

b. Let x and y be measured in inches. Find the total volume of the baseball bat generated by revolving the given curve about the x-axis. Include units on your answer.

c. Suppose that the baseball bat has constant weight density, and that the weight density is 0.6 ounces per cubic inch. Find the total weight of the bat whose volume you found in (b).

d. Because the baseball bat does not have constant cross-sectional area, we see that the amount of weight concentrated at a location x along the bat is determined by the volume of a slice at location x. Explain why we can think about the function $\rho(x) = 0.6\pi f(x)^2$ (where f is the function given at the start of the problem) as being the weight density function for how the weight of the baseball bat is distributed from $x = 0$ to $x = 30$.

e. Compute the center of mass of the baseball bat.

6.4 Physics applications: work, force, and pressure

Motivating Questions

- How do we measure the work accomplished by a varying force that moves an object a certain distance?
- What is the total force exerted by water against a dam?
- How are both of the above concepts and their corresponding use of definite integrals similar to problems we have encountered in the past involving formulas such as "distance equals rate times time" and "mass equals density times volume"?

6.4.1 Introduction

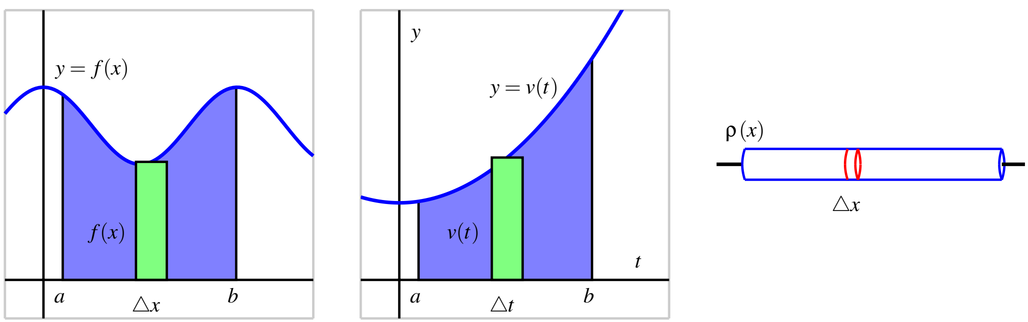


Figure 6.4.1 Three settings where we compute the accumulation of a varying quantity: the area under $y = f(x)$, the distance traveled by an object with velocity $y = v(t)$, and the mass of a bar with density function $y = \rho(x)$.

We have seen several different circumstances where the definite integral enables us to measure the accumulation of a quantity that varies, provided the quantity is approximately constant over small intervals. For instance, to find the area bounded by a nonnegative curve $y = f(x)$ and the x-axis on an interval $[a, b]$, we take a representative slice of width Δx that has area $A_{\text{slice}} = f(x)\Delta x$. As we let the width of the representative slice tend to zero, we find that the exact area of the region is

$$A = \int_a^b f(x)\,dx.$$

In a similar way, if we know the velocity $v(t)$ of a moving object and we wish to know the distance the object travels on an interval $[a, b]$ where $v(t)$ is nonnegative, we can use a definite integral to generalize the fact that $d = r \cdot t$ when the rate, r, is constant. On a short time interval Δt, $v(t)$ is roughly constant, so for a small slice of time, $d_{\text{slice}} = v(t)\Delta t$. As the width of the time interval Δt tends to zero, the exact distance traveled is given by the definite integral

$$d = \int_a^b v(t)\,dt.$$

Finally, if we want to determine the mass of an object of non-constant density, because $M = D \cdot V$ (mass equals density times volume, provided that density is constant), we can consider a small slice of an object on which the density is approximately constant, and a definite integral may be used to determine the exact mass of the object. For instance, if we have a thin rod whose cross sections have constant density, but whose density is distributed along the x axis according to the function $y = \rho(x)$, it follows that for a small slice of the rod that is Δx thick, $M_{\text{slice}} = \rho(x)\Delta x$. In the limit as $\Delta x \to 0$, we then find that the total mass is given by

$$M = \int_a^b \rho(x)\,dx.$$

All three of these situations are similar in that we have a basic rule ($A = l \cdot w$, $d = r \cdot t$, $M = D \cdot V$) where one of the two quantities being multiplied is no longer constant; in each, we consider a small interval for the other variable in the formula, calculate the approximate value of the desired quantity (area, distance, or mass) over the small interval, and then use a definite integral to sum the results as the length of the small intervals is allowed to approach zero. It should be apparent that this approach will work effectively for other situations where we have a quantity that varies.

We next turn to the notion of *work*: from physics, a basic principle is that work is the product of force and distance. For example, if a person exerts a force of 20 pounds to lift a 20-pound weight 4 feet off the ground, the total work accomplished is

$$W = F \cdot d = 20 \cdot 4 = 80 \text{ foot-pounds}.$$

If force and distance are measured in English units (pounds and feet), then the units of work are *foot-pounds*. If we work in metric units, where forces are measured in Newtons and distances in meters, the units of work are *Newton-meters*.

Of course, the formula $W = F \cdot d$ only applies when the force is constant over the distance d. In Preview Activity 6.4.1, we explore one way that we can use a definite integral to compute the total work accomplished when the force exerted varies.

Preview Activity 6.4.1. A bucket is being lifted from the bottom of a 50-foot deep well; its weight (including the water), B, in pounds at a height h feet above the water is given by the function $B(h)$. When the bucket leaves the water, the bucket and water together weigh $B(0) = 20$ pounds, and when the bucket reaches the top of the well, $B(50) = 12$ pounds. Assume that the bucket loses water at a constant rate (as a function of height, h) throughout its journey from the bottom to the top of the well.

(a) Find a formula for $B(h)$.

(b) Compute the value of the product $B(5)\Delta h$, where $\Delta h = 2$ feet. Include units on your answer. Explain why this product represents the approximate work it took to move the bucket of water from $h = 5$ to $h = 7$.

(c) Is the value in (b) an over- or under-estimate of the actual amount of work it took to move the bucket from $h = 5$ to $h = 7$? Why?

(d) Compute the value of the product $B(22)\Delta h$, where $\Delta h = 0.25$ feet. Include units on your answer. What is the meaning of the value you found?

(e) More generally, what does the quantity $W_{\text{slice}} = B(h)\Delta h$ measure for a given value of h and a small positive value of Δh?

(f) Evaluate the definite integral $\int_0^{50} B(h)\,dh$. What is the meaning of the value you find? Why?

6.4.2 Work

Because work is calculated by the rule $W = F \cdot d$ whenever the force F is constant, it follows that we can use a definite integral to compute the work accomplished by a varying force. For example, suppose that a bucket whose weight at height h is given by $B(h) = 12 + 8e^{-0.1h}$ is being lifted in a 50-foot well.

In contrast to the problem in the preview activity, this bucket is not leaking at a constant rate; but because the weight of the bucket and water is not constant, we have to use a definite integral to determine the total work done in lifting the bucket. At a height h above the water, the approximate work to move the bucket a small distance Δh is

$$W_{\text{slice}} = B(h)\Delta h = (12 + 8e^{-0.1h})\Delta h.$$

Hence, if we let Δh tend to 0 and take the sum of all of the slices of work accomplished on these small intervals, it follows that the total work is given by

$$W = \int_0^{50} B(h)\,dh = \int_0^{50} (12 + 8e^{-0.1h})\,dh.$$

While we could evaluate this integral exactly using the First Fundamental Theorem of Calculus, in applied settings such as this one we will typically use computing technology. Here, it turns out that $W = \int_0^{50} (12 + 8e^{-0.1h})\,dh \approx 679.461$ foot-pounds.

Our work in Preview Activity 6.4.1 and in the most recent discussion above employs the following important general principle.

The work done by a varying force.

For an object being moved in the positive direction along an axis with location x by a force $F(x)$, the total work to move the object from a to b is given by

$$W = \int_a^b F(x)\,dx.$$

Activity 6.4.2. Consider the following situations in which a varying force accomplishes work.

(a) Suppose that a heavy rope hangs over the side of a cliff. The rope is 200 feet long and weighs 0.3 pounds per foot; initially the rope is fully extended. How much work is required to haul in the entire length of the rope? (Hint: set up a function $F(h)$ whose value is the weight of the rope remaining over the cliff after h feet have been hauled in.)

(b) A leaky bucket is being hauled up from a 100 foot deep well. When lifted from the water, the bucket and water together weigh 40 pounds. As the bucket is being hauled upward at a constant rate, the bucket leaks water at a constant rate so that it is losing weight at a rate of 0.1 pounds per foot. What function $B(h)$ tells the weight of the bucket after the bucket has been lifted h feet? What is the total amount of work accomplished in lifting the bucket to the top of the well?

(c) Now suppose that the bucket in (b) does not leak at a constant rate, but rather that its weight at a height h feet above the water is given by $B(h) = 25 + 15e^{-0.05h}$. What is the total work required to lift the bucket 100 feet? What is the average force exerted on the bucket on the interval $h = 0$ to $h = 100$?

(d) From physics, *Hooke's Law* for springs states that the amount of force required to hold a spring that is compressed (or extended) to a particular length is proportionate to the distance the spring is compressed (or extended) from its natural length. That is, the force to compress (or extend) a spring x units from its natural length is $F(x) = kx$ for some constant k (which is called the *spring constant*.) For springs, we choose to measure the force in pounds and the distance the spring is compressed in feet. Suppose that a force of 5 pounds extends a particular spring 4 inches (1/3 foot) beyond its natural length.

i. Use the given fact that $F(1/3) = 5$ to find the spring constant k.
ii. Find the work done to extend the spring from its natural length to 1 foot beyond its natural length.
iii. Find the work required to extend the spring from 1 foot beyond its natural length to 1.5 feet beyond its natural length.

6.4.3 Work: Pumping Liquid from a Tank

In certain geographic locations where the water table is high, residential homes with basements have a peculiar feature: in the basement, one finds a large hole in the floor, and in the hole, there is water. For example, in Figure 6.4.2 we see a *sump crock*[1]. A sump crock provides an outlet for water that may build up beneath the basement floor to prevent flooding the basement.

[1]Image credit to www.warreninspect.com/basement-moisture.

In the crock we see a floating pump. This pump is activated by elevation, so when the water level reaches a particular height, the pump turns on and pumps water out of the crock, hence relieving the water buildup beneath the foundation. One of the questions we'd like to answer is: how much work does a sump pump accomplish?

Figure 6.4.2 A sump crock.

Example 6.4.3 Suppose that a sump crock has the shape of a frustum of a cone, as pictured in Figure 6.4.4. The crock has a diameter of 3 feet at its surface, a diameter of 1.5 feet at its base, and a depth of 4 feet. In addition, suppose that the sump pump is set up so that it pumps the water vertically up a pipe to a drain that is located at ground level just outside a basement window. To accomplish this, the pump must send the water to a location 9 feet above the surface of the sump crock. How much work is required to empty the sump crock if it is initially completely full?

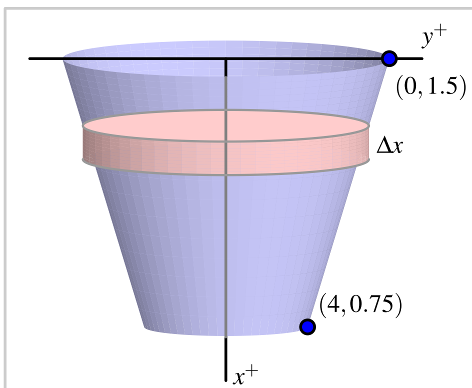


Figure 6.4.4 A sump crock with approximately cylindrical cross-sections.

Solution. It is helpful to think of the depth below the surface of the crock as being the independent variable, so we let the positive x-axis point down, and the positive y-axis to the right, as pictured in the figure. Because the pump sits on the surface of the water, it makes sense to think about the pump moving the water one "slice" at a time, where it takes a thin slice from the surface, pumps it out of the tank, and then proceeds to pump the next slice below.

Each slice of water is cylindrical in shape. We see that the radius of each slice varies according to the linear function $y = f(x)$ that passes through the points $(0, 1.5)$ and $(4, 0.75)$, where x is the depth of the particular slice in the tank; it is a straightforward exercise to find that $f(x) = 1.5 - 0.1875x$. Now we think about the problem in several steps:

a. determining the volume of a typical slice;

b. finding the weight[2] of a typical slice (and thus the force that must be exerted on it);

c. deciding the distance that a typical slice moves;

d. and computing the work to move a representative slice.

Once we know the work it takes to move one slice, we use a definite integral over an appropriate interval to find the total work.

Consider a representative cylindrical slice at a depth of x feet below the top of the crock. The approximate volume of that slice is given by

$$V_{\text{slice}} = \pi f(x)^2 \Delta x = \pi(1.5 - 0.1875x)^2 \Delta x.$$

Since water weighs 62.4 lb/ft^3, the approximate weight of a representative slice is

$$F_{\text{slice}} = 62.4 \cdot V_{\text{slice}} = 62.4\pi(1.5 - 0.1875x)^2 \Delta x.$$

This is also the approximate force the pump must exert to move the slice.

Because the slice is located at a depth of x feet below the top of the crock, the slice being moved by the pump must move x feet to get to the level of the basement floor, and then, as stated in the problem description, another 9 feet to reach the drain at ground level. Hence, the total distance a representative slice travels is

$$d_{\text{slice}} = x + 9.$$

Finally, the work to move a representative slice is given by

$$W_{\text{slice}} = F_{\text{slice}} \cdot d_{\text{slice}} = 62.4\pi(1.5 - 0.1875x)^2 \Delta x \cdot (x + 9).$$

We sum the work required to move slices throughout the tank (from $x = 0$ to $x = 4$), let $\Delta x \to 0$, and hence

$$W = \int_0^4 62.4\pi(1.5 - 0.1875x)^2(x + 9)\, dx.$$

When evaluated using appropriate technology, the integral shows that the total work is $W = 3463.2\pi$ foot-pounds. ◊

The preceding example demonstrates the standard approach to finding the work required to empty a tank filled with liquid. The main task in each such problem is to determine the volume of a representative slice, followed by the force exerted on the slice, as well as the distance such a slice moves. In the case where the units are metric, there is one key difference: in the metric setting, rather than weight, we normally first find the mass of a slice. For instance, if distance is measured in meters, the mass density of water is 1000 kg/m^3. In that setting, we can find the mass of a typical slice (in kg). To determine the force required to move it, we use $F = ma$, where m is the object's mass and a is the gravitational

[2]We assume that the weight density of water is 62.4 pounds per cubic foot.

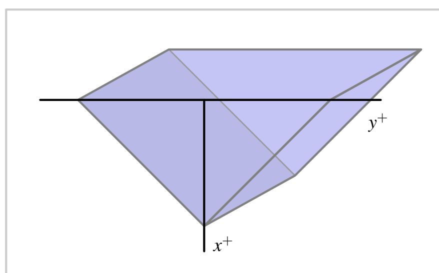
constant $a = 9.81$ N/kg. That is, in metric units, the weight density of water is 9810 N/m^3.

Activity 6.4.3. In each of the following situations, determine the total work required to accomplish the described task. In parts (b) and (c), a key step is to find a formula for a function that describes the curve that forms the side boundary of the tank.

(a) Consider a vertical cylindrical tank of radius 2 meters and depth 6 meters. Suppose the tank is filled with 4 meters of water of mass density 1000 kg/m^3, and the top 1 meter of water is pumped over the top of the tank.

(b) Consider a hemispherical tank with a radius of 10 feet. Suppose that the tank is full to a depth of 7 feet with water of weight density 62.4 pounds/ft^3, and the top 5 feet of water are pumped out of the tank to a tanker truck whose height is 5 feet above the top of the tank.

(c) Consider a trough with triangular ends, as pictured in the following figure where the tank is 10 feet long, the top is 5 feet wide, and the tank is 4 feet deep. Say that the trough is full to within 1 foot of the top with water of weight density 62.4 pounds/ft^3, and a pump is used to empty the tank until the water remaining in the tank is 1 foot deep.



6.4.4 Force due to Hydrostatic Pressure

When building a dam, engineers need to know how much force water will exert against the face of the dam. This force comes from water pressure. The pressure a force exerts on a region is measured in units of force per unit of area: for example, the air pressure in a tire is often measured in pounds per square inch (PSI). Hence, we see that the general relationship is given by

$$P = \frac{F}{A}, \text{ or } F = P \cdot A,$$

where P represents pressure, F represents force, and A the area of the region being considered. Of course, in the equation $F = PA$, we are assuming that the pressure is constant over the entire region A.

We know from experience that the deeper one dives underwater while swimming, the greater the pressure exerted by the water. This is because at a greater depth, there is more water right on top of the swimmer: it is the force that "column" of water exerts that determines the pressure the swimmer experiences. The total water pressure is found by computing the total weight of the column of water that lies above a region of area 1 square foot at a fixed depth. At a depth of d feet, a rectangular column has volume $V = 1 \cdot 1 \cdot d$ ft^3, so the corresponding weight of the water overhead is $62.4d$. This is the amount of force being exerted on a 1 square foot region at a depth d feet underwater, so the pressure exerted by water at depth d is $P = 62.4d$ (lbs/ft^2).

Because pressure is force per unit area, or $P = \frac{F}{A}$, we can compute the total force from a variable pressure by integrating $F = PA$.

Example 6.4.5 Consider a trapezoid-shaped dam that is 60 feet wide at its base and 90 feet wide at its top, and assume the dam is 25 feet tall with water that rises to within 5 feet of the top of its face. Water weighs 62.4 pounds per cubic foot. How much force does the water exert against the dam?

Solution. First, we sketch a picture of the dam, as shown in Figure 6.4.6. Note that, as in problems involving the work to pump out a tank, we let the positive x-axis point down.

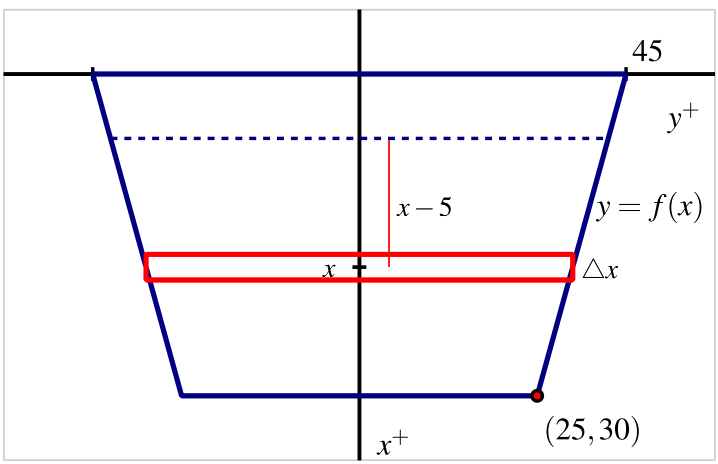


Figure 6.4.6 A trapezoidal dam that is 25 feet tall, 60 feet wide at its base, 90 feet wide at its top, with the water line 5 feet down from the top of its face.

Pressure is constant at a fixed depth, so we consider a slice of water at constant depth on the face, as shown in the figure. The area of this slice is approximately the area of the rectangle pictured. Since the width of that rectangle depends on the variable x, we find a formula for the line that represents one side of the dam. It is straightforward to find that $y = 45 - \frac{3}{5}x$. Hence, the approximate area of a representative slice is

$$A_{\text{slice}} = 2f(x)\Delta x = 2(45 - \frac{3}{5}x)\Delta x.$$

At any point on this slice, the depth is approximately constant, so the pressure can be considered constant. Because the water rises to within 5 feet of the top of the dam, the depth of any point on the representative slice is approximately $(x - 5)$. Now, since pressure is given by $P = 62.4d$, we have that at any point on the slice

$$P_{\text{slice}} = 62.4(x - 5).$$

Knowing both the pressure and area, we can find the force the water exerts on the slice. Using $F = PA$, it follows that

$$F_{\text{slice}} = P_{\text{slice}} \cdot A_{\text{slice}} = 62.4(x - 5) \cdot 2(45 - \frac{3}{5}x)\Delta x.$$

Finally, we use a definite integral to sum the forces over the appropriate range of x-values. Since the water rises to within 5 feet of the top of the dam, we start at $x = 5$ and take slices all the way to the bottom of the dam, where $x = 25$. Hence,

$$F = \int_{x=5}^{x=25} 62.4(x - 5) \cdot 2(45 - \frac{3}{5}x)\, dx.$$

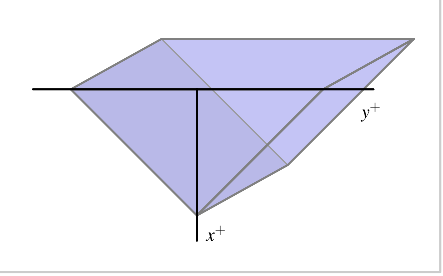
Using technology to evaluate the integral, we find $F = 848640$ pounds. ◊

Activity 6.4.4. In each of the following situations, determine the total force exerted by water against the surface that is described.

(a) Consider a rectangular dam that is 100 feet wide and 50 feet tall, and suppose that water presses against the dam all the way to the top.

(b) Consider a semicircular dam with a radius of 30 feet. Suppose that the water rises to within 10 feet of the top of the dam.

(c) Consider a trough with triangular ends, as pictured in the following figure where the tank is 10 feet long, the top is 5 feet wide, and the tank is 4 feet deep. Say that the trough is full to within 1 foot of the top with water of weight density 62.4 pounds/ft^3. How much force does the water exert against one of the triangular ends?



Although there are many different formulas involving work, force, and pressure, the fundamental ideas behind these problems are similar to others we've encountered in applications of the definite integral. We slice the quantity of interest into more manageable pieces and then use a definite integral to add them up.

6.4.5 Summary

- To measure the work done by a varying force in moving an object, we divide the problem into pieces on which we can use the formula $W = F \cdot d$, and then use a definite integral to sum the work done on each piece.
- To find the total force exerted by water against a dam, we use the formula $F = P \cdot A$ to measure the force exerted on a slice that lies at a fixed depth, and then use a definite integral to sum the forces across the appropriate range of depths.
- Because work is computed as the product of force and distance (provided force is constant), and the force water exerts on a dam can be computed as the product of pressure and area (provided pressure is constant), problems involving these concepts are similar to earlier problems we did using definite integrals to find distance (via "distance equals rate times time") and mass ("mass equals density times volume").

6.4.6 Exercises

1. Work to empty a conical tank. A tank in the shape of an inverted right circular cone has height 6 meters and radius 2 meters. It is filled with 5 meters of hot chocolate. Find the work required to empty the tank by pumping the hot chocolate over the top of the tank. The density of hot chocolate is $\delta = 1080 \text{ kg/m}^3$. Your answer must include the correct .

Work = ______________________________

2. Work to empty a cylindrical tank. A fuel oil tank is an upright cylinder, buried so that its circular top is 12 feet beneath ground level. The tank has a radius of 6 feet and is 18 feet high, although the current oil level is only 15 feet deep. Calculate the work required to pump all of the oil to the surface. Oil weighs 50 lb/ft^3.

Work = ______________________

(include)

3. Work to empty a rectangular pool. A rectangular swimming pool 50 ft long, 15 ft wide, and 8 ft deep is filled with water to a depth of 7 ft. Use an integral to find the work required to pump all the water out over the top. (Take as the density of water $\delta = 62.4 \text{lb/ft}^3$.)

Work = ______________________

(include)

4. Work to empty a cylindrical tank to differing heights. Water in a cylinder of height 11 ft and radius 3 ft is to be pumped out. The density of water is 62.4 lb/ft^3. Find the work required if

(a) The tank is full of water and the water is to be pumped over the top of the tank.

Work = ______________________

(include)

(b) The tank is full of water and the water must be pumped to a height 7 ft above the top of the tank.

Work = ___________________________________

(include)

(c) The depth of water in the tank is 8 ft and the water must be pumped over the top of the tank.

Work = ___________________________________

(include)

5. **Force due to hydrostatic pressure.** A lobster tank in a restaurant is 1.25 m long by 1 m wide by 90 cm deep. Taking the density of water to be $1000\text{kg}/\text{m}^3$, find the water forces

   on the bottom of the tank: Force = ___________________________________

   on each of the larger sides of the tank: Force = ___________________________________

   on each of the smaller sides of the tank: Force = ___________________________________

   (include for each, and use $g = 9.8\ \text{m}/\text{s}^2$)

6. Consider the curve $f(x) = 3\cos(\frac{x^3}{4})$ and the portion of its graph that lies in the first quadrant between the y-axis and the first positive value of x for which $f(x) = 0$. Let R denote the region bounded by this portion of f, the x-axis, and the y-axis. Assume that x and y are each measured in feet.

   a. Picture the coordinate axes rotated 90 degrees clockwise so that the positive x-axis points straight down, and the positive y-axis points to the right. Suppose that R is rotated about the x axis to form a solid of revolution, and we consider this solid as a storage tank. Suppose that the resulting tank is filled to a depth of 1.5 feet with water weighing 62.4 pounds per cubic foot. Find the amount of work required to lower the water in the tank until it is 0.5 feet deep, by pumping the water to the top of the tank.

   b. Again picture the coordinate axes rotated 90 degrees clockwise so that the positive x-axis points straight down, and the positive y-axis points to the right. Suppose that R, together with its reflection across the x-axis, forms one end of a storage tank that is 10 feet long. Suppose that the resulting tank is filled completely with water weighing 62.4 pounds per cubic foot. Find a formula for a function that tells the amount of work required to lower the water by h feet.

   c. Suppose that the tank described in (b) is completely filled with water. Find the total force due to hydrostatic pressure exerted by the water on one end of the tank.

7. A cylindrical tank, buried on its side, has radius 3 feet and length 10 feet. It is filled completely with water whose weight density is $62.4\ \text{lbs}/\text{ft}^3$, and the top of the tank is two feet underground.

   a. Set up, but do not evaluate, an integral expression that represents the amount of work required to empty the top half of the water in the tank to a truck whose tank lies 4.5 feet above ground.

b. With the tank now only half-full, set up, but do not evaluate an integral expression that represents the total force due to hydrostatic pressure against one end of the tank.

6.5 Improper integrals

Motivating Questions

- What are improper integrals and why are they important?
- What does it mean to say that an improper integral converges or diverges?
- What are some typical improper integrals that we can classify as convergent or divergent?

6.5.1 Introduction

Another important application of the definite integral measures the likelihood of certain events. For instance, consider a company that manufactures incandescent light bulbs. Based on a large volume of test results, they have determined that the fraction of light bulbs that fail between times $t = a$ and $t = b$ of use (where t is measured in months) is given by

$$\int_a^b 0.3e^{-0.3t}\,dt.$$

For example, the fraction of light bulbs that fail during their third month of use is given by

$$\begin{aligned}\int_2^3 0.3e^{-0.3t}\,dt &= -e^{-0.3t}\Big|_2^3 \\ &= -e^{-0.9} + e^{-0.6} \\ &\approx 0.1422.\end{aligned}$$

Thus about 14.22% of all lightbulbs fail between $t = 2$ and $t = 3$. Clearly we can adjust the limits of integration to measure the fraction of light bulbs that fail during any time period of interest.

Preview Activity 6.5.1. A company with a large customer base has a call center that receives thousands of calls a day. After studying the data that represents how long callers wait for assistance, they find that the function $p(t) = 0.25e^{-0.25t}$ models the time customers wait in the following way: the fraction of customers who wait between $t = a$ and $t = b$ minutes is given by

$$\int_a^b p(t)\,dt.$$

Use this information to answer the following questions.

(a) Determine the fraction of callers who wait between 5 and 10 minutes.

(b) Determine the fraction of callers who wait between 10 and 20 minutes.

(c) Next, let's study the fraction who wait up to a certain number of minutes:

i. What is the fraction of callers who wait between 0 and 5 minutes?

ii. What is the fraction of callers who wait between 0 and 10 minutes?

iii. Between 0 and 15 minutes? Between 0 and 20?

(d) Let $F(b)$ represent the fraction of callers who wait between 0 and b minutes. Find a formula for $F(b)$ that involves a definite integral, and then use the First FTC to find a formula for $F(b)$ that does not involve a definite integral.

(e) What is the value of $\lim_{b\to\infty} F(b)$? What is the meaning of this limit in the context of the problem?

6.5.2 Improper Integrals Involving Unbounded Intervals

In view of the introductory example and the Preview Activity, we see that we may want to integrate over an interval whose upper limit grows without bound. For example, to find the fraction of light bulbs that fail *eventually*, we wish to find

$$\lim_{b\to\infty} \int_0^b 0.3e^{-0.3t}\,dt,$$

for which we will also use the notation

$$\int_0^\infty 0.3e^{-0.3t}\,dt. \tag{6.5.1}$$

Such an integral can be interpreted as the area of an unbounded region, as pictured at right in Figure 6.5.1.

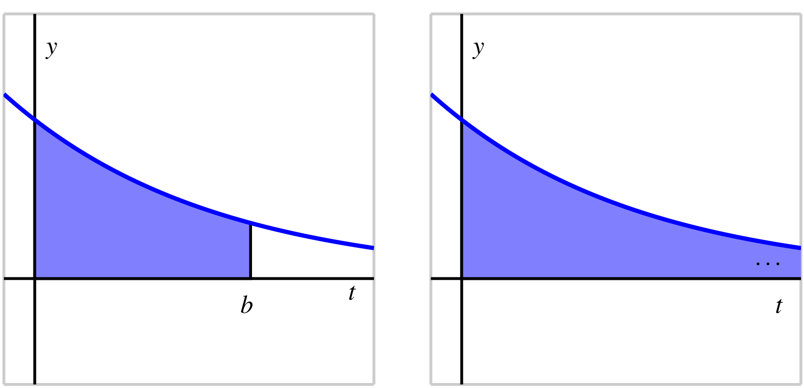


Figure 6.5.1 At left, the area bounded by $p(t) = 0.3e^{-0.3t}$ on the finite interval $[0, b]$; at right, the result of letting $b \to \infty$. By "$\cdots$" in the righthand figure, we mean that the region extends to the right without bound.

We call an integral for which the interval of integration is unbounded *improper*. For instance, the integrals

$$\int_1^\infty \frac{1}{x^2}\,dx, \quad \int_{-\infty}^0 \frac{1}{1+x^2}\,dx, \quad \text{and} \quad \int_{-\infty}^\infty e^{-x^2}\,dx$$

are all improper because they have limits of integration that involve ∞. To evaluate an improper integral we replace it with a limit of proper integrals. That is,

$$\int_0^\infty f(x)\,dx = \lim_{b\to\infty} \int_0^b f(x)\,dx.$$

We first attempt to evaluate $\int_0^b f(x)\,dx$ using the First FTC, and then evaluate the limit. Is it even possible for the area of an unbounded region to be finite? The following activity explores this issue and others in more detail.

Activity 6.5.2. In this activity we explore the improper integrals $\int_1^\infty \frac{1}{x}\,dx$ and $\int_1^\infty \frac{1}{x^{3/2}}\,dx$.

(a) First we investigate $\int_1^\infty \frac{1}{x}\,dx$.

i. Use the First FTC to determine the exact values of $\int_1^{10} \frac{1}{x}\,dx$, $\int_1^{1000} \frac{1}{x}\,dx$, and $\int_1^{100000} \frac{1}{x}\,dx$. Then, use your computational device to compute a decimal approximation of each result.

ii. Use the First FTC to evaluate the definite integral $\int_1^b \frac{1}{x}\,dx$ (which results in an expression that depends on b).

iii. Now, use your work from (ii.) to evaluate the limit given by

$$\lim_{b\to\infty} \int_1^b \frac{1}{x}\,dx.$$

(b) Next, we investigate $\int_1^\infty \frac{1}{x^{3/2}}\,dx$.

i. Use the First FTC to determine the exact values of $\int_1^{10} \frac{1}{x^{3/2}}\,dx$, $\int_1^{1000} \frac{1}{x^{3/2}}\,dx$, and $\int_1^{100000} \frac{1}{x^{3/2}}\,dx$. Then, use your calculator to compute a decimal approximation of each result.

ii. Use the First FTC to evaluate the definite integral $\int_1^b \frac{1}{x^{3/2}}\,dx$ (which results in an expression that depends on b).

iii. Now, use your work from (ii.) to evaluate the limit given by

$$\lim_{b\to\infty} \int_1^b \frac{1}{x^{3/2}}\,dx.$$

(c) Plot the functions $y = \frac{1}{x}$ and $y = \frac{1}{x^{3/2}}$ on the same coordinate axes for the values $x = 0\ldots 10$. How would you compare their behavior as x increases without bound? What is similar? What is different?

(d) How would you characterize the value of $\int_1^\infty \frac{1}{x}\,dx$? of $\int_1^\infty \frac{1}{x^{3/2}}\,dx$? What does this tell us about the respective areas bounded by these two curves for $x \geq 1$?

6.5.3 Convergence and Divergence

Activity 6.5.2 suggests that $\lim_{b\to\infty} \int_1^b f(x)\,dx$ is either finite or infinite (or it doesn't exist). With these possibilities in mind, we introduce the following terminology.

Whether an improper integral converges or diverges.

If $f(x)$ is nonnegative for $x \geq a$, then we say that the improper integral $\int_a^\infty f(x)\,dx$ **converges** provided that

$$\lim_{b\to\infty} \int_a^b f(x)\,dx$$

exists and is finite. Otherwise, we say that $\int_a^\infty f(x)\,dx$ **diverges**.

We will restrict our interest to improper integrals for which the integrand is nonnegative. Also, we require that $\lim_{x\to\infty} f(x) = 0$, for if f does not approach 0 as $x \to \infty$, then it is impossible for $\int_a^\infty f(x)\,dx$ to converge.

Activity 6.5.3. Determine whether each of the following improper integrals converges or diverges. For each integral that converges, find its exact value.

(a) $\int_1^\infty \frac{1}{x^2}\,dx$

(b) $\int_0^\infty e^{-x/4}\,dx$

(c) $\int_2^\infty \frac{9}{(x+5)^{2/3}}\,dx$

(d) $\int_4^\infty \frac{3}{(x+2)^{5/4}}\,dx$

(e) $\int_0^\infty xe^{-x/4}\,dx$

(f) $\int_1^\infty \frac{1}{x^p}\,dx$, where p is a positive real number

6.5.4 Improper Integrals Involving Unbounded Integrands

An integral is also called improper if the integrand is unbounded on the interval of integration. For example, consider

$$\int_0^1 \frac{1}{\sqrt{x}}\,dx.$$

Because $f(x) = \frac{1}{\sqrt{x}}$ has a vertical asymptote at $x = 0$, f is not continuous on $[0, 1]$, and the integral represents the area of the unbounded region shown at right in Figure 6.5.2.

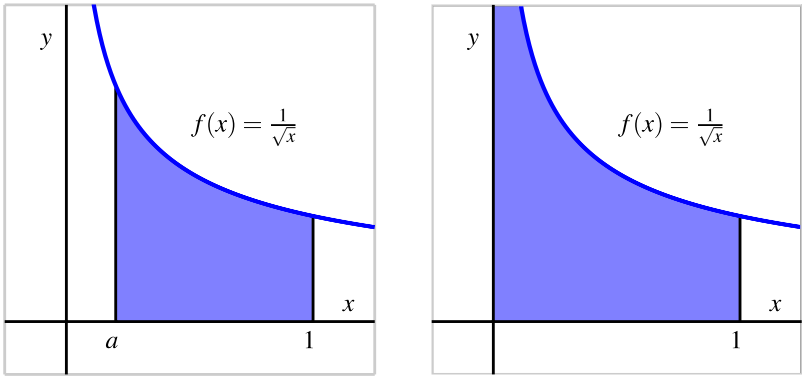


Figure 6.5.2 At left, the area bounded by $f(x) = \frac{1}{\sqrt{x}}$ on the finite interval $[a, 1]$; at right, the result of letting $a \to 0^+$, where we see that the shaded region will extend vertically without bound.

We address the problem of the integrand being unbounded by replacing the improper integral with a limit of proper integrals. For example, to evaluate $\int_0^1 \frac{1}{\sqrt{x}}\,dx$, we replace 0 with a and let a approach 0 from the right. Thus,

$$\int_0^1 \frac{1}{\sqrt{x}}\,dx = \lim_{a\to 0^+} \int_a^1 \frac{1}{\sqrt{x}}\,dx.$$

We evaluate the proper integral $\int_a^1 \frac{1}{\sqrt{x}}\,dx$, and then take the limit. We will say that the improper integral converges if this limit exists, and diverges otherwise. In this example, we observe that

$$\begin{aligned}\int_0^1 \frac{1}{\sqrt{x}}\,dx &= \lim_{a\to 0^+} \int_a^1 \frac{1}{\sqrt{x}}\,dx \\ &= \lim_{a\to 0^+} 2\sqrt{x}\,\Big|_a^1 \\ &= \lim_{a\to 0^+} 2\sqrt{1} - 2\sqrt{a} \\ &= 2,\end{aligned}$$

so the improper integral $\int_0^1 \frac{1}{\sqrt{x}}\,dx$ converges (to the value 2).

We have to be particularly careful with unbounded integrands, for they may arise in ways that may not initially be obvious. Consider, for instance, the integral

$$\int_1^3 \frac{1}{(x-2)^2}\,dx.$$

At first glance we might think that we can simply apply the Fundamental Theorem of Calculus by antidifferentiating $\frac{1}{(x-2)^2}$ to get $-\frac{1}{x-2}$ and then evaluating from 1 to 3. Were we to do so, we would be erroneously applying the FTC because $f(x) = \frac{1}{(x-2)^2}$ fails to be continuous throughout the interval, as seen in Figure 6.5.3.

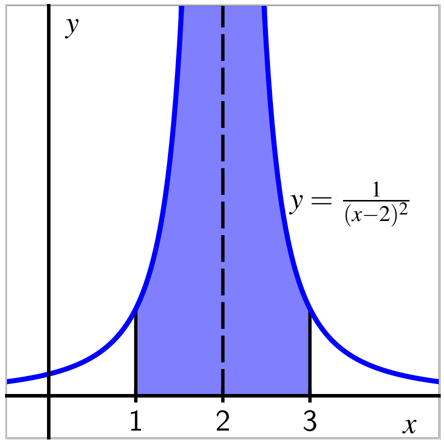


Figure 6.5.3 The function $f(x) = \frac{1}{(x-2)^2}$ on an interval including $x = 2$.

Such an incorrect application of the FTC leads to an impossible result (-2), which would itself suggest that something we did must be incorrect. Instead, we must address the vertical asymptote at $x = 2$ by writing

$$\int_1^3 \frac{1}{(x-2)^2}\,dx = \lim_{a\to 2^-} \int_1^a \frac{1}{(x-2)^2}\,dx + \lim_{b\to 2^+} \int_b^3 \frac{1}{(x-2)^2}\,dx.$$

We then evaluate two separate limits of proper integrals. For instance, doing so for the integral with a approaching 2 from the left, we find

$$\begin{aligned}
\int_1^2 \frac{1}{(x-2)^2}\,dx &= \lim_{a\to 2^-} \int_1^a \frac{1}{(x-2)^2}\,dx \\
&= \lim_{a\to 2^-} -\frac{1}{(x-2)}\bigg|_1^a \\
&= \lim_{a\to 2^-} -\frac{1}{(a-2)} + \frac{1}{1-2} \\
&= \infty,
\end{aligned}$$

since $\frac{1}{a-2} \to -\infty$ as a approaches 2 from the left. Thus, the improper integral $\int_1^2 \frac{1}{(x-2)^2}\,dx$ diverges; similar work shows that $\int_2^3 \frac{1}{(x-2)^2}\,dx$ also diverges. From either of these two results, we can conclude that that the original integral, $\int_1^3 \frac{1}{(x-2)^2}\,dx$ diverges, too.

Activity 6.5.4. For each of the following definite integrals, decide whether the integral is improper or not. If the integral is proper, evaluate it using the First FTC. If the integral is improper, determine whether or not the integral converges or diverges; if the integral converges, find its exact value.

(a) $\int_0^1 \frac{1}{x^{1/3}}\,dx$

(b) $\int_0^2 e^{-x}\,dx$

(c) $\int_1^4 \frac{1}{\sqrt{4-x}}\,dx$

(d) $\int_{-2}^2 \frac{1}{x^2}\,dx$

(e) $\int_0^{\pi/2} \tan(x)\,dx$

(f) $\int_0^1 \frac{1}{\sqrt{1-x^2}}\,dx$

6.5.5 Summary

- An integral $\int_a^b f(x)\,dx$ can be improper if at least one of a or b is $\pm\infty$, making the interval unbounded, or if f has a vertical asymptote at $x = c$ for some value of c that satisfies $a \le c \le b$. One reason that improper integrals are important is that certain probabilities can be represented by integrals that involve infinite limits.

- When we encounter an improper integral, we work to understand it by replacing the improper integral with a limit of proper integrals. For instance, we write

$$\int_a^\infty f(x)\,dx = \lim_{b\to\infty} \int_a^b f(x)\,dx,$$

  and then work to determine whether the limit exists and is finite. For any improper integral, if the resulting limit of proper integrals exists and is finite, we say the improper integral converges. Otherwise, the improper integral diverges.

- An important class of improper integrals is given by

$$\int_1^\infty \frac{1}{x^p}\,dx$$

  where p is a positive real number. We can show that this improper integral converges whenever $p > 1$, and diverges whenever $0 < p \le 1$. A related class of improper integrals is $\int_0^1 \frac{1}{x^p}\,dx$, which converges for $0 < p < 1$, and diverges for $p \ge 1$.

6.5.6 Exercises

1. Calculate the integral below, if it converges. If it does not converge, enter ***diverges*** for your answer.

$\int_2^\infty 2xe^{-x^2}\,dx =$ ______________________

2. Calculate the integral, if it converges. If it diverges, enter ***diverges*** for your answer.

$\int_{-\infty}^3 \frac{e^{2x}}{1+e^{2x}}\,dx =$ ______________________

3. Calculate the integral, if it converges. If it diverges, enter ***diverges*** for your answer.

$$\int_{-1}^{1} \frac{1}{v}\,dv = \text{\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_}$$

4. Find the area under the curve $y = \frac{1}{\cos^2(t)}$ between $t = 0$ and $t = \pi/2$. Enter ***diverges*** if the area is not bounded.

area = ______________________

5. Find what value of c does

$$\int_4^\infty \frac{c}{x^4}dx = 1?$$

Answer: ______________________

6. Compute the value of the following improper integral. If it is divergent, type "Diverges" or "D".

$$\int_0^2 \frac{dx}{x^2 - 5x + 4}$$

Answer: ______________

7. Suppose that a company expects its annual profits t years from now to be $f(t)$ dollars and that interest is considered to be compounded continuously at an annual rate r. Then the present value of all future profits can be shown to be

$$FP = \int_0^\infty e^{-rt} f(t)\mathrm{d}t$$

Find FP if $r = 0.08$ and $f(t) = 100,000 + 1000t$.

The present value is ______________________ dollars.

8. Radioactive substances decay exponentially: The mass at time t is $m(t) = m(0)e^{kt}$, where $m(0)$ is the initial mass and k is a negative constant. The *mean life* M of an atom in the substance is

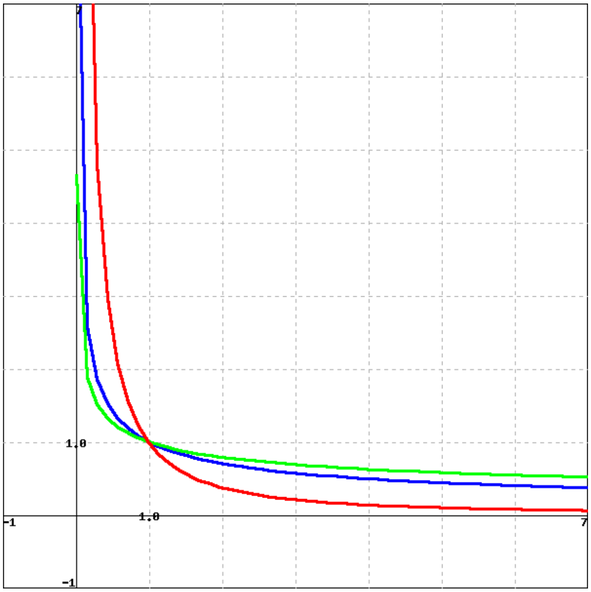
$$M = -k\int_0^\infty te^{kt}\,dt.$$

For the radioactive carbon isotope, ^{14}C, used in radiocarbon dating, the value of k is -0.000121. Find the mean life of a ^{14}C atom.

Make sure that your answer is correct to two decimal places.

$M =$ ______________________________

9. Given the function $f(x) = \dfrac{1}{x^{1/2}}$ (in blue), consider the functions g (in green) and h (in red) graphed below which are continuous on $(0, \infty)$. Assuming the graphs continues in the same way as x goes to infinity, answer the following questions.



(□ choose one □ Converge □ Diverge □ Not sufficient information) ***1.*** Does the improper integral $\int_1^\infty f(x)\,dx$ *converge*, *diverge*, or *not sufficient information*?

(□ choose one □ Converge □ Diverge □ Not sufficient information) ***2.*** Does the improper integral $\int_1^\infty g(x)\,dx$ *converge*, *diverge*, or *not sufficient information*?

(□ choose one □ Converge □ Diverge □ Not sufficient information) ***3.*** Does the improper integral $\int_1^\infty h(x)\,dx$ *converge*, *diverge*, or *not sufficient information*?

Note: *You can click on the graph to enlarge the image.*

Note: *You only have two attempts at this problem.*

10. In electromagnetic theory, the magnetic potential u at a point on the axis of a circular coil is given by

$$u = Ar \int_a^\infty \frac{dx}{(r^2 + x^2)^{(3/2)}}$$

where A, r, a are constants. Compute

$u =$ ______________________________.

11. Determine, with justification, whether each of the following improper integrals converges or diverges.

a. $\displaystyle\int_e^\infty \frac{\ln(x)}{x}\,dx$

b. $\displaystyle\int_e^\infty \frac{1}{x\ln(x)}\,dx$

c. $\displaystyle\int_e^\infty \frac{1}{x(\ln(x))^2}\,dx$

d. $\int_e^\infty \frac{1}{x(\ln(x))^p}\,dx$, where p is a positive real number

e. $\displaystyle\int_0^1 \frac{\ln(x)}{x}\,dx$

f. $\displaystyle\int_0^1 \ln(x)\,dx$

12. Sometimes we may encounter an improper integral for which we cannot easily evaluate the limit of the corresponding proper integrals. For instance, consider $\int_1^\infty \frac{1}{1+x^3}\,dx$. While it is hard (or perhaps impossible) to find an antiderivative for $\frac{1}{1+x^3}$, we can still determine whether or not the improper integral converges or diverges by comparison to a simpler one. Observe that for all $x > 0$, $1 + x^3 > x^3$, and therefore

$$\frac{1}{1+x^3} < \frac{1}{x^3}.$$

It therefore follows that

$$\int_1^b \frac{1}{1+x^3}\,dx < \int_1^b \frac{1}{x^3}\,dx$$

for every $b > 1$. If we let $b \to \infty$ so as to consider the two improper integrals $\int_1^\infty \frac{1}{1+x^3}\,dx$ and $\int_1^\infty \frac{1}{x^3}\,dx$, we know that the larger of the two improper integrals converges. And thus, since the smaller one lies below a convergent integral, it follows that the smaller one must converge, too. In particular, $\int_1^\infty \frac{1}{1+x^3}\,dx$ must converge, even though we never explicitly evaluated the corresponding limit of proper integrals. We use this idea and similar ones in the exercises that follow.

a. Explain why $x^2 + x + 1 > x^2$ for all $x \geq 1$, and hence show that $\int_1^\infty \frac{1}{x^2+x+1}\,dx$ converges by comparison to $\int_1^\infty \frac{1}{x^2}\,dx$.

b. Observe that for each $x > 1$, $\ln(x) < x$. Explain why

$$\int_2^b \frac{1}{x}\,dx < \int_2^b \frac{1}{\ln(x)}\,dx$$

for each $b > 2$. Why must it be true that $\int_2^\infty \frac{1}{\ln(x)}\,dx$ diverges?

c. Explain why $\sqrt{\frac{x^4+1}{x^4}} > 1$ for all $x > 1$. Then, determine whether or not the improper integral

$$\int_1^\infty \frac{1}{x} \cdot \sqrt{\frac{x^4+1}{x^4}}\,dx$$

converges or diverges.

CHAPTER 7

Differential Equations

7.1 An introduction to differential equations

Motivating Questions

- What is a differential equation and what kinds of information can it tell us?
- How do differential equations arise in the world around us?
- What do we mean by a solution to a differential equation?

7.1.1 Introduction

In previous chapters, we have seen that a function's derivative tells us the rate at which the function is changing. The Fundamental Theorem of Calculus helped us determine the total change of a function over an interval from the function's rate of change. For instance, an object's velocity tells us the rate of change of that object's position. By integrating the velocity over a time interval, we can determine how much the position changes over that time interval. If we know where the object is at the beginning of that interval, we have enough information to predict where it will be at the end of the interval.

In this chapter, we introduce the concept of *differential equations*. A differential equation is an equation that provides a description of a function's derivative, which means that it tells us the function's rate of change. Using this information, we would like to learn as much as possible about the function itself, ideally including an algebraic description of the function. As we'll see, this may be too much to ask in some situations, but we will still be able to make accurate approximations.

Preview Activity 7.1.1. The position of a moving object is given by the function $s(t)$, where s is measured in feet and t in seconds. We determine that the object's velocity is $v(t) = 4t + 1$ feet per second.

(a) How much does the position change over the time interval $[0, 4]$?

(b) Does this give you enough information to determine $s(4)$, the position at time $t = 4$? If so, what is $s(4)$? If not, what additional information would you need to know to determine $s(4)$?

(c) Suppose you are told that the object's initial position $s(0) = 7$. Determine $s(2)$, the object's position 2 seconds later.

(d) If you are told instead that the object's initial position is $s(0) = 3$, what is $s(2)$?

(e) If we only know the velocity $v(t) = 4t + 1$, is it possible that the object's position at all times is $s(t) = 2t^2 + t - 4$? Explain how you know.

(f) Are there other possibilities for $s(t)$? If so, what are they?

(g) If, in addition to knowing the velocity function is $v(t) = 4t + 1$, we know the initial position $s(0)$, how many possibilities are there for $s(t)$?

7.1.2 What is a differential equation?

A differential equation is an equation that describes one or more of the derivatives of a function that is unknown to us. For instance, the equation

$$\frac{dy}{dx} = x \sin x$$

describes the derivative of an unknown function $y(x)$.

As many important examples of differential equations involve quantities that change over time, the independent variable in our discussion will frequently be time t. In the preview activity, we considered the differential equation

$$\frac{ds}{dt} = 4t + 1.$$

Knowing the velocity and the starting position of a moving object, we were able to find its position at any later time.

Because differential equations describe the derivative of a function, they give us information about how that function changes. Our goal will be to use this information to predict the value of the function in the future; in this way, differential equations provide us with something like a crystal ball.

Differential equations arise frequently in our everyday world. For instance, you may hear a bank advertising:

> *Your money will grow at a 3% annual interest rate with us.*

This innocuous statement is really a differential equation. Let's translate: $A(t)$ will be amount of money you have in your account at time t. The rate at which your money grows is the derivative $\frac{dA}{dt}$, and we are told that this rate is $0.03A$. This leads to the differential equation

$$\frac{dA}{dt} = 0.03A.$$

This differential equation has a slightly different feel than the previous equation $\frac{ds}{dt} = 4t + 1$. In the earlier example, the rate of change depends only on the independent variable t, and we may find $s(t)$ by integrating the velocity $4t+1$ (with respect to t). In the banking example, however, the rate of change depends on the dependent variable A, so we'll need some new techniques in order to find $A(t)$.

Activity 7.1.2. Express the following statements as differential equations. In each case, you will need to introduce notation to describe the important quantities in the statement, so be sure to clearly state what your notation means.

(a) The population of a town grows continuously at an annual rate of 1.25%.

(b) A radioactive sample loses mass at a rate of 5.6% of its mass every day.

(c) You have a bank account that continuously earns 4% interest every year. At the same time, you withdraw money continually from the account at the rate of $1000 per year.

(d) A cup of hot chocolate is sitting in a 70° room. The temperature of the hot chocolate cools continuously by 10% of the difference between the hot chocolate's temperature and the room temperature every minute.

(e) A can of cold soda is sitting in a 70° room. The temperature of the soda warms continuously at the rate of 10% of the difference between the soda's temperature and the room's temperature every minute.

7.1.3 Differential equations in the world around us

Differential equations give a natural way to describe phenomena we see in physical reality. For instance, physical principles are frequently expressed as a description of how a quantity changes. A good example is Newton's Second Law, which says:

The product of an object's mass and acceleration equals the force applied to it.

For instance, when gravity acts on an object near the earth's surface, it exerts a force equal to mg, the mass of the object times the gravitational constant g. We therefore have

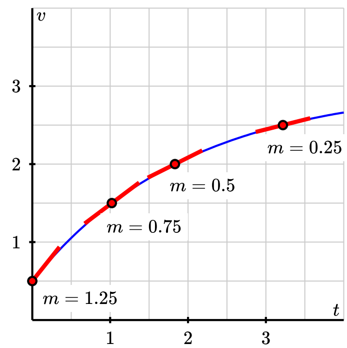
$$ma = mg, \text{ or}$$
$$\frac{dv}{dt} = g,$$

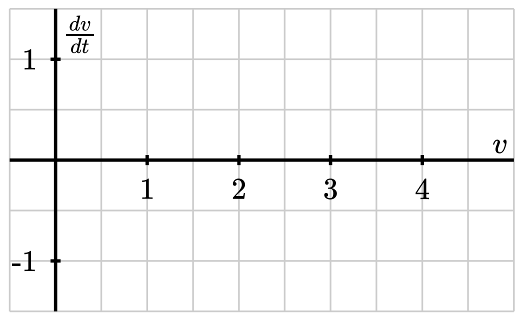
where v is the velocity of the object, and $g = 9.8$ meters per second squared. Notice that this physical principle does not tell us what the object's velocity is, but rather how the object's velocity changes.

Activity 7.1.3. In this activity, we consider information about the velocity of objects falling through the atmosphere: a skydiver in part (a), followed by a meteorite in part (b).

(a) Begin with the skydiver's velocity v given by the graph at left below, which shows how v changes as t changes. Use this graph to find the rate of change dv/dt at the points where the velocity is $v = 0.5, 1.5, 2.0$, and 2.5. Note particularly that several key slopes are provided for you on the given graph.

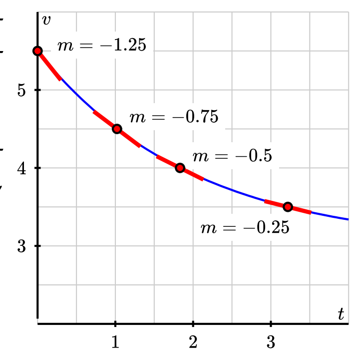
Then, on the axes provided, plot the values of the derivative dv/dt as a function of *velocity*. Think carefully about this: we are focusing on how we can connect the velocity v to the resulting value of dv/dt. For example, in the velocity graph at left, we observe from one of the points that when $v = 2$, $dv/dt = 0.5$.





(b) Next we're going to consider a falling meteorite's velocity.

While the skydiver fell faster and faster, the meteorite falls slower and slower through the atmosphere, as shown in the graph at right. Note the scale on the vertical axis: since the velocity is always greater than $v = 3$, we only show v values from 3 to 6. Reasoning similarly to part (a), use the given graph of the meteorite's velocity to find the rate of change dv/dt at the points where the velocity is $v = 3.5, 4.0, 4.5$, and 5.0. Plot the appropriate resulting points on the same axes used in part (a).



(c) You should find that all of the points you plotted on the axes in (a) lie on a line. Remember that these points show how dv/dt depends on v. Write the equation of this line, being careful to use proper notation for the quantities on the horizontal and vertical axes.

(d) The relationship you just found in (c) is a differential equation. Write a complete sentence that explains its meaning.

(e) Use the differential equation you found in (c) to determine the values of the velocity for which the velocity increases. Write a sentence to explain your thinking.

(f) Similarly, use the differential equation to determine the values of the velocity for which the velocity decreases. Again, explain your thinking.

(g) Finally, determine the value(s) of the velocity for which the velocity remains constant. What do these values mean in the overall context of this activity?

The point of this activity is to demonstrate how differential equations model processes in physical reality. In this example, two factors influence the velocities: gravity and wind resistance. The differential equation describes how these factors influence the rate of change of the velocities.

7.1.4 Solving a differential equation

A differential equation describes one or more derivatives of a function that is unknown to us. By a *solution* to a differential equation, we mean simply a function that satisfies this description.

For instance, the first differential equation we looked at is

$$\frac{ds}{dt} = 4t + 1,$$

which describes an unknown function $s(t)$. We may check that $s(t) = 2t^2 + t$ is a solution because it satisfies this description. Notice that $s(t) = 2t^2 + t + 4$ is also a solution.

If we have a candidate for a solution to a differential equation, it is straightforward to check whether the function is a solution or not. Before we demonstrate how, let's consider the same issue in a simpler context. Suppose we are given the equation $2x^2 - 2x = 2x + 6$ and asked whether $x = 3$ is a solution. To answer this question, we could rewrite the variable x in the equation with the symbol $\square$:

$$2\square^2 - 2\square = 2\square + 6.$$

To determine whether $x = 3$ is a solution, we can investigate the value of each side of the equation separately when the value 3 is placed in $\square$ and see if indeed the two resulting values are equal. Doing so, we observe that

$$2\square^2 - 2\square = 2 \cdot 3^2 - 2 \cdot 3 = 12,$$

and

$$2\square + 6 = 2 \cdot 3 + 6 = 12.$$

Therefore, $x = 3$ is indeed a solution.

We will do the same thing with differential equations. Consider the differential equation

$$\frac{dv}{dt} = 1.5 - 0.5v, \text{ or}$$

$$\frac{d\square}{dt} = 1.5 - 0.5\square.$$

Let's ask whether $v(t) = 3 - 2e^{-0.5t}$ is a solution (for now, don't worry about why we chose this function; we will learn some techniques for finding solutions to differential equations soon). Using this formula for v, observe first that

$$\frac{dv}{dt} = \frac{d\square}{dt} = \frac{d}{dt}[3 - 2e^{-0.5t}] = -2e^{-0.5t} \cdot (-0.5) = e^{-0.5t}$$

and

$$1.5 - 0.5v = 1.5 - 0.5\square = 1.5 - 0.5(3 - 2e^{-0.5t}) = 1.5 - 1.5 + e^{-0.5t} = e^{-0.5t}.$$

Since $\frac{dv}{dt}$ and $1.5 - 0.5v$ agree for all values of t when $v = 3 - 2e^{-0.5t}$, we have indeed found a solution to the differential equation.

Activity 7.1.4. Consider the differential equation

$$\frac{dv}{dt} = 1.5 - 0.5v.$$

Determine, with justification, whether or not each of the following functions is a solution to the differential equation.

(a) $v(t) = 1.5t - 0.25t^2$.

(b) $v(t) = 3 + 2e^{-0.5t}$.

(c) $v(t) = 3$.

(d) $v(t) = 3 + Ce^{-0.5t}$ where C is any constant.

This activity shows us something interesting. As shown by the function in part (d) of the activity, we see that the differential equation $\frac{dv}{dt} = 1.5 - 0.5v$ has infinitely many solutions, which are parametrized by the constant C in $v(t) = 3 + Ce^{-0.5t}$.

Observe further that the value of C is connected to the initial value of the velocity $v(0)$, since $v(0) = 3 + C$. In other words, while the differential equation describes how the velocity changes as a function of the velocity itself, this is not enough information to determine the velocity uniquely: we also need to know the initial velocity. For this reason, differential equations will typically have infinitely many solutions, one corresponding to each initial value. We have seen this phenomenon before: given the velocity of a moving object $v(t)$, we cannot uniquely determine the object's position function unless we also know its initial position.

In Figure 7.1.1, we see the graphs of several different solutions to the differential equation $\frac{dv}{dt} = 1.5 - 0.5v$ for a few values of C, as labeled.

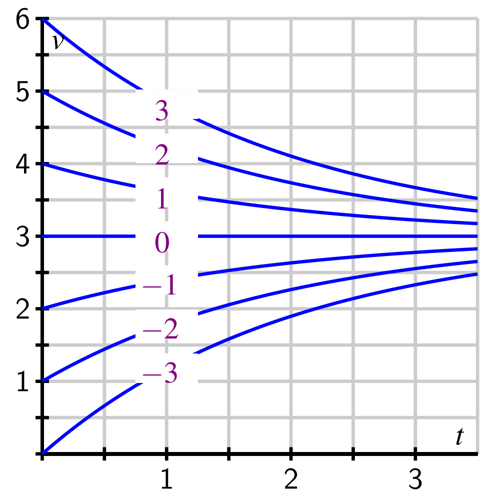


Figure 7.1.1 The family of solutions to the differential equation $\frac{dv}{dt} = 1.5 - 0.5v$.

If we are given a differential equation and an initial value for the unknown function, we say that we have an *initial value problem.* For instance,

$$\frac{dv}{dt} = 1.5 - 0.5v, \ v(0) = 0.5$$

is an initial value problem. In this problem, we know the value of v at one time and we know how v is changing. Consequently, there should be exactly one function v that satisfies the initial value problem.

This demonstrates the following important general property of initial value problems.

Well-behaved initial value problems have unique solutions.

Initial value problems that are "well-behaved" have exactly one solution, which exists in some interval around the initial point.

We won't worry about what "well-behaved" means—it is a technical condition that will be satisfied by all the differential equations we consider.

To close this section, we note that differential equations may be classified based on certain characteristics they may possess. You may see many different types of differential equations in a later course in differential equations. For now, we would like to introduce a few terms that are used to describe differential equations.

A *first-order* differential equation is one in which only the first derivative of the function occurs. For this reason,

$$\frac{dv}{dt} = 1.5 - 0.5v$$

is a first-order equation while

$$\frac{d^2y}{dt^2} = -10y$$

is a second-order equation.

A differential equation is *autonomous* if the independent variable does not appear in the description of the derivative. For instance,

$$\frac{dv}{dt} = 1.5 - 0.5v$$

is autonomous because the description of the derivative dv/dt does not depend on time. The equation

$$\frac{dy}{dt} = 1.5t - 0.5y,$$

however, is not autonomous.

7.1.5 Summary

- A differential equation is simply an equation that describes the derivative(s) of an unknown function.
- Physical principles, as well as some everyday situations, often describe how a quantity changes, which lead to differential equations.
- A solution to a differential equation is a function whose derivative(s) satisfy the equation's description. Differential equations typically have infinitely many solutions, parametrized by the initial values.

7.1.6 Exercises

1. Match the graphs below with the following descriptions by indicating the graph corresponding to each description in the answer blank to the left of the description. *(Note that because this is a matching question, it does not show which parts are incorrect.)*

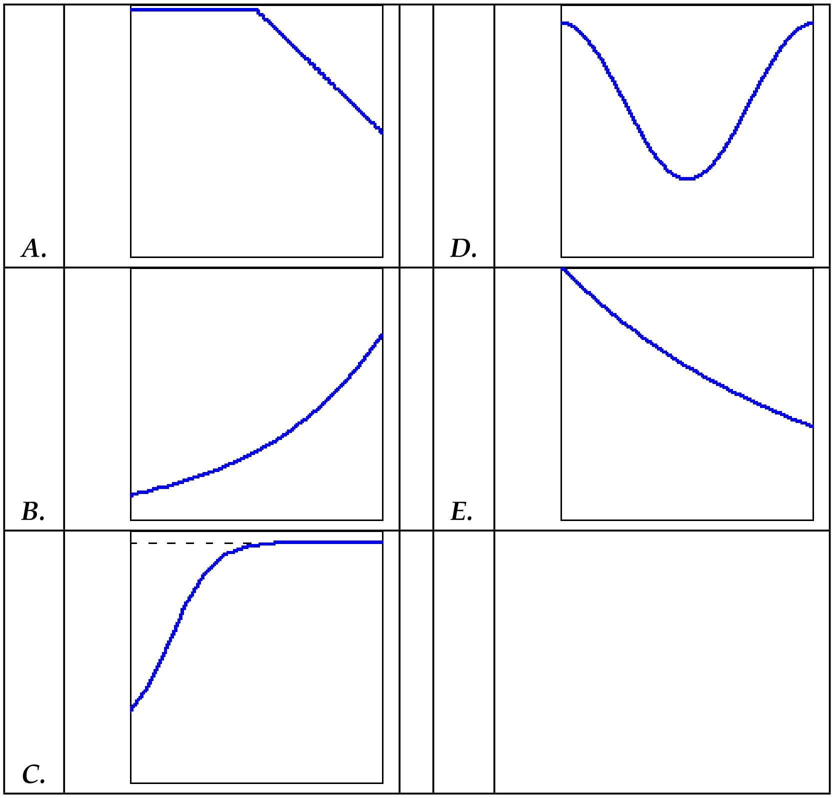
(a) The amount of money in a savings account after an initial investment that compounds continuously over time.

(b) The concentration of tree pollen in the air over the course of a year.

(c) The distance from the finish line of a runner waiting for the start of a race and then running at a constant speed.

(d) The number of bacteria on a nutrient petri dish after a group of bacteria is initially introduced.

(e) The mass of Carbon-14 in a historical specimin.



2. Let A and k be positive constants.

Which of the given functions is a solution to $\frac{dy}{dt} = k(y - A)$?

- ○ $y = -A + Ce^{-kt}$
- ○ $y = A^{-1} + Ce^{-Akt}$
- ○ $y = A + Ce^{kt}$
- ○ $y = -A + Ce^{kt}$
- ○ $y = A + Ce^{-kt}$
- ○ $y = A^{-1} + Ce^{Akt}$

3. The family of functions

$$y = ce^{-2x} + e^{-x}$$

is solution of the equation

$$y' + 2y = e^{-x}.$$

Find the constant c which defines the solution which also satisfies the initial condition $y(-3) = 1$.

$c =$ ____________________

4. The functions

$$y = x^2 + \frac{c}{x^2}$$

are all solutions of equation:

$$xy' + 2y = 4x^2, \ (x > 0).$$

Find the constant c which produces a solution which also satisfies the initial condition $y(4) = 8$.

$c =$ ____________________

5. Match the solutions to the differential equations. If there is more than one solution to an equation, select the answer that includes all solutions.

(a) $\dfrac{dy}{dx} = -5y$

(b) $\dfrac{dy}{dx} = 5y$

(c) $\dfrac{d^2y}{dx^2} = -25y$

(d) $\dfrac{d^2y}{dx^2} = 25y$

A. $y = 5\sin(x)$

B. $y = e^{-5x}$ or $y = e^{5x}$

C. $y = \sin(5x)$ or $y = 5\sin(x)$

D. $y = e^{5x}$

E. $y = e^{-5x}$

F. $y = \sin(5x)$

6. Fill in the missing values in the table given if you know that $dy/dt = 0.6y$. Assume the rate of growth given by dy/dt is approximately constant over each unit time interval and that the initial value of y is 5.

t	0	1	2	3	4
y	5	_____	_____	_____	_____

7. Suppose that $T(t)$ represents the temperature of a cup of coffee set out in a room, where T is expressed in degrees Fahrenheit and t in minutes. A physical principle known as Newton's Law of Cooling tells us that

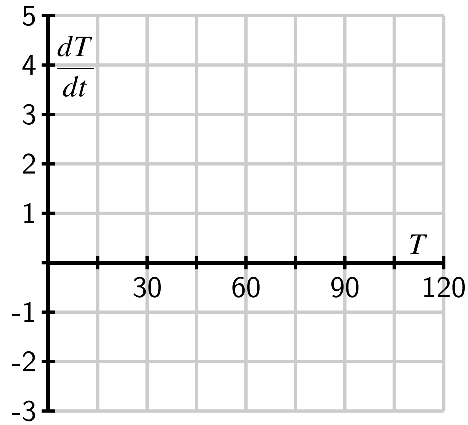
$$\frac{dT}{dt} = -\frac{1}{15}T + 5.$$

a. Supposes that $T(0) = 105$. What does the differential equation give us for the value of $\frac{dT}{dt}|_{T=105}$? Explain in a complete sentence the meaning of these two facts.

b. Is T increasing or decreasing at $t = 0$?

c. What is the approximate temperature at $t = 1$?

d. On the graph below, make a plot of dT/dt as a function of T.



e. For which values of T does T increase? For which values of T does T decrease?

f. What do you think is the temperature of the room? Explain your thinking.

g. Verify that $T(t) = 75 + 30e^{-t/15}$ is the solution to the differential equation with initial value $T(0) = 105$. What happens to this solution after a long time?

8. In this problem, we test further what it means for a function to be a solution to a given differential equation.

a. Consider the differential equation

$$\frac{dy}{dt} = y - t.$$

Determine whether the following functions are solutions to the given differential equation.

i. $y(t) = t + 1 + 2e^t$

ii. $y(t) = t + 1$

iii. $y(t) = t + 2$

b. When you weigh bananas in a scale at the grocery store, the height h of the bananas is described by the differential equation

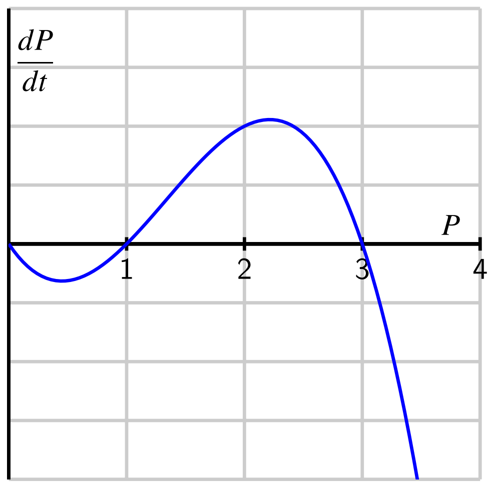
$$\frac{d^2h}{dt^2} = -kh$$

where k is the *spring constant*, a constant that depends on the properties of the spring in the scale. After you put the bananas in the scale, you (cleverly) observe that the height of the bananas is given by $h(t) = 4\sin(3t)$. What is the value of the spring constant?

9. Suppose that the population of a particular species is described by the function $P(t)$, where P is expressed in millions. Suppose further that the population's rate of change is governed by the differential equation

$$\frac{dP}{dt} = f(P)$$

where $f(P)$ is the function graphed below.



a. For which values of the population P does the population increase?

b. For which values of the population P does the population decrease?

c. If $P(0) = 3$, how will the population change in time?

d. If the initial population satisfies $0 < P(0) < 1$, what will happen to the population after a very long time?

e. If the initial population satisfies $1 < P(0) < 3$, what will happen to the population after a very long time?

f. If the initial population satisfies $3 < P(0)$, what will happen to the population after a very long time?

g. This model for a population's growth is sometimes called "growth with a threshold." Explain why this is an appropriate name.

7.2 Qualitative behavior of solutions to differential equations

Motivating Questions

- What is a slope field?
- How can we use a slope field to obtain qualitative information about the solutions of a differential equation?
- What are stable and unstable equilibrium solutions of an autonomous differential equation?

7.2.1 Introduction

In earlier work, we have used the tangent line to the graph of a function f at a point a to approximate the values of f near a. The usefulness of this approximation is that we need to know very little about the function; armed with only the value $f(a)$ and the derivative $f'(a)$, we may find the equation of the tangent line and the approximation

$$f(x) \approx f(a) + f'(a)(x - a).$$

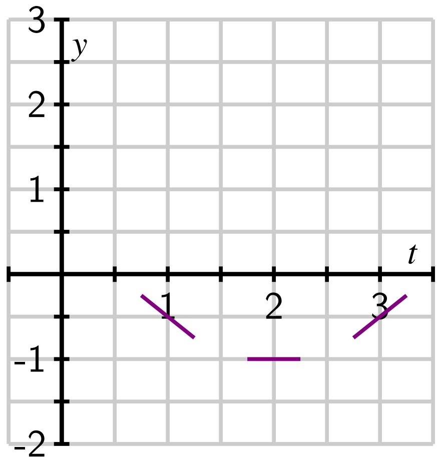
Remember that a first-order differential equation gives us information about the derivative of an unknown function. Since the derivative at a point tells us the slope of the tangent line at this point, a differential equation gives us crucial information about the tangent lines to the graph of a solution. We will use this information about the tangent lines to create a *slope field* for the differential equation, which enables us to sketch solutions to initial value problems with the goal of understanding the solutions qualitatively. That is, we would like to understand the basic nature of solutions, such as their long-range behavior, without precisely determining the value of a solution at a particular point.

In the following Preview Activity, we consider a certain differential equation and explore how we can use the equation to find tangent lines to provide a kind of road map for an approximate solution.

Preview Activity 7.2.1. Let's consider the initial value problem

$$\frac{dy}{dt} = t - 2, \quad y(0) = 1.$$

(a) Use the differential equation to find the slope of the tangent line to the solution $y(t)$ at $t = 0$. Then use the initial value to find the equation of the tangent line at $t = 0$. Sketch this tangent line over the interval $-0.25 \le t \le 0.25$ on the axes provided below.



(b) Also shown in the figure in (a) are the tangent lines to the solution $y(t)$ at the points $t = 1, 2$, and 3 (we will see how to find these later). Use the graph to measure the slope of each tangent line and verify that each agrees with the value specified by the differential equation.

(c) Using the tangent lines from (a) and (b) as a guide, sketch a graph of the solution $y(t)$ over the interval $0 \le t \le 3$ so that the lines are tangent to the graph of $y(t)$.

(d) Use the Fundamental Theorem of Calculus to find $y(t)$, the solution to original initial value problem,

$$\frac{dy}{dt} = t - 2, \quad y(0) = 1.$$

(e) Graph the solution you found in (d) on the axes provided in (a), and compare it to the sketch you made using the tangent lines.

7.2.2 Slope fields

Preview Activity 7.2.1 shows that we can sketch the solution to an initial value problem if we know an appropriate collection of tangent lines. We can use the differential equation to find the slope of the tangent line at any point of interest, and hence generate and plot such a collection.

Let's continue looking at the differential equation $\frac{dy}{dt} = t - 2$. If $t = 0$, this equation says that $dy/dt = 0 - 2 = -2$. Note that this value holds regardless of the value of y. We will therefore sketch tangent lines for several values of y and $t = 0$ with a slope of -2, as shown in Figure 7.2.1. These tangent lines then provide a kind of road map for a solution to follow, starting from a certain initial condition.

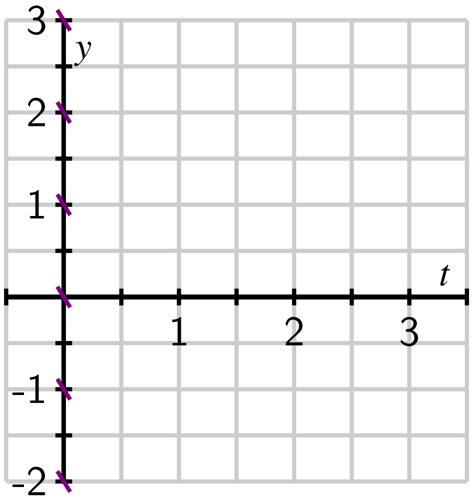


Figure 7.2.1 Tangent lines at points with $t = 0$.

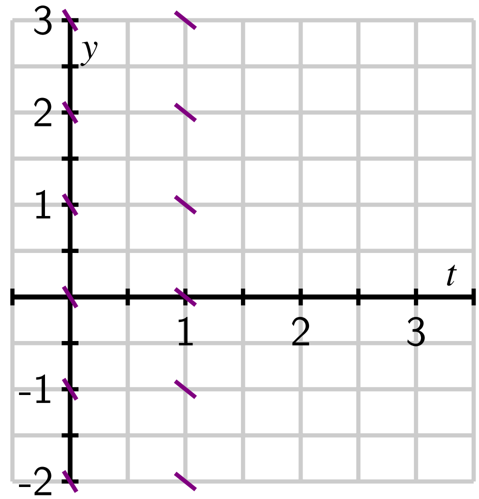


Figure 7.2.2 Adding tangent lines at points with $t = 1$.

Let's continue in the same way: if $t = 1$, the differential equation tells us that $dy/dt = 1 - 2 = -1$, and this holds regardless of the value of y. We now sketch tangent lines for several values of y and $t = 1$ with a slope of -1 in Figure 7.2.2.

Similarly, we see that when $t = 2$, $dy/dt = 0$ and when $t = 3$, $dy/dt = 1$. We may therefore add to our growing collection of tangent line plots to achieve Figure 7.2.3.

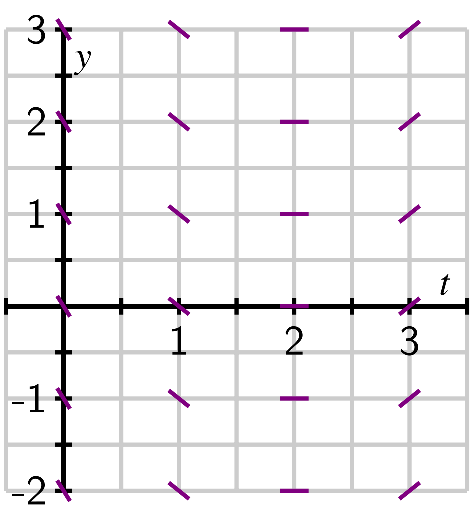


Figure 7.2.3 Adding tangent lines at points with $t = 2$ and $t = 3$.

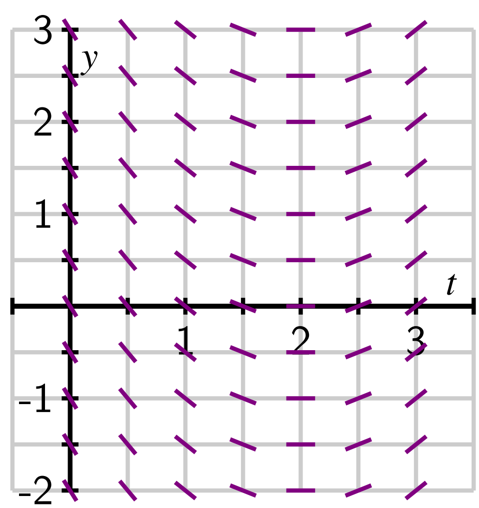


Figure 7.2.4 A completed slope field.

In Figure 7.2.3, we begin to see the solutions to the differential equation emerge. For the sake of even greater clarity, we add more tangent lines to provide the more complete picture shown at right in Figure 7.2.4.

Figure 7.2.4 is called a *slope field* for the differential equation. It allows us to sketch solutions just as we did in the preview activity. We can begin with the initial value $y(0) = 1$ and start sketching the solution by following the tangent line. Whenever the solution passes through a point at which a tangent line is drawn, that line is tangent to the solution. This principle leads us to the sequence of images in Figure 7.2.5.

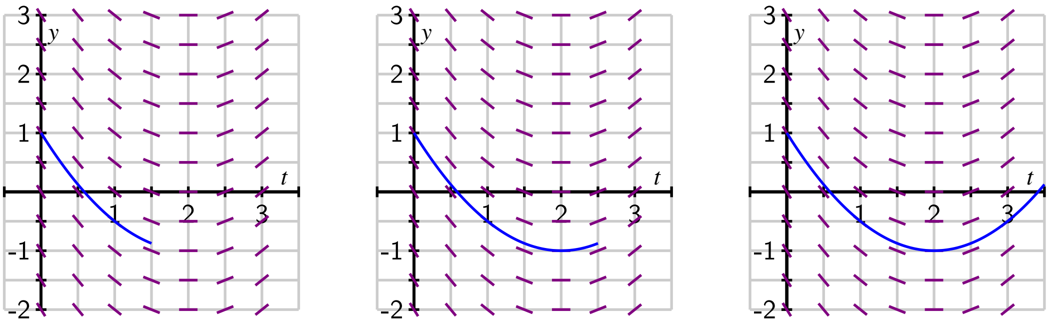


Figure 7.2.5 A sequence of images that show how to sketch the IVP solution that satisfies $y(0) = 1$.

In fact, we can draw solutions for any initial value. Figure 7.2.6 shows solutions for several different initial values for $y(0)$.

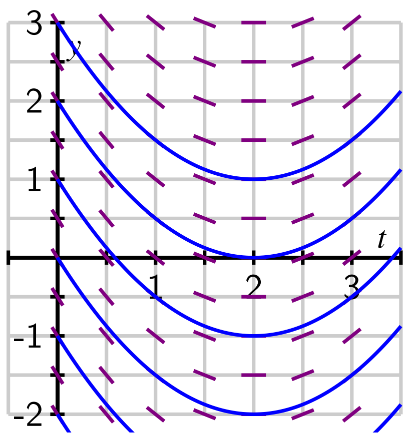


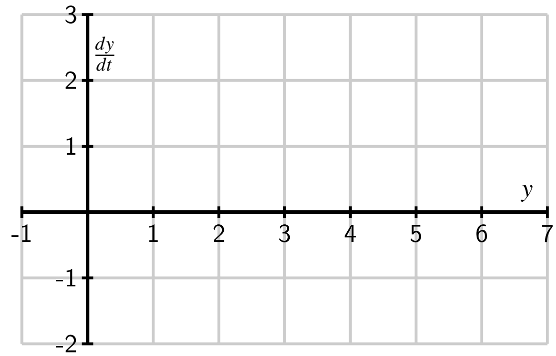
Figure 7.2.6 Different solutions to $\frac{dy}{dt} = t - 2$ that correspond to different initial values.

Just as we did for the equation $\frac{dy}{dt} = t - 2$, we can construct a slope field for any differential equation of interest. The slope field provides us with visual information about how we expect solutions to the differential equation to behave.

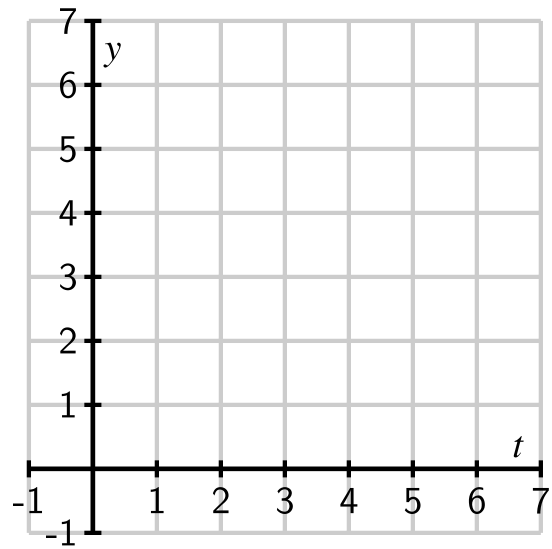
Activity 7.2.2. Consider the autonomous differential equation

$$\frac{dy}{dt} = -\frac{1}{2}(y - 4).$$

(a) Make a plot of $\frac{dy}{dt}$ versus y on the axes provided below. Looking at the graph, for what values of y does y increase and for what values of y does y decrease?



(b) Next, sketch the slope field for this differential equation on the axes provided below.



(c) Use your work in (b) to sketch on the same axes solutions that satisfy $y(0) = 0$, $y(0) = 2$, $y(0) = 4$ and $y(0) = 6$.

(d) Verify that $y(t) = 4 + 2e^{-t/2}$ is a solution to the given differential equation with the initial value $y(0) = 6$. Compare its graph to the one you sketched in (c).

(e) What is special about the solution where $y(0) = 4$?

7.2.3 Equilibrium solutions and stability

As our work in Activity 7.2.2 demonstrates, first-order autonomous equations may have solutions that are constant. These are simple to detect by inspecting the differential equation $dy/dt = f(y)$: constant solutions necessarily have a zero derivative, so $dy/dt = 0 = f(y)$.

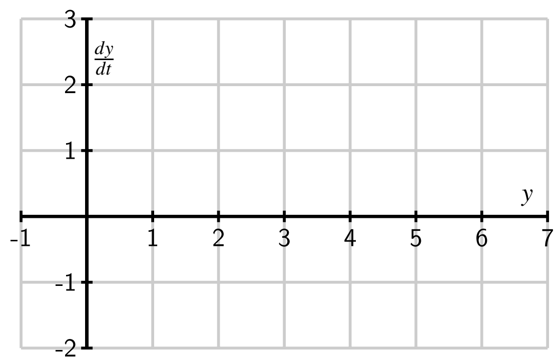
For example, in Activity 7.2.2, we considered the equation $\frac{dy}{dt} = f(y) = -\frac{1}{2}(y - 4)$. Constant solutions are found by setting $f(y) = -\frac{1}{2}(y-4) = 0$, which we immediately see implies that $y = 4$.

Values of y for which $f(y) = 0$ in an autonomous differential equation $\frac{dy}{dt} = f(y)$ are called *equilibrium solutions* of the differential equation.

Activity 7.2.3. Consider the autonomous differential equation

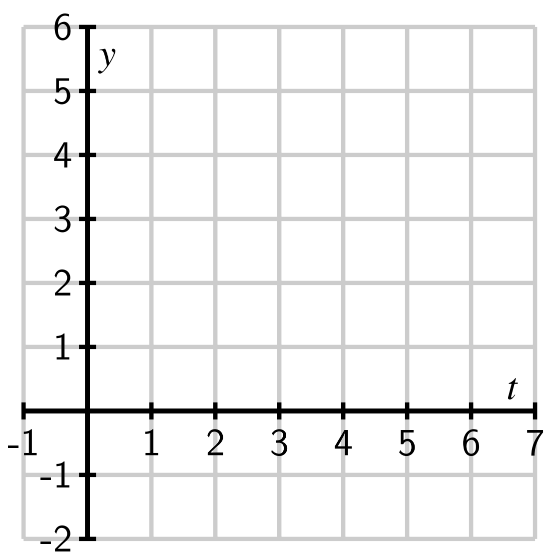
$$\frac{dy}{dt} = -\frac{1}{2}y(y-4).$$

(a) Make a plot of $\frac{dy}{dt}$ versus y on the axes provided. Looking at the graph, for what values of y does y increase and for what values of y does y decrease?



(b) Identify any equilibrium solutions of the given differential equation.

(c) Now sketch the slope field for the given differential equation on the axes provided below.

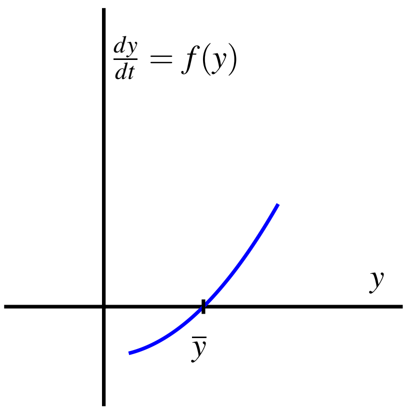


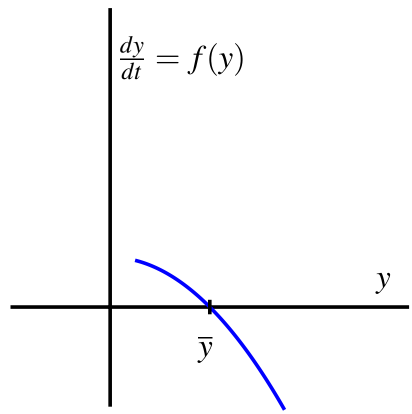
(d) On the slope field in (c), sketch the solutions to the given differential equation that correspond to initial values $y(0) = -1, 0, 1, \dots, 5$.

(e) An equilibrium solution $\overline{y}$ is called *stable* if nearby solutions converge to $\overline{y}$. This means that if the initial condition varies slightly from $\overline{y}$, then $\lim_{t\to\infty} y(t) = \overline{y}$. Conversely, an equilibrium solution $\overline{y}$ is called *unstable* if nearby solutions are pushed away from $\overline{y}$. Using your work above, classify the equilibrium solutions you found in (b) as either stable or unstable.

(f) Suppose that $y(t)$ describes the population of a species of living organisms and that the initial value $y(0)$ is positive. What can you say about the eventual fate of this population?

(g) Now consider a general autonomous differential equation of the form $dy/dt = f(y)$. Remember that an equilibrium solution $\overline{y}$ satisfies $f(\overline{y}) = 0$. If we graph $dy/dt = f(y)$ as a function of y, for which of the differential equations represented below is $\overline{y}$ a stable equilibrium and for which is $\overline{y}$ unstable? Why?





7.2.4 Summary

- A slope field is a plot created by graphing the tangent lines of many different solutions to a differential equation.
- Once we have a slope field, we may sketch the graph of solutions by drawing a curve that is always tangent to the lines in the slope field.
- Autonomous differential equations sometimes have constant solutions that we call equilibrium solutions. These may be classified as stable or unstable, depending on the behavior of nearby solutions.

7.2.5 Exercises

1. Consider the two slope fields shown, in figures 1 and 2 below.

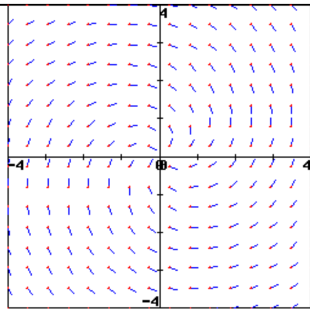


figure 1

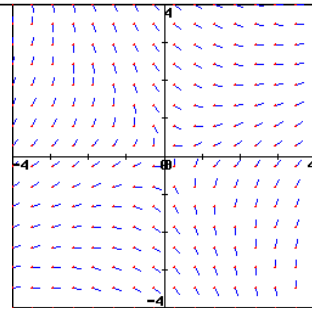


figure 2

On a print-out of these slope fields, sketch for each three solution curves to the differential equations that generated them. Then complete the following statements:

For the slope field in figure 1, a solution passing through the point (3,-1) has a (□ ? □ positive □ negative □ zero □ undefined) slope.

For the slope field in figure 1, a solution passing through the point (-2,2) has a (□ ? □ positive □ negative □ zero □ undefined) slope.

For the slope field in figure 2, a solution passing through the point (1,0) has a (□ ? □ positive □ negative □ zero □ undefined) slope.

For the slope field in figure 2, a solution passing through the point (0,1) has a (□ ? □ positive □ negative □ zero □ undefined) slope.

2. Match the following equations with their direction field. Clicking on each picture will give you an enlarged view. While you can probably solve this problem by guessing, it is useful to try to predict characteristics of the direction field and then match them to the picture.

Here are some handy characteristics to start with -- you will develop more as you practice.

A. Set y equal to zero and look at how the derivative behaves along the x-axis.

B. Do the same for the y-axis by setting x equal to 0

C. Consider the curve in the plane defined by setting $y' = 0$ -- this should correspond to the points in the picture where the slope is zero.

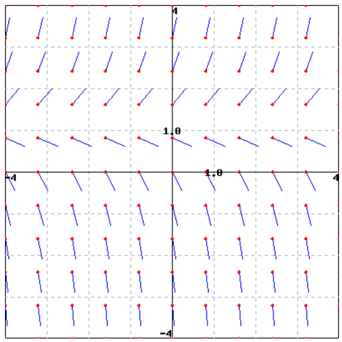
D. Setting y' equal to a constant other than zero gives the curve of points where the slope is that constant. These are called isoclines, and can be used to construct the direction field picture by hand.

(a) $y' = -\dfrac{(2x+y)}{(2y)}$

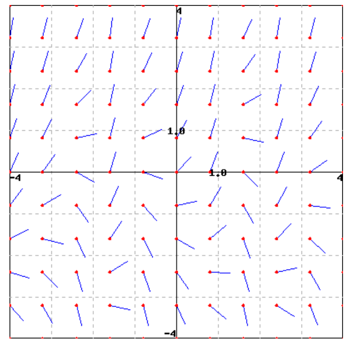
(b) $y' = 2\sin(3x) + 1 + y$

(c) $y' = 2y - 2$

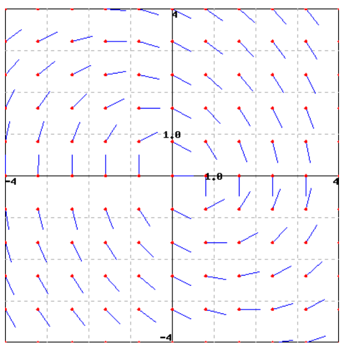
(d) $y' = \dfrac{y}{x} + 3\cos(2x)$



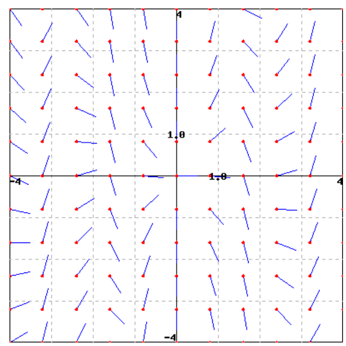
A



B



C



D

3. Given the differential equation $x'(t) = -x^4 + 9x^3 - 12x^2 - 44x + 48$.

Use appropriate computing technology such as sketching the graph[1] to find the equilibrium solutions.

List the equilibrium solutions to this differential equation in increasing order and indicate whether or not these equilibria are stable, semi-stable, or unstable.

__________ (☐ stable ☐ semi-stable ☐ unstable)

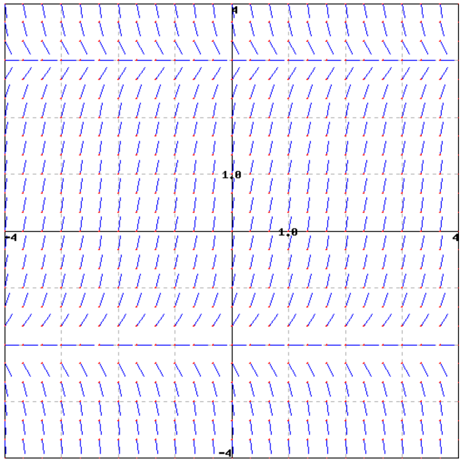
__________ (☐ stable ☐ unstable ☐ semi-stable)

__________ (☐ stable ☐ semi-stable ☐ unstable)

__________ (☐ stable ☐ unstable ☐ semi-stable)

4. Consider the direction field below for a differential equation. Use the graph to find the equilibrium solutions.

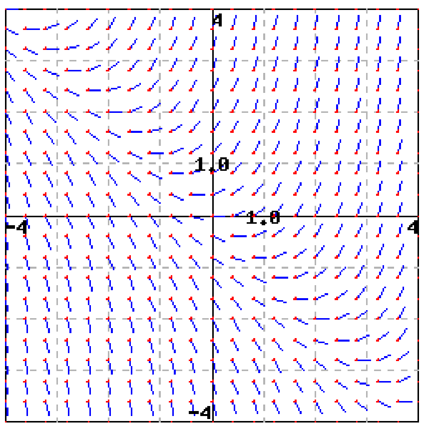
[1] `www.desmos.com/calculator`



Answer (separate by commas): $y =$ ______________________________

Note: *You can click on the graph to enlarge the image.*

5. The slope field for the equation $y' = x + y$ is shown below



On a print out of this slope field, sketch the solutions that pass through the points

(i) (0,0);

(ii) (-3,1); and

(iii) (-1,0).

From your sketch, what is the equation of the solution to the differential equation that passes through (-1,0)? (Verify that your solution is correct by substituting it into the differential equation.)

$y =$ ______________

6. Consider the differential equation

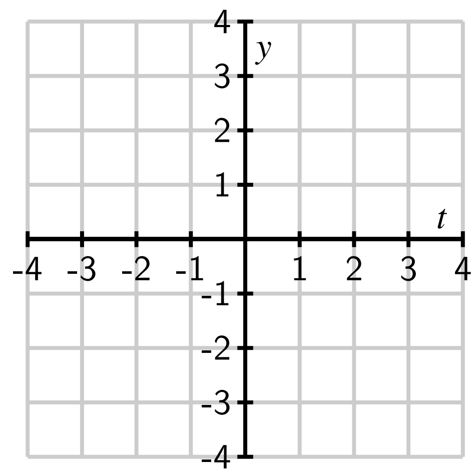
$$\frac{dy}{dt} = t - y.$$

a. Sketch a slope field on the axes at right.

b. Sketch the solutions whose initial values are $y(0) = -4, -3, \ldots, 4$.

c. What do your sketches suggest is the solution whose initial value is $y(0) = -1$? Verify that this is indeed the solution to this initial value problem.

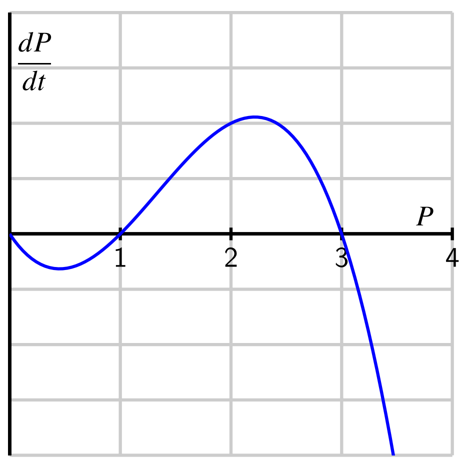
d. By considering the differential equation and the graphs you have sketched, what is the relationship between t and y at a point where a solution has a local minimum?



7. Consider the situation from problem 2 of Section 7.1: Suppose that the population of a particular species is described by the function $P(t)$, where P is expressed in millions. Suppose further that the population's rate of change is governed by the differential equation

$$\frac{dP}{dt} = f(P)$$

where $f(P)$ is the function graphed below.



a. Sketch a slope field for this differential equation. You do not have enough information to determine the actual slopes, but you should have enough information to determine where slopes are positive, negative, zero, large, or small, and hence determine the qualitative behavior of solutions.

b. Sketch some solutions to this differential equation when the initial population $P(0) > 0$.

c. Identify any equilibrium solutions to the differential equation and classify them as stable or unstable.

d. If $P(0) > 1$, what is the eventual fate of the species? if $P(0) < 1$?

e. Remember that we referred to this model for population growth as "growth with a threshold." Explain why this characterization makes sense by considering solutions whose inital value is close to 1.

8. The population of a species of fish in a lake is $P(t)$ where P is measured in thousands of fish and t is measured in months. The growth of the population is described by the differential equation

$$\frac{dP}{dt} = f(P) = P(6 - P).$$

a. Sketch a graph of $f(P) = P(6-P)$ and use it to determine the equilibrium solutions and whether they are stable or unstable. Write a complete sentence that describes the long-term behavior of the fish population.

b. Suppose now that the owners of the lake allow fishers to remove 1000 fish from the lake every month (remember that $P(t)$ is measured in *thousands* of fish). Modify the differential equation to take this into account. Sketch the new graph of dP/dt versus P. Determine the new equilibrium solutions and decide whether they are stable or unstable.

c. Given the situation in part (b), give a description of the long-term behavior of the fish population.

d. Suppose that fishermen remove h thousand fish per month. How is the differential equation modified?

e. What is the largest number of fish that can be removed per month without eliminating the fish population? If fish are removed at this maximum rate, what is the eventual population of fish?

9. Let $y(t)$ be the number of thousands of mice that live on a farm; assume time t is measured in years.[2]

a. The population of the mice grows at a yearly rate that is twenty times the number of mice. Express this as a differential equation.

b. At some point, the farmer brings C cats to the farm. The number of mice that the cats can eat in a year is

$$M(y) = C\frac{y}{2 + y}$$

thousand mice per year. Explain how this modifies the differential equation that you found in part a).

c. Sketch a graph of the function $M(y)$ for a single cat $C = 1$ and explain its features by looking, for instance, at the behavior of $M(y)$ when y is small and when y is large.

d. Suppose that $C = 1$. Find the equilibrium solutions and determine whether they are stable or unstable. Use this to explain the long-term behavior of the mice population depending on the initial population of the mice.

e. Suppose that $C = 60$. Find the equilibrium solutions and determine whether they are stable or unstable. Use this to explain the long-term behavior of the mice population depending on the initial population of the mice.

f. What is the smallest number of cats you would need to keep the mice population from growing arbitrarily large?

[2]This problem is based on an ecological analysis presented in a research paper by C.S. Hollings: The Components of Predation as Revealed by a Study of Small Mammal Predation of the European Pine Sawfly, *Canadian Entomology 91*: 283-320.

7.3 Euler's method

Motivating Questions

- What is Euler's method and how can we use it to approximate the solution to an initial value problem?
- How accurate is Euler's method?

7.3.1 Introduction

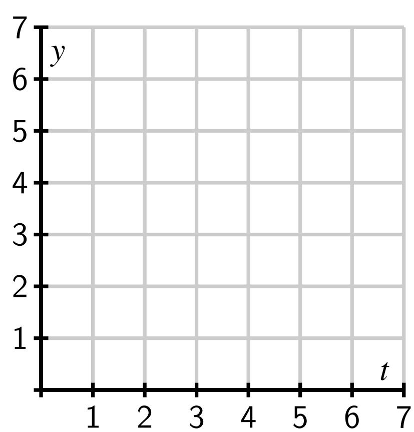
In Section 7.2, we saw how a slope field can be used to sketch solutions to a differential equation. In particular, the slope field is a plot of a large collection of tangent lines to a large number of solutions of the differential equation, and we sketch a single solution by simply following these tangent lines. With a little more care, we can use this same idea to approximate numerically the solutions of a differential equation.

Preview Activity 7.3.1. Consider the initial value problem

$$\frac{dy}{dt} = \frac{1}{2}(y+1), \;\; y(0) = 0.$$

(a) Use the differential equation to find the slope of the tangent line to the solution $y(t)$ at $t = 0$. Then use the given initial value to find the equation of the tangent line at $t = 0$.

(b) Sketch the tangent line you found in (a) on the axes provided on the interval $0 \leq t \leq 2$ and use the tangent line to approximate $y(2)$, the value of the solution at $t = 2$.



(c) Assuming that your approximation for $y(2)$ from (b) is the actual value of $y(2)$, use the differential equation to find the slope of the tangent line to $y(t)$ at $t = 2$ (i.e., at the point $(2, y(2))$). Then, write the equation of the tangent line at $t = 2$.

(d) Add a sketch of this tangent line from (c) on the interval $2 \le t \le 4$ to your plot in (b); use this new tangent line to approximate $y(4)$, the value of the solution at $t = 4$.

(e) Repeat the same step to find an approximation for $y(6)$.

7.3.2 Euler's Method

Preview Activity 7.3.1 demonstrates an algorithm known as Euler's[1] Method, which generates a numerical approximation to the solution of an initial value problem. In this algorithm, we will approximate the solution by taking horizontal steps of a fixed size that we denote by Δt.

Before explaining the algorithm in detail, let's remember how we compute the slope of a line: the slope of a given line is the ratio of the vertical change to the horizontal change, as shown in Figure 7.3.1.

In symbols, $m = \frac{\Delta y}{\Delta t}$. Solving for Δy, we see that the vertical change is the product of the slope and the horizontal change, or

$$\Delta y = m\Delta t.$$

In words, this says that (along a line), the change in y between two points is the product of the slope times the change in x.

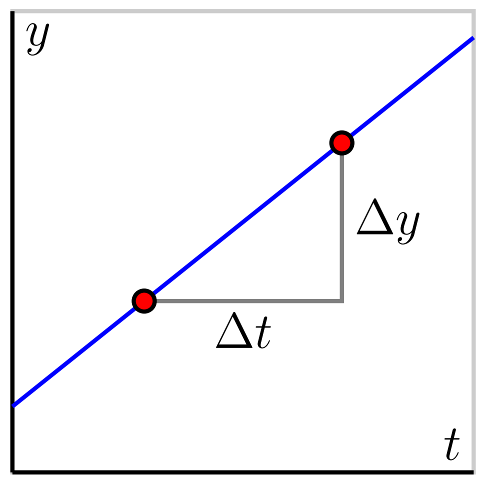


Figure 7.3.1 Visualizing slope.

Now, suppose that we would like to solve the initial value problem

$$\frac{dy}{dt} = t - y, \; y(0) = 1.$$

Using more advanced techniques from differential equations, it's possible to find an algebraic formula for the solution to this initial value problem, and we can check that this solution is $y(t) = t - 1 + 2e^{-t}$. Here we are instead interested in generating an approximate solution by creating a sequence of points (t_i, y_i), where $y_i \approx y(t_i)$, but we'll use the algebraic formula as a way to check the accuracy of our approximation.

For this first example, we choose $\Delta t = 0.2$. Since we know that $y(0) = 1$, we will take the initial point to be $(t_0, y_0) = (0, 1)$ and move horizontally by $\Delta t = 0.2$ to the point (t_1, y_1). Thus, $t_1 = t_0 + \Delta t = 0.2$. Now, the differential equation tells us that the slope of the tangent

[1] "Euler" is pronounced "Oy-ler." Among other things, Euler is the mathematician credited with the famous number e; if you incorrectly pronounce his name "You-ler," you fail to appreciate his genius and legacy.

line at this point is

$$m = \left.\frac{dy}{dt}\right|_{(0,1)} = 0 - 1 = -1,$$

so to move along the tangent line by taking a horizontal step of size $\Delta t = 0.2$, we must also move vertically by

$$\Delta y = m\Delta t = -1 \cdot 0.2 = -0.2.$$

We then have the approximation

$$y(0.2) \approx y_1 = y_0 + \Delta y = 1 - 0.2 = 0.8.$$

At this point, we have executed one step of Euler's method, as seen graphically in Figure 7.3.2.

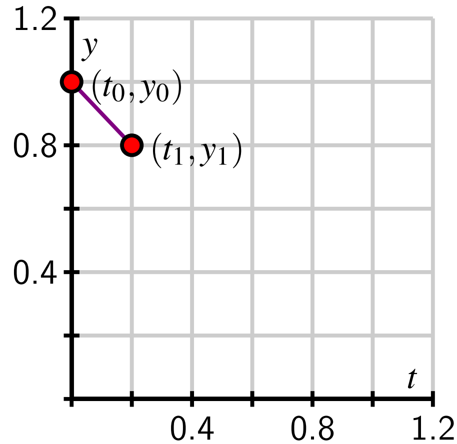


Figure 7.3.2 One step of Euler's method.

Now we repeat this process: at $(t_1, y_1) = (0.2, 0.8)$, the differential equation tells us that the slope is

$$m = \left.\frac{dy}{dt}\right|_{(0.2,0.8)} = 0.2 - 0.8 = -0.6.$$

Since we are moving forward horizontally by Δt to $t_2 = t_1 + \Delta t = 0.4$, we must move vertically by

$$\Delta y = m \cdot \Delta t = -0.6 \cdot 0.2 = -0.12.$$

We consequently arrive at

$$y_2 = y_1 + \Delta y = 0.8 - 0.12 = 0.68,$$

which gives $y(0.2) \approx 0.68$. Now we have completed the second step of Euler's method, as shown in Figure 7.3.3.

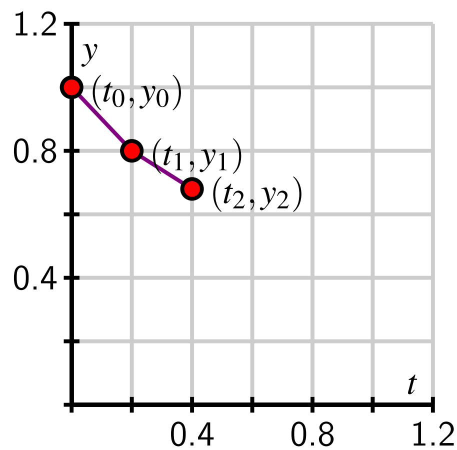


Figure 7.3.3 Two steps of Euler's method.

If we continue in this way, we may generate the points (t_i, y_i) shown in Figure 7.3.4. Because it turns out that the exact to this differential equation is $y(t) = t - 1 + 2e^{-t}$, we can graph $y(t)$ and compare it to the points generated by Euler's method, as shown in Figure 7.3.5.

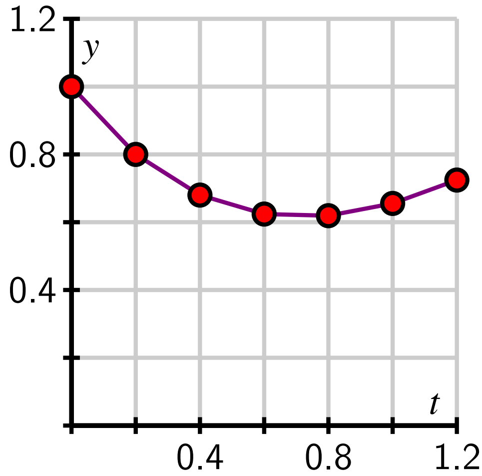


Figure 7.3.4 The points and piecewise linear approximate solution generated by Euler's method.

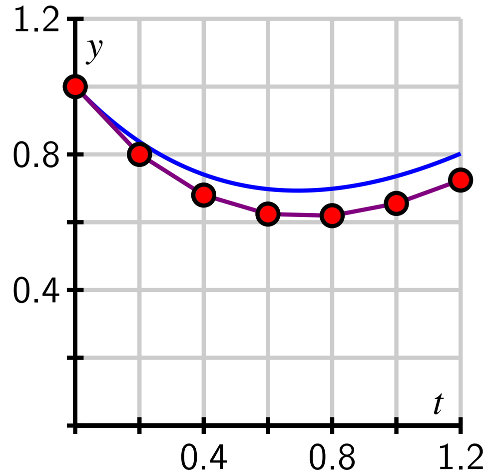


Figure 7.3.5 The approximate solution compared to the exact solution (shown in blue).

Because we need to generate a large number of points (t_i, y_i), it is convenient to organize the implementation of Euler's method in a table. We begin with the given initial data, as shown in Table 7.3.6.

t_i	y_i	dy/dt	Δy
0.0	1.0000		

Table 7.3.6 Starting with given initial data.

t_i	y_i	dy/dt	Δy
0.0	1.0000	−1.0000	−0.2000

Table 7.3.7 The first values of dy/dt and Δy.

From here, we compute the slope of the tangent line $m = dy/dt$ using the formula for dy/dt from the differential equation, and then we find Δy, the change in y, using the rule $\Delta y = m\Delta t$. These two results are shown in the last two columns of Table 7.3.7.

Next, we increase t_i by Δt and y_i by Δy to get the start of the second row in Table 7.3.8.

t_i	y_i	dy/dt	Δy
0.0	1.0000	−1.0000	−0.2000
0.2	0.8000		

Table 7.3.8 Starting the second step of Euler's Method.

t_i	y_i	dy/dt	Δy
0.0	1.0000	−1.0000	−0.2000
0.2	0.8000	−0.6000	−0.1200
0.4	0.6800	−0.2800	−0.0560
0.6	0.6240	−0.0240	−0.0048
0.8	0.6192	0.1808	0.0362
1.0	0.6554	0.3446	0.0689
1.2	0.7243	0.4757	0.0951

Table 7.3.9 Euler's method for 6 steps with $\Delta t = 0.2$.

We can continue the process for however many steps we decide, eventually generating a table like Table 7.3.9, which is the full set of data needed to generate Figure 7.3.4.

Overall, we have developed the following general approach that is known as Euler's Method.

Euler's Method.

Given an initial value problem of the form $\frac{dy}{dt} = f(t, y)$, $y(t_0) = y_0$, we can generate a collection of approximations $y_1, y_2, \ldots$ to the values of $y(t_1), y(t_2), \ldots$ using the following algorithm:

- Choose a value for Δt
- Set $y_0 = y(t_0)$
- For each desired natural number i, set $t_{i+1} = t_i + \Delta t$ and

$$y_{i+1} = y_i + f(t_i, y_i)\Delta y$$

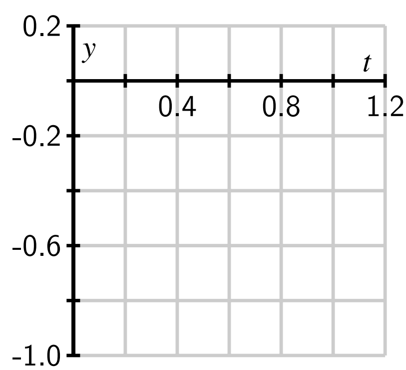
The notation $\frac{dy}{dt} = f(t, y)$ just means "$\frac{dy}{dt}$ depends on y and/or t in the way described by the given differential equation". We have worked most recently with $\frac{dy}{dt} = t - y$, and the formula $f(t, y) = t - y$ was used every time we needed to find the slope at the next point.

Activity 7.3.2. Consider the initial value problem

$$\frac{dy}{dt} = 2t - 1, \ y(0) = 0$$

(a) Use Euler's method with $\Delta t = 0.2$ to approximate the solution at $t_i = 0.2, 0.4, 0.6, 0.8$, and 1.0. Record your work in the following table, and sketch the points (t_i, y_i) on the axes provided.

t_i	y_i	dy/dt	Δy
0.0000	0.0000		
0.2000			
0.4000			
0.6000			
0.8000			
1.0000			



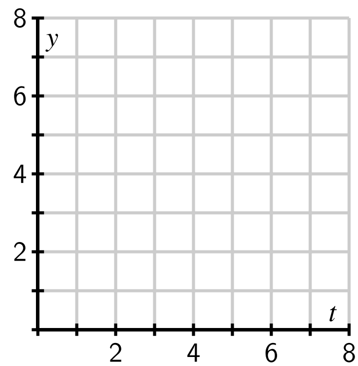
(b) Find the exact solution to the original initial value problem and use this function to find the error in your approximation at each one of the points t_i.

(c) How would your computations differ if the initial value was $y(0) = 1$? What does this mean about different solutions to this differential equation?

(d) Explain why the value y_5 generated by Euler's method for this initial value problem produces the same value as a left Riemann sum for the definite integral $\int_0^1 (2t-1)\,dt$.

Activity 7.3.3. Consider the differential equation $\frac{dy}{dt} = 6y - y^2$.

(a) Sketch the slope field for this differential equation on the axes provided.

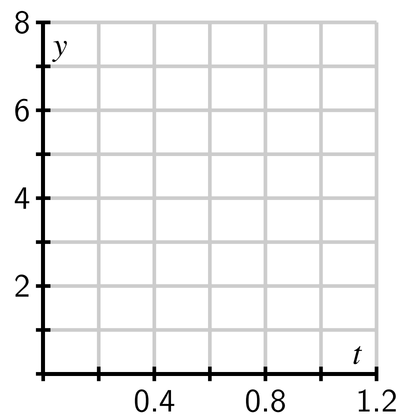


(b) Identify any equilibrium solutions and determine whether they are stable or unstable.

(c) What is the long-term behavior of the solution that satisfies the initial value $y(0) = 1$?

(d) Using the initial value $y(0) = 1$, use Euler's method with $\Delta t = 0.2$ to approximate the solution at $t_i = 0.2, 0.4, 0.6, 0.8$, and 1.0. Record your results in the table and sketch the corresponding points (t_i, y_i) on the axes provided. Note the different horizontal scale on the axes here compared to the axes above.

t_i	y_i	dy/dt	Δy
0.0	1.0000		
0.2			
0.4			
0.6			
0.8			
1.0			



(e) What happens if we apply Euler's method to approximate the solution with $y(0) = 6$?

7.3.3 The error in Euler's method

Since we are approximating the solutions to an initial value problem using tangent lines, we should expect that the error in the approximation will be smaller when the step size is smaller. Consider the initial value problem

$$\frac{dy}{dt} = y, \; y(0) = 1,$$

whose solution we can easily find.

The question posed by this initial value problem is "what function do we know that is the same as its own derivative and has value 1 when $t = 0$?" Observe that the solution is $y(t) = e^t$. We now apply Euler's method to approximate $y(1) = e$ using several values of Δt. These approximations will be denoted by $E_{\Delta t}$, and we'll use them to see how accurate Euler's Method is.

To begin, we apply Euler's method with a step size of $\Delta t = 0.2$. In that case, we find that $y(1) \approx E_{0.2} = 2.4883$. The error is therefore

$$y(1) - E_{0.2} = e - 2.4883 \approx 0.2300.$$

Repeatedly halving Δt gives the following results, expressed in both tabular and graphical form.

Δt	$E_{\Delta t}$	Error
0.200	2.4883	0.2300
0.100	2.5937	0.1245
0.050	2.6533	0.0650
0.025	2.6851	0.0332

Table 7.3.10 Errors that correspond to different Δt values.

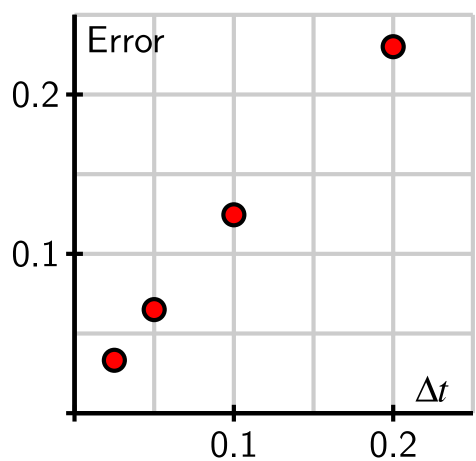


Figure 7.3.11 A plot of the error as a function of Δt.

Notice, both numerically and graphically, that the error is roughly halved when Δt is halved. This example illustrates the following general principle.

If Euler's method is used to approximate the solution to an initial value problem at a point $\bar{t}$, then the error is proportional to Δt. That is,

$$y(\bar{t}) - E_{\Delta t} \approx K\Delta t$$

for some constant of proportionality K.

7.3.4 Summary

- Euler's method is an algorithm for approximating the solution to an initial value problem by following the tangent lines while we take horizontal steps across the t-axis.
- If we wish to approximate $y(\bar{t})$ for some fixed $\bar{t}$ by taking horizontal steps of size Δt, then the error in our approximation is proportional to Δt.

7.3.5 Exercises

1. Consider the solution of the differential equation $y' = -3y$ passing through $y(0) = 1.5$.

A. Sketch the slope field for this differential equation, and sketch the solution passing through the point (0,1.5).

B. Use Euler's method with step size $\Delta x = 0.2$ to estimate the solution at $x = 0.2, 0.4, \ldots, 1$, using these to fill in the following table. *(Be sure **not** to round your answers at each step!)*

$x =$	0	0.2	0.4	0.6	0.8	1.0
$y \approx$	1.5	______	______	______	______	______

C. Plot your estimated solution on your slope field. Compare the solution and the slope field. Is the estimated solution an over or under estimate for the actual solution?

○ over

○ under

D. Check that $y = 1.5e^{-3x}$ is a solution to $y' = -3y$ with $y(0) = 1.5$.

2. Use Euler's method to solve

$$\frac{dB}{dt} = 0.04B$$

with initial value $B = 1200$ when $t = 0$.

A. $\Delta t = 1$ and 1 step: $B(1) \approx$ ____________________

B. $\Delta t = 0.5$ and 2 steps: $B(1) \approx$ ____________________

C. $\Delta t = 0.25$ and 4 steps: $B(1) \approx$ ____________________

D. Suppose B is the balance in a bank account earning interest. Be sure that you can explain why the result of your calculation in part (a) is equivalent to compounding the interest once a year instead of continuously. Then interpret the result of your calculations in parts (b) and (c) in terms of compound interest.

3. Consider the differential equation

$$\frac{dy}{dx} = 4x,$$

with initial condition $y(0) = 2$.

A. Use Euler's method with two steps to estimate y when $x = 1$:

$y(1) \approx$ __________

*(Be sure **not** to round your calculations at each step!)*

Now use four steps:

$y(1) \approx$ __________

*(Be sure **not** to round your calculations at each step!)*

B. What is the solution to this differential equation (with the given initial condition)?

$y =$ ____________________________________

C. What is the magnitude of the error in the two Euler approximations you found?

Magnitude of error in Euler with 2 steps = _______________

Magnitude of error in Euler with 4 steps = _______________

D. By what factor should the error in these approximations change (that is, the error with two steps should be what number times the error with four)?

factor = __________

(How close to this is the result you obtained above?)

4. Consider the differential equation $y' = -x - y$.

Use Euler's method with $\Delta x = 0.1$ to estimate y when $x = 1.4$ for the solution curve satisfying

$y(1) = 1$: Euler's approximation gives $y(1.4) \approx$ ________

Use Euler's method with $\Delta x = 0.1$ to estimate y when $x = 2.4$ for the solution curve satisfying $y(1) = 0$: Euler's approximation gives $y(2.4) \approx$ ________

5. Use Euler's method with step size 0.3 to estimate $y(1.5)$, where $y(x)$ is the solution of the initial-value problem

$$y' = 4x + y^2, \quad y(0) = 0.$$

$y(1.5) =$ __________.

6. Newton's Law of Cooling says that the rate at which an object, such as a cup of coffee, cools is proportional to the difference in the object's temperature and room temperature. If $T(t)$ is the object's temperature and T_r is room temperature, this law is expressed at

$$\frac{dT}{dt} = -k(T - T_r),$$

where k is a constant of proportionality. In this problem, temperature is measured in degrees Fahrenheit and time in minutes.

a. Two calculus students, Alice and Bob, enter a 70° classroom at the same time. Each has a cup of coffee that is 100°. The differential equation for Alice has a constant of proportionality $k = 0.5$, while the constant of proportionality for Bob is $k = 0.1$. What is the initial rate of change for Alice's coffee? What is the initial rate of change for Bob's coffee?

b. What feature of Alice's and Bob's cups of coffee could explain this difference?

c. As the heating unit turns on and off in the room, the temperature in the room is

$$T_r = 70 + 10\sin t.$$

Implement Euler's method with a step size of $\Delta t = 0.1$ to approximate the temperature of Alice's coffee over the time interval $0 \leq t \leq 50$. This will most easily be performed using a spreadsheet such as *Excel*. Graph the temperature of her coffee and room temperature over this interval.

d. In the same way, implement Euler's method to approximate the temperature of Bob's coffee over the same time interval. Graph the temperature of his coffee and room temperature over the interval.

e. Explain the similarities and differences that you see in the behavior of Alice's and Bob's cups of coffee.

7. We have seen that the error in approximating the solution to an initial value problem is proportional to Δt. That is, if $E_{\Delta t}$ is the Euler's method approximation to the solution to an initial value problem at $\bar{t}$, then

$$y(\bar{t}) - E_{\Delta t} \approx K\Delta t$$

for some constant of proportionality K.

In this problem, we will see how to use this fact to improve our estimates, using an idea called *accelerated convergence*.

a. We will create a new approximation by assuming the error is *exactly* proportional to Δt, according to the formula

$$y(\bar{t}) - E_{\Delta t} = K\Delta t.$$

Using our earlier results from the initial value problem $dy/dt = y$ and $y(0) = 1$ with $\Delta t = 0.2$ and $\Delta t = 0.1$, we have

$$\begin{aligned} y(1) - 2.4883 &= 0.2K \\ y(1) - 2.5937 &= 0.1K. \end{aligned}$$

This is a system of two linear equations in the unknowns $y(1)$ and K. Solve this system to find a new approximation for $y(1)$. (You may remember that the exact value is $y(1) = e = 2.71828\ldots$.)

b. Use the other data, $E_{0.05} = 2.6533$ and $E_{0.025} = 2.6851$ to do similar work as in (a) to obtain another approximation. Which gives the better approximation? Why do you think this is?

c. Let's now study the initial value problem

$$\frac{dy}{dt} = t - y, \; y(0) = 0.$$

Approximate $y(0.3)$ by applying Euler's method to find approximations $E_{0.1}$ and $E_{0.05}$. Now use the idea of accelerated convergence to obtain a better approximation. (For the sake of comparison, you want to note that the actual value is $y(0.3) = 0.0408$.)

8. In this problem, we'll modify Euler's method to obtain better approximations to solutions of initial value problems. This method is called the *Improved Euler's method.*

In Euler's method, we walk across an interval of width Δt using the slope obtained from the differential equation at the left endpoint of the interval. Of course, the slope of the solution will most likely change over this interval. We can improve our approximation by trying to incorporate the change in the slope over the interval.

Let's again consider the initial value problem $dy/dt = y$ and $y(0) = 1$, which we will approximate using steps of width $\Delta t = 0.2$. Our first interval is therefore $0 \leq t \leq 0.2$. At $t = 0$, the differential equation tells us that the slope is 1, and the approximation we obtain from Euler's method is that $y(0.2) \approx y_1 = 1 + 1(0.2) = 1.2$.

This gives us some idea for how the slope has changed over the interval $0 \leq t \leq 0.2$. We know the slope at $t = 0$ is 1, while the slope at $t = 0.2$ is 1.2, trusting in the Euler's method approximation. We will therefore refine our estimate of the initial slope to be the average of these two slopes; that is, we will estimate the slope to be $(1+1.2)/2 = 1.1$. This gives the new approximation $y(1) = y_1 = 1 + 1.1(0.2) = 1.22$.

The first few steps look like what is found in Table 7.3.12.

Table 7.3.12 The first several steps of the improved Euler's method

t_i	y_i	Slope at (t_{i+1}, y_{i+1})	Average slope
0.0	1.0000	1.2000	1.1000
0.2	1.2200	1.4640	1.3420
0.4	1.4884	1.7861	1.6372
$\vdots$	$\vdots$	$\vdots$	$\vdots$

a. Continue with this method to obtain an approximation for $y(1) = e$.

b. Repeat this method with $\Delta t = 0.1$ to obtain a better approximation for $y(1)$.

c. We saw that the error in Euler's method is proportional to Δt. Using your results from parts (a) and (b), what power of Δt appears to be proportional to the error in the Improved Euler's Method?

7.4 Separable differential equations

Motivating Questions

- What is a separable differential equation?
- How can we find solutions to a separable differential equation?
- Are some of the differential equations that arise in applications separable?

7.4.1 Introduction

In Sections 7.2 and Section 7.3, we have seen several ways to approximate the solution to an initial value problem. Given the frequency with which differential equations arise in the world around us, we would like to have some techniques for finding explicit algebraic solutions of certain initial value problems. In this section, we focus on a particular class of differential equations (called *separable*) and develop a method for finding algebraic formulas for their solutions.

A *separable differential equation* is a differential equation whose algebraic structure allows the variables to be separated in a particular way. For instance, consider the equation

$$\frac{dy}{dt} = ty.$$

We would like to separate the variables t and y so that all occurrences of t appear on the right-hand side, and all occurrences of y appear on the left, multiplied by dy/dt. For this example, we divide both sides by y so that

$$\frac{1}{y}\frac{dy}{dt} = t.$$

Note that when we attempt to separate the variables in a differential equation, we require that one side is a product in which the derivative dy/dt is one factor and the other factor is solely an expression involving y.

Not every differential equation is separable. For example, if we consider the equation

$$\frac{dy}{dt} = t - y,$$

it may seem natural to separate it by writing

$$y + \frac{dy}{dt} = t.$$

As we will see, this turns out not to be helpful, since the left-hand side is not a product of a function of y with $\frac{dy}{dt}$.

Preview Activity 7.4.1. In this preview activity, we explore whether certain differential equations are separable or not, and then revisit some key ideas from earlier work in integral calculus.

(a) Which of the following differential equations are separable? If the equation is separable, write the equation in the revised form $g(y)\frac{dy}{dt} = h(t)$.

i. $\dfrac{dy}{dt} = -3y$.

ii. $\dfrac{dy}{dt} = ty - y$.

iii. $\dfrac{dy}{dt} = t + 1$.

iv. $\dfrac{dy}{dt} = t^2 - y^2$.

(b) Explain why any autonomous differential equation is guaranteed to be separable.

(c) Why do we include the term "+C" in the expression

$$\int x\,dx = \frac{x^2}{2} + C?$$

(d) Suppose we know that a certain function f satisfies the equation

$$\int f'(x)\,dx = \int x\,dx.$$

What can you conclude about f?

7.4.2 Solving separable differential equations

Before we discuss a general approach to solving a separable differential equation, it is instructive to consider an example.

Example 7.4.1 Find all functions y that are solutions to the differential equation

$$\frac{dy}{dt} = \frac{t}{y^2}.$$

Solution. We begin by separating the variables and writing

$$y^2\frac{dy}{dt} = t.$$

Integrating both sides of the equation with respect to the independent variable t shows that

$$\int y^2 \frac{dy}{dt}\, dt = \int t\, dt.$$

Next, we notice that the left-hand side allows us to change the variable of antidifferentiation[1] from t to y. In particular, $dy = \frac{dy}{dt}\, dt$, so we now have

$$\int y^2\, dy = \int t\, dt.$$

This equation says that two families of antiderivatives are equal to each other. Therefore, when we find representative antiderivatives of both sides, we know they must differ by an arbitrary constant C. Antidifferentiating and including the integration constant C on the right, we find that

$$\frac{y^3}{3} = \frac{t^2}{2} + C.$$

It is not necessary to include an arbitrary constant on both sides of the equation; we know that $y^3/3$ and $t^2/2$ are in the same family of antiderivatives and must therefore differ by a single constant.

Finally, we solve the last equation above for y as a function of t, which gives

$$y(t) = \sqrt[3]{\frac{3}{2}\, t^2 + 3C}.$$

Of course, the term $3C$ on the right-hand side represents 3 times an unknown constant. It is, therefore, still an unknown constant, which we will rewrite as C. We thus conclude that the funtion

$$y(t) = \sqrt[3]{\frac{3}{2}\, t^2 + C}$$

is a solution to the original differential equation for any value of C.

Notice that because this solution depends on the arbitrary constant C, we have found an infinite family of solutions. This makes sense because we expect to find a unique solution that corresponds to any given initial value.

For example, if we want to solve the initial value problem

$$\frac{dy}{dt} = \frac{t}{y^2},\ y(0) = 2,$$

we know that the solution has the form $y(t) = \sqrt[3]{\frac{3}{2}\, t^2 + C}$ for some constant C. We therefore must find the appropriate value for C that gives the initial value $y(0) = 2$. Hence,

$$2 = y(0) = \sqrt[3]{\frac{3}{2}\, 0^2 + C} = \sqrt[3]{C},$$

which shows that $C = 2^3 = 8$. The solution to the initial value problem is then

$$y(t) = \sqrt[3]{\frac{3}{2}t^2 + 8}.$$

◊

The strategy of Example 7.4.1 may be applied to any differential equation of the form

$$\frac{dy}{dt} = g(y) \cdot h(t),$$

and any differential equation of this form is said to be *separable*. We work to solve a separable differential equation by writing

$$\frac{1}{g(y)} \frac{dy}{dt} = h(t),$$

and then integrating both sides with respect to t. After integrating, we try to solve algebraically for y in order to write y as a function of t.

Example 7.4.2 Solve the differential equation

$$\frac{dy}{dt} = 3y.$$

Solution. Following the same strategy as in Example 7.4.1, we have

$$\frac{1}{y} \frac{dy}{dt} = 3.$$

Integrating both sides with respect to t,

$$\int \frac{1}{y} \frac{dy}{dt}\, dt = \int 3\, dt,$$

and thus

$$\int \frac{1}{y}\, dy = \int 3\, dt.$$

Antidifferentiating and including the integration constant, we find that

$$\ln |y| = 3t + C.$$

Finally, we need to solve for y. Here, one point deserves careful attention. By the definition of the natural logarithm function, it follows that

$$|y| = e^{3t+C} = e^{3t} e^{C}.$$

Since C is an unknown constant, e^C is as well, though we do know that it is positive (because e^x is positive for any x). When we remove the absolute value in order to solve for y, however,

[1] This is why we required that the left-hand side be written as a product in which dy/dt is one of the terms.

this constant may be either positive or negative. To account for a possible + or −, we denote this updated constant by C to obtain

$$y(t) = Ce^{3t}.$$

◊

There are two more technical points to make. First, notice that $y = 0$ is an equilibrium solution to this differential equation. In solving the equation above, we begin by dividing both sides by y, which is not allowed if $y = 0$. To be perfectly careful, therefore, we should consider the equilibrium solutions separately. In this case, notice that the final form of our solution captures the equilibrium solution by allowing $C = 0$.

Second, recall the discussion in Section 4.4 immediately following the development of Table 4.4.5 in Activity 4.4.3. When we write

$$\int \frac{1}{u}\, du = \ln(|u|) + C$$

there is a subtle issue with the value of C that is connected to the discontinuity in $\frac{1}{u}$ at $u = 0$. This issue normally gets resolved either by applying a given initial condition to find the value of C or by only being interested in a solution to the differential equation on certain interval of u-values.

Activity 7.4.2. Suppose that the population of a town is growing continuously at an annual rate of 3% per year.

(a) Let $P(t)$ be the population of the town in year t. Write a differential equation that describes the annual growth rate.

(b) Find the solutions of this differential equation.

(c) If you know that the town's population in year 0 is 10,000, find the population $P(t)$.

(d) How long does it take for the population to double? This time is called the *doubling time*.

(e) Working more generally, find the doubling time if the annual growth rate is k times the population.

Activity 7.4.3. Suppose that a cup of coffee is initially at a temperature of 105° F and is placed in a 75° F room. If T is the temperature of the coffee in degrees Fahrenheit at time t in minutes, Newton's law of cooling says that

$$\frac{dT}{dt} = -k(T - 75),$$

where k is a constant of proportionality.

(a) Suppose you measure that the coffee is cooling at one degree per minute at the time the coffee is brought into the room. Use the differential equation to determine the value of the constant k.

(b) Find all the solutions of this differential equation.

(c) What happens to all the solutions as $t \to \infty$? Explain how this agrees with your intuition.

(d) What is the temperature of the cup of coffee after 20 minutes?

(e) How long does it take for the coffee to cool to 80°?

Activity 7.4.4. Solve each of the following differential equations or initial value problems.

(a) $\frac{dy}{dt} - (2-t)y = 2-t$

(b) $\frac{1}{t}\frac{dy}{dt} = e^{t^2-2y}$

(c) $y' = 2y+2, \ y(0) = 2$

(d) $y' = 2y^2, \ y(-1) = 2$

(e) $\frac{dy}{dt} = \frac{-2ty}{t^2+1}, \ y(0) = 4$

7.4.3 Summary

- A separable differential equation is one that may be rewritten with all occurrences of the dependent variable multiplying the derivative and all occurrences of the independent variable on the other side of the equation.
- We may find the solutions to certain separable differential equations by separating variables, integrating with respect to t, and ultimately solving the resulting algebraic equation for y.
- This technique allows us to solve many important differential equations that arise in the world around us. For instance, questions of growth and decay and Newton's Law of Cooling give rise to separable differential equations. Later, we will learn in Section 7.6 that the important logistic differential equation is also separable.

7.4.4 Exercises

1. Find the equation of the solution to $\dfrac{dy}{dx} = x^3 y$ through the point $(x,y) = (1,5)$.

2. Solve the differential equation

$$\frac{dy}{dx} = \frac{x}{49y}.$$

a. Find an implicit solution and put your answer in the following form: ____________________ = constant.

b. Find the equation of the solution through the point $(x, y) = (7, 1)$. Your equation must describe a single curve $y = f(x)$ with the domain of f as large as possible. ____________________

c. Find the equation of the solution through the point $(x, y) = (0, -4)$. Your answer should be of the form $y = f(x)$. ____________________

3. Find the solution to the differential equation

$$\frac{dy}{dt} = 0.9(y - 200)$$

if $y = 75$ when $t = 0$.

$y =$ ______________________________

4. Find the solution to the differential equation

$$\frac{dy}{dt} = y^2(1 + t),$$

$y = 3$ when $t = 1$.

$y =$ ______________________________

5. Solve the separable differential equation for u

$$\frac{du}{dt} = e^{6u+3t}.$$

Use the following initial condition: $u(0) = 15$.

$u =$ ______________________________________.

6. Find an equation of the curve that satisfies

$$\frac{dy}{dx} = 24yx^5$$

and whose y-intercept is 4.

$y(x) =$ ___________________________________.

7. Suppose that we use Euler's method to approximate the solution to the differential equation

$$\frac{dy}{dx} = \frac{x^4}{y}; \qquad y(0.2) = 2.$$

Let $f(x, y) = x^4/y$.

We let $x_0 = 0.2$ and $y_0 = 2$ and pick a step size $h = 0.2$.

Euler's method is the the following algorithm. From x_n and y_n, our approximations to the solution of the differential equation at the nth stage, we find the next stage by computing

$$x_{n+1} = x_n + h, \qquad y_{n+1} = y_n + h \cdot f(x_n, y_n).$$

Complete the following table. Your answers should be accurate to at least seven decimal places.

n	x_n	y_n
0	0.2	2
1	__________	__________
2	__________	__________
3	__________	__________
4	__________	__________
5	__________	__________

The exact solution can also be found using separation of variables. It is

$y(x) =$ ____

Thus the actual value of the function at the point $x = 1.2$

$y(1.2) =$ ____________________.

8. The mass of a radioactive sample decays at a rate that is proportional to its mass.

a. Express this fact as a differential equation for the mass $M(t)$ using k for the constant of proportionality.

b. If the initial mass is M_0, find an expression for the mass $M(t)$.

c. The *half-life* of the sample is the amount of time required for half of the mass to decay. Knowing that the half-life of Carbon-14 is 5730 years, find the value of k for a sample of Carbon-14.

d. How long does it take for a sample of Carbon-14 to be reduced to one-quarter its original mass?

e. Carbon-14 naturally occurs in our environment; any living organism takes in Carbon-14 when it eats and breathes. Upon dying, however, the organism no longer takes in Carbon-14. Suppose that you find remnants of a pre-historic firepit. By analyzing the charred wood in the pit, you determine that the amount of Carbon-14 is only 30% of the amount in living trees. Estimate the age of the firepit.[2]

9. Consider the initial value problem

$$\frac{dy}{dt} = -\frac{t}{y}, \; y(0) = 8$$

a. Find the solution of the initial value problem and sketch its graph.

b. For what values of t is the solution defined?

[2] This approach is the basic idea behind radiocarbon dating.

c. What is the value of y at the last time that the solution is defined?

d. By looking at the differential equation, explain why we should not expect to find solutions with the value of y you noted in (c).

10. Suppose that a cylindrical water tank with a hole in the bottom is filled with water. The water, of course, will leak out and the height of the water will decrease. Let $h(t)$ denote the height of the water. A physical principle called *Torricelli's Law* implies that the height decreases at a rate proportional to the square root of the height.

a. Express this fact using k as the constant of proportionality.

b. Suppose you have two tanks, one with $k = -1$ and another with $k = -10$. What physical differences would you expect to find?

c. Suppose you have a tank for which the height decreases at 20 inches per minute when the water is filled to a depth of 100 inches. Find the value of k.

d. Solve the initial value problem for the tank in part (c), and graph the solution you determine.

e. How long does it take for the water to run out of the tank?

f. Is the solution that you found valid for all time t? If so, explain how you know this. If not, explain why not.

11. The *Gompertz equation* is a model that is used to describe the growth of certain populations. Suppose that $P(t)$ is the population of some organism and that

$$\frac{dP}{dt} = -P \ln\left(\frac{P}{3}\right) = -P(\ln P - \ln 3).$$

a. Sketch a slope field for $P(t)$ over the range $0 \leq P \leq 6$.

b. Identify any equilibrium solutions and determine whether they are stable or unstable.

c. Find the population $P(t)$ assuming that $P(0) = 1$ and sketch its graph. What happens to $P(t)$ after a very long time?

d. Find the population $P(t)$ assuming that $P(0) = 6$ and sketch its graph. What happens to $P(t)$ after a very long time?

e. Verify that the long-term behavior of your solutions agrees with what you predicted by looking at the slope field.

7.5 Modeling with differential equations

Motivating Questions

- How can we use differential equations to describe phenomena in the world around us?
- How can we use differential equations to better understand these phenomena?

7.5.1 Introduction

We have seen several ways that differential equations arise in the world around us, from the growth of a population to the temperature of a cup of coffee. In this section, we look more closely at how differential equations give us a natural way to describe various phenoma. As we'll see, the key is to understand the different factors that cause a quantity to change.

Preview Activity 7.5.1. Any time that the rate of change of a quantity is related to the amount of a quantity, a differential equation naturally arises. In the following prompts, we encouner two such scenarios; for each, we want to develop a differential equation whose solution is the quantity of interest.

(a) Suppose you have a bank account in which money grows at an annual rate of 3%.

If you have $10,000 in the account, at what rate is your money growing?

(b) Suppose that you are withdrawing money from the account at $1,000 per year. What is the rate of change in the amount of money in the account that these withdrawals induce? What are the units on this rate of change?

(c) For a bank account with $\$P$ that earns 3% annual interest and from which you withdraw $1,000 per year, what differential equation does P satisfy?

(d) Now consider a different scenario. Suppose that a water tank holds 100 gallons and that a salty solution, which contains 20 grams of salt in every gallon, enters the tank at 2 gallons per minute.

How many grams of salt enter the tank each minute?

(e) Suppose that initially there are 300 grams of salt in the tank. How many grams of salt are in each gallon at this point in time?

(f) Finally, let $A(t)$ represent the amount of salt in the tank at time t. Suppose that evenly mixed solution is pumped out of the tank at the rate of 2 gallons per minute. How much salt leaves the tank each minute?

(g) What is the total rate of change in the amount of salt in the tank? Use your observations in the preceding questions (that involve the rates at which salt enters and leaves the tank) to write a differential equation satisfied by $A(t)$.

7.5.2 Developing a differential equation

Preview Activity 7.5.1 demonstrates the kind of thinking we will be doing in this section. In each of the two examples we considered, there is a quantity, such as the amount of money in the bank account or the amount of salt in the tank, that is changing due to several factors. The governing differential equation states that the total rate of change is the difference between the rate of increase and the rate of decrease.

Example 7.5.1 In the Great Lakes region, rivers flowing into the lakes carry a great deal of pollution in the form of small pieces of plastic averaging 1 millimeter in diameter. In order to understand how the amount of plastic in Lake Michigan is changing, construct a model for how this type of pollution has built up in the lake.

Solution. First, some basic facts about Lake Michigan.

- The volume of the lake is $5 \cdot 10^{12}$ cubic meters.
- Water flows into the lake at a rate of $5 \cdot 10^{10}$ cubic meters per year. It flows out of the lake at the same rate.
- Each cubic meter flowing into the lake contains roughly $3 \cdot 10^{-8}$ cubic meters of plastic pollution.

Let's denote the amount of pollution in the lake by $P(t)$, where P is measured in cubic meters of plastic and t in years. Our goal is to describe the rate of change of this function; so we want to develop a differential equation describing $P(t)$.

First, we will measure how $P(t)$ increases due to pollution flowing into the lake. We know that $5 \cdot 10^{10}$ cubic meters of water enters the lake every year and each cubic meter of water contains $3 \cdot 10^{-8}$ cubic meters of pollution. Therefore, pollution enters the lake at the rate of

$$\left(5 \cdot 10^{10} \frac{m^3 \text{ water}}{\text{year}}\right)\left(3 \cdot 10^{-8} \frac{m^3 \text{ plastic}}{m^3 \text{ water}}\right) = 1.5 \cdot 10^3 \text{cubic m of plastic per year.}$$

Second, we will measure how $P(t)$ decreases due to pollution flowing out of the lake. If the total amount of pollution is P cubic meters and the volume of Lake Michigan is $5 \cdot 10^{12}$ cubic meters, then the concentration of plastic pollution in Lake Michigan is

$$\frac{P}{5 \cdot 10^{12}} \text{cubic m of plastic per cubic m of water.}$$

Since $5 \cdot 10^{10}$ cubic meters of water flow out each year (and we assume that each cubic meter of water that flows out carries with it the plastic pollution it contains), then the plastic pollution leaves the lake at the rate of

$$\left(\frac{P}{5 \cdot 10^{12}} \frac{m^3 \text{ plastic}}{m^3 \text{ water}}\right)\left(5 \cdot 10^{10} \frac{m^3 \text{ water}}{\text{year}}\right) = \frac{P}{100} \text{cubic m of plastic per year.}$$

The total rate of change of P is thus the difference between the rate at which pollution enters the lake and the rate at which pollution leaves the lake, which is

$$\frac{dP}{dt} = 1.5 \cdot 10^3 - \frac{P}{100}$$

$$= \frac{1}{100}(1.5 \cdot 10^5 - P).$$

We have now found a differential equation that describes the rate at which the amount of pollution is changing. To understand the behavior of $P(t)$, we apply some of the techniques we have recently developed.

Because the differential equation is autonomous, we can sketch dP/dt as a function of P and then construct a slope field, as shown in Figure 7.5.2 and Figure 7.5.3.

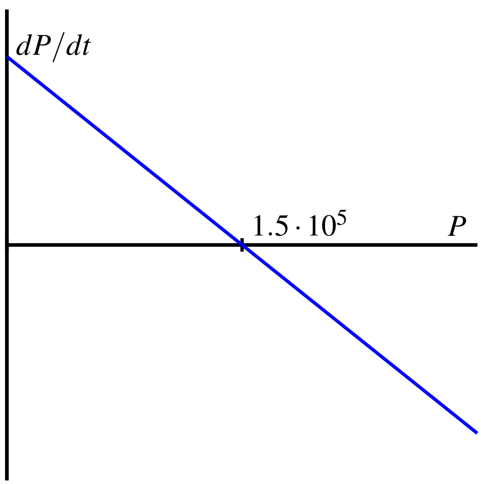


Figure 7.5.2 Plot of $\frac{dP}{dt}$ vs. P for $\frac{dP}{dt} = \frac{1}{100}(1.5 \cdot 10^5 - P)$.

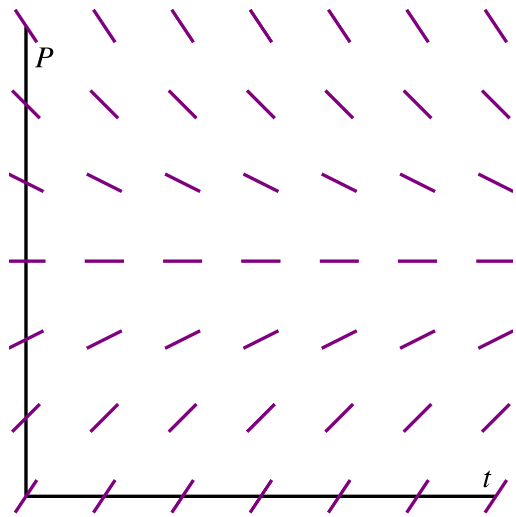


Figure 7.5.3 Plot of the slope field for $\frac{dP}{dt} = \frac{1}{100}(1.5 \cdot 10^5 - P)$.

These plots both show that $P = 1.5 \cdot 10^5$ is a stable equilibrium. Therefore, we should expect that the amount of pollution in Lake Michigan will stabilize near $1.5 \cdot 10^5$ cubic meters of pollution.

Next, assuming that there is initially no pollution in the lake, we will solve the initial value problem

$$\frac{dP}{dt} = \frac{1}{100}(1.5 \cdot 10^5 - P), \; P(0) = 0.$$

Separating variables, we find that

$$\frac{1}{1.5 \cdot 10^5 - P}\frac{dP}{dt} = \frac{1}{100}.$$

Integrating with respect to t, we have

$$\int \frac{1}{1.5 \cdot 10^5 - P}\frac{dP}{dt}\, dt = \int \frac{1}{100}\, dt,$$

and thus changing variables on the left and antidifferentiating on both sides, we find that

$$\int \frac{dP}{1.5 \cdot 10^5 - P} = \int \frac{1}{100}\, dt$$

$$-\ln|1.5 \cdot 10^5 - P| = \frac{1}{100}t + C$$

Finally, multiplying both sides by −1 and using the definition of the logarithm, we find that

$$1.5 \cdot 10^5 - P = Ce^{-t/100}. \tag{7.5.1}$$

This is a good time to determine the constant C. Since $P = 0$ when $t = 0$, we have

$$1.5 \cdot 10^5 - 0 = Ce^0 = C,$$

so $C = 1.5 \cdot 10^5$.

Using this value of C in Equation (7.5.1) and solving for P, we arrive at the solution

$$P(t) = 1.5 \cdot 10^5(1 - e^{-t/100}).$$

Superimposing the graph of P on the slope field we saw in Figure 7.5.3, we see, as shown in Figure 7.5.4

We see that, as expected, the amount of plastic pollution stabilizes around $1.5 \cdot 10^5$ cubic meters.

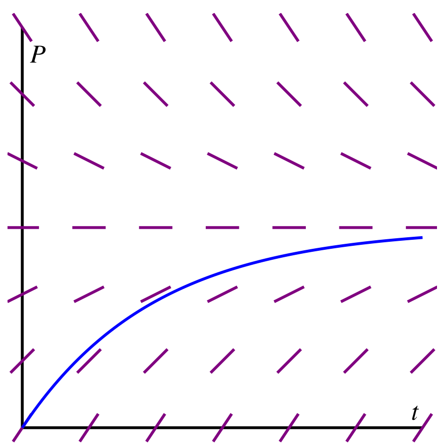


Figure 7.5.4 The solution $P(t)$ and the slope field for the differential equation $\frac{dP}{dt} = \frac{1}{100}(1.5 \cdot 10^5 - P)$.

◊

There are many important lessons to learn from Example 7.5.1. Foremost is how we can develop a differential equation by thinking about the “total rate = rate in - rate out” model. In addition, we note how we can bring together all of our available understanding (plotting $\frac{dP}{dt}$ vs. P, creating a slope field, solving the differential equation) to see how the differential equation describes the behavior of a changing quantity.

We can also explore what happens when certain aspects of the problem change. For instance, let’s suppose we are at a time when the plastic pollution entering Lake Michigan has stabilized at $1.5 \cdot 10^5$ cubic meters, and that new legislation is passed to prevent this type of pollution entering the lake. So, there is no longer any inflow of plastic pollution to the lake. How does the amount of plastic pollution in Lake Michigan now change? For example, how long does it take for the amount of plastic pollution in the lake to halve?

Resetting $t = 0$ at this time, we now have the initial value problem

$$\frac{dP}{dt} = -\frac{1}{100}P, \ P(0) = 1.5 \cdot 10^5.$$

It is a straightforward and familiar exercise to find that the solution to this equation is $P(t) = 1.5 \cdot 10^5 e^{-t/100}$. The time that it takes for half of the pollution to flow out of the lake is given by T where $P(T) = 0.75 \cdot 10^5$. Thus, we must solve the equation

$$0.75 \cdot 10^5 = 1.5 \cdot 10^5 e^{-T/100},$$

or

$$\frac{1}{2} = e^{-T/100}.$$

It follows that

$$T = -100 \ln\left(\frac{1}{2}\right) \approx 69.3 \text{ years}.$$

In the upcoming activities, we explore some other natural settings in which differential equations model changing quantities.

Activity 7.5.2. Suppose you have a bank account that grows by 5% every year. Let $A(t)$ be the amount of money in the account in year t.

(a) What is the rate of change of A with respect to t?

(b) Suppose that you are also withdrawing \$10,000 per year. Write a differential equation that expresses the total rate of change of A.

(c) Sketch a slope field for this differential equation, find any equilibrium solutions, and identify them as either stable or unstable. Write a sentence or two that describes the significance of the stability of the equilibrium solution.

(d) Suppose that you initially deposit \$100,000 into the account. How long does it take for you to deplete the account?

(e) What is the smallest amount of money you would need to have in the account to guarantee that you never deplete the money in the account?

(f) If your initial deposit is \$300,000, how much could you withdraw every year without depleting the account?

Activity 7.5.3. A dose of morphine is absorbed from the bloodstream of a patient at a rate proportional to the amount in the bloodstream.

(a) Write a differential equation for $M(t)$, the amount of morphine in the patient's bloodstream, using k as the constant proportionality.

(b) Assuming that the initial dose of morphine is M_0, solve the initial value problem to find $M(t)$. Use the fact that the half-life for the absorption of morphine is two hours to find the constant k.

(c) Suppose that a patient is given morphine intravenously at the rate of 3 milligrams per hour. Write a differential equation that combines the intravenous administration of morphine with the body's natural absorption.

(d) Find any equilibrium solutions and determine their stability.

(e) Assuming that there is initially no morphine in the patient's bloodstream, solve the initial value problem to determine $M(t)$. What happens to $M(t)$ after a very long time?

(f) To what rate should a doctor reduce the intravenous rate so that there is eventually 7 milligrams of morphine in the patient's bloodstream?

7.5.3 Summary

- Differential equations arise in a situation when we understand how various factors cause a quantity to change.
- We may use the tools we have developed so far—slope fields, Euler's methods, and our method for solving separable equations—to understand a quantity described by a differential equation.

7.5.4 Exercises

1. **Mixing problem.** A tank contains 2100 L of pure water. A solution that contains 0.07 kg of sugar per liter enters the tank at the rate 2 L/min. The solution is mixed and drains from the tank at the same rate.

(a) How much sugar is in the tank at the beginning?

$y(0) =$ _____ (include units)

(b) With S representing the amount of sugar (in kg) at time t (in minutes) write a differential equation which models this situation.

$S' = f(t, S) =$.

Note: Make sure you use a capital S, (and don't use S(t), it confuses the computer). Don't enter units for this function.

(c) Find the amount of sugar (in kg) after t minutes.

$S(t) =$ ____ (function of t)

(d) Find the amount of the sugar after 42 minutes.

$S(42) =$ ________________________ (include `units`)

2. **Mixing problem.** A tank contains 80 kg of salt and 1000 L of water. A solution of a concentration 0.04 kg of salt per liter enters a tank at the rate 8 L/min. The solution is mixed and drains from the tank at the same rate.

(a) What is the concentration of our solution in the tank initially?

concentration = ____________ (kg/L)

(b) Find the amount of salt in the tank after 5 hours.

amount = ____________ (kg)

(c) Find the concentration of salt in the solution in the tank as time approaches infinity.

concentration = ____________ (kg/L)

3. **Population growth problem.** A bacteria culture starts with 800 bacteria and grows at a rate proportional to its size. After 2 hours there will be 1600 bacteria.

(a) Express the population after t hours as a function of t.

population: ________________________ (function of t)

(b) What will be the population after 9 hours?

(c) How long will it take for the population to reach 2330 ?

4. **Radioactive decay problem.** An unknown radioactive element decays into non-radioactive substances. In 900 days the radioactivity of a sample decreases by 38 percent.

(a) What is the half-life of the element?

half-life: ____________________ (days)

(b) How long will it take for a sample of 100 mg to decay to 60 mg?

time needed: ________________________ (days)

5. **Investment problem.** A young person with no initial capital invests k dollars per year in a retirement account at an annual rate of return 0.1. Assume that investments are made continuously and that the return is compounded continuously.

Determine a formula for the sum $S(t)$ -- (this will involve the parameter k):

$S(t) =$ ____

What value of k will provide 3978000 dollars in 41 years?

$k =$ ________________________

6. Congratulations, you just won the lottery! In one option presented to you, you will be paid one million dollars a year for the next 25 years. You can deposit this money in an account that will earn 5% each year.

a. Set up a differential equation that describes the rate of change in the amount of money in the account. Two factors cause the amount to grow—first, you are depositing one millon dollars per year and second, you are earning 5% interest.

b. If there is no amount of money in the account when you open it, how much money will you have in the account after 25 years?

c. The second option presented to you is to take a lump sum of 10 million dollars, which you will deposit into a similar account. How much money will you have in that account after 25 years?

d. Do you prefer the first or second option? Explain your thinking.

e. At what time does the amount of money in the account under the first option overtake the amount of money in the account under the second option?

7. When a skydiver jumps from a plane, gravity causes her downward velocity to increase at the rate of $g \approx 9.8$ meters per second squared. At the same time, wind resistance causes her velocity to decrease at a rate proportional to the velocity.

a. Using k to represent the constant of proportionality, write a differential equation that describes the rate of change of the skydiver's velocity.

b. Find any equilibrium solutions and decide whether they are stable or unstable. Your result should depend on k.

c. Suppose that the initial velocity is zero. Find the velocity $v(t)$.

d. A typical terminal velocity for a skydiver falling face down is 54 meters per second. What is the value of k for this skydiver?

e. How long does it take to reach 50% of the terminal velocity?

8. During the first few years of life, the rate at which a baby gains weight is proportional to the reciprocal of its weight.

a. Express this fact as a differential equation.

b. Suppose that a baby weighs 8 pounds at birth and 9 pounds one month later. How much will he weigh at one year?

c. Do you think this is a realistic model for a long time?

9. Suppose that you have a water tank that holds 100 gallons of water. A briny solution, which contains 20 grams of salt per gallon, enters the tank at the rate of 3 gallons per minute.

At the same time, the solution is well mixed, and water is pumped out of the tank at the rate of 3 gallons per minute.

a. Since 3 gallons enter the tank every minute and 3 gallons leave every minute, what can you conclude about the volume of water in the tank?

b. How many grams of salt enter the tank every minute?

c. Suppose that $S(t)$ denotes the number of grams of salt in the tank in minute t. How many grams are there in each gallon in minute t?

d. Since water leaves the tank at 3 gallons per minute, how many grams of salt leave the tank each minute?

e. Write a differential equation that expresses the total rate of change of S.

f. Identify any equilibrium solutions and determine whether they are stable or unstable.

g. Suppose that there is initially no salt in the tank. Find the amount of salt $S(t)$ in minute t.

h. What happens to $S(t)$ after a very long time? Explain how you could have predicted this only knowing how much salt there is in each gallon of the briny solution that enters the tank.

7.6 Population growth and the logistic equation

Motivating Questions

- How can we use differential equations to realistically model the growth of a population?
- How can we assess the accuracy of our models?

7.6.1 Introduction

The growth of the earth's population has long been an important issue for humankind. Will the population continue to grow? Or will it perhaps level off at some point, and if so, when? In this section, we look at two ways we may use differential equations to help us address these questions.

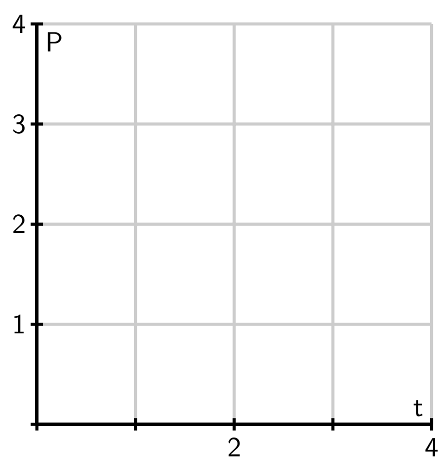
Before we begin, let's consider again two important differential equations that we have seen in earlier work this chapter.

Preview Activity 7.6.1. Recall that one model for population growth states that a population grows at a rate proportional to its size. In symbols: $\frac{dP}{dt} = kP$.

(a) We begin with the differential equation

$$\frac{dP}{dt} = \frac{1}{2}P.$$

Sketch a slope field as well as a few typical solutions on the axes provided.



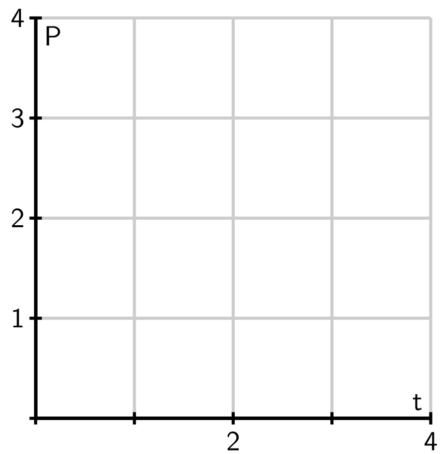
(b) Find all equilibrium solutions of the equation $\frac{dP}{dt} = \frac{1}{2}P$ and classify each as stable or unstable.

(c) If $P(0)$ is positive, describe the long-term behavior of the solution to $\frac{dP}{dt} = \frac{1}{2}P$.

(d) Let's now consider a modified differential equation given by

$$\frac{dP}{dt} = \frac{1}{2}P(3 - P).$$

As before, sketch a slope field as well as a few typical solutions on the following axes provided.



(e) Find any equilibrium solutions and classify them as stable or unstable.

(f) If $P(0)$ is positive, describe the long-term behavior of the solution.

7.6.2 The earth's population

We will now begin studying the earth's population. In Activity 7.6.2, we consider some recent population data and explore an elementary model.

Activity 7.6.2. Using some recent population data for planet Earth, we want to model the growth of the population over time.

Table 7.6.1 Some recent population data (in billions) for planet Earth.

Year	1998	1999	2000	2001	2002	2005	2006	2007	2008	2009	2010
Pop	5.932	6.008	6.084	6.159	6.234	6.456	6.531	6.606	6.681	6.756	6.831

Our first model will be based on the following assumption:

The rate of change of the population is proportional to the population.

On the face of it, this seems pretty reasonable. When there is a relatively small number of people, there will be fewer births and deaths so the rate of change will be small.

When there is a larger number of people, there will be more births and deaths so we expect a larger rate of change. If $P(t)$ is the population t years after the year 2000, we may express this assumption as

$$\frac{dP}{dt} = kP$$

where k is a constant of proportionality.

(a) Use the data in the table to estimate the derivative $P'(0)$ using a central difference. Assume that $t = 0$ corresponds to the year 2000.

(b) What is the population $P(0)$?

(c) Use your results from (a) and (b) to estimate the constant of proportionality k in the differential equation.

(d) Now that we know the value of k, we have the initial value problem

$$\frac{dP}{dt} = kP, \; P(0) = 6.084.$$

Find the solution to this initial value problem.

(e) What does your solution predict for the population in the year 2010? Is this close to the actual population given in the table?

(f) When does your solution predict that the population will reach 12 billion?

(g) What does your solution predict for the population in the year 2500?

Our work in Activity 7.6.2 shows that that the exponential model is fairly accurate for years relatively close to 2000. However, if we go too far into the future, the model predicts increasingly large rates of change, which causes the population to grow arbitrarily large. This does not make much sense since it is unrealistic to expect that the earth would be able to support such a large population.

The constant k in the differential equation has an important interpretation. Let's rewrite the differential equation $\frac{dP}{dt} = kP$ by solving for k, so that we have

$$k = \frac{dP/dt}{P}.$$

We see that k is the ratio of the rate of change to the population; in other words, it is the contribution to the rate of change from a single person. We call this the *per capita growth rate*.

In the exponential model we introduced in Activity 7.6.2, the per capita growth rate is constant. This means that when the population is large, the per capita growth rate is the same as when the population is small. It is natural to think that the per capita growth rate should decrease when the population becomes large, since there will not be enough resources to support so many people. We expect it would be a more realistic model to assume that the per capita growth rate depends on the population P.

In the previous activity, we computed the per capita growth rate in a single year by computing k, the quotient of $\frac{dP}{dt}$ and P (which we did for $t = 0$). If we return to the data in Table 7.6.1 and compute the per capita growth rate over a range of years, we generate the data shown in Figure 7.6.2, which shows how the per capita growth rate is a function of the population, P.

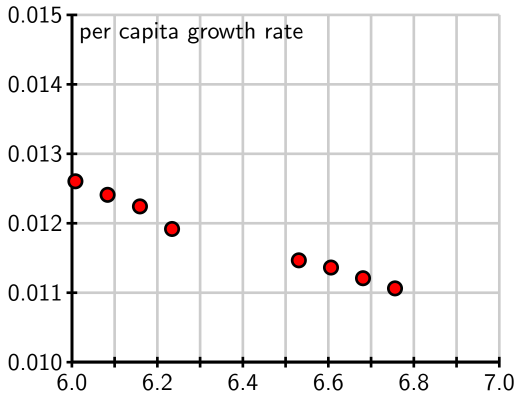


Figure 7.6.2 A plot of per capita growth rate vs. population P.

From the data, we see that the per capita growth rate appears to decrease as the population increases. In fact, the points seem to lie very close to a line, which is shown at two different scales in Figure 7.6.3.

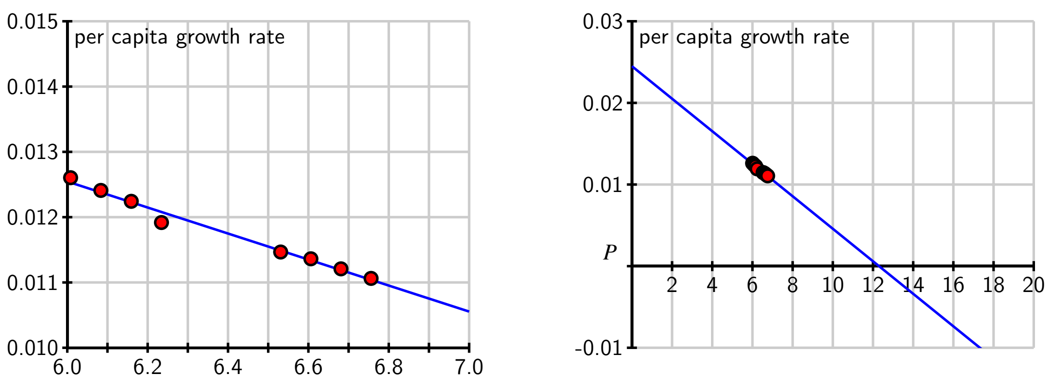


Figure 7.6.3 The line that approximates per capita growth as a function of population, P.

Looking at this line carefully, we can find its equation to be

$$\frac{dP/dt}{P} = 0.025 - 0.002P.$$

If we multiply both sides by P, we arrive at the differential equation

$$\frac{dP}{dt} = P(0.025 - 0.002P).$$

Graphing the dependence of dP/dt on the population P, we see that this differential equation demonstrates a quadratic relationship between $\frac{dP}{dt}$ and P, as shown in Figure 7.6.4.

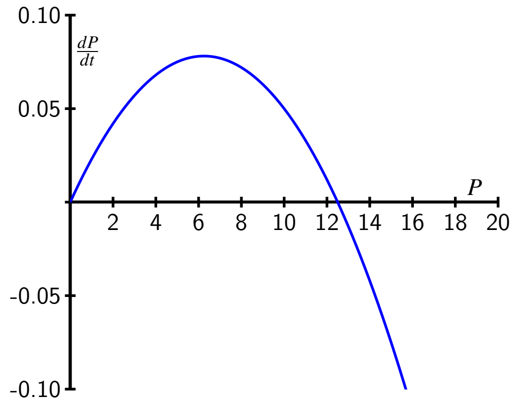


Figure 7.6.4 A plot of $\frac{dP}{dt}$ vs. P for the differential equation $\frac{dP}{dt} = P(0.025 - 0.002P)$.

The equation $\frac{dP}{dt} = P(0.025 - 0.002P)$ is an example of the *logistic equation*, and is the second model for population growth that we will consider. This model should be more realistic, because the per capita growth rate is a decreasing function of the population.

Indeed, the graph in Figure 7.6.4 shows that there are two equilibrium solutions, $P = 0$, which is unstable, and $P = 12.5$, which is a stable equilibrium. The graph shows that any solution with $P(0) > 0$ will eventually stabilize around 12.5. Thus, our model predicts the world's population will eventually stabilize around 12.5 billion.

A prediction for the long-term behavior of the population is a valuable conclusion to draw from our differential equation. We would, however, also like to answer some quantitative questions. For instance, how long will it take to reach a population of 10 billion? To answer this question, we need to find an explicit solution of the equation.

7.6.3 Solving the logistic differential equation

Since we would like to apply the logistic model in more general situations, we state the logistic equation in its more general form,

$$\frac{dP}{dt} = kP(N - P). \tag{7.6.1}$$

The equilibrium solutions here are $P = 0$ and $1 - \frac{P}{N} = 0$, which shows that $P = N$. The equilibrium at $P = N$ is called the *carrying capacity* of the population for it represents the stable population that can be sustained by the environment.

We now solve the logistic equation (7.6.1). The equation is separable, so we separate the variables

$$\frac{1}{P(N-P)}\frac{dP}{dt} = k,$$

and integrate to find that

$$\int \frac{1}{P(N-P)}\,dP = \int k\,dt.$$

To find the antiderivative on the left, we use the partial fraction decomposition

$$\frac{1}{P(N-P)} = \frac{1}{N}\left[\frac{1}{P} + \frac{1}{N-P}\right].$$

Now we are ready to integrate, with

$$\int \frac{1}{N}\left[\frac{1}{P} + \frac{1}{N-P}\right]\, dP = \int k\, dt.$$

On the left, observe that N is constant, so we can remove a factor of $\frac{1}{N}$ and antidifferentiate to find that

$$\frac{1}{N}(\ln|P| - \ln|N-P|) = kt + C.$$

Multiplying both sides of this last equation by N and using a rule of logarithms, we next find that

$$\ln\left|\frac{P}{N-P}\right| = kNt + C.$$

From the definition of the logarithm, replacing e^C with C, and letting C absorb the $\pm$ that arises from the absolute value, we now know that

$$\frac{P}{N-P} = Ce^{kNt}.$$

At this point, all that remains is to determine C and solve algebraically for P.

If the initial population is $P(0) = P_0$, then it follows that $C = \frac{P_0}{N-P_0}$, so

$$\frac{P}{N-P} = \frac{P_0}{N-P_0}e^{kNt}.$$

We will solve this equation for P by multiplying both sides by $(N-P)(N-P_0)$ to obtain

$$\begin{aligned} P(N-P_0) &= P_0(N-P)e^{kNt} \\ &= P_0Ne^{kNt} - P_0Pe^{kNt}. \end{aligned}$$

Solving for P_0Ne^{kNt}, expanding, and factoring, it follows that

$$\begin{aligned} P_0Ne^{kNt} &= P(N-P_0) + P_0Pe^{kNt} \\ &= P(N-P_0+P_0e^{kNt}). \end{aligned}$$

Dividing to solve for P, we see that

$$P = \frac{P_0Ne^{kNt}}{N-P_0+P_0e^{kNt}}.$$

Finally, we choose to multiply the numerator and denominator by $\frac{1}{P_0}e^{-kNt}$ to obtain

$$P(t) = \frac{N}{\left(\frac{N-P_0}{P_0}\right)e^{-kNt} + 1}.$$

While that was a lot of algebra, notice the result: we have found an explicit solution to the logistic equation.

Solution to the Logistic Equation.

The solution to the initial value problem

$$\frac{dP}{dt} = kP(N-P),\ P(0) = P_0,$$

is

$$P(t) = \frac{N}{\left(\frac{N-P_0}{P_0}\right)e^{-kNt} + 1}. \tag{7.6.2}$$

For the logistic equation describing the earth's population that we worked with earlier in this section, we have

$$k = 0.002, N = 12.5, \text{ and } P_0 = 6.084.$$

This gives the solution

$$P(t) = \frac{12.5}{1.0546e^{-0.025t} + 1},$$

whose graph is shown in Figure 7.6.5.

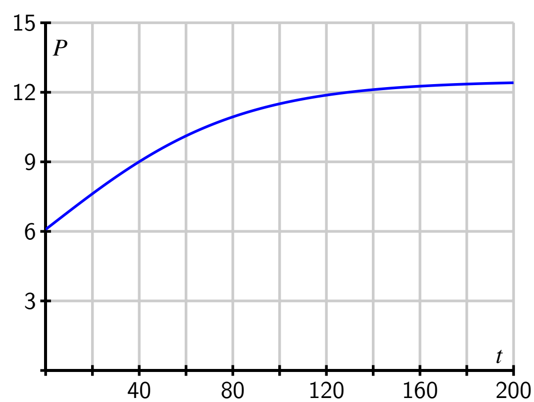


Figure 7.6.5 The solution to the logistic equation modeling the earth's population.

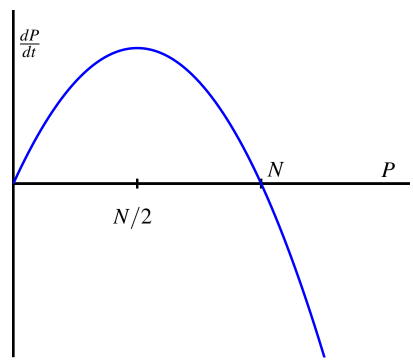
The graph shows the population leveling off at 12.5 billion, as we expected, and that the population will be around 10 billion in the year 2050. These results, which we have found using a relatively simple mathematical model, agree fairly well with predictions made using a much more sophisticated model developed by the United Nations.

The logistic equation is good for modeling any situation in which limited growth is possible. For instance, it could model the spread of a flu virus through a population contained on a cruise ship, the rate at which a rumor spreads within a small town, or the behavior of an animal population on an island. Through our work in this section, we have completely solved the logistic equation, regardless of the values of the constants N, k, and P_0. Anytime we encounter a logistic equation, we can apply the formula we found in Equation (7.6.2).

Activity 7.6.3. Consider the logistic equation

$$\frac{dP}{dt} = kP(N - P)$$

with the graph of $\frac{dP}{dt}$ vs. P shown below.



(a) At what value of P is the rate of change greatest?

(b) Consider the model for the earth's population that we recently created: $\frac{dP}{dt} = P(0.025 - 0.002P)$. At what value of P is the rate of change greatest? How does that compare to the population in recent years?

(c) According to the model we developed (recall we found that $P = \frac{12.5}{1.0546e^{-0.025t}+1}$), what will the population be in the year 2100?

(d) According to the model we developed, when will the population reach 9 billion?

(e) Now consider the general solution to the general logistic initial value problem that we found, given by

$$P(t) = \frac{N}{\left(\frac{N-P_0}{P_0}\right) e^{-kNt} + 1}.$$

Verify algebraically that $P(0) = P_0$ and that $\lim_{t\to\infty} P(t) = N$.

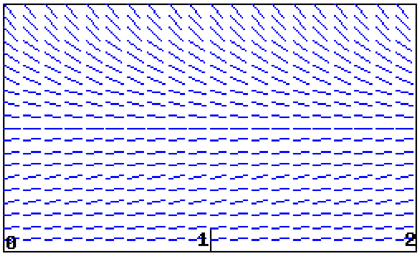
7.6.4 Summary

- If we assume that the rate of growth of a population is proportional to the population, we are led to a model in which the population grows without bound and at a rate that grows without bound.

- By assuming that the per capita growth rate decreases as the population grows, we are led to the logistic model of population growth, which predicts that the population will eventually stabilize at the carrying capacity.

7.6.5 Exercises

1. **Analyzing a logistic equation.** The slope field for a population P modeled by $dP/dt = 1P - 2P^2$ is shown in the figure below.



(a) On a print-out of the slope field, sketch three non-zero solution curves showing different types of behavior for the population P. Give an initial condition that will produce each:

$P(0) =$ ________________,

$P(0) =$ ________________, and

$P(0) =$ ________________.

(b) Is there a stable value of the population? If so, give the value; if not, enter ***none***:

Stable value = ________________

(c) Considering the shape of solutions for the population, give any intervals for which the following are true. If no such interval exists, enter ***none***, and if there are multiple intervals, give them as a list. *(Thus, if solutions are increasing when P is between 1 and 3, enter (1,3) for that answer; if they are decreasing when P is between 1 and 2 or between 3 and 4, enter (1,2),(3,4). Note that your answers may reflect the fact that P is a population.)*

P is increasing when P is in __________________________

P is decreasing when P is in __________________________

Think about what these conditions mean for the population, and be sure that you are able to explain that.

In the long-run, what is the most likely outcome for the population?

$P \to$ __________________________

(Enter ***infinity*** *if the population grows without bound.)*

Are there any inflection points in the solutions for the population? If so, give them as a comma-separated list (e.g., 1,3); if not, enter ***none***.

Inflection points are at $P =$ ________________

Be sure you can explain what the meaning of the inflection points is for the population.

(d) Sketch a graph of dP/dt against P. Use your graph to answer the following questions.

When is dP/dt positive?

When P is in ________________

When is dP/dt negative?

When P is in ________________

(Give your answers as intervals or a list of intervals.)

When is dP/dt zero?

When $P =$ ________________

(If there is more than one answer, give a list of answers, e.g., ***1,2****.)*

When is dP/dt at a maximum?

When $P =$ ________________

Be sure that you can see how the shape of your graph of dP/dt explains the shape of solution curves to the differential equation.

2. **Analyzing a logistic model.** The table below gives the percentage, P, of households with a VCR, as a function of year.

Year	1978	1979	1980	1981	1982	1983	1984
P	0.3	0.5	1.1	1.8	3.1	5.5	10.6
Year	1985	1986	1987	1988	1989	1990	1991
P	20.8	36.0	48.7	58.0	64.6	71.9	71.9

(a) A logistic model is a good one to use for these data. Explain why this might be the case: logically, how large would the growth in VCR ownership be when they are first introduced? How large can the ownership ever be?

We can also investigate this by estimating the growth rate of P for the given data. Do this at the beginning, middle, and near the end of the data:

$P'(1980) \approx$ ____________

$P'(1985) \approx$ ____________

$P'(1990) \approx$ ____________

Be sure you can explain why this suggests that a logistic model is appropriate.

(b) Use the data to estimate the year when the point of inflection of P occurs.

The inflection point occurs approximately at ________.

(Give the year in which it occurs.)

What percent of households had VCRs then? $P =$ __________

What limiting value L does this point of inflection predict (note that if the logistic model is reasonable, this prediction should agree with the data for 1990 and 1991)?

$L =$ __________

(c) The best logistic equation (solution to the logistic differential equation) for these data turns out to be the following.

$$P = \frac{75}{1 + 316.75e^{-0.699t}}.$$

What limiting value does this predict?

$L =$ __________

3. **Finding a logistic function for an infection model.** The total number of people infected with a virus often grows like a logistic curve. Suppose that 15 people originally have the virus, and that in the early stages of the virus (with time, t, measured in weeks), the number of people infected is increasing exponentially with $k = 1.7$. It is estimated that, in the long run, approximately 4000 people become infected.

(a) Use this information to find a logistic function to model this situation.

$P =$ ____________________________________

(b) Sketch a graph of your answer to part (a). Use your graph to estimate the length of time until the rate at which people are becoming infected starts to decrease. What is the vertical coordinate at this point?

vertical coordinate = ____________________________________

4. **Analyzing a population growth model.** Any population, P, for which we can ignore immigration, satisfies

$$\frac{dP}{dt} = \text{Birth rate} - \text{Death rate}.$$

For organisms which need a partner for reproduction but rely on a chance encounter for meeting a mate, the birth rate is proportional to the square of the population. Thus, the population of such a type of organism satisfies a differential equation of the form

$$\frac{dP}{dt} = aP^2 - bP \quad \text{with } a, b > 0.$$

This problem investigates the solutions to such an equation.

(a) Sketch a graph of dP/dt against P. Note when dP/dt is positive and negative.

$dP/dt < 0$ when P is in __________________________

$dP/dt > 0$ when P is in __________________________

(Your answers may involve a and b. Give your answers as an interval or list of intervals: thus, if dP/dt is less than zero for P between 1 and 3 and P greater than 4, enter ***(1,3),(4,infinity)****.)*

(b) Use this graph to sketch the shape of solution curves with various initial values: use your answers in part (a), and where dP/dt is increasing and decreasing to decide what the shape of the curves has to be. Based on your solution curves, why is $P = b/a$ called the threshold population?

If $P(0) > b/a$, what happens to P in the long run?

$P \longrightarrow$ ____________________

If $P(0) = b/a$, what happens to P in the long run?

$P \longrightarrow$ ____________________

If $P(0) < b/a$, what happens to P in the long run?

$P \longrightarrow$ ____________________

5. The logistic equation may be used to model how a rumor spreads through a group of people. Suppose that $p(t)$ is the fraction of people that have heard the rumor on day t. The equation

$$\frac{dp}{dt} = 0.2p(1-p)$$

describes how p changes. Suppose initially that one-tenth of the people have heard the rumor; that is, $p(0) = 0.1$.

a. What happens to $p(t)$ after a very long time?

b. Determine a formula for the function $p(t)$.

c. At what time is p changing most rapidly?

d. How long does it take before 80% of the people have heard the rumor?

6. Suppose that $b(t)$ measures the number of bacteria living in a colony in a Petri dish, where b is measured in thousands and t is measured in days. One day, you measure that there are 6,000 bacteria and the per capita growth rate is 3. A few days later, you measure that there are 9,000 bacteria and the per capita growth rate is 2.

a. Assume that the per capita growth rate $\frac{db/dt}{b}$ is a linear function of b. Use the measurements to find this function and write a logistic equation to describe $\frac{db}{dt}$.

b. What is the carrying capacity for the bacteria?

c. At what population is the number of bacteria increasing most rapidly?

d. If there are initially 1,000 bacteria, how long will it take to reach 80% of the carrying capacity?

7. Suppose that the population of a species of fish is controlled by the logistic equation

$$\frac{dP}{dt} = 0.1P(10-P),$$

where P is measured in thousands of fish and t is measured in years.

a. What is the carrying capacity of this population?

b. Suppose that a long time has passed and that the fish population is stable at the carrying capacity. At this time, humans begin harvesting 20% of the fish every year. Modify the differential equation by adding a term to incorporate the harvesting of fish.

c. What is the new carrying capacity?

d. What will the fish population be one year after the harvesting begins?

e. How long will it take for the population to be within 10% of the carrying capacity?

CHAPTER 8

Taylor Polynomials and Taylor Series

8.1 Extending local linearization

Motivating Questions

- How well does the tangent line at $a = 0$ approximate the function $f(x) = e^x$ near $a = 0$?
- Can we find higher degree polynomials that approximate $f(x) = e^x$ near $a = 0$ more effectively than the approximation generated by the tangent line?
- How does the degree of the polynomial impact the accuracy of the approximation of $f(x) = e^x$?

8.1.1 Introduction

Early in our study of calculus in Section 1.8, we learned that if a function f has a derivative at a fixed value $x = a$, when we zoom in on its graph near $(a, f(a))$, the function looks linear. Indeed, such a function is differentiable, and we know that near a fixed input value a,

$$f(x) \approx L(x) = f(a) + f'(a)(x - a),$$

where L is the tangent line approximation to f at a.

In this section, we use the function $f(x) = e^x$ as a case study to investigate how we can use other basic functions to better approximate the value of $f(x)$ near $a = 0$.

Preview Activity 8.1.1. Consider the function $f(x) = e^x$ near $a = 0$. We know that $f'(x) = e^x$, so $f'(0) = 1$; along with the fact that $f(0) = 1$, it follows that the tangent line approximation is

$$L(x) = f(0) + f'(0)(x - 0) = 1 + 1(x - 0) = 1 + x.$$

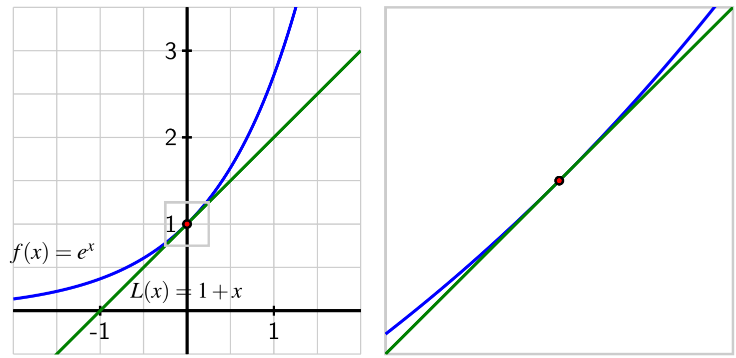


Figure 8.1.1 The function $f(x) = e^x$ and its tangent line $L(x) = 1 + x$ near the point $(0, f(0))$.

(a) Build a spreadsheet that computes the difference between $f(x)$ and $L(x)$ for x-values between -1 and 1, spaced 0.1 units apart. ***Note:*** we will revisit this spreadsheet for additional computations in Activity 8.1.4, so it would be ideal if you save your work for later reference.

Your spreadsheet should start like the one shown in Table 8.1.2.

Table 8.1.2 Comparing $f(x) = e^x$ and $L(x) = 1 + x$ near $a = 0$.

Δx	x	$f(x) = e^x$	$L(x) = 1 + x$
0.1	-1.0	0.36788	0.0
0.1	-0.9	0.40657	0.1
0.1	-0.8	0.44933	0.2

(b) Next, add a column to your spreadsheet that computes $|f(x) - L(x)|$.

How do the differences between $f(x)$ and $L(x)$ change as you move across the interval from $x = -1$ to $x = 1$?

(c) For approximately what values of x is it true that $|f(x) - L(x)| < 0.1$?

(d) Notice that the curvature in $f(x) = e^x$ is what makes the linear approximation $L(x)$ lose accuracy. What kind of simple (non-exponential) function might do a better job approximating e^x than a linear one?

8.1.2 Finding a quadratic approximation

In Preview Activity 8.1.1, we found that the error in the tangent line approximation of $f(x) = e^x$ at $a = 0$ grows significantly as we consider x-values further and further from 0. This is due to the fact that the tangent line is straight while the function $f(x) = e^x$ has some curvature. To hopefully improve the approximation, we are going to try to find a quadratic function whose curvature matches that of $f(x) = e^x$ at the point of tangency.

While we have usually used the notation "$L(x)$" for the tangent line, in what follows we will instead write "$T_1(x)$", and think of this as "the degree 1 approximation". In a similar way, we will write "$T_2(x)$" for the quadratic (degree 2) approximation.

Recall that for any function f that has a derivative at $a = 0$, its tangent line approximation at $a = 0$ is

$$T_1(x) = f(0) + f'(0)(x - 0).$$

Moreover, the functions T_1 and f have two exact values in common. First, their function values agree at the point of tangency: $T_1(0) = f(0)$. And second, since $T_1(x)$ is a linear function whose slope is $f'(0)$, it is also true that their derivative values agree at the point of tangency: $T_1'(0) = f'(0)$.

To generate a quadratic function that approximates f near $a = 0$, we choose to have this quadratic function not only share the same function value and derivative value as f at $a = 0$, but also the same second derivative value[1] at $a = 0$ in order to match the concavity or curvature of f. In other words, we are adding a term to the linear approximation that gives the same amount of curvature as the function f.

We can state these requirements more formally as follows.

The quadratic approximation of $f(x) = e^x$.

To extend the linear approximation of $f(x) = e^x$ to a quadratic approximation, we seek a function $T_2(x)$ of the form

$$T_2(x) = b_0 + b_1 x + b_2 x^2$$

that satisfies

- $T_2(0) = f(0)$, so T_2 and f share the same height at $a = 0$;
- $T_2'(0) = f'(0)$, so T_2 and f share the same slope at $a = 0$;
- $T_2''(0) = f''(0)$, so T_2 and f share the same concavity at $a = 0$.

In Activity 8.1.2, we explore how these three requirements determine b_0, b_1, and b_2 in $T_2(x)$ for the function $f(x) = e^x$.

Activity 8.1.2. Let $f(x) = e^x$ and $T_2(x) = b_0 + b_1 x + b_2 x^2$. We seek numerical values for the constants b_0, b_1, and b_2 so that $f(0) = T_2(0)$, $f'(0) = T_2'(0)$, and $f''(0) = T_2''(0)$.

(a) Note that since b_0, b_1, and b_2 are constants, if we take the derivative of the quadratic function T_2 using the sum and constant multiple rules, it follows that $T_2'(x) = b_1 + 2b_2 x$.
What is $T_2''(x)$?

(b) Recall that $f(x) = e^x$. Determine $f'(x)$ and $f''(x)$.

[1]Here we are implicitly assuming that the function $f(x)$ has a second derivative at $a = 0$, which is a property that holds for $f(x) = e^x$.

(c) Enter the formulas you've determined for $f'(x)$, $f''(x)$, and $T_2''(x)$ to fill in the blanks below.

$f(x) = e^x$	$T_2(x) = b_0 + b_1 x + b_2 x^2$
$f'(x) =$	$T_2'(x) = b_1 + 2b_2 x$
$f''(x) =$	$T_2''(x) =$

(d) Next, observe that since $T_2(x) = b_0 + b_1 x + b_2 x^2$, it follows that $T_2(0) = b_0$. Reason similarly to determine the values of $T_2'(0)$ and $T_2''(0)$, as well as those of $f(0)$, $f'(0)$, and $f''(0)$ and enter these values appropriately in the blanks below.

$f(0) =$	$T_2(0) = b_0$
$f'(0) =$	$T_2'(0) =$
$f''(0) =$	$T_2''(0) =$

(e) Now, recall that we want the function values, first derivative values, and second derivative values of f and T_2 to match at $a = 0$. What does $T_2(0) = f(0)$ tell us about the value of b_0, and what is its value? What does $T_2'(0) = f'(0)$ imply the value of b_1 is? How can we reason similarly to find b_2?

(f) Having now determined the numerical values of b_0, b_1, and b_2, use appropriate computing technology to plot the function $T_2(x) = b_0 + b_1 x + b_2 x^2$ along with $f(x) = e^x$ and $T_1(x) = 1 + x$ in the same window as that shown in Figure 8.1.3.

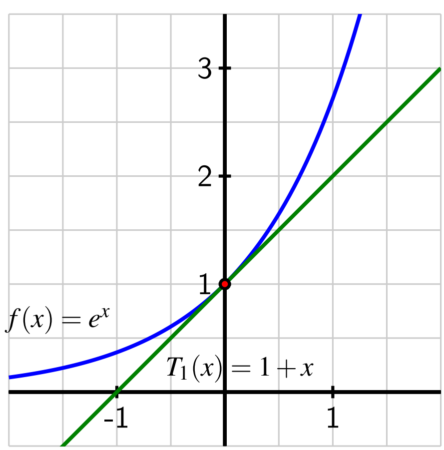


Figure 8.1.3 The function $f(x) = e^x$ and its tangent line $T_1(x) = 1 + x$ near the point $(0, f(0))$.

What do you notice? For about which values of x is $|f(x) - T_2(x)| < 0.1$?

8.1.3 Over and over again

A remarkable feature of mathematics is that when a process effectively generates an approximation, doing that same process again (perhaps with some slight modifications) often improves the approximation. In Activity 8.1.2, we found a quadratic approximation of $f(x) = e^x$ near the point $(0, f(0))$ that results in an improvement over the linear approximation of f. It is reasonable to hope that a degree 3 polynomial approximation of $f(x) = e^x$ will be even better.

To investigate, we seek a degree 3 polynomial $T_3(x)$ of the form

$$T_3(x) = c_0 + c_1 x + c_2 x^2 + c_3 x^3$$

that satisfies

- the same conditions we imposed on $T_2(x)$:

  $$T_3(0) = f(0), T_3'(0) = f'(0), \text{ and } T_3''(0) = f''(0)$$

  so that T_3 and f share the same function value, first derivative value, and second derivative value at $a = 0$,

- plus the additional condition that

  $$T_3'''(0) = f'''(0),$$

  so T_3 and f share the same third derivative value[2] at $a = 0$.

We are using the unknown constants c_0, c_1, c_2, and c_3 for $T_3(x)$ instead of the constants b_0, b_1, and b_2 that we considered for $T_2(x)$ since we don't yet know whether the first three values of c_i will be the same as those of b_i or not.

Like in our work with T_2, we observe that since T_3 is a polynomial, its derivatives are straightforward to compute. For instance,

$$T_3'(x) = c_1 + 2c_2 x + 3c_3 x^2.$$

We continue our investigation of this new approximation of $f(x) = e^x$ in Activity 8.1.3, where we work to determine the values of c_0, c_1, c_2, and c_3 plus explore how well $T_3(x)$ approximates $f(x)$ near $a = 0$.

Activity 8.1.3. Let $f(x) = e^x$ and $T_3(x) = c_0 + c_1 x + c_2 x^2 + c_3 x^3$.

(a) By computing the third derivative of $f(x)$ and the second and third derivatives of $T_3(x)$ and evaluating the relevant functions at $x = 0$, fill in the blanks below.

$$f(x) = e^x \qquad T_3(x) = c_0 + c_1 x + c_2 x^2 + c_3 x^3$$
$$f'(x) = e^x \qquad T_3'(x) = c_1 + 2c_2 x + 3c_3 x^2$$

[2] Here we are assuming that the original function f has a third derivative at $a = 0$, which is valid since $f'''(x) = e^x$.

$$
\begin{array}{ll}
f''(x) = e^x & T_3''(x) = \\
f'''(x) = & T_3'''(x) =
\end{array}
$$

$$
\begin{array}{ll}
f(0) = 1 & T_3(0) = \\
f'(0) = 1 & T_3'(0) = \\
f''(0) = 1 & T_3''(0) = \\
f'''(0) = & T_3'''(0) =
\end{array}
$$

(b) Next, recall that we want f and T_3 to share the same function and derivative values at $a = 0$ up to and including the third derivative. For instance, one of the four needed equations is $T_3'(0) = f'(0)$. Use the four equations your work in the preceding question to determine the values of c_0, c_1, c_2, and c_3.

(c) Having now determined the numerical values of c_0, c_1, c_2, and c_3, use appropriate computational technology to plot the function $T_3(x) = c_0 + c_1x + c_2x^2 + c_3x^3$ along with $f(x) = e^x$, $T_1(x) = 1 + x$, and $T_2(x) = 1 + x + \frac{1}{2}x^2$ in the same window as shown in Figure 8.1.4.

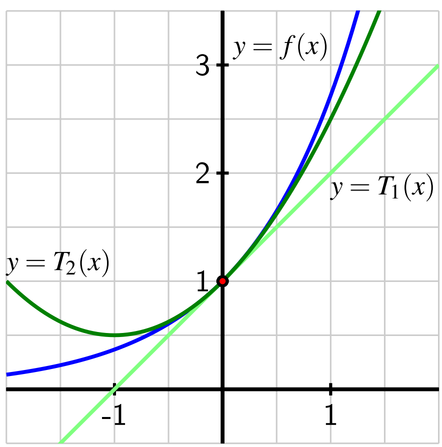


Figure 8.1.4 The function $f(x) = e^x$, its tangent line $T_1(x) = 1 + x$, and the quadratic approximation $T_2(x) = 1 + x + \frac{1}{2}x^2$ near the point $(0, f(0))$.

What do you notice? For approximately which values of x is $|f(x) - T_3(x)| < 0.1$?

(d) What if we wanted a degree-4 polynomial approximation to $f(x) = e^x$ near $a = 0$? Based on the patterns you've observed in T_1, T_2, and T_3, conjecture values for the constants $d_0, \ldots, d_4$ for a function T_4 of the form

$$T_4(x) = d_0 + d_1x + d_2x^2 + d_3x^3 + d_4x^4$$

that satisfies $T_4(0) = f(0)$, $T_4'(0) = f'(0)$, ..., $T_4^{(4)}(0) = f^{(4)}(0)$. Add this function T_4 to your plot in part (c) that includes $f(x)$ and the lower-degree polynomial approximations. What do you notice?

8.1.4 As the degree of the approximation increases

Our work so far with the case study function $f(x) = e^x$ suggests that as we find degree n polynomial approximations, T_n, that satisfy

$$T_n(0) = f(0), T_n'(0) = f'(0), T_n''(0) = f_n''(0), \ldots, T_n^{(n)}(0) = f^{(n)}(0),$$

increasing the value of n improves the accuracy of the approximation.

In the next activity, we introduce the idea of the **error** of a polynomial approximation and investigate explicitly how the error varies for approximations of $f(x) = e^x$ as we vary n and vary x.

Activity 8.1.4. We continue to work with $f(x) = e^x$ and the four approximations of degree 1, 2, 3, and 4 given by $T_1(x)$, $T_2(x)$, $T_3(x)$, and $T_4(x)$.

(a) In Preview Activity 8.1.1, we built a spreadsheet that computed the differences between $f(x)$ and $T_1(x)$ for x-values between -1 and 1, spaced 0.1 units apart. Your spreadsheet started like the one shown in the table in Preview Activity 8.1.1.

Next, we build an updated version of this spreadsheet that computes similar differences between f and the three higher degree approximations we have found. In particular, we now want to have columns for Δx, x, $f(x)$, $T_1(x)$, $T_2(x)$, $T_3(x)$, and $T_4(x)$, plus the absolute differences $|f(x) - T_1(x)|$, $|f(x) - T_2(x)|$, $|f(x) - T_3(x)|$, and $|f(x) - T_4(x)|$. ***Hint:*** when building your entries, note that you can think of $T_2(x)$ as $T_2(x) = T_1(x) + \frac{1}{2}x^2$, and similarly view $T_3(x)$ as "$T_2(x)$ plus one more term".

Include at least 5 digits of accuracy beyond the decimal. The first seven columns of your spreadsheet might start like this:

Table 8.1.5 Comparing $f(x) = e^x$ and its degree 1, 2, 3, 4 and approximations near $a = 0$.

Δx	x	$f(x)$	$T_1(x)$	$T_2(x)$	$T_3(x)$	$T_4(x)$
0.1	-1.0	0.36787	0.00000	0.50000	0.33333	0.37500
0.1	-0.9	0.40657	0.10000	0.50500	0.38350	0.41083

The next four columns of your spreadsheet should begin as follows:

Table 8.1.6 The absolute error between $f(x) = e^x$ and its degree 1, 2, 3, and 4 approximations at $x = -1$ and $x = -0.9$.

$\lvert f(x) - T_1(x)\rvert$	$\lvert f(x) - T_2(x)\rvert$	$\lvert f(x) - T_3(x)\rvert$	$\lvert f(x) - T_4(x)\rvert$
0.36787	0.13212	0.03454	0.00712
0.30657	0.09843	0.02307	0.00426

(b) We call the value of $|f(x) - T_2(x)|$ the **absolute error of the quadratic approximation of f at the value x**. What is the absolute error of the quadratic approximation at $x = -1$? at $x = 1$?

(c) What is the absolute error of the cubic (degree 3) approximation, $T_3(x)$, at $x = -1$? at $x = 1$?

(d) What is the absolute error of the quartic (degree 4) approximation, $T_4(x)$, at $x = -1$? at $x = 1$?

(e) Study your spreadsheet for trends that you notice as the value of x changes or the degree n of the approximation changes. What are your observations?

(f) Investigate the errors in the various approximations for a wider interval of x-values. For example, you might consider starting at $x = -2$ with $\Delta x = 0.2$. What do you notice?

One important application of our work so far is that these polynomial approximations provide a way to approximate values of the function $f(x) = e^x$. For example, since we've shown that

$$e^x \approx 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4,$$

it follows that

$$e^{\frac{1}{2}} \approx 1 + \frac{1}{2} + \frac{1}{2}\left(\frac{1}{2}\right)^2 + \frac{1}{6}\left(\frac{1}{2}\right)^3 + \frac{1}{24}\left(\frac{1}{2}\right)^4 = \frac{211}{128} = 1.6484375.$$

In fact, this approach through polynomial approximation is one way that computers determine the value of $e^{\frac{1}{2}}$, which is approximately 1.64872127, to whatever accuracy is needed: by using even better polynomial approximations than the degree-4 one that we found, computers are able to generate the approximate value 1.64872127 simply by the basic computations of addition and multiplication with enough terms.

Throughout this section, we have focused on $f(x) = e^x$. One of the characteristics that makes $f(x) = e^x$ special is the fact that its derivative is itself; indeed, the n^{th} derivative of f is $f^{(n)}(x) = e^x$ for every natural number n, which in turn implies that $f^{(n)}(0) = 1$ for every value of n. This will ultimately help to find patterns in the coefficients of the degree n polynomial approximation, $T_n(x)$, and be able to easily write down a formula for any value of n.

It is natural to think that we can find similar approximations of other functions, especially ones such as $\sin(x)$ and $\cos(x)$ that also exhibit repeating patterns in their derivatives. In Section 8.2, we will develop a general approach to finding the coefficient of x^n in the degree n approximation of any function with n derivatives and learn how to find a general expression for the degree n approximation.

8.1.5 Summary

- For the function $f(x) = e^x$, which bends considerably as we move away from $a = 0$ (especially for $x > 0$), the tangent line, $T_1(x)$, is not a very good approximation for x-values that satisfy $|x| > 0.5$. For example, $|e^{0.5} - T_1(0.5)| \approx 0.148721$, so the linear approximation has an absolute error of more than 0.1 at $x = 0.5$.

- Using the strategy of finding a higher degree polynomial whose function and derivative values match at the selected point of tangency, we are able to find higher degree polynomials that much more effectively approximate $f(x) = e^x$ near $a = 0$ than the approximation generated by the tangent line. For example, using the degree 3 approximation $T_3(x) = 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3$, we see that $|f(x) - T_3(x)| < 0.01$ for all x that satisfy $|x| < 0.6$.

- It appears that the degree of the polynomial impacts the accuracy of the approximation of $f(x) = e^x$ in at least two ways: if we fix an x-value, the higher the degree of the polynomial, the more accurate the approximation. In addition, raising the degree of the polynomial approximation appears to widen the interval on which the approximation is effective.

8.1.6 Exercises

1. Let $f(x) = \cos(x)$ and let $T_2(x) = b_0 + b_1x + b_2x^2$. We find the values of b_0, b_1 and b_2 that make the functions $\cos(x)$ and $T_2(x)$ have the same height, slope, and concavity at $x = 0$.

(a) Complete the table below by computing derivatives of $T_2(x)$ and $\cos(x)$ and evaluating them at $x = 0$ as directed.

$f(x) =$	$\cos(x)$	$T_2(x) =$	$b_0 + b_1x + b_2x^2$
$f'(x) =$	________	$T_2'(x) =$	________
$f''(x) =$	________	$T_2''(x) =$	________
$f(0) =$	________	$T_2(0) =$	b_0
$f'(0) =$	________	$T_2'(0) =$	________
$f''(0) =$	________	$T_2''(0) =$	________

(b) Now, since we want $T_2(x)$ and $\cos(x)$ to have the same height at $x = 0$, set $T_2(0) = f(0)$.

This implies $b_0 =$ ________.

Since we want $T_2(x)$ and $\cos(x)$ to have the same slope at $x = 0$, set $T_2'(0) = f'(0)$.

This implies $b_1 =$ ________.

Finally, since we want $T_2(x)$ and $\cos(x)$ to have the same concavity at $x = 0$, set $T_2''(0) = f''(0)$.

This implies $b_2 =$ ________.

Putting everything together, what is the resulting formula for $T_2(x) = b_0 + b_1x + b_2x^2$?

$T_2(x) =$ ____________________

2. Let $f(x) = \ln(1 + x)$ and let $T_2(x) = b_0 + b_1x + b_2x^2$. We find the values of b_0, b_1 and b_2 that make the functions $\ln(1 + x)$ and $T_2(x)$ have the same height, slope, and concavity at $x = 0$.

(a) Complete the table below by computing derivatives of $T_2(x)$ and $\ln(1+x)$ and evaluating them at $x = 0$ as directed.

$f(x) =$	$\ln(1+x)$	$T_2(x) =$	$b_0 + b_1x + b_2x^2$
$f'(x) =$	________	$T_2'(x) =$	________
$f''(x) =$	________	$T_2''(x) =$	________
$f(0) =$	________	$T_2(0) =$	b_0
$f'(0) =$	________	$T_2'(0) =$	________
$f''(0) =$	________	$T_2''(0) =$	________

(b) Now, since we want $T_2(x)$ and $\ln(1+x)$ to have the same height at $x = 0$, set $T_2(0) = f(0)$.

This implies $b_0 =$ ________.

Since we want $T_2(x)$ and $\ln(1+x)$ to have the same slope at $x = 0$, set $T_2'(0) = f'(0)$.

This implies $b_1 =$ ________.

Finally, since we want $T_2(x)$ and $\ln(1+x)$ to have the same concavity at $x = 0$, set $T_2''(0) = f''(0)$.

This implies $b_2 =$ ________.

Putting everything together, what is the resulting formula for $T_2(x) = b_0 + b_1x + b_2x^2$?

$T_2(x) =$ ________________

3. Let $f(x) = e^{-2x}$ and let $T_3(x) = c_0 + c_1x + c_2x^2 + c_3x^3$. We find the values of $c_0, \ldots, c_3$ that make the functions e^{-2x} and $T_3(x)$ have the same height, slope, concavity, and third derivative value at $x = 0$.

(a) Complete the table below by computing derivatives of $T_3(x)$ and e^{-2x} and evaluating them at $x = 0$ as directed.

$f(x) =$	e^{-2x}	$T_3(x) =$	$c_0 + c_1x + c_2x^2 + c_3x^3$
$f'(x) =$	________	$T_3'(x) =$	________
$f''(x) =$	________	$T_3''(x) =$	________
$f'''(x) =$	________	$T_3'''(x) =$	________
$f(0) =$	________	$T_3(0) =$	c_0
$f'(0) =$	________	$T_3'(0) =$	________
$f''(0) =$	________	$T_3''(0) =$	________
$f'''(0) =$	________	$T_3'''(0) =$	________

(b) Now, since we want $T_3(x)$ and e^{-2x} to have the same height at $x = 0$, set $T_3(0) = f(0)$.

This implies $c_0 =$ ________.

Since we want $T_3(x)$ and e^{-2x} to have the same slope at $x = 0$, set $T_3'(0) = f'(0)$.

This implies $c_1 =$ ________.

Since we want $T_3(x)$ and e^{-2x} to have the same concavity at $x = 0$, set $T_3''(0) = f''(0)$.

This implies $c_2 =$ __________.

Finally, since we want $T_3(x)$ and e^{-2x} to have the same third derivative value at $x = 0$, set $T_3'''(0) = f'''(0)$.

This implies $c_3 =$ __________.

Putting everything together, what is the resulting formula for $T_3(x) = c_0 + c_1x + c_2x^2 + c_3x^3$?

$T_3(x) =$ ____________________

4. Let $f(x) = 2x^2 + x + 4$ and let $T_2(x) = b_0 + b_1x + b_2x^2$. We find the values of b_0, b_1 and b_2 that make the functions $f(x)$ and $T_2(x)$ have the same height, slope, and concavity at $x = 0$.

(a) Complete the table below by computing derivatives of $T_2(x)$ and $2x^2 + x + 4$ and evaluating them at $x = 0$ as directed.

$f(x) =$	$2x^2 + x + 4$	$T_2(x) =$	$b_0 + b_1x + b_2x^2$
$f'(x) =$	__________	$T_2'(x) =$	__________
$f''(x) =$	__________	$T_2''(x) =$	__________
$f(0) =$	__________	$T_2(0) =$	b_0
$f'(0) =$	__________	$T_2'(0) =$	__________
$f''(0) =$	__________	$T_2''(0) =$	__________

(b) Now, since we want $T_2(x)$ and $2x^2 + x + 4$ to have the same height at $x = 0$, set $T_2(0) = f(0)$.

This implies $b_0 =$ __________.

Since we want $T_2(x)$ and $2x^2 + x + 4$ to have the same slope at $x = 0$, set $T_2'(0) = f'(0)$.

This implies $b_1 =$ __________.

Finally, since we want $T_2(x)$ and $2x^2 + x + 4$ to have the same concavity at $x = 0$, set $T_2''(0) = f''(0)$.

This implies $b_2 =$ __________.

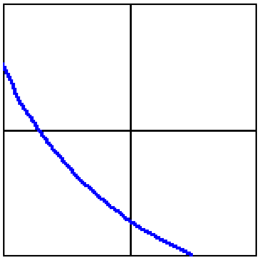
Putting everything together, what is the resulting formula for $T_2(x) = b_0 + b_1x + b_2x^2$?

$T_2(x) =$ ____________________

What does your answer mean about the quadratic approximation to a quadratic function?

5. In the following problem, note that "second degree Taylor polynomial" is the same as the quadratic approximation we've been studying, and we'll see this terminology again in the following section.

Suppose that $T_2(x) = c_0 + c_1x + c_2x^2$ is the second degree Taylor polynomial for the function f about $x = 0$. What can you say about the signs of c_0, c_1, c_2 if f has the graph given below?



(For each, enter + if the term is positive, and − if it is negative. Note that because this is essentially multiple choice problem it will not show which parts of your answer are correct or incorrect.)

c_0 is ___

c_1 is ___

c_2 is ___

6. Throughout our work in Section 8.1, we have focused on approximating the function $f(x) = e^x$. In this exercise, we change the function of interest to $f(x) = \frac{1}{3}x^3 + \frac{1}{4}x^2 - 2x - 1$, and consider the linear and quadratic approximations to f near $a = 0$.

a. Determine $f'(x)$ and $f''(x)$ and enter their formulas below.

$$f(x) = \frac{1}{3}x^3 + \frac{1}{4}x^2 - 2x - 1$$
$$f'(x) = \underline{\qquad\qquad}$$
$$f''(x) = \underline{\qquad\qquad}$$

b. Next, compute $f(0)$, $f'(0)$, and $f''(0)$ and enter those values below.

$$f(0) = \underline{\qquad}$$
$$f'(0) = \underline{\qquad}$$
$$f''(0) = \underline{\qquad}$$

c. Use your work so far to determine the formula for $T_1(x)$, the tangent line approximation to $f(x)$ at $a = 0$ (which satisfies $T_1(0) = f(0)$ and $T_1'(0) = f'(0)$).

d. Let $T_2(x)$ be the quadratic approximation to $f(x)$ near $a = 0$ that satisfies $T_2(0) = f(0)$, $T_2'(0) = f'(0)$, and $T_2''(0) = f''(0)$. You might start by letting $T_2(x) = c_0 + c_1x + c_2x^2$, and creating an updated table like the one shown below.

$f(x) = x^3/3 + x^2/4 - 2x - 1$		$T_2(x) = c_0 + c_1 x + c_2 x^2$	
$f'(x) =$		$T_2'(x) = c_1 + 2c_2 x$	
$f''(x) =$		$T_2''(x) =$	
$f(0) =$		$T_2(0) =$	
$f'(0) =$		$T_2'(0) =$	
$f''(0) =$		$T_2''(0) =$	

Use your work in in the table above to find the formula for $T_2(x)$ that satisfies $T_2(0) = f(0)$, $T_2'(0) = f'(0)$, and $T_2''(0) = f''(0)$.

e. Plot $f(x)$, $T_1(x)$, and $T_2(x)$ on the same axes, centered at $a = 0$. What do you notice?

7. In this exercise, we extend our work in Exercise 8.1.6.6. We continue to consider the function $f(x) = \frac{1}{3}x^3 + \frac{1}{4}x^2 - 2x - 1$, but now build the cubic (degree 3) approximation to f near $a = 0$.

(a) Let $T_3(x) = k_0 + k_1 x + k_2 x^2 + k_3 x^3$ and determine $T_3'(x)$, $T_3''(x)$, and $T_3'''(x)$. In addition, determine $f'''(x)$. Record your results below, along with the values of each of these functions at $a = 0$.

$f(x) = x^3/3 + x^2/4 - 2x - 1$		$T_3(x) = k_0 + k_1 x + k_2 x^2 + k_3 x^3$	
$f'(x) =$		$T_3'(x) =$	
$f''(x) =$		$T_3''(x) =$	
$f'''(x) =$		$T_3'''(x) =$	
$f(0) =$		$T_3(0) =$	
$f'(0) =$		$T_3'(0) =$	
$f''(0) =$		$T_3''(0) =$	
$f'''(0) =$		$T_3'''(0) =$	

(b) Use your work in above to find the formula for $T_3(x)$ that satisfies $T_3(0) = f(0)$, $T_3'(0) = f'(0)$, $T_3''(0) = f''(0)$, and $T_3'''(0) = f'''(0)$.

(c) Plot $f(x)$, $T_1(x)$, $T_2(x)$, and $T_3(x)$ on the same axes, centered at $a = 0$. What do you notice?

(d) What do you expect the degree 4 approximation to $f(x)$, $T_4(x)$ to be? Why?

(e) What do you expect will happen if we find the degree 6 approximation to any function $f(x)$ that is itself a degree 6 polynomial?

8.2 Taylor polynomials

Motivating Questions

- For functions such as $\sin(x)$, $\cos(x)$, and $\ln(1+x)$ that, like $f(x) = e^x$, have n derivatives at $a = 0$ for any choice of n, how can we use patterns in those derivatives to find a general formula for the degree n polynomial approximation to each?
- How are the coefficients of the polynomial approximation to a function $f(x)$ near $a = 0$ completely determined by the values of the various derivatives of f, evaluated at $a = 0$?
- How do the polynomial approximations to a given function f change when we center the approximation at a point other than $a = 0$?

8.2.1 Introduction

In Activity 8.1.3, we used $f(x) = e^x$ as a case study to investigate polynomial approximations to $f(x)$ near near $a = 0$. For the degree 3 approximation, we chose the conditions $T_3(0) = f(0)$, $T_3'(0) = f'(0)$, $T_3''(0) = f''(0)$, and $T_3'''(0) = f'''(0)$. Starting with $T_3(x) = c_0 + c_1x + c_2x^2 + c_3x^3$ and $f(x) = e^x$, we found the first three derivatives of f and T_3 and evaluated them at $a = 0$, which led to the results below.

$$\begin{aligned} f(x) &= e^x & T_3(x) &= c_0 + c_1x + c_2x^2 + c_3x^3 \\ f'(x) &= e^x & T_3'(x) &= c_1 + 2c_2x + 3c_3x^2 \\ f''(x) &= e^x & T_3''(x) &= 2c_2 + 6c_3x \\ f'''(x) &= e^x & T_3'''(x) &= 6c_3 \end{aligned}$$

$$\begin{aligned} f(0) &= 1 & T_3(0) &= c_0 \\ f'(0) &= 1 & T_3'(0) &= c_1 \\ f''(0) &= 1 & T_3''(0) &= 2c_2 \\ f'''(0) &= 1 & T_3'''(0) &= 6c_3 \end{aligned}$$

Equating the function and derivative values of f and T_3 at $a = 0$, it follows

$$c_0 = 1, c_1 = 1, 2c_2 = 1, 6c_3 = 1$$

and therefore

$$c_0 = 1, c_1 = 1, c_2 = \frac{1}{2}, c_3 = \frac{1}{6}$$

so that

$$T_3(x) = 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3.$$

Moreover, plotting f and T_3 near $a = 0$, we see in Figure 8.2.1 that the interval of accuracy for a tolerance of 0.1 is about $-1.2 \leq x \leq 1.2$.

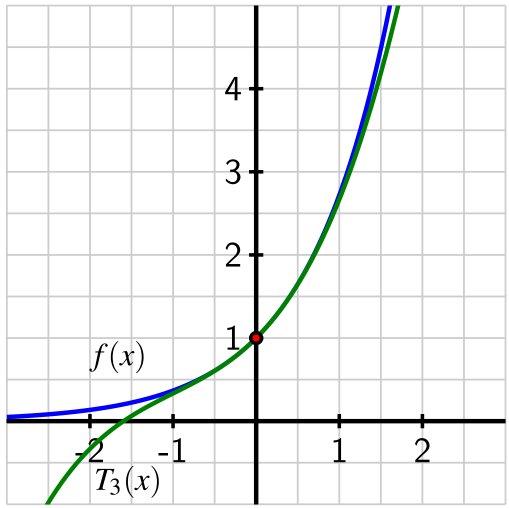


Figure 8.2.1 The function $f(x) = e^x$ and its degree 3 Taylor approximation $T_3(x) = 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3$ near the point $(0, f(0))$.

One important pattern we observed in our work with $f(x) = e^x$ is that for every natural number n, the n^{th} derivative's value at $a = 0$ is $f^{(n)}(0) = 1$. As we will see, this information will ultimately help us find a general formula for c_n, the coefficient of x^n in the degree n polynomial approximation of $f(x) = e^x$.

In this section, we will learn how we can more systematically find degree n approximations for functions that have at least n derivatives, as well as how to center the approximation at a value other than $a = 0$.

Preview Activity 8.2.1. Let $f(x) = \sin(x)$ and let $T_3(x) = c_0 + c_1x + c_2x^2 + c_3x^3$. Our goal is to find the values of $c_0, \ldots, c_3$ that make the sine function and its derivative values agree with those of the cubic polynomial T_3 at $a = 0$ and to study the resulting degree 3 approximation of the sine function.

(a) As in previous work, the derivatives of $T_3(x)$ and their respective values at $a = 0$ are those shown below.

Compute the first three derivatives of $f(x) = \sin(x)$ and evaluate them at $a = 0$ accordingly, recording your results in the blanks provided below.

$$f(x) = \sin(x) \qquad T_3(x) = c_0 + c_1x + c_2x^2 + c_3x^3$$
$$f'(x) = \_\_\_\_ \qquad T_3'(x) = c_1 + 2c_2x + 3c_3x^2$$
$$f''(x) = \_\_\_\_ \qquad T_3''(x) = 2c_2 + 6c_3x$$
$$f'''(x) = \_\_\_\_ \qquad T_3'''(x) = 6c_3$$

$$f(0) = \_\_\_\_ \qquad T_3(0) = c_0$$
$$f'(0) = \_\_\_\_ \qquad T_3'(0) = c_1$$

$$f''(0) = \boxed{} \qquad T_3''(0) = 2c_2$$

$$f'''(0) = \boxed{} \qquad T_3'''(0) = 6c_3$$

(b) Now, set $T_3(0) = f(0)$, $T_3'(0) = f'(0)$, $T_3''(0) = f''(0)$, and $T_3'''(0) = f'''(0)$. What are the resulting values of $c_0, \ldots, c_3$? What is the resulting formula for $T_3(x)$?

(c) Recall that the tangent line approximation T_1 to $f(x) = \sin(x)$ at $a = 0$ is $T_1(x) = x$. Use appropriate computing technology to plot the cubic approximation $T_3(x)$ you found in (b) along with $f(x)$ and $T_1(x)$ in the same window shown in Figure 8.2.2.

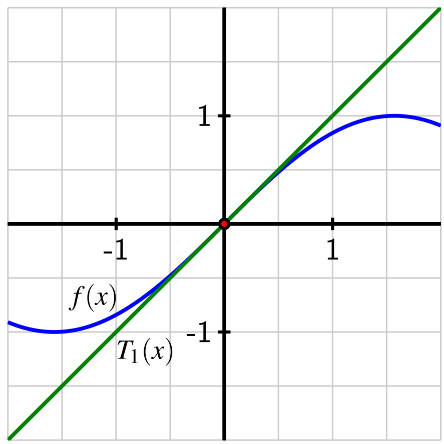


Figure 8.2.2 The function $f(x) = \sin(x)$ and its tangent line approximation $T_1(x) = x$ near the point $(0, f(0))$.

(d) What do you observe about the approximation of $f(x)$ by $T_3(x)$ compared to the approximation of $f(x)$ by $T_1(x)$? For example, how do $f(1)-T_1(1)$ and $f(1)-T_3(1)$ compare?

(e) In our work so far, we've seen that raising the degree of polynomial improves the approximation of the original function $f(x)$. If we wanted to find a degree 9 approximation to $f(x) = \sin(x)$, we'd need to find a polynomial whose value and first 9 derivatives match those of the sine function at $a = 0$.

Fill in the blanks below to determine the next six derivatives of $f(x) = \sin(x)$, up to $f^{(9)}(x)$.

$$\begin{aligned} f(x) &= \sin(x) & f^{(5)}(x) &= \boxed{} \\ f'(x) &= \cos(x) & f^{(6)}(x) &= \boxed{} \\ f''(x) &= -\sin(x) & f^{(7)}(x) &= \boxed{} \\ f'''(x) &= -\cos(x) & f^{(8)}(x) &= \boxed{} \\ f^{(4)}(x) &= \boxed{} & f^{(9)}(x) &= \boxed{} \end{aligned}$$

What pattern do you notice in the derivatives of $f(x) = \sin(x)$?

8.2.2 Taylor polynomials

In our work so far in Chapter 8, we have found several different approximations of two important functions: e^x and $\sin(x)$. In Section 8.1, we saw that near $a = 0$

- $e^x \approx 1 + x$ (the degree 1 approximation);
- $e^x \approx 1 + x + \frac{1}{2}x^2$ (the degree 2 approximation);
- $e^x \approx 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3$ (the degree 3 approximation); and
- $e^x \approx 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4$ (the degree 4 approximation).

In Preview Activity 8.2.1, we saw further that

- $\sin(x) \approx x$ (the degree 1 approximation); and
- $\sin(x) \approx x - \frac{1}{6}x^3$ (the degree 3 approximation).

We've also observed that as the degree of the approximation increases, the polynomial approximation gets more accurate by being closer to the original function $f(x)$ at each fixed value of x as well as on a wider interval. To find better and better approximations of any function with a sufficient number of derivatives, we naturally want to find approximations of arbitrary degree n. We thus define the **Taylor polynomial of degree n centered at $a = 0$.**

Definition 8.2.3 Let n be a natural number and let f be a function with at least n derivatives at $a = 0$. The **degree n Taylor polynomial of f centered at** $a = 0$ is the function

$$T_n(x) = c_0 + c_1x + c_2x^2 + \cdots + c_nx^n$$

that satisfies

$$T_n(0) = f(0), T_n'(0) = f'(0), T_n''(0) = f''(0), \ldots, T_n^{(n)}(0) = f^{(n)}(0).$$

◊

By definition, T_n is the polynomial whose function value and first n derivative values at $a = 0$ match the function value and all n derivative values of f at $a = 0$.

The $n + 1$ equations

$$T_n(0) = f(0), T_n'(0) = f'(0), T_n''(0) = f''(0), \ldots, T_n^{(n)}(0) = f^{(n)}(0)$$

enable us to determine the coefficients $c_0, c_1, \ldots, c_n$ in terms of the values of the various derivatives of f. First, we take n derivatives of $T_n(x)$, and assemble those below. As we do so, we choose not to combine products of numbers that arise in order to see certain patterns in the coefficients.

$$\begin{aligned}
T_n(x) &= c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + \cdots + c_nx^n \\
T_n'(x) &= c_1 + 2c_2x + 3c_3x^2 + 4c_4x^3 + \cdots + nc_nx^{n-1} \\
T_n''(x) &= 2c_2 + (3 \cdot 2)c_3x + (4 \cdot 3)c_4x^2 + \cdots + n(n-1)c_nx^{n-2}
\end{aligned}$$

$$\begin{aligned}
T_n'''(x) &= (3\cdot 2\cdot 1)c_3 + (4\cdot 3\cdot 2)c_4 x + \cdots + n(n-1)(n-2)c_n x^{n-3}\\
T_n^{(4)}(x) &= (4\cdot 3\cdot 2\cdot 1)c_4 + \cdots + n(n-1)(n-2)(n-3)c_n x^{n-4}\\
&\vdots\\
T_n^{(n)}(x) &= \left[n(n-1)(n-2)(n-3)\cdots 3\cdot 2\cdot 1\right]c_n
\end{aligned}$$

Next, we evaluate each of the derivatives of $T_n(x)$ at $a = 0$ and set each result equal to the corresponding derivative value of f evaluated at $a = 0$, which ultimately enables us to determine the coefficients $c_0, c_1, \ldots, c_n$. These two steps are summarized in Table 8.2.4. Note how we use the index variable, k, to track the various derivatives of T_n and f.

Table 8.2.4 Using the defining properties of a degree n Taylor polynomial to find equations involving the coefficients c_k.

k	$T_n^{(k)}(0)$	what follows from $T_n^{(k)}(0) = f^{(k)}(0)$
0	$T_n(0) = c_0$	$c_0 = f(0)$
1	$T_n'(0) = c_1$	$c_1 = f'(0)$
2	$T_n''(0) = 2c_2$	$2c_2 = f''(0)$
3	$T'''(0) = (3\cdot 2\cdot 1)c_3$	$(3\cdot 2\cdot 1)c_3 = f'''(0)$
4	$T^{(4)}(0) = (4\cdot 3\cdot 2\cdot 1)c_4$	$(4\cdot 3\cdot 2\cdot 1)c_4 = f^{(4)}(0)$
$\vdots$	$\vdots$	$\vdots$
n	$T^{(n)}(0) = \left[n(n-1)\cdots 3\cdot 2\cdot 1\right]c_n$	$\left[n(n-1)\cdots 3\cdot 2\cdot 1\right]c_n = f^{(n)}(0)$

We see a natural pattern that results from taking the k^{th} derivative of the kth power of x, x^k. For example, the repeated derivatives of x^4 are $4x^3$, $(4\cdot 3)x^2$, $(4\cdot 3\cdot 2)x$, and finally $4\cdot 3\cdot 2\cdot 1$. By the time we get to the fourth derivative of x^4, only a constant remains, and that constant is the factorial[1]

$$\frac{d^4}{dx^4}\left[x^4\right] = 4\cdot 3\cdot 2\cdot 1 = 4!$$

From the rightmost column of Table 8.2.4, we now see how the values of $c_0, c_1, \ldots, c_n$ are determined by the values of the various derivatives evaluated at $a = 0$, each scaled by a corresponding factorial. In particular, solving each equation in the rightmost column of Table 8.2.4 for c_k, we see that

$$c_0 = f(0), c_1 = f'(0), c_2 = \frac{f''(0)}{2!}, c_3 = \frac{f'''(0)}{3!}, \ldots, c_n = \frac{f^{(n)}(0)}{n!}$$

This enables us to find the degree n Taylor polynomial for any function f by finding the values of $f(0), f'(0), f''(0), \ldots, f^{(n)}(0)$ and using these numbers to determine $c_0, c_1, c_2, \ldots, c_n$. We summarize our recent work as follows.

[1]For any positive whole number n, its factorial, $n!$, is the product of all of the positive whole numbers less than or equal to n: $n! = n\cdot(n-1)\cdot(n-2)\cdot\cdots\cdot 3\cdot 2\cdot 1$.

Finding the degree n Taylor polynomial of f centered at $a = 0$.

If f is a function with at least n derivatives at $a = 0$, then the degree n Taylor polynomial of f centered at $a = 0$, $T_n(x)$, is

$$T_n(x) = c_0 + c_1 x + c_2 x^2 + \cdots + c_n x^n$$

where each coefficient c_k is given by

$$c_k = \frac{f^{(k)}(0)}{k!}.$$

The Taylor polynomial is an approximation of f near $a = 0$. In particular,

$$f(x) \approx T_n(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n.$$

Example 8.2.5 The degree 5 Taylor polynomial of $f(x) = \ln(1+x)$. Find the degree 5 Taylor polynomial centered at $a = 0$ for the function $f(x) = \ln(1 + x)$.

Solution. To find the degree 5 Taylor polynomial, we need to compute $f(0)$, $f'(0)$, $f''(0)$, ..., $f^{(5)}(0)$, so we first find the first through fifth derivatives of f in the first several rows of Table 8.2.6, and then evaluate those derivatives at $a = 0$ in the last several rows.

Table 8.2.6 The derivatives of $f(x) = \ln(1 + x)$ at $a = 0$.

$k = 0$	$f(x) = \ln(1 + x)$
$k = 1$	$f'(x) = \frac{1}{1+x} = (1 + x)^{-1}$
$k = 2$	$f''(x) = (-1)(1 + x)^{-2}$
$k = 3$	$f'''(x) = (-2)(-1)(1 + x)^{-3}$
$k = 4$	$f^{(4)}(x) = (-3)(-2)(-1)(1 + x)^{-4}$
$k = 5$	$f^{(5)}(x) = (-4)(-3)(-2)(-1)(1 + x)^{-5}$
—	—
$k = 0$	$f(0) = \ln(1) = 0$
$k = 1$	$f'(0) = (1)^{-1} = 1$
$k = 2$	$f''(0) = (-1)(1)^{-2} = -1$
$k = 3$	$f'''(0) = (-2)(-1)(1)^{-3} = (-2)(-1)$
$k = 4$	$f^{(4)}(0) = (-3)(-2)(-1)(1)^{-4} = (-3)(-2)(-1)$
$k = 5$	$f^{(5)}(0) = (-4)(-3)(-2)(-1)(1)^{-5} = (-4)(-3)(-2)(-1)$

When finding the coefficients of a Taylor polynomial, it is often helpful to *not* combine products such as $(-3)(-2)(-1)$ and $(-4)(-3)(-2)(-1)$ into a single number, in order to better observe patterns; indeed, by not combining the constants that arise in higher derivatives of $f(x) = \ln(1 + x)$, we see patterns of alternating signs and factorials that arise. From the last six rows of Table 8.2.6 and the fact that $c_k = \frac{f^{(k)}(0)}{k!}$, we find that

$$\begin{aligned} c_0 &= f(0) = 0 \\ c_1 &= f'(0) = 1 \\ c_2 &= \frac{1}{2!}f''(0) = \frac{-1}{2!} = -\frac{1}{2} \end{aligned}$$

$$c_3 = \frac{1}{3!}f'''(0) = \frac{(-2)(-1)}{3!} = \frac{1}{3}$$
$$c_4 = \frac{1}{4!}f^{(4)}(0) = \frac{(-3)(-2)(-1)}{4!} = -\frac{1}{4}$$
$$c_5 = \frac{1}{5!}f^{(5)}(0) = \frac{(-4)(-3)(-2)(-1)}{5!} = \frac{1}{5}$$

so that the degree 5 Taylor approximation of $f(x) = \ln(1 + x)$ at $a = 0$ is

$$T_5(x) = 1x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \frac{1}{5}x^5.$$

Thus, we have found the approximation

$$\ln(1 + x) \approx 1x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \frac{1}{5}x^5,$$

and by plotting $T_5(x)$ along with $f(x)$ and $T_1(x)$ in Figure 8.2.7, we see how much better the degree 5 approximation is than the tangent line approximation.

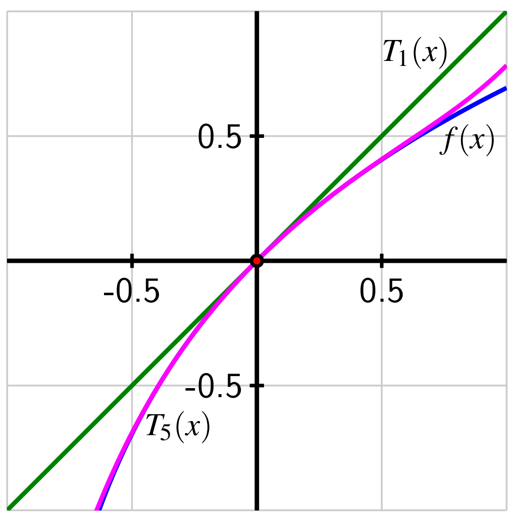


Figure 8.2.7 The function $f(x) = \ln(1 + x)$ and its degree 5 Taylor approximation $T_5(x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \frac{1}{5}x^5$ near the point $(0, f(0))$, along with $T_1(x) = x$.

◊

From our work in Example 8.2.5, we see the pattern that arises in the various derivatives of $f(x) = \ln(1 + x)$. We therefore find that the general degree n Taylor polynomial centered at $a = 0$ for $f(x) = \ln(1 + x)$ is

$$T_n(x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \cdots + (-1)^{n+1}\frac{1}{n}x^n.$$

For many familiar functions, a pattern emerges in their derivatives that enables us to find the general form of the degree n Taylor polynomial.

Activity 8.2.2. Let $f(x) = \cos(x)$. Through the questions that follow, we seek to find the degree n Taylor polynomial for $f(x)$ centered at $a = 0$.

(a) Determine the first 8 derivatives of $f(x) = \cos(x)$ and evaluate each at $a = 0$. Summarize your work by filling in all the blanks below.

$k = 0$	$f(x) = \cos(x)$	$f(0) = \cos(0) = 1$
$k =$	$f'(x) =$	$f'(0) =$
$k =$	$f''(x) =$	$f''(0) =$
$k =$	$f'''(x) =$	$f'''(0) =$
$k =$	$f^{(4)}(x) =$	$f^{(4)}(0) =$
$k =$	$f^{(5)}(x) =$	$f^{(5)}(0) =$
$k =$	$f^{(6)}(x) =$	$f^{(6)}(0) =$
$k =$	$f^{(7)}(x) =$	$f^{(7)}(0) =$
$k =$	$f^{(8)}(x) =$	$f^{(8)}(0) =$

(b) Use your work in (a) along with the fact that the coefficients of the Taylor polynomial are determined by $c_k = \frac{f^{(k)}(0)}{k!}$ to find formulas for $T_2(x)$, $T_4(x)$, $T_6(x)$, and $T_8(x)$.

(c) Based on the patterns you observe in your prior work, what do you expect to be the formula for $T_{10}(x)$?

(d) Use appropriate computing technology to plot $T_4(x)$ and $T_6(x)$ along with $f(x) = \cos(x)$ and $T_2(x)$ in the same window as shown in Figure 8.2.8.

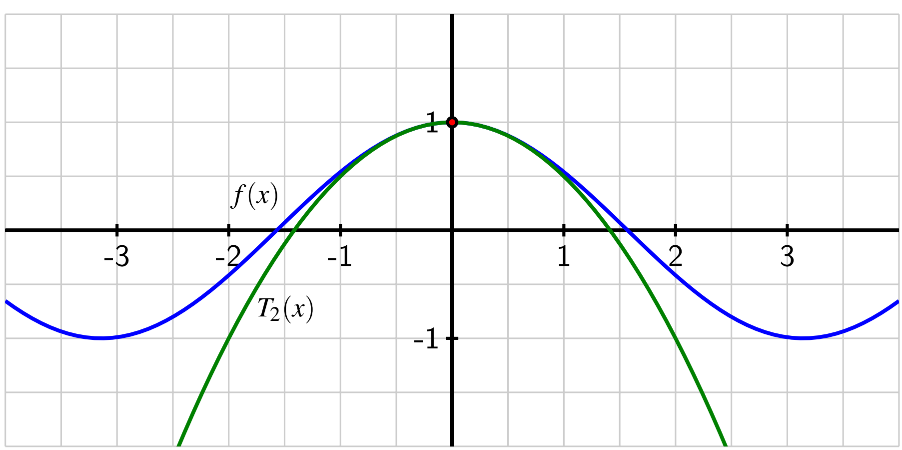


Figure 8.2.8 The function $f(x) = \cos(x)$ and its degree 2 Taylor approximation $T_2(x) = 1 - \frac{1}{2}x^2$ near the point $(0, f(0))$.

What do you notice?

(e) Build a spreadsheet similar to the one in Table 8.1.5 and Table 8.1.6 from Activity 8.1.4, but do so using $\Delta x = 0.2$, a start value of $x = -2$, and the functions $f(x) = \cos(x)$, $T_2(x)$, $T_4(x)$, and $T_6(x)$. The first six columns of your spreadsheet should begin as shown in Table 8.2.9.

Table 8.2.9 Comparing $f(x) = \cos(x)$ and its degree 2, 4, and 6 approximations near $a = 0$.

Δx	x	$f(x)$	$T_2(x)$	$T_4(x)$	$T_6(x)$
0.2	−2.0	−0.41615	−1.00000	−0.33333	−0.42222
0.2	−1.8	−0.22720	−0.62000	−0.18260	−0.22984

and the last three columns of your spreadsheet should begin as follows:

Table 8.2.10 The absolute error between $f(x) = \cos(x)$ and its degree 2, 4, and 6 approximations.

$\lvert f(x) - T_2(x)\rvert$	$\lvert f(x) - T_4(x)\rvert$	$\lvert f(x) - T_6(x)\rvert$
0.58385	0.08281	0.00608
0.39280	0.04460	0.00263

For about what interval of x-values is it true that $|f(x) - T_2(x)| < 0.1$? How does the interval of x-values change if we instead consider where $|f(x) - T_4(x)| < 0.1$? $|f(x) - T_6(x)| < 0.1$?

Our work so far with the functions e^x, $\sin(x)$, $\cos(x)$, and $\ln(1 + x)$ has revealed patterns in their derivatives that enable us to find even higher degree Taylor polynomials easily. Furthermore, these higher degree polynomials provide outstanding approximations that only require the use of addition and multiplication.

For example, in Activity 8.2.2, we found that the degree 8 Taylor approximation of the cosine function at $a = 0$ is

$$T_8(x) = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \frac{1}{8!}x^8,$$

so

$$\cos(x) \approx 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \frac{1}{8!}x^8.$$

The pattern of alternating signs and even numbers in the factorials and powers of x lets us see that we could easily write down, say, $T_{20}(x)$. We can reason similarly to extend what we found in Preview Activity 8.2.1 and observe that

$$\sin(x) \approx T_7(x) = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7,$$

the degree 7 Taylor approximation of the sine function at $a = 0$.

When we ask a computational device to find numerical estimates for quantities such as e^2, $\sin(1)$, and $\cos(1.2)$, high degree Taylor polynomials are one approach[2] that can be used to

[2]For computing values of trigonometric functions, a method known as CORDIC (that remarkably doesn't even use multiplication) is employed by many calculating devices.

generate the results. For example,

$$\sin(1) \approx T_7(1) = 1 - \frac{1}{3!}1^3 + \frac{1}{5!}1^5 - \frac{1}{7!}1^7 = \frac{4241}{5040} \approx 0.84146825$$

is a surprisingly accurate estimate of $\sin(1) = 0.84147098\ldots$ that only involves the sum of four rational numbers.

8.2.3 Taylor polynomial approximations centered at an arbitrary value a

In all of our work so far in Chapter 8, we have focused on approximating functions such as e^x, $\sin(x)$, $\ln(1+x)$, and $\cos(x)$ near $a = 0$. But we could instead be interested in the behavior of some function f near $a = 5$, or be interested in a function f that wasn't even defined at $a = 0$. Thus, we next generalize our earlier work to Taylor polynomial approximations centered at any value a.

From our early studies in Section 1.8, we know that at any input value $x = a$ where a function f has a first derivative, f has a tangent line approximation

$$L(x) = f(a) + f'(a)(x-a)$$

that satisfies $f(x) \approx L(x)$ for x values near a. Provided that f has a second derivative at $x = a$, we can build a quadratic approximation near a for f, similar to the one we found at $a = 0$ for $f(x) = e^x$ in Activity 8.1.3. In addition, as long as f has a third derivative at $x = a$, we can even find a cubic approximation (just as we did at $a = 0$ in Activity 8.1.4), and so on.

In developing such approximations centered at any value $x = a$, our guiding principle is the same as with our work at $a = 0$: we'll require that at the input value a, the original function's output and its derivatives' outputs match the corresponding approximation's output and derivatives' output.

Building on the form of the tangent line approximation, which we now denote $T_1(x)$,

$$T_1(x) = f(a) + f'(a)(x-a),$$

it is natural for us to consider quadratic and cubic approximations of form

$$T_2(x) = c_0 + c_1(x-a) + c_2(x-a)^2, \text{ and,}$$

a cubic approximation of form

$$T_3(x) = c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3,$$

and so on. We define the more general degree n Taylor Polynomial centered at a as follows.

Definition 8.2.11 Let n be a natural number and let f be a function with at least n derivatives at a. The **degree n Taylor polynomial of f centered at a** is the function

$$T_n(x) = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots + c_n(x-a)^n$$

that satisfies $T_n(a) = f(a), T_n'(a) = f'(a), T_n''(a) = f''(a), \ldots, T_n^{(n)}(a) = f^{(n)}(a)$. ◊

Similar to the situation when $a = 0$, it follows that we can find the coefficients c_k of the Taylor polynomial in terms of the various derivatives of f evaluated at a.

Finding the degree n Taylor polynomial of f centered at a.

If f is a function with at least n derivatives at a, then the degree n Taylor polynomial of f centered at a, $T_n(x)$, is

$$T_n(x) = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots + c_n(x-a)^n$$

where each coefficient c_k is given by

$$c_k = \frac{f^{(k)}(a)}{k!}.$$

As with approximations centered at $a = 0$, the Taylor polynomial now provides us with an approximation of f near a. In particular,

$$f(x) \approx T_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

Activity 8.2.3. Let $f(x) = \ln(x)$, and recall that f is only defined for $x > 0$. As such, we can't consider the tangent line (or any other) approximation at $a = 0$. Instead, we choose to work with an approximation to $f(x) = \ln(x)$ centered at $a = 1$ and will find the degree 4 Taylor polynomial approximation

$$T_4(x) = c_0 + c_1(x-1) + c_2(x-1)^2 + c_3(x-1)^3 + c_4(x-1)^4.$$

(a) Determine $f'(x)$, $f''(x)$, $f'''(x)$, and $f^{(4)}(x)$, and then compute $f'(1)$, $f''(1)$, $f'''(1)$, and $f^{(4)}(1)$. Enter your results in the provided blanks below.

$k = 0$	$f(x) = \ln(x)$	$f(1) = \ln(1) = 0$
$k =$	$f'(x) =$	$f'(1) =$
$k =$	$f''(x) =$	$f''(1) =$
$k =$	$f'''(x) =$	$f'''(1) =$
$k =$	$f^{(4)}(x) =$	$f^{(4)}(1) =$

(b) Use your work in (a) along with the fact that the coefficients of the Taylor polynomial are determined by $c_k = \frac{f^{(k)}(a)}{k!}$ to determine

$$T_4(x) = c_0 + c_1(x-1) + c_2(x-1)^2 + c_3(x-1)^3 + c_4(x-1)^4.$$

(c) Use appropriate technology to plot $f(x) = \ln(x)$, its tangent line, $T_1(x) = x - 1$, and $T_4(x)$ in the same window shown in Figure 8.2.12.

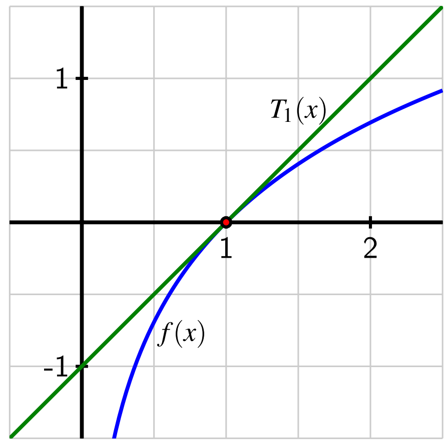


Figure 8.2.12 The function $f(x) = \ln(x)$ and its degree 1 Taylor approximation $T_1(x) = x - 1$ near the point $(1, f(1))$.

What do you notice?

(d) Compute $|f(x) - T_4(x)|$ for several different x values (you might find it helpful to use a slider in *Desmos* in the variable b to experiment with $|f(b) - T_4(b)|$); for approximately what values of x is it true that $|f(x) - T_4(x)| < 0.1$?

(e) Use the patterns you observe in your work in parts (a) and (b) to conjecture formulas for $T_5(x)$ and $T_6(x)$.

For approximately what interval of x-values is it true that $|f(x) - T_5(x)| < 0.1$? What about $|f(x) - T_6(x)| < 0.1$? How is this situation different from what we observed with $f(x) = \cos(x)$ in Activity 8.2.2?

Activity 8.2.4. This activity builds on Activity 8.2.3, and only changes one key thing: the location where the approximation is centered. Again, we let $f(x) = \ln(x)$, and recall that f is only defined for $x > 0$. Here, we choose to work with an approximation centered at $a = 2$, and find the degree 4 Taylor polynomial approximation

$$T_4(x) = c_0 + c_1(x-2) + c_2(x-2)^2 + c_3(x-2)^3 + c_4(x-2)^4.$$

(a) We recall $f'(x)$, $f''(x)$, $f'''(x)$, and $f^{(4)}(x)$ from our work in Activity 8.2.3, and then compute $f'(2)$, $f''(2)$, $f'''(2)$, and $f^{(4)}(2)$. Enter the updated results in the blanks below.

$k = 0$	$f(x) = \ln(x)$	$f(2) =$
$k = 1$	$f'(x) = x^{-1}$	$f'(2) =$
$k = 2$	$f''(x) = (-1)x^{-2}$	$f''(2) =$
$k = 3$	$f'''(x) = (-2)(-1)x^{-3}$	$f'''(2) =$
$k = 4$	$f^{(4)}(x) = (-3)(-2)(-1)x^{-4}$	$f^{(4)}(2) =$

(b) Use your work in (a) along with the fact that the coefficients of the Taylor polynomial are determined by $c_k = \frac{f^{(k)}(a)}{k!}$ to determine $T_4(x) = c_0 + c_1(x-2) + c_2(x-2)^2 + c_3(x-2)^3 + c_4(x-2)^4$.

(c) Use appropriate technology to plot $f(x) = \ln(x)$, its tangent line, $T_1(x) = \ln(2) + \frac{1}{2}(x-2)$, and $T_4(x)$ on the same axes in the window shown Figure 8.2.13.

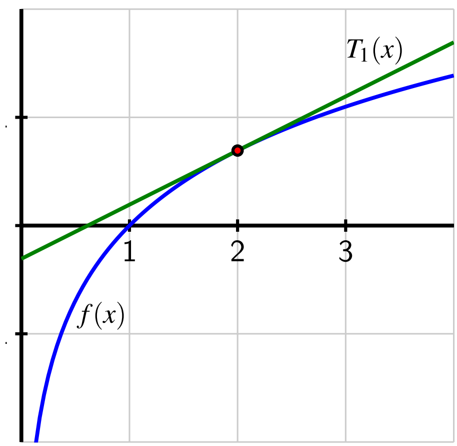


Figure 8.2.13 The function $f(x) = \ln(x)$ and its degree 1 Taylor approximation $T_1(x) = \ln(2) + \frac{1}{2}(x-2)$ near the point $(2, f(2))$.

What do you notice?

(d) Compute $|f(x) - T_4(x)|$ for several different x values (you might find it helpful to use a slider in *Desmos*); for approximately what values of x is it true that $|f(x) - T_4(x)| < 0.1$?

(e) Use the patterns you observe in parts (a) and (b) to conjecture formulas for $T_5(x)$ and $T_6(x)$.

For about what interval of x-values is it true that $|f(x) - T_5(x)| < 0.1$? What about $|f(x) - T_6(x)| < 0.1$? How is this different from what we observed with the Taylor approximations centered at $a = 1$ in Activity 8.2.3? How is it similar?

8.2.4 Summary

- Provided that a function $f(x)$ has n derivatives at a selected input value $x = a$, we can find a degree n polynomial $T_n(x)$ that approximates $f(x)$ near a by requiring that $T_n(a) = f(a), T_n'(a) = f'(a), T_n''(a) = f''(a), \ldots, T_n^{(n)}(a) = f^{(n)}(a)$.

- When $a = 0$, the degree n polynomial approximation, $T_n(x)$, to a function $f(x)$, centered at $a = 0$, is a polynomial of the form

$$T_n(x) = c_0 + c_1 x + c_2 x^2 + \cdots + c_n x^n$$

and it follows that the coefficients c_k are determined by the values of the various derivatives of $f(x)$ evaluated at 0 according to the formula

$$c_k = \frac{f^{(k)}(0)}{k!}.$$

Therefore, for x near $a = 0$,

$$f(x) \approx T_n(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n.$$

- Just as we can consider any function f that has n derivatives at $a = 0$ and find approximations centered there, we can also consider any input value a at which those n derivatives exist, and find a polynomial approximation that satisfies $T_n(a) = f(a)$, $T_n'(a) = f'(a)$, $T_n''(a) = f''(a)$, ..., $T_n^{(n)}(a) = f^{(n)}(a)$.

  At such a value a, the degree n Taylor polynomial of f centered at a has form

  $$T_n(x) = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots + c_n(x-a)^n$$

  and it follows that the coefficients c_k are determined by the values of the various derivatives of $f(x)$ evaluated at a according to the formula

  $$c_k = \frac{f^{(k)}(a)}{k!}.$$

  Thus, for x near a,

  $$f(x) \approx T_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

8.2.5 Exercises

1. Find the Taylor polynomials of degree n approximating $\cos(x)$ for x near 0:

For $n = 2$, $T_2(x) =$ ____________________

For $n = 4$, $T_4(x) =$ ____________________

For $n = 6$, $T_6(x) =$ ____________________

2. Calculate the Taylor polynomials $T_2(x)$ and $T_3(x)$ centered at $x = \frac{\pi}{2}$ for $f(x) = \sin(x)$.

3. Calculate the Taylor polynomials $T_2(x)$ and $T_3(x)$ centered at $x = 3$ for $f(x) = \ln(x+1)$.

$T_2(x) =$ ____________________

$T_3(x) = T_2(x) +$ ____________________

4. Compute $T_2(x)$ at $x = 0.4$ for $y = e^x$ and use a calculator to compute the error $|e^x - T_2(x)|$ at $x = 1.1$.

$T_2(x) =$ ____________________

$|e^x - T_2(x)| =$ ____________________

5. Find the second-degree Taylor polynomial for $f(x) = 4x^2 - 6x + 5$ about $x = 0$.

$T_2(x) =$ ______________________________

What do you notice about your polynomial?

6. Suppose g is a function which has continuous derivatives, and that $g(6) = -4, g'(6) = -5, g''(6) = 5, g'''(6) = 2$.

(a) What is the Taylor polynomial of degree 2 for g near 6?

$T_2(x) =$ ______________________________

(b) What is the Taylor polynomial of degree 3 for g near 6?

$T_3(x) =$ ______________________________

(c) Use the two polynomials that you found in parts (a) and (b) to approximate $g(6.1)$.

With T_2, $g(6.1) \approx$ __________

With T_3, $g(6.1) \approx$ __________

7. Suppose we know the following information about a function f:

$$f(0) = 2, f'(0) = -3, f''(0) = -1, f'''(0) = 0, f^{(4)}(0) = -3.$$

a. Determine $T_4(x)$, the degree 4 Taylor polynomial of f that is centered at $a = 0$.

b. Use $T_4(x)$ to estimate $f(0.5)$.

c. State each of the Taylor polynomials $T_3(x)$, $T_2(x)$, and $T_1(x)$.

8. In Exercise 8.2.5.5, we found that the degree 2 Taylor polynomial centered at $a = 0$ of a quadratic function *is* the quadratic function itself. In this exercise, we explore how changing the center of the approximation offers additional insight into the function.

Let $f(x) = \frac{1}{2}x^2 - 2x + 5$, and let $a = 2$ be the center at which we will find a degree 2 Taylor polynomial approximation of f.

a. By finding $f'(x)$, $f''(x)$, $f(2)$, $f'(2)$, and $f''(2)$, determine $T_2(x)$, the degree 2 Taylor polynomial approximation of f that is centered at $a = 2$.

b. Plot both $f(x)$ and $T_2(x)$ on the same axes. What do you observe?

c. What does the algebraic form of $T_2(x)$ tell you about the original function $f(x)$?

d. What is the tangent line approximation to $f(x)$ at $a = 2$? What is special about the function's behavior at this input value?

9. Recall that we found in Preview 8.2.1 and subsequent work that $\sin(x) \approx x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7$, which is the degree 7 Taylor approximation centered at 0. And in Activity 8.2.2, we found that the degree 6 Taylor approximation centered at $a = 0$ for $\cos(x)$ is $\cos(x) \approx 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6$.

In this exercise, we investigate Taylor polynomial approximations of $f(x) = \sin(x)$ centered at $a = \frac{\pi}{2}$.

a. By finding the appropriate derivatives of $f(x) = \sin(x)$ and evaluating them at $a = \frac{\pi}{2}$, determine the degree 6 Taylor polynomial approximation of $\sin(x)$ centered at $a = \frac{\pi}{2}$.

b. How is your result in (a) similar to the degree 6 Taylor polynomial of $\cos(x)$ that is centered at $a = 0$?

c. Recall the trigonometric identity that states $\sin(x) = \cos(x - \frac{\pi}{2})$. How does this identity help explain what you found in (a) and (b)?

10. In Example 8.2.5, we found that

$$f(x) = \ln(1+x) \approx T_5(x) = 1x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \frac{1}{5}x^5,$$

where T_5 is the degree 5 Taylor approximation of $\ln(1+x)$ centered at $a = 0$.

In Activity 8.2.3, we found that

$$g(x) = \ln(x) \approx P_5(x) = 1(x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 - \frac{1}{4}(x-1)^4 + \frac{1}{5}(x-1)^5,$$

where P_5 is the degree 5 Taylor approximation of $\ln(x)$ centered at $a = 1$. (Here we are using "T_5" and "P_5" to distinguish between these two degree 5 polynomial approximations of the two different functions $f(x) = \ln(1+x)$ and $g(x) = \ln(x)$, centered at two different values.)

a. Note that $f(0.5) = \ln(1.5)$. Use $T_5(x)$ appropriately to find an estimate of $\ln(1.5)$.

b. Observe that $g(1.5) = \ln(1.5)$. Use $P_5(x)$ to estimate $\ln(1.5)$.

c. Are the estimates of $\ln(1.5)$ generated by $T_5(x)$ and $P_5(x)$ the same or different? Why do you think this happens?

8.3 Geometric sums

Motivating Questions

- What is a finite geometric sum and how can we quickly find its value, no matter how many terms are in the sum?
- How can a finite geometric sum be extended to an infinite geometric series? In what circumstances can we quickly find the value of an infinite geometric series?
- How are finite and infinite geometric series connected to Taylor polynomials?

8.3.1 Introduction

In our work in Section 8.2, we learned how to find a degree n polynomial approximation centered at a value a for a given function f with at least n derivatives. By working with several different functions and n-values, we've seen that increasing the degree of the polynomial improves the approximation, and also often helps us to see a pattern in the coefficients of the Taylor polynomials.

For example, the degree 5 Taylor approximation of $f(x) = \ln(1+x)$ at $a = 0$ is

$$T_5(x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \frac{1}{5}x^5,$$

and we see a pattern in the coefficients that allows us to easily generate $T_6(x)$, $T_{10}(x)$, or indeed $T_n(x)$ for any n. Note that if we want to use $T_{10}(x)$ to estimate $\ln(\frac{3}{2}) = \ln(1+\frac{1}{2})$, we need to compute the sum of 10 terms given by

$$\ln\left(\frac{3}{2}\right) \approx T_{10}\left(\frac{1}{2}\right) = \left(\frac{1}{2}\right) - \frac{1}{2}\left(\frac{1}{2}\right)^2 + \frac{1}{3}\left(\frac{1}{2}\right)^3 - \cdots - \frac{1}{10}\left(\frac{1}{2}\right)^{10}.$$

This computation suggests at least two questions: is there an easy way (without a computer) to determine the exact value of the 10-term sum

$$\left(\frac{1}{2}\right) - \frac{1}{2}\left(\frac{1}{2}\right)^2 + \frac{1}{3}\left(\frac{1}{2}\right)^3 - \frac{1}{4}\left(\frac{1}{2}\right)^4 + \cdots - \frac{1}{10}\left(\frac{1}{2}\right)^{10},$$

and is there a way we can make sense of continuing the sum indefinitely,

$$\left(\frac{1}{2}\right) - \frac{1}{2}\left(\frac{1}{2}\right)^2 + \frac{1}{3}\left(\frac{1}{2}\right)^3 - \frac{1}{4}\left(\frac{1}{2}\right)^4 + \cdots - \frac{1}{10}\left(\frac{1}{2}\right)^{10} + \cdots?$$

In this section, we investigate a special collection of similar sums that are called **geometric**.

Preview Activity 8.3.1. Let's explore sums of the form

$$S_n = 1 + \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \cdots + \left(\frac{1}{2}\right)^{n-1}.$$

Observe that we can think of the first term, 1, as $(\frac{1}{2})^0$, and since the powers of $\frac{1}{2}$ include every whole number from 0 to $n-1$, there are exactly n terms in the sum given by S_n.

(a) Note that $S_1 = 1$, $S_2 = S_1 + \frac{1}{2} = 1 + \frac{1}{2} = \frac{3}{2}$, and $S_3 = S_2 + \frac{1}{4} = \frac{3}{2} + \frac{1}{4} = \frac{7}{4}$. Using the fact that each subsequent value of S_n can be computed by adding one additional term to the preceding sum, complete Table 8.3.1 with the exact (fractional) value of each sum.

Table 8.3.1 Values of S_n for increasing values of n

$n = 1$	$S_1 = 1$
$n = 2$	$S_2 = \frac{3}{2}$
$n = 3$	$S_3 = \frac{7}{4}$
$n = 4$	$S_4 =$
$n = 5$	$S_5 =$
$n = 6$	$S_6 =$
$n = 7$	$S_7 =$

(b) What do you expect will be the exact fractional value of S_8? S_9?

(c) Conjecture a simple formula for S_n that is given by a single fraction where the numerator and/or denominator depend on n.

(d) What happens to the value of S_n as $n \to \infty$? Why?

(e) How do the results change if we instead consider

$$S_n = 1 + \frac{1}{4} + \left(\frac{1}{4}\right)^2 + \left(\frac{1}{4}\right)^3 + \cdots + \left(\frac{1}{4}\right)^{n-1}?$$

8.3.2 Finite Geometric Series

Sums such as

$$1 + \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \cdots + \left(\frac{1}{2}\right)^{n-1},$$

$$1 + \frac{1}{4} + \left(\frac{1}{4}\right)^2 + \left(\frac{1}{4}\right)^3 + \cdots + \left(\frac{1}{4}\right)^{n-1},$$

and

$$2 - \frac{4}{3} + \frac{8}{9} - \frac{16}{27} + \cdots + 2\cdot\left(\frac{-2}{3}\right)^{n-1}$$

all share a similar structure: each has exactly n terms, and the next term in each sum is found by multiplying the last term by the same number. In the first sum, each subsequent term is found by multiplying by $\frac{1}{2}$; in the second sum, by multiplying by $\frac{1}{4}$; in the third sum, the multiplier is $\frac{-2}{3}$. These sums each have the form

$$a + ar + ar^2 + \cdots + ar^{n-1}, \tag{8.3.1}$$

which we call a **finite geometric series** with ratio r. It turns out that the value of each sum that has this form can be computed quickly without having to add all of the individual terms.

Activity 8.3.2. Let $a = 1$ and $r = \frac{2}{5}$ be real numbers and consider the corresponding finite geometric series

$$S_n = 1 + \frac{2}{5} + \left(\frac{2}{5}\right)^2 + \cdots + \left(\frac{2}{5}\right)^{n-2} + \left(\frac{2}{5}\right)^{n-1}. \tag{8.3.2}$$

Our goal in this activity is to find a shortcut formula for S_n that can be written as a single fraction that does not involve a sum of n terms.

(a) Multiply both sides of Equation (8.3.2) by $r = \frac{2}{5}$. Write the new equation in the form

$$\frac{2}{5} \cdot S_n = \underline{\qquad\qquad}. \tag{8.3.3}$$

(b) Now subtract Equation (8.3.3) from Equation (8.3.2), and explain why it follows that

$$S_n - \frac{2}{5} \cdot S_n = 1 - \left(\frac{2}{5}\right)^n. \tag{8.3.4}$$

(c) Solve Equation (8.3.4) for S_n to find a simple formula for S_n that does not involve adding n terms.

(d) How would your work above change if instead of the original geometric sum S_n, we considered the situation with $a = 7$,

$$S_n = 7 + 7 \cdot \frac{2}{5} + 7 \cdot \left(\frac{2}{5}\right)^2 + \cdots + 7 \cdot \left(\frac{2}{5}\right)^{n-1} ?$$

The ideas in Activity 8.3.2 can be extended to the general case for any value of a and any value of $r \neq 1$. In particular, replacing 1 with a and $\frac{2}{5}$ with r, our work shows that by finding $r \cdot S_n$ and then subtracting that quantity from S_n, we get

$$S_n - rS_n = (a + ar + \cdots + ar^{n-2} + ar^{n-1}) - (ar + ar^2 + \cdots + ar^{n-1} + ar^n),$$

so that

$$S_n(1 - r) = a - ar^n,$$

and therefore

$$S_n = \frac{a - ar^n}{1 - r}.$$

We summarize this result formally as follows.

The value of a finite geometric series.

A finite geometric series S_n is a sum of the form

$$S_n = a + ar + ar^2 + \cdots + ar^{n-1}, \tag{8.3.5}$$

where a and r are real numbers such that $r \neq 1$. The exact value of the finite geometric series S_n can be computed directly as

$$S_n = \frac{a(1-r^n)}{1-r}. \tag{8.3.6}$$

Example 8.3.2 The value of a finite geometric series. Use the shortcut formula in Equation (8.3.6) to find the exact value of the finite geometric series

$$1 + \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \cdots + \left(\frac{1}{2}\right)^9.$$

Solution. We can view the given sum as the finite geometric series S_{10} that has $a = 1$ and $r = \frac{1}{2}$. By Equation (8.3.6), it follows that

$$S_{10} = \frac{1 - 1 \cdot \left(\frac{1}{2}\right)^{10}}{1 - \frac{1}{2}} = \frac{1 - \frac{1}{1024}}{\frac{1}{2}} = \frac{2047}{1024} = 1.9990234375.$$

◊

Example 8.3.3 The value of another finite geometric series. Use the shortcut formula in Equation (8.3.6) to find the exact value of the finite geometric series

$$2 - \frac{8}{3} + \frac{32}{9} - \frac{128}{27} + \cdots + 2 \cdot \left(-\frac{4}{3}\right)^7.$$

Solution. We can view the given sum as the finite geometric series S_8 that has $a = 2$ and $r = -\frac{4}{3}$. By Equation (8.3.6), it follows that

$$S_8 = \frac{2 - 2 \cdot \left(-\frac{4}{3}\right)^8}{1 + \frac{4}{3}} = -\frac{16850}{2187} \approx -7.704618198.$$

◊

Example 8.3.4 The value of one more finite geometric series. Use the shortcut formula in Equation (8.3.6) to find the exact value of the finite geometric series

$$\frac{1}{3} + 1 + 3 + 9 + \cdots + 6561.$$

Solution. Note that $6561 = 3^8$, so since this sum begins with $\frac{1}{3} = 3^{-1}$, this is a finite geometric series with 10 terms. Hence we can view the given sum as the finite geometric series S_{10} that has $a = \frac{1}{3}$ and $r = 3$:

$$S_{10} = \frac{1}{3} \cdot 3^0 + \frac{1}{3} \cdot 3^1 + \frac{1}{3} \cdot 3^2 + \cdots + \frac{1}{3} \cdot 3^9.$$

By Equation (8.3.6), we find that

$$S_{10} = \frac{\frac{1}{3} - \frac{1}{3} \cdot 3^{10}}{1-3} = \frac{29524}{3}.$$

◊

8.3.3 Infinite Geometric Series

Our initial introduction to sums with many terms came from Taylor polynomial approximations such as

$$e^1 \approx T_5(1) = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \frac{1}{5!}.$$

Because the approximation gets better as we add more terms, it's natural to think about the possibility of the sum extending forever. We begin by asking this question for a finite geometric series such as

$$S_n = 1 + \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \cdots + \left(\frac{1}{2}\right)^{n-1}:$$

what happens if we let the sum continue indefinitely?

Example 8.3.5 An infinite geometric series. Can we find the value of

$$1 + \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \cdots,$$

where the sum never terminates?

Solution. In Preview Activity 8.3.1, we saw that for the finite geometric series

$$S_n = 1 + \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \cdots + \left(\frac{1}{2}\right)^{n-1},$$

$S_1 = 1$, $S_2 = \frac{3}{2} = 1.5$, $S_3 = \frac{7}{4} = 1.75$, $S_4 = \frac{15}{8} = 1.875$, $S_5 = \frac{31}{16} = 1.9375$, and indeed

$$S_n = \frac{1 - \left(\frac{1}{2}\right)^n}{1 - \frac{1}{2}} = \frac{1 - \left(\frac{1}{2}\right)^n}{\frac{1}{2}}.$$

If we now multiply both the numerator and denominator by 2^n, we find that

$$S_n = \frac{2^n - 1}{2^{n-1}}. \tag{8.3.7}$$

We can view S_n as a "partial sum" (indeed, the sum of the first n terms) of the infinite geometric series

$$1 + \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \cdots$$

whose value we seek. Plotting these partial sums on a number line, we see evidence that the value of the nth partial sum is approaching 2.

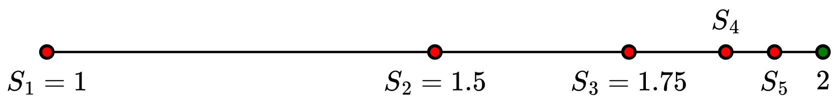


Figure 8.3.6 A plot of S_1, S_2, S_3, S_4, S_5, and the number 2 on the interval $[1, 2]$.

Indeed, we observe that each partial sum lies halfway between the preceding partial sum and the number 2: $S_2 = 1.5$ is halfway between $S_1 = 1$ and 2; $S_3 = 1.75$ is halfway between $S_2 = 1.5$ and 2; and so on. This shows that the partial sums S_n are approaching 2 as n increases without bound.

We can see this more formally in Equation (8.3.7) if we divide the two terms in the numerator of S_n by the denominator. Doing so, an equivalent formula for S_n is

$$S_n = 2 - \frac{1}{2^{n-1}}. \tag{8.3.8}$$

In Equation (8.3.8), if we let $n \to \infty$, we have

$$\lim_{n\to\infty} S_n = \lim_{n\to\infty} \left(2 - \frac{1}{2^{n-1}}\right) = 2,$$

since $\frac{1}{2^{n-1}} \to 0$ as n increases without bound.

Thus, it makes sense to say that the sum of the infinite geometric series

$$S = 1 + \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \cdots$$

is finite and that $S = 2$. ◊

Example 8.3.5 demonstrates the general principle that we use to determine if any infinite series has a finite value: we consider the **partial sum**, S_n, which is the finite sum of the first n terms, and then investigate whether the partial sums converge to a single value as n increases without bound. For geometric series, determining whether the partial sums converge or not is straightforward.

The value of an infinite geometric series.

An infinite geometric series S is a sum of the form

$$S = a + ar + ar^2 + \cdots + ar^{n-1} + \cdots, \tag{8.3.9}$$

where a and r are real numbers. If $|r| < 1$, then the infinite geometric series converges to the finite value

$$S = \frac{a}{1-r}. \tag{8.3.10}$$

If $|r| \geq 1$, then the infinite geometric series does not converge to a finite value.

Equation (8.3.10) holds because when $|r| < 1$, it follows that $r^n \to 0$ as $n \to \infty$. Indeed, when $|r| < 1$, we see by taking the limit as $n \to \infty$ in Equation (8.3.6) that

$$\lim_{n\to\infty} S_n = \lim_{n\to\infty} \frac{a(1-r^n)}{1-r} = \frac{a}{1-r}.$$

In Exercise 8.3.6.7, we will explore why the infinite geometric series does not converge when $|r| = 1$ and when $|r| > 1$.

As in our work in Chapter 4 with Riemann sums, we can use sigma notation (see also Subsection 4.2.2) to express a sum in convenient shorthand. For instance, it's equivalent to write

$$S_n = a + ar + ar^2 + \cdots + ar^{n-1} = \sum_{k=0}^{n-1} ar^k.$$

Furthermore, we can use sigma notation for a more concise way to express an infinite geometric series. For example, it is equivalent to write

$$1 + \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \cdots = \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k.$$

Activity 8.3.3. For each of the following infinite geometric series, determine the values of a and r, compute the partial sums S_5 and S_{10} exactly (writing each as a fraction), and if the infinite geometric series converges, find its value.

(a) $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \cdots$

(b) $4 - 2 + 1 - \frac{1}{2} + \frac{1}{4} - \cdots$

(c) $2 + \frac{8}{3} + \frac{32}{9} + \frac{128}{27} + \cdots$

(d) $\sum_{k=0}^{\infty} 5 \cdot \left(\frac{3}{4}\right)^k$

(e) $\sum_{k=1}^{\infty} -2 \cdot \left(-\frac{2}{3}\right)^k$

8.3.4 How geometric series naturally connect to Taylor polynomials

If we take $a = 1$ in Equation (8.3.9) and Equation (8.3.10), we see that

$$1 + r + r^2 + \cdots + r^{n-1} + r^n + \cdots = \frac{1}{1-r}, \tag{8.3.11}$$

provided that $|r| < 1$. To study this equation further, we are going to let r vary and thus we introduce the function $f(x) = \frac{1}{1-x}$ and replace r by x in Equation (8.3.11) to have

$$f(x) = \frac{1}{1-x} = 1 + x + x^2 + \cdots + x^{n-1} + x^n + \cdots, \tag{8.3.12}$$

which is valid for $|x| < 1$. One reason this equation is interesting is that we have a function $f(x)$ that can be represented in two different ways: as the rational function $\frac{1}{1-x}$, and as the infinite polynomial function $1 + x + x^2 + \cdots$.

For x such that $|x| < 1$, we know from Equation (8.3.12) that the infinite geometric series $1 + x + x^2 + \cdots + x^{n-1} + x^n + \cdots$ converges. Thus, it follows that partial sums of the series will approximate its value, which means

$$f(x) = \frac{1}{1-x} \approx 1 + x + x^2 + \cdots + x^{n-1} + x^n. \tag{8.3.13}$$

Since the series converges for $|x| < 1$, this also means the approximate equality in (8.3.13) holds for x near $a = 0$. This result reminds us of approximations generated by Taylor polynomials of degree n.

Moreover, because the infinite series converges, the larger the value of n, the better the approximation will be. Plotting the function $f(x) = \frac{1}{1-x}$ along with several of the polynomials that arise for different choices of n, say $P_1(x) = 1 + x$, $P_4(x) = 1 + x + \cdots + x^4$, and $P_7(x) = 1 + x + \cdots + x^7$, we see the impact of increasing the degree n in Figure 8.3.7.

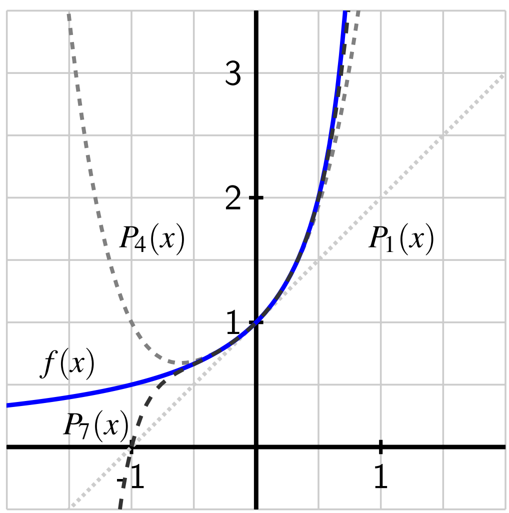


Figure 8.3.7 A plot of $f(x) = \frac{1}{1-x}$ (blue, no dashes), $P_1(x) = 1 + x$ (light gray, small dashes), $P_4(x) = 1 + x + \cdots + x^4$ (medium gray, medium dashes), and $P_7(x) = 1 + x + \cdots + x^7$ (dark gray, long dashes).

In particular, we observe that as the degree of the polynomial approximation increases, the polynomial not only appears closer to $f(x) = \frac{1}{1-x}$, but does so on a wider interval of x-values. Since the infinite series only converges when $|x| < 1$ and the function $f(x) = \frac{1}{1-x}$ is undefined when $x = 1$, we also expect the approximations to only be accurate on an interval that lies within $-1 < x < 1$. This also reminds of our earlier work with Taylor polynomials where in images such as Figure 8.2.7, increasing the degree of the Taylor polynomial similarly improves the approximation.

In our recent preceding work, we arrived at the function $f(x) = \frac{1}{1-x}$ by starting with the infinite geometric series

$$1 + x + x^2 + \cdots + x^{n-1} + x^n + \cdots$$

and exploring its partial sums. Next, we change perspective and start with function $f(x) = \frac{1}{1-x}$ and determine the Taylor polynomial approximations to f that are centered at $a = 0$ in order to see an interesting connection.

Activity 8.3.4. Consider the function $f(x) = \frac{1}{1-x}$. Find the degree n Taylor polynomial approximation centered at $a = 0$ for f.

(a) To begin finding $T_n(x)$, do the usual work of computing the various derivatives of f and their respective values at $a = 0$; note that it's helpful to view $f(x)$ in the form $f(x) = (1-x)^{-1}$ so that we can easily compute the derivatives of f using the chain rule. For instance,

$$f'(x) = (-1)(1-x)^{-2}(-1)$$

where the first "−1" arises from the power rule, while the second "−1" results from the chain rule, since $\frac{d}{dx}[1-x] = -1$. In order to see key patterns that arise, it's helpful not to combine the products of numbers that arise in the various derivatives. Record the first five derivatives of $f(x)$ in the spaces provided below.

$$\begin{aligned}
f(x) &= \frac{1}{1-x} = (1-x)^{-1}\\
f'(x) &= (-1)(1-x)^{-2}(-1)\\
f''(x) &= \\
f'''(x) &= \\
f^{(4)}(x) &= \\
f^{(5)}(x) &=
\end{aligned}$$

(b) Next, evaluate the derivatives you determined in (a) at $a = 0$ and use these to find the values of the coefficients of the Taylor polynomial centered at $a = 0$. Record your work in the blank spaces provided.

$k = 0$	$f(0) = \frac{1}{1-0} = 1$	$c_0 = f(0) = 1$
$k = 1$	$f'(0) = \frac{(-1)}{(1-0)^2} \cdot (-1) = 1$	$c_1 = \frac{f'(0)}{1!} = \frac{1}{1!} = 1$
$k =$	$f''(0) =$	$c_2 = \frac{f''(0)}{2!} =$
$k =$	$f'''(0) =$	$c_3 =$
$k =$	$f^{(4)}(0) =$	$c_4 =$
$k =$	$f^{(5)}(0) =$	$c_5 =$

(c) What pattern do you observe in the value of c_k? State the degree 5 Taylor polynomial, $T_5(x)$, as well as the formula you expect for the general degree n Taylor polynomial, $T_n(x)$.

(d) By identifying the value of r, explain why the degree n Taylor polynomial $T_n(x)$ can be thought of as a finite geometric series.

(e) What is the Taylor series centered at $a = 0$ for $f(x) = \frac{1}{1-x}$? (As we will see in the next section, the "Taylor series" of a function is the infinite series that results by simply extending the Taylor polynomials indefinitely.)

8.3.5 Summary

- A finite geometric series S_n is a sum of the form

$$S_n = a + ar + ar^2 + \cdots + ar^{n-1},$$

where a and r are real numbers such that $r \neq 1$. The value of the finite geometric series S_n can be computed directly as

$$S_n = \frac{a(1-r^n)}{1-r}.$$

- An infinite geometric series S is a sum of the form

$$S = a + ar + ar^2 + \cdots + ar^{n-1} + ar^n + \cdots,$$

where a and r are real numbers. If $|r| < 1$, then the infinite geometric series converges to the finite value

$$S = \frac{a}{1-r}.$$

If $|r| \geq 1$, then the infinite geometric series does not converge to a finite sum.

- If we consider the infinite geometric series with $a = 1$ and $|r| < 1$ and replace r with x, we can say that

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots + x^n + \cdots.$$

As we found in Activity 8.3.4, the Taylor series centered at $a = 0$ for $f(x) = \frac{1}{1-x}$ is precisely the infinite geometric series

$$T(x) = 1 + x + x^2 + x^3 + \cdots + x^n + \cdots$$

and we know this series is equal to $f(x)$ for $|x| < 1$.

8.3.6 Exercises

1. For each of the following, decide if the given series is a geometric series.

A. $10 + 10x + 20x + 30x + 40x + \cdots$:

Is this a geometric series?

- ◯ Yes
- ◯ No

If this is a geometric series, enter

the first term: _______________

and the ratio between successive terms: _______________

*(Enter **na** for the first term and ratio if this is not a geometric series.)*

B. $1 + 10x + 20x + 30x + 40x + \cdots$:

Is this a geometric series?

- ◯ Yes
- ◯ No

If this is a geometric series, enter

the first term: _______________

and the ratio between successive terms: _______________

*(Enter **na** for the first term and ratio if this is not a geometric series.)*

C. $10x^5 + 10x^6 + 10x^7 + 10x^8 + \cdots$:

Is this a geometric series?

- ◯ Yes
- ◯ No

If this is a geometric series, enter

the first term: _______________

and the ratio between successive terms: _______________

*(Enter **na** for the first term and ratio if this is not a geometric series.)*

2. For each of the finite geometric series given below, indicate the number of terms in the sum and find the sum. For the value of the sum, enter an expression that gives the exact value, rather than entering an approximation.

A. $5 + 5(0.2) + 5(0.2)^2 + \cdots + 5(0.2)^{15}$

number of terms = _____

value of sum = ______________________

B. $5(0.2)^4 + 5(0.2)^5 + 5(0.2)^6 + \cdots + 5(0.2)^{11}$

number of terms = _____

value of sum = ______________________

3. Find the sum of each of the geometric series given below. For the value of the sum, enter an expression that gives the exact value, rather than entering an approximation.

A. $-10 + 5 - \frac{5}{2} + \frac{5}{2^2} - \frac{5}{2^3} + \cdots + \frac{5}{2^{12}} =$ ______________________

B. $\sum_{n=4}^{\infty} \left(\frac{1}{3}\right)^n =$ ______________________

4. Find the sum of the infinite geometric series

$$1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \cdots$$

Enter *DNE* if the sum does not exist.

Sum = ______________________

5. Find the sum of the series

$-6 + \dfrac{-6}{4} + \dfrac{-6}{16} + \ldots + \dfrac{-6}{4^{n-1}} + \ldots.$

Answer: ______________

6. Determine whether the series converges, and if so find its sum.

$\displaystyle\sum_{k=1}^{\infty} \frac{2^{k+3}}{7^{k-1}} =$ ______________________

(Enter DNE if the sum does not exist.)

7. In Equation (8.3.10), we learned that for an infinite geometric series

$$S = a + ar + ar^2 + \cdots + ar^{n-1} + \cdots,$$

if $|r| < 1$, then

$$S = \frac{a}{1-r}.$$

This result provides a quick way to determine the value of an infinite geometric series that converges.

In this exercise, we explore why the infinite geometric series fails to converge when $r = 1$, $r = -1$, and $|r| > 1$.

a. If we consider the finite geometric series that results when $a = 1$ and $r = 1$, we get the sum

$$S_n = 1 + 1 + \cdots + 1,$$

where there are n terms in the sum.

i. Compute S_2, S_3, and S_4. What is the general formula for S_n?

ii. Explain why the sequence S_n does not converge to a finite value. Contrast this with, for example, our work in Example 8.3.5.

b. Next, consider the finite geometric series that results when $a = 1$ and $r = -1$, which is the sum

$$S_n = 1 - 1 + 1 - 1 + \cdots + (-1)^{n-1},$$

where there are n terms in the sum. (We view the first "1" as resulting from "$1 \cdot (-1)^0$".)

i. Compute S_2, S_3, S_4, and S_5. What do you observe?

ii. Explain why the sequence S_n does not converge to a finite value. Contrast this with, for example, our work in Example 8.3.5.

c. Now consider the geometric sum with $a = 1$ and $r = 2$, so

$$S_n = 1 + 2 + 4 + \cdots + 2^{n-1}.$$

i. Compute S_2, S_3, and S_4. What is the general formula for S_n?

ii. What do you observe happens to these partial sums as n increases without bound? What does this tell us about the infinite geometric series $1 + 2 + 4 + \cdots + 2^{n-1} + \cdots$?

8. Suppose you drop a golf ball onto a hard surface from a height h. The collision with the ground causes the ball to lose energy and so it will not bounce back to its original height. The ball will then fall again to the ground, bounce back up, and continue. Assume that at each bounce the ball rises back to a height $\frac{3}{4}$ of the height from which it dropped. Let h_n be the height of the ball on the nth bounce, with $h_0 = h$. In this exercise we will determine the distance traveled by the ball and the time it takes to travel that distance.

a. Determine a formula for h_1 in terms of h.

b. Determine a formula for h_2 in terms of h.

c. Determine a formula for h_3 in terms of h.

d. Determine a formula for h_n in terms of h.

e. Write an infinite series that represents the total distance traveled by the ball. Then determine the value of this series.

f. Next, let's determine the total amount of time the ball is in the air.

i) When the ball is dropped from a height H, if we assume the only force acting on it is the acceleration due to gravity, then the height of the ball at time t is given by

$$H - \frac{1}{2}gt^2.$$

Use this formula to determine the time it takes for the ball to hit the ground after being dropped from height H.

ii) Use your work in the preceding item, along with that in (a)-(e) above to determine the total amount of time the ball is in the air.

9. It is important to understand the power of geometric growth compared to linear growth. Suppose you are hired for a job that will take you 30 days to complete and are offered two options for how you'll be compensated.

Option 1. You can be paid \$500 per day, or

Option 2. You can be paid 1 cent the first day, 2 cents the second day, 4 cents the third day, 8 cents the fourth day, and so on, doubling the amount you are paid each day.

a. How much will you be paid for the job in total under Option 1?

b. Complete Table 8.3.8 to determine the pay you will receive under Option 2 for the first 10 days.

Table 8.3.8 Option 2 payments

Day	Pay on this day	Total amount paid to date
1	\$0.01	\$0.01
2	\$0.02	\$0.03
3		
4		
5		
6		
7		
8		
9		
10		

c. Find a formula for the amount paid *on* day n, as well as for the total amount paid *by* day n. Use this formula to determine which option (1 or 2) you should take.

8.4 Taylor series

Motivating Questions

- What is the Taylor series centered at a of a function f?
- What does it mean for an infinite Taylor series to converge to a finite sum, and how can we determine for which x-values the series converges?
- What are the Taylor series centered at $a = 0$ for $\frac{1}{1-x}$, $\ln(1+x)$, $\sin(x)$, $\cos(x)$, and e^x, and for which x-values do these series converge?

8.4.1 Introduction

In Activity 8.3.4, we investigated several Taylor polynomials centered at 0 for the function $f(x) = \frac{1}{1-x}$ and found that

$$T_n(x) = 1 + x + x^2 + x^3 + \cdots + x^n.$$

Moreover, we saw that it is natural to extend the degree n Taylor polynomial to an infinite Taylor *series* (which will be formally defined in Definition 8.4.1). For $f(x) = \frac{1}{1-x}$, its corresponding Taylor series is

$$T(x) = 1 + x + x^2 + x^3 + \cdots x^n + \cdots.$$

Because $T(x)$ is an infinite geometric series with $a = 1$ and $r = x$, whenever $|r| = |x| < 1$, the series converges and by Equation (8.3.10) its sum is

$$T(x) = 1 + x + x^2 + x^3 + \cdots x^n + \cdots = \frac{1}{1-x}. \tag{8.4.1}$$

Remarkably, this shows that the original function, $f(x) = \frac{1}{1-x}$, is *equal to its Taylor series* for all values of x that satisfy $|x| < 1$. So, not only do Taylor polynomials provide increasingly better approximations to a function as the degree n increases, but it appears that if we let n increase without bound we arrive at a new representation of the original function through its Taylor series. We note that this new representation may only be valid for a limited set of x-values, such as $|x| < 1$ in the case of $f(x) = \frac{1}{1-x}$.

In Preview Activity 8.4.1, we consider an infinite geometric series and explore the function to which the series is related and that function's Taylor polynomials.

Preview Activity 8.4.1. Let

$$T(x) = 3 - x + \frac{1}{3}x^2 - \frac{1}{9}x^3 + \cdots + 3 \cdot \left(-\frac{1}{3}\right)^n x^n + \cdots.$$

(a) Explain why $T(x)$ is a geometric series and identify the values of a and r.

(b) Recall that the sum of an infinite geometric series with first term a and constant ratio r is $S = \frac{1}{1-r}$, provided $|r| < 1$. Use this result to find a single fraction that represents the value of $T(x)$. For which values of x is this representation valid?

(c) Consider the function $f(x) = 9(3+x)^{-1}$. Determine $f(0)$, $f'(0)$, $f''(0)$, and $f'''(0)$. What is the degree 3 Taylor polynomial centered at 0 for the function $f(x)$?

(d) How is your work in (c) connected to the original given series $T(x)$?

8.4.2 Taylor series and the Ratio Test

From our work in Activity 8.3.4 and Preview Activity 8.4.1, we have seen two examples where the Taylor polynomials of a given function lead to a corresponding infinite Taylor series that is actually equal to the original function itself. This suggests that one reason Taylor series are important is that they give rise to polynomial-like representations of non-polynomial functions.

Next we formally define the Taylor series of a function and introduce a tool for determining where its Taylor series converges.

Definition 8.4.1 Let f be a function such that $f^{(n)}(a)$ exists for every natural number n. The **Taylor series for f centered at** $x = a$ is the series $T_f(x)$ defined by

$$\begin{aligned} T_f(x) &= \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}(x-a)^k \\ &= f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \cdots. \end{aligned}$$

◊

In the special case where $a = 0$ in Definition 8.4.1, the Taylor series is also sometimes called the **Maclaurin series** for f.

From Activity 8.3.4, we know the degree n Taylor polynomial centered at 0 for the function $f(x) = \frac{1}{1-x}$ is

$$T_n(x) = 1 + x + x^2 + \cdots + x^n,$$

and thus the Maclaurin series for $f(x) = \frac{1}{1-x}$ is

$$T_f(x) = \sum_{k=0}^{\infty} x^k.$$

As we noted earlier in Equation (8.4.1), because this particular series is geometric (with common ratio $r = x$), we are able to easily find a shortcut formula for its value and understand the values of x for which the infinite sum makes sense. In particular, for $|x| < 1$,

$$T_f(x) = \sum_{k=0}^{\infty} x^k = \frac{1}{1-x},$$

which shows that the Taylor series is equal to the original function $f(x)$ for this set of x values.

It turns out that while most Taylor series are not geometric series, we can compare any Taylor series to a geometric series in order to determine an interval of x-values for which the Taylor series converges. To see how to approach this issue, we consider the following example with a particular function with a known Taylor series and determine the x-values for which its Taylor series should converge.

Example 8.4.2 Let $f(x) = \ln(1+x)$. By investigating how close the terms of the Taylor series of f of resemble a geometric series, determine the x-values for which the Taylor series appears to converge.

Solution. In Example 8.2.5, we found that the general degree n Taylor polynomial centered at $a = 0$ for $f(x) = \ln(1 + x)$ is

$$T_n(x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \cdots + (-1)^{n-1}\frac{1}{n}x^n.$$

This pattern thus allows us to see that the Taylor series for $f(x)$, centered at $a = 0$, is

$$\begin{aligned} T_f(x) &= \sum_{k=1}^{\infty}(-1)^{k-1}\frac{1}{k}x^k \\ &= x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \cdots + (-1)^{n-1}\frac{1}{n}x^n + \cdots. \end{aligned}$$

Unlike the Taylor series for the function $\frac{1}{1-x}$, the Taylor series for $f(x) = \ln(1 + x)$ turns out to not be geometric since there is not a common ratio between the terms, so we introduce some new ideas to think about the x-values for which we expect the series to converge.

For a series to be geometric, we must be able to construct the next term in the series by multiplying the prior term by a constant ratio. In $T_f(x)$, the general n^{th} term is

$$a_n(x) = (-1)^{n-1}\frac{1}{n}x^n$$

and the $(n + 1)^{\text{st}}$ term is

$$a_{n+1}(x) = (-1)^n\frac{1}{n+1}x^{n+1}.$$

If we consider the ratio of these consecutive terms and call that quantity $r_n(x)$, we find that

$$r_n(x) = \frac{a_{n+1}(x)}{a_n(x)} = \frac{(-1)^n\frac{1}{n+1}x^{n+1}}{(-1)^{n-1}\frac{1}{n}x^n}.$$

We can rewrite this fraction algebraically and simplify it considerably. Grouping the similar terms between the numerator and denominator, we see

$$r_n(x) = \frac{(-1)^n}{(-1)^{n-1}} \cdot \frac{n}{n+1} \cdot \frac{x^{n+1}}{x^n},$$

which allows us to simplify and conclude that

$$r_n(x) = (-1) \cdot \frac{n}{n+1} \cdot x.$$

This tells us the series $T_f(x)$ is not geometric since the ratio of consecutive terms depends on n. For example, if we take $n = 2$, we have

$$r_2(x) = -\frac{2}{3}x,$$

which is the ratio between the third and second terms of $T_f(x)$, while

$$r_3(x) = -\frac{3}{4}x$$

is the ratio between the fourth and third terms. Since the ratio $r_n(x)$ depends on n, the series lacks a common ratio between terms.

However, as n gets large, the series $T_f(x)$ is *almost* geometric. This is because the quantity $\frac{n}{n+1}$ approaches 1 as n increases without bound, and therefore for large n,

$$r_n(x) \approx -x.$$

This suggests that the series $T_f(x)$ converges for $|r_n(x)| \approx |-x| = |x| < 1$, since the series is very close to being geometric with a common ratio $-x$.

Graphical evidence supports the conclusion that for $|x| < 1$, the series $T_f(x)$ converges in the same way that a geometric one would, and also that the interval of convergence is $-1 < x < 1$. In Figure 8.4.3, we plot $f(x) = \ln(1 + x)$, $T_{10}(x) = x - \frac{1}{2}x^2 + \cdots - \frac{1}{10}x^{10}$, and $T_{20}(x) = x - \frac{1}{2}x^2 + \cdots - \frac{1}{20}x^{20}$.

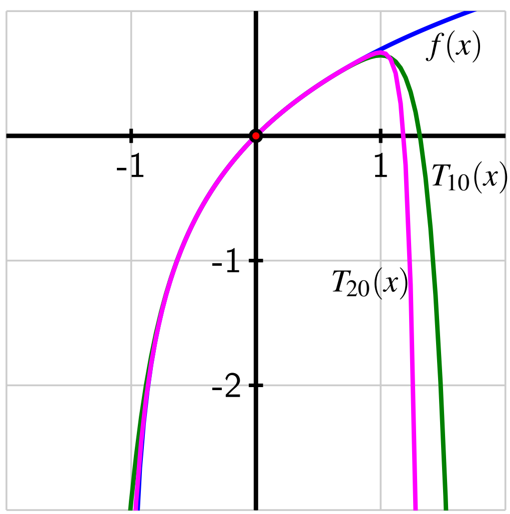


Figure 8.4.3 The function $f(x) = \ln(1 + x)$ and its degree 10 and 20 Taylor polynomial approximations $T_{10}(x)$ and $T_{20}(x)$ near the point $(0, f(0))$.

In Figure 8.4.3 we see that on the interval $-0.8 < x < 0.8$, $f(x)$, $T_{10}(x)$, and $T_{20}(x)$ are nearly indistinguishable. Furthermore, we see that if we widen the interval to $-0.9 < x < 0.9$, $T_{20}(x)$ remains a very good approximation of $f(x)$. For example, we can do some computations and find that

$$f(-0.8) - T_{10}(-0.8) \approx -0.030563, \; f(-0.8) - T_{20}(-0.8) \approx -0.001091,$$

and

$$f(-0.9) - T_{10}(-0.9) \approx -0.183837, f(-0.9) - T_{20}(-0.9) \approx -0.032564.$$

Increasing the degree of the approximation further allows us to do a bit better, but since $f(x) = \ln(1+x)$ is undefined with a vertical asymptote at $x = -1$, no polynomial function can agree with $f(x)$ at $x = -1$. Thus, it appears to be the case that $T_f(x)$ converges for x-values that satisfy $-1 < x < 1$, and on that interval the Taylor series converges to $f(x) = \ln(1+x)$. $\diamond$

From our most recent work in Example 8.4.2, we see a striking result: it appears that for x-values where the Taylor series $T_f(x)$ of $f(x) = \ln(1+x)$ converges to a finite value, the Taylor series actually equals the function f. Said differently, it appears that

$$\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \cdots + (-1)^{n-1}\frac{1}{n}x^n + \cdots$$

for x-values that satisfy $-1 < x < 1$. (It also turns out this equality holds when $x = 1$ for reasons that we'll consider in Section 8.6.) This result is analogous to Equation (8.4.1), but is different in the sense that the Taylor series for $\ln(1+x)$ is not geometric.

We next introduce a more formal way to test "how geometric" a Taylor series is in an effort to understand where the Taylor series is guaranteed to converge.

The Ratio Test.

Consider an infinite series of the form $T(x) = \sum_{k=0}^{\infty} c_k(x-a)^k$, where a is any real number. Let $r_n(x) = \frac{c_{n+1}(x-a)^{n+1}}{c_n(x-a)^n}$ be the ratio of consecutive terms of the series and let

$$r(x) = \lim_{n\to\infty} r_n(x) = \lim_{n\to\infty} \frac{c_{n+1}(x-a)^{n+1}}{c_n(x-a)^n}. \tag{8.4.2}$$

Then,

- for any real number x such that $|r(x)| < 1$, the series $T(x)$ converges;
- for any real number x such that $|r(x)| > 1$, the series $T(x)$ does not converge;
- and for any x such that $|r(x)| = 1$, the series $T(x)$ may or may not converge.

In essence, the Ratio Test is a way to find x values for which $T(x)$ behaves like a geometric series with ratio $r(x)$ such that $|r(x)| < 1$; for those x-values, the infinite series converges to a finite value. There is a subtle issue that arises for x-values for which $|r(x)| = 1$: it's possible for the infinite series to either converge or diverge at such x-values, but we will normally not concern ourselves with this issue.

After we evaluate the limit

$$r(x) = \lim_{n\to\infty} \frac{c_{n+1}(x-a)^{n+1}}{c_n(x-a)^n}$$

and set $|r(x)| < 1$, we can usually do algebra to write the resulting inequality in the form

$$|x-a| < R.$$

We call R the **radius of convergence** and say that the set of all x-values such that $|x - a| < R$ is the **open interval of convergence**. Again, it is possible that the Taylor series may either converge or diverge when $|x - a| = R$, but we will normally not concern ourself with this issue at the endpoints.

In light of the formal statement of the Ratio Test, we now see that in our initial investigation of the Taylor series for $f(x) = \ln(1 + x)$, we applied the Ratio Test to find an open interval of x-values for which its Taylor series converges. In particular, for

$$T_f(x) = \sum_{k=1}^{\infty} (-1)^{k-1} \frac{1}{k} x^k$$

we found that

$$r_n(x) = \frac{(-1)^n \frac{1}{n+1} x^{n+1}}{(-1)^{n-1} \frac{1}{n} x^n} = (-1) \cdot \frac{n}{n+1} \cdot x$$

and therefore

$$r(x) = \lim_{n \to \infty} r_n(x) = \lim_{n \to \infty} (-1) \cdot \frac{n}{n+1} \cdot x = -x.$$

By the Ratio Test, it follows that if $|r(x)| = |-x| = x < 1$, then the series converges; that is, $-1 < x < 1$ is the open interval of convergence, and the radius of convergence of this series is $R = 1$. We also know that if $|r(x)| = x > 1$, the series does not converge. When $x = \pm 1$, we don't know if the series converges or diverges, but we will normally not be concerned with this endpoint behavior because of many subtle and complicated issues that can arise.

Activity 8.4.2. Consider the infinite Taylor series given by

$$\begin{aligned} T(x) &= \sum_{k=1}^{\infty} \frac{1}{k \cdot 2^k} (x-1)^k \\ &= \frac{1}{1 \cdot 2}(x-1) + \frac{1}{2 \cdot 4}(x-1)^2 + \frac{1}{3 \cdot 8}(x-1)^3 + \cdots + \frac{1}{n \cdot 2^n}(x-1)^n + \cdots \end{aligned}$$

(a) As described in the statement of the Ratio Test, let $r_n(x)$ be the ratio of the $(n+1)^{\text{st}}$ term of $T(x)$ to the n^{th} term of $T(x)$. Find the simplest formula that you can for $r_n(x)$.

(b) Let $r(x) = \lim_{n \to \infty} r_n(x)$. Evaluate this limit to find the simplest formula you can for $r(x)$.

(c) For what values of x is $|r(x)| < 1$? What is the open interval of convergence for $T(x)$?

(d) Let $T_{10}(x)$ be the sum of the first 10 terms of $T(x)$, and let $f(x) = \ln(2) - \ln(3 - x)$. Plot $f(x)$ and $T_{10}(x)$ on the same coordinate axes in a window centered around $x = 1$. What do you notice? What does this suggest about the series $T(x)$?

8.4.3 Taylor series of several important functions

So far, we have established that the Taylor series of two important functions, $\frac{1}{1-x}$ and $\ln(1+x)$, converge on an open interval of x-values and on that interval converge to the respective functions. For these functions,

- if $|x| < 1$, then
$$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k = 1 + x + x^2 + x^3 + \cdots$$
- if $|x| < 1$, then[1]
$$\ln(1+x) = \sum_{k=1}^{\infty} (-1)^{n-1} \cdot \frac{1}{n} \cdot x^n = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \cdots.$$

Three other important functions that we have studied — $\sin(x)$, $\cos(x)$, and e^x — have similar representations, and so we build on our work from Section 8.1 and Section 8.2 with their respective Taylor polynomials in order to investigate and understand their Taylor series.

Following Activity 8.2.2, we noted some of the higher degree Taylor polynomials centered at $a = 0$ for $\sin(x)$, $\cos(x)$, and e^x. For the sine function, we observed that

$$\sin(x) \approx T_7(x) = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7.$$

In the next example, we determine the Taylor series of $f(x) = \sin(x)$ and investigate the x-values for which the series converges.

Example 8.4.4 Explain why the Taylor series centered at $a = 0$ for $f(x) = \sin(x)$ is

$$T_f(x) = \sum_{k=0}^{\infty} (-1)^k \frac{1}{(2k+1)!} x^{2k+1}$$

and find the interval of x-values for which this Taylor series converges. Investigate whether or not the Taylor series converges to $f(x) = \sin(x)$.

Solution. We observed in Section 8.2 that the derivatives of the sine function cycle and repeat every 4 derivatives, since

$$f(x) = \sin(x),\ f'(x) = \cos(x),\ f''(x) = -\sin(x),\ f'''(x) = -\cos(x),$$

and so

$$f^{(4)}(x) = \sin(x),\ f^{(5)}(x) = \cos(x),\ f^{(6)}(x) = -\sin(x),\ f^{(7)}(x) = -\cos(x).$$

More generally, $f^{(4k)}(x) = \sin(x)$ and $f^{(4k+2)}(x) = -\sin(x)$ for every natural number k. This shows that $f^{(4k)}(0) = \sin(0) = 0$ and $f^{(4k+2)}(0) = -\sin(0) = 0$, from which it follows that every even power of x in the Taylor series of the sine function has a coefficient of 0.

[1]It turns out that this representation is also valid when $x = 1$ because the series $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$ converges, and does so to the exact value $\ln(2)$. This means that while the *open* interval of convergence is $-1 < x < 1$ for this series, the complete *interval of convergence* is actually $-1 < x \leq 1$.

For odd values of n, we can see that since $f'(x) = \cos(x)$, we also have $f^{(5)}(x) = \cos(x)$, $f^{(9)}(x) = \cos(x)$, and so on. From this, $f'(0) = \cos(0) = 1$, and similarly, $f^{(5)}(0) = 1$, $f^{(9)}(x) = 1$, and indeed $f^{(4k+1)}(0) = 1$ for every nonnegative whole number k. Finally, since $f'''(x) = -\cos(x)$, we have $f'''(0) = -\cos(0) = -1$, and so $f'''(0) = -1$, $f^{(7)}(0) = -1$, and thus $f^{(4k+3)}(0) = -1$ for every nonnegative whole number k.

Putting these observations together about the various derivatives of $f(x) = \sin(x)$ evaluated at $a = 0$ and using the fact that the Taylor series of $f(x)$ centered at 0 has form

$$T_f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k$$

we see that

$$\begin{aligned} T_f(x) &= x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \cdots + (-1)^n \frac{1}{(2n+1)!}x^{2n+1} + \cdots \\ &= \sum_{k=0}^{\infty} (-1)^k \frac{1}{(2k+1)!} x^{2k+1} \end{aligned} \tag{8.4.3}$$

Next, we use the Ratio Test to investigate the values of x where $T_f(x)$ converges. We therefore consider $r_n(x)$, which is the ratio of consecutive nonzero terms, and have

$$r_n(x) = \frac{(-1)^{n+1} \frac{1}{(2(n+1)+1)!} x^{2(n+1)+1}}{(-1)^n \frac{1}{(2n+1)!} x^{2n+1}}.$$

Expanding the expression $2(n + 1)$, combining like terms, and slightly rearranging how we write the fraction, we see that

$$\begin{aligned} r_n(x) &= \frac{(-1)^{n+1}}{(-1)^n} \cdot \frac{\frac{1}{(2n+3)!}}{\frac{1}{(2n+1)!}} \cdot \frac{x^{2n+3}}{x^{2n+1}} \\ &= \frac{(-1)^{n+1}}{(-1)^n} \cdot \frac{(2n+1)!}{(2n+3)!} \cdot \frac{x^{2n+3}}{x^{2n+1}} \end{aligned} \tag{8.4.4}$$

We now consider the three fractions in Equation (8.4.4) one at a time and simplify each. First, we observe that

$$\frac{(-1)^{n+1}}{(-1)^n} = -1$$

since the numerator and denominator are each always either 1 or -1 and always of opposite signs. Next, we expand the factorials and see that

$$(2n+1)! = (2n+1)(2n)(2n-1)\cdots(2)(1)$$

and

$$(2n+3)! = (2n+3)(2n+2)(2n+1)(2n)(2n-1)\cdots(2)(1)$$

so

$$\frac{(2n+1)!}{(2n+3)!} = \frac{1}{(2n+3)(2n+2)}.$$

Finally, we observe that

$$\frac{x^{2n+3}}{x^{2n+1}} = x^{(2n+3)-(2n+1)} = x^2.$$

Putting all of this work together, we can update Equation (8.4.4) to the simpler form

$$r_n(x) = (-1) \cdot \frac{1}{(2n+3)(2n+2)} \cdot x^2.$$

From here, we compute $r(x)$, the limit of $r_n(x)$ as n increases without bound. Since $\frac{1}{(2n+3)(2n+2)} \to 0$ as $n \to \infty$, it follows that for any fixed value of x,

$$r(x) = \lim_{n\to\infty} r_n(x) = \lim_{n\to\infty} (-1) \cdot \frac{1}{(2n+3)(2n+2)} \cdot x^2 = -1 \cdot 0 \cdot x^2 = 0.$$

The Ratio Test tells us that for any x for which $|r(x)| < 1$, the series $T_f(x)$ converges. Here, since $r(x) = 0$ for every value of x, this means that $T_f(x)$ converges for every real number x.

Finally, we investigate whether the Taylor series appears to converge to $f(x) = \sin(x)$ or not. In Figure 8.4.5, we see the Taylor polynomials of degree 5 and 9, and the significant improvement of the approximation as the degree increases.

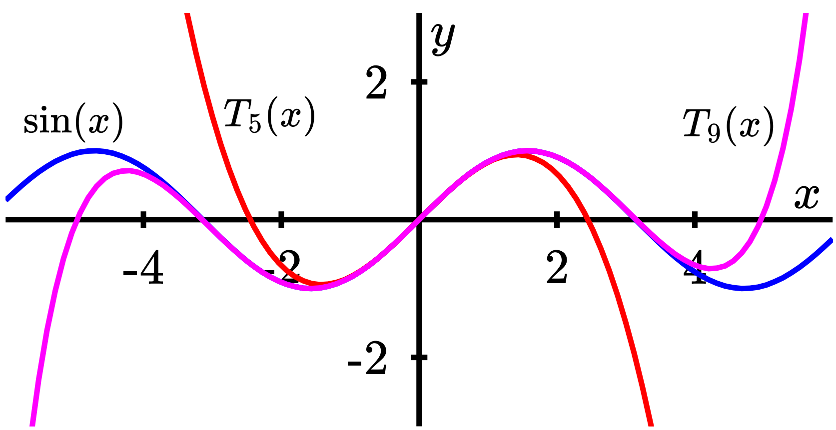


Figure 8.4.5 The degree 5 and 9 Taylor polynomials centered at $x = 0$ for $f(x) = \sin(x)$, along with the sine function.

For the degree 9 approximation, we see that $T_9(x)$ and $f(x) = \sin(x)$ are almost indistinguishable on $-4 < x < 4$, which is much wider than the interval $-2 < x < 2$ on which the $T_5(x)$ approximation is accurate. Experimenting with even higher degree Taylor polynomials shows that increasing the degree of the approximation appears to widen the interval on which the approximation is accurate and seems to do so without bound, which suggests that the Taylor series of $f(x) = \sin(x)$ in fact converges to $f(x) = \sin(x)$. ◊

In Example 8.4.4, we found that the Taylor series for the sine function appears to converge to the sine function, and to do so for every real number x. This result can be proven formally using a famous theorem called the Lagrange Error Bound, a theorem that quantifies how accurate a Taylor polynomial approximation is on an interval, which we will study in Section 8.6. To summarize, we now know the remarkable result that the sine function is equal to its Taylor series for every value of x, so

$$\sin(x) = \sum_{k=0}^{\infty} (-1)^k \frac{1}{(2k+1)!} x^{2k+1} = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \cdots \tag{8.4.5}$$

Nearly identical reasoning shows a similar result for the cosine function: that its Taylor series centered at $a = 0$ converges for every real number x and converges to the cosine function itself. Thus, for any real number x,

$$\cos(x) = \sum_{k=0}^{\infty} (-1)^k \frac{1}{(2k)!} x^{2k} = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \cdots \tag{8.4.6}$$

In the next activity, we conduct a similar investigation for $f(x) = e^x$.

> **Activity 8.4.3.** Let $f(x) = e^x$. Our goal is to understand why the Taylor series for $f(x)$ converges for every real number x and see that the Taylor series converges to e^x.
>
> **(a)** For $f(x) = e^x$, explain why $f^{(k)}(0) = 1$ for every natural number k.
>
> **(b)** State the Taylor series, $T_f(x)$ centered at $a = 0$ for $f(x) = e^x$. Write $T_f(x)$ in both sigma notation and as an expanded sum.
>
> **(c)** Let $r_n(x)$ be the ratio of the $(n+1)^{\text{st}}$ term to the n^{th} term of $T_f(x)$. Find the simplest expression you can for $r_n(x)$.
>
> **(d)** Let $r(x) = \lim_{n\to\infty} r_n(x)$. Evaluate this limit, and then apply the Ratio Test to say what you can conclude about the x-values for which $T_f(x)$ converges.
>
> **(e)** Use a computational device to graph $f(x) = e^x$, $T_{10}(x)$, and $T_{20}(x)$ on the same axes. What do you observe?

While there is considerable technical detail involved in finding Taylor series and deciding where they converge, it's important not to miss the big picture. For certain "nice" functions that are infinitely differentiable, such as $f(x) = \ln(1+x)$ and $f(x) = \sin(x)$, not only can we find the Taylor series for each and determine an open interval of x-values at which the Taylor series converges, but for those values *the Taylor series converges to the function itself*. This means that for certain x-values (and sometimes *all* x-values), we can represent a non-polynomial function as being equal to an infinite polynomial.

A function such as $f(x) = \sin(x)$ that is infinitely differentiable and has a power series that converges on some interval to the function itself is said to be **analytic**. Analytic functions are amazing and among the nicest functions in all of mathematics: they are completely determined by what happens at a single point.

Almost all functions are not analytic. For example, if a function measures your location as a function of time on a highway, that function cannot be analytic, since what happens at 12:01 is independent of what happens at 12:00 due to other circumstances such as traffic or construction. At the same time, the mathematical universe is better to us than we deserve in that many important, common functions are analytic, including several we've mentioned ($\frac{1}{1-x}$, $\ln(1+x)$, $\sin(x)$, $\cos(x)$, and e^x) and most of those represented by buttons on a calculator.

For the function $f(x) = e^x$ in particular, we now understand that we may write

$$e^x = \sum_{k=0}^{\infty} \frac{1}{k!} x^k = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \cdots + \frac{1}{n!}x^n + \cdots,$$

and this equation is true for every possible value of x. This equation is completely determined by the values of $f(x) = e^x$ and its derivatives only at the value $a = 0$. When we use a computational device like *Desmos* or a calculator and enter e^2, this is how the device can find the decimal representation of the number. Since

$$e^2 = 1 + 2 + \frac{1}{2!}2^2 + \frac{1}{3!}2^3 + \cdots + \frac{1}{n!}2^n + \cdots$$

if we take enough terms in the sum, we get a highly accurate approximation of e^2. For instance,

$$e^2 \approx 1 + 2 + \frac{1}{2!}2^2 + \frac{1}{3!}2^3 + \cdots + \frac{1}{12!}2^{12} = \frac{691}{93555} = 7.3890545668\ldots,$$

which an accurate representation of e^2 to 5 decimal places, and more terms in the sum can be used in order to generate any desired level of accuracy.

The five Taylor series that we've found in our work to date are extremely important and will be used extensively in Section 8.5, so we summarize them in the following table.

Important Taylor series representations.

Table 8.4.6 Taylor series and the open intervals of x-values where they converge for 5 important functions.

$f(x)$	Taylor series, T_f, centered at 0	$f(x) = T_f(x)$
$\frac{1}{1-x}$	$\sum_{k=0}^{\infty} x^k = 1 + x + x^2 + x^3 + \cdots$	if $\lvert x\rvert < 1$
$\ln(1+x)$	$\sum_{k=1}^{\infty} (-1)^{k-1}\frac{x^k}{k} = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \cdots$	if $\lvert x\rvert < 1$
$\sin(x)$	$\sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!} = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \cdots$	for all real x
$\cos(x)$	$\sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!} = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \cdots$	for all real x
e^x	$\sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \cdots$	for all real x

8.4.4 Summary

- Provided that f is a function for which every derivative of f exists at $x = a$, the Taylor series centered at a of f is the series $T_f(x)$ defined by

$$\begin{aligned} T_f(x) &= \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}(x-a)^k \\ &= f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \cdots. \end{aligned}$$

- We know that an infinite geometric series converges whenever its common ratio, r, satisfies $|r| < 1$. The Ratio Test enables us to consider a Taylor series and determine an open interval of x-values on which the series converges, essentially by comparing the Taylor series to a geometric series. By letting $r_n(x)$ be the ratio of consecutive terms in the series and taking the limit

$$r(x) = \lim_{n\to\infty} r_n(x),$$

we can say that for all x for which $|r(x)| < 1$, the Taylor series converges to a finite value.

- We have found the Taylor series for five important functions, determined an open interval of x-values on which each converges, and seen the striking result that where each Taylor series converges, it converges to original function $f(x)$ that we used to generate the Taylor series. These results are summarized in Table 8.4.6.

8.4.5 Exercises

1. Write out the first four terms of the Maclaurin series of $f(x)$ if

$$f(0) = -2, \qquad f'(0) = 7, \qquad f''(0) = 13, \qquad f'''(0) = -15$$

$f(x) =$ ____________________ $+ \cdots$

2. Find the first four terms of the Taylor series for the function $\frac{3}{x}$ about the point $a = 3$. (Your answers should include the variable x when appropriate.)

$\frac{3}{x} =$ _______________ + _______________ + _______________ + _______________ + ...

3. Find the first five terms of the Taylor series for the function $f(x) = \ln(x)$ about the point $a = 7$. (Your answers should include the variable x when appropriate.)

$\ln(x) =$ _______________ + _______________ + _______________ + _______________ + _______________ + ...

4. Find the first four terms of the Taylor series for the function $4\cos(x)$ about the point $a = -\pi/4$. (Your answers should include the variable x when appropriate.)

$4\cos(x) =$ ____________________ + ____________________ + ____________________ + ____________________ + ...

5. For each of the following, solve exactly for the variable x.

(a) $1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots = 0.6$

$x =$ _________________________

(b) $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = 4$

$x =$ _________________________

6. By recognizing each series below as a Taylor series evaluated at a particular value of x, find the sum of each convergent series.

A. $1 - \frac{2^2}{2!} + \frac{2^4}{4!} - \frac{2^6}{6!} + \cdots + \frac{(-1)^n 2^{2n}}{(2n)!} + \cdots =$ __________

B. $1 + \frac{1}{7} + (\frac{1}{7})^2 + (\frac{1}{7})^3 + (\frac{1}{7})^4 + \cdots + (\frac{1}{7})^n + \cdots =$ __________

7. Use the ratio test to find the open interval of convergence of

$$\sum_{n=1}^{\infty} \frac{n^2 x^{3n}}{3^{3n}}.$$

interval of convergence = ______________

(Enter your answer as an interval: thus, if the interval of convergence were $-3 < x < 5$, you would enter ***(-3,5)****. Use* ***Inf*** *for any endpoint at infinity.)*

8. The examples we have considered so far in this section have all been for Taylor polynomials and series centered at 0, but Taylor polynomials and series can be centered at any value of a.

a. Let $f(x) = \cos(x)$. Find the Taylor polynomials up through order four of f centered at $a = \frac{\pi}{2}$. Then find the Taylor series for $f(x)$ centered at $a = \frac{\pi}{2}$. Why is the result not surprising?

b. Let $f(x) = \frac{1}{1+x} = (1+x)^{-1}$. Find the Taylor polynomials up through order four of f centered at $a = 1$. Then find the Taylor series for $f(x)$ centered at $a = 1$.

9. As we will see in more detail in the next section, we can use known Taylor series to obtain other Taylor series, and we preview that idea in this exercise.

a. Calculate the first four derivatives of $\sin(x^2)$ and hence find the fourth order Taylor polynomial for $\sin(x^2)$ centered at $a = 0$.

b. Part (a) demonstrates the direct approach to finding Taylor polynomials and series. Next we utilize a known Taylor series to make the process simpler. Recall that the Taylor series centered at 0 for $f(x) = \sin(x)$ is

$$T(x) = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}. \tag{8.4.7}$$

i. Substitute x^2 for x in the Taylor series $T(x)$ in Equation (8.4.7). Write out the first several terms and compare to your work in part (a). Explain why the substitution in this problem should result in the Taylor series for $\sin(x^2)$ centered at 0.

ii. For what interval of x-values should we expect the Taylor series for $\sin(x^2)$ to converge?

8.5 Finding and using Taylor series

Motivating Questions

- Is it possible to find the Taylor series expansion for a given function f without computing and evaluating various derivatives of f?
- Can we differentiate or integrate a Taylor series in the same way that we differentiate or integrate a polynomial? If so, how does doing so affect the values of x for which the Taylor series converges?

8.5.1 Introduction

So far, we have focused our attention on a collection of five basic functions functions — $\frac{1}{1-x}$, $\ln(1+x)$, $\sin(x)$, $\cos(x)$, and e^x — and their Taylor series centered at $a = 0$. One of the reasons we were able to find these Taylor series is the patterns that arise in the derivatives of each of these functions. While we can always use Definition 8.4.1 to find the first few terms of the Taylor series, for most functions it is much more challenging to find a pattern among the various derivatives that allows us to state the general n^{th} term of the series.

In Preview Activity 8.5.1, we explore this issue and investigate a different approach to finding the Taylor series of a given function.

Preview Activity 8.5.1. Let

$$g(x) = \frac{1}{1+x^2}.$$

(a) Note that we can write $g(x) = (1+x^2)^{-1}$. Determine $g'(x)$, $g''(x)$, and $g'''(x)$.

(b) In (a), you found that

$$g'''(x) = \frac{24x - 24x^3}{(1+x^2)^4}.$$

What derivative rules would be needed to find $g^{(4)}(x)$?

(c) Find $g(0)$, $g'(0)$, $g''(0)$, and $g'''(0)$, and use the results to determine $T_3(x)$.

(d) Since the derivatives of g become so complicated, we consider a different approach to finding the Taylor series centered at $a = 0$ for $g(x) = \frac{1}{1+x^2}$. We take advantage of the fact that $f(x) = \frac{1}{1-x}$ has a Taylor series expansion we already know, and observe that f has an algebraic structure similar to g.

We introduce a new variable, u, and recall that

$$f(u) = \frac{1}{1-u} = 1 + u + u^2 + u^3 + \cdots + u^n + \cdots \qquad (8.5.1)$$

for $|u| < 1$.

Let $u = -x^2$. What equation results from evaluating $f(-x^2)$ in Equation (8.5.1)?

(e) How does our work in (d) compare to the formula you found for $P_3(x)$ and suggest the Taylor series centered at $a = 0$ for $g(x) = \frac{1}{1+x^2}$? For what values of x do you expect this series will converge? Why?

8.5.2 Using substitution and algebra to find new Taylor series expressions

The substitution technique we used in Preview Activity 8.5.1 can be used to find the Taylor series for any function whose structure is similar to that of a Taylor series we already know.

Example 8.5.1 Find the Taylor series expansion for $g(x) = x^4 \cos(x^3)$ and determine the set of all x-values for which the series converges.

Solution. Because $\cos(x^3)$ is part of the function whose Taylor series we seek, we start with the familiar series for $\cos(x)$. We know that

$$f(u) = \cos(u) = 1 - \frac{1}{2!}u^2 + \frac{1}{4!}u^4 - \cdots + (-1)^n \frac{1}{(2n)!}u^{2n} + \cdots \tag{8.5.2}$$

for all real numbers u.

By letting $u = x^3$ in Equation (8.5.2), it follows that

$$\cos(x^3) = 1 - \frac{1}{2!}(x^3)^2 + \frac{1}{4!}(x^3)^4 - \cdots + (-1)^n \frac{1}{(2n)!}(x^3)^{2n} + \cdots,$$

so

$$\cos(x^3) = 1 - \frac{1}{2!}x^6 + \frac{1}{4!}x^{12} - \cdots + (-1)^n \frac{1}{(2n)!}x^{6n} + \cdots. \tag{8.5.3}$$

Then, multiplying both sides of Equation (8.5.3) by x^4, we find that

$$g(x) = x^4 \cos(x^3) = x^4 - \frac{1}{2!}x^{10} + \frac{1}{4!}x^{16} - \cdots + (-1)^n \frac{1}{(2n)!}x^{6n+4} + \cdots.$$

Since Equation (8.5.2) is valid for every real number u, letting $u = x^3$ tells us that Equation (8.5.3) is valid for every real number x. The Ratio Test can be used to show that multiplying every term of a Taylor series by the same power of x does not change the set of x-values for which the series converges, so the Taylor series for $g(x)$ also converges for every value of x. ◊

Because the approaches in Preview Activity 8.5.1 and Example 8.5.1 each require us to use a known Taylor series, we restate the Taylor series we've established so far for 5 important functions in Table 8.5.2. These 5 Taylor series will also be valuable when we soon learn another approach for finding series representations of other more complicated functions.

Important Taylor series representations.

Table 8.5.2 Taylor series and the open intervals of x-values where they converge for 5 important functions.

$f(x)$	Taylor series, T_f, centered at 0	$f(x) = T_f(x)$
$\frac{1}{1-x}$	$\sum_{k=0}^{\infty} x^k = 1 + x + x^2 + x^3 + \cdots$	if $\lvert x\rvert < 1$
$\ln(1+x)$	$\sum_{k=1}^{\infty} (-1)^{k-1} \frac{x^k}{k} = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \cdots$	if $\lvert x\rvert < 1$
$\sin(x)$	$\sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!} = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \cdots$	for all real x
$\cos(x)$	$\sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!} = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \cdots$	for all real x
e^x	$\sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \cdots$	for all real x

In Activity 8.5.2, we find several Taylor series using substitution and algebraic techniques.

Activity 8.5.2. Use known Taylor series and substitution and algebraic techniques to find a Taylor series representation for each of the following functions. In addition, state the interval of x-values for which you expect each Taylor series to converge.

(a) $g(x) = x^3 \sin(x^2)$

(b) $h(x) = e^{-x^4}$

(c) $p(x) = \dfrac{1}{1+5x}$

(d) $q(x) = x^2 \ln(1 + x^4)$

(e) $r(x) = \dfrac{e^{3x} - 1}{3x}$

8.5.3 Differentiating and integrating Taylor series

In Chapter 5, we discussed the challenge posed by definite integrals such as

$$\int_0^1 \sin(x^2)\, dx.$$

Because we are unable to find a simple algebraic antiderivative for the function $\sin(x^2)$, we cannot use the First Fundamental Theorem of Calculus to evaluate the integral exactly. We learned in Section 5.2 that the Second Fundamental Theorem of Calculus provides us with an antiderivative of a given function by using an integral function: one antiderivative of $f(x) = \sin(x^2)$ is

$$F(x) = \int_0^x \sin(u^2)\, du.$$

Our recent work with Taylor series now suggests another way to find an antiderivative $F(x)$ for $f(x) = \sin(x^2)$, and this approach also provides new options for finding additional Taylor series. In Activity 8.5.2, as part of our work in finding a Taylor series for $x^3 \sin(x^2)$ we found that

$$\sin(x^2) = x^2 - \frac{1}{3!}x^6 + \frac{1}{5!}x^{10} - \cdots + (-1)^{n-1}\frac{1}{(2n-1)!}x^{4n-2} + \cdots \tag{8.5.4}$$

and that this representation of $\sin(x^2)$ is valid for every value of x. This infinite series representation suggests that we could find an antiderivative $F(x)$ of $\sin(x^2)$ by using Equation (8.5.4) to actually evaluate the integral $\int_0^x \sin(u^2)\,du$. Doing so, we see

$$\begin{aligned}
F(x) &= \int_0^x \sin(u^2)\,du \\
&= \int_0^x \left(u^2 - \frac{1}{3!}u^6 + \frac{1}{5!}u^{10} + \cdots + (-1)^{n+1}\frac{1}{(2n-1)!}u^{4n-2} + \cdots\right) du \\
&= \frac{1}{3}u^3 - \frac{1}{7\cdot 3!}u^7 + \frac{1}{11\cdot 5!}u^{11} + \cdots + \frac{(-1)^{n+1}}{(4n-1)(2n-1)!}u^{4n-1} + \cdots \Big|_0^x \\
&= \left(\frac{1}{3}x^3 - \frac{1}{7\cdot 3!}x^7 + \frac{1}{11\cdot 5!}x^{11} + \cdots + \frac{(-1)^{n+1}}{(4n-1)(2n-1)!}x^{4n-1} + \cdots\right) - 0 \\
&= \frac{1}{3}x^3 - \frac{1}{7\cdot 3!}x^7 + \frac{1}{11\cdot 5!}x^{11} + \cdots + \frac{(-1)^{n+1}}{(4n-1)(2n-1)!}x^{4n-1} + \cdots
\end{aligned}$$

Hence we have found that

$$F(x) = \frac{1}{3}x^3 - \frac{1}{7\cdot 3!}x^7 + \frac{1}{11\cdot 5!}x^{11} + \cdots + \frac{(-1)^{n+1}}{(4n-1)(2n-1)!}x^{4n-1} + \cdots$$

is an antiderivative of $f(x) = \sin(x^2)$.

It turns out that integrating a Taylor series has no effect on the open[1] interval of convergence of a Taylor series, nor does differentiating such a series. This fact is stated formally in a result called The Power Series Differentiation and Integration Theorem. A **power series** is any series of the form

$$f(x) = \sum_{k=0}^{\infty} c_k(x-a)^k.$$

Every Taylor series is a power series, and a famous result called Borel's Theorem tells us that every power series is in fact the Taylor series of a related function.

The Power Series Differentiation and Integration Theorem.

If $f(x) = \sum_{k=0}^{\infty} c_k(x-a)^k$ converges on the open interval $|x-a| < R$ for a positive real number R, then

$$f'(x) = \sum_{k=1}^{\infty} k\cdot c_k(x-a)^{k-1}$$

[1]It is possible for the convergence status at the endpoints of the interval to change, but we are normally not concerned with those specific x-values.

and

$$\int f(x)\,dx = C + \sum_{k=0}^{\infty} \frac{c_k}{k+1}(x-a)^{k+1}$$

and both of these series converge on the same open interval, $|x-a| < R$.

Note how the index of representation of $f'(x)$ in sigma notation changes. This is because if we originally have

$$f(x) = c_0 + c_1(x-1) + c_2(x-a)^2 + c_3(x-a)^3 + \cdots + c_k(x-a)^k + \cdots$$

and we differentiate, we get

$$f'(x) = c_1 + 2c_2(x-a) + 3c_3(x_a)^2 + \cdots + k \cdot c_k(x-a)^{k-1} + \cdots$$

and so to have sigma notation for $f'(x)$, we start with $k = 1$ to accommodate the fact that the general term is $k \cdot c_k(x-a)^{k-1}$ in order to ensure all powers of x are nonnegative.

Furthermore, this result shows that differentiating or integrating a power series has no effect on radius of convergence of the power series. Stated more informally, the Power Series Differentiation and Integration Theorem tells us that when it comes to differentiating or integrating a Taylor series, we can do so just as if they were finite polynomials: we can differentiate or integrate the Taylor series term-wise following the Power Rule for differentiating or integrating x^n. Moreover, doing so doesn't change the open interval on which the Taylor series converges[2]. Since polynomials are the easiest of all functions to differentiate and integrate, we can now find many more Taylor series of interesting functions, and even find Taylor series of some familiar functions in different ways.

Example 8.5.3 Use the familiar Taylor series for $g(x) = \frac{1}{1-x}$ to develop the Taylor series for $h(x) = \ln(1+x)$ using the Power Series Differentiation and Integration Theorem.

Solution. In Example 8.2.5, we developed the Taylor polynomials centered at $a = 0$ for $h(x) = \ln(1+x)$, using the definition. And in Example 8.4.2, we considered the Taylor series for $h(x) = \ln(1+x)$ that we deduced from patterns in the Taylor polynomials.

Here, we take a different approach to finding the Taylor series for $h(x) = \ln(1+x)$ that starts with the familiar geometric series expansion of $g(x) = \frac{1}{1-x}$.

The key insight in this approach is the fact that

$$\frac{d}{dx}\left[\ln(1+x)\right] = \frac{1}{1+x}.$$

We thus first find a Taylor series representation for $\frac{1}{1+x}$, and then integrate to find the Taylor series for $\ln(1+x)$.

We know that for $|x| < 1$,

$$g(x) = \frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots + x^n + \cdots.$$

[2]Differentiating or integrating can change the convergence status at the endpoints of the interval, but we again will not concern ourselves with that issue .

Using the variable u instead (in anticipation of a change of variables), we have

$$g(u) = \frac{1}{1-u} = 1 + u + u^2 + u^3 + \cdots + u^n + \cdots.$$

Next, letting $u = -x$, it follows that

$$\begin{aligned} g(-x) = \frac{1}{1+x} &= 1 + (-x) + (-x)^2 + (-x)^3 + \cdots + (-x)^n + \cdots \\ &= 1 - x + x^2 - x^3 + \cdots + (-1)^n x^n + \cdots, \end{aligned}$$

which also converges for $|x| < 1$.

Now, since $\frac{1}{1+x}$ is the derivative of $\ln(1+x)$ and $\ln(1) = 0$, we can use the Second FTC to write

$$\ln(1+x) = \int_0^x \frac{1}{1+t}\, dt,$$

which combined with our earlier series representation for $\frac{1}{1+x}$ shows that

$$\begin{aligned} \ln(1+x) &= \int_0^x \frac{1}{1+t}\, dt \\ &= \int_0^x 1 - t + t^2 - t^3 + \cdots \, dt \\ &= t - \frac{1}{2}t^2 + \frac{1}{3}t^3 - \frac{1}{4}t^4 + \cdots \Big|_0^x \\ &= x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \cdots \end{aligned}$$

and this series is guaranteed to converge on the same open interval as the series we started with $(-1 < x < 1)$. This is also the same series we found for $h(x) = \ln(1+x)$ working directly from the definition of a Taylor polynomial in Example 8.2.5. ◊

Activity 8.5.3. In this activity we determine the Taylor series expansion for $\sin(x)$ in a different way and then also find the Taylor series for $\arctan(x)$.

(a) To find the Taylor series for $\sin(x)$ without using the definition, we consider the function $f(x) = \cos(x)$ and recall that $f'(x) = -\sin(x)$. Then we use the Power Series Differentiation Theorem on the known series for $f(x) = \cos(x)$ to find the Taylor series for $f'(x) = -\sin(x)$.

(i) Consider $f(x) = \cos(x)$ and recall that its Taylor series in sigma notation is

$$f(x) = \cos(x) = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!}.$$

Write out the first four nonzero terms of the Taylor series in expanded form.

(ii) Using the expanded form of the Taylor series for $f(x) = \cos(x)$, apply the Power Series Differentiation Theorem to find the Taylor series for $f'(x) =$

$-\sin(x)$ in expanded form, being sure to simplify each constant coefficient in the expansion. Show the first three nonzero terms.

(iii) Now recall the series expansion of $\cos(x)$ in sigma notation. What is the derivative of the general term of the series,

$$\frac{d}{dx}\left[(-1)^k \frac{x^{2k}}{(2k)!}\right],$$

in its most simplified form?

(iv) Finally, state the Taylor series for $f'(x) = -\sin(x)$ in sigma notation. Be sure to decide whether the series should be indexed starting with $k = 0$ or $k = 1$ and to justify your choice.

(v) How can you now use your work in part (iv) to find the Taylor series for $\sin(x)$?

(b) Consider the function $g(x) = \arctan(x)$.

(i) What is $g'(x)$?

(ii) Recall that in Preview Activity 8.5.1, we found the Taylor series expansion for

$$f(x) = \frac{1}{1+x^2}.$$

State the Taylor series for $f(x)$ that you found in the preview activity.

(iii) Explain why we can think of $g(x)$ as being given by

$$g(x) = \int_0^x f(u)\, du.$$

Use this relationship to find a power series expansion for $g(x)$.

(iv) For what interval of x-values is the Taylor series for $g(x) = \arctan(x)$ guaranteed to converge?

Activity 8.5.4. In this activity we will determine the Taylor series expansion for the famous *error function*,

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\, dt,$$

that arises in probability theory and other important applications.

(a) Use the Taylor series for e^x to find the Taylor series for e^{-t^2}.

(b) Next, evaluate the integral $\int_0^x e^{-t^2}\, dt$ by replacing e^{-t^2} with its Taylor series.

(c) Use your work in (b) to state the Taylor series for $\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\, dt$

(d) For what interval of x-values will the Taylor series for $\mathrm{erf}(x)$ converge? Why?

(e) In probability theory, $\text{erf}(x)$ is important because of its connection to the *normal distribution*, which is represented by a bell curve. Indeed, $\text{erf}(x)$ represents the fraction of a normally distributed characteristic in a population that lies between 0 and x. How can you use your result in (c) to estimate erf(0.5)?

Near the end of Section 8.4, we noted the important big-picture perspective that for familiar basic functions that are infinitely differentiable, such as $f(x) = \sin(x)$, not only can we find the function's Taylor series and determine the open interval of x-values for which the Taylor series converges, but *the Taylor series converges to the function itself*. Furthermore, we noted that these representations play a key role in how computers provide decimal approximations to quantities such as e^1, which can be represented as

$$e^1 = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \cdots.$$

Our most recent work with Taylor series shows that the news is better still: now we can easily represent even more complicated functions with Taylor series, such as $f(x) = e^{-x^2}$ and $g(x) = \sin(x^2)$, and determine their antiderivatives using their infinite Taylor series and treating that representation just like a polynomial. In Section 8.6, we investigate further how certain infinite series of numbers can be easily and accurately approximated using partial sums that result from evaluating Taylor series, and how we can even quantify how much error is present in certain partial sum approximations.

8.5.4 Summary

- Through substitution, we can use any known Taylor series to find the Taylor series of a related function. For example, we know that if $|u| < 1$,

  $$\ln(1+u) = u - \frac{1}{2}u^2 + \frac{1}{3}u^3 - \frac{1}{4}u^4 + \cdots$$

  Letting $u = 4x^2$, it follows that

  $$\ln(1+4x^2) = 4x^2 - \frac{1}{2}\cdot 16x^4 + \frac{1}{3}\cdot 64x^6 - \frac{1}{4}\cdot 256x^8 + \cdots$$

  and that this representation converges if $|4x^2| < 1$, so for x such that $|x| < \frac{1}{2}$.

- The Power Series Differentation and Integration Theorem tells us that we can differentiate or integrate a Taylor series in the natural way and that doing so has essentially no impact on the set of x-values for which the series converges. For instance, we might note that if we first found the Taylor series for $\sin(x)$, which is

  $$\sin(x) = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \cdots$$

  and converges for every value of x, it follows by differentating that

  $$\cos(x) = \frac{d}{dx}[\sin(x)]$$

$$= \frac{d}{dx}[x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \cdots]$$
$$= 1 - \frac{1}{3!} \cdot 3x^2 + \frac{1}{5!} \cdot 5x^4 - \frac{1}{7!} \cdot 7x^6 + \cdots$$
$$= 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \cdots$$

which is precisely the Taylor series for $\cos(x)$ that we found by taking derivatives and applying Definition 8.4.1.

8.5.5 Exercises

1. If $y = \sum_{k=0}^{\infty} (k+1)\, x^{k+3}$ then $y' = \sum_{k=0}^{\infty}$ ____________________________

2.

(a) *Part 1.*

Note that when the constant in the denominator is not 1, we still can use the geometric series, but we have to multiply by a form of 1 that helps us. We'll use $\frac{\frac{1}{4}}{\frac{1}{4}}$.

$$\frac{1}{(4+x)} = \frac{\frac{1}{4}}{\frac{1}{4}(4+x)} = \frac{\frac{1}{4}}{1+\frac{1}{4}x} = \frac{1}{4} \cdot \frac{1}{1+\frac{1}{4}x}$$

And now we can use the geometric series expression to get

$$\frac{1}{(4+x)} = \frac{1}{4} \cdot \frac{1}{1+\frac{1}{4}x} = \frac{1}{4}\sum_{n=0}^{\infty}\left(\frac{-1}{4}x\right)^n = \frac{1}{4}\sum_{n=0}^{\infty}\left(\frac{-1}{4}\right)^n x^n$$

Bringing the coefficient inside the sum means that

$$\frac{1}{(4+x)} = \sum_{n=0}^{\infty} \frac{(-1)^n}{4^{n+1}} x^n$$

Differentiate both sides of the equation above and use the result to express the following function as a power series centered at $x = 0$.

$$f(x) = \frac{1}{(4+x)^2}$$

$f(x) = \sum_{n=1}^{\infty}$ ________________________

(b) *Part 2.*

Use your answer above and more differentiation to now express the following function as a power series centered at $x = 0$.

$$g(x) = \frac{1}{(4+x)^3}$$

$g(x) = \sum_{n=2}^{\infty}$ ____________________

(c) *Part 3.*

Use your answers above to now express the function as a power series centered at $x = 0$.

$$h(x) = \frac{x^2}{(4+x)^3}$$

$h(x) = \sum_{n=2}^{\infty}$ ____________________

3. For the following indefinite integral, find the full power series centered at $t = 0$ and then give the first 5 nonzero terms of the power series and the open interval of convergence.

$$f(t) = \int \frac{t}{1+t^5}\, dt$$

$f(t) = C + \sum_{n=0}^{\infty}$ ____________________

$f(t) = C+$ __________ $+$ __________ $+$ __________ $+$ __________ $+$ __________ $+\cdots$

The open interval of convergence is: ____________________ (Give your answer in .)

4. For the following indefinite integral, find the full power series centered at $x = 0$ and then give the first 5 nonzero terms of the power series and the open interval of convergence.

$$f(x) = \int x^2 \ln(1+x)\, dx$$

$f(x) = C + \sum_{n=1}^{\infty}$ ____________________

$f(x) = C+$ __________ $+$ __________ $+$ __________ $+$ __________ $+$ __________ $+\cdots$

The open interval of convergence is: ____________________ (Give your answer in .)

5. You can interpret "power series" in the following problem statement as "Taylor series".

Represent the function $\dfrac{7}{(1-10x)}$ as a power series $f(x) = \sum_{n=0}^{\infty} c_n x^n$

$c_0 =$ ____________________

$c_1 =$ ____________________

$c_2 =$ ____________________

$c_3 =$ ____________________

$c_4 =$ ____________________

Find the radius of convergence $R =$ ____________________ .

6. You can interpret "power series" in the following problem statement as "Taylor series". You might also want to recall the result of Activity 8.5.3 where we found the series expansion for $\arctan(x)$.

The function $f(x) = 8x \arctan(5x)$ is represented as a power series

$$f(x) = \sum_{n=0}^{\infty} c_n x^n.$$

Find the first few coefficients in the power series.

$c_0 =$ ____________________

$c_1 =$ ____________________

$c_2 =$ ____________________

$c_3 =$ ____________________

$c_4 =$ ____________________

Find the radius of convergence R of the series.

$R =$ ____________________

7. In this exercise we find the Taylor series representation for two famous functions, the Fresnel integral functions

$$C(x) = \int_0^x \cos(t^2)\, dt$$

and

$$S(x) = \int_0^x \sin(t^2)\, dt.$$

The Fresnel integral functions are important in optics and are used in the design of Fresnel lenses such as those found in lighthouses along the Lake Michigan shore.

a. Use the Taylor series for $\cos(x)$ to find the Taylor series for $\cos(t^2)$ and hence write $C(x)$ as a Taylor series.

b. For what interval of x-values will the Taylor series for $C(x)$ converge? Why?

c. Apply your result from (a) to estimate $C(0.5)$ to within 0.001.

d. Similarly, use the Taylor series for $\sin(x)$ to find the Taylor series for $\sin(t^2)$ and hence write $S(x)$ as a Taylor series.

e. For what interval of x-values will the Taylor series for $S(x)$ converge? Why?

f. Apply your result from (d) to estimate $S(0.8)$ to within 0.001.

8. The fact that we can differentiate or integrate a Taylor series reveals other important ways we can think about functions such as e^x.

a. Perhaps the most important property of the function $h(x) = e^x$ is that $h'(x) = e^x$; that is, the function e^x is its own derivative. Suppose that we didn't yet know the coefficients of the Taylor series expansion for e^x, so we just said

$$e^x = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots + a_n x^n + \cdots . \tag{8.5.5}$$

Let $x = 0$ in Equation (8.5.5); what does this tell us about the value of a_0?

b. Take the derivative of both sides of Equation (8.5.5) and call your resulting equation for e^x "Equation 2". Why do Equation (8.5.5) and Equation 2 together tell us that $a_1 = a_0$? Combine this observation with your conclusion in (b) and note that you now know the numerical value of both a_0 and a_1.

c. Why do Equation (8.5.5) and Equation 2 together tell us that $a_2 = \frac{1}{2}a_1$?

d. Continue reasoning similarly to find the value of a_3, a_4, and a_5. What do you observe?

9. Taylor series also provide an alternate way to evaluate indeterminate limits.

a. Find the Taylor series for $\sin(2t)$ and use it to evaluate the indeterminate limit given by

$$\lim_{t \to 0} \frac{\sin(2t)}{t}.$$

Compare your result to what follows from L'Hôpital's Rule (see Section 3.2 as needed).

b. Consider the indeterminate limit given by

$$\lim_{x \to 0} \frac{e^x - 1 - x}{\cos(2x) - 1}.$$

Find the Taylor series representations for $f(x) = e^x - 1 - x$ and $g(x) = \cos(2x) - 1$ and use them to evaluate the given limit. How does your result compare to using L'Hôpital's Rule?

8.6 Quantifying the accuracy of approximations

Motivating Questions

- For an infinite series that converges, how much accuracy do we lose when we truncate the series after a finite number of terms to approximate its value with a finite sum? Said differently, to get a good approximation with a finite sum, how many terms are enough?
- What is an alternating series and how can we determine whether or not it converges? How do the partial sums of a converging alternating series accurately estimate its exact sum?
- What is the Lagrange Error Bound and how can we use it to understand how accurate a function's degree-n Taylor polynomial approximation is?

8.6.1 Introduction

This section is all about understanding precisely how accurate certain approximations are. Through Taylor series, we now know that we can represent certain functions and their values exactly through infinite sums. For example, we know that for any value of x,

$$e^x = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \cdots + \frac{1}{n!}x^n + \cdots$$

so it follows that

$$e^{-1/2} = 1 + \left(\frac{-1}{2}\right) + \frac{1}{2!}\left(\frac{-1}{2}\right)^2 + \cdots + \frac{1}{n!}\left(\frac{-1}{2}\right)^n + \cdots. \tag{8.6.1}$$

In fact, Equation (8.6.1) is one way[1] that a calculator or computer can respond to the human prompt "$e^{-1/2}$". Note that the right-hand side of the equation only uses addition and multiplication, and that the special number e makes no appearance on the right. A natural question is: how does a calculator or computer know that the decimal value it returns is as accurate as the decimal representation it displays?

For instance, if we decide to estimate the exact value of $e^{-1/2}$ using the first 5 terms of this infinite sum, it turns out to be possible to use both the fact that we chose "5" and properties of $f(x) = e^x$ to quantify the maximum error in the estimate

$$e^{-1/2} \approx 1 + \left(\frac{-1}{2}\right) + \frac{1}{2!}\left(\frac{-1}{2}\right)^2 + \frac{1}{3!}\left(\frac{-1}{2}\right)^3 + \frac{1}{4!}\left(\frac{-1}{2}\right)^4.$$

It also turns out to be possible to determine how many terms are needed in order to find an approximation that meets a desired level of accuracy. These questions and issues are what this section is about.

[1] Mathematicians have devised other clever and efficient methods for computing values of transcendental functions, such as sin(1) and ln(2). For example, a method known as CORDIC (that remarkably doesn't even use multiplication) is employed by many calculating devices to compute values of trigonometric functions, such as sin(1).

As we have seen, most of the infinite series that arise (such as the one in Equation (8.6.1)) are not geometric. But since geometric series are among the easiest infinite series to understand and evaluate, we use a geometric series as the starting point for our analysis of approximation errors. In Preview Activity 8.6.1, we investigate an example of a convergent geometric series and explore certain patterns that arise when we compare partial sums the exact sum of the series.

Preview Activity 8.6.1. Consider the alternating geometric series

$$S = \sum_{k=0}^{\infty} (-1)^k \left(\frac{4}{5}\right)^k = 1 - \frac{4}{5} + \frac{16}{25} - \cdots + (-1)^{n-1} \left(\frac{4}{5}\right)^{n-1} + \cdots.$$

We want to explore how the partial sums of the series compare to and approximate the exact sum of the series, which is

$$S = \frac{a}{1-r} = \frac{1}{1-\left(\frac{-4}{5}\right)} = \frac{1}{\frac{9}{5}} = \frac{5}{9}.$$

(a) Recall that the nth partial sum, S_n, is the sum of the first n terms of the infinite geometric series S. This means that

$$S_n = \sum_{k=0}^{n-1} (-1)^k \left(\frac{4}{5}\right)^k = 1 - \frac{4}{5} + \frac{16}{25} - \cdots + (-1)^{n-1} \left(\frac{4}{5}\right)^{n-1}.$$

Note that the exact fractional values of $S_1, S_2, \ldots, S_6$, have been recorded below along with their decimal representations. Your task is to use this information to do some related computations that help us understand the behavior of the series and its partial sums.

First, by computing the differences between S_n and S for several different values of n (recalling that $S = \frac{5}{9} = 0.\overline{5}$), fill in the first column of blank spaces provided below. Then, by viewing the series $S = 1 - \frac{4}{5} + \frac{16}{25} - \cdots + (-1)^{n-1} \left(\frac{4}{5}\right)^{n-1} + \cdots$ as being in the form $S = a_0 + a_1 + a_2 + \cdots$, fill in the last column of blank spaces.

$n = 1$	$S_1 = 1$	$S_1 - S =$	$a_1 = -0.8$
$n = 2$	$S_2 = \frac{1}{5} = 0.2$	$S_2 - S =$	$a_2 = 0.64$
$n = 3$	$S_3 = \frac{21}{25} = 0.84$	$S_3 - S =$	$a_3 =$
$n = 4$	$S_4 = \frac{41}{125} = 0.328$	$S_4 - S =$	$a_4 =$
$n = 5$	$S_5 = \frac{461}{625} = 0.7376$	$S_5 - S =$	$a_5 =$
$n = 6$	$S_6 = \frac{1281}{3125} = 0.40992$	$S_6 - S =$	$a_6 =$

(b) What do you notice about the differences between S_n and S as the value of n increases? There are at least two important things you can say.

(c) What do you notice about how the differences between S_n and S compare to the value of a_n? There are at least two important things you can say.

8.6.2 Alternating series of real numbers

In several situations we've encountered, series whose terms alternate in sign arise naturally. For instance, consider the definite integral

$$\int_0^1 e^{-x^2}\,dx,$$

which is related to the well-known error function, erf(x). While we are unable to find an elementary algebraic antiderivative of e^{-x^2}, if we use the Taylor series

$$e^{-x^2} = 1 - x^2 + \frac{1}{2!}x^4 - \frac{1}{3!}x^6 + \cdots + (-1)^n\frac{1}{n!}x^{2n} + \cdots$$

and apply the Fundamental Theorem of Calculus to the series representation of e^{-x^2}, it turns out that

$$\int_0^1 e^{-x^2}\,dx = 1 - \frac{1}{3} + \frac{1}{2!\cdot 5} - \frac{1}{3!\cdot 7} + \cdots + \frac{(-1)^n}{n!\cdot(2n+1)} + \cdots \tag{8.6.2}$$

Like the geometric series $\sum_{k=0}^{\infty}(-1)^k\left(\frac{4}{5}\right)^k$ that we encountered in Preview Activity 8.6.1, the infinite series in Equation (8.6.2) is an example of an **alternating series** of real numbers. It turns out to be straightforward to determine whether or not an alternating series converges and to estimate the value of a convergent alternating series.

Definition 8.6.1 An **alternating series** is a series of the form

$$\sum_{k=0}^{\infty}(-1)^k a_k,$$

where $a_k > 0$ for each k. ◊

Note that a_k just represents the non-alternating part of the series. For example, using the series from Preview Activity 8.6.1,

$$\sum_{k=0}^{\infty}(-1)^k\left(\frac{4}{5}\right)^k = 1 - \frac{4}{5} + \frac{16}{25} - \frac{64}{125} + \cdots,$$

the expression a_k corresponds to $a_k = \left(\frac{4}{5}\right)^k$.

In Preview Activity 8.6.1, we investigated the partial sums, S_n, of the convergent alternating geometric series $S = \sum_{k=0}^{\infty}(-1)^k\left(\frac{4}{5}\right)^k$, whose sum is $S = \frac{5}{9}$. For instance, S_3 is the sum of the first 3 terms of the infinite series.

If we plot the partial sums as ordered pairs of the form (n, S_n), as shown in Figure 8.6.2, we see important patterns in how the partial sums both approach and differ from the exact value of the infinite sum.

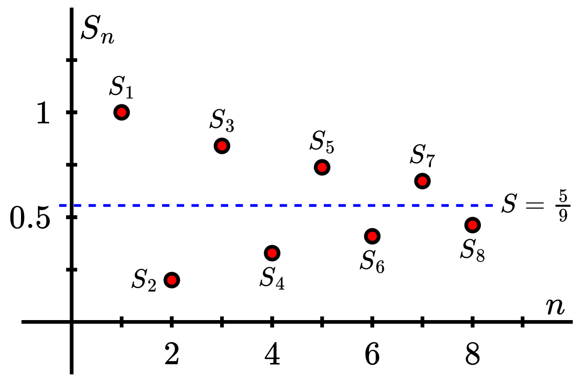


Figure 8.6.2 Plotting the partial sums $S_1, S_2, \ldots, S_8$. The dashed horizontal line represents the exact sum of the infinite series, $S = \frac{5}{9}$.

In both our work in the Preview Activity and in Figure 8.6.2, we see how consecutive partial sums oscillate back and forth above and below the exact sum of the infinite series, $\frac{5}{9}$, and moreover how the absolute difference between S and S_n decreases as n increases.

The quantity $|S - S_n|$ is the error in the partial sum approximation of the infinite series; we can study how this error compares to terms in the series itself. For instance,

$$|S - S_5| = \left|\frac{5}{9} - S_5\right| \approx 0.1820. \tag{8.6.3}$$

Observe that while S_5 is greater than S, S_6 falls below S, so

$$|S - S_5| < |S_6 - S_5|. \tag{8.6.4}$$

Remember that S_5 is the partial sum of the first 5 terms of the series, and likewise S_6 is simply the sum of the first 6 terms. This means that $|S_6 - S_5|$ is the absolute value of the 6^{th} term of the series, or $|S_6 - S_5| = \left(\frac{4}{5}\right)^5 = 0.32768$. Combining (8.6.3) and (8.6.4), we have

$$\left|\frac{5}{9} - S_5\right| \approx 0.1820 < \left(\frac{4}{5}\right)^5 = 0.32768$$

Making this argument for an arbitrary value of n shows that the next term in the alternating sum always provides a bound on the error that arises from a given partial sum.

Our work with the geometric series $S = \sum_{k=0}^{\infty}(-1)^k \left(\frac{4}{5}\right)^k$ suggests two general results that are true for any alternating series.

The Alternating Series Test.

Given an alternating series

$$\sum_{k=0}^{\infty}(-1)^k a_k,$$

if the positive terms a_k decrease to 0 as $k \to \infty$, then the alternating series converges.

If we were to compute the partial sums S_n of any alternating series whose terms decrease to zero and plot the points (n, S_n) as we did in Figure 8.6.2, we would see a similar picture: the partial sums alternate above and below the value to which the infinite alternating series converges. In addition, because the terms go to zero, the amount a given partial sum deviates from the total sum is at most the next term in the series. This result is formally stated as the Alternating Series Estimation Theorem.

Alternating Series Estimation Theorem.

If the alternating series $S = \sum_{k=0}^{\infty}(-1)^k a_k$ has positive terms a_k that decrease to zero as $k \to \infty$, and $S_n = \sum_{k=0}^{n-1}(-1)^k a_k$ is the nth partial sum of the alternating series, then

$$|S - S_n| = \left|\sum_{k=0}^{\infty}(-1)^k a_k - S_n\right| < a_n.$$

Again, this result simply says: if we use a partial sum to estimate the exact sum of a convergent alternating infinite series, the absolute error of the approximation is less than the next term in the series.

Example 8.6.3 Determine how well the 100$^{\text{th}}$ partial sum S_{100} of

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{k+1}$$

approximates the value of the converging alternating series.

Solution. If we let S be the value of the series $\sum_{k=0}^{\infty} \frac{(-1)^k}{k+1}$, then we know that

$$|S - S_{100}| < a_{100}.$$

Now

$$a_{100} = \frac{1}{101} \approx 0.0099,$$

so the 100$^{\text{th}}$ partial sum is within 0.0099 of the exact value of the series. In addition, it turns out that $S_{100} \approx 0.688172$ and $S = \ln(2) \approx 0.69314$, so we see that the actual difference between S_{100} and S is

$$|S - S_{100}| \approx |\ln(2) - 0.688172| \approx 0.004975,$$

which is indeed less than the error bound of 0.0099 provided by the Alternating Series Estimation Theorem. ◊

Activity 8.6.2. In this activity we approximate the values of several different alternating series using the Alternating Series Estimation Theorem.

(a) Use the fact that $\sin(x) = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \cdots$ to estimate $\sin(1)$ to within 0.0001. Do so without entering "sin(1)" on a computational device. After you find your estimate, enter "sin(1)" on a computational device and compare the results.

(b) Recall our recent work with $\int_0^1 e^{-x^2}\,dx$ which can be expressed as the series

$$\int_0^1 e^{-x^2}\,dx = 1 - \frac{1}{3} + \frac{1}{2!\cdot 5} - \frac{1}{3!\cdot 7} + \cdots + \frac{(-1)^n}{n!\cdot(2n+1)} + \cdots.$$

Use this series representation to estimate $\int_0^1 e^{-x^2}\,dx$ to within 0.0001. Then, compare what a computational device reports when you use it to estimate the definite integral.

(c) Find the Taylor series for $\cos(x^2)$ and then use the Taylor series and to estimate the value of $\int_0^1 \cos(x^2)\,dx$ to within 0.0001. Compare your result to what a computational device reports when you use it to estimate the definite integral.

(d) Recall we know that if $|x| < 1$, then

$$\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \cdots + (-1)^{n-1}\frac{1}{n}x^n + \cdots.$$

What happens if $x = 1$?

Explain why the series $1 - \frac{1}{2}\cdot 1^2 + \frac{1}{3}\cdot 1^3 - \cdots + (-1)^{n-1}\frac{1}{n}\cdot 1^n + \cdots$ must converge and estimate its sum to within 0.01. What is the exact sum of this series?

8.6.3 Error Approximations for Taylor Polynomials

In the same way that the next term in an alternating series reveals the maximum error of a finite approximation, it turns out that a quantity related to the next term in a Taylor series determines the maximum error found in a Taylor polynomial approximation. This will enable us to know the degree of the Taylor polynomial that is needed in order to achieve a given accuracy tolerance on a chosen interval.

Throughout what follows, we assume that $f(x)$ is a function with at least $n+1$ derivatives at $a = 0$, and let $T_n(x)$ be its degree n Taylor polynomial centered at $a = 0$. We define the error function of the degree n approximation, $E_n(x)$, by

$$E_n(x) = f(x) - T_n(x).$$

Our overall goal is to understand how much error there is in the approximation $f(c) \approx T_n(c)$ for some fixed value of c; in other words, we'd like to know the maximum possible value of $|E_n(c)|$.

This leads us to maximize the error function, $E_n(x)$, on the interval $[0, c]$. We observe several important properties of $E_n(x)$:

- $E_n(0) = E_n'(0) = E_n''(0) = \cdots = E_n^{(n)}(0) = 0$, since $E_n(x) = f(x) - T_n(x)$ and f and T_n share the same function value and same first n derivative values at $a = 0$;
- $E_n^{(n+1)}(x) = f^{(n+1)}(x)$, since T_n is a degree n polynomial, which means $T_n^{(n+1)}(x) = 0$ for every x;

- If we assume that $|f^{(n+1)}(x)| \leq M$ for some positive real number M for every x in the interval $[0, c]$, then
$$\left|E_n^{(n+1)}(x)\right| \leq M \tag{8.6.5}$$
since $f^{(n+1)}(x) = E_n^{(n+1)}(x)$.

Taking Inequality (8.6.5), writing it in the form $-M < E_n^{(n+1)}(t) \leq M$, and integrating all three terms in the inequality from $t = 0$ to $t = x$ and doing so $n+1$ times, it can be shown that
$$-M\frac{x^{n+1}}{(n+1)!} \leq E_n(x) \leq M\frac{x^{n+1}}{(n+1)!}$$
It follows that
$$|E_n(x)| \leq M\frac{|x|^{n+1}}{(n+1)!} \leq M\frac{|c|^{n+1}}{(n+1)!}$$
since $|c|$ is the maximum value of $|x|$ on the interval $0 \leq x \leq c$.

A similar argument works if we center the Taylor polynomial approximation at any real number a and leads to the following result, known as the **Lagrange Error Bound**.

The Lagrange Error Bound for $T_n(x)$.

Let f be a continuous function with $n+1$ continuous derivatives. Suppose that M is a positive real number such that $\left|f^{(n+1)}(x)\right| \leq M$ on the interval $[a, c]$. If $T_n(x)$ is the degree n Taylor polynomial for $f(x)$ centered at $x = a$, then
$$\left|f(c) - T_n(c)\right| \leq M\frac{|c-a|^{n+1}}{(n+1)!}.$$

The Lagrange Error Bound shows that the total error in the degree n Taylor polynomial approximation is determined by a combination of:

- the maximum value of the $(n+1)^{\text{st}}$ derivative of f on the interval of interest (which is a measure of how much polynomial approximation is being missed by using the degree n approximation);
- the quantity $|c-a|$ (which measures how far we've moved away from where the approximation is centered);
- and the degree n of the approximation, captured in two parts of the term $\frac{|c-a|^{n+1}}{(n+1)!}$ (which makes sense since we've learned that the higher the degree, the better the approximation).

In practice, the quantity $M\frac{|c-a|^{n+1}}{(n+1)!}$ provides a straightforward way to find a bound on the maximum error of a polynomial approximation of a non-polynomial function.

Example 8.6.4 Determine the maximum error possible when using the degree 10 Taylor polynomial centered at $a = 0$ for $\sin(x)$ to approximate the value of $\sin(2)$.

Solution. To approximate the value of $\sin(2)$ using the degree 10 Taylor polynomial, we will use $f(x) = \sin(x)$, $c = 2$, $a = 0$, and $n = 10$ in the Lagrange Error Bound formula. We also need to find a value for M so that $\left|f^{(11)}(x)\right| \leq M$ on the interval $[0, 2]$.

Since the derivatives of $f(x) = \sin(x)$ are all one of $\pm\sin(x)$ or $\pm\cos(x)$, and the maximum absolute values of these functions on any interval is 1, we have

$$\left|f^{(n+1)}(x)\right| \leq 1$$

for any n and any x. Therefore, we let $M = 1$. By the Lagrange Error Bound formula, we find that

$$\left|f(2) - T_{10}(2)\right| \leq 1 \cdot \frac{|2-0|^{11}}{(11)!} = \frac{2^{11}}{(11)!} \approx 0.00005130671797.$$

So $T_{10}(2)$ approximates $\sin(2)$ to within at most 0.00005130671797. Using a computer algebra system, we can find that

$$T_{10}(2) \approx 0.9093474427 \quad \text{and} \quad \sin(2) \approx 0.9092974268$$

with an actual difference of about 0.0000500159. Note that the computer algebra system itself is using a related approach (possibly with more terms) to generate the fact that $\sin(2) \approx 0.9092974268$ in such a way that every decimal the computer reports is accurate. $\lozenge$

Activity 8.6.3. In this activity we apply the Lagrange Error Bound to quantify the accuracy of several different approximations.

(a) Use the degree 10 Taylor polynomial (centered at $a = 0$) of $f(x) = e^x$ to estimate the value of e^2. What is the maximum error of your estimate, according to the Lagrange Error Bound? How does this compare to the actual error between e^2 and $T_{10}(2)$ as reported by a computer algebra system?

(b) Use a degree n Taylor polynomial (centered at $a = 0$) of $f(x) = \cos(x)$ to estimate the value of $\cos(1)$ within an accuracy of 0.00000001. According to the Lagrange Error Bound, what value of n is needed to achieve this accuracy? What is the resulting approximate value of $\cos(1)$?

(c) Recall that for $f(x) = \ln(1 + x)$, its Taylor series centered at $a = 0$ is given by

$$\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \cdots + (-1)^{n-1}\frac{1}{n}x^n + \cdots$$

and that the n^{th} derivative of $f(x) = \ln(1 + x)$ is given by

$$f^{(n)}(x) = \frac{(-1)^{n-1}(n-1)!}{(1+x)^n}.$$

If we want to estimate $f(0.5) = \ln(1.5)$ to within an accuracy of 0.0001, what value of n is needed to achieve this accuracy from computing $T_n(0.5)$, according to the Lagrange Error Bound?

8.6.4 Summary

- If we approximate an infinite series with one of its partial sums, we can often quantify the maximum error present in the truncated sum. Two ways the error can be measured are through the Alternating Series Estimation Theorem (if the original series is alternating) and through the Lagrange Error Bound (if the original series can be viewed as a Taylor series).

- An alternating series is one whose terms alternate in sign, often represented by

$$\sum_{k=0}^{\infty}(-1)^k a_k$$

where $a_k > 0$ for all values of k. Any alternating series whose terms a_k approach zero as $k \longrightarrow \infty$ is guaranteed to converge. Moreover, the Alternating Series Estimation Theorem tells us that we can estimate the exact value of a converging alternating series by using a partial sum, and the error of that approximation is at most the next term in the series. That is,

$$\left|\sum_{k=0}^{\infty}(-1)^k a_k - \left(a_0 - a_1 + a_2 - a_3 + \cdots + (-1)^{n-1}a_{n-1}\right)\right| < a_n.$$

- The Lagrange Error Bound quantifies the accuracy when we a Taylor polynomial to approximate a function. Specifically, if $T_n(x)$ is the degree n order Taylor polynomial for f centered at $x = a$ and if $\left|f^{(n+1)}(x)\right| \leq M$ for some real number M on the interval $[a, c]$, then

$$\left|f(c) - T_n(c)\right| \leq M\frac{|c-a|^{n+1}}{(n+1)!}.$$

8.6.5 Exercises

1. For the following alternating series,

$$\sum_{n=1}^{\infty} a_n = 0.6 - \frac{(0.6)^3}{3!} + \frac{(0.6)^5}{5!} - \frac{(0.7)^7}{7!} + \ldots$$

how many terms do you have to compute in order for your approximation (your partial sum) to be within 0.0000001 from the convergent value of that series?

2. For the following alternating series,

$$\sum_{n=1}^{\infty} a_n = 1 - \frac{1}{10} + \frac{1}{100} - \frac{1}{1000} + \ldots$$

how many terms do you have to go for your approximation (your partial sum) to be within 1e-10 from the convergent value of that series?

3. Consider the function $f(x) = \frac{1}{x^2}$.

Let $T_n(x)$ be the n^{th} degree Taylor approximation of $f(x)$ about $x = 1$.

Use 4 decimal places in your two answers below, but make sure you carry all decimals when performing calculations.

Calculate:

$T_1(1.02) =$ __________

$T_2(1.02) =$ __________

$T_2(1.02)$ is an (□ over □ under) estimate of $f(1.02)$.

Using the Lagrange Error Bound for $T_n(x)$, calculate:

$\left|f(1.02) - T_2(1.02)\right| \leq$ __________

4. Consider the function $f(x) = \sqrt{x+1}$.

Let $T_n(x)$ be the n^{th} degree Taylor approximation of $f(x)$ about $x = 8$.

Use 3 decimal places in your two answers below, but make sure you carry all decimals when performing calculations.

Calculate:

$T_1(10) =$ _______________

$T_2(10) =$ _______________

$T_2(10)$ is an (□ over □ under) estimate of $f(10)$.

Using the Lagrange Error Bound for $T_n(x)$, calculate:

$\left|f(10) - T_2(10)\right| \leq$ __________

5. Consider the function $f(x) = x\ln(x)$.

Let $T_n(x)$ be the n^{th} degree Taylor approximation of $f(x)$ about $x = 1$.

Use 3 decimal places in your three answers below, but make sure you carry all decimals when performing calculations.

Calculate:

$T_1(2) =$ ____________________

$T_2(2) =$ ____________________

$T_3(2) =$ ____________________

$T_3(2)$ is an (□ over □ under) estimate of $f(2)$.

Using the Lagrange Error Bound for $T_n(x)$, calculate:

$\left|f(2) - T_3(2)\right| \leq$ __________

6. Use series to approximate the definite integral to within the indicated accuracy:

$$\int_0^1 \sin(x^3)\,dx, \text{ with an error } < 10^{-4}$$

Note: The answer you derive here should be the partial sum of an appropriate series (the number of terms determined by an error estimate). This number is not necessarily the correct value of the integral truncated to the correct number of decimal places.

7. Use series to approximate the definite integral to within the indicated accuracy:

$$\int_0^{0.4} e^{-x^3}\,dx, \text{ with an error } < 10^{-4}$$

Note: The answer you derive here should be the partial sum of an appropriate series (the number of terms determined by an error estimate). This number is not necessarily the correct value of the integral truncated to the correct number of decimal places.

8. In this exercise we consider the definite integral

$$\int_0^1 \frac{4}{1+x^2}\,dx$$

from two different perspectives.

a. First, find the Taylor series for $\frac{4}{1+x^2}$ and use it to evaluate

$$\int_0^1 \frac{4}{1+x^2}\,dx$$

as an infinite series of real numbers.

b. Observe that your result in (a) is an alternating series. Estimate the value of that alternating series to within 0.01. How many terms of the series are needed to do so?

c. Recall that $\frac{d}{dx}[\arctan(x)] = \frac{1}{1+x^2}$. Use this fact and the First Fundamental Theorem of Calculus to evaluate

$$\int_0^1 \frac{4}{1+x^2}\,dx$$

exactly.

d. How are your results in (a) and (c) connected? What famous number can we now estimate using an alternating series? How many terms of the series were needed to ensure the first two digits of the famous number are accurate?

APPENDIX A

A Short Table of Integrals

a. $$\int \frac{du}{a^2+u^2} = \frac{1}{a}\arctan\left(\frac{u}{a}\right) + C$$

b. $$\int \frac{du}{\sqrt{u^2 \pm a^2}} = \ln\left|u + \sqrt{u^2 \pm a^2}\right| + C$$

c. $$\int \sqrt{u^2 \pm a^2}\, du = \frac{u}{2}\sqrt{u^2 \pm a^2} \pm \frac{a^2}{2}\ln\left|u + \sqrt{u^2 \pm a^2}\right| + C$$

d. $$\int \frac{u^2 du}{\sqrt{u^2 \pm a^2}} = \frac{u}{2}\sqrt{u^2 \pm a^2} \mp \frac{a^2}{2}\ln\left|u + \sqrt{u^2 \pm a^2}\right| + C$$

e. $$\int \frac{du}{u\sqrt{u^2+a^2}} = -\frac{1}{a}\ln\left|\frac{a+\sqrt{u^2+a^2}}{u}\right| + C$$

f. $$\int \frac{du}{u\sqrt{u^2-a^2}} = \frac{1}{a}\operatorname{arcsec}\left(\frac{u}{a}\right) + C$$

g. $$\int \frac{du}{\sqrt{a^2-u^2}} = \arcsin\left(\frac{u}{a}\right) + C$$

h. $$\int \sqrt{a^2-u^2}\, du = \frac{u}{2}\sqrt{a^2-u^2} + \frac{a^2}{2}\arcsin\left(\frac{u}{a}\right) + C$$

i. $$\int \frac{u^2}{\sqrt{a^2-u^2}}\, du = -\frac{u}{2}\sqrt{a^2-u^2} + \frac{a^2}{2}\arcsin\left(\frac{u}{a}\right) + C$$

j. $$\int \frac{du}{u\sqrt{a^2-u^2}} = -\frac{1}{a}\ln\left|\frac{a+\sqrt{a^2-u^2}}{u}\right| + C$$

k. $$\int \frac{du}{u^2\sqrt{a^2-u^2}} = -\frac{\sqrt{a^2-u^2}}{a^2 u} + C$$

Index

Colophon

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